

Lectures on Symplectic Geometry

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1 Introduction

Hamiltonian systems appear in conservative problems of mechanics as in celestial mechanics but also in statistical mechanics governing the motion of particles and molecules in fluid. A mechanical system of N planets (particles) is modeled by a Hamiltonian function $H(x)$ where $x = (q, p)$, $q = (q_1, \dots, q_N)$, $p = (p_1, \dots, p_N)$ with $(q_i, p_i) \in \mathbb{R}^d \times \mathbb{R}^d$ being the position and the momentum of the i -th particle. The Hamiltonian's equations of motion are

$$(1.1) \quad \dot{q} = H_p(q, p), \quad \dot{p} = -H_q(q, p),$$

which is of the form

$$(1.2) \quad \dot{x} = \bar{J} \nabla_x H(x), \quad \bar{J} = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix},$$

where $n = dN$ and I_n denotes the $n \times n$ identity matrix. The equation (1.2) is an ODE that possesses a unique solution for every initial data x_0 provided that we make some standard assumptions on H . If we denote such a solution by $\phi_t(x_0) = \phi(t, x_0)$, then ϕ enjoys the group property

$$\phi_t \circ \phi_s = \phi_{t+s}, \quad t, s \in \mathbb{R}.$$

The ODE (1.1) is a system of $2n = 2dN$ unknowns. Such a typically large system can not be solved explicitly. A reduction of such a system is desirable and this can be achieved if we can find some conservation laws associated with our system. To find such conservation laws systematically, let us look at a general ODE of the form

$$(1.3) \quad \frac{dx}{dt} = b(x),$$

with the corresponding flow denoted by ϕ_t , and study $u(x, t) = T_t f(x) = f(\phi_t(x))$. A celebrated theorem of Liouville asserts that the function u satisfies

$$(1.4) \quad \frac{\partial u}{\partial t} = \mathcal{L}u = b \cdot u_x.$$

Recall that a function $f(x)$ is *conserved* if $\frac{d}{dt} f(\phi_t(x)) = 0$. From (1.4) we learn that a function f is conserved if and only if

$$(1.5) \quad b \cdot f_x = 0.$$

In the case of a Hamiltonian system, $b = (H_p, -H_q)$ and the equation (1.5) becomes

$$(1.6) \quad \{f, H\} := f_q \cdot H_p - f_p \cdot H_q = 0.$$

As an obvious choice, we may take $f = H$ in (1.6). In general, we may have other conservation laws that are not so obvious to be found. *Noether's principle* tells us how to find a conservation law using a symmetry of the ODE (1.3). With the aid of the symmetries, we may reduce our system to a simpler one that happens to be another Hamiltonian-type system.

Liouville discovered that for a Hamiltonian system of Nd -degrees of freedom ($2Nd$ unknowns) we only need Nd conserved functions in order to solve the system completely by means of quadratures. Such a system is called *completely integrable* and unfortunately hard to come by. Recently there has been a revival of the theory of completely integrable systems because of several infinite dimensional examples (Korteweg–deVries equation, nonlinear Schrödinger equation, etc.).

As we mentioned before, the conservation laws can be used to simplify a Hamiltonian system by reducing its size. To get more information about the solution trajectories, we may search for other conserved quantities. For example, imagine that we have a flow ϕ_t associated with (1.3) and we may wonder how the volume of $\phi_t(A)$ changes with time for a given measurable set A . For this, imagine that there exists a density function $\rho(x, t)$ such that

$$(1.7) \quad \int g(\phi_t(x))\rho^0(x)dx = \int g(x)\rho(x, t)dx,$$

for every bounded continuous function g . This is equivalent to saying that for every nice set A ,

$$(1.8) \quad \int_{\phi_{-t}(A)} \rho(x, 0)dx = \int_A \rho(x, t)dx,$$

where $\rho^0(x) = \rho(x, 0)$. In words, the ρ^0 -weighted volume of $\phi_{-t}(A)$ is given by the $\rho(\cdot, t)$ -weighted volume of A . Using (1.5), it is not hard to see that in fact ρ satisfies the (dual) Liouville's equation

$$(1.9) \quad \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho b) = 0.$$

As a result, the measure $\rho^0(x)dx$ is invariant for the flow ϕ_t if and only if

$$\operatorname{div}(\rho^0 b) = 0.$$

In particular, if $\operatorname{div} b = 0$, then the Lebesgue measure is invariant. In the case of a Hamiltonian system $b = J\nabla H$, we do have $\operatorname{div} b = 0$, and as a consequence,

$$(1.10) \quad \operatorname{vol}(\phi_t(A)) = \operatorname{vol}(A),$$

for every measurable set A .

In our search for other invariance properties, let us now look for vector fields $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$(1.11) \quad \frac{d}{dt} \int_{\phi_t(\gamma)} F \cdot dx \equiv 0,$$

for every closed curve γ . Such an invariance property is of interest in (for example) fluid mechanics because $\int_{\gamma} F \cdot dx$ measures the *circulation* of the velocity field F around γ . To calculate the left-hand side of (1.11), observe

$$\int_{\phi_t(\gamma)} F \cdot dx = \int_{\gamma} F(\phi_t(x)) D\phi_t(x) \cdot dx$$

where $D\phi_t$ denotes the derivative of ϕ_t in x and we regard F as a row vector. Set $F = (F^1, \dots, F^k)$, $\phi = (\phi^1, \dots, \phi^k)$, $u = (u^1, \dots, u^k)$, $u(x, t) = T_t F(x) = F \circ \phi_t(x) D\phi_t(x)$, so that

$$u^j(x, t) = \sum_i F^i(\phi_t(x)) \frac{\partial \phi_t^i}{\partial x_j}(x).$$

To calculate the time derivative, we write

$$\begin{aligned} u(x, t+h) &= F(\phi_{t+h}) D\phi_{t+h} = F(\phi_t \circ \phi_h) D\phi_t \circ \phi_h D\phi_h \\ &= u(\phi_h(x), t) D\phi_h(x), \end{aligned}$$

so that

$$\begin{aligned} \left. \frac{d}{dh} u^j(x, t+h) \right|_{h=0} &= \sum_i \left[(\nabla u^i \cdot b) \delta_{ij} + u^i \frac{\partial b^i}{\partial x_j} \right] \\ &= \nabla u^j \cdot b + \sum_i u^i \frac{\partial b^i}{\partial x_j} \\ &= (u \cdot b)_{x_j} + \sum_i (u_{x_i}^j - u_{x_j}^i) b^i. \end{aligned}$$

In summary,

$$(1.12) \quad \frac{\partial u}{\partial t} = \nabla(u \cdot b) + \mathcal{C}(u)b,$$

where $\mathcal{C}(u)$ is the matrix $[u_{x_j}^i - u_{x_i}^j]$. In particular,

$$(1.13) \quad \frac{d}{dt} \int_{\phi_t(\gamma)} F \cdot dx = \int_{\gamma} \mathcal{C}(u)b \cdot dx,$$

for every closed curve γ . Recall that we would like to find vector fields F for which (1.11) is valid. For this it suffices to have $\mathcal{C}(F)b$ a gradient. Indeed if $\mathcal{C}(F)b$ is a gradient, then

$$\begin{aligned} \frac{d}{dt} \int_{\phi_t(\gamma)} F \cdot dx &= \frac{d}{dh} \int_{\phi_{t+h}(\gamma)} F \cdot dx \Big|_{h=0} \\ &= \frac{d}{dh} \int_{\phi_h(\phi_t(\gamma))} F \cdot dx \Big|_{h=0} \\ &= \int_{\phi_t(\gamma)} \mathcal{C}(F)b \cdot dx = 0. \end{aligned}$$

Let us examine this for some examples.

Example 1.1. (i) Assume that $k = 2n$ with $x = (q, p) = (q_1, \dots, q_n, p_1, \dots, p_n)$. Let $b(q, p) = (H_p, -H_q)^* = J\nabla H$ for a Hamiltonian $H(q, p)$. Choose $F(q, p) = (p, 0)$. We then have $\mathcal{C}(F) = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix} = J$, and $\mathcal{C}(F)b = JJ\nabla H = -\nabla H$. This and (1.13) imply that for a Hamiltonian flow ϕ_t and closed γ ,

$$(1.14) \quad \frac{d}{dt} \int_{\phi_t(\gamma)} p \cdot dq = 0,$$

which was discovered by Poincaré originally.

(ii) Assume $k = 2n + 1$ with $x = (q, p, t)$ and $b(q, p, t) = (H_p, -H_q, 1)^*$ where H is now a time-dependent Hamiltonian function. Define

$$F(q, p, t) = (p, 0, -H(q, p, t)).$$

We then have

$$(1.15) \quad \mathcal{C}(F) = \left[\begin{array}{c|c|c} 0 & I_n & H_q^* \\ \hline -I_n & 0 & H_p^* \\ \hline -H_q & -H_p & 0 \end{array} \right] = \left[\begin{array}{c|c} J & (\nabla H)^* \\ \hline -\nabla H & 0 \end{array} \right].$$

Since $\mathcal{C}(F)b = 0$, we deduce that for any closed (q, p, t) -curve γ ,

$$(1.14) \quad \frac{d}{ds} \int_{\phi_s(\gamma)} (p \cdot dq - H(q, p, t)dt) = 0,$$

proving a result of *Poincaré* and *Cartan*. Note that if γ has no t -component in (1.16), then (1.16) becomes (1.14).

(iii) Assume $n = 3$. Then $\mathcal{C}(F) = \begin{bmatrix} 0 & -\alpha_3 & \alpha_2 \\ \alpha_3 & 0 & -\alpha_1 \\ -\alpha_2 & \alpha_1 & 0 \end{bmatrix}$ with $(\alpha_1, \alpha_2, \alpha_3) = \nabla \times F$. Now if $b = \nabla \times F$, then $\mathcal{C}(F)b = (\nabla \times F) \times b$ and $\frac{d}{dt} \int_{\phi_t(\gamma)} F \cdot dx = 0$. In words, the F -circulation of a curve moving with velocity field $\nabla \times F$ is preserved with time. $\square$

Example 1.2. A Hamiltonian system (1.2) simplifies if we can find a function $w(q, t)$ such that $p(t) = w(q(t), t)$. If such a function w exists, then $q(t)$ solves

$$(1.17) \quad \frac{dq}{dt} = H_p(q, w(q, t), t).$$

The equation $\dot{p}$ gives us the necessary condition for the function w :

$$\begin{aligned} \dot{p} &= w_q \dot{q} + w_t = w_q \cdot H_p(q, w, t) + w_t, \\ \dot{p} &= -H_q(q, w, t). \end{aligned}$$

Hence $w(q, t)$ must solve,

$$(1.18) \quad w_t + w_q \cdot H_p(q, w, t) + H_q(q, w, t) = 0.$$

For example, if $H(q, p, t) = \frac{1}{2}|p|^2 + V(q, t)$, then (1.18) becomes

$$(1.19) \quad w_t + w_q w + V_q(q, t) = 0.$$

The equation (1.17) simplifies to

$$(1.20) \quad \frac{dq}{dt} = w(q, t)$$

in this case. If the flow of (1.20) is denoted by ψ_t , then $\phi_t(q, w(q, 0)) = (\psi_t(q), w(\psi_t(q), t))$. Now (1.14) means that for any closed q -curve η ,

$$\frac{d}{dt} \int_{\psi_t(\eta)} w(q, t) \cdot dq = 0.$$

This is the celebrated *Kelvin's circulation theorem*. $\square$

We may use Stokes' theorem to rewrite (1.14) as

$$(1.21) \quad \frac{d}{dt} \int_{\phi_t(\Gamma)} \bar{\omega} := \frac{d}{dt} \int_{\phi_t(\Gamma)} dp \wedge dq = 0$$

for every two-dimensional surface Γ . In words, the 2-form $\bar{\omega}$ is invariant under the Hamiltonian flow ϕ_t . In summary, we have found various invariance principles for Hamiltonian flows:

- The conserved functions f satisfying (1.6) is an example of an invariance principle for 0-forms.
- The Liouville's theorem (1.10) is an example of an invariance principle of an n -form.
- Poincaré's theorem (1.21) is an instance of an invariance principle involving a 2-form.

In fact (1.21) implies (1.10) because the invariance of $\bar{\omega}$ implies the invariance of the $k = 2n$ form $\bar{\omega}^n = \bar{\omega} \wedge \cdots \wedge \bar{\omega}$ which is a constant multiple of the volume form. More generally, we may take an arbitrary l -form ω and evolve it by the flow ϕ_t of a velocity field b . If we write $\omega(t)$ for $\phi_t^* \omega$:

$$\int_{\Gamma} \omega(t) = \int_{\Gamma} \phi_t^* \omega = \int_{\phi_t(\Gamma)} \omega,$$

then by a formula of Cartan,

$$\frac{d\omega}{dt} = \mathcal{L}_b \omega := d(i_b \omega) + i_b(d\omega)$$

where $\mathcal{L}_b \omega$ denotes the *Lie derivative*.

The configuration space of a system with constraints is a manifold. Also, when we use conservation laws to reduce our Hamiltonian system, we obtain a Hamiltonian system on a manifold. If the configuration space is a n -dimensional differentiable manifold N , and $L : TN \rightarrow \mathbb{R}$ is a differentiable *Lagrangian function*, then $p = \frac{\partial L}{\partial \dot{q}}$ is a cotangent vector. The cotangent bundle $M = T^*N$ is an example of a *symplectic manifold* because it possesses a natural closed non-degenerate form $\bar{\omega}$ which is simply $\sum_1^n dp_i \wedge dq_i$, in local coordinates. More generally we may study an even dimensional manifold M , equipped with a non-degenerate closed 2-form ω , and construct vector fields X_H associated with scalar functions H such that $i_{X_H}(\omega) = -dH$. The vector field X_H is the analog of $J\nabla H$ in the Euclidean case $M = \mathbb{R}^{2n}$. By the non-degeneracy of ω , such X_H exists for every differentiable Hamiltonian function H .

A celebrated theorem of Darboux asserts that any symplectic manifold is locally equivalent to an Euclidean space with its standard symplectic structure. As a result, the most important questions in symplectic geometry are the global ones.

Consider the Euclidean space $(\mathbb{R}^{2n}, \bar{\omega})$. If the hypersurface $\Gamma = H^{-1}(c)$ is a compact energy level set with $\nabla H \neq 0$ on Γ , then the unparametrized orbits on Γ of the Hamiltonian vector field $X_H = J\nabla H$ are independent of the choice of H . One can therefore wonder what hypersurfaces carry a periodic orbit. P. Rabinowitz showed that every star-like hypersurface carries a periodic orbit. Later, Viterbo showed that the same holds more generally for hypersurfaces of *contact type*, establishing affirmatively a conjecture of A. Weinstein.

Consider two compact connected domains U_1 and U_2 in $\mathbb{R}^{2n}$ with smooth boundaries. If U_1 and U_2 are diffeomorphic and $\text{volume}(U_1) = \text{volume}(U_2)$, then we can find a diffeomorphism

between U_1 and U_2 that is also volume preserving (Dacorogna–Moser). We may wonder whether or not there exists a symplectic diffeomorphism between U_1 and U_2 . *Gromov’s squeezing theorem* shows that the symplectic transformations are more rigid; if there exists a symplectic embedding from the ball

$$B_R(0) = \{(q, p) : |q|^2 + |p|^2 < R^2\}$$

into the cylinder

$$Z_r(0) = \{(q, p) : q_1^2 + p_1^2 < r^2\},$$

then we must have $r \geq R$! Motivated by this, Gromov defines the *symplectic radius* $r(M)$ of a symplectic manifold (M, ω) as the largest r for which there exists a symplectic embedding from $B_r(0)$ into M . The Gromov’s radius is an example of a *symplectic capacity* that is a symplectic invariant. Since the discovery of the Gromov radius, new capacities have been discovered. The existence of some of these capacities can be used to prove various global properties of Hamiltonian systems such as Viterbo’s existence of periodic orbits.

Another rigidity of symplectic transformation is illustrated in an important result of Eliashberg and Gromov: If $\{f_m\}$ is a sequence of symplectic transformation that converges uniformly to a differentiable function f , then f is also symplectic. The striking aspect of this result is that our definition of a symplectic function f involves the first derivative of f . As a result, we should expect to have a definition of symplecticity that does not involve any derivative. This should be compared to the definition of a volume preserving transformation that can be formulated with or without using derivative.

2 Quadratic Hamiltonians and Linear Symplectic Geometry

In this section, we discuss several central concepts and fundamental results of symplectic geometry in linear setting. More specifically, we establish Darboux Theorem for *symplectic vector spaces*, define *symplectic spectrum* for quadratic Hamiltonian functions, construct *linear symplectic capacities* for ellipsoids, establish *symplectic rigidity* for linear symplectic maps, and analyze *complex structures* that are compatible with a symplectic form.

A symplectic vector space (V, ω) is a pair of finite dimensional real vector space V and a bilinear form $\omega : V \times V \rightarrow \mathbb{R}$ which is antisymmetric and non-degenerate. That is, $\omega(a, b) = -\omega(b, a)$ for all $a, b \in V$, and that $\forall a \in V$ with $a \neq 0$, $\exists b \in V$ such that $\omega(a, b) \neq 0$. The non-degeneracy is equivalent to saying that the transformation $a \mapsto \omega(a, \cdot)$ is a linear isomorphism between V and its dual V^* . Clearly $(\mathbb{R}^k, \bar{\omega})$ is an example of a symplectic vector space when $k = 2n$ and $\omega(a, b) = \bar{J}a \cdot b$, where $\bar{J}$ was defined in (1.2). More generally, given a k by k matrix C , the bilinear form $\omega(a, b) = Ca \cdot b$ is symplectic if C is invertible and skew-symmetric. Note that since

$$\det C = \det C^* = \det(-C) = (-1)^n \det C,$$

necessarily $k = 2n$ is even. Observe that the eigenvalues of C are of the form $\pm i\lambda_j$, with $\lambda_1 \leq \dots \leq \lambda_k$. Moreover, we can find eigenvectors of the form $a_j \pm ib_j : j = 1, \dots, k$, such that

$$Ca_j = -\lambda_j b_j, \quad Cb_j = \lambda_j a_j, \quad a_j \cdot a_\ell = b_j \cdot b_\ell = \delta_{j\ell}, \quad a_j \cdot b_\ell = 0,$$

for all $j, \ell \in \{1, \dots, k\}$. As a consequence,

$$\omega(a_j, a_\ell) = \omega(b_j, b_\ell) = 0, \quad \omega(b_j, a_\ell) = \lambda_j \delta_{j\ell}.$$

Given a symplectic (V, ω) , then we say a and b are ω -orthogonal and write $a \amalg b$ if $\omega(a, b) = 0$. If W is a linear subspace of V , then

$$W^\amalg = \{a \in V : a \amalg W\}.$$

In our first result we state a linear version of Darboux's theorem and some elementary facts about symplectic vector spaces. Darboux's theorem in Euclidean setting asserts that for every invertible skew-symmetric matrix C we can find an invertible matrix T such that $C = T^* \bar{J} T$.

Proposition 2.1 *Let (V, ω) be a symplectic linear space of dimension $k = 2n$ and W be a subspace of V .*

- (i) $\dim W + \dim W^\Pi = \dim V$.
- (ii) $(W^\Pi)^\Pi = W$.
- (iii) (W, ω) is symplectic iff $W \oplus W^\Pi = V$.
- (iv) If W is a symplectic subspace, then W^Π is also symplectic.
- (v) There exist a linear invertible $T : V \rightarrow \mathbb{R}^k$ such that $\omega(a, b) = \bar{\omega}(Ta, Tb)$. Equivalently, there exists a basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$ such that $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$ and $\omega(f_i, e_j) = \delta_{ij}$.

Proof (i) Assume that $\dim V = k$ and $\dim W = m$. Choose a basis $\{a_1, \dots, a_m\}$ for W . Then by non-degeneracy $a_1^*, \dots, a_m^*$ are independent where $a_j^*(b) = \omega(a_j, b)$. Since $W^\Pi = \{a : a_j^*(a) = 0 \text{ for } j = 1, \dots, m\}$, with a_j^* independent, we have that $\dim W^\Pi = n - m$.

(ii) Evidently $W \subseteq (W^\Pi)^\Pi$. Since $\dim W + \dim W^\Pi = \dim W^\Pi + \dim (W^\Pi)^\Pi$, we deduce that $W = (W^\Pi)^\Pi$.

(iii) By definition, (W, ω) is symplectic iff $W \cap W^\Pi = \{0\}$. Since $\dim W + \dim W^\Pi = \dim V$, we have that $W \oplus W^\Pi = V$.

(iv) If W is symplectic, then $V = W \oplus W^\Pi = (W^\Pi)^\Pi \oplus W^\Pi$, which implies that in fact W^Π is symplectic.

(v) Inductively, we construct a collection $\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\}$, with $\ell \leq n$, such that

$$(2.1) \quad \omega(e_i, e_j) = \omega(f_i, f_j) = 0, \quad \omega(f_j, e_i) = \delta_{ij}.$$

We start from $\ell = 1$ (assuming $\dim V \geq 2$). Let e_1 be a non-zero vector of V . Since ω is non-degenerate, we can find $f_1 \in V$ such that $\omega(f_1, e_1) = 1$. Clearly f_1, e_1 are linearly independent, and (2.1) holds for $\ell = 1$.

Assume that (2.1) holds for some $\ell < n$. Set

$$W_\ell = \text{span}\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\}.$$

Clearly, W_ℓ is symplectic, and by parts (iii) and (iv), W^Π is also symplectic, and

$$V = W_\ell \oplus W_\ell^\Pi.$$

Using the case $\ell = 1$, we can find $e_{\ell+1}, f_{\ell+1} \in W_\ell^\Pi$, such that $\omega(f_{\ell+1}, e_{\ell+1}) = 1$. Clearly the collection

$$\{e_1, \dots, e_\ell, f_1, \dots, f_\ell\},$$

satisfies (2.1) where ℓ is replaced with $\ell + 1$. We continue this until we reach the index n . $\square$

We now turn our attention to quadratic Hamiltonian functions and ellipsoids. By a quadratic Hamiltonian we mean a function $H(x) = \frac{1}{2}Bx \cdot x$ for a symmetric matrix B . We are particularly interested in the case $B \geq 0$. We note that for such quadratic Hamiltonians, the corresponding Hamiltonian vector field $X(x) = \bar{J}Bx$ is linear. Since the flow of

$$(2.2) \quad \dot{x} = \bar{J}Bx,$$

preserves H , we also study the level sets of nonnegative quadratic functions. By an *ellipsoid* we mean a set E of the form

$$E = \{x : H(x) \leq 1\}$$

where $H(x) = \frac{1}{2}Bx \cdot x$ with $B \geq 0$. Note that if $B > 0$, the ellipsoid E is a bounded set. Otherwise, the set E is unbounded and may be also called a *cylinder* or *cylindrical ellipsoid*. Our goal is to show that we can make a change of coordinates to turn the ODE to a simpler Hamiltonian system for which B is a diagonal matrix. Before embarking on this, let us first review some well-known facts about symmetric matrices, which is the symmetric counterpart of what we will discuss for symplectic matrices.

To begin, let us recall that the standard Euclidean inner product is preserved by a matrix A if A is orthogonal. That is

$$Aa \cdot Ab = a \cdot b \quad \text{for all } a, b \in \mathbb{R}^k \Leftrightarrow A^{-1} = A^*.$$

Let us write $O(k)$ for the space of $k \times k$ orthogonal matrices. We also write $S(k)$ for the space of symmetric matrices. A quadratic function $H : \mathbb{R}^k \rightarrow \mathbb{R}$ is defined by $H(x) = \frac{1}{2}Bx \cdot x$ with $B \in S(k)$.

Proposition 2.2 *Let H_1 and H_2 be two quadratic functions associated with the symmetric matrices B_1 and B_2 . Then there exists $A \in O(k)$ such that $H_1 \circ A = H_2$ if and only if B_1 and B_2 have the same spectrum.*

As a consequence, if $H(x) = \frac{1}{2}Bx \cdot x$ and B has eigenvalues $\lambda(H) = (\lambda_1, \dots, \lambda_k)$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$, then there exists $A \in O(k)$ such that $H(Ax) = \frac{1}{2} \sum_{j=1}^k \lambda_j x_j^2$. In particular, if $B \geq 0$, then λ_j 's are nonnegative and we may define *radii* $\mathbf{R}(H) = (R_1(H), \dots, R_k(H)) \in (0, \infty]^k$ by $R_i^2 = R_i^2(H) = \frac{2}{\lambda_i}$ so that $0 < R_1(H) \leq R_2(H) \leq \dots \leq R_k(H)$ and

$$H(A(x)) = \sum_{j=1}^k \frac{x_j^2}{R_j^2}.$$

If E is the corresponding ellipsoid,

$$E = \{x : H(x) \leq 1\},$$

then we write

$$\mathbf{R}(E) = (R_1(E), \dots, R_k(E))$$

for $\mathbf{R}(H)$ and refer to its coordinates as the radii of E . We now rephrase Proposition 2.1 as

Corollary 2.1 *Let E_1 and E_2 be two ellipsoids. Then there exists $A \in O(k)$ such that $A(E_1) = E_2$ if and only if $\mathbf{R}(E_1) = \mathbf{R}(E_2)$.*

We next discuss the monotonicity of $\mathbf{R}$.

Proposition 2.3 (i) *Let H_1 and H_2 be two quadratic functions. Then $H_1 \circ A \leq H_2$ for some $A \in O(k)$ if and only if $\lambda(H_1) \leq \lambda(H_2)$.*

(ii) *Let E_1 and E_2 be two ellipsoids. Then $A(E_2) \subseteq E_1$ for some $A \in O(k)$ if and only if $\mathbf{R}(E_2) \leq \mathbf{R}(E_1)$.*

Proof We note that (i) implies (ii) because if $E_r = \{x : H_r(x) \leq 1\}$ for $r = 1$ and 2, then

$$E_2 \subseteq A^{-1}E_1 \Leftrightarrow H_1 \circ A \leq H_2.$$

As for the proof of (i), observe that if $\lambda = \lambda(H_1) \leq \lambda' = \lambda(H_2)$, then we can find A_1 and $A_2 \in O(k)$ such that

$$H_1(A_1x) = \frac{1}{2} \sum_{j=1}^k \lambda_j x_j^2 \leq \frac{1}{2} \sum_{j=1}^k \lambda'_j x_j^2 = H_2(A_2x),$$

proving the “if” part of (i). The “only if” is an immediate consequence of Courant–Hilbert Minimax Principle that will be stated in Lemma 2.1 below. $\square$

Lemma 2.1 (Courant–Hilbert) *Let $B \in S(k)$ with eigenvalues $\mu_1 \leq \mu_2 \leq \dots \leq \mu_k$. Then*

$$(2.3) \quad \mu_j = \inf_{\dim V=j} \sup_{x \in V \setminus \{0\}} \frac{Bx \cdot x}{|x|^2},$$

$$(2.4) \quad \mu_j = \sup_{\dim V=j-1} \inf_{x \in V^\perp \setminus \{0\}} \frac{Bx \cdot x}{|x|^2},$$

where V denotes a linear subspace of $\mathbb{R}^k$.

Proof Let us write $\hat{\mu}_j$ for the right-hand side of (2.3). Let $u_1, u_2, \dots, u_k$ be an orthonormal basis with $Bu_j = \mu_j u_j$, $j = 1, \dots, k$. Note that

$$\sup \left\{ \frac{Bx \cdot x}{|x|^2} : x \in \text{span}\{u_1, \dots, u_j\}, x \neq 0 \right\} = \sup_{c_1, \dots, c_j} \frac{\sum_1^j \mu_l c_l^2}{\sum_1^j c_l^2} \leq \mu_j,$$

proving $\hat{\mu}_j \leq \mu_j$. For $\hat{\mu}_j \geq \mu_j$, pick a linear subspace V of dimension j and choose non-zero $x \in V$ such that $x \perp u_1, \dots, u_{j-1}$. Such x exists because $\dim V = j$ and we are imposing $j - 1$ many conditions. Since we can write $x = \sum_{l=j}^k c_l u_l$, we have

$$(2.5) \quad \frac{Bx \cdot x}{|x|^2} = \frac{\sum_j^k \mu_l c_l^2}{\sum_j^k c_l^2} \geq \mu_j.$$

As a result, $\hat{\mu}_j \geq \mu_j$ and this completes the proof of (2.3).

As for (2.4), note that if $x \perp u_1, \dots, u_{j-1}$, $x \neq 0$, then $\frac{Bx \cdot x}{|x|^2} \geq \mu_j$ by (2.5). Hence, if $\bar{\mu}_j$ denotes the right-hand side of (2.3), then $\bar{\mu}_j \geq \mu_j$ by choosing $V = \text{span}\{u_1, \dots, u_{j-1}\}$. For $\mu_j \geq \bar{\mu}_j$, let V be any linear space of dimension $j - 1$ and pick a non-zero $x \in \text{span}\{u_1, \dots, u_j\} \cap V^\perp$. For such a vector x we have $x = \sum_1^j c_l u_l$, $(c_1, \dots, c_j) \neq 0$, and $\frac{Bx \cdot x}{|x|^2} \leq \mu_j$. This implies that $\mu_j \geq \bar{\mu}_j$. $\square$

We would like to develop a theory similar to what we have seen in this section but now for the bilinear form $\bar{\omega}$. The following table summarizes our main results:

	Symmetric	Antisymmetric
Form	$a \cdot b$	$\bar{\omega}(a, b) = Ja \cdot b$
Invariant matrix	$A \in O(k) : A^{-1} = A^*$	$T \in Sp(n) : T^{-1} = -\bar{J}T^*\bar{J}$
Vector field	$\nabla H(x) = Bx, B \in S(k)$	$\bar{J}\nabla H(x) = \bar{J}Bx; \bar{J}B \in \text{Ham}(n)$
Spectral theorem	Proposition 2.2	Weierstrauss Theorem (Theorem 2.1)
Monotonicity	Courant–Hilbert Minimax (Lemma 2.1)	Theorem 2.2, Lemma 2.2

We say a matrix T is *symplectic* if $\bar{\omega}(Ta, Tb) = \bar{\omega}(a, b)$. Equivalently $T^*\bar{J}T = \bar{J}$ or $T^{-1} = -\bar{J}T^*\bar{J}$. The set of $2n \times 2n$ symplectic matrices is denoted by $Sp(n)$. We say a matrix C is Hamiltonian if $C = \bar{J}B$ for a symmetric matrix B . The space of $2n \times 2n$ Hamiltonian matrices is denoted by $\text{Ham}(n)$. We have

$$C \in \text{Ham}(n) \Leftrightarrow \bar{J}C + C^*\bar{J} = 0 \Leftrightarrow C^* = \bar{J}C\bar{J}.$$

We note that if $H(x) = \frac{1}{2}Bx \cdot x$ with $B \in S(2n)$, then $\bar{J}\nabla H(x) = \bar{J}Bx$ with $\bar{J}B \in \text{Ham}(n)$.

We are now ready to state Weierstrass Theorem which allows us to diagonalize a Hamiltonian matrix using a symplectic change of variable.

Theorem 2.1 *Let B be a positive matrix. Then the matrix $\hat{B} = \bar{J}B$ has purely imaginary eigenvalues of the form $\pm i\lambda_1, \dots, \pm i\lambda_n$ with $\lambda_1 \geq \dots \geq \lambda_n > 0$. Moreover there exists $T \in Sp(n)$ such that the quadratic Hamiltonian function $H(x) = \frac{1}{2}Bx \cdot x$ can be represented as*

$$H \circ T(x) = \sum_{j=1}^n \frac{\lambda_j}{2} (q_j^2 + p_j^2),$$

where $x = (q_1, p_1, \dots, q_n, p_n)$.

Proof (Step 1) Let $\mu \pm i\lambda$ be an eigenvalue of $\hat{B}$ associated with the (non-zero) eigenvector $a \pm ib$ (withoust loss of generality, we may assume that $\lambda \geq 0$). As a result,

$$\hat{B}a = \mu a - \lambda b, \quad \hat{B}b = \lambda a + \mu b \quad \text{or} \quad Ba = \lambda Jb - \mu Ja, \quad Bb = -\lambda Ja - \mu Jb.$$

Hence

$$Ba \cdot b = \mu \bar{\omega}(b, a), \quad Bb \cdot a = -\mu \bar{\omega}(b, a), \quad Ba \cdot a = Bb \cdot b = \lambda \bar{\omega}(b, a).$$

From this, $B = B^*$, $\mu + i\lambda \neq 0$, and $a + ib \neq 0$, we deduce that $\mu = 0$, $\bar{\omega}(b, a) \neq 0$, and $Ba \cdot b = 0$. Since by assumption $\lambda > 0$, we must have $\bar{\omega}(b, a) > 0$. As a result, for every eigenvalue $i\lambda$, we can find an eigenvector $a + ib$, such that

$$(2.6) \quad \begin{aligned} Ba \cdot a = Bb \cdot b = \lambda, \quad Ba \cdot b = 0, \quad \bar{\omega}(b, a) = 1, \\ \hat{B}a = -\lambda b, \quad \hat{B}b = \lambda a. \end{aligned}$$

(Step 2) Let λ_1 be the largest number such that $\pm i\lambda_1$ is an eigenvalue of $\hat{B}$. By Step 1, we can find (2.6) find a_1 and b_1 such that

$$Ba_1 \cdot a_1 = Bb_1 \cdot b_1 = \lambda_1, \quad Ba_1 \cdot b_1 = 0, \quad \hat{B}a = -\lambda_1 b, \quad \hat{B}b = \lambda_1 a.$$

As a result, if $V_1 = \text{span}\{a_1, b_1\}$, then $\hat{B}V_1 \subseteq V_1$, and for q_1 and $p_1 \in \mathbb{R}$,

$$H(q_1 a_1 + p_1 b_1) = q_1^2 H(a_1) + p_1^2 H(b_1) + q_1 p_1 Ba_1 \cdot b_1 = \frac{\lambda_1}{2} (q_1^2 + p_1^2).$$

By Proposition 2.1, the spaces V_1 and V_1^Π are symplectic and $\mathbb{R}^{2n} = V_1 \oplus V_1^\Pi$. We now claim

$$(2.7) \quad a \in V_1, \quad b \in V_1^\Pi \Rightarrow Ba \cdot b = 0, \quad \text{and} \quad \hat{B}V_1^\Pi \subseteq V_1^\Pi.$$

Indeed if $a \in V_1$, $b \in V_1^\Pi$, then $\hat{B}a \in V_1$, and

$$\bar{\omega}(\hat{B}b, a) = \bar{J}\hat{B}b \cdot a = -Bb \cdot a = -b \cdot Ba = -\bar{J}b \cdot \hat{B}a = -\bar{\omega}(b, \hat{B}a) = 0,$$

which proves both claims in (2.7) because we just showed that $\hat{B}b \in V_1^\Pi$.

(*Final Step*) From (2.7) we learn that if $a \in V_1$ and $b \in V_1^\Pi$, then

$$H(a + b) = H(a) + H(b).$$

Let us look at the restriction of H to the symplectic vector space $(V_1^\Pi, \bar{\omega})$, which is a quadratic function. By Proposition 2.1, this pair is isomorphic to $(\mathbb{R}^{2n-2}, \bar{\omega})$. As a result, we may repeat the above argument to assert that there exists a pair of vectors $a_2, b_2 \in V_1^\Pi$ with $\bar{\omega}(b_2, a_2) = 1$, $H(a_2) = H(b_2) = \lambda_2/2$ and $Ba_2 \cdot b_2 = 0$. Continuing this process would yield a basis $(a_1, b_1, a_2, b_2, \dots, a_n, b_n)$ such that

$$\begin{aligned} \bar{\omega}(a_i, a_j) &= \bar{\omega}(b_i, b_j) = 0, & \bar{\omega}(b_i, a_j) &= \delta_{ij}, \\ H(a_j) &= H(b_j) = \lambda_j/2, & Ba_j \cdot b_j &= 0. \end{aligned}$$

From this we learn that the linear map $T : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ defined by

$$T(q_1, p_1, \dots, q_n, p_n) = \sum_1^n q_j a_j + p_j b_j$$

is symplectic, and

$$H(T(x)) = \sum_1^n \frac{\lambda_j}{2} (q_j^2 + p_j^2).$$

□

Given $H(x) = \frac{1}{2}Bx \cdot x$ with $B \geq 0$, let us write $\frac{1}{2}\lambda_j = \frac{1}{r_j^2}$ so that $r_j = r_j(H) \in (0, \infty]$ satisfy

$$0 < r_1(H) \leq r_2(H) \leq \dots \leq r_n(H).$$

We also write $\mathbf{r}(H) = (r_1(H), \dots, r_n(H))$ and if E is the corresponding ellipsoid, we write $\mathbf{r}(E)$ for $\mathbf{r}(H)$. We may rephrase Theorem 2.1 as follows:

- Corollary 2.2** (i) *If H_1 and H_2 are two positive definite quadratic forms, then $\mathbf{r}(H_1) = \mathbf{r}(H_2)$ if and only if $H_2 = H_1 \circ T$ for some $T \in Sp(n)$.*
- (ii) *Let E_1 and E_2 be two ellipsoid. Then $T(E_2) = E_1$ for $T \in Sp(n)$ if and only if $\mathbf{r}(E_1) = \mathbf{r}(E_2)$.*

Example 2.1(i) In Theorem 2.1, we cannot drop the assumption $B > 0$. For example, if

$$B = \begin{bmatrix} \mu_1 & 0 \\ 0 & \mu_2 \end{bmatrix},$$

with $\mu_1\mu_2 < 0$, then the eigenvalues of $\hat{B}$ are $\pm\sqrt{-\mu_1\mu_2}$.

(ii) Let $n = 1$ and $H(q_1, p_1) = \frac{q_1^2}{R_1^2} + \frac{p_1^2}{R_2^2}$ so that $\mathbf{R}(H) = (R_1, R_2)$. Here $H(x) = \frac{1}{2}Bx \cdot x$ with

$$B = \begin{bmatrix} \frac{2}{R_1^2} & 0 \\ 0 & \frac{2}{R_2^2} \end{bmatrix}.$$

We have

$$C = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} B = \begin{bmatrix} 0 & \frac{2}{R_2^2} \\ -\frac{2}{R_1^2} & 0 \end{bmatrix}.$$

The matrix C has eigenvalues $\pm i\frac{2}{R_1R_2}$. Hence $\mathbf{r}(H) = (r_1(H))$ with $r_1(H) = \sqrt{R_1R_2}$ and there exists $T \in Sp(n)$ such that $H \circ T(q_1, p_1) = \frac{q_1^2 + p_1^2}{R_1R_2}$. $\square$

As our next corollary to Theorem 2.1, we solve (2.2) with the aid of a symplectic change of coordinates:

Corollary 2.3 *Let H , T and $(\lambda_1, \dots, \lambda_n)$ be as in Theorem 2.1. Let $x(t)$ be a solution of (2.2) and define $y(t) = T^{-1}x(t)$. Then $\dot{y} = \bar{J}B_0y$, where B_0 is a diagonal matrix that has the entries*

$$\lambda_1, \dots, \lambda_n, \lambda_1, \dots, \lambda_n,$$

on its main diagonal.

Proof From (2.2), we learn that $T\dot{y} = \bar{J}BTy$. Also, by Theorem 2.1 we know that $(BTx) \cdot (Tx) = B_0x \cdot x$, which means that $T^*BT = B_0$. As a result

$$\dot{y} = T^{-1}\bar{J}BTy = -\bar{J}T^*\bar{J}\bar{J}BTy = \bar{J}T^*BTy = \bar{J}B_0y.$$

$\square$

Remark 2.1(i) If we write $\phi_t(y)$ for the flow of $\dot{y} = \bar{J}B_0y$ and use the complex notation $y = (z_1, \dots, z_n)$ with $z_j = q_j + ip_j =: \rho_j e^{i\theta_j}$, then

$$\phi_t(z_1, \dots, z_n) = (e^{-i\lambda_1 t} z_1, \dots, e^{-i\lambda_n t} z_n).$$

What we have is known as a *quasi-periodic* motion. We note that $(\rho_j = |z_j| : j = 1, \dots, n)$ are conserved, and in terms of the angles, we have, $(\theta_1, \dots, \theta_n) \mapsto (\theta_1 - t\lambda_1, \dots, \theta_n - t\lambda_n)$. In particular, for each $z_j \neq 0$, $\lambda_j \neq 0$, if we set $\hat{z}_j$ for the complex vector that has z_j for the j -coordinate and 0 for the other coordinates, then the orbit $(\phi_t(\hat{z}_j) : t \in \mathbb{R})$ is periodic of period $2\pi/\lambda_j = \pi r_j^2$.

The periods $(c_j = \pi r_j(E)^2 : 1 \leq j \leq n)$ are closely related to the symplectic capacities of the ellipsoid E . Indeed, if we form the collection

$$\{mc_j : m \in \mathbb{N}^*, j \in \mathbb{N}\},$$

and rewrite them in nondecreasing order as $\mu_0 = 0 < \mu_1 \leq \mu_2 \leq \dots$ (with repetition), then μ_ℓ equals to $\mathbf{c}_\ell^{EH}(E)$, the ℓ -th *Ekeland-Hofer capacity* of E . $\square$

We now turn to the question of monotonicity.

Theorem 2.2 (i) *If H_1 and H_2 are two non-negative definite quadratic forms, then $\mathbf{r}(H_1) \geq \mathbf{r}(H_2)$ if and only if there exists $T \in Sp(n)$ such that $H_1 \circ T \leq H_2$.*
(ii) *Let E_1 and E_2 be two ellipsoids. Then $\mathbf{r}(E_2) \leq \mathbf{r}(E_1)$ if and only if $T(E_2) \subseteq E_1$ for some $T \in Sp(n)$.*

Proof As before (i) implies (ii). By approximation, it suffices to establish (i) when H_1 and H_2 are positive definite. In this case, (ii) is an immediate consequence of a variational formula we obtain for $r_j(H)$ in Lemma 2.2 below. $\square$

Lemma 2.2 *Let H be a positive definite quadratic function of $\mathbb{R}^{2n}$. Then*

$$(2.8) \quad \frac{1}{2}r_j^2(H) = \inf_{\dim V = 2n+2j} \sup_{[x,y]^* \in V \setminus \{0\}} \frac{\bar{\omega}(x,y)^+}{H(x) + H(y)}.$$

Here V is for linear subspace of $\mathbb{R}^{4n}$.

Proof Recall that $\pm i2r_j^{-2}$ are the eigenvalues of $C = JB$ where $H(x) = \frac{1}{2}Bx \cdot x$. Hence $\pm \frac{i}{2}r_j^2$ are the eigenvalues of C^{-1} . If $a_j + ib_j$ denotes the corresponding eigenvector, then $C^{-1}(a_j + ib_j) = \frac{i}{2}r_j^2(a_j + ib_j)$. This means

$$(2.9) \quad C^{-1}a_j = -\frac{r_j^2}{2}b_j, \quad C^{-1}b_j = \frac{r_j^2}{2}a_j.$$

This suggests looking at the $4n \times 4n$ matrix

$$D = \begin{bmatrix} 0 & C^{-1} \\ -C^{-1} & 0 \end{bmatrix}.$$

From (2.9) we readily deduce

$$\begin{aligned} D \begin{bmatrix} a_j \\ b_j \end{bmatrix} &= \frac{r_j^2}{2} \begin{bmatrix} a_j \\ b_j \end{bmatrix}, \quad D \begin{bmatrix} b_j \\ -a_j \end{bmatrix} = \frac{r_j^2}{2} \begin{bmatrix} b_j \\ -a_j \end{bmatrix}, \\ D \begin{bmatrix} b_j \\ a_j \end{bmatrix} &= -\frac{r_j^2}{2} \begin{bmatrix} b_j \\ a_j \end{bmatrix}, \quad D \begin{bmatrix} -a_j \\ b_j \end{bmatrix} = -\frac{r_j^2}{2} \begin{bmatrix} -a_j \\ b_j \end{bmatrix}. \end{aligned}$$

Note that since $a_j + ib_j \neq 0$, the vectors $\begin{bmatrix} a_j \\ b_j \end{bmatrix}, \begin{bmatrix} b_j \\ -a_j \end{bmatrix}$ are linearly independent. Hence $\pm \frac{i}{2} r_j^2$ produces eigenvalue $\pm \frac{r_j^2}{2}$ of multiplicity 2 for D . We would like to apply Courant–Hilbert minimax principle to D , except that D is not symmetric with respect to the dot product of $\mathbb{R}^{4n}$. However if we define an inner product

$$\langle [x, y]^*, [x', y']^* \rangle = Bx \cdot x' + By \cdot y',$$

with corresponding norm

$$\|[x, y]^*\| = 2H(x) + 2H(y),$$

then D is $\langle \cdot, \cdot \rangle$ -symmetric. Indeed,

$$\begin{aligned} \left\langle D \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x' \\ y' \end{bmatrix} \right\rangle &= \left\langle \begin{bmatrix} C^{-1}y \\ -C^{-1}x \end{bmatrix}, \begin{bmatrix} x' \\ y' \end{bmatrix} \right\rangle \\ &= BC^{-1}y \cdot x' - BC^{-1}x \cdot y' \\ &= \bar{J}x' \cdot y + \bar{J}x \cdot y', \end{aligned}$$

which is symmetric. Moreover,

$$\left\langle D \begin{bmatrix} x \\ y \end{bmatrix}, \begin{bmatrix} x \\ y \end{bmatrix} \right\rangle = 2\bar{\omega}(x, y).$$

We are now in a position to apply Lemma 2.1 to obtain (2.8) with $\bar{\omega}$ instead of $\bar{\omega}^+$ in the numerator. (Note that $\frac{1}{2}r_j^2(H)$ is the $2n + 2j$ -th eigenvalue of D .) Finally we need to replace $\bar{\omega}$ with $\bar{\omega}^+$. This is plausible because the left-hand side is positive. $\square$

An immediate consequence of Theorem 2.2 is a linear version of Gromov’s non-squeezing theorem. More precisely, if we define

$$B_R = \{x : |x| \leq R\}, \quad Z_R = \{x : q_1^2 + p_1^2 \leq R^2\},$$

then $\mathbf{r}(B_R) = (R, R, \dots, R)$ and $\mathbf{r}(Z_R) = (R, \infty, \infty, \dots, \infty)$. By Theorem 2.2(ii) if for some $T \in Sp(n)$, we have $T(B_r) \subseteq Z_R$, then $r \leq R$. We now slightly improve this and give a direct proof of it.

Proposition 2.4 *Suppose that for some $T \in Sp(n)$ and $z^0 \in \mathbb{R}^{2n}$, $T(B_r) \subseteq z^0 + Z_R$. Then $r \leq R$.*

Proof Write $z^0 = (q_1^0, \dots, q_n^0, p_1^0, \dots, p_n^0)$ and let $(s_1, \dots, s_n, t_1, \dots, t_n)$ denote the rows of T . By assumption

$$(x \cdot s_1 - q_1^0)^2 + (x \cdot t_1 - p_1^0)^2 \leq R^2$$

for x satisfying $|x| \leq r$. Hence

$$(2.10) \quad (x \cdot s_1)^2 + (x \cdot t_1)^2 - 2x \cdot (q_1^0 s_1 + p_1^0 t_1) \leq R^2.$$

On the other hand, since T^* is symplectic,

$$(2.11) \quad \bar{\omega}(s_1, t_1) = \bar{\omega}(T^* \bar{e}_1, T^* \bar{f}_1) = \bar{\omega}(\bar{e}_1, \bar{f}_1) = -1$$

where $\{\bar{e}_1, \dots, \bar{e}_n, \bar{f}_1, \dots, \bar{f}_n\}$ denote the standard symplectic basis for $\mathbb{R}^{2n}$, i.e., $\bar{e}_j \cdot x = q_j$ and $\bar{f}_j \cdot x = p_j$ for $x = (q_1, \dots, q_n, p_1, \dots, p_n)$. From (2.11) we learn that

$$1 = |\bar{\omega}(s_1, t_1)| = |J s_1 \cdot t_1| \leq |s_1| |t_1|.$$

So either $|s_1| \geq 1$ or $|t_1| \geq 1$. Both cases can be treated similarly, so let us assume that for example $|t_1| \geq 1$. We then choose $x = \pm r \frac{t_1}{|t_1|}$ in (2.10). We select $+$ or $-$ for x so that $x \cdot (q_1^0 s_1 + p_1^0 t_1) \leq 0$. This would allow us to deduce $r^2 \leq R^2$ from (2.10) and $|t_1| \geq 1$, and this completes the proof. $\square$

Remark 2.2 Note that if we consider $Z'_R = \{x : q_1^2 + q_2^2 \leq R^2\}$ instead, then $\mathbf{r}(Z'_R) = (\infty, \dots, \infty)$ and we can embed B_r symplectically inside Z'_R no matter how large r is. This is because the map $T(q, p) = (\varepsilon q, \varepsilon^{-1} p)$ is symplectic (use $\omega(q, p, q', p') = p \cdot q' - q \cdot p'$ to check this), and $T(B_r)$ consists of points (q, p) such that $\varepsilon^{-2}|q|^2 + \varepsilon^2|p|^2 \leq r^2$. $\square$

As our next topic, we address the issue of symplectic rigidity for linear transformations. Note that the condition $|\det A| = 1$ for a matrix A is equivalent to the claim that the sets E and $A(E)$ have the same Euclidean volume. To be able to establish Eliashberg-Gromov rigidity, we would like to have a similar criterion for symplectic maps. Since a symplectic change of variables does not change symplectic radii, the volume must be replaced with suitable linear capacities that are defined in terms of the symplectic radii. Though as in the case of volume, the orientation could be reversed when a symplectic capacity is preserved. So instead of $T \in Sp(k)$, what we really have is

$$(2.12) \quad |\bar{\omega}(Ta, Tb)| = |\bar{\omega}(a, b)|.$$

We set $S'(k)$ to be the set of matrices T for which (2.12) is valid for all $a, b \in \mathbb{R}^k$. We also say that matrix T is *anti-symplectic* if $\bar{\omega}(Tx, Ty) = -\bar{\omega}(x, y)$, or equivalently $T^* \bar{J} T = -\bar{J}$. It is not hard to show that $T \in S'(k)$ iff T is either symplectic or anti-symplectic (see Exercise 2.1). On the other hand, it is straightforward to check that a linear map T is anti-symplectic iff $T \circ \tau \in S(k)$, where $\tau(q, p) = (p, q)$. From this we learn that indeed in Theorem 2.1 and Corollary 2.2 apply to anti-symplectic transformations as well. In summary,

Proposition 2.5 (i) *Let H be a positive definite quadratic function. Then there exists $T \in S'(k)$ such that $H \circ T(x) = \sum_1^n \frac{q_j^2 + p_j^2}{r_j^2}$, where $r_j = r_j(H)$.*

- (ii) Let E_1 and E_2 be two ellipsoids. Then $T(E_2) = E_1$ for some $T \in S'(k)$ if and only if $\mathbf{r}(E_2) = \mathbf{r}(E_1)$.

We are now ready for a converse to Proposition 2.5, which will be used for the proof of Eliashberg's theorem in Section 6.

Theorem 2.3 *Let T be an invertible $2n \times 2n$ matrix. Then $T \in S'(k)$ iff $r_1(E) = r_1(T(E))$ for every ellipsoid E .*

Proof Given a pair of vectors (a, b) with $\bar{\omega}(a, b) \neq 0$, let us define

$$Z(a, b) = \{x : (x \cdot a)^2 + (x \cdot b)^2 \leq 1\},$$

which is a cylinder. We claim that in fact $Z(a, b)$ is a (degenerate) ellipsoid with

$$(2.13) \quad r_1(Z(a, b)) = |\bar{\omega}(a, b)|^{-1/2}, \quad r_j(Z(a, b)) = \infty, \quad \text{for } j \geq 2.$$

Once we establish this, we are done: If $r_1(Z(a, b)) = r_1(T(Z(a, b)))$ for every a and b with $\bar{\omega}(T^*a, T^*b)\bar{\omega}(a, b) \neq 0$, then using $Z(a, b) = T(Z(T^*a, T^*b))$, we deduce

$$|\bar{\omega}(a, b)| = |\bar{\omega}(T^*a, T^*b)|,$$

whenever $\bar{\omega}(a, b), \bar{\omega}(T^*a, T^*b) \neq 0$. As a result

$$A := \{(a, b) : |\bar{\omega}(a, b)| \neq |\bar{\omega}(T^*a, T^*b)|\} \subseteq A' := \{(a, b) : \bar{\omega}(a, b)\bar{\omega}(T^*a, T^*b) = 0\}.$$

Since the set A is open and the set A' is the union of two linear sets of codimension 1, we must have $A = \emptyset$, which in turn implies that $T^* \in S'(k)$. From this, we can readily show that $T \in S'(k)$.

It remains to verify (2.13). First observe that if $r = |\bar{\omega}(a, b)|^{-1/2}$ and $(a_1, b_1) = r(a, b)$, then $|\bar{\omega}(a_1, b_1)| = 1$, and

$$Z(a_1, b_1) = \{x : H(x) := [(x \cdot a_1)^2 + (x \cdot b_1)^2] / r^2 \leq 1\}.$$

Without loss of generality, let us assume that in fact $\bar{\omega}(a_1, b_1) = -1$. We then build a (symplectic) basis $\{a_1, \dots, a_n, b_1, \dots, b_n\}$ such that

$$\bar{\omega}(b_j, a_i) = \delta_{i,j}, \quad \bar{\omega}(a_i, a_j) = \bar{\omega}(b_i, b_j) = 0,$$

for all i and j . Let us write $\{\bar{e}_1, \dots, \bar{e}_n, \bar{f}_1, \dots, \bar{f}_n\}$ for the standard basis, in other words, $\bar{e}_i$ and $\bar{f}_i$ satisfy $\bar{e}_i \cdot x = q_i$ and $\bar{f}_i \cdot x = p_i$. We then choose a map $\hat{T}$ so that $\hat{T}^*a_i = \bar{e}_i$ and $\hat{T}^*b_i = \bar{f}_i$. We have

$$H \circ \hat{T}(x) = [(\hat{T}x \cdot a_1)^2 + (\hat{T}x \cdot b_1)^2] / r^2 = [(x \cdot \bar{e}_1)^2 + (x \cdot \bar{f}_1)^2] / r^2 = (q_1^2 + p_1^2) / r^2.$$

From this we deduce (2.13) by definition. $\square$

As we observed in Remark 2.1, sometimes it is beneficiary to identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ and use complex number. More precisely, we may write $z_j = q_j + ip_j$, so that if $a = (z_1, \dots, z_n)$ and $b = (z'_1, \dots, z'_n)$ are in $\mathbb{C}^n$, then $\bar{J}a = (-iz_1, \dots, -iz_n)$, and

$$a \cdot b = \operatorname{Re} \sum_{j=1}^n z_j \bar{z}'_j, \quad \bar{\omega}(a, b) = \operatorname{Im} \sum_{j=1}^n z_j \bar{z}'_j.$$

More generally, given a symplectic vector space (V, ω) , we may try to express ω as

$$(2.14) \quad \omega(a, b) = g(Ja, b),$$

where g is an inner product on V and $J : V \rightarrow V$ is a linear map satisfying $J^2 = -I$. When (2.14) holds, we say that the pair (g, J) is *compatible* with ω . Let us write $\mathcal{I}(\omega)$ for the space of compatible pairs (g, J) . We also define

$$\mathcal{G}(\omega) := \{g : (g, J) \in \mathcal{I}(\omega), \text{ for some } J\}.$$

Note that if $(g, J) \in \mathcal{I}(\omega)$, then

$$g(Ja, Jb) = \omega(a, Jb) = -\omega(Jb, a) = g(b, a) = g(a, b),$$

which means that $J^*J = I$, where J^* is the g -adjoint or transpose of J . From this and $J^2 = -I$, we learn that for every $(g, J) \in \mathcal{I}(\omega)$, we have $J^* = -J$. Define $T^*g(a, b) = g(Ta, Tb)$ and $T^*\omega(a, b) = \omega(Ta, Tb)$.

Proposition 2.6 *For an invertible linear map $T : V \rightarrow V$, we have $T^*(\mathcal{G}(\omega)) = \mathcal{G}(T^*\omega)$.*

Proof Set $\hat{T}(J) = T^{-1}JT$. If $(g, J) \in \mathcal{I}(\omega)$, then

$$(T^*\omega)(a, b) = \omega(Ta, Tb) = g(JTa, Tb) = \omega(T\hat{T}(J)a, Tb) = (T^*\omega)(\hat{T}(J)a, b).$$

From this we deduce that if $(g, J) \in \mathcal{I}(\omega)$, then $(T^*g, \hat{T}(J)) \in \mathcal{I}(T^*\omega)$. As a result, $T^*(\mathcal{G}(\omega)) \subseteq \mathcal{G}(T^*\omega)$. Similarly, $(T^{-1})^*\mathcal{G}(T^*\omega) \subseteq \mathcal{G}(\omega)$. Hence $T^*(\mathcal{G}(\omega)) = \mathcal{G}(T^*\omega)$. $\square$

Example 2.2 Identifying a metric $g(a, b) = Ga \cdot b$ with the matrix $G > 0$, one can readily show

$$(2.15) \quad \mathcal{I}(\bar{\omega}) = \{(G, G^{-1}\bar{J}) : G > 0, G \in Sp(2n)\}.$$

To see this, observe

$$(G^{-1}\bar{J})^2 = -I \Leftrightarrow G^{-1}\bar{J}G^{-1} = \bar{J} \Leftrightarrow G^{-1} \in Sp(2n) \Leftrightarrow G \in Sp(2n).$$

□

When $g \in \mathcal{G}(\omega)$, we know how to calculate the area of the parallelogram associated with two vectors a and b , namely

$$A_g(a, b) = (\|a\|_g^2 \|b\|_g^2 - g(a, b)^2)^{\frac{1}{2}},$$

where $\|a\|_g = g(a, a)^{1/2}$. Of course $\omega(a, b)$ offers the symplectic area of the same parallelogram. In the next proposition, we compare these two areas.

Proposition 2.7 *For every $g \in \mathcal{G}(\omega)$, we have*

$$(2.16) \quad \omega(a, b) \leq A_g(a, b) \leq \frac{1}{2} (\|a\|_g^2 + \|b\|_g^2).$$

Moreover we have equality iff $b = Ja$.

Proof The second inequality is obvious and the first inequality is also obvious when $g(a, b) = 0$, because

$$\omega(a, b)^2 = g(Ja, b)^2 \leq g(Ja, Ja)g(b, b) = \|a\|_g^2 \|b\|_g^2.$$

Given arbitrary a and b , with $a \neq 0$, set $t = g(a, b)/g(a, a)$ so that $g(a, b') = 0$, for $b' = b - ta$ (ta is the orthogonal projection of b onto a). We certainly have

$$\begin{aligned} \omega(a, b)^2 &= \omega(a, b')^2 \leq g(Ja, Ja)g(b', b') = g(a, a)g(b', b') = g(a, a)g(b', b) \\ &= g(a, a)g(b, b) - tg(a, a)g(a, b) = A_g(a, b), \end{aligned}$$

with equality iff $Ja = \theta b'$ for some $\theta \geq 0$. However, if we require $2\omega(a, b) = \|a\|_g^2 + \|b\|_g^2$, then

$$\|Ja - b\|_g^2 = \|a\|_g^2 + \|b\|_g^2 - 2g(Ja, b) = \|a\|_g^2 + \|b\|_g^2 - 2\omega(a, b) = 0.$$

□

Exercise 2.1

- (i) Let V be a vector space with $\dim V = 2n$. Then a 2-form $\omega : V \times V \rightarrow \mathbb{R}$ is non-degenerate if and only if $\omega^n = \underbrace{\omega \wedge \cdots \wedge \omega}_{n \text{ times}} \neq 0$.
- (ii) Use part (i) to deduce that if $T \in Sp(n)$, then $\det T = 1$.
- (iii) Recall that an invertible matrix T can be written as $T = PO$ with $P > 0$ symmetric and O orthogonal, and that this decomposition is unique (polar decomposition). Show that if $T \in Sp(n)$, then P and $O \in Sp(n)$.

(iv) Show that if $T \in Sp(n) \cap O(2n)$, then $T = \begin{bmatrix} X & -Y \\ Y & X \end{bmatrix}$ with X, Y two $n \times n$ matrices such that $X + iY$ is a unitary matrix.

(v) Let V be a vector space with $\dim V = 2n + 1$. Assume β is an antisymmetric 2-form on V with

$$\dim\{v \in V : \beta(v, a) = 0 \text{ for all } a \in V\} = 1.$$

Then there exists a basis $\{e_1, \dots, e_n, f_1, \dots, f_n, \bar{a}\}$ such that $\beta(e_i, e_j) = \beta(f_i, f_j) = 0$, $\beta(f_j, e_i) = \delta_{ij}$, and $\beta(\bar{a}, f_j) = \beta(\bar{a}, e_j) = 0$.

(vi) If $T \in Sp(n)$ and $A \in Ham(n)$, then $T^{-1}AT \in Ham(n)$.

(vii) An invertible T maps the flows of $\frac{dx}{dt} = Ax$ to the flows of $\frac{dx}{dt} = Bx$ iff $B = TAT^{-1}$.

(viii) If $C_1, C_2 \in Ham(n)$ and $r \in \mathbb{R}$, then $C_1 + C_2, [C_1, C_2] = C_1C_2 - C_2C_1, C_1^t, rC_1 \in Ham(n)$.

(ix) If $T_1, T_2 \in Sp(n)$, then $T_1^{-1}, T_1^*, T_1T_2 \in Sp(n)$.

(x) Show that if (2.12) is true for all a and b , then T is either symplectic or anti-symplectic.

(xi) If $Z(t_0, t)$ denotes the fundamental solution of $\dot{x} = JB(t)x$ with $B : [t_0, \infty) \rightarrow S(2n)$ a C^1 -function, then $Z(t_0, t) \in Sp(n)$ for every $t \geq t_0$.

(xii) If $e^{tC} \in Sp(n)$ for every t , then $C \in Ham(n)$.

(xiii) If $C \in Ham(n)$, and $p_C(\lambda) = \det(\lambda I - C)$, then $p_C(\lambda) = p_C(-\lambda)$.

(xiv) If $T \in Sp(n)$, then $p_T(\lambda^{-1}) = \lambda^{-2n}p_T(\lambda)$.

(xv) Let T be an invertible matrix and assume that $n \geq 2$. Show that if $r_2(T(E)) = r_2(E)$ for every ellipsoid E , then $T \in S'(2n)$.

(xvi) Let (V, ω) be a symplectic vector space, and assume that $(g, J) \in \mathcal{I}(\omega)$. Then

$$\omega^k(v_1, \dots, v_{2k}) \leq k! \|v_1 \wedge \dots \wedge v_{2k}\|_g,$$

where $\|v_1 \wedge \dots \wedge v_{2k}\|_g$ denotes the $2k$ -volume associated with the metric g . Show that the equality holds iff $W = span(v_1, \dots, v_{2k})$ is J -complex (i.e. $JW = W$).

(xvii) Assume that W is a $\bar{J}$ -complex subspace of $\mathbb{R}^{2n}$ of dimension $2k$. Let $T \in Sp(n)$. Show

$$Vol_{2k}(W \cap T(B^{2n})) \leq Vol_{2k}(B^{2k}).$$

Here B^r denotes the unit ball in $\mathbb{R}^r$, and Vol_r is the r -dimensional volume.

□

3 Symplectic Manifolds and Darboux's Theorem

Before discussing symplectic manifolds, let us review some useful facts about our basic example $(\mathbb{R}^{2n}, \bar{\omega})$ with $\bar{\omega}(a, b) = \bar{J}a \cdot b$. We write $Sp(\mathbb{R}^{2n})$ for the space of differentiable functions φ such that $\varphi^*\bar{\omega} = \bar{\omega}$. This means

$$\varphi^*\bar{\omega}(a, b) = \bar{\omega}(\varphi'(x)a, \varphi'(x)b) = \bar{\omega}(a, b),$$

for every $a, b, x \in \mathbb{R}^{2n}$. Here $\varphi'(x)$ denotes the derivative of φ . A function $\varphi \in Sp(\mathbb{R}^{2n})$ is called *symplectic*. Note that $\varphi \in Sp(\mathbb{R}^{2n})$ iff $\varphi'(x) \in Sp(2n)$ for every x . Hence for a symplectic transformation φ , we have

$$\varphi'(x)^* \bar{J} \varphi'(x) = \bar{J}.$$

Evidently $\bar{\omega} = \sum_{i=1}^n dp_i \wedge dq_i = d\bar{\lambda}$ where $\bar{\lambda} = \sum_{i=1}^n p_i dq_i = p \cdot dq$. Let us define

$$(3.1) \quad A(\gamma) = \int_{\gamma} \bar{\lambda}.$$

Clearly $\varphi \in Sp(\mathbb{R}^{2n})$ iff $d(\varphi^*\bar{\lambda} - \bar{\lambda}) = 0$. Hence $\varphi \in Sp(\mathbb{R}^{2n})$ is equivalent to saying

$$(3.2) \quad A(\varphi \circ \gamma) = A(\gamma)$$

for every closed curve γ . It is worth mentioning that if γ is parametrized by $\theta \mapsto x(\theta)$, $\theta \in [0, T]$, then

$$(3.3) \quad A(\gamma) = \int_0^T p \cdot \dot{q} d\theta = \frac{1}{2} \int_0^1 (p \cdot \dot{q} - q \cdot \dot{p}) d\theta = \frac{1}{2} \int_0^1 (\bar{J}x \cdot \dot{x}) d\theta.$$

Given a scalar-valued (0-form) function H , we may use non-degeneracy of $\bar{\omega}$ to define a vector-field X_H such that

$$\bar{\omega}(X_H(x), a) = -dH(x)a = -\nabla H(x) \cdot a,$$

which means that $X_H = J\nabla H$. We write $\phi_t = \phi_t^H$ for the corresponding flow:

$$(3.4) \quad \begin{cases} \frac{d}{dt}\phi_t(x) = X_H(\phi_t(x)), \\ \phi_0(x) = x. \end{cases}$$

Our interest in symplectic transformation stems from two important facts. Firstly, $\phi_t \in Sp(\mathbb{R}^{2n})$ if ϕ_t is a Hamiltonian flow. We have seen this in Section 1 and will be proved later in this section for general symplectic manifolds. Secondly a symplectic change of coordinates preserve Hamiltonian structure (see Proposition 3.1 below). More precisely, if $\varphi \in Sp(\mathbb{R}^{2n})$,

and ϕ_t is the flow of X_H , then $\psi_t = \varphi^{-1} \circ \phi_t \circ \varphi$ is the flow of a Hamiltonian system. To guess what the Hamiltonian function of ψ_t is, observe

$$\begin{aligned} \left. \frac{d}{dt} \psi_t \right|_{t=0} &= (\varphi^{-1})' \circ \varphi X_H \circ \varphi = (\varphi')^{-1} \bar{J} \nabla H \circ \varphi \\ &= -\bar{J}(\varphi')^* \bar{J} \bar{J} \nabla H \circ \varphi = \bar{J}(\varphi')^* \nabla H \circ \varphi = \bar{J} \nabla (H \circ \varphi). \end{aligned}$$

A pair (M, ω) is called a *symplectic manifold* if M is an even dimensional manifold and ω is a closed non-degenerate 2-form on M . This implies that for each $x \in M$, the pair $(T_x M, \omega_x)$ is a symplectic vector space. Also, by Exercise 2.1(i) we know that if $\dim(M) = 2n$, then the form ω^n is a volume form. Hence M is an orientable manifold. In fact if M is a compact symplectic manifold without boundary, then ω is never exact. This is because if $\omega = d\lambda$, then $\Omega := \omega^n = d(\lambda \wedge \omega^{n-1})$. But by Stokes' theorem $\int_M \Omega = \int_M d(\lambda \wedge \omega^{n-1}) = 0$, which contradicts the non-degeneracy of Ω . Note however that $(\mathbb{R}^{2n}, \bar{\omega})$ is an example of a non-compact symplectic manifold with $\bar{\omega} = d\bar{\lambda}$.

Example 3.1 (i) Any orientable 2-dimensional manifold is symplectic where ω is chosen to be any volume form.

(ii) The sphere S^{2n} with $n > 1$ is not symplectic because any closed 2-form is exact, hence degenerate.

(iii) The classical example $(\mathbb{R}^{2n}, \bar{\omega})$ has a natural generalization that is relevant for models in classical mechanics: Every cotangent bundle T^*M can be equipped with a symplectic $\omega = d\lambda$ where λ is a *standard* 1-form that can be defined in two ways; using local charts and pull-backing $\bar{\lambda}$ to λ , or giving a chart-free description. We start with the former. Let us assume that M is an n -dimensional C^2 manifold and choose an atlas $\mathcal{A}$ of charts (U, h) of M such that U is an open subset of M and $h : U \rightarrow h(U) := V \subseteq \mathbb{R}^n$ is a diffeomorphism. This induces a C^1 transformation $dh : TU \rightarrow TV = V \times \mathbb{R}^n$. We next define a natural transformation $\bar{h} : T^*U \rightarrow T^*V = V \times \mathbb{R}^n$. To construct $\bar{h}$, take the standard basis $\{e_1, \dots, e_n\}$ of $\mathbb{R}^n$ and define $\hat{e}_j(q) = (d\phi)_q(e_j)$, where $\phi = h^{-1}$. (We may define $T_q M$ as the equivalence classes of curves $\gamma : (-\delta, \delta) \rightarrow M$ with $\gamma(0) = q$, and two such curves γ_1 and γ_2 are equivalent if $(h \circ \gamma_1)'(0) = (h \circ \gamma_2)'(0)$. We may then define $\hat{e}_j(q)$ as the equivalent class of $\gamma^j(\theta) = h^{-1}(h(q) + \theta e_j)$.) Certainly $\{\hat{e}_1(q), \dots, \hat{e}_n(q)\}$ defines a basis for $T_q M$. We now define a basis for $T_q^* L$ by taking dual vectors $e_1^*, \dots, e_n^*$ that are defined $e_i^*(q) \left(\sum_{j=1}^n v_j \hat{e}_j(q) \right) = v_i$. We now define $\bar{h}$ by

$$\bar{h} \left(q, \sum_{j=1}^n p_j e_j^*(q) \right) = (h(q), (p_1, \dots, p_n)).$$

We finally define λ as the unique 1-form such that for each chart (U, h) , the restriction of λ to T^*U is given by $\lambda = \bar{h}^* \bar{\lambda}$. For an alternative chart-free description, observe that if

$\hat{\pi} : T^*\mathbb{R}^n \rightarrow \mathbb{R}^n$ is the projection $\hat{\pi}(q, p) = q$, then

$$d\hat{\pi} : T(T^*\mathbb{R}^n) \rightarrow T\mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$$

is simply given by

$$d\hat{\pi}_{(q,p)}(\alpha, \beta) = \alpha.$$

Hence we may write $\bar{\lambda}_{(q,p)}(\alpha, \beta) = p \cdot \alpha = p \cdot (d\hat{\pi})_{(q,p)}(\alpha, \beta)$. Going back to M , let us also define $\pi : T^*M \rightarrow M$ to be the projection onto the base point, i.e., $\pi(q, p) = q$ with $q \in M$ and $p \in T_q^*M$. Since the following diagram commutes

$$\begin{array}{ccc} T^*U & \xrightarrow{\pi} & U \\ \bar{h} \downarrow & & \downarrow h \\ \mathbb{R}^{2n} & \xrightarrow{\hat{\pi}} & \mathbb{R}^n \end{array}$$

we also have that their derivatives

$$\begin{array}{ccc} T(T^*U) & \xrightarrow{d\pi} & TU \\ d\bar{h} \downarrow & & \downarrow dh \\ T\mathbb{R}^{2n} & \xrightarrow{d\hat{\pi}} & \mathbb{R}^{2n} \end{array}$$

commute. Hence we also have

$$\lambda_{(q,p)}(a) = p(d\pi_{(q,p)}(a)),$$

which gives the desired chart-free description of λ . □

A differentiable map $f : (M_1, \omega^1) \rightarrow (M_2, \omega^2)$ between two symplectic manifolds is called *symplectic* if $f^*\omega^2 = \omega^1$. This means

$$(3.5) \quad \omega_{f(x)}^2(df(x)a, df(x)b) = \omega_x^1(a, b)$$

for $x \in M_1$ and $a, b \in T_x M_1$. We write $Sp(M_1, M_2)$ for the space of symplectic transformations. When $M_1 = M_2 = M$ and $\omega^1 = \omega^2 = \omega$, we simply write $Sp(M)$ for $Sp(M_1, M_2)$. We note that if $f \in Sp(M_1, M_2)$, then $df(x)$ is injective by (3.5) and non-degeneracy of ω_x^1 . Hence, if such f exists, then $\dim M_1 \leq \dim M_2$.

Let (M, ω) be a symplectic manifold and assume that X is sufficiently nice vector field for which the ODE $\dot{x} = X(x, t)$ is well defined. The flow of this vector field is denoted by $\phi_t = \phi_t^X$. We wish to find conditions on X to guarantee that $\phi_t^*\omega = \omega$. To prepare for this let us take an arbitrary ℓ -form α and evolve it with the flow; set $\alpha(t) = \phi_t^*\alpha$. We would like to derive an evolution equation for $\alpha(t)$. Let's examine some examples.

Example 3.2 Assume that $(M, \omega) = (\mathbb{R}^{2n}, \bar{\omega})$.

- (i) If $\alpha = f$ is a 0-form, then $\alpha(t)$ is a function u that is given by $u(x, t) = f(\phi_t(x))$. By differentiating u at $t = 0$ and using the group property of the flow (see the proof of Proposition 3.1 below), we can readily show

$$u_t = X \cdot u_x.$$

- (ii) If $\alpha = m(x) dx_1 \dots dx_k$, with $k = 2n$, then $\alpha(t) = m(\phi_t(x)) \det(\phi'_t(x)) dx_1 \dots dx_k$. In this case, differentiating in t yields,

$$\rho_t = \rho_x \cdot X + \rho \operatorname{div} X = \operatorname{div}(\rho X),$$

because for small t , we have $\det(\phi'_t(x)) = \det(I + tX'(x) + o(t)) = 1 + t \operatorname{div} X + o(t)$.

- (iii) If $\omega = F \cdot dx$ is a 1-form, then

$$\alpha(t) = \phi_t^* \alpha = F(\phi_t(x)) \cdot \phi'_t(x) dx = \phi'_t(x)^* F(\phi_t(x)) \cdot dx.$$

Hence, if we set $u(x, t) = \phi'_t(x)^* F(\phi_t(x))$, then

$$(3.6) \quad u_t(x, t) = X_x(x, t)^* u(x, t) + u_x(x, t) X(x, t).$$

If $u = (u^1, \dots, u^k)$ and $X = (X^1, \dots, X^k)$, then the i -th component of the right-hand side of (3.6) equals

$$\sum_j \left(X_{x_i}^j u^j + u_{x_j}^i X^j \right) = \left(\sum_j X^j u^j \right)_{x_i} + \sum_j (u_{x_j}^i - u_{x_i}^j) X^j.$$

As a result,

$$(3.7) \quad u_t = (X \cdot u)_x + \mathcal{C}(u)X.$$

□

As Example 3.2(iii) indicates, a simple manipulation of the right-hand side of (3.6) leads to the compact expression of the right-hand side of (3.7), that may be recognized as

$$[(X \cdot u)_x + \mathcal{C}(u)X] \cdot dx = d(i_X \alpha(t)) + i_X d\alpha(t).$$

More generally we have the following useful result of Cartan:

Proposition 3.1 (i) *Let X be a vector field with flow ϕ_t and let α be a ℓ -form. Then*

$$(3.8) \quad \frac{d}{dt} \phi_t^* \alpha = \mathcal{L}_X \phi_t^* \alpha = \phi_t^* \mathcal{L}_X \alpha$$

with $\mathcal{L}_X = i_X \circ d + d \circ i_X$.

(ii) *Let $X(\cdot, t)$ be a possibly time dependent vector field and denote its flow by $\phi_{s,t}$. If $\phi_t = \phi_{0,t}$, then*

$$\frac{d}{dt} \phi_t^* \alpha = \phi_t^* \mathcal{L}_{X(\cdot, t)} \alpha.$$

Proof (i) Let us define

$$\mathcal{L}\beta = \lim_{h \rightarrow 0} \frac{1}{h} (\phi_h^* \beta - \beta)$$

whenever the limit exists. Since

$$(\phi_{t+h}^* - \phi_t^*)\alpha = \phi_t^*(\phi_h^* \alpha - \alpha) = \phi_h^*(\phi_t^* \alpha) - \phi_t^* \alpha,$$

it suffices to show that $\mathcal{L} = \mathcal{L}_X$. Let us study some properties of $\mathcal{L}$. From $\phi_t^*(\alpha \wedge \beta) = \phi_t^* \alpha \wedge \phi_t^* \beta$, we learn

$$\phi_t^*(\alpha \wedge \beta) - \alpha \wedge \beta = (\phi_t^* \alpha - \alpha) \wedge \phi_t^* \beta + \alpha \wedge (\phi_t^* \beta - \beta).$$

From this we deduce

$$(3.9) \quad \mathcal{L}(\alpha \wedge \beta) = \alpha \wedge \mathcal{L}\beta + \mathcal{L}\alpha \wedge \beta.$$

From $\phi_t^* \circ d = d \circ \phi_t^*$, we deduce

$$(3.10) \quad \mathcal{L} \circ d = d \circ \mathcal{L}.$$

We can readily show that $\mathcal{L}_X$ satisfy (3.9) and (3.10) as well. Since locally every form can be built from 0-th forms using the operations $\wedge$ and d , we only need to check that $\mathcal{L} = \mathcal{L}_X$ on 0-forms. That is, if $f : M \rightarrow \mathbb{R}$, then $\mathcal{L}f = i_X \circ df = df(X)$. This is trivially verified because $\phi_h(x) = x + hX(x) + o(h)$.

(ii) For $h > 0$, we write

$$(\phi_{t+h}^* - \phi_t^*)\alpha = \phi_t^*(\phi_{t,t+h}^* \alpha - \alpha).$$

A repetition of the proof of Part (i) leads to

$$\lim_{h \rightarrow 0} \frac{1}{h} (\phi_{t,t+h}^* \beta - \beta) = \mathcal{L}_{X(\cdot, t)} \beta.$$

This readily implies Part (ii). □

Armed with (3.8), we can readily find necessary and sufficient conditions on a vector field X such that the flow of X preserves ω . In view of Proposition 3.1,

$$(\phi_t^X)^* \omega = \omega \quad \text{for all } t \quad \Leftrightarrow \quad \mathcal{L}_X \omega = d(i_X \omega) = 0.$$

This leads to two definitions:

Definition 3.1

- We call a vector field X *symplectic* iff $i_X \omega$ is exact.

- Given a differentiable $H : M \rightarrow \mathbb{R}$, we can find a unique vector field $X = X_H = X_H^\omega$ such that

$$(i_{X_H}\omega) = \omega(X_H, \cdot) = -dH.$$

(Note that the non-degeneracy of ω guarantees the existence X_H .) The vector field X_H is called *Hamiltonian* and its corresponding flow is denoted by ϕ_t^H .

Example 3.2 If $\omega_x(v_1, v_2) = C(x)v_1 \cdot v_2$ is a symplectic form in $\mathbb{R}^{2n}$, then $X_H = -C^{-1}\nabla H$. $\square$

As in the case of $\bar{\omega}$, a change of coordinates turn a Hamiltonian flow to another Hamiltonian flow as our next Proposition demonstrates.

Proposition 3.2 *Let (M, ω) be a symplectic manifold. If $\varphi : N \rightarrow M$ is a diffeomorphism and $H : M \rightarrow \mathbb{R}$ is a smooth Hamiltonian, then $X_{H \circ \varphi}^{\varphi^*\omega} = (d\varphi)^{-1}X_H^\omega \circ \varphi$. In other words, $\varphi^*X_H^\omega = X_{H \circ \varphi}^{\varphi^*\omega}$, and*

$$\varphi^{-1} \circ \phi_t^{X_H^\omega} \circ \varphi = \phi_t^{\varphi^*X_H^\omega} = \phi_t^{X_{H \circ \varphi}^{\varphi^*\omega}},$$

where ϕ_t^X denotes the flow of the vector field X .

Proof By Lemma 10.2 of Appendix A, ψ_t is the flow of $\varphi^*X_H^\omega$. Furthermore, for $X = X_H^\omega$ and $\hat{X} = (d\varphi)^{-1}X \circ \varphi$,

$$\begin{aligned} (\varphi^*\omega)_x(\hat{X}(x), v) &= \omega_{\varphi(x)}\left((d\varphi)_x\hat{X}(x), (d\varphi)_xv\right) = \omega_{\varphi(x)}(X(\varphi(x)), (d\varphi)_xv) \\ &= -(dH)_{\varphi(x)}((d\varphi)_xv) = -(\varphi^*(dH))_x(v) = -d(H \circ \varphi)_x(v), \end{aligned}$$

for every v . Hence $\hat{X} = X_{H \circ \varphi}^{\varphi^*\omega}$, as desired. $\square$

We now turn to the question of the equivalence of two symplectic manifolds or the embedding of one symplectic manifold inside another symplectic manifold. As a warm-up, let us discuss the analogous question for volume forms.

Theorem 3.1 (Moser) *Let M be a connected oriented compact manifold with no boundary. Assume that α and β are two volume forms. Then there exists a diffeomorphism φ such that $\varphi^*\alpha = \beta$ iff $\int_M \alpha = \int_M \beta$.*

Proof Evidently if for some diffeomorphism φ , we have $\varphi^*\alpha = \beta$, then

$$\int_M \beta = \int_M \varphi^*\alpha = \int_{\varphi(M)} \alpha = \int_M \alpha.$$

As for the converse, assume that $c = \int_M \alpha = \int_M \beta$. Without loss of generality, we may assume that $c = 1$, and regard α and β as two mass (or probability) distributions on M . We may interpret φ as a plan of transportation; a unit mass at x is transported to $\phi(x)$ so that after this transportation is performed for all points, the mass distribution changes from α to β . With this interpretation in mind, we may design a route for our transportation so that this change of transportation is carried out in one unit of time. Equivalently, we may search for a (possibly time dependent) vector field X such that if ϕ_t denotes its flow, then $\varphi = \phi_1$. Note that if we set $\alpha(t) = \phi_t^* \alpha$, then $\alpha(0) = \alpha$ and $\alpha(1) = \beta$. This scheme of finding φ has a chance to work only if there exists a path of volume forms connecting α to β . So for achieving our goal, let us first such a path. We now argue that in fact the path $\alpha(t) = t\beta + (1-t)\alpha$ would do the job. Since α and β are volume forms, locally $\alpha = a dx_1 \wedge \cdots \wedge dx_k$ and $\beta = b dx_1 \wedge \cdots \wedge dx_k$ with a and b non-zero and continuous. Since $\int_M \alpha = \int_M \beta$, the functions a and b must have the same sign. Hence $t\beta + (1-t)\alpha$ is never zero, and as a result, $\alpha(t)$ is never degenerate. We next search for a vector field $X(\cdot, t)$ such that its flow ϕ_t satisfies $\phi_t^* \alpha = t\beta + (1-t)\alpha$. Differentiating both sides with respect to t yields

$$\beta - \alpha = \frac{d}{dt} \alpha(t) = \mathcal{L}_X \alpha(t) = d(i_{X(\cdot, t)} \alpha(t)).$$

by Proposition 3.1. But $\int_M (\beta - \alpha) = 0$ implies that $\beta - \alpha = d\gamma$ for some $k-1$ -form γ (See Lemma A1 of the Appendix). The existence of $X(\cdot, t)$ with $i_{X(\cdot, t)} \alpha(t) = \gamma$ follows from the non-degeneracy of $\alpha(t)$. $\square$

Remark 3.1. (i) As a consequence of Theorem 3.1, if M and N are two oriented compact closed manifolds with volume forms α and η respectively, then there exists a diffeomorphism $\phi : N \rightarrow M$ with $\phi^* \alpha = \eta$ iff M and N are diffeomorphic and $\int_M \alpha = \int_N \eta$. Indeed if $\psi : M \rightarrow N$ is a diffeomorphism, then α and $\beta = \psi^* \eta$ are two volume forms on M for which Theorem 3.1 applies: there exists a diffeomorphism $\varphi : M \rightarrow M$, such that $\phi^* \alpha = \psi^* \eta$. We then set $\phi = \varphi \circ \psi^{-1}$ to deduce that $\phi^* \alpha = \eta$.

(ii) If $M = \mathbb{T}^k$ and $\alpha = \hat{\alpha}(x) dx_1 \wedge \cdots \wedge dx_k$, $\beta = \hat{\beta}(x) dx_1 \wedge \cdots \wedge dx_k$, then $\varphi^* \alpha = \beta$ means that $\hat{\alpha}(\varphi(x)) \det \varphi'(x) = \hat{\beta}(x)$. If $\varphi = \nabla w$ for some $w : \mathbb{T}^k \rightarrow \mathbb{R}$, then $\hat{\alpha}(\nabla w) \det(D^2 w) = \hat{\beta}$. This is the celebrated Monge–Ampère’s equation. However the function φ in Moser’s proof is not a gradient. In fact the vector field X in the proof of Theorem 3.1 can be constructed by the following recipe: First find a vector field Y such that $\text{div } Y = \hat{\beta} - \hat{\alpha}$, so that if $\gamma = \sum_1^k (-1)^{j-1} Y^j dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_k$, then $d\gamma = \beta - \alpha$. Such a vector field Y exists because $\int (\hat{\beta} - \hat{\alpha}) dx = 0$. We then set

$$X(x, t) = - \frac{Y(x)}{t\hat{\beta}(x) + (1-t)\hat{\alpha}(x)}.$$

In fact for the existence of Y , we may try a gradient $Y = \nabla u$ so that the scalar-valued function u satisfies $\Delta u = \hat{\beta} - \hat{\alpha}$. Again this equation has a solution because $\int (\hat{\beta} - \hat{\alpha}) dx = 0$.

(iii) There has been new developments for Moser's theorem. In fact the transformation φ in Theorem 3.1 is by no means unique. However, one may wonder whether or not a "nice" φ exists. More precisely, let (M, g) be a Riemannian manifold. Assume that M is compact, connected with no boundary. Using g we can talk about a Riemannian distance. More precisely, let $d(x, y)$ be the length of the geodesic distance between two points x and y . We have a natural volume form Ω that is expressed by $(\det[g_{ij}])^{1/2} dx_1 \wedge \cdots \wedge dx_k$ in local coordinates. Consider two forms $\alpha = a\Omega$ and $\beta = b\Omega$ with $a, b > 0$ and $\int_M \alpha = \int_M \beta = 1$. Set

$$S(\alpha, \beta) = \{f : M \rightarrow M \text{ with } \varphi^* \alpha = \beta\}.$$

By Moser's theorem, this set is non-empty. Monge's problem searches for a function $f \in S(\alpha, \beta)$ which minimizes the cost function

$$\mathcal{I}(f) = \int_M c(x, f(x)) \beta,$$

with $c(x, y)$ a suitable function of $M \times M$. If we choose $c(x, y) = \frac{1}{2}(d(x, y))^2$, then the minimizer φ is of the form $\varphi = \nabla w$ for a *convex* function w . This was shown by Brenier (1987, 1991) in the Euclidean case and by McCann in (2001) in the case of a Riemannian manifold. Brenier observed that such a minimizer can be used to find a *non-linear polar decomposition*. To explain this, let $F : U \rightarrow \mathbb{R}^k$ be an invertible integrable function with $\alpha = (F^{-1})^* \beta$ where $\beta = dx_1 \wedge \cdots \wedge dx_k$ and α is a volume form. According to Brenier's theorem, there exists a convex function ψ such that $(\nabla \psi)^* \alpha = \beta$. If we write $\rho = (\nabla \psi)^{-1} \circ F$, then $\rho^* \beta = F^* (\nabla \psi)^{-1*} \beta = F^* \alpha = \beta$. As a consequence, any function F can be decomposed as $F = \nabla \psi \circ \rho$ with ψ convex and ρ volume preserving. It turns out that polar decomposition implies the Hodge decomposition. To see this assume that $F^\epsilon(x) = x + \epsilon f(x)$ with ϵ small. We then expect to have $\psi^\epsilon(x) = \frac{1}{2}|x|^2 + \epsilon \varphi(x) + o(\epsilon)$ and $\rho^\epsilon(x) = x + \epsilon m(x) + o(\epsilon)$. Hence

$$\begin{aligned} x + \epsilon f(x) &= x + \epsilon \nabla \varphi(x + \epsilon m(x)) + \epsilon m(x) + o(\epsilon) \\ (3.11) \quad &= x + \epsilon (\nabla \varphi(x) + m(x)) + o(\epsilon). \end{aligned}$$

On the other hand, since ρ^ϵ is volume preserving,

$$1 = \det(id + \epsilon m') = 1 + \epsilon \operatorname{div}(m) + o(\epsilon).$$

From this and (3.11) we learn

$$f(x) = \nabla \varphi(x) + m(x) \text{ with } \operatorname{div} m = 0.$$

(iv) Theorem 3.1 can be used to show that the total volume is the only invariant of volume preserving diffeomorphisms. More precisely, if

$$\mathcal{V}(M) = \{\alpha : \alpha \text{ is a volume form on } M\}$$

with M compact and closed, and

$$c : \mathcal{V}(M) \rightarrow \mathbb{R}$$

is a function such that $c(\alpha) = c(\beta)$ whenever $\varphi^*\alpha = \beta$ for some diffeomorphism φ , then c must be a function of $\int_M \alpha$. $\square$

Given a manifold M with two symplectic forms α and β , we may wonder whether or not for some diffeomorphism φ , we have $\varphi^*\alpha = \beta$. By Theorem 3.1, this would be the case if $\dim M = 2$ and $\int_M \alpha = \int_M \beta$. However, when $\dim M = 2n \geq 4$, we may have non-isomorphic symplectic forms α and β with $\int_M \alpha^n = \int_M \beta^n$. To see how the proof of Theorem 3.1 breaks down, we note that we may fail to find a path of symplectic forms $\alpha(t)$ that connects α to β . Even if such a path exists, the equation $\mathcal{L}_X \alpha(t) = d i_X \alpha(t) = d\alpha(t)/dt$ requires that $d\alpha(t)/dt$ to be exact for $t \in [0, 1]$. Though both of these issues can be handled locally as the next theorem demonstrate.

Theorem 3.2 (*Darboux*) *Let (M, ω) be a symplectic manifold of dimension $2n$ and take $x^0 \in M$. Then there exists an open set $U \subseteq \mathbb{R}^{2n}$ with $0 \in U$ and a diffeomorphism $\varphi : U \rightarrow M$ such that $\varphi(0) = x^0$ and $\varphi^*\omega = \bar{\omega}$.*

Proof Since M is locally diffeomorphic to an open subset of $\mathbb{R}^{2n}$, we may assume that $M = U_0 \subseteq \mathbb{R}^{2n}$ and $x^0 = 0 \in U$. In view of Proposition 2.1, we may also assume that $\omega_0 = \bar{\omega}$. Our goal is finding an open set $U \subseteq U_0$ with $0 \in U$, and a diffeomorphism $\varphi : U \rightarrow U$ with $\varphi(0) = 0$ and $\varphi^*\omega = \bar{\omega} = \omega_0$. Indeed if $\omega_x(v_1, v_2) = C(x)v_1 \cdot v_2$ and $\omega(t) = \bar{\omega} + t(\omega - \bar{\omega})$, for $t \in [0, 1]$, then $\omega(t)_x(v_1, v_2) = C(x, t)v_1 \cdot v_2$, with $C(x, t) = \bar{J} + t(C(x) - \bar{J})$ and $C(0) = \bar{J}$. Clearly, we can find an open neighborhood $U = B_r(0)$ of 0 such that for $x \in U$ we have that $\|C(x) - \bar{J}\| < \|\bar{J}\| = 1$, which in turn guarantees that $C(x, t)$ is invertible for $(x, t) \in U \times [0, 1]$. This means that in U , the form $\omega(t)$ is symplectic for all $t \in [0, 1]$. We then search for a time-dependent vector field $X(\cdot, t)$ such that its flow ϕ_t satisfies

$$\phi_t^* \omega = \omega(t) \quad \text{for } t \in [0, 1].$$

From differentiating both sides with respect to t and using Proposition 3.1, we learn

$$(3.12) \quad \omega - \bar{\omega} = \mathcal{L}_{X(\cdot, t)} \omega(t) = d i_{X(\cdot, t)} \omega(t).$$

In the ball $U = B_r(0)$, we can express $\omega - \bar{\omega} = d\alpha$ for a 1-form α such that $\alpha_0 = 0$. Hence (3.12) would follow if we can find a time-dependent vector field X such that $i_{X(\cdot, t)} \omega(t) = \alpha$. By nondegeneracy of $\omega(t)$, such a vector field X exists with $X(0, t) = 0$ for every $t \in [0, 1]$. We are done. $\square$

Remark 3.2 If we write $\omega = u \cdot dx$ and $\bar{\omega} = \bar{u} \cdot dx$ near the origin, then X constructed in the proof has the form

$$X(x, t) = \mathcal{C}(tu(x) + (1 - t)\bar{u}(x))^{-1}(u(x) - \bar{u}(x)) = C(x, t)^{-1}(u(x) - \bar{u}(x)).$$

□

As an immediate consequence of Darboux's theorem we learn that any symplectic manifold M of dimension $2n$ has an atlas consisting of pairs $\{(U_j, h_j)\}$ with $h_j : U_j \rightarrow \mathbb{R}^{2n}$, such that

$$h_i \circ h_j^{-1} : h_j(U_i \cap U_j) \rightarrow h_i(U_i \cap U_j)$$

is symplectic for every i and j . The family $\{(U_j, h_j)\}$ is an example of a *symplectic atlas* that always exists by Darboux's theorem.

Definition 3.2(i) Let (M, ω) be a symplectic manifold and assume that L is a submanifold of M . then we say that L is a *Lagrangian submanifold* if the restriction of ω to L is 0.

(ii) Note that if $\omega = d\lambda$ for a 1-form λ , then the submanifold L is a Lagrangian iff the restriction of λ to L is closed. We say that L is an *exact Lagrangian* if the restriction of λ to L is exact. That is $\lambda = du$ on L for some C^1 function $u : L \rightarrow \mathbb{R}$. □

Exercise 3.1 (i) Show that $\mathcal{L}_X = d \circ i_X + i_X \circ d$ satisfy

$$\mathcal{L}_X \circ d = d \circ \mathcal{L}_X, \quad \mathcal{L}_X(\alpha \wedge \beta) = (\mathcal{L}_X \alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X \beta).$$

(ii) Let (M, ω) be a $2n$ -dimensional symplectic manifold. Assume that M is compact with no boundary. Then for every $j \in \{1, \dots, n\}$, there exists a closed $2j$ -form which is not exact.

(iii) Let (M_1, ω^1) and (M_2, ω^2) be two symplectic manifolds. Define $(M_1 \times M_2, \omega^1 \times \omega^2)$ with

$$(\omega^1 \times \omega^2)_{(x,y)}((a_1, a_2), (b_1, b_2)) = \omega_x^1(a_1, b_1) + \omega_y^2(a_2, b_2).$$

Show that $(M_1 \times M_2, \omega^1 \times \omega^2)$ is symplectic.

(iv) Let (M, ω) be a symplectic manifold. Show that if $\varphi^* \omega = f \omega$ for some diffeomorphism $\varphi : M \rightarrow M$, and C^1 scalar function f , then either $\dim M = 2$ or f is a constant function.

(v) Use polar coordinates in $\mathbb{R}^{2n}$ to write $q_i = r_i \cos \theta_i$, $p_i = r_i \sin \theta_i$, and let e_i (respectively f_i) denote the vector for which the q_i -th coordinate (respectively p_i -th coordinate) is 1 and any other coordinate is 0. Set

$$e_i(\theta_i) = (\cos \theta_i) e_i + (\sin \theta_i) f_i, \quad f_i(\theta_i) = -(\sin \theta_i) e_i + (\cos \theta_i) f_i.$$

Given a vector field u , we may write

$$u = \sum_{i=1}^n (a^i e_i(\theta_i) + b^i f_i(\theta_i)).$$

The form $\alpha = u \cdot dx$ can be written as

$$\alpha = \sum_{i=1}^n (a^i dr_i + r_i b^i d\theta_i) =: \sum_{i=1}^n (a^i dr_i + B^i d\theta_i).$$

Assume that all a^i 's and b^i 's depend on $r = (r_1, \dots, r_n)$ only. What are the necessary and sufficient conditions on a and B in order for $\omega = d\alpha$ to be symplectic.

(vi) Let U_1 and $U_2 \subset B_r$ be two planar open sets with smooth boundaries that are diffeomorphic to a disc. Show that if $\text{area}(U_1) = \text{area}(U_2)$, then there is an area preserving diffeomorphism $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\psi(U_1) = U_2$ and $\psi(x) = x$ for $x \notin B_r$. Here $B_r = \{x : |x| < r\}$.

Hint: Construct ψ from three diffeomorphisms ψ_1, ψ_2 and ψ_3 with the following properties:

- $\psi_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $\psi_1(x) = x$, for $x \notin B_r$, $\psi_1(U_1) = U_2$, and ψ_1 is area-preserving on a small neighborhood of ∂U_1 .
- $\psi_2 : \bar{U}_1 \rightarrow \bar{U}_2$ is area preserving and $\psi_2(x) = \psi_1(x)$ for x near ∂U_1 .
- $\psi_3 : B_r \setminus \bar{U}_1 \rightarrow B_r \setminus \bar{U}_2$ is area-preserving and $\psi_3(x) = \psi_1(x)$ for x near $\partial U_1 \cup \partial B_r$.

□

4 Contact Manifolds and Weinstein Conjecture

In this section we will give two motivations for studying contact manifolds. As our first motivation we observe that we can construct *exotic* symplectic forms on $\mathbb{R}^4$ from certain contact forms in $\mathbb{R}^3$. Our second motivation is the *Weinstein's conjecture* that predicts every compact nonsingular contact level set of a Hamiltonian function carries at least one periodic orbit of the corresponding Hamiltonian vector field.

We first give a simple recipe for constructing a symplectic form on $\hat{M} = M \times \mathbb{R}$ from certain 1-forms on a manifold M : Given a 1-form α on M , define $\hat{\alpha}_{(x,s)}(v, \tau) = e^s \alpha_x(v)$ for every $x \in M$, $v \in T_x M$, and $s, \tau \in \mathbb{R}$. There is nothing special about e^s and our approach is applicable if we replace e^s with a strictly increasing or decreasing C^1 function of s . The question is whether or not $\hat{\omega} = d\hat{\alpha}$ is symplectic.

Proposition 4.1 *Let M be any manifold of odd dimension and α any 1-form on M . Then the form $d\hat{\alpha}$ is symplectic iff $\ell_x(d\alpha) \cap \xi_x(\alpha) = \{0\}$ for every $x \in M$, where*

$$\begin{aligned}\ell_x &= \ell_x(\omega) = \{v \in T_x M : \omega_x(v, w) = 0 \text{ for every } w \in T_x M\}, \\ \xi_x &= \xi_x(\alpha) = \ker \alpha_x = \{v \in T_x M : \alpha_x(v) = 0\}.\end{aligned}$$

Proof We certainly have $d\hat{\alpha} = e^s(d\alpha - \alpha \wedge ds)$. As a result,

$$(d\hat{\alpha})_{(x,s)}((v, \tau), (w, \tau')) = e^s((d\alpha)_x(v, w) - \alpha_x(v)\tau' + \alpha_x(w)\tau).$$

From this we can readily show that if $v \in \ell_x \cap \xi_x$, then

$$(d\hat{\alpha})_{(x,s)}((v, 0), (w, \tau')) = 0,$$

for every $(w, \tau') \in T_x M \times \mathbb{R}$.

Conversely, suppose that $\ell_x(\alpha) \cap \xi_x(\alpha) = \{0\}$, and that for some $(v, \tau) \in T_x M \times \mathbb{R}$, we have

$$(4.1) \quad (d\alpha)_x(v, w) - \alpha_x(v)\tau' + \alpha_x(w)\tau = 0,$$

for every $(w, \tau') \in T_x M \times \mathbb{R}$. We wish to show that $(v, \tau) = 0$. To see this, first vary τ' in (4.1), to deduce that $\alpha_x(v) = 0$, or $v \in \xi_x$. Hence we now have

$$(4.2) \quad (d\alpha)_x(v, w) + \alpha_x(w)\tau = 0,$$

for every $w \in T_x M$. If $\tau = 0$, then (4.2) means that $v \in \ell_x$. As a result, $v \in \ell_x \cap \xi_x = \{0\}$ and we are done. On the other hand, if $\tau \neq 0$, choose any non-zero $w \in \ell_x$ in (4.2) to deduce that $\alpha_x(w) = 0$, or $w \in \xi_x$, which contradicts our assumption $\ell_x \cap \xi_x = \{0\}$. (Note that since the dimension of M is odd, $\ell_x \neq \{0\}$.) This completes the proof. $\square$

Note that since $\dim M = 2n - 1$ is odd, the dimension of ℓ_x is at least 1. Hence the condition $\ell_x \cap \xi_x = \{0\}$ implies that $\dim \ell_x = 1$ and $\dim \xi_x = 2n - 2$. Equivalently,

$$(4.3) \quad T_x M = \ell_x \oplus \xi_x,$$

for every $x \in M$.

Definition 4.1(i) Let M be a C^1 manifold of dimension $2n - 1$, α be a 1-form, and ω be a closed 2-form on M . The triplet (M, α, ω) is called a *framed Hamiltonian manifold* if $\ell_x(\omega) \cap \xi_x(\alpha) = \{0\}$ for every $x \in M$. The corresponding *Reeb vector field* $R = R^\alpha$ is the unique $R \in \ell_x$ such that $\alpha(R) = 1$. The R -projection onto ξ is denoted by $\pi = \pi^\alpha$; $\pi_x(v) = v - \alpha_x(v)R(x)$.

(ii) The pair (M, α) of a manifold M and 1-form α is called a *contact manifold* if $(M, \alpha, d\alpha)$ is a framed Hamiltonian manifold. $\square$

One of the main interest in contact manifold is the following conjecture:

Weinstein Conjecture The Reeb vector field of a closed contact manifold has a closed (periodic) orbit. $\square$

We next discuss the analog of Hamiltonian vector fields for contact manifolds.

Proposition 4.2 Let (M, α) be a contact manifold, and let $H : M \rightarrow \mathbb{R}$ be a C^1 function. Set $\hat{H}(x, s) = e^s H(x)$. If $X_{\hat{H}}^{d\hat{\alpha}} = (Z, V)$, then the vector field Z and the function $V : M \rightarrow \mathbb{R}$ depend on x and are uniquely determined by the equations

$$(4.4) \quad i_Z d\alpha = dH(R^\alpha)\alpha - dH, \quad \alpha(Z) = H, \quad V = -dH(R^\alpha).$$

Moreover, $\mathcal{L}_Z \alpha = -V\alpha$.

Proof By definition

$$i_{(Z,V)} d\hat{\alpha} = e^s (i_Z d\alpha - \alpha(Z) ds + V\alpha) = -e^s (H ds + dH).$$

This implies that $\alpha(Z) = H$ and $i_Z d\alpha + V\alpha = -dH$. By evaluating both sides at R^α we learn that $V = -dH(R^\alpha)$, completing the proof of (4.4). The vector field Z is uniquely determined by the first two equations of (4.4) because the restriction of $d\alpha$ to ξ is symplectic, and $\pi(Z)$ satisfies

$$i_{\pi(Z)} d\alpha = i_Z d\alpha = dH(R^\alpha)\alpha - dH := \nu,$$

with $\nu(R) = 0$. Finally,

$$\mathcal{L}_Z \alpha = i_Z d\alpha + d(\alpha(Z)) = i_Z d\alpha + dH = dH(R^\alpha)\alpha = -V\alpha,$$

by (4.4). $\square$

Definition 4.2 Given a contact manifold (M, α) , we say that a vector field Z is an α -contact, if $\mathcal{L}_Z \alpha + V \alpha = 0$, for some function $V : M \rightarrow \mathbb{R}$. Moreover, given a C^1 Hamiltonian function H , the unique α -contact vector field Z and scalar-valued function V satisfying (4.4) are denoted by $Z_H = Z_H^\alpha$ and $V_H^\alpha = V_H$ respectively. $\square$

Remark 4.1 (i) Note that if the flow of an α -contact vector field Z is denoted by ϕ_t , then

$$\frac{d}{dt} \phi_t^* \alpha = \phi_t^* \mathcal{L}_Z \alpha = -\phi_t^* (V \alpha) = -(V \circ \phi_t) \phi_t^* \alpha,$$

which implies

$$(\phi_t^* \alpha)_x = e^{-\int_0^t \phi_\theta(x) d\theta} \alpha_x.$$

(ii) Since $\mathcal{L}_{R^\alpha} \alpha = 0$, the Reeb vector field is an example of an α -contact vector field. Indeed, when $H = 1$ in Proposition 4.2, then $Z = R^\alpha$, and $V = 0$. In fact the flow of R^α preserves α . $\square$

Example 4.1 When $M = \mathbb{R}^3$, the form $\alpha = u \cdot dx$ is contact if and only if $\rho = (\nabla \times u) \cdot u$ is never 0. Indeed,

$$d\alpha(v_1, v_2) = ((\nabla \times u) \times v_1) \cdot v_2 =: [(\nabla \times u), v_1, v_2]$$

which implies that when $(\nabla \times u)(x) \neq 0$, the set ℓ_x is the line spanned by $(\nabla \times u)(x)$. In this case the Reeb vector field is given by $R = \rho^{-1}(\nabla \times u)$, and

$$(4.5) \quad \alpha \wedge d\alpha = \rho \, dx_1 \wedge dx_2 \wedge dx_3,$$

is a volume form. Moreover, if we write $\mathcal{L}_Z(u \cdot dx) = (\mathcal{L}'_Z u) \cdot dx$, then

$$(4.6) \quad \mathcal{L}'_Z u = \nabla(u \cdot Z) + (\nabla \times u) \times Z.$$

The contact vector field associated with H is given by

$$(4.7) \quad \rho X_H = u \times \nabla H + H(\nabla \times u).$$

More generally, if $M = \mathbb{R}^{2n-1}$ with $n \geq 2$, we may express a 1-form α as $\alpha = u \cdot dx$ for a vector field u . Moreover

$$\beta_x(v_1, v_2) = d\alpha_x(v_1, v_2) = C(u)v_1 \cdot v_2,$$

where $C(u) = Du - (Du)^*$. The form α is contact iff u never vanishes, and the null space of $C(u)(x)$ is not orthogonal to u . The set ℓ_x is one dimensional and R^α is the unique vector in ℓ_x such that $u(x) \cdot R(x) \equiv 1$. Writing $u^\perp$ and $R^\perp$ for the space of vectors perpendicular to u and R respectively, then $\xi = u^\perp$, and we may define a matrix $C'(u)$ which is not exactly the inverse of $C(u)$ (because $C(u)$ is not invertible), but it is specified uniquely by two requirements:

- (i) $C'(u)$ restricted to $R^\perp$ is the inverse of $C(u) : u^\perp \rightarrow R^\perp$.
- (ii) $C'(u)u = 0$.

The α -contact vector field associated with H is given by

$$(4.8) \quad X_H = -C'(u)\nabla H + HR.$$

□

As Example 4.1 and (4.5) indicates, in dimension 3, a form α is contact iff $\alpha \wedge d\alpha$ is a volume form. More generally we have the following elementary result.

Proposition 4.3 *Let α be a 1-form and ω be a closed 2-form on a manifold M with $\dim M = 2n - 1$. Then (M, α, ω) is a framed Hamiltonian manifold iff $\alpha \wedge \omega^{n-1}$ is a volume form.*

Proof. Assume that (M, α, ω) is framed Hamiltonian manifold and set $\gamma = \omega^{n-1}$. Since the restriction of ω to ξ_x is non-degenerate and $\dim \xi_x = 2(n - 1)$, we may use Exercise 2.1(i) to assert that ω_x^{n-1} is a volume form on ξ_x . More specifically, we may choose a symplectic basis $\{e_1, \dots, e_{n-1}, f_1, \dots, f_{n-1}\}$ for ξ_x so that $\omega_x = \sum_i f_i^* \wedge e_i^*$, where

$$\{e_1^*, \dots, e_{n-1}^*, f_1^*, \dots, f_{n-1}^*\} =: \{dx_1, \dots, dx_{2n-2}\}$$

denote the duals of $\{e_1, \dots, f_{n-1}\}$. From this we deduce

$$\gamma_x = (n - 1)! \, dx_1 \wedge \dots \wedge dx_{2n-2}.$$

To show that $\alpha \wedge \gamma$ is a volume form, observe

$$(\alpha \wedge \gamma)(v_1, \dots, v_{2n-1}) = \sum_{i=1}^{2n-1} (-1)^{i-1} \alpha(v_i) \gamma(v_1, \dots, \hat{v}_i, \dots, v_{2n-1})$$

which means that if dz denotes the dual of the vector R , then

$$\alpha \wedge \gamma = (n - 1)! \, dz \wedge dx_1 \wedge \dots \wedge dx_{2n-2}.$$

The converse is left as an exercise. (See Exercise 4.1(ii).)

□

If (N, ω) is a symplectic manifold and M is a closed co-dimension 1 submanifold of N , we may wonder whether or not we can find a contact form α on M , $\delta > 0$, and a diffeomorphism $\psi : \hat{M} = M \times (-\delta, \delta) \rightarrow U \subseteq M$, such that $\psi^* \omega = \hat{\omega}$, where $\hat{\omega} = d\hat{\alpha}$, for $\hat{\alpha} = e^s \alpha$. In words, a neighborhood U of N in M is isomorphic to $(\hat{M}, \hat{\omega})$. As we will see below that this is

possible whenever M is compact and there is a vector field X that plays the role of $\frac{\partial}{\partial s}$. The point is that if the flow of the vector field $\frac{\partial}{\partial s}$ is denoted by ϕ_t , then $\phi_t(x, s) = (x, s + t)$ and

$$\phi_t^* \hat{\alpha} = e^{s+t} \alpha = e^t \hat{\alpha}.$$

Hence $\phi_t^* \hat{\omega} = e^{s+t} d\alpha = e^t \hat{\omega}$, which means

$$(\psi \circ \phi_t \circ \psi^{-1})^* \omega = e^t \omega.$$

As a result, if

$$X = \psi_* \left(\frac{\partial}{\partial s} \right),$$

then

$$(4.9) \quad (\phi_t^X)^* \omega = e^t \omega.$$

In fact the existence of a vector field X that satisfies (4.9) is exactly what we need for ω to be isomorphic to $\hat{\omega}$ in a neighborhood of M .

Definition 4.3 Let X be a vector field on a symplectic manifold (M, ω) . We say X is an ω -Liouville vector field if $\mathcal{L}_X \omega = di_X \omega = \omega$. Equivalently, X is ω -Liouville if (4.9) is valid. $\square$

Remark 4.2 Note that if a Liouville vector field X exists, then we can define a 1-form $\alpha' = i_X \omega$ which satisfies $d\alpha' = \omega$. Moreover, since $\alpha'(X) = \omega(X, X) = 0$, we also have

$$(4.10) \quad \mathcal{L}_X \alpha' = i_X d\alpha' = i_X \omega = \alpha', \quad (\phi_t^X)^* \alpha' = e^t \alpha'.$$

$\square$

Theorem 4.1 Let (N, ω) be a symplectic manifold of dimension $2n$, and let M be compact closed submanifold of N with $\dim M = 2n - 1$. The following statements are equivalent:

- (i) There exists a neighborhood U of M in N and an ω -Liouville vector field X such that X is transverse to M .
- (ii) There exists a contact form α on M such that $d\alpha$ is the restrictions of ω to M .
- (iii) There exists a contact form α on M , $\delta > 0$, and a diffeomorphism $\psi : \hat{M} = M \times (-\delta, \delta) \rightarrow U \subseteq N$, with $\psi(x, 0) = 0$, such that if $\hat{\alpha} = e^s \alpha$ and $\alpha' = (\psi^{-1})^* \hat{\alpha}$, then $d\alpha' = \omega$.

Proof Suppose that (i) is true. In other words, there exists a Liouville vector field X such that $X(x) \notin T_x M$ for every $x \in M$. Set $\alpha' = i_X \omega$ so that $d\alpha' = \omega$. Write α for the restriction of α' to M . Set $\xi_x = \ker \alpha$, and

$$\ell_x := \{v \in T_x M : \omega_x(v, w) = 0 \text{ for all } w \in T_x M\},$$

for $x \in M$. We wish to show that $\ell_x \cap \xi_x = \{0\}$. Pick any $v \in \ell_x \cap \xi_x$. By definition, $\omega_x(v, w) = 0$ for all $w \in T_x M$. On the other hand $\omega_x(X(x), v) = \alpha_x(v) = 0$. Hence $\omega_x(v, a) = 0$ for all $v \in T_x N$ because X transverses to M . Since ω_x is symplectic, we deduce that $v = 0$. Thus α is contact. In summary (i) implies (ii).

The converse is carried out in several steps.

Step1. Starting from a contact form α on M with $d\alpha = \omega|_M$, we wish extend α to α' with $d\alpha' = \omega$, and construct a Liouville vector field X in a neighborhood of M . First we construct X on M . Recall that $R = R^\alpha$ denotes the Reeb vector field and $\xi_x = \ker \alpha_x \subset T_x M$ is of dimension $2n - 2$. Since we hope to find X satisfying $\alpha = i_X \omega$ on M , the restriction of such X to M must satisfy $\omega_x(X(x), v) = \alpha(v)$ for every $v \in T_x M$. As a result, for every $x \in M$ and $v \in \xi_x$, we must have

$$(4.11) \quad \omega_x(X(x), R(x)) = 1, \quad \omega_x(X(x), v) = 0.$$

It is not hard to construct a vector field X that satisfies (4.11). For example, we may use Proposition 4.2 to take an almost complex structure J and a Riemannian metric g so that $\omega(v_1, v_2) = g(Jv_1, v_2)$. We then search for a vector field X of the form

$$X(x) = f(x)J_x R(x) + Y(x), \quad Y(x) \in \xi_x,$$

with $f : M \rightarrow \mathbb{R}$ a scalar-valued function. To satisfy (4.11), we need

$$f = -g(R, R)^{-1}, \quad i_Y \omega = -f(i_{JR} \omega).$$

The latter equation has a unique solution $Y \in \xi$ because $d\alpha|_\xi$ is symplectic. We note that the first requirement in (4.11) implies that $X(x) \notin T_x M$ for every $x \in M$. In summary, we have constructed a vector field X that transverses to M and satisfies $i_X \omega = \alpha$ on M .

Step2. So far our vector field $X(x)$ is defined only on M . Since M is compact and X transverses to M , we can use the g -exponential map to define $\varphi(x, s) = \exp_x(sX(x))$, so that $\varphi : M \times (-\delta, \delta) \rightarrow U \subseteq N$ is a diffeomorphism. Here $\varphi(x, s) = \gamma(s)$, where γ is the unique geodesic with $\gamma(0) = x$ and $\dot{\gamma}(0) = X(x)$. (Alternatively, extend X to a vector field $\hat{X}$ that is defined on a neighborhood of M , and set $\varphi(x, s) = \phi_s^{\hat{X}}(x)$.) This map has the

desired property on M ;

$$\begin{aligned}
(\varphi^*\omega)_{(x,0)}((v_1, \tau_1), (v_2, \tau_2)) &= \omega_x(d\varphi_{(x,0)}(v_1, \tau_1), d\varphi_{(x,0)}(v_2, \tau_2)) \\
&= \omega_x(v_1 + \tau_1 X(x), v_2 + \tau_2 X(x)) \\
&= \omega_x(v_1, v_2) + \tau_1 \omega_x(X(x), v_2) + \tau_2 \omega_x(v_1, X(x)) \\
&= \omega_x(v_1, v_2) + \tau_1 \alpha(v_2) - \tau_2 \alpha(v_1) \\
&= (\omega + ds \wedge \alpha)_{(x,0)}((v_1, \tau_1), (v_2, \tau_2)),
\end{aligned}$$

because $\varphi_{(x,0)}(v, \tau) = \tau X(x) + v$. Hence, if we define $\hat{\alpha} = e^s \alpha$ and $\hat{\omega} = d\hat{\alpha}$ on $\hat{M} = M \times (-\delta, \delta)$, then $\varphi^* \omega = \hat{\omega}$ on $M = M \times \{0\}$. As a result, if we set $\omega' = \varphi^* \omega$, then $\eta := \omega' - \hat{\omega}$ is a closed form on $\hat{M}$ such that $\eta|_M = 0$. As we will see in Lemma 4.1 below, the form η is necessarily exact. In fact, there exists a 1-form β such that $\beta|_M = 0$ and $d\beta = \eta$. Hence, $\omega' = d(\hat{\alpha} + \beta)$ with $(\hat{\alpha} + \beta)|_M = \alpha$. From this we learn that if $\alpha' = (\varphi^{-1})^*(\hat{\alpha} + \beta)$, then $d\alpha' = \omega$, and $\alpha'|_M = \alpha$.

Step 3. So far we now that there exists a neighborhood U of M , and a 1-form α' that is defined in U and satisfies

$$\alpha'|_M = \alpha, \quad d\alpha' = \omega.$$

Since ω is symplectic, we can find a vector field X in U such that $i_X \omega = \alpha'$. This vector field is a Liouville vector field because $di_X \omega = d\alpha' = \omega$. Moreover, since $\omega_x(X(x), v) = \alpha_x(v)$, for every $x \in M$ and $v \in T_x M$, we learn that $X(x) \notin T_x M$ for every $x \in M$; otherwise $\alpha_x(R(x)) = 0$ which is impossible. This completes the proof of **(i)**.

It remains to show the equivalence of **(i)** and **(iii)**. If **(i)** is true, simply define $\psi(x, s) = \phi_s^X(x)$. Note that by (4.10), we have $\phi_s^* \alpha' = e^s \alpha'$. Since

$$\hat{\phi}_t(x, s) := (\psi^{-1} \circ \phi_t \circ \psi)(x, s) = (x, t + s),$$

we deduce

$$\psi^* X = \frac{\partial}{\partial s}.$$

On the other hand, if $\hat{\alpha} = \psi^* \alpha'$, then $(\hat{\phi}_s)^* \hat{\alpha} = e^s \hat{\alpha}$, which implies that $\hat{\alpha}_{(x,s)}(v, \tau) = e^s \alpha_x(v)$, because $\hat{\phi}$ is the flow of $\partial/\partial s$. We are done.

Conversely, if **(iii)** is true, then the Liouville vector field is given by

$$X = (\psi^{-1})^* \frac{\partial}{\partial s}.$$

□

We continue with the proof of a Poincaré-type lemma that was used in the proof of Theorem 4.1.

Lemma 4.1 *Let η be a closed l -form on $\hat{M} = M \times (-\delta, \delta)$ with $\eta|_{M \times \{0\}} = 0$. Then there exists an $(l-1)$ -form β on $\hat{M}$ such that $d\beta = \eta$ and $\beta|_{M \times \{0\}} = 0$.*

Proof Define $\Phi_\theta(x, s) = (x, e^\theta s)$ for $y = (x, s) \in \hat{M}$ and $\theta \in \mathbb{R}$. Note that $\frac{d}{d\theta}\Phi_\theta(y) = Y(\Phi_\theta(y))$, for $Y(x, s) = (0, s)$. Let us simply write M for $M \times \{0\}$. Since $\Phi_{-\infty}(x, s) = (x, 0)$, we have $\Phi_{-\infty}^*\eta = \eta|_M = 0$ by our assumption. We now write

$$\begin{aligned} \eta &= \Phi_0^*\eta - \Phi_{-\infty}^*\eta = \int_{-\infty}^0 \frac{d}{d\theta} \Phi_\theta^*\eta \, d\theta = \int_{-\infty}^0 \Phi_\theta^* \mathcal{L}_Y \eta \, d\theta \\ &= d \left[\int_{-\infty}^0 \Phi_\theta^* i_Y \eta \, d\theta \right] =: d\beta. \end{aligned}$$

For example, when $l = 2$,

$$\begin{aligned} \beta_{(x,s)}(v, \tau) &= \int_{-\infty}^0 \eta_{(x,e^\theta s)}((0, e^\theta s), (v, e^\theta \tau)) \, d\theta \\ &= \int_{-\infty}^0 \eta_{(x,e^\theta s)}((0, s), (v, e^\theta \tau)) \, e^\theta d\theta \\ &= \int_0^1 \eta_{(x,\theta s)}((0, s), (v, \theta \tau)) \, d\theta. \end{aligned}$$

From this we learn that β is well-defined and that $\beta|_M = 0$. The case of general l can be treated in the same way. $\square$

Definition 4.2 If the assumptions of Theorem 4.1 are satisfied, we say that the submanifold M is of *contact type*. $\square$

Remark 4.3 An important consequence of Theorem 4.1 is that a neighborhood of a submanifold M of contact type is isomorphic to $(\hat{M}, \hat{\omega})$. This means that such a neighborhood can be foliated into submanifolds that are, in some sense isomorphic to M . More precisely, if ϕ_t is the flow of the corresponding Liouville vector field X , then the hypersurfaces $(M^s = \phi_s(M) : s \in (-\delta, \delta))$ are all of contact type. In fact if we write $\ell(M^s)$ for the corresponding line bundle,

$$\ell_x(M^s) = \{v \in T_x M^s : \omega_x(v, w) = 0 \text{ for all } w \in T_x M^s\},$$

and $R^s(x)$ for the Reeb vector field associated with $(M^s, \alpha'_{|M^s})$, then we have the following identities:

$$(4.12) \quad (d\phi_s)_x(\ell_x(M^0)) = \ell_{\phi_s(x)}(M^s), \quad (d\phi_s)_x(R^0(x)) = e^s R^s(\phi_s(x)).$$

The second equation is a consequence of (4.10):

$$e^s = e^s \alpha_x(R^0(x)) = e^s \alpha'_x(R^0(x)) = (\phi_s^* \alpha')_x(R^0(x)) = \alpha'_{\phi_s(x)}((d\phi_s)_x(R^0(x))).$$

We may also define a Hamiltonian function $K = K_M : U \rightarrow (-\delta, \delta)$ with $K(M^s) = e^s$. In other words, if ψ is as in Theorem 4.1(iii) and $\hat{H} : \hat{M} \rightarrow (-\delta, \delta)$ is defined by $\hat{H}(x, s) = e^s$, then

$$(4.13) \quad K = \hat{H} \circ \psi^{-1}.$$

We can readily show

$$X_{\hat{H}}^{\omega}(x, s) = (R(x), 0).$$

From this we deduce

$$X_K^{\omega}(\phi_s(x)) = (d\psi)_{(x,s)}(R(x), 0) = (d\phi_s)_x(R(x)) = e^s R^s(\phi_s(x)),$$

because by Proposition 3.2 $X_K^{\omega} \circ \psi = (d\psi)X_{\hat{H}}^{\omega}$. In summary,

$$(4.14) \quad (X_K^{\omega})|_{M^s} = e^s R^s.$$

Hence the periodic orbits of K coincide with the closed orbits of $(R^s : s \in (-\delta, \delta))$. $\square$

As we learned from Remark 4.2, the hypersurfaces of contact-type may be regarded as the level sets of a Hamiltonian function for which the corresponding Hamiltonian vector field is closely related to the Reeb vector field. Now imagine that we start with a $2n$ dimensional symplectic manifold (N, ω) and a Hamiltonian vector field X_H^{ω} , and wonder whether or not level sets of H carry periodic orbits. We note that when the Hamiltonian function H is independent of time, then the level sets of H are conserved because

$$\frac{d}{dt} (H \circ \phi_t^H) = dH(X_H^{\omega} \circ \phi_t^H) = -\omega(X_H^{\omega}, X_H^{\omega}) \circ \phi_t^H = 0.$$

Let us write $M(c) = \{x : H(x) = c\}$, and set $\ell_x(c) := (T_x M(c))^{\perp}$. We call $M = M(c)$ *regular* if $dH_x \neq 0$ for all $x \in M(c)$. Since $\dim T_x M(c) = 2n - 1$ for a regular $M(c)$, we use Proposition 2.1(i) to assert

$$\dim \ell_x(c) = 1.$$

On the other hand, since

$$T_x M(c) = \{w : (dH)_x(w) = 0\},$$

we deduce that indeed $X_H(x) \in \ell_x(c)$, for $c = H(x)$ because for every $w \in T_x M(c)$,

$$\omega_x(X_H(x), w) = -(dH)_x(w) = 0.$$

This means that the line l_x is parallel $X_H(x)$. What we learn from this is that the existence of a periodic orbit of X_H on $M = M(c)$ is a property of M and ω and does not depend on H . In other words, if M possesses a periodic orbit of X_H and if $M = \{x : H'(x) = c'\}$ for another regular H' , then $X_{H'}$ possesses a periodic orbit as well. Evidently $l_x = (T_x M)^\perp$ offers an H -independent candidate for the tangent lines to the orbit. More precisely, if M is a closed hypersurface of a symplectic manifold (N, ω) , define $\ell_x(M) = (T_x M)^\perp$ which is a line. We always have $\ell_x(M) \subseteq T_x M$ because if $v \in \ell_x(M)$ but $v \notin T_x M$, then $v \perp (\ell_x(M) \oplus T_x M) = T_x N$ which implies $v = 0$ because ω is symplectic. As a result, $L_M = \bigsqcup_{x \in M} \ell_x$ is a line bundle of M , and we can express our question of the existence of periodic orbits purely in terms of this line bundle. The existence of periodic orbits is now reduced to the existence of closed characteristics of the line bundle L_M . It turns out that there are hypersurfaces M and symplectic structures ω such that the corresponding line bundle has no closed characteristics. However Weinstein's conjecture asserts that if M is of contact type, then such a closed characteristic can be found. In view of (4.14), the existence of a closed orbit of the Reeb vector field follows from the existence of a periodic orbit of the vector field X_K where K was defined by (4.13). From the preceding discussion and (4.14) we learn that we only need to find a periodic orbit in a neighborhood of a hypersurface of contact type:

Proposition 4.4 *Let M be a compact hypersurface of contact type of the symplectic manifold (M, ω) . Let $K = K_M : U \rightarrow \mathbb{R}$ be the corresponding Hamiltonian function that is defined for a neighborhood U of M as in (4.13). If the vector field X_K^ω has a periodic orbit in U , then the Reeb vector field of M has a closed orbit.*

Remark 4.4 As we mentioned in the introduction, Weinstein's conjecture has been established for the hypersurfaces of $(\mathbb{R}^{2n}, \bar{\omega})$ by Viterbo. As we will see in Section 6 below, we will use *Hofer-Zehnder Capacity* to give a rather straightforward proof of Viterbo's theorem. The point is that if Hofer-Zehnder Capacity of a neighborhood U of a contact type hypersurface M is positive, then by very definition of this capacity, a periodic orbit would exist for every Hamiltonian vector field with compact support inside U . From this it is not hard to deduce that X_{K_M} also possesses periodic orbits. We then use Proposition 4.4 to conclude that M carries a closed orbit. It is not known how to use the same line of reasoning to settle Weinstein's conjecture for general (N, ω) ; it is not known how to show the positivity of Hofer-Zehnder Capacity of open sets of general symplectic manifolds. Though the Weinstein's conjecture has now been established for all closed 3-dimensional manifolds by Taubes (2007) employing a variant of *Seiberg-Witten Floer homology*. $\square$

Starting from a symplectic manifold (N, ω) with $\dim N = 2n$, and a Hamiltonian function $H : N \rightarrow \mathbb{R}$ for which the level set $M = \{x : H(x) = c\}$ is compact and regular, we have seen that the line bundle $\ell_x(c) = (T_x M)^\perp$ is well-defined. If we already know that $\omega = d\alpha'$ for a 1-form α' , in a neighborhood U of M , then using the fact that ω is symplectic, we learn

that there exists a unique vector field X such that $i_X \omega = \alpha'$. On the other hand

$$i_{X_H^\omega} \omega = -dH, \quad \text{and} \quad i_X \omega = \alpha' \Rightarrow \alpha'(X_H^\omega) = dH(X).$$

Hence

$$X(x) \notin T_x M \text{ for all } x \in M \Leftrightarrow \alpha = \alpha'_M \text{ is of contact type.}$$

If $M = M(c)$ is of contact type, then we also have a natural volume form $\alpha \wedge (d\alpha)^{n-1}$ that is defined on M and we expect it to be invariant under the flow of X_H^ω . Indeed, if $(\hat{M}, \hat{\omega})$ with $\hat{\omega} = d\hat{\alpha}$ is as in Theorem 4.1, then

$$\hat{\Omega} := (n!)^{-1} \hat{\omega}^n = (n!)^{-1} e^{ns} ds \wedge \alpha \wedge (d\alpha)^{n-1} = d\hat{H} \wedge m,$$

where $\hat{H} = e^s$, and

$$m = (n!)^{-1} e^{(n-1)s} \alpha \wedge (d\alpha)^{n-1}.$$

Using ψ as in Theorem 4.1, we can write

$$(4.15) \quad \Omega := (n!)^{-1} \omega^n = dH \wedge \mu,$$

where $H = \hat{H} \circ \psi^{-1}$, and $\mu = (\psi^{-1})^* m$. In particular, $\mu|_M$ is a volume form for M when M is of contact type. We now assert that in general we can always find a $2n-1$ -form on M that satisfies (4.15).

Proposition 4.5 *Let (N, ω) be a symplectic manifold with $\dim N = 2n$, and assume that $H : N \rightarrow \mathbb{R}$ is a function for which the level set $M = \{x : H(x) = c\}$ is compact and regular.*

(i) *There exists a $(2n-1)$ -form μ such that the volume form $\Omega = (n!)^{-1} \omega^n$ can be expressed as*

$$(4.16) \quad \Omega = dH \wedge \mu,$$

in a neighborhood of M .

(ii) *The form $\mu|_M$ is uniquely determined by (4.16): If $\omega = dH \wedge \mu = dH \wedge \mu'$ then $\mu|_M = \mu'|_M$.*

(ii) *$(\phi_t^H)^* \mu|_M = \mu|_M$. In words, μ is invariant for ϕ_t^H , restricted to M .*

Proof (i) The existence of μ is an immediate consequence of the non-degeneracy of $dH \neq 0$ on M . (See Exercise 4.1(vi).)

(ii) Write $j : M \rightarrow N$ for the inclusion map. We certainly have $dH \wedge (\mu - \mu') = 0$. From Exercise 4.1(vii) below we deduce that for some $2n-2$ form γ , $\mu - \mu' = dH \wedge \gamma$. As a result, $\mathcal{I}(\mu - \mu') = \mathcal{I}(dH \wedge \gamma) = (\mathcal{I}dH) \wedge \mathcal{I}\gamma = 0$ because $\mathcal{I}dH = d(H \circ j) = 0$.

(iii) Write ϕ_t for ϕ_t^H . By (4.16),

$$\Omega = \phi_t^* \Omega = (\phi_t^* dH) \wedge \phi_t^* \mu = d(H \circ \phi_t) \wedge \phi_t^* \mu = dH \wedge \phi_t^* \mu.$$

By uniqueness result of (ii), we deduce $\mathcal{I}\phi_t^* \mu = \mathcal{I}\mu$, or $(\phi_t \circ j)^* \mu = \mathcal{I}\mu$. We may write $\phi_t \circ j = j \circ \phi_t$ where the second $\phi_t : M \rightarrow M$ is the restriction of ϕ_t to M . Hence $\Phi_t^* \mathcal{I}\mu = \mathcal{I}\mu$, as desired. $\square$

Remark 4.4 When $(N, \omega) = (\mathbb{R}^{2n}, \bar{\omega})$, we can describe the form μ of (4.16) explicitly. To

ease the notation, let us write $k = 2n$. Note that if $\{g_j : j = 1, \dots, k\}$ is a collection of smooth functions with $\sum_{j=1}^k g_j H_{x_j} (-1)^{j-1} \equiv 1$, then we may choose for μ ,

$$\mu = \sum_{j=1}^k g_j dx_1 \wedge \dots \wedge \widehat{dx_j} \wedge \dots \wedge dx_k.$$

For example $g_j = (-1)^{j-1} H_{x_j} / |\nabla H|^2$ would do the job.) We now claim

$$(4.17) \quad \int_{\Gamma} \alpha = c_0 \int_{\Gamma} |\nabla H|^{-1} d\sigma,$$

for a constant c_0 . To see this, let us assume $H_{x_k} \neq 0$ so that locally we can find a function $f(x_1, \dots, x_{k-1})$ with

$$H(x_1, \dots, x_{k-1}, f(x_1, \dots, x_{k-1})) = c.$$

This means that the graph of f gives a parametrization of M and that $f_{x_j} = -\frac{H_{x_j}}{H_{x_k}}$ for $j = 1, \dots, k-1$. On the other we can use the function f to express the volume measure on M that is induced by the standard volume of $\mathbb{R}^k$. More precisely, consider the vector $a = (a_1, \dots, a_k)$ with

$$a_j = \frac{\partial(x_1, \dots, \hat{x}_j, \dots, x_k)}{\partial(x_1, \dots, x_{k-1})} = (-1)^{k-j} \frac{H_{x_j}}{H_{x_k}},$$

for $1 \leq j \leq k$. If $\Gamma = \{(x_1, \dots, x_{k-1}, f(x_1, \dots, x_{k-1})) : (x_1, \dots, x_k) \in U\}$, then

$$\int_{\Gamma} \mu = \int_U \left(\sum_{j=1}^k a_j g_j \right) dx_1, \dots, dx_{k-1}.$$

On the other hand

$$\bar{a} = \left(\sum_{j=1}^k a_j^2 \right)^{1/2} = \frac{|\nabla H|}{|H_{x_k}|}.$$

Since locally either $H_{x_k} > 0$ or $H_{x_k} < 0$, we have

$$\begin{aligned} \int_{\Gamma} \mu &= \int_U (-1)^{k-1} \frac{1}{H_{x_k}} dx_1, \dots, dx_{k-1} \\ &= \pm \int_U \frac{1}{|\nabla H|} \bar{a} dx_1, \dots, dx_{k-1} = \pm \int_{\Gamma} \frac{1}{|\nabla H|} d\sigma, \end{aligned}$$

proving (4.17). Note that this is consistent with the coarea formula

$$\int f dx_1, \dots, dx_k = \int_{-\infty}^{\infty} \left[\int_{H=c} f \frac{d\sigma}{|\nabla H|} \right] dc.$$

□

Given a symplectic manifold (N, ω) , and a Hamiltonian function $H : N \rightarrow \mathbb{R}$, assume that $M = M(c)$ is a regular closed level set of H . With a slight abuse of notation, we write ω for the restriction of ω to M . We claim that we can always find a 1-form α on M such that (M, α, ω) is a framed Hamiltonian manifold, and that the Hamiltonian vector field X_H is the corresponding Reeb vector field. For this, pick an almost complex structure J on N that is compatible with ω , and set $\xi'_x = T_x M \cap J_x T_x M$. Note that $X_H \notin \xi'_x$ because for $v \in T_x M$,

$$0 = dH(v) = -\omega(X_H, v) = -\omega(JX_H, v) = g(X_H, v),$$

where g is the corresponding metric. From this it is clear that $\xi_x \cap \ell_x(\omega) = \{0\}$. We then choose α so that $\xi(\alpha) = \xi'$, and $\alpha(X_H) = 1$.

We now give examples of contact type hypersurfaces.

Example 4.2(i) Let $N = T^*Q$ for an n -dimensional manifold Q , and let $\omega = d\lambda$ denotes its standard symplectic form as was defined in Example 3.1. The level set M of $H : T^*Q \rightarrow \mathbb{R}$ is of contact type if

$$(4.18) \quad \lambda_{(q,p)}(X_H(q,p)) = p((d\pi)_{(p,q)}(X_H(q,p))) \neq 0,$$

on M . When $Q = \mathbb{R}^n$, then (4.19) means that $p \cdot H_p \neq 0$ on M .

(ii) Let M be as in **(i)** and assume that $H : T^*Q \rightarrow \mathbb{R}$ is homogeneous of degree $l > 0$ in the p -variable: for every $r > 0$,

$$(4.19) \quad H(q, rp) = r^l H(q, p).$$

We claim that this condition is equivalent to

$$(4.20) \quad i_{X_H} \lambda = lH.$$

The proof of the equivalence of (4.19) and (4.20) is straightforward when $Q = \mathbb{R}^n$: By differentiating both sides of (4.19) we can show that H is homogeneous of degree $l > 0$ iff $p \cdot H_p = lH$, which is exactly (4.20). The proof of general Q is similar. Use the Darboux charts we defined in Example 3.1, namely take $h : U \rightarrow \mathbb{R}^n$ as a chart of Q and use this to construct $\bar{h} : T^*U \rightarrow \mathbb{R}^{2n}$ so that $\lambda = \bar{h}^*\bar{\lambda}$. Define $\bar{H} : h(U) \times \mathbb{R}^n \rightarrow \mathbb{R}$ so that $\bar{H} \circ \bar{h} = H$. By the construction,

$$\bar{h}(q, p) = (h(q), \bar{p}) \Rightarrow \bar{h}(q, rp) = (h(q), r\bar{p}),$$

for every $r > 0$. This implies that H is homogeneous of degree $l > 0$ iff $\bar{H}$ is homogeneous of degree $l > 0$. Similarly,

$$i_{X_H} \lambda = l\bar{H} \Leftrightarrow \bar{h}^* \left(i_{X_{\bar{H}}} \bar{\lambda} \right) = l\bar{H} \circ \bar{h} \Leftrightarrow i_{X_H} \lambda = lH.$$

This completes the proof of the equivalence of (4.19) and (4.20). From this and (4.18) we deduce that $M = \{x : H(x) = c\}$ is always of contact type for every homogeneous H and non-zero c . $\square$

Remark 4.5 We note that the Hamiltonian vector field X_H on T^*Q preserves the 1-form λ if and only if $H(q, p)$ is (positively) homogeneous of degree 1 in p . Indeed by Proposition 3.1, the form λ is preserved by the flow of X_H iff

$$0 = \mathcal{L}_{X_H} \lambda = d \circ i_{X_H} \lambda + i_{X_H} \circ d\lambda = d(i_{X_H} \lambda - H).$$

This is equivalent to assert that $i_{X_H} \lambda - H$ is a constant. Since adding a constant does not change X_H , we may assume that the constant is zero so that the condition now is $i_{X_H} \lambda = H$. In view of the equivalence of (4.19) and (4.20), we deduce that $\mathcal{L}_{X_H} \lambda = 0$ is equivalent to the degree-1 p -homogeneity of H . $\square$

The positions of a conservative system are points in an n -dimensional manifold M that is known as the configuration space. The phase space in Lagrangian formulation is the tangent bundle TQ . The motion is determined by a Lagrangian $L : TQ \rightarrow \mathbb{R}$. In the Hamiltonian formulation we use a Hamiltonian function $H : M = T^*Q \rightarrow \mathbb{R}$ to determine the motion of the system. In our next example we study the level set of classical Hamiltonians.

Example 4.3 Observe that the Riemannian metric g on a manifold Q , allows us to define an operator $\sharp : T^*Q \rightarrow TQ$ that maps 1-forms to vector fields by requiring

$$g_q((\sharp p)_q, v) = p_q(v),$$

for every vector $v \in T_q M$. This duality also induces a metric on T^*Q by

$$G_q(p, p') = g_q(\sharp p, \sharp p').$$

Given a smooth potential energy $V : Q \rightarrow \mathbb{R}$, we define

$$(4.21) \quad H(q, p) = \frac{1}{2}G_q(p, p) + V(q).$$

We now claim

$$(4.22) \quad \lambda(X_H^\omega) = G_q(p, p),$$

for H given by (4.21). The proof is very similar to the equivalence of (4.19) and (4.20); using h and $\bar{h}$ as in Examples 3.1 and 4.2, we can find a metric $\bar{G}$ on $h(U)$ and a function $\bar{V} : h(U) \rightarrow \mathbb{R}$ such that $V = \bar{V} \circ h$, and $H = \bar{H} \circ \bar{h}$, where

$$\bar{H}(q, p) = \frac{1}{2}\bar{G}_q(p, p) + \bar{V}(q).$$

Writing $\bar{G}_q(p, p) = A(q)p \cdot p$ with $A > 0$, we deduce that $X_{\bar{H}}^\omega = (S(q)p, -\nabla V(q))$. From this we can readily deduce

$$\bar{\lambda}(X_{\bar{H}}^\omega) = \bar{G}_q(p, p).$$

We then apply $\bar{h}^*$ to both sides to arrive at (4.22). Using the elementary Lemma 4.10, we can now readily show that if

$$M_E = \{(q, p) : H(q, p) = E\}$$

is compact and regular, and if $E > \max V$, then M_E is of contact type simply because for such E we always have $p \neq 0$, whenever $(q, p) \in M_E$. It turns out that M_E is of contact type even when $E \leq \max V$ (See [HZ]). It is worth mentioning that when $E > \max V$, then we can define a new Riemannian metric

$$\hat{G}_q(p, p) = \frac{G_q(p, p)}{E - V(q)},$$

that is known as the *Jacobi metric*. We then have

$$M_E = \{(q, p) : H(q, p) = E\} = \left\{ (q, p) : \frac{1}{2}\hat{G}_q(p, p) = 1 \right\}.$$

The Hamiltonian $\hat{H}(q, p) = \frac{1}{2}\hat{G}_q(p, p)$ induces a Hamiltonian vector field $X_{\hat{H}}$. It is simply related to X_H by

$$(4.23) \quad X_H(q, p) = G_q(p, p)X_{\hat{H}}(q, p), \quad (q, p) \in S.$$

Hence a periodic orbit exists on M_E for X_H if the same is true for $X_{\hat{H}}$. Geometrically, $X_{\hat{H}}$ generates the geodesic flow defined by the Jacobi metric $\hat{G}$ on $M = T^*Q$. $\square$

We next discuss some examples of non-contact types hypersurfaces.

Example 4.3 (Zehnder [Z])

(i) Let $\hat{M} = \mathbb{T}^3 \times I$ where $\mathbb{T}^3$ is the 3-dimensional torus and I is an open interval. We write $\theta = (\theta_1, \theta_2, \theta_3) \in \mathbb{T}^3$ for points in $\mathbb{T}^3$ and define $H : \hat{M} \rightarrow I$ by $H(\theta, s) = s$. The level sets of H are $M_s = \mathbb{T}^3 \times \{s\}$ for $s \in I$. Writing $(x_1, \dots, x_4)$ for $(\theta_1, \theta_2, \theta_3, s)$, any constant skew-symmetric $C = [c_{ij}]$, define a 2-form $\omega = \sum_{i < j} c_{ij} dx_i \wedge dx_j$. Note that each $d\theta_i$ is closed (but not exact), and ds is exact. Hence ω is always closed because C is constant. As a result, ω is symplectic iff C is invertible. Given a vector $\zeta = (\zeta_1, \zeta_2, \zeta_3)$ with $\zeta_3 \neq 0$, we define C by

$$-C^{-1} = \hat{C} := \begin{bmatrix} 0 & 1 & 0 & \zeta_1 \\ -1 & 0 & 0 & \zeta_2 \\ 0 & 0 & 0 & \zeta_3 \\ -\zeta_1 & -\zeta_2 & -\zeta_3 & 0 \end{bmatrix}.$$

We further assume that ζ is *irrational* in the following sense:

$$a \in \mathbb{Z}^3, \quad a \cdot \zeta = 0 \quad \Rightarrow \quad a = 0.$$

Evidently, $X_H = X_H^\omega = \hat{C} \nabla H$ that in our case leads to $X_H = (\xi, 0)$. Hence $\phi_t^H(\theta, s) = (\theta + t\zeta, s)$ where $\theta + s\zeta$ is understood as a *(mod 1)* summation; here we identify $\mathbb{T}$ as the interval $[0, 1]$ with 0 and 1 regarded as the same point. The irrationality ζ guarantees that X_H has no periodic orbit. Hence $\omega|_{M_s}$ does not induce a contact structure.

(ii) Let us write $x = (q_1, q_2, p_1, p_2)$ for angles in a 4-dimensional torus $\mathbb{T}^4$ that may be defined as $[0, 1]^4$ with 0 = 1. Take a 1-periodic function $f : \mathbb{R} \rightarrow \mathbb{R}$ that can be regarded as a function on the circle $\mathbb{T}$. Let $\omega = dp_1 \wedge dq_1 + dp_2 \wedge dq_2 + \zeta dp_1 \wedge dq_2$ for an irrational number ζ . Take $H(x) = f(p_1)$ so that $X_H^\omega(x) = (f'(p_1), \zeta f'(p_1), 0, 0)$. The corresponding flow is

$$\phi_t^H(x) = (q_1 + t f'(p_1), q_2 + \zeta f'(p_1), p_1, p_2),$$

with *(mod 1)* additions. Again since ζ is irrational, $\phi_t^H(x)$ is never periodic whenever $f'(p_1) \neq 0$. $\square$

As we mentioned in the beginning of this section, contact structures in $\mathbb{R}^3$ can be used to produce exotic structures. The process of going from a contact manifold (M, α) to the symplectic manifold $(\hat{M}, \hat{\omega})$ as in Proposition 4.1 is called the *symplectization*. For example we equip

$$\mathbb{R}^{2n-1} = \{(q, p, z) : q, p \in \mathbb{R}^{n-1}, z \in \mathbb{R}\},$$

with the so-called *standard contact form* $\bar{\alpha} = p \cdot q + dz$, then the corresponding symplectization $(\mathbb{R}^{2n}, \hat{\omega})$ is isomorphic to the standard $(\mathbb{R}^{2n}, \bar{\omega})$. However if we equip $\mathbb{R}^3$ with an *overtwisted* contact form, the symplectization yields a symplectic form on $\mathbb{R}^4$ that is not equivalent to the standard symplectic structure. Before giving the definition of overtwisted contact structures,

let us observe that if (M, α) is contact, then $\xi = \ker \alpha$ is never integrable. The reason is that the integrability of ξ is equivalent to $\alpha \wedge d\alpha = 0$ (see Lemma A.4 in Appendix), whereas the contact assumption requires $\alpha \wedge (d\alpha)^{n-1}$ to be a volume form by Proposition 4.3. In fact we have a following results for integrals of ξ :

Proposition 4.6 *Let (M, α) be a contact manifold of dimension $2n - 1$. If Γ is an l dimensional submanifold of M with $T_x\Gamma \subseteq \xi_x = \ker \alpha_x$ for every $x \in \Gamma$, then $\dim \Gamma \leq n - 1$.*

Proof Using Lemma A.3 of the Appendix, we can readily show that if X and Y are two vector fields that are tangent to Γ , then $d\alpha(X, Y) = 0$. Simply because for $X, Y \in \xi$, we have $\alpha(X) = \alpha(Y) = 0$, and from $X, Y \in T\Gamma$ we know that $[X, Y] \in T\Gamma$ by Frobenius' theorem. This means that if we use the symplectic form $\omega = d\alpha|_\xi$ for orthogonality, then $T_x\Gamma \subseteq (T_x\Gamma)^\perp$, which by Proposition 2.1 implies

$$2n - 2 = \dim T_x\Gamma + \dim(T_x\Gamma)^\perp \geq 2 \dim T_x\Gamma.$$

This clearly implies that $\dim T_x\Gamma \leq n - 1$. □

Definition 4.3 We call a submanifold Γ of a $2n - 1$ dimensional contact manifold (M, α) Legendrian if $T_x\Gamma \subset \xi_x$ and $\dim \Gamma = n - 1$. □

Let us now examine contact forms on $\mathbb{R}^3$. If α is a contact form on $\mathbb{R}^3$ and $\xi = \ker \alpha$ is the corresponding contact structure, we expect that the plan field ξ experience some twists as we move along a hypersurface because ξ is not integrable. Given a 2 dimensional surface Γ , we may define a plane field $\gamma_x = T_x\Gamma \cap \xi_x$. For generic points of Γ , γ_x is a line and the integrals of this line bundle have been used to classify the contact structures in $\mathbb{R}^3$ or even arbitrary 3-dimensional manifolds. The simplest way to produce examples of exotic contact structures in $\mathbb{R}^3$ is using cylindrical coordinates; $q = r \cos \theta, p = r \sin \theta$ so that if $x = (q, p, z)$, then

$$dq = \cos \theta \, dr - \sin \theta \, r d\theta, \quad dp = \sin \theta \, dr + \cos \theta \, r d\theta.$$

Example 4.4(i) If $\alpha = qdp - pdq - dz = r^2 \, d\theta - dz$, then for $x = (q, p, z)$,

$$\xi_x = \{(\hat{q}, \hat{p}, \hat{z}) : q\hat{p} - p\hat{q} = \hat{z}\}.$$

Observe that for $\Gamma = \{z = 0\}$, the set $\gamma_x = T_x\Gamma \cap \xi_x$ is a line only when $x \neq 0$. More precisely,

$$\gamma_x = \{(\hat{q}, \hat{p}, 0) : q\hat{p} - p\hat{q} = 0\},$$

is the ray $\{(sq, sp, 0) : s \in \mathbb{R}\}$ if $x \neq 0$, whereas $\xi_0 = \Gamma$. So the line bundle $\{\gamma_x : x \in \Gamma\}$ has a singularity at 0. For example, on the $\{(s, 0, 0) : s > 0\}$, the vector $u = (-p, q, -1)$ twists from $-\pi/2$ to 0 as s goes from 0 to ∞ .

(ii) Let $\beta = \sin r \, rd\theta + \cos r \, dz$, so that

$$\xi_x = \{(\hat{q}, \hat{p}, \hat{z}) : (r \tan r)\hat{\theta} = \hat{z}\},$$

where $(1 + \tan^2 \theta)\hat{\theta} = q^{-2}(qdp - pdq)$. Note that the curve $\{r = \pi, z = 0\}$ is Legendrian. If we set

$$\Gamma_\varepsilon = \{(q, p, \varepsilon) : r \leq \pi\},$$

then the interior Γ_ε has no singular point for $\varepsilon > 0$ and $\partial\Gamma$ is Legendrian. We say a contact form is overtwisted if such $\Gamma = \Gamma_\varepsilon$ exists for β . Note that for this particular example, if we move along a ray emanating from the origin, the plane ξ_x make complete turns infinitely many times. $\square$

Given a framed Hamiltonian manifold (M, α, ω) , we may write

$$T_{(x,s)}\hat{M} = \zeta_x(\omega) \oplus \xi_x(\alpha),$$

where $\zeta_x = \zeta_x(\omega)$ is the 2-dimensional space that is spanned by R_x and ∂_s . We may search for almost complex structures $\hat{J} : T\hat{M} \rightarrow T\hat{M}$ that respects this splitting. In other words, $\hat{J}(a, b) = (J^0 a, Jb)$ for $a \in \zeta$ and $b \in \xi$, with $J^0 : \zeta \rightarrow \zeta$ and $J : \xi \rightarrow \xi$. We say that $\hat{J}$ is compatible with (α, ω) if $g(a, b) = \omega(a, Jb)$ is a metric on ξ , and $g^0(a, b) = \omega(a, J^0 b)$ is a metric on ζ .

Exercise 4.1

(i) Verify (4.6)-(4.8).

(ii) Let M be a manifold of dimension $2n - 1$ and let α be a 1-form on M . Show that if $\alpha \wedge (d\alpha)^{n-1}$ is a volume form, then α is contact.

(iii) Consider $(\mathbb{R}^{2n}, \bar{\omega})$ and let S be the boundary of a set A that is star-shaped with respect to the origin. Assume that each ray emanating from the origin intersect S at exactly one point and $X(x) = x/2$ traverses to S . Show that S is of contact type. Determine the corresponding contact form. **Hint:** Show that X is a Liouville vector field.

(iv) Consider the contact manifold $(\mathbb{R}^{2n-1}, \bar{\alpha})$. Let $H(q, p, z)$ be a smooth function. Show that the corresponding contact vector field Z_H is given by

$$(H_p, -H_q + pH_z, -p \cdot H_p + H).$$

(v) Let S^{2n-1} be a unit sphere in $\mathbb{R}^{2n}$. The form $\bar{\lambda}$ induces a 1-form $\bar{\mu}$ on S^{2n-1} . Show that $(S^{2n-1}, \bar{\mu})$ is not a contact manifold. Define $\mu = \frac{1}{2}(p \cdot dq - q \cdot dp)$. Show that (S^{2n-1}, μ) is a contact manifold. Find its Reeb flow.

- (vi) Let Ω be a volume form and let τ be a non-degenerate 1-form. Show that there exists a form β such that $\Omega = \tau \wedge \beta$.
- (vii) Let η be an l -form and τ a 1-form with $\tau \wedge \eta = 0$. Show that $\eta = \tau \wedge \gamma$ for some $(l - 1)$ -form γ .
- (viii) Verify (4.22). □

5 Variational Principle and Convex Hamiltonian

In this section, we use variational techniques to prove the following result of A. Weinstein:

Theorem 5.1 *Assume that the hypersurface $S \subseteq \mathbb{R}^{2n}$ is the smooth boundary of a compact strictly convex region. Then the Reeb's vector field on S has a periodic orbit.*

In fact Theorem 5.1 allows us to define some kind of a *symplectic width* or *capacity* of convex sets K with nonempty interiors:

$$(5.1) \quad \mathbf{c}_0(K) := \inf \{ |A(\gamma)| : \gamma \text{ is a Reeb characteristic of } \partial K \} \\ = \inf \{ T : \text{There is a } T\text{-periodic orbit of the Reeb's vector field on } \partial K \},$$

where the *symplectic action* $A(\gamma)$ of γ was defined by (3.1). Of course Theorem 5.1 guarantees that $\mathbf{c}_0(K) < \infty$ for every bounded convex set K of nonempty interior. Our strategy for proving Theorem 5.1 is by figuring out an alternative *dual* like variational principle of (5.1). Once this dual problem is formulated, we can readily establish its finiteness.

Remark 5.1 Let us write

$$\bar{\alpha} = \frac{1}{2}(p \cdot dq - q \cdot dp) \quad \text{or} \quad \bar{\alpha}_x(v) = \frac{1}{2} \bar{J}x \cdot v,$$

so that $d\bar{\alpha} = \bar{\omega}$. Let $K = \{x : H(x) \leq r\}$ be a bounded convex set, where H is a Hamiltonian function and $r \in \mathbb{R}$ is a constant. Assume that $x : \mathbb{R} \rightarrow \mathbb{R}^{2n}$ is a non-constant T -periodic function with $H(x) = r$ and $\dot{x} = \bar{J}\nabla H(x)$. To turn x into an $\bar{\alpha}$ -Reeb orbit, set $\eta(t) = \frac{1}{2} \int_0^t x \cdot \nabla H(x) ds$, and choose τ with $(\tau \circ \eta)(t) = t$. If $\gamma = x \circ \tau$, then γ is periodic of period $T' = \eta(T) = A(x)$, and

$$\bar{\alpha}_\gamma(\dot{\gamma}) = \frac{1}{2} \bar{J}\gamma \cdot \dot{\gamma} = \dot{\tau}(\bar{J}x \circ \tau) \cdot (\bar{J}\nabla H(x \circ \tau)) = \dot{\tau}(\dot{\eta} \circ \tau) = 1.$$

We have

$$A(x) = A(\gamma) = \int_\gamma \bar{\alpha} = T'.$$

□

Example 5.1 Assume that U is an ellipsoid with $0 \in U$. We learned in Chapter 2 that there are radii $r_1 \leq \dots \leq r_n$ and a linear symplectic T such that $T(U) = E$, where $E = \{x : H(x) \leq 1\}$, with

$$H(x) = \sum_{j=1}^n \frac{q_j^2 + p_j^2}{r_j^2}.$$

The corresponding Hamiltonian flow is $\phi_t(z_1, \dots, z_n) = (e^{-i\lambda_1 t} z_1, \dots, e^{-i\lambda_n t} z_n)$, where $z_j = q_j + ip_j$, and $\lambda_j = 2/r_j^2$. For $y = (z_1, \dots, z_n)$ and $\gamma : [0, T] \rightarrow \mathbb{R}^{2n}$, defined by $\gamma(t) = \phi_t(y)$, we have

$$\begin{aligned} A(\gamma) &= \frac{1}{2} \int_0^T \bar{\omega}(\gamma(s), \dot{\gamma}(s)) \, ds = \frac{1}{2} \int_0^T \sum_j \operatorname{Im} (e^{-i\lambda_j s} z_j \, i\lambda_j e^{i\lambda_j s} \bar{z}_j) \, ds \\ &= \frac{1}{2} T \sum_j \lambda_j |z_j|^2 = TH(\gamma(0)). \end{aligned}$$

If this γ lies on ∂E , we must have $H(\gamma(0)) = 1$ which means that $A(\gamma) = T$ for such γ . Now if γ is a periodic orbit of period $T > 0$, then we must have

$$z_j \neq 0 \quad \Rightarrow \quad \lambda_j T = 2k\pi, \quad \text{for some } k \in \mathbb{N}.$$

This implies $T \geq 2\pi/\lambda_1 = \pi r_1^2$. Hence, $\mathbf{c}_0(E) \geq \pi r_1^2$. On the other hand, if we choose $(z_1, \dots, z_n)$ satisfying $|z_1| = r_1$, and $z_j = 0$ for $j \neq 1$, we have that $A(\gamma) = T = \pi r_1^2$ for the corresponding γ . From all this we deduce that for an ellipsoid E ,

$$(5.2) \quad \mathbf{c}_0(E) = \pi r_1(E)^2.$$

□

Before embarking on finding the dual problem associated with (5.1), let us first review the Lagrangian formulation of Hamiltonian systems. As we will see below, this formulation is not well-suited for finding periodic characteristics on the boundary of a convex set. Nonetheless we will learn some fruitful ideas from it that will play essential role later in showing the finiteness of $\mathbf{c}_0(U)$.

To study Newton's equation with constraints, Lagrange initiated variational formulation of conservative mechanical problems. Let $L : TQ \rightarrow \mathbb{R}$ be a C^1 -function. Given $q^0, q^1 \in Q$, we define $\mathcal{B} : \Gamma_T(q^0, q^1) \rightarrow \mathbb{R}$ with

$$\begin{aligned} \Gamma_T(q^0, q^1) &= \{\gamma : [0, T] \rightarrow Q \text{ is } C^1 \text{ and } \gamma(0) = q^0, \gamma(T) = q^1\}, \\ \mathcal{B}(\gamma) &= \int_0^T L(\gamma(t), \dot{\gamma}(t)) dt. \end{aligned}$$

Let us denote the argument of L by (q, v) with $v \in T_q Q$ and write L_q and L_v for the partial derivatives of L .

We now claim that if $q(\cdot)$ is a critical point of $\mathcal{B}$, then q solves the Euler–Lagrange–Newton's equation

$$(5.3) \quad \frac{d}{dt} L_v(q, \dot{q}) = L_q(q, \dot{q}).$$

Indeed if we use the L^2 -inner product, then the derivative of $\mathcal{B}$ is given by

$$(5.4) \quad \partial \mathcal{B}(\gamma) = L_q(\gamma, \dot{\gamma}) - \frac{d}{dt} L_v(\gamma, \dot{\gamma}),$$

for every $\gamma \in \Gamma_T(q^0, q^1)$. To see this, take any $\eta = \eta(\theta, t)$ with $\theta \in (-\delta, \delta)$ and $t \in [0, T]$, such that $\eta_\theta(0, t) = \tau(t)$, and $\eta(0, \cdot) = \gamma(\cdot)$. Now

$$\begin{aligned} \langle \partial \mathcal{B}(\gamma), \tau \rangle &:= \left. \frac{d}{d\theta} \mathcal{B}(\eta(\theta, \cdot)) \right|_{\theta=0} \\ &= \int_0^T [L_q(\gamma, \dot{\gamma}) \cdot \tau + L_v(\gamma, \dot{\gamma}) \cdot \dot{\tau}] dt \\ &= \int_0^T \left[L_q(\gamma, \dot{\gamma}) - \frac{d}{dt} L_v(\gamma, \dot{\gamma}) \right] \cdot \tau dt. \end{aligned}$$

Here we used the fact that $\tau(0) = \tau(T) = 0$ which follows from $\eta(\theta, 0) = q^0$, $\eta(\theta, T) = q^1$ for all θ . As an example, let $L(q, v) = \frac{m}{2}|v|^2 - V(q)$ in $\mathbb{R}^n$. Then the equation (5.3) reads as $m\ddot{q} = -\nabla V(q)$.

To explain the connection between Lagrangian formulation with the the Hamiltonian, let us assume that $Q = \mathbb{R}^n$ and that for $p \in \mathbb{R}^n = T_q^*Q$, we can solve the relationship

$$p = L_v(q, v)$$

uniquely for v . Denoting the solution by $v(q, p)$, and setting

$$H(q, p) = p(v(q, p)) - L(q, v(q, p)),$$

we learn

$$\begin{aligned} H_p &= v + pv_p - v_p L_v = v + pv_p - pv_p = v, \\ H_q &= pv_q - L_q(q, v) - L_v(q, v)v_q = -L_q(q, v). \end{aligned}$$

Hence

$$(5.5) \quad p = L_v(q, v) \Leftrightarrow H_p(q, p) = v,$$

and

$$(5.6) \quad H_q(q, p) = -L_q(q, v).$$

If we set $p(t) = L_v(q(t), \dot{q}(t))$, then by (5.3), (5.5), and (5.6),

$$\dot{q} = H_p(q, p), \quad \dot{p} = -H_q(q, p).$$

The inversion (5.5) is possible if we assume that L is strictly convex in v . In this case H can be constructed from L by *Legendre transform*:

$$H(q, p) = \sup_v (p \cdot v - L(q, v)).$$

More generally, if $L : TQ \rightarrow \mathbb{R}$ is convex in v , we may define $H : T^*Q \rightarrow \mathbb{R}$ by

$$H(q, p) = \sup_v (p(v) - L(q, v)).$$

This relationship may be inverted to yield

$$L(q, v) = \sup_p (p(v) - H(q, p)).$$

If we assume that L has superlinear growth in variable v as $|v| \rightarrow \infty$, then H is finite and the supremum is attained at $v = v(q, p)$ so that (5.5) is valid.

For a v -convex L , we may find solutions to (5.1) by minimizing $\mathcal{B}$. For example, if for some constants $c_1, c_2 > 0$ and $\alpha > 1$,

$$(5.7) \quad L(q, v) \geq c_1 |v|^\alpha - c_2,$$

for every $v \in T_q N$ and $q \in N$, and if γ_l is a sequence in $\Gamma_T(q^0, q^1)$ such that $\lim_{l \rightarrow \infty} \mathcal{B}(\gamma_l) = A = \inf \mathcal{B}$, then by (5.7) we have the bound

$$\sup_l \int_0^T |\dot{\gamma}_l(t)|^\alpha dt < \infty.$$

This bound allows us to extract a subsequence of γ_l which converges weakly with respect to the topology of Sobolev space $W^{1,\alpha}$. It turns out that $\mathcal{B}$ is lower semicontinuous because L is convex in v . This allows us to deduce that for any limit point $q(\cdot)$ of the sequence $\gamma_l(\cdot)$, we have that $\mathcal{B}(q) \leq A$. Since $A = \inf \mathcal{B}$, we learn that $\mathcal{B}(q) = A$ and that the infimum is achieved.

In spite of the appeal of the above argument, it is not clear how to use it to prove Theorem 5.1. Recall that we are searching for a periodic solution on a given energy surface. Of course we could have chosen $q^0 = q^1$ so that our solution satisfies $q(0) = q(T)$. But to lie on the surface S we need to make sure that $(q(0), p(0)) \in S$. In fact the solution we have found may not even be periodic in (q, p) coordinates because we cannot guarantee $p(0) = p(T)$.

To this end let us define another functional $\mathcal{A}$ of which critical points solve the Hamiltonian system. Set $\mathcal{A} : \Gamma_T \rightarrow \mathbb{R}$ to be

$$\begin{aligned} \mathcal{A}(x(\cdot)) &= \int_{x(\cdot)} [p \cdot dq - H(x)] dt \\ &= \int_0^T \left[\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right] dt, \end{aligned}$$

where

$$\Gamma_T = \{x : \mathbb{R} \rightarrow \mathbb{R}^{2n} \text{ is } C^1 \text{ and } T\text{-periodic}\}.$$

We have

$$(5.8) \quad \partial \mathcal{A}(x(\cdot)) = -\bar{J}\dot{x} - \nabla H(x),$$

because

$$\begin{aligned} \langle \partial \mathcal{A}(x(\cdot)), \tau(\cdot) \rangle &= \left. \frac{d}{d\delta} \mathcal{A}(x + \delta\tau) \right|_{\delta=0} \\ &= \int_0^T \left[\frac{1}{2} \bar{J}\tau \cdot \dot{x} + \frac{1}{2} \bar{J}x \cdot \dot{\tau} - \nabla H(x) \cdot \tau \right] dt \\ &= - \int_0^T (\bar{J}\dot{x} + \nabla H(x)) \cdot \tau dt \end{aligned}$$

for every $x, \tau \in \Gamma_T$. From (5.8) we learn that $\partial \mathcal{A}(x(\cdot)) \equiv 0$ iff x solves

$$(5.9) \quad \dot{x} = \bar{J}\nabla H(x).$$

Note that $\mathcal{A}$ involves H explicitly and no additional assumption such as convexity is needed. However typically the critical points of $\mathcal{A}$ are saddle points and it is helpless to search for (local) maximizers or minimizers. Because of this, finding critical points for $\mathcal{A}$ is far more challenging. Before discovering a remedy for this, let us show that a hypersurface S as in Theorem 5.1 can be realized as a level set of a convex homogeneous Hamiltonian function.

Definition 5.1 Let $U \subseteq \mathbb{R}^{2n}$ be an open convex set with $0 \in U$. Set $K = \bar{U}$ for its closure.

(i) The *gauge function* associated with K is defined by

$$g_K(x) = \|x\|_K = \inf\{r > 0 : x/r \in K\} = \inf\{r > 0 : x/r \in U\}.$$

(ii) The *polar set* associated with K is defined by

$$K^o := \{x : x \cdot y \leq 1 \text{ for all } y \in K\}.$$

(iii) The *support function* h_K associated with K is defined by

$$(5.10) \quad h_K(x) := \sup_y \{x \cdot y : y \in K\}.$$

(iii) The indicator function of K is a convex function that is defined by

$$\chi_K(x) := \begin{cases} 0 & \text{if } x \in K; \\ \infty & \text{otherwise,} \end{cases}$$

□

Proposition 5.1 *Let $U \subseteq \mathbb{R}^{2n}$ be an open convex set with $0 \in U$ and $K = \bar{U}$.*

(i) *The gauge function $\|\cdot\|_K$ is a norm. Its Legendre transform is given by*

$$\sup_y (x \cdot y - \|y\|_K) = \chi_{K^\circ}(x).$$

(ii) *We have*

$$(5.11) \quad K = \{y : x \cdot y \leq h_K(x) \text{ for all } x\}.$$

(iii) *We have $K^{\circ\circ} = K$. Moreover, $x \cdot y \leq \|x\|_K \|y\|_{K^\circ}$, and*

$$(5.12) \quad \|x\|_{K^\circ} = h_K(x), \quad \|x\|_K = h_{K^\circ}(x).$$

(iv) *If $H_K(x) = \frac{1}{2}\|x\|_K^2$, then its Legendre transform is given by $H_K^* = \frac{1}{2}h_K^2 = H_{K^\circ}$.*

Proof The proof of (i) and (ii) are left as an exercise. As for (iii), observe

$$\begin{aligned} \|x\|_{K^\circ} &= \inf\{\rho > 0 : x/\rho \in K^\circ\} = \inf\{\rho : x \cdot y \leq \rho \text{ for every } y \in K\} \\ &= \inf\{\rho : h_K(x) \leq \rho\} = h_K(x). \end{aligned}$$

For the second equality, observe that since $x \cdot y \leq \|x\|_K$ for every $y \in K^\circ$, we have

$$h_{K^\circ}(x) = \sup\{x \cdot y : y \in K^\circ\} \leq \|x\|_K.$$

On the other hand, by (5.11),

$$\begin{aligned} \|x\|_K &= \inf\{\rho : x/\rho \in K\} = \inf\{\rho : \rho^{-1}(x \cdot z) \leq h_K(z) \text{ for every } z\} \\ &= \inf\{\rho : (x \cdot z/h_K(z)) \leq \rho \text{ for every } z\} = \sup_z (x \cdot z/h_K(z)) \\ &\leq \sup_{y \in K^\circ} (x \cdot y) = h_{K^\circ}(x), \end{aligned}$$

because $z/h_K(z) \in K^\circ$ always. This completes the proof of second equality in (5.12). Note that two equalities of (5.12) imply that $\|x\|_K = \|x\|_{K^{\circ\circ}}$. As an immediate consequence we conclude that $K = K^{\circ\circ}$.

We now turn to (iv). We certainly have

$$H_K^*(x) = \sup_y \left(x \cdot y - \frac{1}{2}\|y\|_K^2 \right) = \sup_y \sup_{t \geq 0} \left(tx \cdot y - \frac{t^2}{2}\|y\|_K^2 \right).$$

If $x \cdot y > 0$, then the t -supremum is attained at $t = (x \cdot y)/\|y\|_K^2$; otherwise the t -supremum is 0. As a result,

$$\begin{aligned} 2H_K^*(x) &= \sup_y \frac{((x \cdot y)^+)^2}{\|y\|_K^2} = \sup_y \left[\left(x \cdot \frac{y}{\|y\|_K} \right)^+ \right]^2 \\ &= \left[\sup \{x \cdot y : \|y\|_K = 1\} \right]^2 = \left[\sup \{x \cdot y : \|y\|_K \leq 1\} \right]^2 \\ &= \left[\sup \{x \cdot y : y \in K\} \right]^2 = h_K(x)^2, \end{aligned}$$

as desired. $\square$

We next address the regularity of the function H_K .

Lemma 5.1 *Let $U \subseteq \mathbb{R}^{2n}$ be an open strictly convex set with $0 \in U$, $S = \partial U$, and $K = \bar{U}$. If S is C^2 , then H_K is C^2 and strictly convex.*

Proof Write $g(x) = g_K(x) := \|x\|_K$, $2H = 2H_K = g^2$. Note that g is not differentiable at 0. When $x \neq 0$, the derivative $Dg(x)$ is uniquely determined from

$$Dg|_{T_x S} = 0, \quad x \cdot \nabla g(x) = g(x).$$

The latter follows from the homogeneity of g . On the other hand,

$$(5.13) \quad D^2 g(x)|_{T_x S} > 0, \quad D^2 g(x)x = 0$$

for $x \in S$. Again the latter is a consequence of the homogeneity of g . To see the first claim in (5.13), let us look at the curvatures of S . Note that $n(x) = \frac{\nabla g(x)}{|\nabla g(x)|}$ is the unit normal to S and

$$Dn(x) = |\nabla g(x)|^{-1} D^2 g(x) + \nabla g(x) \otimes \nabla(|\nabla g(x)|^{-1}).$$

As a result,

$$Dn(x)a \cdot a = |\nabla g(x)|^{-1} D^2 g(x)a \cdot a + (\nabla g(x) \cdot a)(\nabla(|\nabla g(x)|^{-1}) \cdot a).$$

As a result, if $a \in T_x S$, then

$$(5.14) \quad Dn(x)a \cdot a = |\nabla g(x)|^{-1} D^2 g(x)a \cdot a.$$

Since S is strictly convex, we have $Dn(x) > 0$. This and (5.14) imply (5.13).

Evidently H is C^1 and C^2 off the origin. Since H is homogeneous of degree 2,

$$\begin{aligned} \nabla H &= g \nabla g, \\ D^2 H &= \nabla g \otimes \nabla g + g D^2 g. \end{aligned}$$

Hence,

$$D^2H(x)a \cdot a = (\nabla g(x) \cdot a)^2 + g(x)D^2g(x)a \cdot a.$$

Note that if $a = b + c$ with $b \parallel x$ and $c \in T_x S$, then $(\nabla g(x) \cdot a)^2 > 0$ if $b \neq 0$ and $D^2g(x)a \cdot a > 0$ if $b = 0$ and $c \neq 0$. Hence $D^2H(x) > 0$ for all $x \neq 0$. $\square$

Let us set $H = H_K$ and study the corresponding functional $\mathcal{A}$. Note that $\mathcal{A} = \mathcal{A}_T$ is defined for T -periodic functions whereas for Theorem 5.1 we need a periodic orbit on $S = \{H = 1/2\}$ of some period. Of course if $x(\cdot)$ is such a periodic orbit of period T , then $y(t) = x(Tt)$ is 1-periodic and

$$(5.15) \quad \dot{y} = T\bar{J}\nabla H(y).$$

In view of the form of functional $\mathcal{A}_T$, perhaps we should fix the period to be 1 always and now insist that $y(\cdot)$ solves (5.15) for some T . As a result, we now want to find a critical point of

$$T^{-1} \int_0^1 \left[\frac{1}{2} \bar{J}y \cdot \dot{y} - TH(y) \right] dt = \frac{1}{2T} \int_0^1 \bar{J}y \cdot \dot{y} dt - \int_0^1 H(y) dt$$

with y a 1-periodic path and some $T \in \mathbb{R}$. The scalar $\eta = T^{-1}$ resembles a Lagrange multiplier that now is employed for a functional defined on an infinite dimensional space. Motivated by this, define $\mathcal{C} : \Lambda \rightarrow \mathbb{R}$ by

$$\mathcal{C}(x(\cdot)) = \int_0^1 H(x(t)) dt$$

with

$$\Lambda = \left\{ x : \mathbb{R} \rightarrow \mathbb{R}^{2n} \text{ is } C^1, \text{ 1-periodic and } \int_0^1 \bar{J}x \cdot \dot{x} dt = 1 \right\}.$$

It is not hard to show that $\Lambda \neq \emptyset$.

Lemma 5.2 (i) *Let $y(\cdot)$ be a non-constant critical point of $\mathcal{C} : \Lambda \rightarrow \mathbb{R}$. Then either $\nabla H(y) \equiv 0$, or there exists a constant T such that $\dot{y} = T\bar{J}\nabla H(y)$.*

(ii) *If $H = H_K$, then any critical $y(\cdot)$ satisfies $\dot{y} = T\bar{J}\nabla H(y)$ for some $T > 0$. Moreover, $z(t) = \sqrt{T}y(t/T)$ is a T -periodic solution of $\dot{z} = \bar{J}\nabla H(z)$, such that $A(z(\cdot)) = T/2$ and $2H(z) \equiv 1$.*

Proof First, let us determine the space $T_{y(\cdot)}\Lambda$. Take a path $z : (-\delta, \delta) \rightarrow \Lambda$ with $z(\cdot, 0) = y(\cdot)$ and $z_\theta(\cdot, 0) = \tau(\cdot)$. The condition

$$\int_0^1 \bar{J}z(\cdot, \theta) \cdot z_t(\cdot, \theta) dt = 1$$

can be differentiated with respect to θ to yield

$$0 = \int_0^1 [\bar{J}y \cdot \dot{\tau} + \bar{J}\tau \cdot \dot{y}] dt = -2 \int_0^1 \bar{J}\dot{y} \cdot \tau dt.$$

In other words, for $\tau \in T_{y(\cdot)}\Lambda$, we always have

$$(5.16) \quad \int_0^1 \bar{J}\dot{y} \cdot \tau dt = 0.$$

The converse is also true; if a 1-periodic function τ satisfies (5.16), then the path $z(t, \theta) = y(t) + \theta\tau(t)$ lies in Λ and satisfies $z_\theta(\cdot, 0) = \tau(\cdot)$.

Now, if $y(\cdot)$ is critical for $\mathcal{C}$, then $\frac{d}{d\theta}\mathcal{C}(z(\cdot, \theta))|_{\theta=0} = 0$. This means that $y(\cdot)$ is critical iff

$$(5.17) \quad \int_0^1 \bar{J}\dot{y} \cdot \tau dt = 0 \Rightarrow \int_0^1 \nabla H(y) \cdot \tau dt = 0.$$

In particular if $y \in \Lambda$ satisfies $\nabla H(y) \equiv 0$, it is critical. Assume that $\nabla H(y)$ is not identically 0. We then use (5.17) to deduce that for every 1-periodic τ satisfying (5.16), and every $T \in \mathbb{R}$, we have

$$(5.18) \quad \int_0^1 (\bar{J}\dot{y} + T\nabla H(y)) \cdot \tau dt = 0.$$

We wish to select $\tau = \bar{J}\dot{y} + T\nabla H(y)$ to deduce that $\bar{J}\dot{y} + T\nabla H(y) = 0$. For the admissibility of such selection, we need

$$\int_0^1 |\dot{y}|^2 dt + T \int_0^1 \bar{J}\dot{y} \cdot \nabla H(y) dt = 0.$$

This can be solved for T if

$$(5.19) \quad \int_0^1 \bar{J}\dot{y} \cdot \nabla H(y) dt \neq 0.$$

In fact if (5.19) were not the case, then we could choose $\tau = \nabla H(y)$ in (5.17) to deduce that $\nabla H(y) = 0$, which contradicts our assumption. Hence y must satisfy $\dot{y} = T\bar{J}\nabla H(y)$.

We now turn to the proof of (ii). For a critical $y \in \Lambda$, we know that $\dot{y} = T\bar{J}\nabla H(y)$. On the other hand, since H is homogeneous of degree 2, we also know that $y \cdot \nabla H(y) = 2H(y)$. Hence

$$(5.20) \quad 1 = \int_0^1 \bar{J}y \cdot \dot{y} dt = 2T \int_0^1 H(y) dt = 2TH(y),$$

because $H(y)$ is a constant function. Since y cannot be identically 0 by the constraint, we deduce that $T > 0$ and that $H(y) = (2T)^{-1}$. Now if $z(t) = \sqrt{T}y(t/T)$, then obviously z is T -periodic with $H(z) = 1/2$ by the homogeneity of H . On the other hand,

$$\dot{z}(t) = T^{-1/2} \dot{y}(t/T) = \sqrt{T} \bar{J} \nabla H(y(t/T)) = \bar{J} \nabla H(\sqrt{T}y(t/T)) = \bar{J} \nabla H(z(t)).$$

Finally

$$A(z) = \frac{1}{2} \int_0^T \bar{J}z \cdot \dot{z} \, dt = \frac{T}{2} \int_0^T \bar{J}y(t/T) \cdot \dot{y}(t/T) T^{-1} \, dt = \frac{T}{2} \int_0^1 \bar{J}y(t) \cdot \dot{y}(t) \, dt = \frac{T}{2}.$$

This completes the proof of (ii). $\square$

On account of Lemma 5.2, we only need to find critical points for $\mathcal{C}$ on Λ . Since H is convex, we may wonder whether or not a minimum provides us with a critical point. It turns out that $\sup_{\Lambda} \mathcal{C} = +\infty$ and $\inf_{\Lambda} \mathcal{C} = 0$. Note that for H as in Lemma 5.2 infimum is not achieved because if $\mathcal{C}(x(\cdot)) = 0$, then $x(\cdot) \equiv 0$ and $\frac{1}{2} \int_0^1 \bar{J}x \cdot \dot{x} \neq 1$ for such $x(\cdot)$. Let us study an example to see why the infimum is not achieved.

Example 5.2 Assume that $n = 1$ and $H : \mathbb{R}^2 \rightarrow \mathbb{R}$ is given by $H(x) = \pi|x|^2$. We already know how to solve the corresponding system $\dot{x} = \bar{J}\nabla H(x)$; the solutions are given by $x(t) = (q(t), p(t)) = r(\sin 2\pi t, \cos 2\pi t)$, $t \geq 0$. The set $\{x : H(x) = 1/2\}$ carries the periodic orbit $x(t) = (2\pi)^{-1/2}(\sin 2\pi t, \cos 2\pi t)$. The set Λ consists of 1-periodic y with $\int_0^1 \bar{J}y \cdot \dot{y} \, dt = 1$. If y is a simple curve with coordinates q and p , then $\int_0^1 \bar{J}y \cdot \dot{y} \, dt$ is twice the area enclosed by y . We can easily construct a sequence $y_l(t)$ in Λ with $\mathcal{C}(y_l) \rightarrow \infty$ as $l \rightarrow \infty$. For example, set $y_l(t) = \left(2\sqrt{l} \sin 2\pi t, \frac{1}{\pi\sqrt{l}} \cos 2\pi t\right)$, so that $y_l \in \Lambda$ and $\mathcal{C}(y_l) = 2\pi l + \frac{1}{2\pi l}$. This confirms $\sup_{\Lambda} \mathcal{C} = +\infty$. As for the infimum, let us choose a sequence $z_l \in \Lambda$ of high oscillation, say $z_l(t) = (2\pi l)^{-1/2}(\sin 2\pi l t, \cos 2\pi l t)$. Since $H(z_l) = (2l)^{-1}$, we learn that $\inf_{\Lambda} \mathcal{C} = 0$. $\square$

From Example 5.2 it is clear that a control on $\mathcal{C}(y_l)$ for $y_l \in \Lambda$ guarantees no control on $\dot{y}_l$ and this results in $\inf_{\Lambda} \mathcal{C} = 0$. We now follow an idea of Clarke to switch to a new functional $\mathcal{D}$ which involves the derivative. To motivate the definition, observe that if

$$(5.21) \quad \dot{y} = T \bar{J} \nabla H(y),$$

for some $T > 0$, then for any constant $c \in \mathbb{R}^{2n}$,

$$\frac{d}{dt} \bar{J}(c - y) = T \nabla H(y).$$

Writing $w = \bar{J}(c - y)$, we deduce that for $H = H_K$ and $G = H_{K^\circ}$,

$$(5.22) \quad \nabla G(w) = T(\bar{J}w + c).$$

because ∇H is homogeneous of degree 1 and $(\nabla H)^{-1} = \nabla G$. Here we are using Proposition 5.1 to assert that $H^* = G$. The condition $\int_0^1 \bar{J}y \cdot \dot{y} \, dt = 1$ becomes

$$1 = \int_0^1 \bar{J}(\bar{J}w + c) \cdot \bar{J}\dot{w} \, dt = \int_0^1 \bar{J}w \cdot \dot{w} \, dt.$$

Motivated by this and (5.21), set

$$\Lambda' = \left\{ w \in \Lambda : \int_0^1 w \, dt = 0 \right\},$$

and let us define $\mathcal{D} : \Lambda' \rightarrow \mathbb{R}$ by

$$\mathcal{D}(y) = \int_0^1 G(\dot{y}(t)) \, dt.$$

Lemma 5.3 *If $w(\cdot)$ is a C^1 critical point of $\mathcal{D} : \Lambda \rightarrow \mathbb{R}$ such that $\int_0^1 w \, dt = 0$, then w solves (5.22) for some constants $T > 0$ and $c \in \mathbb{R}^{2n}$. Moreover if $y = \bar{J}w + c$, then $y \in \Lambda$, y satisfies (5.21), $2TH(y(\cdot)) \equiv 1$, and $\int_0^1 G(\dot{y}) \, dt = T/2$.*

Proof As in the proof of Lemma 5.2, we can show that if w is a critical point for $\mathcal{D}$ on Λ , then

$$\int_0^1 \bar{J}w \cdot \dot{\tau} \, dt = 0, \quad \int_0^1 \tau \, dt = 0 \quad \Rightarrow \quad \int_0^1 \nabla G(\dot{w}) \cdot \dot{\tau} \, dt = 0,$$

for every periodic C^1 path τ . Hence

$$(5.23) \quad \int_0^1 (T\bar{J}w + c' - \nabla G(\dot{w})) \cdot \dot{\tau} \, dt = 0,$$

for every $T \in \mathbb{R}$ and $c' \in \mathbb{R}^{2n}$. We wish to choose τ so that

$$(5.24) \quad \dot{\tau} = T\bar{J}w + c' - \nabla G(\dot{w}), \quad \int_0^1 \tau \, dt = 0.$$

For this, we need to satisfy two conditions:

$$\int_0^1 (T\bar{J}w + c' - \nabla G(\dot{w})) \, dt = 0, \quad \int_0^1 (T\bar{J}w + c' - \nabla G(\dot{w})) \cdot \bar{J}w \, dt = 0.$$

Since $\int_0^1 w \, dt = 0$, we simply have

$$c' = \int_0^1 \nabla G(\dot{w}) \, dt, \quad T \int_0^1 |w|^2 \, dt = \int_0^1 \nabla G(\dot{w}) \cdot \bar{J}w \, dt.$$

The second equation can be solved for T because w is not zero. Now choosing τ satisfying (5.24) in (5.23) implies that $\dot{\tau}$ is identically 0, which in turn implies

$$(5.25) \quad \nabla G(\dot{w}) = T\bar{J}w + c'.$$

Since G is homogeneous of degree 2, we know that $\dot{w} \cdot \nabla G(\dot{w}) = 2G(\dot{w})$. From this and (5.25) we deduce

$$\int_0^1 2G(\dot{w}) \, dt = \int_0^1 \dot{w} \cdot \nabla G(\dot{w}) \, dt = T \int_0^1 \bar{J}w \cdot \dot{w} \, dt = T.$$

Hence $T > 0$ because $G > 0$ away from 0. We then choose c so that $cT = c'$. Substituting this in (5.25) yields (5.22). We can readily show that if $y = \bar{J}w + c$, then $y \in \Lambda$ and y satisfies (5.21). Clearly (5.21) implies that $2TH(y) = 1$ by (5.20). We are done. $\square$

We are now ready for the proof of Theorem 5.1.

Proof of Theorem 5.1 First we obtain some useful properties of G . Note that by homogeneity of G ,

$$G(x) = |x|^2 G\left(\frac{x}{|x|}\right).$$

As a result, there are positive constants c_1 and c_2 , such that

$$(5.26) \quad c_1 \frac{|x|^2}{2} \leq G(x) \leq c_2 \frac{|x|^2}{2}$$

for all $x \in \mathbb{R}^{2n}$. From the homogeneity of G , we also know that $\nabla G(a) = |a| \nabla G\left(\frac{a}{|a|}\right)$. This in turn implies that for some constant c_3 ,

$$(5.27) \quad |\nabla G(a)| \leq c_3 |a|.$$

We then remark that even though we have worked with the space C^1 , our Lemma 5.3 is valid for a larger space $\mathcal{H}^1$. The space $\mathcal{H}^1$ consists of 1-periodic functions $w : \mathbb{R} \rightarrow \mathbb{R}^{2n}$ which are weakly differentiable with the weak derivative $\dot{w} \in L^2$. That is, there exists $v \in L^2$ such that for every $\zeta \in C^1$,

$$\int_0^1 w \cdot \dot{\zeta} \, dt = - \int_0^1 v \cdot \zeta \, dt.$$

We simply write $\dot{w}$ for the weak derivative v . Note that we can define $\mathcal{D} : \bar{\Lambda} \rightarrow \mathbb{R}$ where

$$\bar{\Lambda} = \left\{ w \in \mathcal{H}^1 : \int_0^1 \bar{J}w \cdot \dot{w} \, dt = 1, \int_0^1 w \, dt = 0 \right\}.$$

In Lemma 5.3, we may replace Λ with $\bar{\Lambda}$.

On account of our extension of Lemma 5.3, it suffices to find a critical point of $\mathcal{D} : \bar{\Lambda} \rightarrow \mathbb{R}$. This can be achieved by showing the existence of a minimizer of $\mathcal{D}$.

Set $a = \inf_{\bar{\Lambda}} \mathcal{D}$ and choose a sequence $w_l \in \bar{\Lambda}$ such that $w_l(0) = 0$ and $\mathcal{D}(w_l) \rightarrow a$. In view of (5.24), we certainly have

$$(5.28) \quad c_4 := \sup_l \int_0^1 |\dot{w}_l|^2 dt < \infty.$$

From this and Exercise 5.1(ii) we know

$$|w_l(t) - w_l(s)| \leq \sqrt{c_4} |t - s|^{1/2}.$$

This implies that $w_l - w_l(0)$ has a convergent subsequence with respect to the uniform topology. If h is the limit of this sequence, then

$$\int_0^1 h dt = \lim_{l \rightarrow \infty} \int_0^1 (w_l - w_l(0)) dt = - \lim_{l \rightarrow \infty} w_l(0).$$

As a result, w_l also has a convergent subsequence with respect to the uniform topology. On account of (5.28) we may choose a subsequence such that

$$\begin{aligned} w_l &\rightarrow w \text{ uniformly,} \\ \dot{w}_l &\rightarrow v \text{ weakly,} \end{aligned}$$

for some $v \in L^2$ and continuous w . We now assert that in fact w is weakly differentiable and its weak derivative is v . Indeed if $\zeta \in C^1$, then

$$\int_0^1 w \cdot \dot{\zeta} dt = \lim_{l \rightarrow \infty} \int_0^1 w_l \cdot \dot{\zeta} dt = - \lim_{l \rightarrow \infty} \int_0^1 \dot{w}_l \cdot \zeta dt = - \int_0^1 v \cdot \zeta dt.$$

It remains to show that $w \in \bar{\Lambda}$ and

$$(5.29) \quad a = \mathcal{D}(w).$$

For the former observe

$$\begin{aligned} \int_0^1 \bar{J}w \cdot \dot{w} dt &= \int_0^1 \bar{J}w \cdot (\dot{w} - \dot{w}_l) dt + \int_0^1 \bar{J}(w - w_l) \cdot \dot{w}_l dt + \int_0^1 \bar{J}w_l \cdot \dot{w}_l dt \\ &=: \Omega_1 + \Omega_2 + \Omega_3. \end{aligned}$$

We certainly have that $\lim_{l \rightarrow \infty} \Omega_1 = 0$ and $\Omega_3 = 1$. Moreover

$$|\Omega_2| \leq \|w - w_l\|_{L^2} \|\dot{w}_l\|_{L^2} \rightarrow 0$$

as $l \rightarrow \infty$ by (5.28) and uniform convergence of w_l to w . This shows that $w \in \bar{\Lambda}$.

It remains to establish (5.29). Since $a = \inf \mathcal{D}$, it suffices to show that $\mathcal{D}(w) \leq a$. This follows from the lower semi-continuity of the functional $\mathcal{D}$ which is a consequence of the convexity of G . Indeed by convexity of G ,

$$G(\dot{w}) + \nabla G(\dot{w}) \cdot (\dot{w}_l - \dot{w}) \leq G(\dot{w}_l).$$

Hence,

$$\int_0^1 G(\dot{w}) \, dt + \int_0^1 \nabla G(\dot{w}) \cdot (\dot{w}_l - \dot{w}) \, dt \leq \int_0^1 G(\dot{w}_l) \, dt.$$

We now send $l \rightarrow \infty$. Since $\nabla G(\dot{w}) \in L^2$ by (5.17) and $\dot{w} \in L^2$, we know that the second term on the left-hand side goes to 0. As a result,

$$\int_0^1 G(\dot{w}) \, dt \leq \liminf_{l \rightarrow \infty} \int_0^1 G(\dot{w}_l) \, dt = a.$$

Using $\mathcal{H}^1$ version of Lemma 5.3, we know that for some $y \in \mathcal{H}^1$, and $T > 0$, we have $\dot{y} = T\bar{J}\nabla H(y)$. Since the right-hand side is continuous, we use Exercise 5.6(i) to deduce that for almost all t ,

$$y(t) = y(0) + T \int_0^t \bar{J}\nabla H(y(s)) \, ds.$$

From this we deduce that in fact $y \in C^1$. As in Lemma 5.2, we may use y to construct a solution x to $\dot{x} = \bar{J}\nabla H(x)$ with $2H(x) = 1$ to complete the proof. $\square$

As a corollary to the proof of Theorem 5.1, we derive three useful expressions for the symplectic capacity that was defined in (5.1).

Theorem 5.2 *For every bounded convex set K with the origin in its interior, we have $\mathbf{c}_0(K) = \mathbf{c}'_0(K) = \mathbf{c}''_0(K)$, where*

$$(5.30) \quad \begin{aligned} \mathbf{c}'_0(K) &:= \inf_{w \in \Lambda} \frac{1}{2} \int_0^1 h_K^2(\dot{w}) \, dt = \inf_{w \in \bar{\Lambda}} \frac{1}{2} \int_0^1 h_K^2(\dot{w}) \, dt \\ \mathbf{c}''_0(K) &:= \inf \left\{ \frac{T}{2} : \text{There exists a } T\text{-periodic orbit of } X_{H_K} \text{ on } \partial K \right\}. \end{aligned}$$

Proof Note that infimum over Λ can be replaced with an infimum over $\bar{\Lambda}$ because adding a constant to w does not change $h_K(\dot{w})$ or $A(w)$. Let w be a 1-periodic function in Λ that minimizes $\mathcal{D}$. By Lemma 5.3, we can find $T > 0$ and $c \in \mathbb{R}^{2n}$ such that

$$(5.31) \quad \dot{y} = T\bar{J}\nabla H(y), \quad \int_0^1 \bar{J}y \cdot \dot{y} \, dt = 1, \quad 2TH(y) = 1, \quad \int_0^1 G(\dot{w}) \, dt = \frac{T}{2},$$

for $y = \bar{J}w + c$. We then build z out of y by $z(t) = \sqrt{T}y(t/T)$, so that z is T -periodic and

$$(5.32) \quad \dot{z} = \bar{J}\nabla H(z), \quad A(z(\cdot)) = \frac{T}{2}, \quad 2H(z) = 1,$$

by Lemma 5.2. Hence $z(\cdot)$ is a periodic orbit of X_H that lies on $S = \partial K$. This and Remark 5.1 immediately imply

$$(5.33) \quad \mathbf{c}_0(K) \leq \mathbf{c}'_0(K), \quad \mathbf{c}''_0(K) \leq \mathbf{c}'_0(K).$$

For the reverse inequalities, let z be a periodic orbit of the Reeb vector field. By reversing the orientation if necessary, we may assume that $A(z) > 0$. After a reparametrization, we may assume that z is a T -periodic orbit of the vector field X_H with $H = H_K$, that lies on the level set $H = 1/2$. Note that since H satisfies $z \cdot \nabla H(z) = 2H$, we have

$$\frac{1}{2} \int_0^T \bar{J}z \cdot \dot{z} \, dt = \frac{1}{2} \int_0^T \bar{J}z \cdot \bar{J}\nabla H(z) \, dt = \frac{1}{2} \int_0^T 2H(z) \, dt = \frac{T}{2}.$$

This implies that $\mathbf{c}_0 = \mathbf{c}''_0$. Also this z (a T -periodic orbit of the vector field X_H) satisfies (5.32). Defining $y(t) = T^{-1/2}z(Tt)$ yields a 1-periodic function which satisfies the first three equations in (5.31). We then take any constant $c \in \mathbb{R}^{2n}$ and set $w = \bar{J}(c - y)$ so that (5.22) is valid, which in turn can be used as in the proof of Lemma 5.3 to derive the last equation of (5.31). As in the calculation right after (5.22), we can use (5.32) to assert that $y \in \Lambda$. From this and the last equation of (5.31) we deduce that $\mathbf{c}'_0(K) \leq \mathbf{c}_0(K) = \mathbf{c}''_0(K)$. This and (5.33) complete the proof. $\square$

Remark 5.2(i) In (5.1), we defined $\mathbf{c}_0(K)$ for a convex set with smooth boundary. However the equality $\mathbf{c}_0(K) = \mathbf{c}'_0(K)$ gives us an expression that is well defined for arbitrary convex sets with nonempty interior. In fact if w is a minimizer for $c'(K)$ and z is the corresponding orbit as in Lemmas 5.2 and 5.3, then z still solves $\dot{z} = \bar{J}\nabla H(z)$ in some generalized sense. Given any convex function H , we may define its *subdifferential* $\partial H(z)$ as the set of vectors v such that

$$H(a) - H(z) \geq v \cdot (a - z),$$

for all $a \in \mathbb{R}^{2n}$. Now the corresponding Hamiltonian ODE reads as

$$\dot{z}^\pm \in J\partial H(z),$$

where $\dot{z}^\pm$ denotes the left and right derivatives. Similarly, the line bundle ℓ_x for $x \in \partial K$ consists of lines in the direction Jv , where $v \in \partial g_K(x)$. So a Reeb orbit will be tangent to a single line in the direction of $J\nabla g_K(x)$ if it passes through a point $x \in \partial K$ at which $\nabla g_K(x)$ exists; otherwise there is a cone of directions $\{Jv : v \in \partial g_K(x)\}$ that represents ℓ_x . When K has a smooth boundary, $\nabla g(x)$ exists at every $x \in \partial K$ and belongs to ∂K° .

(ii) When $H = H_K$, then a T -periodic orbit $z(\cdot)$ on ∂K yields periodic orbits on the other level sets of the same period. Indeed if $\lambda > 0$ and $u(t) = \sqrt{2\lambda}z(t)$, then u still solves $\dot{u} = \bar{J}\nabla H(u)$, simply because ∇H is homogeneous of degree 1. What we learn from this is that if T is the smallest period that a periodic orbit of X_H can have on ∂K , then this T is the smallest period that a non-constant periodic orbit can have anywhere in $\mathbb{R}^{2n}$. Moreover, since $H(u) = 2\lambda H(z) = \lambda$, then

$$\mathbf{c}_0(\{H \leq \lambda\}) = T\lambda.$$

It is this formula that generalizes to the *Hofer-Zehnder capacity* in Chapter 6. $\square$

Example 5.3 Let V and W be two open convex subsets of $\mathbb{R}^n$ with $0 \in V$ and W . Set $K = A \times B$, where $A = \bar{V}$ and $B = \bar{W}$. This convex set does not have a smooth boundary. Nonetheless we can use $\mathbf{c}'(K)$ to define its symplectic capacity as we discussed in Remark 5.1. We certainly have

$$\partial K = (\partial V \times W) \cup (V \times \partial W) \cup (\partial V \times \partial W).$$

It is the set $\partial U \times \partial V$ that is responsible for the non-smoothness of ∂K . (For example if $n = 1$, then K is a rectangle and $\partial U \times \partial V$ consists of its 4 corners.) To figure out how a Reeb orbit looks like, let us use the Hamiltonian function g_K for which K is the level set $\{g_K = 1\}$. In this case $g_K(q, p) = \max\{g_A(q), g_B(p)\}$, and $\partial g_K(x) = \{\nabla g_K(x)\}$ is singleton only in $(\partial V \times W) \cup (V \times \partial W)$:

$$\nabla g_K(q, p) = \begin{cases} (0, \nabla g_B(p)) & \text{if } (q, p) \in V \times \partial W \\ (\nabla g_A(q), 0) & \text{if } (q, p) \in \partial V \times W. \end{cases}$$

The corresponding Hamiltonian vector field is given by

$$X_{g_K}(q, p) = \begin{cases} (\nabla g_B(p), 0) & \text{if } (q, p) \in V \times \partial W \\ (0, -\nabla g_A(q)) & \text{if } (q, p) \in \partial V \times W. \end{cases}$$

We now describe an orbit $(q(t), p(t))$ of X_{g_K} on ∂K with $(q(0), p(0)) = (q_0, p_0) \in V \times \partial W$:

- (i) While in $V \times \partial W$, $p(t)$ stays put and $q(t)$ travels with velocity $\nabla g_B(p_0)$;
- (ii) At some time t_0 , the position $q(t)$ reaches the boundary ∂V . At this point $q(t_0) =: q_1$, the velocity $\nabla g_B(p_0)$ is pointing outward with respect to the convex set V ;

$$\nabla g_B(p_0) \cdot \nabla g_A(q_1) < 0.$$

Observe that now the vector $-\nabla g_A(q_1)$ is inward at the point p_0 with respect to the convex set W ;

$$(-\nabla g_A(q_1)) \cdot \nabla g_B(p_0) > 0.$$

Then the particle enters the set $\partial V \times W$ through the point $(q_1, p_0) \in \partial V \times \partial W$. While in $\partial V \times W$, $q(t) = q(t_0) = q_1$ stays put and $p(t)$ travels in W with velocity $-\nabla g_A(q(t_0))$.

(iii) At some time t_1 , the momentum $p(t)$ reaches the boundary ∂W . At $p_1 = p(t_1)$, the vector $-\nabla g_A(q(t_0))$ is pointing outward;

$$(-\nabla g_A(q_1)) \cdot \nabla g_B(p_1) < 0.$$

Then it enters the set $V \times \partial W$ through the point $(q(t_0), p(t_1)) \in \partial V \times \partial W$. Then (i) is repeated with (q_0, p_0) replaced with $(q(t_0), p(t_1))$.

The orbit we described in (i)-(iii) is a trajectory of a *Minkowski Billiard*. To see why we have a billiard, let us describe the orbit when $W = B_0(1)$ is the unit ball in $\mathbb{R}^n$: Given $q_0 \in V$ and p_0 of length 1, we have the following orbit:

(i) A particle starts at q_0 and travels with velocity p_0 in V .

(ii) As q reaches the boundary at $q_1 = q(t_0)$, it enters $\{q(t_0)\} \times B_0(1)$. Then $q(t_0)$ stays put while p_0 travels with velocity $-n(q_1) = -\nabla g_A(q_1)$ which is the inward normal to ∂V at q_1 .

(iii) As $p(t)$ reaches the boundary of the unit ball at time t_1 , we simply have

$$p_1 = p(t_1) = p_0 - 2|n_1|^{-2}(n_1 \cdot p_0)n_1,$$

where $n_1 = n(q_1)$. So, the relationship between p_0 and p_1 is that of a *specular reflection*.

From the preceding description, it is clear that if we ignore the time spent in $\partial V \times W$, what we have is a billiard trajectory. The part of dynamics spent in $\partial V \times W$ is for transforming the incident velocity $p(t_{2i})$ to the reflecting velocity $p(t_{2i+1})$. If we calculate the symplectic action of an orbit $\int_0^T p \cdot dq$, the part of the orbit inside $\partial V \times W$ does not contribute because $q(t)$ stays put. From this we learn that for the capacity of K , we are dealing with a billiard trajectory that is defined in the following way:

(i) We start from a point $q_0 \in V$ that has a velocity $\nabla g_A(p_0)$ with $p_0 \in \partial W$. The q -point travels according to its velocity until it reaches the boundary.

(ii) At the boundary point $q_1 = q(t_0)$, the velocity changes from $\nabla g_A(p_0)$ to $\nabla g_A(p_1)$, where $p_1 = \eta_B(p_0, q_1)$. The function η_B depends on the incoming velocity and the normal at the boundary point.

Now a periodic Reeb orbit corresponds to a periodic Minkowski billiard trajectory. If we write $q_0, q_1, \dots, q_l \in \partial V$ with $q_l = q_1$, for the hitting locations of a periodic Reeb orbit γ of the boundary, then

$$A(\gamma) = (q_1 - q_0) \cdot p_0 + \dots + (q_l - q_{l-1}) \cdot p_{l-1},$$

where $p_0, p_1, \dots, p_l$ are the corresponding momenta with $p_l = p_1$, and $p_{i+1} = \eta_B(q_i, p_i)$. In the case of a standard billiard (when $B = B_0(1)$ is the unit ball), we have that $p_i =$

$(q_{i+1} - q_i)/|q_{i+1} - q_i|$, and $A(\gamma)$ is the total length of the billiard trajectory. So, $\mathbf{c}_0(A \times B_0(1))$ is simply the length of the shortest periodic billiard trajectory in the convex set A . $\square$

In Example 5.3, we related $\mathbf{c}_0(A \times B)$ to the the shortest (with respect to B) periodic billiard trajectories in A . However in general it is hard to calculate the capacity of a convex body. We now state a conjecture that would allow us to compare the capacity of a set with its volume.

Conjecture 5.1 (Viterbo) If B^{2n} denotes the unit ball in $\mathbb{R}^{2n}$, then

$$(5.34) \quad \frac{\mathbf{c}_0(K)}{\mathbf{c}_0(B^{2n})} \leq \left[\frac{\text{Vol}(K)}{\text{Vol}(B^{2n})} \right]^{\frac{1}{n}},$$

for every bounded convex set $K \subset \mathbb{R}^{2n}$. $\square$

Of course we know that $\mathbf{c}_0(B^{2n}) = \pi$ and $\text{Vol}(B^{2n}) = \pi^n/n!$. Also we have equality in (5.34) if K is a ball.

Recently Artstein-Avidan, Karasev and Ostrover have observed that Viterbo's conjecture implies an old conjecture of Mahler in Convex Geometry that was formulated in 1939.

Conjecture 5.2 (Mahler) For every centrally symmetric convex set A of $\mathbb{R}^n$ with 0 in its interior,

$$(5.35) \quad \text{Vol}(A) \text{Vol}(A^\circ) \geq 4^n/n!.$$

To see how (5.34) implies (5.35), observe that if we apply (5.34) to the set $A \times A^\circ$, then (5.34) reads as

$$\frac{\mathbf{c}_0^n(A \times A^\circ)}{\pi^n} \leq \frac{n! \text{Vol}(A) \text{Vol}(A^\circ)}{\pi^n},$$

which implies (5.34) if we can show that $\mathbf{c}_0(A \times A^\circ) = 4$. This was established by Artstein-Avidan et al. in [AKO] (see Exercise 5.1(iv) below). According to [AKO],

$$\mathbf{c}_0(A \times B) = 4 \max\{r > 0 : rB^\circ \subset A\}.$$

Remark 5.3(i) Conjecture 5.1 asserts that among all convex sets of a given volume, the Euclidean ball has the largest possible capacity. Indeed if $B_{2n}(R)$ is a ball of $\text{Vol}(K)$, then the right-hand side of (5.34) is R^2 .

(ii) Several weaker versions of (5.34) have been already established. Viterbo showed

$$\frac{\mathbf{c}_0(K)}{\mathbf{c}_0(B^{2n})} \leq \gamma_n \left[\frac{\text{Vol}(K)}{\text{Vol}(B^{2n})} \right]^{\frac{1}{n}},$$

where $\gamma_n = 2n$ for a centrally symmetric K and $\gamma_n = 32n$ for general convex K . According to Arestein-Avidan Milman and Ostrove (2007), we can choose the constant γ_n to be independent of n .

(iii) According to Gluskin and Ostrover [GO],

$$(5.36) \quad \bar{\omega}(K^\circ)^{-1} \leq \mathbf{c}_0(K),$$

where $\bar{\omega}(K^\circ) = \sup_{a,b \in K^\circ} \bar{\omega}(a,b)$ and K is any bounded convex set that has the origin in its interior. The proof follows from the fact that $\bar{\omega}(K^\circ)$ bounds the Lipschitz constant of a K -characteristic with respect to the g_K -norm. More precisely, if $\dot{\gamma} = J\nabla g_K(\gamma)$ and $0 \leq s \leq t \leq T$, then

$$(5.37) \quad g_K(\gamma(t) - \gamma(s)) \leq \int_s^t g_K(\dot{\gamma}(\theta)) d\theta = \int_s^t h_{K^\circ}(J\nabla g_K(\gamma)) d\theta \leq (t-s)\bar{\omega}(K^\circ),$$

because $\nabla g_K \in \partial K^\circ$. On the other hand

$$g_K(\gamma(t) - \gamma(s)) = h_{K^\circ}(\gamma(t) - \gamma(s)) \geq (\gamma(t) - \gamma(s)) \cdot \nabla g_K(\gamma(t)) = 1 - \gamma(s) \cdot \nabla g_K(\gamma(t)).$$

This and (5.37) yields

$$1 \leq \gamma(s) \cdot \nabla g_K(\gamma(t)) + (t-s)\bar{\omega}(K^\circ).$$

Choose $s = 0$ and integrate this over $t \in [0, T]$, where T is the period;

$$T \leq \int_0^T \gamma(s) \cdot \nabla g_K(\gamma(t)) dt + \bar{\omega}(K^\circ)T^2/2 = \bar{\omega}(K^\circ)T^2/2,$$

because $\int_0^T J\nabla g_K(\gamma(t)) dt = \int_0^T \dot{\gamma} dt = 0$. Hence $2 \leq \bar{\omega}(K^\circ)T$, which proves (5.36).

In the case of a centrally symmetric convex K , we also have

$$\mathbf{c}_0(K) \leq 4(\bar{\omega}(K^\circ))^{-1},$$

as was shown in [GO]. □

In (5.30), we can replace L^2 norm with the L^p norm. Observe,

$$\begin{aligned} \mathbf{c}_0(K) &= \inf \left\{ \frac{1}{2} \int_0^1 h_K^2(\dot{w}) dt : w \in \Lambda_1, 2A(w) = 1 \right\} \\ &= \inf \left\{ \frac{1}{4} \int_0^1 h_K^2(\dot{w}) dt : w \in \Lambda_1, A(w) = 1 \right\}. \end{aligned}$$

Given $p \geq 1$, define

$$(5.38) \quad \mathbf{c}_{0,p}(K) = \inf \left\{ \frac{1}{4} \left(\int_0^1 h_K^p(\dot{w}) dt \right)^{2/p} : w \in \Lambda_1, A(w) = 1 \right\}.$$

We can show that $\mathbf{c}_{0,p} = \mathbf{c}_0$. Moreover, if we define

$$(5.39) \quad \begin{aligned} \hat{\mathbf{c}}_{0,p}(K) &= \left[\sup \left\{ 2 \int_0^1 \bar{J}w \cdot \dot{w} \, dt : \int_0^1 h_K^p(\dot{w}) \, dt \leq 1 \right\} \right]^{-1} \\ &= \left[\sup \left\{ 2 \int_0^1 \bar{J}w \cdot \dot{w} \, dt : \int_0^1 h_K^p(\dot{w}) \, dt = 1 \right\} \right]^{-1}, \end{aligned}$$

then

$$(5.40) \quad \mathbf{c}_{0,p}(K) = \hat{\mathbf{c}}_{0,p}(K),$$

In particular, when $p = \infty$, we simply have

$$(5.41) \quad \begin{aligned} \mathbf{c}_0(K) &= \hat{\mathbf{c}}_{0,p}(K) = \hat{\mathbf{c}}_{0,\infty}(K) \\ &= \left[\sup \left\{ \int_0^1 \bar{J}w \cdot \dot{w} \, dt : \frac{1}{2} h_K(\dot{w}) \leq 1 \right\} \right]^{-1} \\ &= \left[\sup \left\{ \frac{1}{2} \int_0^1 \bar{J}w \cdot \dot{w} \, dt : \dot{w} \in K^\circ \right\} \right]^{-1}, \end{aligned}$$

for every $p \geq 1$.

Exercise 5.1 (i) Verify Parts **(i)** and **(ii)** of Proposition 5.1.

(ii) Let $w \in \mathcal{H}^1$. Show that $w(t) = \int_0^t \dot{w}(\theta) d\theta$ for almost all t . Also show

$$|w(t) - w(s)| \leq \|\dot{w}\|_{L^2} |t - s|^{\frac{1}{2}}.$$

for $s, t \in [0, 1]$ and $L^2 = L^2[0, 1]$.

(iii) Prove that if $y \in \mathcal{H}^1$ and $\int_0^1 y \, dt = 0$, then

$$\|y\|_{L^2} \leq \frac{1}{2\pi} \|\dot{y}\|_{L^2}.$$

(iv) Show that if A is a centrally symmetric convex subset $\mathbb{R}^n$ with 0 in its interior, then $\mathbf{c}_0(A \times A^\circ) \leq 4$. (*Hint:* Show that if the Hamiltonian flow starts from $(0, \nabla g_A(q))$ for some $q \in \partial A$ then it reaches the boundary at q and reflect backs on itself to hit ∂A at $-q$ for the next hitting location. Such a periodic orbit has the value 4 for its action.)

(v) Verify (5.40). Show that the function $p \mapsto c_{0,p}$ is non-decreasing.

(vi) Verify (5.41). (*Hint:* Show that if w is a maximizer for $\hat{\mathbf{c}}_{0,1}$, then a reparametrization of w can serve as a maximizer for $\hat{\mathbf{c}}_{0,\infty}$.)

(vii) For any convex sets K and K' in $\mathbb{R}^{2n}$, show

$$\mathbf{c}_0(K + K')^{1/2} \geq \mathbf{c}_0(K)^{1/2} + \mathbf{c}_0(K')^{1/2},$$

where

$$K + K' = \{x + x' : x \in K, x' \in K'\}.$$

(*Hint:* use $\mathbf{c}_0 = \mathbf{c}_{0,1}$.)

□

6 Capacities and Their Applications

In this chapter we assume the existence of a capacity for the symplectic manifolds and deduce several properties of Hamiltonian systems. As we mentioned in the introduction the non-squeezing theorem of Gromov inspired the search for symplectic capacities.

In (5.1), we assign a positive number $\mathbf{c}_0(K)$ to every convex set with nonempty interior. By Theorem 5.2, we have an alternative variational expression $\mathbf{c}'_0(K)$ for $\mathbf{c}_0(K)$ that allows us to easily verify two important properties of $\mathbf{c}_0(\cdot)$:

$$(6.1) \quad K \subseteq K' \Rightarrow \mathbf{c}_0(K) \leq \mathbf{c}_0(K'), \quad \mathbf{c}_0(\lambda K) = \lambda^2 \mathbf{c}_0(K),$$

for every $\lambda > 0$. Moreover, (5.1) allows us to show that $\mathbf{c}_0(K)$ is invariant under a symplectic change of coordinates. More precisely, if ψ is a symplectic diffeomorphism, and both K and $\psi(K)$ are convex, then

$$(6.2) \quad \mathbf{c}_0(K) = \mathbf{c}_0(\psi(K)),$$

because $(d\psi)_x T_x(\partial K) = T_{\psi(x)}(\partial\psi(K))$, which implies

$$(d\psi)_x(\ell_x(\partial K)) = \ell_{\psi(x)}(\psi(\partial K)),$$

where the line $\ell_x(A)$ denotes the kernel of $\bar{\omega}|_A$. (See also (4.12) and Proposition 4.4.) Clearly the finiteness of $\mathbf{c}_0(K)$ is exactly Weinstein's conjecture for convex subsets of $\mathbb{R}^{2n}$. We now wonder whether $\mathbf{c}_0(\cdot)$ has an extension to all subsets of $\mathbb{R}^{2n}$. The finiteness of $\mathbf{c}_0(A)$ for a bounded A would lead to the existence of a periodic orbit of the Reeb vector field on ∂A as we discussed in Remark 4.4. As a naive guess we may try $\mathbf{c}_0(K) = \inf |A(\gamma)|$, with infimum over the periodic characteristic of the line bundle $\ell_x(\partial K)$, which is always well-defined because $\bar{\omega}$ is symplectic. But this cannot be a useful extension for our purposes because many smooth hypersurfaces carry no periodic orbit as we saw in Example 4.3. We will offer shortly several legitimate extensions of (5.1). Let us first formulate our wish-list for what an extension should satisfy.

Definition 6.1 Write $\mathcal{P}(\mathbb{R}^{2n})$ for the set of subsets of $\mathbb{R}^{2n}$. Then $\mathbf{c} : \mathcal{P}(\mathbb{R}^{2n}) \rightarrow [0, \infty]$ is called a *weak Euclidean capacity* if the following conditions hold:

- (i) If $\psi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is a symplectomorphism (i.e., a symplectic diffeomorphism with $\psi(\mathbb{R}^{2n}) = \mathbb{R}^{2n}$) and $\psi(A) \subseteq B$ then $\mathbf{c}(A) \leq \mathbf{c}(B)$.
- (ii) $\mathbf{c}(\lambda A) = \lambda^2 \mathbf{c}(A)$ for $\lambda > 0$.
- (iii) $\mathbf{c}(B^{2n}(1)) > 0$ and $\mathbf{c}(Z^{2n}(1)) < \infty$, where $B^{2n}(1)$ is the Euclidean ball of radius 1, and $Z^{2n}(1)$ is the cylinder $\{(q, p) : q_1^2 + p_1^2 \leq 1\}$.

We say $\mathbf{c}$ is a *strong Euclidean capacity* if in place of **(iii)**, we require the stronger assumption

$$\mathbf{c}(B^{2n}(1)) = \mathbf{c}(Z^{2n}(1)) = \pi.$$

□

Definition 6.2 Write $\mathcal{SM}$ for the set of all symplectic manifolds. Then $\mathbf{c} : \mathcal{SM} \rightarrow [0, \infty]$ is a *weak symplectic capacity* if the following conditions hold:

- (i) If $\psi : (M_1, \omega_1) \rightarrow (M_2, \omega_2)$ is a symplectic embedding and $\dim M_1 = \dim M_2$, then $\mathbf{c}(M_1, \omega_1) \leq \mathbf{c}(M_2, \omega_2)$.
- (ii) $\mathbf{c}(M, \lambda\omega) = \lambda\mathbf{c}(M, \omega)$ for $\lambda > 0$.
- (iii) $\mathbf{c}(B^{2n}(1), \bar{\omega}) > 0$, and $\mathbf{c}(Z^{2n}(1), \bar{\omega}) < \infty$.

We say c is a *strong symplectic capacity* if in place of **(iii)**, we require the stronger assumption

$$(6.3) \quad \mathbf{c}(B^{2n}(1), \bar{\omega}) = \mathbf{c}(Z^{2n}(1), \bar{\omega}) = \pi.$$

□

Note that the condition $\mathbf{c}(B^{2n}(1), \bar{\omega}) > 0$, guarantees that $\mathbf{c} \equiv 0$ is not a capacity. The requirement $\mathbf{c}(Z^{2n}(1)) < \infty$, disqualifies $\mathbf{c}(M, \omega) = |\int_M \omega^n|^{1/n}$, $n = \dim(M)$, to be a capacity. A strong capacity for symplectic manifolds can be used to define a strong Euclidean capacity by

$$(6.4) \quad \mathbf{c}(A) = \inf \{ \mathbf{c}(U, \bar{\omega}) : A \subseteq U, U \text{ open in } \mathbb{R}^{2n} \}.$$

The proof of this is left to the reader (Exercise 6.1(i)).

We now define four capacities.

Definition 6.3(i) (*Gromov Width*) For a symplectic manifold (M, ω) with $\dim(M) = 2n$,

$$\underline{\mathbf{c}}(M, \omega) = \sup \{ \pi r^2 : \text{There exists a symplectic embedding from } (B^{2n}(r), \bar{\omega}) \text{ into } (M, \omega) \},$$

$$\bar{\mathbf{c}}(M, \omega) = \inf \{ \pi r^2 : \text{There exists a symplectic embedding from } (M, \omega) \text{ into } (Z^{2n}(r), \bar{\omega}) \}$$

(ii) (*Hofer-Zehnder Capacity*) Given a symplectic (M, ω) , we first set $\mathcal{H}(M)$ to denote the set C^1 functions $H : M \rightarrow [0, \infty)$ such that $H = \max H$ outside a compact subset of M and $H = \min H = 0$ on some non-empty open set. We then set

$$\hat{\mathcal{H}}(M, \omega) = \{ H \in \mathcal{H}(M) : X_H \text{ has no periodic orbit of period } T \in (0, 1] \}.$$

We now define

$$(6.5) \quad \mathbf{c}_{HZ}(M, \omega) = \sup_{H \in \hat{\mathcal{H}}(M, \omega)} \max H.$$

(iii) (*Displacement Energy*) First we define it for open subsets $U \subseteq \mathbb{R}^{2n}$. Given a time dependent Hamiltonian function $H : \mathbb{R}^{2n} \times [0, 1] \rightarrow \mathbb{R}$, define

$$\|H\|_{\infty, 1} := \int_0^1 \left(\sup_x H(x, t) - \inf_x H(x, t) \right) dt.$$

We then define the *displacement energy* of a set U by

$$\mathbf{e}(U) := \inf \{ \|H\|_{\infty, 1} : \phi_1^H(U) \cap U = \emptyset, H \text{ is of compact support} \}.$$

As in (6.4), we may also define $\mathbf{e}(A)$ for arbitrary subsets of $\mathbb{R}^{2n}$. More generally, if (M, ω) is symplectic and U is an open subset of M , then

$$\mathbf{e}(U; M, \omega) := \inf \{ \|H\|_{\infty, 1} : \phi_1^{H, \omega}(U) \cap U = \emptyset, H \text{ is of compact support} \}.$$

□

Remark 6.1(i) The motivation behind the definition of $\underline{\mathbf{c}}$ and $\bar{\mathbf{c}}$ is Gromov's non-squeezing theorem. Later we give a minimax-type expression for $\underline{\mathbf{c}}$ and $\bar{\mathbf{c}}$ that is based on Gromov's proof of non-squeezing theorem and involves *pseudo-holomorphic curves*.

(ii) Let us write

$$T^{\min}(H; \omega) = T^{\min}(H) = \inf \{ T : X_H^\omega \text{ has a non-constant periodic orbit of period } T \}.$$

We may then write

$$\mathbf{c}_{HZ}(M, \omega) = \sup_{H \in \mathcal{H}(M)} \{ \max H : T^{\min}(H; \omega) \geq 1 \}.$$

Obviously,

$$\mathbf{c}_{HZ}(M, \omega) \leq \sup_{H \in \mathcal{H}(M)} (T^{\min}(H; \omega) \max H).$$

Since a T -periodic orbit $x(\cdot)$ of X_H yields a T/λ -periodic orbit $y(t) = x(t\lambda)$ of $X_{\lambda H}$, we learn

$$(6.6) \quad T^{\min}(\lambda H) = \lambda^{-1} T^{\min}(H),$$

for every $\lambda > 0$. As a result, if $H \in \mathcal{H}(M)$ and $\bar{T} = T^{\min}(H)$, then $T^{\min}(\bar{T}H) = 1$ and $\bar{T}H \in \hat{\mathcal{H}}(M, \omega)$, which in turn implies that $\bar{T} \max H \leq \mathbf{c}_{HZ}(M, \omega)$. In summary, for every H of compact support,

$$(6.7) \quad T^{\min}(H; \omega) \leq \frac{\mathbf{c}_{HZ}(M, \omega)}{\max H}, \quad \text{and} \quad \mathbf{c}_{HZ}(M, \omega) = \sup_{H \in \mathcal{H}(M)} (T^{\min}(H; \omega) \max H).$$

When $M = U$ is a convex open subset of $\mathbb{R}^{2n}$ with $0 \in U$ and $K = \bar{U}$, then we may set $H = \min\{H_K, 1/2\}$ so that $H = 1/2$ in the complement of K and $H(0) = 0$. In some sense H is the Hamiltonian for which the supremum in (6.9) is attained. Since $H \notin \mathcal{H}(U)$, we choose a sequence $H_l \in \mathcal{H}(U)$ so that $H_l \rightarrow H$ as $l \rightarrow \infty$. This implies that $\mathbf{c}_{HZ}(U) = \mathbf{c}_{HZ}(\bar{U}) \geq \mathbf{c}(\bar{U})$. The details are left to the reader (see Exercise 6.1(**v**)).

(iii) If $X_H = X_H^{\bar{\omega}}$ has a periodic orbit x of period T , then $y(t) = x(Tt)$ is 1-periodic and solves $\dot{y} = T\bar{J}\nabla H(y)$. By differentiating we obtain $\ddot{y} = T\bar{J}D^2H(y)\dot{y}$, which in turn yields the bound

$$\|\ddot{y}\|_{L^2} \leq aT\|\dot{y}\|_{L^2} \leq \frac{aT}{2\pi}\|\ddot{y}\|_{L^2},$$

where $a = \max |D^2H|$ and for the last inequality we used Exercise 5.1(iii). From this we learn

$$T^{\min}(H) \geq \frac{2\pi}{\max |D^2H|}.$$

However, the capacity $\mathbf{c}_{HZ}(M, \omega)$, provides us with an upper-bound on the period that depends on $\|H\|$ only. Note that if we write $\mathcal{T}(H) = \mathcal{T}(H; \omega)$ for $1/T^{\min}(H; \omega)$, then in some sense, $\mathcal{T}(H)$ measures the size of the function H in some symplectic sense. In fact, for $H \in \mathcal{H}(U)$ and $\lambda > 0$,

$$(6.8) \quad \mathcal{T}(\lambda H) = \lambda \mathcal{T}(H), \quad \mathbf{c}_{HZ}(U)^{-1} \max H \leq \mathcal{T}(H) \leq (2\pi)^{-1} \max |D^2H|.$$

(iv) From (6.8) we deduce the bound

$$\|H\| \leq (2\pi)^{-1} \mathbf{c}_{HZ}(U) \max |D^2H|,$$

for every H with compact support in U . In fact it is elementary to show

$$\|H\| \leq \text{diam}(U) \max |\nabla H|, \quad \max |\nabla H| \leq \text{diam}(U) \max |D^2H|,$$

where $\text{diam}U$ denotes the diameter of U . So, we have maximum-type-principle inequality of the form

$$\|H\| \leq \text{diam}(U)^2 \max |D^2H|.$$

According to *Alexandroff-Bakelman-Pucci (ABP) Maximum Principle*,

$$\max_U u \leq \max_{\partial U} u + \frac{\text{diam}(U)}{\text{Vol}(B^{2n})^{1/n}} \left[\int_U |\det D^2u| \, dx \right]^{\frac{1}{n}}.$$

In Viterbo [V], a symplectic geometry proof of ABP is given by using Hofer's theorem for the displacement energy (Theorem 6.2 (iv) below). $\square$

Here are some straightforward properties of capacities.

Proposition 6.1 *Suppose that c is a strong capacity.*

(i) *For every ellipsoid E , we have $\mathbf{c}(E, \bar{\omega}) = \pi r_1(E)^2$.*

(ii) *For every symplectic manifold (M, ω) ,*

$$\underline{\mathbf{c}}(M, \omega) \leq \mathbf{c}(M, \omega) \leq \bar{\mathbf{c}}(M, \omega).$$

Proof By Corollary 2.2, the ellipsoid E is symplectomorphic to the ellipsoid

$$E' = \left\{ x : \sum_1^n r_j^{-2} (q_j^2 + p_j^2) \leq 1 \right\}.$$

Hence $\mathbf{c}(E, \bar{\omega}) = \mathbf{c}(E', \bar{\omega})$. On the other hand,

$$B^{2n}(r_1) \subseteq E \subseteq Z^{2n}(r_1).$$

As a result, $\pi r_1^2 \leq \mathbf{c}(E', \bar{\omega}) = \mathbf{c}(E, \bar{\omega}) \leq \pi r_1^2$. This completes the proof of (i). The proof of (ii) is left as an exercise. $\square$

In Chapter 5 we had a candidate for the symplectic capacity of a convex set. It is conjectured that all capacities coincide on convex sets.

Conjecture 6.1 (*Viterbo*) For every convex subset K of $\mathbb{R}^{2n}$ with nonempty interior, $\underline{\mathbf{c}}(K) = \bar{\mathbf{c}}(K)$. $\square$

In Chapter 7 we will show that $\mathbf{c}_{HZ}$ is a capacity.

Theorem 6.1 *The Hofer-Zehnder function $\mathbf{c}_{HZ}$ is a strong capacity.*

Remark 6.2 As we will see in Exercise 6.1(iv), it is not hard to show that $\mathbf{c}_{HZ}(B^{2n}(1)) \geq \pi$. Chapter 7 is devoted to the proof of $\mathbf{c}_{HZ}(Z^{2n}(1)) \leq \pi$. Let us explain how the *coarea formula* can be used to prove the latter bound for $n = 1$. In fact we can even show that $\mathbf{c}_{HZ}(U) \leq \text{area}(U)$ for every connected open set in the plane. According to the coarea formula,

$$\int f |\nabla H| \, dx = \int_{-\infty}^{\infty} \left(\int_{\{H=r\}} f \, d\ell \right) dr,$$

where dx and $d\ell$ denote the area and the length integration (see Proposition 4.5 and Remark 4.4, or [EG] for the coarea formula). Hence if $H \in \mathcal{H}(M)$ with support in an open set U ,

$$(6.9) \quad \text{area}(U \cap \{\nabla H \neq 0\}) = \int_0^{\max H} \left(\int_{U \cap \{H=r\}} \mathbb{1}(\nabla H \neq 0) \frac{d\ell}{|\nabla H|} \right) dr.$$

Now if $x(\cdot)$ is a periodic orbit of period T that lies on the level set $\{H = r\}$, then its arc length is given by

$$d\ell = |\dot{x}(t)|dt = |\nabla H(x(t))|dt.$$

Hence

$$(6.10) \quad \int_{U \cap \{H=r\}} \mathbb{1}(\nabla H \neq 0) \frac{d\ell}{|\nabla H|} \geq T$$

If $\hat{T}$ is the smallest possible period for the non-constant periodic orbits of X_H , then by (6.9) and (6.10), we have

$$(\max H)\hat{T} \leq \text{area}(U).$$

This in turn implies that $\mathbf{c}_{HZ}(U) \leq \text{area}(U)$. For the reverse inequality $\mathbf{c}_{HZ}(U) \geq \text{area}(U)$, we need to find $H : U \rightarrow [0, \infty)$ so that each nonzero level set $\{H = r\}$ consists of exactly one periodic orbit of period very close to $\text{area}(U)$. In fact when U is convex with $0 \in U$, then a slight modification of $H_{\bar{U}}$ would do the job as we demonstrated this in Remark 5.1(ii). If U is simply connected, we can find an area preserving φ such that $\varphi(B^{2n}(r)) = U$ with $\pi r^2 = \text{area}(U)$. Then a modification of $H(x) = |x|^2$ yields a Hamiltonian in $\mathcal{H}(B^{2n}(r))$, and this will be pushed forward to U by φ to yield the desired Hamiltonian. $\square$

We now discuss four fundamental results in symplectic geometry that will be established in this chapter with the aid of Theorem 6.1.

Theorem 6.2 (i) (Gromov) *If there exists a symplectic diffeomorphism $\psi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ with $\psi(B^{2n}(r)) \subseteq Z^{2n}(R)$, then $r \geq R$.*

(ii) (Gromov, Eliashberg) *If $\psi_k : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ are symplectomorphisms such that $\psi_k \rightarrow \psi$ uniformly with ψ a diffeomorphism, then ψ is symplectic.*

(iii) (Viterbo) *The Reeb vector field of a closed hypersurface S of contact type in $\mathbb{R}^{2n}$ has a periodic orbit.*

(iv) (Hofer-Zehnder) *Let $H \in \mathcal{H}(\mathbb{R}^{2n})$. Then the set*

$$W := \{r \in \mathbb{R} : H^{-1}(r) \text{ carries no periodic orbit of } X_H\},$$

is of zero Lebesgue measure.

(v) (Hofer) *For every open set U , we have $\mathbf{c}_{HZ}(U) \leq \mathbf{e}(U)$.*

Remark 6.3(i) As we will see later, parts (i) and (ii) are consequences of the existence a strong symplectic capacity. Though (iii)-(v) rely on the form of $\mathbf{c}_{HZ}$.

(ii) In view of Theorems 6.1-6.2, Proposition 6.1, and Exercise 6.1(vi), we deduce that $\underline{c}, c_{HZ}, e$ and $\bar{c}$ are all strong capacities and

$$\underline{c} \leq c_{HZ} \leq e \leq \bar{c}.$$

□

Proof of Theorem 6.2(i) If ψ is a symplectomorphism with $\psi(B^{2n}(r)) \subseteq Z^{2n}(R)$, and c is a strong capacity, then

$$r^2\pi = c(rB^{2n}(1)) = c(B^{2n}(r)) = c(\psi(B^{2n}(r))) \leq c(Z^{2n}(R)) = R^2\pi,$$

as desired.

Proof of Theorem 6.2(iii) On account of Proposition 4.4, we only need to show that if $K = K_S$, then the Hamiltonian vector field X_K has a periodic orbit in a bounded neighborhood V of S . Recall that there exists an open set U with $U = \cup_{t \in (-\delta, \delta)} S_t$, and $K(S_s) = e^s$. To turn K to a Hamiltonian $H \in \mathcal{H}(U)$, take a smooth function $g : [e^{-\delta}, e^{\delta}] \rightarrow [0, 1]$ such that

$$g(r) = 0 \text{ for } r \leq e^{-\delta/2}, \quad g(r) = 1 \text{ for } r \geq e^{\delta/2}, \quad g'(r) > 0 \text{ for } e^{-\delta/2} < r < e^{\delta/2}.$$

If we $H = g(K)$, then $H \in \mathcal{H}(U)$. By Theorem 6.1, we know that $c_{HZ}(U) < \infty$. From this and (6.9), we deduce that X_H has a non-constant periodic orbit in U . This periodic orbit must lie in

$$V = \cup \{S_t : t \in (-\delta/2, \delta/2)\}.$$

We are done by Proposition 4.4. □

Proof of Theorem 6.2(iv) Fix an open ball D such that the support of H is contained in D . Suppose to the contrary $|W| > 0$, where $|W|$ represents to one-dimensional Lebesgue measure of the set W . Select a closed set $W' \subset W$ such that $|W'| > 0$. Then we can find a sequence of open sets $V_k \subset \mathbb{R}$ such that $V_{k+1} \subset V_k$ for every $k \in \mathbb{N}$, and

$$\cap_{k=1}^{\infty} V_k = W'.$$

Fix $\delta > 0$, and set

$$U_k = \{x \in D : H(x) \in V_k\}.$$

If every U_k carries a periodic orbit $x_k(\cdot)$ of X_H of period $T_k \geq \delta$, then this sequence is equicontinuous. If $(x(\cdot), T)$ is a limit point of the sequence $(x_k(\cdot), T_k)_{k \in \mathbb{N}}$, then $x(\cdot)$ is a periodic orbit of period $T \geq \delta$ such that $x(\cdot)$ such that $H(x(t)) \in W'$ for every t . Since this contradicts the definition of W' , we deduce that for sufficiently large $k = k(\delta)$, U_k carries no periodic orbit of X_H of period $T \geq \delta$ □

We now turn to the proof of Theorem 6.2(ii). The key idea of the proof is an equivalent criterion for the symplecticity of a transformation that does not involve any derivative. This should be compared to the notion of a volume preserving transformations. We can say that a transformation is volume preserving if its Jacobian is one, or equivalently it preserves the volume. The latter criterion is more useful when we want to show that a limit of a sequence of measure preserving transformations is again measure preserving.

Theorem 6.3 (Ekeland-Hofer) *Let $\mathbf{c}$ be a strong capacity and let $\psi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a diffeomorphism. Then the following statements are equivalent:*

- (i) *For every small ellipsoid E , we have $\mathbf{c}(\psi(E)) = \mathbf{c}(E)$.*
- (ii) *Either $\psi^*\bar{\omega} = \bar{\omega}$ or $\psi^*\bar{\omega} = -\bar{\omega}$.*

Proof Evidently if $\psi^*\bar{\omega} = \bar{\omega}$, then (i) is true. If $\psi^*\bar{\omega} = -\bar{\omega}$, then $\hat{\psi}^*\bar{\omega} = \bar{\omega}$ for $\hat{\psi} = \psi \circ \tau$, where τ is given by $\tau(q, p) = (p, q)$. We then have that for an ellipsoid E , $\mathbf{c}(\psi(\tau(E))) = \mathbf{c}(E)$. On the other hand, if E is a *standard ellipsoid* given by

$$\sum_{i=1}^n r_i^{-2}(q_i^2 + p_i^2) \leq 1,$$

then $\tau(E) = E$. Hence for such ellipsoids, $\mathbf{c}(\psi(E)) = \mathbf{c}(E)$. Since for any symplectic φ we have $(\psi \circ \varphi)^*\bar{\omega} = -\bar{\omega}$, we also have

$$\mathbf{c}(\varphi(E)) = \mathbf{c}(E) = \mathbf{c}(\psi(\varphi(E))),$$

for any standard ellipsoid E . Now any ellipsoid can be represented as $\varphi(E)$ for some linear symplectic φ and a standard E . This completes the proof of (ii) $\Rightarrow$ (i).

For the converse, assume (i) is true and define

$$\phi_k(x) = k(\psi(a + k^{-1}x) - \psi(a)).$$

We have that $\phi_k(x) \rightarrow \psi'(a)x$ locally uniformly as $k \rightarrow \infty$. Note that if E is an ellipsoid and ψ satisfies (i), then

$$\begin{aligned} \mathbf{c}(\phi_k(E)) &= \mathbf{c}(k\psi(a + k^{-1}E) - k\psi(a)) \\ &= k^2\mathbf{c}(\psi(a + k^{-1}E)) \\ &= k^2\mathbf{c}(k^{-1}E) = \mathbf{c}(E). \end{aligned}$$

On the other hand, it follows from Lemma 6.1 below that $\lim \phi_k(\cdot) = \psi'(a)$ also satisfies (i). We finally use Theorem 2.3 to deduce that the matrix $A = \psi'(a)$ satisfies $A^*\bar{\omega} = \bar{\omega}$ or $A^*\bar{\omega} = -\bar{\omega}$. By continuity we have $\psi(a)'*\bar{\omega} = \bar{\omega}$ for all a or $\psi(a)'*\bar{\omega} = -\bar{\omega}$ for all a . $\square$

To complete the proof of Theorem 6.3, we still need to show that the property (i) of Theorem 2.3 is preserved under a uniform limit. In Lemma 6.1 we prove a stronger variant that does not assume that the functions $\{\psi_k\}_{k \in \mathbb{N}}$ are diffeomorphism.

Lemma 6.1 *Let $\{\psi_k\}$ be a sequence of continuous functions for which (i) of Theorem 6.3 is valid. If $\psi_k \rightarrow \psi$ locally uniformly and ψ is a homeomorphism, then ψ satisfies (i) as well.*

Proof Imagine that we can prove this: For every $\lambda \in (0, 1)$ and every 0-centered ellipsoid E , there exists k_0 such that for $k > k_0$ we have

$$(6.11) \quad \psi_k(\lambda E) \subseteq \psi(E) \subseteq \psi_k(\lambda^{-1}E).$$

Then we are done because

$$\begin{aligned} \mathbf{c}(\psi(E)) &\leq \mathbf{c}(\psi_k(\lambda^{-1}E)) = \mathbf{c}(\lambda^{-1}E) = \lambda^{-2}\mathbf{c}(E), \\ \mathbf{c}(\psi(E)) &\geq \mathbf{c}(\psi_k(\lambda E)) = \mathbf{c}(\lambda E) = \lambda^2\mathbf{c}(E) \end{aligned}$$

and yields property (i) for ψ by sending λ to 1.

To establish (6.11), let us write $\varphi_k = \psi^{-1} \circ \psi_k$. Clearly the sequence φ_k converges to the identity, locally uniformly in the large k limit. From this it is clear that for large k ,

$$\psi^{-1} \circ \psi_k(\lambda E) \subseteq E,$$

establishing the first inclusion in (6.11).

If ψ_k is a homeomorphism for each k , then the second inclusion in (6.11) can be established in the same way. It remains to show that for large k ,

$$(6.12) \quad E \subseteq \varphi_k(\lambda^{-1}E),$$

even when ψ_k 's are not homeomorphism. If (6.12) fails, then there exists a sequence $k_l \rightarrow \infty$ and $y_l \in E$ such that $y_l \notin \varphi_{k_l}(\lambda^{-1}E)$. This allows us to define

$$F_l(x) = \frac{\varphi_{k_l}(x) - y_l}{|\varphi_{k_l}(x) - y_l|},$$

for $x \in \lambda^{-1}E =: E_\lambda$. It follows from Lemma A.1 of the Appendix that $\deg f_l = 0$ where $f_l : \partial E_\lambda \rightarrow S^{2n-1}$ is the restriction of F_l to ∂E_λ . On the other hand, we may define $g_l : \partial E_\lambda \rightarrow \partial S^{2n-1}$ by $g_l(x) = \frac{\varphi_{k_l}(x)}{|\varphi_{k_l}(x)|}$. The function g_l is well-defined for large l because φ_{k_l} is uniformly close to identity over the set ∂E_λ . The function g_l has $\deg 1$ simply because g_l is uniformly close to $x \mapsto \frac{x}{|x|}$ which has degree 1. To arrive at a contradiction, it suffices to show that g_l is homotopic to f_k . (By Lemma A.1 homotopic transformations have the same degree.) For homotopy, define

$$\Phi_l(x, t) = \frac{\varphi_{k_l}(x) - ty_l}{|\varphi_{k_l}(x) - ty_l|},$$

for $x \in \partial E_\lambda$, $t \in [0, 1]$. Again, since $\varphi_k \rightarrow id$ and $ty_l \in E$ the homotopy is well-defined. $\square$

With Theorem 6.3 and Lemma 6.1 at our disposal, we can now give a straightforward proof for Theorem 6.2(iii).

Proof of Theorem 6.2(iii) Let ψ_k be a sequence of symplectic transformations such that $\psi_k \rightarrow \psi$ locally uniformly. Assume that ψ is a diffeomorphism. By Lemma 6.1, ψ preserves the capacity of ellipsoids and by Theorem 6.3, either $\psi^*\bar{\omega} = \bar{\omega}$ or $\psi^*\bar{\omega} = -\bar{\omega}$. We now need to rule out the second possibility. Indeed if $\psi^*\bar{\omega} = -\bar{\omega}$, and we define

$$\phi_n = \psi_n \times id : \mathbb{R}^{4n} \rightarrow \mathbb{R}^{4n}, \quad \omega = \bar{\omega} \times \bar{\omega},$$

then $\phi_n^*\omega = \omega$, and $\phi_n \rightarrow \phi := \psi \times id$ locally uniformly. But $\phi^*\omega = \psi^*\bar{\omega} \times \bar{\omega} = (-\bar{\omega}) \times \bar{\omega} \neq \pm\omega$. As a result, we must have $\psi^*\bar{\omega} = \bar{\omega}$. $\square$

Motivated by Theorem 6.3, we may define the notion of symplecticity for homeomorphism. Note that if $\psi : \mathbb{R}^{4n} \rightarrow \mathbb{R}^{4n}$ is a diffeomorphism with $\psi^*\bar{\omega} = -\bar{\omega}$, then $\psi^*\bar{\omega}^n = (-1)^n\bar{\omega}^n$, and if n is odd then ψ can not be orientation preserving. Based on this we have the following definition.

Definition 6.3 Let $\psi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a homeomorphism. We say ψ is a symplectic homeomorphism if either n is odd and ψ is an orientation preserving transformation for which $\mathbf{c}(\psi(E)) = \mathbf{c}(E)$ for every ellipsoid E . Or n is even and $\psi \times id : \mathbb{R}^{2n+2} \rightarrow \mathbb{R}^{2n+2}$ is a symplectic homeomorphism.

From the proof of Theorem 6.2(iii) it is clear that if ψ_k is a sequence of symplectic homeomorphism such that $\psi_k \rightarrow \psi$ locally uniformly, then ψ is also a symplectic homeomorphism.

Remark 6.4 Observe that if $\psi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is C^1 , with

$$\psi(x) = \psi(x_1, \dots, x_{2n}) = (X_1(x), \dots, X_{2n}(x)),$$

and $\{u_1, \dots, u_{2n}\}$ is the standard basis for $\mathbb{R}^{2n}$, then

$$(d\psi)\bar{J}(d\psi)^*u_i \cdot u_j = \bar{J}(d\psi)^*u_i \cdot (d\psi)^*u_j = \bar{J}\nabla X_i \cdot \nabla X_j =: \{X_i, X_j\}.$$

In other words

$$(d\psi)\bar{J}(d\psi)^* = [\{X_i, X_j\}]_{i,j=1}^{2n}.$$

We may wonder whether or not we can make sense of $(d\psi)\bar{J}(d\psi)^*$ in some weak sense when ψ is only a continuous function. Equivalently, given a pair of continuous function $f, g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$, we wish to make sense of $\{f, g\}$. Observe

$$\{f, g\} = \bar{J}\nabla f \cdot \nabla g = \nabla \cdot (f\bar{J}\nabla g), \quad \int \{f, g\} \zeta \, dx = - \int f \{g, \zeta\} \, dx,$$

for any smooth $\zeta : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ of compact support. Hence, it suffice to find a weak interpretation of $f\nabla g$. It turns out that this is possible of $(f, g) \in C^\alpha \times C^\beta$, when $\alpha + \beta > 1$, where C^α represents the space of α -Hölder continuous functions. Indeed, we can talk about suitable negative Hölder space $C^{\beta-1}$, and a continuous bilinear operator

$$\mathcal{B} : C^\alpha \times C^\beta \rightarrow C^{\beta-1},$$

such that $\mathcal{B}(f, g) = f\nabla g$ if $g \in C^1$. From this we learn that if $\psi \in C^\alpha$, with $\alpha > 1/2$, then we can make sense of $(d\psi)\bar{J}(d\psi)^*$ as a Schwartz distribution in $C^{\alpha-2}$, and such a ψ is symplectic if $(d\psi)\bar{J}(d\psi)^* = \bar{J}$. The operator $\mathcal{B}$ was constructed by Young in dimension 1. Indeed, for $f, g : \mathbb{R} \rightarrow \mathbb{R}$, with $(f, g) \in C^\alpha \times C^\beta$, we can find a candidate for $fg' \in C^{\beta-1}$, if

$$h(x) = \int_0^x f(y)g'(y) dy =: \int_0^x f dg,$$

can be defined as a function in C^β . Young discovered that the standard Riemann-Stieltjes approximation is convergent even when $(f, g) \in C^\alpha \times C^\beta$, for some α and β satisfying $\alpha + \beta > 1$. We refer to [R] for more details. $\square$

We now turn our attention to the energy displacement and Theorem 6.2(iv). The proof of Hofer's theorem will be discussed in Chapter 8. Alternatively, we may use Theorem 6.1 or Theorem 6.2 (i) to establish a weaker version of Theorem 6.2(iv) but in a more general setting. A slight variant of Theorem 6.4 below is due to Lalonde-McDuff (see [DS]).

Theorem 6.4 *Let $\mathbf{c} : \mathcal{SM} \rightarrow [0, \infty)$ be a capacity (satisfying (i) and (ii) of Definition 6.2) such that for any symplectic manifold (M, ω) , and any open planar set V , we have*

$$(6.13) \quad \min \{ \mathbf{c}(M, \omega), |V| \} \leq \mathbf{c}(M \times V, \omega \otimes \bar{\omega}) \leq |V|,$$

with $|V| = \text{area}(V)$. Then for any symplectic manifold (M, ω) , and any open subset U of M , we have

$$(6.14) \quad \mathbf{c}(U, \omega) \leq 2\mathbf{e}(U; M, \omega).$$

As a preparation, we start with a straightforward proposition:

Proposition 6.2 *Let (M, ω) be a symplectic manifold, and $H : M \times \mathbb{R} \rightarrow \mathbb{R}$ be a time dependent Hamiltonian function. Set*

$$\begin{aligned} \tilde{M} &:= M \times \mathbb{R}^2 = \{(x, h, t) : x \in M, h, t \in \mathbb{R}\}, & \tilde{\omega} &:= \omega + dt \wedge dh, \\ \psi(x, h, t) &:= (\phi_t(x), h + H(\phi_t(x), t), t), \end{aligned}$$

with $\phi_t = \phi_t^{H, \omega}$. Then the transformation ψ is $\tilde{\omega}$ -symplectic.

Proof Let us write d for differentiation with respect to $x \in M$, and $\hat{d}$ for the differentiation with respect to if $z = (x, h, t)$. If $\hat{z} = (\hat{x}, \hat{h}, \hat{t}) \in T_x M \times \mathbb{R}^2$, then

$$(\hat{d}\psi)_z \hat{z} = \left((d\phi_t)_x \hat{x} + \hat{t} X_H(\phi_t(x)), \hat{h} + \hat{t} V(x, t) + d(H \circ \phi_t)_x(\hat{x}), \hat{t} \right),$$

where $V = V(x, t) = H_t(\phi_t(x), t) + (dH)_{\phi_t(x)}(X_H(\phi_t(x)))$. As a result,

$$(\psi^* \tilde{\omega})_z(\hat{z}, \hat{z}') = \tilde{\omega}_{\psi(z)}((d\psi)_z(\hat{z}), (d\psi)_z(\hat{z}')),$$

equals

$$\begin{aligned} & (\phi_t^* \omega)_x(\hat{x}, \hat{x}') + \hat{t} \omega_{\phi_t(x)}(X_H(\phi_t(x)), (d\phi_t)_x \hat{x}') - \hat{t}' \omega_{\phi_t(x)}(X_H(\phi_t(x)), (d\phi_t)_x \hat{x}) \\ & + (dt \wedge dk) \left(\left((\hat{h} + \hat{t} V(x, t) + d(H \circ \phi_t)_x(\hat{x})), \hat{t} \right), \left((\hat{h}' + \hat{t}' V(x, t) + d(H \circ \phi_t)_x(\hat{x}')), \hat{t}' \right) \right) \\ & = \omega_x(\hat{x}, \hat{x}') + (\phi_t^* (i_{X_H} \omega))_x(\hat{t} \hat{x}' - \hat{t}' \hat{x}) + (dt \wedge dk) \left((\hat{h}, \hat{t}), (\hat{h}', \hat{t}') \right) \\ & + (dt \wedge dk) \left((d(H \circ \phi_t)_x \hat{x}, \hat{t}), (d(H \circ \phi_t)_x \hat{x}', \hat{t}') \right) \\ & = \tilde{\omega}_z(\hat{z}, \hat{z}') - (\phi_t^* dH)_x(\hat{t} \hat{x}' - \hat{t}' \hat{x}) + (dt \wedge dk) \left((d(H \circ \phi_t)_x \hat{x}, \hat{t}), (d(H \circ \phi_t)_x \hat{x}', \hat{t}') \right) \\ & = \tilde{\omega}_z(\hat{z}, \hat{z}'), \end{aligned}$$

as desired. $\square$

Remark 6.5(i) In the case of $M = \mathbb{R}^{2n}$, simply write $\phi_t(q, p) = (Q, P)$, and $k = h + H(\phi_t^H(x), t)$, and observe

$$\begin{aligned} \sum_i dP^i \wedge dQ^i + dt \wedge dk &= \sum_i dp_i \wedge dq_i + \sum_i [P_t^i dt \wedge dQ^i + Q_t^i dP^i \wedge dt] + dt \wedge dh \\ &+ \sum_i [H_{q_i}(Q, P, t) dt \wedge dQ^i + H_{p_i}(Q, P, t) dt \wedge dP^i] \\ &= \sum_i dp_i \wedge dq_i + dh \wedge dt. \end{aligned}$$

(ii) Observe that if $\alpha = p \cdot dq - H(x, t)dt$, then $d\alpha = \bar{\omega} + dt \wedge dh$, where $h = H(x, t)$. Regarding h as a new coordinate we get $\tilde{\omega} = \bar{\omega} + dt \wedge dh$. It is worth mentioning that if $\hat{H}(x, h, t) = H(x, t) - h$, and $H^t(x, s) := H(x, t + s)$, then

$$\phi_s^{\hat{H}, \tilde{\omega}}(x, h, t) = \left(\phi_s^{H^t, \bar{\omega}}(x), H \left(\phi_s^{H^t, \bar{\omega}}(x), t + s \right) - H(x, t) + h, t + s \right),$$

which resembles ψ . $\square$

Proof of Theorem 6.4 To ease the notation, we write ϕ for ϕ_1^H . Before embarking on the proof (6.19), let us explain the idea behind the proof. Observe that the condition $\phi_1^H(U) \cap U =$

$\emptyset$ is a property of the flow at time 1, whereas the right-hand side of (6.19) involves the Hamiltonian function H . Because of this, we switch from ϕ_1^H to ψ of Proposition 6.2 that explicitly involves the Hamiltonian function. Let us define

$$h^+(t) = \max_x H(x, t), \quad h^-(t) = \min_x H(x, t), \quad \Gamma = \psi(U \times \{0\} \times [0, 1]).$$

The set Γ is a hypersurface in $\tilde{M}$, and $\Gamma \subseteq M \times E_0$, where

$$E_0 = \{(k, t) : t \in [0, 1], k \in [h^-(t), h^+(t)]\},$$

In other words, ψ embeds $U \times \{0\} \times [0, 1]$ into a cylinder like set $M \times E_0$, and we wish to show

$$(6.15) \quad \mathbf{c}(U, \omega) \leq 2 \operatorname{area}(E_0).$$

This indeed has the same flavor as our Theorem 6.2(iii) (Gromov's non-squeezing theorem), though $U \times [0, 1]$ is not a symplectic manifold. To rectify this, we replace $[0, 1]$ with a planar set A and use ψ instead of ψ_0 . More specifically, the set A is defined as,

$$A = ([-a, a] \times [-2a, 0]) \cup ([-\varepsilon, \varepsilon] \times [0, 1]) \cup ([-a, a] \times [1, 1 + 2a]) =: A^- \cup A^0 \cup A^+.$$

As the first step, let us assume that $H(x, t) = 0$ for t close to 0 or 1. This allows us to extend H for all times by setting $H(x, t) = 0$ for $t < 0$ and $t > 1$. Note

- $\psi(x, h, t) = (x, h, t)$ for $t \leq 0$,
- $\psi(x, h, t) = (\phi(x), h, t)$ for $t \geq 1$.

As a result, if we write $W^\pm = U \times A^\pm$, $W^0 = U \times A^0$, and $W = U \times A$, then

$$\psi(W^-) = W^-, \quad \psi(W^+) = \phi(U) \times A^+, \quad \psi(W^0) \subset M \times E(\varepsilon),$$

where $E(\varepsilon)$ is a small neighborhood of E_0 for small ε . We have the embedding

$$\psi : U \times A \hookrightarrow M \times (A^- \cup E(\varepsilon) \cup A^+) =: Z.$$

We now want to use $\phi^H(U) \cap U = \emptyset$ to replace Z with a slimmer cylinder. To achieve this, take a symplectic covering map $\lambda : \mathbb{R}^2 \rightarrow [\mathbb{R} \times S^1] \cup [\mathbb{R} \times (-\infty, 0)]$ such that

$$\lambda(\mathbb{R} \times [0, 1]) = \mathbb{R} \times S^1, \quad \lambda(h, t) = \begin{cases} (h, 1 - t), & \text{for } t \geq 1 \\ (h, t), & \text{for } t \leq 0. \end{cases}$$

Since $\phi^H(U) \cap U = \emptyset$, we have that the map

$$\hat{\psi} = (id \times h) \circ \psi,$$

is a symplectic diffeomorphism. This time

$$(6.16) \quad \hat{\psi} : U \times A \hookrightarrow M \times (A^- \cup E'(\varepsilon)) := M \times E'',$$

where $E'(\varepsilon)$, is the set we obtain from $E(\varepsilon)$ by setting $0 = 1$ for the t -coordinate. Evidently $\text{area}(E'(\varepsilon)) \rightarrow \|H\|_{\infty,1}$ as $\varepsilon \rightarrow 0$. From (6.16), we learn

$$(6.17) \quad \mathbf{c}(U \times A, \tilde{\omega}) \leq \mathbf{c}(M \times E'', \tilde{\omega}),$$

where $\hat{\omega} = \omega + d\theta \wedge dt$, where $d\theta$ is the standard 1-form on $\mathbb{S}^1$. This and (6.13) imply

$$(6.18) \quad \min \{ \mathbf{c}(M, \omega), |A| \} \leq |E''|.$$

By sending ε to 0 we obtain

$$\min \{ \mathbf{c}(M, \omega), 2(2a)^2 \} \leq (2a)^2 + \|H\|_{\infty,1}.$$

We now choose a so that $\mathbf{c}(M, \omega) = 2(2a)^2$, to deduce (6.19).

It remains to show that without loss of generality, we can always assume that $H = 0$ for t near 0 and 1. For this, first observe that since the definition of $\mathbf{c}_{HZ}$ involves Hamiltonian function with compact support inside U , we may find a sequence of nested open sets U_k so that $\cup_k U_k = U$, and that if δ_k denotes the distance between $M \setminus U$ and U_k , then $\delta_k \rightarrow 0$ in large k -limit. Because of this, it suffices to establish (6.19) for U_k in place of U . We now take a Hamiltonian H with support in U_k and mollify it in the following way: Take a smooth function $\chi = \chi_\delta$ such that $0 \leq \chi \leq 1$, $\chi(t) = 1$, for $t \in [\delta, 1 - \delta]$, and $\chi(t) = 0$ for $t \notin (0, 1)$, and set $H'(x, t) = \chi(t)H(x, t)$. We certainly have that $\|H\|_{\infty,1} - \|H'\|_{\infty,1} = O(\delta)$, and that there exists $\bar{\delta} > 0$ such that if $0 < \delta < \bar{\delta}$, then $\phi'(U_k) \cap U = \phi'(U_k) \cap U_k = \emptyset$ for $\phi' = \phi_1^{H', \omega}$. The constant $\bar{\delta}$ many depend on C^2 -norm of H but that would not cause any difficulty because our bound (6.19) depends only on $\|H'\|_{\infty,1}$. We finally send $\delta \rightarrow 0$ to arrive at our bound for U_k . $\square$

Note that the condition $\mathbf{c}_{HZ}(M \times B^2(r)) \leq \pi r^2$ is valid for $M = \mathbb{R}^{2n}$. Given a symplectic manifold (M, ω) , and $H \in \mathcal{H}(M)$, we set

$$S(H; \omega) = (\max H) T^{\min}(H; \omega).$$

Proposition 6.3 *Let (M, ω) and (M', ω') be a pair of symplectic manifolds.*

(i) *If $H \in \mathcal{H}(M)$, $H' \in \mathcal{H}(M')$, and $(H \otimes H')(x, x') = H(x)H'(x')$, then*

$$S(H \otimes H'; \omega \oplus \omega') = \min \{ S(H; \omega), S(H'; \omega') \}.$$

(ii) *We have*

$$(6.19) \quad \mathbf{c}_{HZ}(M \times M', \omega \oplus \omega') \geq \min \{ \mathbf{c}_{HZ}(M, \omega), \mathbf{c}_{HZ}(M', \omega') \}.$$

Proof(i) Without loss of generality, we may assume that $H, H' \neq 0$. Clearly

$$X_{H \otimes H'}(x, x') = (H'(x')X_H(x), H(x)X_{H'}(x')).$$

Choose $a \in M$ and $a' \in M'$ such that $m := \max H = H(a)$ and $m' := \max H' = H'(a')$. Let x (respectively x') be a T -periodic (respectively T' -periodic) orbit of X_H (respectively $X_{H'}$). Then $z(t) = (x(tm'), a')$ and $z'(t) = (a, x'(tm))$ are two orbits of $X_{H \otimes H'}$ with periods T/m' and T'/m respectively. From this we learn

$$\begin{aligned} S(H \otimes H'; \omega \oplus \omega') &= mm' T^{\min}(H \otimes H'; \omega \oplus \omega') \\ &\leq mm' \min \{T^{\min}(H; \omega)/m', T^{\min}(H'; \omega')/m\} \\ &= \min \{m T^{\min}(H; \omega), m' T^{\min}(H'; \omega')\} \\ &= \min \{S(H; \omega), S(H'; \omega')\}. \end{aligned}$$

For the reverse inequality, let $z = (x, x')$ be a non-constant T -periodic orbit. Hence $\dot{z} = X_{H \otimes H'}(z)$, or

$$\dot{x} = H'(x')X_H(x), \quad \dot{y} = H(x)X_{H'}(x').$$

Clearly

$$\frac{d}{dt}H(x) = \frac{d}{dt}H'(x') = 0.$$

Since z is a T -periodic orbit with $\lambda = H(x)$ and $\lambda' = H'(x')$, then $\hat{x}(t) = x(t/\lambda')$ is a $T\lambda'$ periodic orbit of X_H and $\hat{x}(t) = x(t/\lambda)$ is a $T\lambda$ periodic orbit of $X_{H'}$. Since z is non-constant, either $\hat{x}$ or $\hat{x}'$ is non-constant. Without loss of generality, we may assume that $\hat{x}$ is non-constant. Hence

$$\lambda'T \geq T^{\min}(H, \omega).$$

This in turn implies

$$mm'T \geq m\lambda'T \geq m T^{\min}(H, \omega) = S(H, \omega).$$

From this we can readily deduce

$$S(H \otimes H'; \omega \oplus \omega') \geq S(H, \omega) \geq \min \{S(H, \omega), S(H', \omega')\}.$$

(ii) We have

$$\begin{aligned} \min \{\mathbf{c}_{HZ}(M, \omega), \mathbf{c}_{HZ}(M', \omega')\} &= \min \left\{ \sup_{H \in \mathcal{H}(M)} S(H; \omega), \sup_{H' \in \mathcal{H}(M')} S(H'; \omega') \right\} \\ &\leq \sup_{H \in \mathcal{H}(M)} \sup_{H' \in \mathcal{H}(M')} \min \{S(H; \omega), S(H'; \omega')\} \\ &= \sup_{H \in \mathcal{H}(M)} \sup_{H' \in \mathcal{H}(M')} S(H \otimes H'; \omega \otimes \omega') \\ &\leq \sup_{K \in \mathcal{H}(M \times M')} S(K; \omega \otimes \omega') = \mathbf{c}_{HZ}(M \times M', \omega \oplus \omega'), \end{aligned}$$

as desired. □

Exercise 6.1(i) Show that $\mathbf{c}$ given by (6.4) is a strong Euclidean capacity.

(ii) Verify properties **(i)** and **(ii)** of Definition 6.2 for $\underline{\mathbf{c}}$, $\bar{\mathbf{c}}$, $\mathbf{c}_{HZ}$, and Euclidean $\mathbf{e}$. (**Hint:** For $\mathbf{e}$, use $\hat{H}(x) = \lambda^2 H(x/\lambda)$.)

(iii) Verify Proposition 6.1**(ii)**.

(iv) Show that $\mathbf{c}_{HZ}(B^{2n}(1), \bar{\omega}) \geq \pi$.

(v) Show that $\mathbf{c}_{HZ}(K, \bar{\omega}) \geq \mathbf{c}_0(K)$ for every bounded convex subset of $\mathbb{R}^{2n}$ with nonempty interior.

(vi) Show that $\mathbf{e}(U) = \text{area}(U)$ for every simply connected open subset $U \subset \mathbb{R}^2$. Use this to show $\mathbf{e}(Z^{2n}(1)) \leq \pi$. **Hint:** Use Exercise 3.1**(vi)** to reduce the problem to the case of $U = [0, a]^2$.

(vii) Show that if H is of compact support and its support is contained in the set U , then

$$T^{\min}(H) \leq \frac{\mathbf{c}_{HZ}(M, \omega)}{\|H\|_{\infty}},$$

where $\|H\|_{\infty} = \max |H|$.

(viii) Show

$$\frac{\underline{\mathbf{c}}(U)}{\underline{\mathbf{c}}(B^{2n})} \leq \left(\frac{\text{Vol}(U)}{\text{Vol}(B^{2n})} \right)^{1/n},$$

for every bounded open set $U \subset \mathbb{R}^{2n}$. From this deduce that if $\underline{\mathbf{c}}(K) = \mathbf{c}_0(K)$, for a convex set K , then (5.34) holds. □

7 Hofer-Zehnder Capacity

This section is devoted to the proof of Theorem 6.1. In fact we have the following result of Hofer and Zehnder.

Theorem 7.1 *For every convex set K of nonempty interior, $\mathbf{c}_{HZ}(K) = \mathbf{c}_0(K)$.*

On account of Exercise 6.1(v), we only need to show that $\mathbf{c}_{HZ}(K) \leq \mathbf{c}_0(K)$. Similarly, for Theorem 6.1, we only need to verify $\mathbf{c}_{HZ}(Z^{2n}(1), \bar{\omega}) \leq \pi$. Equivalently, we need to show that for every $H_1 \in \mathcal{H}(Z^{2n})$, and $\lambda > 1$,

$$1 \geq \lambda^{-1} \pi^{-1} (\max H) T^{\min}(H) = T^{\min} \left(\frac{\lambda \pi H}{\max H} \right) =: T^{\min}(H_0).$$

Since $\max H_0 > \pi$, it suffices to prove that such a Hamiltonian function has a non-constant periodic orbit.

Theorem 7.2 *Assume $H_0 \in \hat{\mathcal{H}}(Z^{2n}(1))$ with $\sup H_0 > \pi$. Then the Hamiltonian flow of H_0 has a non-constant periodic orbit of period 1.*

Let $H_0 \in \mathcal{H}(Z^{2n}(1))$. In fact we may assume that H_0 vanishes near the origin. This is because we may replace H with $H \circ \psi$ for a symplectic $\psi : Z(1) \rightarrow Z(1)$ that satisfies $\psi(x) = x$ outside a compact subset of $Z(1)$. Indeed we may choose $\psi = \phi_1^h$ where h is a suitable Hamiltonian function that vanishes outside a compact subset of $Z(1)$. For example, first choose $h_0(x) = \bar{J}x \cdot a$ for a fix vector a , so that $\phi_1^{h_0}(x) = x + a$, then set $h = h_0\chi$, where χ is a function of compact support in $Z(1)$ that is identically 1 in a neighborhood of the line segment connecting the origin to a . Using such ψ , we may shift a minimizer of H to the origin. Since the flows of X_H and $X_{H \circ \psi}$ are conjugated, it suffices to find the desired 1-periodic orbit for $H_0 \circ \psi$. From now on we assume that H_0 in Theorem 7.2 vanishes near the origin.

As our next step we extend H_0 to a Hamiltonian function $H : \mathbb{R}^{2n} \rightarrow \mathbb{R}$. For this, we take an ellipsoid

$$E^0 = \left\{ x \in \mathbb{R}^{2n} : Q(x) = q_1^2 + p_1^2 + \frac{1}{l^2} \sum_{j=2}^n (q_j^2 + p_j^2) < 1 \right\}.$$

Since $H_0 \in \mathcal{H}(Z^{2n}(1))$, we have that $H_0 = \max H_0$ for $x \notin K$ where K is a compact subset of $Z := Z^{2n}(1)$. Choose l sufficiently large so that $K \subseteq E^0$. We now pick $\epsilon > 0$ so that $\max H_0 > \pi + \epsilon$ and pick a smooth function $f : [0, \infty) \rightarrow [0, \infty)$ such that $f(r) = \max H$ for

$r \in [0, 1]$, $f(r) = (\pi + \epsilon)r$ for large r , $f(r) \geq (\pi + \epsilon)r$ for all r , and $0 < f'(r) \leq \pi + \epsilon$ for $r > 1$. We now define

$$(7.1) \quad H(x) = \begin{cases} H_0(x) & \text{if } x \in E^0, \\ f(Q(x)) & \text{if } x \notin E^0. \end{cases}$$

We note that $H(x) = (\pi + \epsilon)Q(x)$ for large x . Also note that if Q is quadratic (or even homogeneous of degree 2), then its action for a 1-periodic orbit is always 0:

$$\mathcal{A}(x(\cdot)) = \frac{1}{2} \int_0^1 (\bar{J}x \cdot \bar{J}\nabla Q(x) - 2Q(x)) \, dt = 0.$$

This suggests that a periodic orbit of X_H of positive action must be periodic orbit of the original X_{H_0} .

Lemma 7.1 *Assume that $x(\cdot)$ is a 1-periodic solution of $\dot{x} = \bar{J}\nabla H(x)$ with*

$$(7.2) \quad \mathcal{A}(x(\cdot)) = \int_0^1 \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) dt > 0.$$

Then $x(t) \in E^0$ for all t and $x(\cdot)$ is non-constant.

Proof Evidently $X_H = 0$ on ∂E^0 . Hence, all points on ∂E^0 are equilibrium points and if $x(t) \equiv a \in \partial E^0$, then $\mathcal{A}(x(\cdot)) = -H(a) \leq 0$ which contradicts (7.2). As a result, either $x(t) \in E^0$ for all t , or $x(t) \notin E^0$ for all t . It remains to rule out the latter possibility.

If $x(t) \notin E^0$ for all t , then

$$\begin{aligned} \dot{x} &= \bar{J}\nabla H(x) = f'(Q(x))\bar{J}\nabla Q(x), \\ \frac{d}{dt}Q(x) &= f'(Q(x))\nabla Q(x) \cdot \bar{J}\nabla Q(x) = 0. \end{aligned}$$

Hence, for such $x(\cdot)$ we have that $Q(x(\cdot)) = Q^0$, and

$$\begin{aligned} \mathcal{A}(x(\cdot)) &= \int_0^1 \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) dt \\ &= \int_0^1 \left[\frac{1}{2} f'(Q^0) \nabla Q(x) \cdot x - f(Q^0) \right] dt \\ &= f'(Q^0)Q^0 - f(Q^0) \leq (\pi + \epsilon)Q^0 - (\pi + \epsilon)Q^0 = 0, \end{aligned}$$

which contradicts (7.2). Here we used the fact that $2Q(x) = \nabla Q(x) \cdot x$. □

In view of Lemma 7.1, we only need to find a critical point of $\mathcal{A}$ which satisfies (7.2). Let us first observe that $\mathcal{A}$ is not bounded from below or above. Indeed if $y_k(t) = (\cos 2\pi kt)a + (\sin 2\pi kt)\bar{J}a$, for some $a \in \mathbb{R}^{2n}$, then

$$\int_0^1 |y_k(t)|^2 dt = |a|^2, \quad \int_0^1 \bar{J}y_k(t) \cdot \dot{y}_k(t) dt = 2\pi k|a|^2,$$

which in particular implies that $\lim_{k \rightarrow \pm\infty} \mathcal{A}(y_k) = \pm\infty$, whenever $a \neq 0$. Because of this, we search for saddle-type critical points of $\mathcal{A}$. A standard way of locating such critical points is by using the celebrated *minimax* principle.

To prepare for this, let us first extend the domain of definition of $\mathcal{A}$ from C^1 to the largest possible Sobolev space which turns out to be the space of function with “half” a derivative. We begin with $\mathcal{H}^0 = L^2$ which consists of measurable functions

$$x(t) = \sum_{k \in \mathbb{Z}} e^{2\pi kt \bar{J}} x_k = \sum_{k \in \mathbb{Z}} [(\cos 2\pi kt)I + (\sin 2\pi kt)\bar{J}]x_k$$

with $x_k \in \mathbb{R}^{2n}$ and $\|x\|_0 = \sum_k |x_k|^2 < \infty$. Here we are using the Fourier expansion of $x(\cdot)$ where instead of $i = \sqrt{-1}$ we use $-\bar{J}$. We write

$$\langle x, y \rangle_0 = \int_0^1 x(t) \cdot y(t) dt,$$

for the standard inner product of $\mathcal{H}^0$. Note that if

$$x(t) = \sum_k e^{2\pi kt \bar{J}} x_k \text{ and } y(t) = \sum_k e^{2\pi kt \bar{J}} y_k,$$

then $\langle x, y \rangle_0 = \sum_k x_k \cdot y_k$ and $\|x\|_0^2 = \langle x, x \rangle_0$. We note that $\int_0^1 H(x(t)) dt$ is well defined for every $x \in \mathcal{H}^0$ because $H(x) = Q(x)$ for large x . However, to make sense of $\int_0^1 \bar{J}x \cdot \dot{x} dt$, we need to assume that $x(\cdot)$ possesses half a derivative. To see this first observe that if $x \in C^1$, with $x = \sum_k e^{2\pi kt \bar{J}} x_k$, then

$$\int_0^1 \frac{1}{2} \bar{J}x \cdot \dot{x} dt = \pi \sum_k k |x_k|^2.$$

This suggests defining

$$\mathcal{H}^{1/2} = \left\{ x \in \mathcal{H}^0 : \sum_k |k| |x_k|^2 < \infty \right\}.$$

We turn $\mathcal{H}^{1/2}$ into a Hilbert space by defining

$$\langle x, y \rangle = \langle x, y \rangle_{1/2} = x_0 \cdot y_0 + 2\pi \sum_{k \in \mathbb{Z}} |k| (x_k \cdot y_k).$$

More generally, we define

$$\langle x, y \rangle_s = x_0 \cdot y_0 + (2\pi)^{2s} \sum_{k \in \mathbb{Z}} |k|^{2s} (x_k \cdot y_k),$$

for every $s > 0$ and $\mathcal{H}^s$ consists of function $x \in \mathcal{H}^0$ such that $\|x\|_s^2 = \langle x, x \rangle_s < \infty$. Observe that if $x \in C^1$, then $\int_0^1 |\dot{x}(t)|^2 dt = \sum_k (2\pi k)^2 |x_k|^2$ and that in general $x \in \mathcal{H}^1$ iff x has a weak derivative in L^2 .

So far we know that our functional $\mathcal{A}$ is defined on the Hilbert space $\mathcal{H}^{1/2}$. Let us take an arbitrary Hilbert space $\mathcal{E}$ and a function $F : \mathcal{E} \rightarrow \mathbb{R}$, and explain the idea of minimax principle for such a function.

Definition 7.1(i) We say that F is continuously differentiable with a derivative ∇F if $\nabla F : \mathcal{E} \rightarrow \mathcal{E}$ is a continuous function such that for all x and a ,

$$F(x) = F(a) + \langle \nabla F(a), x - a \rangle + o(\|x - a\|).$$

We say that x is a critical point of F if $\nabla F(x) = 0$.

(ii) We say that $F \in C^1(\mathcal{E}; \mathbb{R})$ satisfies *Palais–Smale* (PS) condition if the conditions

$$(7.3) \quad \sup_l |F(x_l)| < \infty, \quad \lim_{l \rightarrow \infty} \nabla F(x_l) = 0,$$

for a sequence $\{x_l\}$ imply that $\{x_l\}$ has a convergent subsequence. □

Given a family $\mathcal{F}$ of subsets of $\mathcal{E}$, define

$$\alpha(F, \mathcal{F}) = \inf_{A \in \mathcal{F}} \sup_{x \in A} F(x) \in [-\infty, +\infty].$$

Theorem 7.3 (*Minimax Principle*). *Let $F : \mathcal{E} \rightarrow \mathbb{R}$ be a Palais–Smale function and assume that the flow Φ_t of the gradient ODE $\dot{x} = -\nabla F(x)$ is well-defined for all $t \in \mathbb{R}^+$. If $\Phi_t(A) \in \mathcal{F}$ for $A \in \mathcal{F}$ and that $\alpha = \alpha(F, \mathcal{F}) \in \mathbb{R}$, then there exists $x^* \in \mathcal{E}$ such that*

$$(7.4) \quad \nabla F(x^*) = 0 \text{ and } F(x^*) = \alpha(F, \mathcal{F}).$$

Proof It suffices to show

$$(7.5) \quad \inf \{ \|\nabla F(x)\| : \alpha^- < F(x) < \alpha^+ \} = 0,$$

for every pair of constants α^+ and α^- such that $\alpha^- < \alpha < \alpha^+$. Indeed if (7.5) is true, then for every $l \in \mathbb{N}$, we can find x_l such that

$$\alpha - l^{-1} < F(x_l) < \alpha + l^{-1}, \quad \|\nabla F(x_l)\| \leq l^{-1}.$$

We then use the Palais–Smale property of F to assert that $\{x_l\}$ has a convergent subsequence that converges to x^* . Since $F \in C^1$, we deduce that $\nabla F(x^*) = 0$.

To establish (7.5), we argue by contradiction. Suppose to the contrary,

$$(7.6) \quad \inf \{ \|\nabla F(x)\| : \alpha^- < F(x) < \alpha^+ \} = \varepsilon > 0.$$

By the definition of α , we can find a set $A \in \mathcal{F}$ such that

$$\sup_{x \in A} F(x) < \alpha^+.$$

To get a contradiction, we use the flow of Φ_t to come up with another set $\hat{A} = \Phi_t(A) \in \mathcal{F}$ for which $\sup_{\hat{A}} F \leq \alpha^-$ for sufficiently large t . Indeed, if $x \in A$ and $F(\Phi_t(x)) > \alpha^-$, then

$$\alpha^- < F(\Phi_s(x)) < \alpha^+,$$

for every $s \in [0, t]$. From this and (7.6) we deduce

$$\alpha^- < F(\Phi_t(x)) = F(x) - \int_0^t \|\nabla F(\Phi_s(x))\|^2 ds < \alpha^+ - t\varepsilon^2,$$

which is impossible if $t > (\alpha^+ - \alpha^-)/\varepsilon^2$. Hence we must have $F(\Phi_t(x)) \leq \alpha^-$, for such large t . This in turn implies

$$\sup_{\Phi_t(A)} F \leq \alpha^-,$$

which contradicts our assumption $\alpha^- < \alpha$. Thus (7.5) must be true. $\square$

Example 7.1 Let $F \in C^1(\mathbb{R}^d; \mathbb{R})$ be a function which satisfies the Palais–Smale condition. Assume that F is bounded from below. Then F has a minimizer x^* , i.e., $F(x^*) = \inf F$. This can be shown using Theorem 7.3 by taking $\mathcal{F} = \{\{x\} : x \in \mathbb{R}^d\}$. Similarly, when F is bounded above, take $\mathcal{F} = \{\mathcal{E}\}$ to deduce that the function F has a maximizer. $\square$

We now give two applications of Theorem 7.3.

Proposition 7.1 *Let $F \in C^1(\mathcal{E})$ be a Palais–Smale function.*

(i) *(Mountain Pass Lemma of Ambrosetti and Rabinowitz) Assume that $R \subseteq \mathcal{E}$ is a mountain range relative to F in the following sense:*

- $\mathcal{E} \setminus R$ is not connected,
- $\inf_R F =: \beta > -\infty$,
- If A is a connected component of $\mathcal{E} \setminus R$, then $\inf_A F < \beta$.

Then F has a critical value α satisfying $\alpha \geq \beta$.

(ii) Let Γ and Σ be two bounded subsets of $\mathcal{E}$ such that $\inf_{\Gamma} F = \beta > -\infty$, $\Phi_t(\Sigma) \cap \Gamma \neq \emptyset$ for all $t \geq 0$, and $\sup_{\Sigma} F < \infty$. Then F has a critical point x^* such that

$$F(x^*) = \inf_{t \geq 0} \sup_{x \in \Phi_t(\Sigma)} F(x) \geq \beta.$$

(As in Theorem 7.3, Φ_t denotes the flow of $-\nabla F$.)

Proof (i) Let $\mathcal{E}^1$ and $\mathcal{E}^2$ be two connected components of $\mathcal{E} \setminus R$ and set $\hat{\mathcal{E}}^i = \{x \in \mathcal{E}^i : F(x) < \beta\}$, for $i = 1$ and 2 . We now define

$$\mathcal{F} = \{\gamma[0, 1] \text{ such that } \gamma : [0, 1] \rightarrow \mathcal{E} \text{ is continuous with } \gamma(0) \in \hat{\mathcal{E}}^1 \text{ and } \gamma(1) \in \hat{\mathcal{E}}^2\}.$$

We set $\alpha = \alpha(F, \mathcal{F})$ and would like to apply Theorem 7.2. Note that if $A = \gamma[0, 1] \in \mathcal{F}$, then $A \cap R \neq \emptyset$ and $\sup_A F \geq \beta$. Hence $\alpha \geq \beta$. Evidently $\alpha < \infty$ because $A \in \mathcal{F}$ is compact. On the other hand, if $A = \gamma[0, 1] \in \mathcal{F}$ with $\gamma(0) = a_1 \in \hat{\mathcal{E}}^1$ and $\gamma(1) = a_2 \in \hat{\mathcal{E}}^2$, then $\Phi_t(a_j) \in \hat{\mathcal{E}}^j$ for $j = 1, 2$ and $t \geq 0$, because

$$F(\Phi_t(a_j)) \leq F(a_j) < \beta$$

and $\Phi_t(a_j) \notin R$ by $\inf_R F = \beta$.

(ii) We simply take $\mathcal{F} = \{\Phi_t(\Sigma) : t \geq 0\}$. Evidently $\alpha(F, \mathcal{F}) \leq \sup_{\Sigma} F < \infty$. Moreover, since $\Phi_t(\Sigma) \cap \Gamma \neq \emptyset$, we have

$$\sup_{t \in \Phi_t(\Sigma)} F(x) \geq \beta$$

for every $t \geq 0$. We can now apply Theorem 7.3 to complete the proof. $\square$

Our goal is proving Theorem 7.2 with the aid of Lemma 7.1 and Proposition 7.1 for $F = \mathcal{A}$ and $\mathcal{E} = \mathcal{H}^{1/2}$. Let us write $\mathcal{A} = \mathcal{A}_0 - \mathcal{C}$ where

$$\mathcal{A}_0(x) = \frac{1}{2} \int_0^1 \bar{J}x \cdot \dot{x} dt, \quad \mathcal{C}(x) = \int_0^1 H(x) dt.$$

To differentiate $\mathcal{A}$, we need the following operators: given $x = \sum_k e^{2\pi k t \bar{J}} x_k$, set

$$P^{\pm}x = \sum_{\pm k > 0} e^{2\pi k t \bar{J}} x_k, \quad P^0x = x_0,$$

$$\mathcal{I}(x) = x_0 + \sum_{k \neq 0} \frac{1}{2\pi|k|} e^{2\pi k t \bar{J}} x_k.$$

When we differentiate $x(\cdot)$, the k -th Fourier coefficient is multiplied by $2\pi k\bar{J} = -2\pi ki$, where as in the definition of $\mathcal{I}(x)$, the k -th coefficient is divided by $2\pi|k|$. In some sense $\mathcal{I}$ is an integration operator and it is easy to see

$$(7.7) \quad \|\mathcal{I}(x)\|_1 = \|x\|_0.$$

As we have seen before, if we consider the Hilbert space $\mathcal{H}^0 = L^2$, then the gradient of $\mathcal{A}$ with respect to the L^2 inner product is of the form

$$\partial\mathcal{A}(x) = -\bar{J}\dot{x} - \nabla H(x).$$

However, if we write $\nabla\mathcal{A}$ for the derivative of $\mathcal{A}$ with respect to $\mathcal{E} = \mathcal{H}^{1/2}$, then

$$\nabla\mathcal{A} = \mathcal{I}\partial\mathcal{A}.$$

Proposition 7.2 (i) *The function $\mathcal{A}$ is C^1 with*

$$(7.8) \quad \nabla\mathcal{A}(x) = P^+x - P^-x - \nabla\mathcal{C}(x) = P^+x - P^-x - \mathcal{I}(\nabla H(x)).$$

(ii) *The operator $\mathcal{C}$ is a compact operator with*

$$\|\nabla\mathcal{C}(x) - \nabla\mathcal{C}(y)\|_{1/2} \leq c_0\|x - y\|,$$

where $c_0 = \sup \|D^2H\|$.

(iii) *The flow for the vector field $-\nabla\mathcal{A}$ is well-defined for all times.*

(iv) *If $\nabla\mathcal{A}(x) = 0$, then x is C^1 and $\dot{x} = \bar{J}\nabla H(x)$.*

(v) *If $\nabla\mathcal{A}(x_l) \rightarrow 0$ in $\mathcal{H}^{1/2}$, then the sequence $\{x_l\}$ has a convergent subsequence. In particular, the function $\mathcal{A}$ satisfies the (PS) condition.*

Proof (i) Note that $\mathcal{A}_0$ is quadratic and therefore smooth. In fact for $x = \sum_k e^{2\pi kt\bar{J}}x_k$,

$$\mathcal{A}_0(x) = \sum_k \pi k |x_k|^2 = \frac{1}{2}\|P^+x\|^2 - \frac{1}{2}\|P^-x\|^2,$$

which in turn implies

$$(7.9) \quad \nabla\mathcal{A}_0(x) = (P^+ - P^-)x.$$

We now turn to the functional $\mathcal{C}$. Of course $\mathcal{C}$ is differentiable with respect to L^2 -inner product and its derivative is given by $\nabla H(x)$. More precisely, since $\sup \|D^2H\| = c_0 < \infty$, we have

$$|H(x+h) - H(x) - \nabla H(x) \cdot h| \leq \frac{c_0}{2}|h|^2.$$

As a result,

$$|\mathcal{C}(x+h) - \mathcal{C}(x) - \langle \nabla H(x), h \rangle_0| \leq \frac{c_0}{2} \|h\|_0^2 \leq \frac{c_0}{2} \|h\|_{1/2}^2$$

which implies $\nabla \mathcal{C}(x) = \mathcal{I}(\nabla H(x))$, because

$$\langle \nabla H(x), h \rangle_0 = \langle \mathcal{I}(\nabla H(x)), h \rangle_{1/2},$$

completing the proof of (7.7).

(ii) The operator $x(\cdot) \rightarrow \nabla H(x(\cdot))$ maps bounded subsets of $\mathcal{H}^0 = L^2$ to bounded subset of $\mathcal{H}^0$ because $|\nabla^2 H(x)| \leq c_0$. The operator $\mathcal{I}$ maps bounded subset of $\mathcal{H}^0$ to bounded subsets of $\mathcal{H}^1$ by (7.7). By Exercise 7.1(iv) below, bounded subsets of $\mathcal{H}^1$ are precompact in $\mathcal{H}^{1/2}$. Hence $\nabla \mathcal{C}$ is a compact operator.

For Lipschitzness of $\nabla \mathcal{C}$, observe that by (7.7)

$$\begin{aligned} \|\nabla \mathcal{C}(x) - \nabla \mathcal{C}(y)\|_{1/2} &= \|\mathcal{I}(\nabla H(x) - \nabla H(y))\|_{1/2} \leq \|\mathcal{I}(\nabla H(x) - \nabla H(y))\|_1 \\ &= \|\nabla H(x) - \nabla H(y)\|_0 \leq c_0 \|x - y\|_0 \leq c_0 \|x - y\|_{1/2}. \end{aligned}$$

(iii) Since $\nabla \mathcal{C}$ is Lipschitz and $\nabla \mathcal{A}_0$ is linear, we learn that $\nabla \mathcal{A}$ is Lipschitz. This guarantees that the gradient flow is well-defined.

(iv) Since $\nabla \mathcal{A}(x) = 0$, we have that $P^+x - P^-x = \mathcal{I}(\nabla H(x))$. If

$$x = \sum_k e^{2\pi k t \bar{J}} x_k, \quad \nabla H(x) = \sum_k e^{2\pi k t \bar{J}} a_k,$$

then we deduce that $a_0 = 0$ and $\text{sgn}(k)x_k = (2\pi|k|)^{-1}a_k$ for $k \neq 0$, or $2\pi k x_k = a_k$ for all k . Since $\nabla H(x) \in \mathcal{H}^0 = L^2$, we learn that $\sum_k |k|^2 |x_k|^2 < \infty$, or $x \in \mathcal{H}^1$. From $2\pi k x_k = a_k$, we can readily deduce that $\dot{x} = \bar{J} \nabla H(x)$ weakly. Since the right-hand side is continuous by $\mathcal{H}^1 \subseteq C(S^1)$, we learn that $x \in C^1(S^1)$.

As an alternative proof, observe that if $\nabla \mathcal{A}(x) = 0$, then for every $\zeta \in \mathcal{H}^{1/2}$,

$$0 = \langle \nabla \mathcal{A}(x), \zeta \rangle_{1/2} = \langle \partial \mathcal{A}(x), \zeta \rangle_0,$$

which implies $\dot{x} = \bar{J} \nabla H(x)$ weakly.

(v) *Step 1.* Take a sequence $\{x_l\}$ such that

$$(7.10) \quad \nabla \mathcal{A}(x_l) = P^+x_l - P^-x_l - \nabla \mathcal{C}(x_l) \rightarrow 0,$$

as $l \rightarrow \infty$. We first prove the boundedness of $\{x_l\}$. Assume to the contrary

$$\lim_{l \rightarrow \infty} \|x_l\|_{1/2} = \infty.$$

Observe that if $y_l = \frac{x_l}{\|x_l\|_{1/2}}$, then by (7.8),

$$(7.11) \quad P^+ y_l - P^- y_l - \mathcal{I} \left(\frac{1}{\|x_l\|_{1/2}} \nabla H(x_l) \right) \rightarrow 0.$$

Now we use

$$\left\| \frac{\nabla H(x_l)}{\|x_l\|_{1/2}} \right\|_0 \leq c_0 \frac{\|x_l\|_0}{\|x_l\|_{1/2}} \leq c_0,$$

to deduce that the sequence

$$\mathcal{I} \left(\frac{1}{\|x_l\|_{1/2}} \nabla H(x_l) \right),$$

has a convergent subsequence in $\mathcal{H}^{1/2}$. This and (7.11) implies that the sequence $\{P^+ y_l - P^- y_l\}$ has a convergent subsequence. Without loss of generality, we may assume that the sequence $\{P^+ y_l - P^- y_l\}$ is convergent. Since P^+ and P^- project onto the positive and negative frequencies, we deduce that both sequences $\{P^+ y_l\}$ and $\{P^- y_l\}$ are convergent. Since the sequence $\{y_l\}$ is bounded, the sequence $\{P^0 y_l\}$ is bounded in $\mathbb{R}^{2n}$. Hence, by switching to a subsequence if necessary, we may assume that $y_l = P^+ y_l + P^- y_l + P^0 y_l$ is convergent. Let us continue to use $\{y_l\}$ for such a subsequence and write y for its limit. Recall that for $z \in \mathbb{R}^{2n}$ with large $|z|$,

$$H(z) = (\pi + \epsilon)Q(z) =: \hat{Q}(z)$$

where $Q(z) = q_1^2 + p_1^2 + l^{-2} \sum_{j=2}^n (q_j^2 + p_j^2)$. We now argue

$$(7.12) \quad \lim_{l \rightarrow \infty} \left\| \frac{\nabla H(x_l)}{\|x_l\|_{1/2}} - \nabla \hat{Q}(y) \right\|_0 = 0.$$

To see this, observe

$$\left\| \frac{\nabla H(x_l)}{\|x_l\|_{1/2}} - \nabla \hat{Q}(y) \right\|_0 \leq \|\nabla H(x_l) - \nabla \hat{Q}(x_l)\| \|x_l\|_{1/2}^{-1} + \|\nabla \hat{Q}(x_l) - \nabla \hat{Q}(y)\|_0.$$

Now (7.12) follows because $|\nabla H - \nabla \hat{Q}|$ is uniformly bounded and $\|y_l - y\|_{1/2} \rightarrow 0$. From (7.12) and (7.7) we deduce

$$\lim_{l \rightarrow \infty} \frac{1}{\|x_l\|_{1/2}} \mathcal{I}(\nabla H(x_l)) = \mathcal{I}(\nabla \hat{Q}(y)),$$

in $\mathcal{H}^{1/2}$. From this, $\lim_{l \rightarrow \infty} y_l = y$, and (7.11) we deduce

$$P^+ y - P^- y - \mathcal{I}(\nabla \hat{Q}(y)) = 0,$$

for y satisfying $\|y\| = 1$. This means that y is a critical point of

$$\mathcal{A}_1(y) = \int_0^1 \left[\frac{1}{2} \bar{J}y \cdot \dot{y} - \hat{Q}(y) \right] dt.$$

We then apply part **(iv)** where H is replaced with $\hat{Q}$. As a result y is C^1 and $\dot{y} = \bar{J}\nabla\hat{Q}(y)$. Hence

$$y(t) = \left(e^{-2(\pi+\epsilon)it} a_1, e^{-2\frac{(\pi+\epsilon)}{i^2}it} a_2, \dots, e^{-2\frac{(\pi+\epsilon)}{i^2}it} a_n \right).$$

This is 1-periodic only if $y \equiv 0$, contradicting $\|y\| = 1$. Hence the sequence $\{x_l\}$ must be bounded.

Step 2. For a sequence $\{x_l\}$ satisfying (7.10), we know that $\{x_l\}$ is bounded. This implies the precompactness of $\{\nabla\mathcal{C}(x_l)\}$ by the compactness of the operator $\nabla\mathcal{C}$. This and (7.10) imply that $x_l^+ - x_l^-$ has a convergent subsequence. As in *Step 1*, we learn that $\{P^+x_l + P^-x_l\}$ has a convergent subsequence. From this we deduce that $\{x_l\}$ has a convergent subsequence because $\{P^0x_l\}$ is also bounded. $\square$

We are now ready to establish Theorem 7.2. On the account of Lemma 7.1, we need to find a critical point of $\mathcal{A}$ with $\mathcal{A}(x) > 0$. Note that our Hamiltonian H is supposed to vanish on some neighborhood of the origin.

Proof of Theorem 7.2 *Step 1.* To ease the notation, let us write $x^\pm = P^\pm x$, and $x^0 = P^0 x$. We also write

$$\mathcal{H}^{1/2} = \mathcal{E} = \mathcal{E}^- \oplus \mathcal{E}^0 \oplus \mathcal{E}^+,$$

where $\mathcal{E}^\pm$ and $\mathcal{E}^0$ denote the ranges of the operators $P^\pm$ and P^0 respectively. Let us use the *variation of constants formula* to derive a nice representation for the flow Φ_t of the vector field $-\nabla\mathcal{A}$:

$$(7.13) \quad \Phi_t(x) = e^t x^- + x^0 + e^{-t} x^+ + K(x, t),$$

for a function $K : \mathcal{E} \times \mathbb{R} \rightarrow \mathcal{E}$ that is continuous and compact. To see this, observe that the flow $\bar{\Phi}_t$ of the vector field $-\nabla\mathcal{A}_0 = P^- - P^+$ is simply given by

$$\bar{\Phi}_t(x) = e^t x^- + x^0 + e^{-t} x^+.$$

From this, we can readily deduce (7.13) with K given by

$$\begin{aligned} K(x, t) &= \int_0^t (e^{t-s} P^- + P^0 + e^{s-t} P^+) \nabla\mathcal{C}(\Phi_s(x)) ds \\ &= \mathcal{I} \int_0^t (e^{t-s} P^- + P^0 + e^{s-t} P^+) \nabla H(\Phi_s(x)) ds. \end{aligned}$$

Here we are using the fact that $P^\pm$ and P^0 commute with $\mathcal{I}$. Now the compactness of K follows from the facts that the flow Φ_s maps bounded sets to bounded sets (see Exercise 7.1(iii)) and that $\mathcal{I}$ is a compact operator.

Step 2. To establish Theorem 7.2 with the aid of Proposition 7.1(ii), we need to come up with suitable candidates for the sets Γ and Σ . Define

$$\begin{aligned}\Gamma &= \Gamma(r) = \{x \in \mathcal{E}^+ : \|x\| = r\}, \\ \Sigma &= \Sigma(\theta) = \{x : x = x^- + x^0 + se^+ : \|x^- + x^0\| \leq \theta, 0 \leq s \leq \theta\}\end{aligned}$$

where $e^+(t) = e^{2\pi t \bar{J}} e_1$ with $e_1 = (1, 0, \dots, 0) \in \mathbb{R}^{2n}$. Evidently $\sup_\Sigma \mathcal{A} < \infty$, because

$$\mathcal{A}(x) \leq \frac{1}{2} \|P^+ x\|_{1/2}^2 \leq \frac{\theta^2}{2} \|e^+\|_{1/2}^2.$$

Let us check that indeed $\inf_{\Gamma(r)} \mathcal{A} = \beta > 0$, for $r > 0$ sufficiently small. Recall that $|H(z)| \leq c_1 |z|^2$ for a constant c_1 . This however doesn't do the job and we need to use the fact that H vanishes near the origin. This property implies that for every $p \in (2, \infty)$, we can find a constant $c_1(p)$ such that $|H(z)| \leq c_1(p) |z|^p$. On the other hand, by a well-known Sobolev-type inequality,

$$\|x\|_{L^p} = \left(\int_0^1 |x(t)|^p dt \right)^{\frac{1}{p}} \leq c_0(p) \|x\|_{1/2},$$

for a universal constant $c_0(p)$ (see Appendix B). As a result, if $x \in \Gamma(r)$, and $p > 2$, then

$$\begin{aligned}\mathcal{A}(x) &= \frac{1}{2} \|x^+\|_{1/2}^2 - \int_0^1 H(x(t)) dt \geq \frac{1}{2} \|x^+\|_{1/2}^2 - c_1(p) \int_0^1 |x(t)|^p dt \\ &\geq \frac{1}{2} \|x\|_{1/2}^2 - c_1(p) c_0(p)^p \|x\|_{1/2}^p = \frac{1}{2} r^2 - c_1(p) c_0(p)^p r^p =: \beta.\end{aligned}$$

Evidently $\beta > 0$, if r is sufficiently small.

Step 3. On account of Proposition 7.1(ii), it suffices to show that if θ is sufficiently large, then $\Phi_t(\Sigma(\theta)) \cap \Gamma \neq \emptyset$ for all $t \geq 0$. Note that this is true for $t = 0$ whenever $\theta > r(2\pi)^{-1/2}$:

$$\Sigma(\theta) \cap \Gamma = \{(2\pi)^{-1} r e^+\}.$$

As we evolve points in Σ with time, this intersection point also evolves with time. To achieve our goal, it suffices to make sure that this single intersection point does not leave $\Phi_t(\Sigma(\theta))$. If the intersection point runs off, at some point must cross the boundary of $\Phi_t(\Sigma(\theta))$. To rule out such possibility, it suffices to show

$$(7.14) \quad \Phi_t(\partial\Sigma) \cap \Gamma = \emptyset,$$

for every $t \geq 0$. The idea is that in some sense $\partial\Sigma$ and Γ *link* with respect to Φ_t . That is, $\Phi_t(\partial\Sigma)$ can not cross the sphere Γ as t increases, so Γ must intersect the “frame” $\Phi_t(\Sigma)$.

If fact for (7.14) the requirement $H(x) = (\pi + \varepsilon)Q(x)$ with $\pi + \varepsilon > \frac{1}{2}\|e^+\|_{1/2}$ is used in an essential way. This is the only place that the condition of $\max_Z H_0 > \pi$ of Theorem 7.2 is used. Since $\inf_\Gamma \mathcal{A} = \beta > 0$, it suffices to show

$$\sup_{t \geq 0} \sup_{\Phi_t(\partial\Sigma)} \mathcal{A} \leq 0.$$

Since $\frac{d}{dt}\mathcal{A}(\Phi_t(x)) \leq 0$, it suffices to show

$$(7.15) \quad \sup_{\partial\Sigma} \mathcal{A} \leq 0.$$

We write $\partial\Sigma = \partial_1\Sigma \cup \partial_2\Sigma$ where $\partial_1\Sigma = \{x \in \partial\Sigma : x = x^- + x^0\}$. In the case of $x \in \partial_1\Sigma$, we have

$$\mathcal{A}(x) = -\frac{1}{2}\|x^-\|^2 - \int_0^1 H(x) dt \leq 0$$

because $H \geq 0$. It remains to show that $\sup_{\partial_2\Sigma} \mathcal{A} \leq 0$, for sufficiently large θ . Recall that there exists a constant c_1 such that $H(x) \geq (\pi + \epsilon)Q(x) - c_1$. Hence, for $x = x^- + x^0 + se^+$,

$$\begin{aligned} \mathcal{A}(x) &= \frac{1}{2}s^2\|e^+\|^2 - \frac{1}{2}\|x^-\|^2 - \int_0^1 H(x) dt \\ &\leq \pi s^2 - \frac{1}{2}\|x^-\|^2 - (\pi + \epsilon) \int_0^1 Q(x^- + x^0 + se^+) dt + c_1 \\ &= \pi s^2 - \frac{1}{2}\|x^-\|^2 - (\pi + \epsilon) \left[\int_0^1 Q(x^-) dt + \int_0^1 Q(x^0) dt + \int_0^1 Q(se^+) dt \right] + c_1 \\ &\leq -\frac{1}{2}\|x^-\|^2 - (\pi + \epsilon) \left[\int_0^1 Q(x^-) dt + Q(x^0) \right] - \epsilon s^2 + c_1 \\ &\leq c_1 - c_2(\|x^- + x^0\|^2 + \|se^+\|^2), \end{aligned}$$

where $c_2 = \min\{1/2, (\pi + \epsilon)/l^2\}$, for the first inequality we used $H \geq (\pi + \epsilon)Q - c_1$ and $\|e^+\|^2 = 2\pi$, and for the second equality we used the fact that Q is quadratic and that $\mathcal{E}^+, \mathcal{E}^-, \mathcal{E}^0$ are orthogonal with respect to L^2 -inner product. It is now clear that if either $\|x^- + x^0\| = \theta$ or $s = \theta$ with θ sufficiently large, then $\mathcal{A}(x) \leq 0$, proving $\sup_{\partial_2\Sigma} \mathcal{A} \leq 0$. This completes the proof of (7.15) which in turn implies (7.14).

Step 3. To show that $\Phi_t(\Sigma) \cap \Gamma \neq \emptyset$ for $t \geq 0$, we need to find $x \in \Sigma$ such that $\|\Phi_t(x)\| = r$ and $(P^- + P^0)\Phi_t(x) = 0$. The latter means

$$e^t x^- + x^0 + (P^- + P^0)K(x, t) = 0,$$

or equivalently,

$$(7.16) \quad x^0 + P^0 K(x, t) = 0, \quad x^- + e^{-t} P^- K(x, t) = 0.$$

To combine this with the former condition $\|\Phi_t(x)\| = r$, define

$$L(x, t) = (e^{-t} P^- + P^0) K(x, t) + P^+ \{(\|\Phi_t(x)\| - r)e^+ - x\}.$$

We can readily show that $\Phi_t(\Sigma) \cap \Gamma \neq \emptyset$ is equivalent to finding $x \in \Sigma$ such that

$$(7.17) \quad x + L(x, t) = 0.$$

We wish to find a solution of (7.17) in the interior of Σ , which is an open bounded subset of

$$\hat{\mathcal{E}} = \mathcal{E}^- \oplus \mathcal{E}^0 \oplus \mathbb{R}e^+.$$

Note that $L : \hat{\mathcal{E}} \times \mathbb{R} \rightarrow \hat{\mathcal{E}}$ is a compact operator simply because K is compact and $\mathcal{E}^+$ part of $\hat{\mathcal{E}}$ is one-dimensional.

Step 4. We will use Leray-Schauder degree theory to solve (7.17). (We refer to Appendix C, for a review of Degree Theory.) Note that by (7.14),

$$0 \notin (I + L(\cdot, t))(\partial\Sigma).$$

Because of this, $\deg_0(I + L(\cdot, t))$ is well-defined. Observe that $(I + L(\cdot, s) : s \in [0, t])$ defines a homotopy, which implies

$$(7.18) \quad \deg_0(I + L(\cdot, t)) = \deg_0(I + L(\cdot, 0)).$$

From the definition of K given in (7.13), we know that $K(\cdot, 0) \equiv 0$. As a result,

$$L(x, 0) = P^+ \{(\|x\| - r)e^+ - x\}.$$

To calculate the right-hand side of (7.18), let us define

$$L^\alpha(x) = P^+ \{(\alpha\|x\| - r)e^+ - \alpha x\},$$

for $\alpha \in [0, 1]$. We claim

$$(7.19) \quad 0 \notin (I + L^\alpha)(\partial\Sigma).$$

Indeed if $(I + L^\alpha)(x) = 0$ for some $x \in \partial\Sigma$, then $x = se^+$ for some $s \in \{0, \theta\}$ and $s + \alpha s\|e^+\| - r - \alpha s = 0$, or $s((1 - \alpha) + \alpha\sqrt{2\pi}) = r$ because $\|e^+\| = \sqrt{2\pi}$. Evidently $s \neq 0$ because $r \neq 0$. To rule out $s = \theta$, observe that we may take θ large enough to have $\theta > r$.

But $r = s((1 - \alpha) + \sqrt{2\pi}\alpha) > s$ which implies that $\theta > s$. In summary $x + L^\alpha(x) = 0$ has no solution in $\partial\Sigma$, or equivalently (7.19) holds.

By (7.19), $\deg_0(I + L^\alpha)$ is well-defined and by the homotopy invariance of degree,

$$\begin{aligned}\deg_0(I + L(\cdot, 0)) &= \deg_0(I + L^1) = \deg_0(I + L^0) \\ &= \deg_0(I - re^+) = \deg_{re^+}(I) = 1,\end{aligned}$$

provided that $re^+ \in \Sigma$, which is true by our assumption $r < \theta$. From this and (7.18) we deduce that (7.17) has a solution $x \in \Sigma$, for which $\mathcal{A}(x) \geq \beta > 0$. This completes the proof of Theorem 7.2. $\square$

Proof of Theorem 7.1 (Sketch) The proof of Theorem 7.1 is similar to the proof of Theorem 7.2. We refer to [HZ] for details and only give a list of adjustments that we need to make to the proof of Theorem 7.2:

(Step 1) Take a bounded strictly convex set K with 0 in its interior. Given $H_0 \in \mathcal{H}(K)$ with $\max H_0 > \mathbf{c}_0(K)$, it suffices to show that X_{H_0} has a non-constant 1-periodic orbit. As before, we may assume that $H_0 = 0$ in a neighborhood of the origin. We then pick a constant $m \in (\mathbf{c}_0(K), \max H)$, and define

$$H(x) = \begin{cases} H_0(x) & \text{if } x \in K, \\ f(Q(x)) & \text{if } x \notin K. \end{cases}$$

where $Q = g_K^2$, and $f : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth function such that $f(r) = \max H$ for $r \in [0, 1]$, $f(r) = mr$ for large r , $f(r) \geq mr$ for all r , and $0 < f'(r) \leq m$ for $r > 1$. We then have the analog of Lemma 7.1 for H provided that the ellipsoid E is replaced with K .

(Step 2) As we verify PS condition, let y be as in the proof of Proposition 7.2(v). We need to get a contradiction from the conditions $\|y\|_{1/2} = 1$ and $\dot{y} = m\bar{J}\nabla Q(y)$, where $Q = g_K^2$. Note that y is a 1-periodic solution of X_{mQ} . On the other hand,

$$T^{\min}(mQ) = (2m)^{-1} T^{\min}(H_K) = (2m)^{-1} 2\mathbf{c}_0(K) < 1,$$

where we used Remark 5.1(ii) for the second equality. This of course does not lead to a contradiction. However, it is possible to find a new compact set K' near K so that K' has no 1-periodic orbit. We then choose $Q = g_{K'}^2$ in part (i) in our extension. To ease the notation, we write K for K' .

(Step 3) We choose the same $\Gamma = \Gamma(r)$ as in the proof of Theorem 7.2. Given $p > 2$, we can find a constant $c_1(p)$ such that $H(x) \leq c_1(p)|x|^p$. Fix $p > 2$, and a repetition of our proof in Step 2 of the proof of Theorem 7.2 yields

$$\inf_{\Gamma(r)} \mathcal{A}_H =: \beta > 0,$$

for sufficiently small r .

(Step 4) Our choice of $\Sigma = \Sigma(\theta)$ is as before, expect that our candidate for e^+ is related to the minimizing orbit in the definition of $\mathbf{c}_0(K)$. Set $\bar{m} = \mathbf{c}_0(K)$ and consider the Hamiltonian equation $\dot{x} = \bar{m}\bar{J}\nabla Q(x)$. This corresponds to the Hamiltonian function $\bar{m}Q$ with

$$T^{min}(\bar{m}Q) = (2\bar{m})^{-1} T^{min}(H_K) = 1,$$

which means that the vector field $\bar{m}X_Q$ has a 1-periodic orbit $\bar{x}$ that lies on ∂K . We then set $e^+ = P^+\bar{x}$. We note that $A(\bar{x}) = \bar{m} > 0$ which implies that $e^+ \neq 0$, because $2A(\bar{x}) = \|P^+\bar{x}\|^2 - \|P^-\bar{x}\|^2$. We note for sufficiently large θ ,

$$\Gamma \cap \Sigma(\theta) = \{s_0 e^+\},$$

where

$$s_0 = \frac{r}{\sqrt{A(e^+)}}.$$

Repeating the proof of Theorem 7.1, it remains to verify

$$(7.20) \quad \sup_{\partial_2 \Sigma} \mathcal{A} \leq 0.$$

Observe that there exists a constant c_1 such that

$$H(x) \geq mQ(x) - c_1 = \bar{m}Q(x) + (m - \bar{m})Q(x) - c_1.$$

Hence,

$$(7.21) \quad \mathcal{A}_H(x) \leq \mathcal{A}_{\bar{m}Q}(x) - (m - \bar{m}) \int_0^1 Q(x) dt + c_1.$$

Let us write

$$\mathcal{E}_s = \mathcal{E}^- \oplus \mathcal{E}^0 \oplus s e^+.$$

We claim

$$(7.22) \quad \sup_{\mathcal{E}_s} \mathcal{A}_{\bar{m}Q} \leq 0.$$

Assuming this, we can use (7.21) to assert

$$\begin{aligned} \sup_{\partial_2 \Sigma} \mathcal{A} &\leq - (m - \bar{m}) \inf_{x \in \Sigma} \int_0^1 Q(x) dt + c_1 \leq -\delta(m - \bar{m}) \inf_{x \in \partial_2 \Sigma} \int_0^1 |x|^2 dt + c_1 \\ &\leq c_1 - \delta(m - \bar{m}) \inf \left\{ \int_0^1 [|x^-|^2 + |x^0|^2 + s^2 |e^+|^2] dt : \|x^- + x^0\| = \theta, \text{ or } s = \theta \right\}, \end{aligned}$$

where $\delta > 0$ is chosen so that $Q(x) \geq \delta|x|^2$. From this we readily deduce (7.20).

(Step 5) It remains to verify (7.22). By strict convexity of K , we can readily show that the supremum on the left-hand side of (7.22) is attained at a unique $\bar{y} \in \mathcal{E}_s$. This $\bar{y}$ must satisfy

$$(P^- \oplus P^0)(\nabla \mathcal{A}_{\bar{m}Q})(\bar{y}) = 0.$$

On the other hand, $(\nabla \mathcal{A}_{\bar{m}Q})(\bar{x}) = 0$, which implies $(\nabla \mathcal{A}_{\bar{m}Q})(s\bar{x}) = 0$. Hence $\bar{y} = s\bar{x}$. On the other hand, since Q is homogeneous of degree 2, and $\bar{y}$ solves the Hamiltonian ODE associated with sQ , we deduce that $\mathcal{A}_{\bar{m}Q}(\bar{y}) = 0$. This implies (7.22). $\square$

Exercise 7.1(i) Assume $F \in C^1(\mathbb{R}^d; \mathbb{R})$ satisfies $\lim_{|x| \rightarrow \infty} F(x) = \infty$ and that F possesses two distinct relative minima x_1 and x_2 . Show that F has a third critical point x_3 that is different from x_1 and x_2 . **Hint:** Use paths connecting x_1 to x_2 for the members of $\mathcal{F}$ in Theorem 7.2.

(ii) Show that $F(x, y) = e^{-x} - y^2$ does not satisfy Palais–Smale condition. Let

$$E^\pm = \{(x, y) : F(x, y) \leq 0, \pm y \geq 0\}$$

and set

$$\mathcal{F} = \{\gamma[0, 1] : \gamma : [0, 1] \rightarrow \mathbb{R}^2, \gamma(0) \in E^-, \gamma(1) \in E^+ \text{ and } \gamma \text{ is continuous}\}.$$

Show that $\alpha(F, \mathcal{F}) = 0$ but there is no x^* with $F(x^*) = 0$ and that F has no critical point.

(iii) Let $G : \mathcal{E} \rightarrow \mathcal{E}$ be a Lipschitz function and let Φ_t denote the flow of the ODE $\dot{x} = G(x)$. Show that for every $l > 0$,

$$\sup_{0 \leq t \leq l} \sup_{\|x\| \leq l} \|\Phi_t(x)\| < \infty.$$

(iv) Let $\mathcal{H}^s = \{x \in L^2 : \|x\|_s < \infty\}$. Show that if $s < t$, then a bounded subset of H^t is precompact in H^s .

(v) Show that if $x \in \mathcal{H}^\theta$ for $\theta > 1/2$, and $\theta \neq 3/2$, then x is Hölder continuous with

$$|x(t) - x(s)| \leq c\|x\|_\theta |t - s|^\alpha$$

with $\alpha = \min(1, \theta - \frac{1}{2})$. **Hint:** Given $x = \sum_k e^{2\pi kt\bar{J}} x_k$, write $x = y + z$ with $z = \sum_{|k| \leq N} e^{2\pi kt\bar{J}} x_k$. Estimate $\sup_{|t-s| < \delta} |y(t) - y(s)|$ and $\sup_t |z(t)|$ in terms of δ and N . This yields a bound for $|x(t) - x(s)|$ that can be minimized with respect to N . When $\theta = 3/2$, a logarithmic correction is needed:

$$|x(t) - x(s)| \leq c\|x\|_\theta |t - s| |\log |t - s||^{1/2}.$$

$\square$

8 Hofer Geometry

As we can see from (6.7) and Exercise 6.1(vii) of Chapter 6, a uniform lower bound on a Hamiltonian function yields an upper bound on the shortest non-constant periodic orbit of the corresponding Hamiltonian vector field. In this chapter we explore deeper connections between the norm $\|H\|_{\infty,1}$ and the corresponding flow ϕ_t^H . In particular we will be able to define a metric on the space of Hamiltonian functions with compact support that is invariant with respect to any symplectic change of coordinates. This metric was defined by Hofer and is closely related to the displacement energy and the capacity $\mathbf{e}$.

Let us write $\mathcal{H}$ (respectively $\mathcal{H}_0$) for the set of smooth functions $H : \mathbb{R}^{2n} \times [0, 1] \rightarrow \mathbb{R}$ (respectively $H : \mathbb{R}^{2n} \rightarrow \mathbb{R}$) of compact support. The time-one map $\phi^H = \phi_1^H$ defines a map from $\mathcal{H}$ into the space of $\bar{\omega}$ -symplectic maps. The image of this map is denoted by $\mathcal{D}$. Note that the map $H \mapsto \phi^H$ is not injective. We may define an equivalence relation $\sim$ on $\mathcal{H}$ by $H \sim K$ iff $\phi^H = \phi^K$. The set of equivalence classes will be denoted by $\tilde{\mathcal{H}}$. We map $H \mapsto \phi^H$ induces a bijection between $\tilde{\mathcal{H}}$ and $\mathcal{D}$. We wish to equip $\mathcal{D}$, or equivalently $\tilde{\mathcal{H}}$ with a metric. This metric is closely related to the norm

$$\|H\|_{\infty,1} = \int_0^1 \|H(\cdot, t)\| dt,$$

where $\|K\| = \max K - \min K$. This norm induces a notion of “size” on the set of equivalence classes $\tilde{\mathcal{H}}$ by

$$E([H]) = \inf\{\|K\|_{1,\infty} : K \sim H\}.$$

Alternatively, we define the expression

$$E(\psi) = \inf\{\|H\|_{1,\infty} : \phi^H = \psi\},$$

for $\psi \in \mathcal{D}$. In fact we may regard $E(\psi)$ as the distance of ψ from identity. We can turn this into a metric for $\mathcal{D}$, because $\mathcal{D}$ is a group. More precisely, the expression

$$(8.1) \quad D(\psi, \varphi) := E(\psi^{-1} \circ \varphi),$$

defines a metric on $\mathcal{D}$. Before verifying the metric properties of D , let us examine the group properties of $\mathcal{D}$ and $\tilde{\mathcal{H}}$ in details.

Proposition 8.1 *Let $H, K \in \mathcal{H}$ and $\varphi \in \mathcal{D}$. (i) $\phi_t^{H\sharp K} = \phi_t^H \circ \phi_t^K$, where*

$$(H\sharp K)(x, t) := H(x, t) + K\left((\phi_t^H)^{-1}(x), t\right).$$

(ii) $(\phi_t^H)^{-1} = \phi_t^{\bar{H}}$, where $\bar{H}(x, t) = -H(\phi_t^H(x), t)$.

Proof Part (ii) follows from Part (i), because $H\sharp\bar{H} = 0$. For Part (i), observe that if $\psi_t = \phi_t^H \circ \phi_t^K$, and $\eta_t = (\phi_t^H)^{-1}$, then

$$\begin{aligned} \frac{d\psi_t}{dt} &= \frac{d\phi_t^H}{dt} \circ \phi_t^K + (d\phi_t^H) \circ \phi_t^K \frac{d\phi_t^K}{dt} = X_H \circ \phi_t^H \circ \phi_t^K + (d\phi_t^H) \circ \phi_t^K X_K \circ \phi_t^K \\ &= X_H \circ \psi_t + (d\phi_t^H) \circ \eta_t \circ \psi_t X_K \circ \eta_t \circ \psi_t \\ &= (X_H + (d\eta_t)^{-1} X_K \circ \eta_t) \circ \psi_t = (X_H + X_{K \circ \eta_t}) \circ \psi_t \\ &= X_{H\sharp K} \circ \psi_t, \end{aligned}$$

where we used Proposition 3.2 for the fifth equality. It is straightforward to show that $H\sharp K \in \mathcal{H}$ because $\phi_t^H(x) = x$ outside a compact set. We are done. $\square$

To ease the notation, we simply write $\varphi\psi$ for $\varphi \circ \psi$ when there is no danger of confusion.

Proposition 8.2 (i) *The function D defined by (8.1) is a metric on $\mathcal{D}$.*

(ii) $D(\varphi, \psi) = D(\varphi\eta, \psi\eta) = D(\eta\varphi, \eta\psi)$, for every $\eta \in \mathcal{D}$.

(iii) $D(\varphi_1 \dots \varphi_k, \psi_1 \dots \psi_k) \leq D(\varphi_1, \psi_1) + \dots + D(\varphi_k, \psi_k)$.

This is an immediate consequence of the following result.

Proposition 8.3 *For $\varphi, \psi, \eta, \varphi_1, \dots, \varphi_k, \psi_1, \dots, \psi_k \in \mathcal{D}$, we have*

(i) $E(\varphi) = E(\varphi^{-1})$,

(ii) $E(\varphi\psi) \leq E(\varphi) + E(\psi)$,

(iii) $E(\eta^{-1}\varphi\eta) = E(\varphi)$,

(iv) if $\varphi \neq id$, then $E(\varphi) > 0$,

(v) $E((\varphi_1 \dots \varphi_k)(\psi_k \dots \psi_1)) \leq \sum_{i=1}^k E(\varphi_i \psi_i)$.

Proof (i) This is an immediate consequence of the fact that $\varphi = \phi^H$ iff $\varphi^{-1} = \phi^{\bar{H}}$, and that $\|H\|_{\infty,1} = \|\bar{H}\|_{\infty,1}$.

(ii) It suffices to check that $\|H\sharp K\|_{\infty,1} \leq \|H\|_{\infty,1} + \|K\|_{\infty,1}$. Indeed, using the fact that $\|A + B\| \leq \|A\| + \|B\|$ for every $A, B \in \mathcal{H}_0$, we have

$$\begin{aligned} \|H\sharp K\|_{\infty,1} &\leq \int_0^1 [\|H(\cdot, t)\| + \|K(\eta_t(\cdot), t)\|] dt = \int_0^1 [\|H(\cdot, t)\| + \|K(\cdot, t)\|] dt \\ &= \|H\|_{\infty,1} + \|K\|_{\infty,1}, \end{aligned}$$

where $\eta_t = (\phi_t^H)^{-1}$.

(iii) The Proposition 3.2 implies that $\eta^{-1} \circ \phi_t^H \circ \eta = \phi^{H \circ \eta}$. This and $\|H\|_{\infty,1} = \|H \circ \eta\|_{\infty,1}$ imply the desired equality.

(iv) If $\varphi \neq id$, then for some $a \in \mathbb{R}^{2n}$, we have that $\varphi(a) \neq a$. As a result, there exists $r > 0$, such that $\varphi(B_r(a)) \cap B_r(a) = \emptyset$. We then use Theorem 6.2(vi) to assert

$$E(\varphi) \geq \mathbf{c}_{HZ}(B_r(a)) = \pi r^2.$$

Alternatively, we may use Theorem 6.5 to show that $E(\varphi) \geq \pi r^2/2$. In any case, we deduce that $E(\varphi) > 0$, as desired.

(v) We certainly have $(\varphi_1 \dots \varphi_k)(\psi_k \dots \psi_1)$ equals to

$$(\varphi_1 \psi_1) [\psi_1^{-1} (\varphi_2 \psi_2) \psi_1] [(\psi_2 \psi_1)^{-1} (\varphi_3 \psi_3) (\psi_2 \psi_1)] \dots [(\psi_{k-1} \dots \psi_1)^{-1} (\varphi_k \psi_k) (\psi_{k-1} \dots \psi_1)].$$

From this and Parts (ii) and (iii), we deduce the desired inequality. $\square$

One possible interpretation of Hofer metric is that we are turning the infinite dimensional group $\mathcal{D}$ into a *Finsler Manifold*. More precisely, since paths of symplectic paths in $\mathcal{D}$ are generated by Hamiltonian vector fields, we may define the tangent fibers of $\mathcal{D}$ by

$$T_\varphi \mathcal{D} = \{X_H \circ \varphi : H \in \mathcal{H}\},$$

for every $\varphi \in \mathcal{D}$. Identifying this space with the vector space $\mathcal{H}$, we equip the space $T_\varphi \mathcal{D}$ with the norm $\|X_H \circ \varphi\| = \|H\|_{\infty,1}$. We may define $\mathcal{L} : T\mathcal{D} \rightarrow \mathbb{R}$ by

$$\mathcal{L}(\varphi, \nu) = \|H\|_{\infty,1},$$

where H is uniquely defined by $\nu \circ \varphi^{-1} = X_H$. Now

$$(8.2) \quad D(\varphi, \psi) = \inf \left\{ \int_0^1 \mathcal{L}(\gamma(t), \dot{\gamma}(t)) dt : \gamma : [0, 1] \rightarrow \mathcal{D}, \gamma(0) = \varphi, \gamma(1) = \psi \right\}.$$

Remark 8.1 When $n = 1$, the set $\mathcal{D}$ consists of area preserving diffeomorphisms that coincide with the identity map outside a compact set. Also

$$T\mathcal{D} = \{(\varphi, u \circ \varphi) : \varphi \in \mathcal{D}, \operatorname{div} u = 0\}.$$

If we equip $T_\varphi \mathcal{D}$ with the Riemannian metric

$$g_\varphi(u \circ \varphi, v \circ \varphi) = \int (u \circ \varphi) \cdot (v \circ \varphi) dx = \int u \cdot v dx,$$

then we obtain a very different geometry that is related to the *incompressible Euler Equation*, as was observed by Arnold (in other words, we replace the L^∞ -norm of H with the L^2 -norm of ∇H). More generally, if $\mathcal{V}$ denotes the space of volume preserving diffeomorphism of $\mathbb{R}^d$, then

$$T_\varphi \mathcal{V} = \{u \circ \varphi : \operatorname{div} u = 0\},$$

and a geodesic $\varphi(\cdot, t)$ with respect the metric g will satisfy

$$\dot{\varphi} \perp T_\varphi \mathcal{V} = \{\nabla P \circ \varphi : P : \mathbb{R}^d \rightarrow \mathbb{R} \text{ smooth}\}.$$

If we define $u(\cdot, t)$ by $u \circ \varphi = \dot{\varphi}$, then the divergence-free vector field $u(\cdot, t)$ satisfies

$$u_t + (Du)u + \nabla P = 0,$$

for a scalar valued function $P(x, t)$. □

In view of Remark 8.1 and the interpretation (8.2), we may wonder what the geodesic of $(\mathcal{D}, D)$ are.

Definition 8.1(i) Let I be an interval. We say a smooth path $\varphi : I \rightarrow \mathcal{D}$ is *regular* if $\dot{\varphi}(t) \neq 0$ for every $t \in I$. Equivalently, if $\dot{\varphi} = X_H(\varphi)$, then φ is regular, if $\|H(\cdot, t)\| \neq 0$ for every $t \in I$.

(ii) A regular smooth path $\varphi : I \rightarrow \mathcal{D}$ with $\dot{\varphi} = X_H(\varphi)$ is called *minimal geodesic* if

$$D(\varphi(\cdot, a), \varphi(\cdot, b)) = \int_a^b \|H(\cdot, t)\| dt,$$

for every $a, b \in I$ with $a < b$.

(iii) A regular smooth path $\varphi : I \rightarrow \mathcal{D}$ with $\dot{\varphi} = X_H(\varphi)$ is called *geodesic* if for every s in the interior of I , there exists an interval $(s - \delta, s + \delta) \subset I$ such that the restriction of φ to $(s - \delta, s + \delta)$ is minimal geodesic. □

We have a complete classifications of the geodesic paths in $\mathcal{D}$. As a warm-up, we first consider the flows of time-independent Hamiltonian functions.

Theorem 8.1 (Hofer) Let $H \in \mathcal{H}$ be a non-zero time-independent Hamiltonian function. Then $E(\phi_t^H) = t\|H\|$ for $t < T^{\min}(H)$, where $T^{\min}$ is the shortest period of the non-constant periodic orbits of X_H .

Remark 8.2(i) To appreciate the significance of Theorem 8.1, assume that $0 < T_0 = T^{\min}(H) < 1$ for a nonzero $H \in \mathcal{H}_0$ so that the system $\dot{x} = X_H(x)$ has a periodic orbit $(\phi_t^H(x_0) : t \in \mathbb{R})$ of period T_0 . We note that if $\varphi = \phi_1^H$, then

$$\varphi(y) = \phi_1^H(y) = \phi_{1-T_0}^H(y) = \phi_1^{(1-T_0)H}(y),$$

for every y on the orbit

$$\gamma = \{ \phi_t^H(x_0) : t \in \mathbb{R} \},$$

of x_0 . Hence, on the set γ , we can lower the energy $\|H\|$ by switching from H to $(1 - T_0)H$. From this we expect to lower the energy for φ by modifying H near γ . In other words, if $T^{\min}(H) < 1$, we may have $E(\varphi) < \|H\|$ for $\varphi = \phi^H$, because of the presence of a (fast) periodic orbit of period $T_0 < 1$. What we learn from Theorem 8.2 is that such fast periodic orbits are the only feature of the system $\dot{x} = X_H(x)$ to watch out for; if they are absent, then we do have $E(\phi^H) = \|H\|$.

(ii) As an immediate consequence of Theorem 8.1 we learn that for any non-zero time-independent Hamiltonian function $H \in \mathcal{H}$, the path $\varphi(t) = \phi_t^H$ with $t \in (0, T^{\min}(H))$ is minimal geodesic. Consequently $\varphi : (0, \infty) \rightarrow \mathcal{D}$ is geodesic because $\varphi : (t, t + T^{\min}(H)) \rightarrow \mathcal{D}$ is minimal geodesic. $\square$

Before embarking on the proof of Theorem 8.1 and discuss the analog of Theorem 8.1 for the time dependent Hamiltonian functions, we need to discuss two important notions of *generating functions* and *action spectrum*. These two notions would prepare us for the construction of an auxiliary function $\mathcal{U} : \mathcal{D} \rightarrow [0, \infty)$, that would play an essential role in proving Theorem 8.1.

Consider a symplectic manifold (M, ω) with $\omega = d\lambda$ for a 1-form λ . Let us write Γ^T for the set of smooth paths $x : [0, T] \rightarrow M$. We also write $\Gamma_T \subset \Gamma^T$ for the T -periodic loops. Given a smooth Hamiltonian function $H : M \times [0, T] \rightarrow \mathbb{R}$, we define $\mathcal{A}_H : \Gamma^T \rightarrow \mathbb{R}$ by

$$\mathcal{A}_H(x) = \int_{x(\cdot)} \lambda - \int_0^T H(x(t), t) dt.$$

The restriction of $\mathcal{A}_H$ to the set of T -periodic x is denoted by $\mathcal{A}_H^T$. We note that as in (5.8), we can show

$$(8.3) \quad \partial \mathcal{A}_H^T(x(\cdot)) = -\bar{J}\dot{x}(\cdot) - \nabla H(x(\cdot), \cdot),$$

if $(M, \omega) = (\mathbb{R}^{2n}, \bar{\omega})$ and $\lambda = \bar{\lambda}$.

We define $\Lambda_H^T : M \rightarrow \mathbb{R}$, by

$$(8.4) \quad \Lambda_H^T(x_0) = \mathcal{A}_H(\eta_T^{x_0}), \quad \text{where} \quad \eta_T^{x_0}(t) = \phi_t^H(x_0) \text{ for } t \in [0, T].$$

We now claim that Λ is a generating function for ϕ^H .

Proposition 8.4 *For every $T \geq 0$ and any Hamiltonian H , we have*

$$(8.5) \quad d\Lambda_H^T = (\phi_T^H)^* \lambda - \lambda.$$

Proof I Set

$$A(x_0) = \int_{\eta_T^{x_0}} \lambda, \quad B(x_0) = \int_0^T H(\eta_T^{x_0}(t), t) dt.$$

Take any $(\tau(\theta) : 0 \leq \theta \leq h)$ with $\tau(0) = x_0$ and $\dot{\tau}(0) = v \in T_{x_0}M$. Set $y(t, \theta) = \phi_H^t(\tau(\theta))$,

$$\Theta = \{y(t, \theta) : 0 \leq t \leq T, 0 \leq \theta \leq h\},$$

and use Stokes' theorem to assert

$$\begin{aligned} h^{-1} \int_0^h \int_0^T \omega_y(y_t, y_\theta) dt d\theta &= h^{-1} \int_\Theta d\lambda = h^{-1} \left[\int_{\eta^{\tau(0)}} \lambda - \int_{\eta^{\tau(h)}} \lambda + \int_{\varphi \circ \tau(\cdot)} \lambda - \int_{\tau(\cdot)} \lambda \right], \\ h^{-1} \int_0^h \int_0^T (i_{X_H} \omega)_y(y_\theta) dt d\theta &= h^{-1} \left[\int_{\eta^{\tau(0)}} \lambda - \int_{\eta^{\tau(h)}} \lambda + \int_{\tau(\cdot)} (\varphi^* \lambda - \lambda) \right], \end{aligned}$$

where $\varphi = \phi_T^H$. Sending $h \rightarrow 0$ yields

$$-(dB)_{x_0}(v) = -(dA)_{x_0}(v) + (\varphi^* \lambda - \lambda)_{x_0}(v).$$

This is exactly (8.5).

Proof II Let us write $H^t(x) = H(x, t)$. By Proposition 3.1,

$$\begin{aligned} (\phi_T^H)^* \lambda - \lambda &= \int_0^T \frac{d}{dt} (\phi_t^H)^* \lambda dt = \int_0^T (\phi_t^H)^* \mathcal{L}_{X_H(\cdot, t)} \lambda dt \\ &= \int_0^T (\phi_t^H)^* [(i_{X_H(\cdot, t)} \circ d) \lambda + (d \circ i_{X_H(\cdot, t)}) \lambda] dt \\ &= \int_0^T (\phi_t^H)^* d[-H^t + \lambda(X_H(\cdot, t))] dt \\ &= d \int_0^T (\phi_t^H)^* [-H^t + \lambda(X_H(\cdot, t))] dt := d\hat{\Lambda}. \end{aligned}$$

On the other hand, if we write $\eta(t) = \phi_t^H(x_0)$ for $t \in [0, T]$, then

$$\hat{\Lambda}(x_0) = \int_0^T \lambda(X_H(\eta(t), t)) dt - \int_0^T H(\eta(t), t) dt = \Lambda^T(x_0),$$

because $\dot{\eta} = X_H \circ \eta$. We are done. $\square$

From now on we assume $M = \mathbb{R}^{2n}$ and that $\lambda = \bar{\lambda}$. We also write Λ_H for Λ_H^1 . We note that if $H \in \mathcal{H}$, then $\Lambda_H(x_0) = 0$ if $|x_0|$ is sufficiently large. From this and the identity (8.5)

we learn that if $\phi^H = \phi^K$, then $\Lambda_H = \Lambda_K$. This allows us to define Λ on $\tilde{\mathcal{H}}$ or $\mathcal{D}$. More precisely, for every $\varphi \in \mathcal{D}$, with $\varphi = \phi^H$, we set

$$\Lambda_\varphi := \Lambda_H.$$

We also note that if

$$\hat{H}(x, t) = p \cdot H_p(x, t) - H(x, t),$$

then

$$\Lambda_H(x_0) = \int_0^1 \hat{H}(\phi_t^H(x_0), t) dt.$$

Remark 8.3(i) We note that by integration by parts,

$$\int \hat{H}(x, t) dx = -(n+1) \int H(x, t) dx.$$

From this and the fact that ϕ_t^H is volume preserving we learn that if $\varphi = \phi^H$, then

$$\int \Lambda_\varphi dx = \int \int_0^1 \hat{H}(x, t) dt dx = -(n+1) \int \int_0^1 H(x, t) dt dx.$$

Motivated by this, we may define

$$Cal(\varphi) := -\frac{1}{n+1} \int \Lambda_\varphi dx = \int \int_0^1 H(x, t) dt dx.$$

This defines a map $Cal : \mathcal{D} \rightarrow \mathbb{R}$, that is known as *Calabi Homomorphism*. It is a group homomorphism because $Cal(\varphi \circ \psi) = Cal(\varphi) + Cal(\psi)$. One way to see this is to use

$$d\Lambda_{\varphi \circ \psi} = (\varphi \circ \psi)^* \lambda - \lambda = \psi^* d\Lambda_\varphi + \psi^* \lambda - \lambda = d(\Lambda_\varphi \circ \psi + \Lambda_\psi),$$

to deduce

$$\Lambda_{\varphi \circ \psi} = \Lambda_\varphi \circ \psi + \Lambda_\psi.$$

Alternatively, we may use $\phi^{H\sharp K} = \phi^H \circ \phi^K$ and

$$\int (H\sharp K) dx = \int H dx + \int K dx.$$

(ii) Set $\mathcal{H}_{c_0}$ for the set of Hamiltonian function with support in the set

$$\{x : |x| \leq c_0\} \times [0, 1].$$

We write $\mathcal{D}_{c_0}$ for the set of symplectic maps associated with $H \in \mathcal{H}_{c_0}$. We may define

$$\hat{D}_{p,c_0}(id, \varphi) := \inf \left\{ \int_0^1 \int |\nabla H(x, t)|^p dx dt : \phi^H = \varphi \text{ for some } H \in \mathcal{H}_{c_0} \right\}.$$

We also set

$$\hat{D}_p(id, \varphi) = \inf_{c_0} \hat{D}_{p,c_0}(id, \varphi).$$

When $n = 1$, and $p = 2$, then the corresponding metric coincides with the Arnold's metric that was discussed in Remark 8.1. We note that since

$$\varphi(x) - x = \int_0^1 X_H(\phi_t^H(x), t) dt,$$

and ϕ_t^H is volume preserving, we have

$$\int |\varphi(x) - x|^p dx \leq \int \int_0^1 |X_H(x, t)|^p dt dx = \int \int_0^1 |\nabla H(x, t)|^p dt dx.$$

This implies

$$\int |\varphi(x) - x|^p dx \leq \hat{D}_p(id, \varphi).$$

There is also an interesting lower bound in terms of $Cal(\varphi)$ on $\hat{D}_1$ that was discovered by Eliashberg and Ratiu. To explain their lower bound, observe that if $H \in \mathcal{H}_{c_0}$, and we pick some a with $|a| = c_0$, then

$$|\hat{H}(x, t)| = |p \cdot H_p(x, t) - H(x, t) + H(a, t)| \leq c_0 \left[|\nabla H(x, t)| + \int_0^1 |\nabla H((1 - \theta)x + \theta a)| d\theta \right],$$

for any x with $|x| \leq c_0$. As a result,

$$\int_{|x| \leq c_0} |\Lambda_\varphi(x)| dx \leq 2c_0 \int_{|x| \leq c_0} \int_0^1 |\nabla H(x, t)| dt dx,$$

because ϕ_t^H is volume preserving. Since $\Lambda_\varphi(x) = 0$ for $|x| \geq c_0$, we deduce

$$|Cal(\varphi)| \leq \frac{1}{(d+1)} \int |\Lambda_\varphi(x)| dx \leq \frac{2c_0}{d+1} \hat{D}_{1,c_0}(id, \varphi).$$

From this inequality we can readily show that the diameter of the set $\mathcal{D}_{c_0}$ is infinite, a fact that was discovered by Shnirelman in the case of $n = 1$.

(iii) Note that Λ_H is well-defined for any Hamiltonian function H . When $H(q, p)$ is convex in p , then

$$\hat{H}(q, p) = L(q, H_p(q, p)),$$

where $L(q, v)$ denotes the Legendre Transform of H in p -variable. Moreover, if $x(t) = (q(t), p(t)) = \phi_t^H(x_0)$, then

$$\Lambda_H(x_0) = \int_0^1 L(q(t), H_p(q(t), p(t))) dt = \int_0^1 L(q(t), \dot{q}(t)) dt.$$

(iv) Let $H : \mathbb{R}^{2d} \rightarrow \mathbb{R}$ be a time-independent Hamiltonian. Given $x : \mathbb{R}^2 \rightarrow \mathbb{R}^{2d}$, and writing $\dot{x} = x_t$ for the time derivative and x_s for the s -derivative, we have

$$\begin{aligned} \frac{d}{ds} \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) &= \frac{1}{2} (\bar{J}x_s \cdot \dot{x} + \bar{J}x \cdot \dot{x}_s) - \nabla H(x) \cdot x_s \\ &= \frac{d}{dt} \left(\frac{1}{2} \bar{J}x \cdot x_s \right) - (\bar{J}\dot{x} + \nabla H(x)) \cdot x_s, \\ \frac{d^2}{ds^2} \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) &= \frac{d}{dt} \left(\frac{1}{2} \bar{J}x \cdot x_{ss} \right) - (\bar{J}\dot{x} + \nabla H(x)) \cdot x_{ss} \\ &\quad - (\bar{J}\dot{x}_s + D^2 H(x)x_s) \cdot x_s \end{aligned}$$

Hence, if $x(t) = x(t, 0)$ is a Hamiltonian orbit and $\hat{x}(t) = x_s(t, 0)$, $\tilde{x}(t) = x_{ss}(t, 0)$, then

$$\begin{aligned} \frac{d}{ds} \int_0^T \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) dt &= \frac{1}{2} (\bar{\omega}(x(T), \hat{x}(T)) - \bar{\omega}(x(0), \hat{x}(0))), \\ \frac{d^2}{ds^2} \int_0^T \left(\frac{1}{2} \bar{J}x \cdot \dot{x} - H(x) \right) dt &= \frac{1}{2} (\bar{\omega}(x(T), \tilde{x}(T)) - \bar{\omega}(x(0), \tilde{x}(0))) \\ &\quad - \int_0^T \left(\bar{J} \frac{d\hat{x}}{dt} + D^2 H(x) \hat{x} \cdot \hat{x} \right) dt. \end{aligned}$$

These equations correspond to the first and second variations of the energy in Riemannian Geometry, when $H(q, p) = \frac{1}{2} \sum_{i,j} g^{ij}(q) p_i p_j$ is quadratic in p -variable. $\square$

Definition 8.2 We define

$$Fix(\varphi) = \{x_0 : \varphi(x_0) = x_0\}.$$

The *action spectrum* $\sigma(\varphi)$ of $\varphi \in \mathcal{D}$ is a subsets of $\mathbb{R}$ that is defined by

$$\sigma(\varphi) = \{\Lambda_\varphi(x_0) : x_0 \in Fix(\varphi)\}.$$

Similarly, for any Hamiltonian function H , we define

$$\sigma(H) = \{\Lambda_H(x_0) : x_0 \in Fix(\phi^H)\},$$

so that $\sigma(\phi^H) = \sigma(H)$. $\square$

Proposition 8.5 (i) *The set $\sigma(\varphi)$ is compact and nowhere dense.*

(ii) *For every $\varphi \in \mathcal{D}$ and symplectic map $\tau : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$, we have $\sigma(\tau^{-1} \circ \varphi \circ \tau) = \sigma(\varphi)$.*

Proof(i) It is not hard to show that if $H \in C^{r+2}$, then $\Lambda_H \in C^r$. In particular Λ is smooth if H is smooth. Since the set of fixed points of φ is closed, we deduce that $\sigma(\varphi)$ is closed. Also, since Λ_φ is of compact support, we learn that the set $\sigma(\varphi)$ is bounded.

It remains to show that the set $\sigma(\varphi)$ is nowhere dense. Recall that if Γ_1 denotes the space of 1-periodic loops, and we regard $\mathcal{A}_H^1 : \Gamma_1 \rightarrow \mathbb{R}$, then

$$\partial \mathcal{A}_H^1(x(\cdot)) = -\bar{J}\dot{x}(\cdot) - \nabla H(x(\cdot), \cdot),$$

as we saw in (8.5). We now find a smooth function $G : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ such that $G(x_0) = \Lambda_\varphi(x_0)$, whenever $\varphi(x_0) = x_0$, and

$$\sigma(\varphi) \subseteq \{G(x) : \nabla G(x) = 0\}, \quad .$$

This and the celebrated Sard's theorem imply that the set $\sigma(\varphi)$ is nowhere dense.

Indeed we may define G by $G(x) = \mathcal{A}_H^1(\psi(x, \cdot))$, where

$$\psi(x, t) = (1 - f(t))\phi_t^H(x) + f(t)\phi_t^H\left((\phi_1^H)^{-1}(x)\right),$$

with $f : [0, 1] \rightarrow [0, 1]$ any smooth function such that $f(0) = 0$ and $f(1) = 1$. Clearly $\psi(x_0, t) = \phi_t^H(x_0)$ for $x_0 \in \text{Fix}(\varphi)$ and $\psi(x, \cdot) \in \Gamma_1$ for every $x \in \mathbb{R}^{2n}$. We are done.

(ii) Assume that $\phi^H = \varphi$ so that by Proposition 3.2, we have $\phi^{H \circ \tau} = \tau^{-1} \circ \varphi \circ \tau := \hat{\varphi}$. Now, if $x_0 \in \text{Fix}(\varphi)$, and $\Lambda_H(x_0) \in \sigma(\varphi)$, then we wish to show that $\Lambda_H(x_0) \in \sigma(\hat{\varphi})$. We show this by verifying

$$\Lambda_{H \circ \tau}(\hat{x}_0) = \Lambda_H(x_0).$$

for $\hat{x}_0 = \tau^{-1}(x_0) \in \text{Fix}(\hat{\varphi})$. Indeed if

$$\eta(t) = \phi_t^H(x_0), \quad \hat{\eta}(t) = \phi_t^{H \circ \tau}(\hat{x}_0) = \tau^{-1}(\eta(t)).$$

As a result, since η is a closed curve, we have

$$\Lambda_{H \circ \tau}(\hat{x}_0) = \int_{\hat{\eta}} \bar{\lambda} - \int_0^1 H(\tau \circ \hat{\eta}(t), t) dt = \int_{\eta} \bar{\lambda} - \int_0^1 H(\eta(t), t) dt = \Lambda_H(x_0),$$

as desired. □

We are now ready to state a theorem that is a generalization of Theorem 8.1 for the time dependent Hamiltonian functions. As it turns out the nature of the local maximizer and minimizer of the Hamiltonian function would play a pivotal role on the geodesic property. Here is the relevant definition.

Definition 8.3(i) A smooth Hamiltonian function $H : \mathbb{R}^d \times I \rightarrow \mathbb{R}$ is called *quasi-autonomous* if there exists $x^\pm \in \mathbb{R}^{2n}$ such that

$$(8.6) \quad \max_x H(x, t) = H(x^+, t), \quad \min_x H(x, t) = H(x^-, t),$$

for all $t \in I$.

(ii) A smooth Hamiltonian function $H : \mathbb{R}^d \times I \rightarrow \mathbb{R}$ is called *locally quasi-autonomous* if for every t in the interior of I , we can find small $\delta > 0$ such that the restriction H to $\mathbb{R}^{2n} \times (t - \delta, t + \delta)$ is quasi-autonomous.

(iii) Let $\varphi : [a, b] \rightarrow \mathcal{D}$ be a regular smooth path associated with a quasi-Hamiltonian function H . Its *bifurcation diagram* is defined by

$$\Gamma(\varphi) := \{(s, h) \in (a, b] \times \mathbb{R} : h \in \sigma(\varphi(s) \circ \varphi(a)^{-1})\}.$$

(iv) Let φ be as in part **(iii)** and set

$$\gamma^\pm(t) = - \int_a^t H(x^\pm, s) ds,$$

with $x^\pm$ as in (8.3). We say that a bifurcation diagram $\Gamma(\varphi)$ is simple if there is no continuous curve in $\Gamma(\varphi)$ that connects a point of the form $(t, \gamma^\pm(t))$, $t > a$, to a points on the line $s = b$. $\square$

Theorem 8.2 (Bialy-Polterovich) **(i)** A regular smooth path $\varphi : I \rightarrow \mathcal{D}$ with $\dot{\varphi} = X_H(\varphi)$ is geodesic iff H is locally quasi-autonomous.

(ii) If a regular smooth path $\varphi : I \rightarrow \mathcal{D}$ is associated with a quasi-autonomous Hamiltonian that has a simple bifurcation diagram, then φ is minimal geodesic.

As a consequence of Theorem 8.2, one can show that locally $\mathcal{D}$ is flat. For example, near identity, we can use *generating functions* to represent elements of $\mathcal{D}$ and their distances to the identity map can be readily expressed it terms of their generating functions.

We have already seen an example of a generating function. Namely, if $\varphi = \phi^H$, then $d\Lambda_H = \varphi^* \bar{\lambda} - \bar{\lambda}$. We regard $\Lambda = \Lambda_H$ as a generating function for φ . If we write $\varphi(x) = \varphi(q, p) = (Q, P)$, then

$$(8.7) \quad d\Lambda = P \cdot dQ - p \cdot dq,$$

which is compatible with the fact that φ is symplectic so that $d(P \cdot dQ - p \cdot dq) = 0$. However because of the very form (8.7), it is desirable to have a function dS with $S = S(q, Q)$ on the left-hand side of (8.7). That is,

$$(8.8) \quad dS = P \cdot dQ - p \cdot dq,$$

This is possible if we can solve the equation $Q = Q(q, p)$ for $p = p(q, Q)$ for every pair $(q, Q) \in \mathbb{R}^{2n}$. We refer to Proposition 9.1 of Chapter 9 for conditions that would guarantee the existence of S , that is called (*Type I*) *generating function*. Note that (8.8) implies

$$(8.9) \quad p = -S_q(q, Q), \quad P = S_Q(q, Q).$$

There are other possibilities for the definition of generating functions. For example, if we set $\hat{\Lambda} = \Lambda + p \cdot q$, then the equation (8.7) reads as

$$d\hat{\Lambda} = P \cdot dQ + q \cdot dp.$$

We now search for a function $W(Q, p)$ such that

$$(8.10) \quad dW = P \cdot dQ + q \cdot dp.$$

If such a function W exists, we call it a *generating function of Type II*. By (8.10),

$$(8.11) \quad W_Q(Q, p) = P, \quad W_p(Q, p) = q.$$

The next Proposition explains the original motivation behind the formulation of the *Hamilton-Jacobi* PDE.

Proposition 8.6 (i) *Assume that the flow ϕ_t^H has a generating function $S(q, Q, t)$ of Type I for $t \in [a, b]$. Then S satisfies the Hamilton-Jacobi PDE*

$$(8.12) \quad S_t(q, Q, t) + H(Q, S_Q(q, Q, t), t) = 0,$$

for $t \in (a, b)$.

(ii) *If the flow ϕ_t^H has a generating function $W(Q, p, t)$ of Type II for $t \in [a, b]$, then W satisfies the Hamilton-Jacobi PDE*

$$(8.13) \quad W_t(Q, p, t) + H(Q, W_Q(Q, p, t), t) = 0,$$

for $t \in (a, b)$.

Proof(i) Since $S(q, Q, t) = \Lambda_t(q, p)$, for $p = S_Q(q, Q, t)$, we may write

$$S(q(0), q(t), t) = \int_0^t [p(t) \cdot \dot{q}(t) - H(q(t), p(t), t)] dt,$$

where $(q(t), p(t)) = \phi_t^H(q(0), p(0))$. Differentiating both sides with respect to t yields

$$S_t(q(0), q(t), t) + S_Q(q(0), q(t), t) \cdot \dot{q}(t) = p(t) \cdot \dot{q}(t) - H(q(t), p(t), t).$$

This implies (8.12) because $S_Q(q(0), q(t), t) = p(t)$.

(ii) Since $W = \Lambda + q \cdot p$, we have

$$W(q(t), p(0), t) = p(0) \cdot q(0) + \int_0^t [p(t) \cdot \dot{q}(t) - H(q(t), p(t), t)] dt.$$

Differentiating both sides with respect to t yields

$$W_t(q(t), p(0), t) + W_Q(q(t), p(0), t) \cdot \dot{q}(t) = p(t) \cdot \dot{q}(t) - H(q(t), p(t), t).$$

This implies (8.13) because $W_Q(q(t), p(0), t) = p(t)$. □

Note that when $H \in \mathcal{H}$, and $\varphi = \phi^H \in \mathcal{D}$ has a generating function of Type II, then we may write

$$W(Q, p) = Q \cdot p + u(Q, p),$$

for a function $u : \mathbb{R}^{2n} \rightarrow \mathbb{R}$, that is of compact support. In this case we can write

$$(8.14) \quad \varphi(Q + u_p(Q, p), p) = (Q, p + u_Q(Q, p)).$$

We write $\|u\| = \max u - \min u$ and refer to u as the generating function of φ .

Theorem 8.3 (*Bialy-Polterovich*) *If φ and $\psi \in \mathcal{D}$ have generating functions u and u' , then $D(\varphi, \psi) = \|u - u'\|$.*

Remark 8.4 Since the identity map has a generating function of the form $W(Q, p) = Q \cdot p$, it is not hard to show that any $\varphi \in \mathcal{D}$ that is sufficiently close to identity has a generating function of type II with the corresponding u of compact support. In fact the formula (8.14) allows us to define a map $\Phi(u) = \varphi$ that maps a small neighborhood of the origin in $C_c(\mathbb{R}^{2n})$, onto a small neighborhood of identity in $\mathcal{D}$. We then have the formula $D(id, \Phi(u)) = \|u\|$. □

We now design a strategy for tackling Theorems 8.1-8.3. As a warm-up, we try to come up with a direct proof of $E(\varphi) > 0$ for $\varphi \neq id$. (This was verified in Proposition 8.1 with the aid of Theorem 6.2(vi).) In other words, we wish to find non-trivial lower bounds for $E(\varphi)$. In fact it is easier to divide our job into two parts; if we define

$$(8.15) \quad E^\pm(\varphi) = \inf \left\{ \int_0^1 \sup_x (\pm H(x, t)) dt : \phi^H = \varphi \right\},$$

then we always have where

$$(8.16) \quad E(\varphi) \geq E^+(\varphi) + E^-(\varphi),$$

and we may try to find non-trivial lower bounds for $E^\pm$ separately.

As we learned from Proposition 8.4, the set $\sigma(H)$ depends on ϕ^H , and could be useful for finding such lower bounds. Based on our experience in Chapter 7, we may try the Minimax Principle to pick a point in $\sigma(H)$ to serve as a lower bound. In other words, we try to find family $\mathcal{F}$ of subsets of $\mathcal{E}$ such that if

$$(8.17) \quad \mathcal{U}(H) = \sup_{A \in \mathcal{F}} \inf_{x \in A} \mathcal{A}_H(x),$$

then

$$\mathcal{U}(H) \leq \mathcal{U}(0) - \int_0^1 \sup_x (-H(x, t)) dt,$$

which maybe used to find a lower bound for E^- provided that $\mathcal{U}(0) = 0$. Evidently, $\mathcal{U}(0)$ is equivalent to saying

$$\sup_{A \in \mathcal{F}} \inf_{x \in A} \int_{x(\cdot)} p \cdot dq = 0.$$

We now state a result of Hofer that would guarantee the existence of a good $\mathcal{F}$.

Theorem 8.4 (Hofer) *There exists a universal family $\mathcal{F}$ of subsets of $\mathcal{E}$ such that $\mathcal{E}^+ \in \mathcal{F}$, and for the corresponding $\mathcal{U}$, we have the following properties:*

- (1) $\mathcal{U}(0) = 0$ (or equivalently (8.8) holds),
- (2) $\mathcal{U}(H)$ is a critical value (or equivalently $\mathcal{U}(H) \in \sigma(H)$).

We now assert that $\mathcal{U}$ of Theorem 8.5 induces a map on $\mathcal{D}$.

Proposition 8.7 *Let $\mathcal{F}$ be as in Theorem 8.4. Then $\mathcal{U}(H) = \mathcal{U}(K)$, whenever $\phi^H = \phi^K$. (Hence $\mathcal{U}(\phi^H) := \mathcal{U}(H)$ is well-defined.)*

Proof First observe that $\mathcal{A}_H$ is well-defined for a continuous function H . Moreover, $\mathcal{A}_H$ is continuous in H ; we have the bound

$$|\mathcal{A}_H(x) - \mathcal{A}_K(x)| \leq \|H - K\|_{\infty,1}.$$

This yields

$$(8.18) \quad |\mathcal{U}(H) - \mathcal{U}(K)| \leq \|H - K\|_{\infty,1}.$$

Since $\mathcal{U}$ is continuous by (8.18), and the set $\sigma(H) = \sigma(\phi^H)$ is nowhere dense, we have the following property: If $s \mapsto H^s$ is a continuous path in $\mathcal{H}$, such that $\phi^{H^s} = \varphi$ is independent of s , then $\mathcal{U}(H^s) \in \sigma(\varphi)$ is independent of s . This suggest a two-steps strategy for the proof of Proposition:

(1) Design a map $\mathcal{K} : \mathcal{D} \rightarrow \mathcal{H}$ such that $\varphi = \phi^{\mathcal{K}(\varphi)}$.

(2) For every H with $\phi^H = \varphi$, find a continuous family $(H^s : s \in I)$ in $\mathcal{H}$ that would connect H and $\mathcal{K}(\varphi)$.

Once (1) and (2) are materialized, we would deduce that $\mathcal{U}(H) = \mathcal{U}(\mathcal{K}(\phi^H))$ depends on ϕ^H only, which is exactly what we wish to prove.

In fact our candidate for $\mathcal{K}(H) = K$ is the unique Hamiltonian function for which the flow ϕ_t^K is given by

$$\varphi_t(x) := \phi_t^K(x) = \begin{cases} t\varphi\left(\frac{x}{t}\right) & \text{if } t \neq 0, \\ x & \text{if } t = 0. \end{cases},$$

where $\varphi = \phi_1^H$. Observe that if $\varphi(x) = x$ for $|x| \geq c$, then $\varphi_t(x) = x$ for $|x| \geq ct$. Hence φ_t is continuous in $t \geq 0$. One can readily show φ_t is also differentiable for $t \geq 0$, and that

$$\frac{d}{dt}\phi_t^K(x) = X(\phi_t^K(x), t),$$

where

$$X(x, t) = \begin{cases} \hat{X}(x/t) & \text{if } t \neq 0, \\ \varphi(0) & \text{if } t = 0, x = 0, \\ 0 & \text{if } t = 0, x \neq 0, \end{cases}$$

with

$$\hat{X}(x) = x - (d\varphi)_{\varphi^{-1}(x)}(\varphi^{-1}(x)).$$

We note that $\hat{X}$ is of compact support because $d\varphi$ is of compact support and $\varphi = \varphi^{-1} = id$ outside a large ball. Note however,

$$\lim_{t \rightarrow 0^+} X(x, t) = \begin{cases} 0 & \text{if } x \neq 0, \\ -(d\varphi)_{\varphi^{-1}(0)}(\varphi^{-1}(0)) & \text{if } x = 0. \end{cases}$$

Hence X is of compact support but not continuous at $t = 0$ and $x = 0$. Nonetheless, we can find a function K such that $X = \bar{J}\nabla K$, with

$$K(x, t) := \begin{cases} t\hat{K}(x/t) & \text{if } t \neq 0, \\ 0 & \text{if } t = 0, \end{cases}$$

where $\hat{K}$ is chosen to be of compact support and satisfying

$$\bar{J}\nabla\hat{K}(x) = \hat{X}(x).$$

We note $K(x, t) = 0$ for $|x| \geq ct$. This in particular implies the smoothness of K everywhere except at $x = 0$ and $t = 0$. As a result, $\mathcal{U}(K)$ is well-defined.

Now take any $H \in \mathcal{H}$ with $\phi^H = \varphi$. It remains to find a continuous family of Hamiltonian functions that would connect H to $\mathcal{K}(\varphi)$.

We first choose two smooth functions

$$\tau : [0, 1) \times [0, 1] \rightarrow [0, 1], \quad \beta : [0, 1]^2 \rightarrow [0, 1],$$

such that if we set $\tau^s(t) = \tau(s, t)$, and $\beta^s(t) = \beta(s, t)$, then $\tau^0(t) = t$, $\beta^1(t) = t$, and

$$\begin{aligned} s < 1 &\Rightarrow \tau^s(0) = 0, & \beta^s(1) &= 1, \\ t \geq 1 - s &\Rightarrow \tau^s(t) = 1, \\ t \leq 1 - s &\Rightarrow \beta^s(t) = 0. \end{aligned}$$

We then set $\gamma^{s,\theta}(t) = 1 - \theta + \theta\beta^s(t)$, and define

$$\psi_t^{s,\theta}(x) = \gamma^{s,\theta}(t) \phi_{\tau^s(t)}^H \left(\frac{x}{\gamma^{s,\theta}(t)} \right),$$

for $(s, \theta) \in [0, 1) \times [0, 1]$. The idea behind the construction of $\psi_t^{s,\theta}$ is that if we consider only $\psi_t^{s,0}$, then we have a family of flows that satisfies $\psi_t^{s,0} = \varphi$ for $t \geq 1 - s$, with $s < 1$. Ideally we wish to have a flow that depends on φ only. We are tempted to send $s \rightarrow 1$ but this is not allowed because at $s = 1$ we get a

$$\psi_t^{1,0} = \begin{cases} \varphi & \text{if } t > 0, \\ id & \text{if } t = 0, \end{cases}$$

which is not continuous at $t = 0$. To rectify the discontinuity of $\psi_t^{1,0}$, we switch to $\theta > 0$ that would yield a continuous flow even at $s = 1$. To explain this in details, note that, $\psi_t^{0,0} = \phi_t^H$. On the other hand,

$$t \geq 1 - s \Rightarrow \psi_t^{s,1}(x) = \beta^s(t) \varphi \left(\frac{x}{\beta^s(t)} \right).$$

From this we learn

$$\psi_t^{1,1}(x) := \lim_{s \rightarrow 1} \psi_t^{s,1}(x) = \begin{cases} t\varphi \left(\frac{x}{t} \right) & \text{if } t > 0, \\ x & \text{if } t = 0, \end{cases}.$$

which is $\phi_t^K(x)$. Hence, we have a continuous family of paths in $\mathcal{D}$ that connects the flow of X_H to the flow of $X_{\mathcal{K}(\varphi)}$. We note that for each (s, θ) , we have $\psi_0^{s,\theta}(x) = x$, because $\tau^s(0) = 0$ and $\phi_0^H(x) = x$. Moreover, $\psi_1^{s,\theta} = \varphi$ because $\tau^s(1) = \beta^s(1) = 1$ and $\phi_1^H = \varphi$. We are done. $\square$

We continue with an application of Theorem 8.4 and Proposition 8.7 to demonstrate that the function $\mathcal{U}$ does serve our purposes.

Proposition 8.8 *Let $\mathcal{U}$ be as in Proposition 8.7, then*

(i) *For every $H \in \mathcal{H}$, we have $\mathcal{U}(H) \geq 0$.*

(ii) *We have $E^-(\varphi) \geq \mathcal{U}(\varphi)$, and*

$$(8.19) \quad E(\varphi) \geq \mathcal{U}(\varphi) + \mathcal{U}(\varphi^{-1}).$$

More generally,

$$(8.20) \quad -E^+(\varphi) \leq \mathcal{U}(\psi \circ \varphi) - \mathcal{U}(\psi) \leq E^-(\varphi).$$

(iii) *If $H \geq 0$, then $\mathcal{U}(H) = 0$. Moreover, if $H \leq 0$ and $H \neq 0$, then $\mathcal{U}(H) > 0$.*

(iv) *If H is independent of t , $1 < T^{\min}(H)$, then $\mathcal{U}(H) = -\inf H$.*

Proof(i) For every $a \in \mathbb{R}^{2n}$ and consider $H^a(x, t) = H(x + a, t)$. Since $H^a = H \circ \tau^a$, for the symplectic map $\tau^a(x) = x + a$, we may apply Proposition 8.5(ii) to assert that $\sigma(H) = \sigma(H^a)$. Since $a \mapsto \mathcal{U}(H^a) \in \sigma(H)$ is continuous in a and the set $\sigma(H)$ is nowhere dense, we deduce that $\mathcal{U}(H) = \mathcal{U}(H^a)$. For every a . Since H is of compact support, we may choose a with $|a|$ sufficiently large so that $H^a(x, t) \leq \pi|x|^2 =: H_0$ for all $(x, t) \in \mathbb{R}^{2n} \times [0, 1]$. For such a ,

$$\mathcal{U}(H) = \mathcal{U}(H^a) \geq \mathcal{U}(H_0) \geq \inf_{x \in \mathcal{E}^+} \mathcal{A}_{H_0}(x) = \inf_{x \in \mathcal{E}^+} \left[\frac{1}{2} \|x^+\|_{1/2}^2 - \int_0^1 \pi|x|^2 dt \right] \geq 0,$$

because $2\pi\|x\|_0^2 \leq \|x\|_{1/2}^2$. This completes the proof of Part (i).

(ii) First we show how (8.20) implies (8.19). Indeed by choosing $\psi = id$ in (8.20), we have $\mathcal{U}(\varphi) \leq E^-(\varphi)$. (this can be established directly using the definition of $\mathcal{U}$ and the property $\mathcal{U}(0) = 0$.) As a result,

$$\mathcal{U}(\varphi) + \mathcal{U}(\varphi^{-1}) \leq E^-(\varphi) + E^-(\varphi^{-1}) = E^-(\varphi) + E^+(\varphi) \leq E(\varphi),$$

where we use Proposition 8.1(ii) and its consequence $E^-(\varphi^{-1}) = E^+(\varphi)$.

For the proof of (8.20), recall that if $\varphi = \phi^H$ and $\psi = \phi^K$, then $\psi \circ \phi = \phi^{K\sharp H}$. As a result, (8.20) follows if we can show

$$-\int_0^1 \sup_x H(x, t) dt \leq \mathcal{A}_{K\sharp H} - \mathcal{A}_K \leq \int_0^1 \sup_x (-H(x, t)) dt.$$

This is an immediate consequence of the equality

$$(\mathcal{A}_{K\sharp H} - \mathcal{A}_K)(x(\cdot)) = -\int_0^1 \int H((\phi_t^K)^{-1}(x(t)), t) dt.$$

(iii) Since

$$H \leq K \Rightarrow \mathcal{A}_H \geq \mathcal{A}_K,$$

we immediately deduce

$$(8.21) \quad H \leq K \Rightarrow \mathcal{U}(H) \geq \mathcal{U}(K).$$

As a result, if $H \geq 0$, then $\mathcal{U}(H) \leq 0$, which in turn implies $\mathcal{U}(H) = 0$ by Part (i).

(iv) Suppose that $H \in \mathcal{H}$ is independent of time. Note that by the definition of $\mathcal{U}$ and $\mathcal{U}(0) = 0$ we always have

$$(8.22) \quad \mathcal{U}(tH) = \mathcal{U}(\phi^{tH}) \leq \mathcal{U}(0) + t \int_0^1 \sup_x (-H(x)) dt = -t \inf H.$$

To prove the opposite inequality, we first assert that $\mathcal{U}(tH) \geq -t \min H$, for t sufficiently small. To see this, we make a translation so that $\min H = H(0)$. This will not change the value of $\mathcal{U}(tH)$ by our Proof of Part (i). We then choose t_0 sufficiently small so that $t_0 \max |D^2 H| \leq 2\pi$. As a result, if $t \in (0, t_0]$, then

$$tH(x) \leq H_1(x) := tH(0) + \pi|x|^2 =: tH(0) + H_0(x)$$

From this, Part (i), and (8.21) we deduce

$$\mathcal{U}(tH) \geq \mathcal{U}(H_1) \geq -tH(0) + \mathcal{U}(H_0) \geq -t \min H.$$

This and (8.22) imply that $\mathcal{U}(tH) = -t \inf H$, for $t \leq t_0$.

We now want to argue that $\mathcal{U}(tH) = -t \min H$ even if $t < T^{\min}(H)$. Note that for such t , the flow ϕ^{tH} has no non-constant 1-periodic orbit. This implies

$$\sigma(\phi^{tH}) = \{-tH(y) : \nabla H(y) = 0\} =: tA_0,$$

for every $t \in (0, T^{\min}(H))$. This implies that $t^{-1}\mathcal{U}(\phi^{tH}) \in A_0$ for every $t \in (0, T^{\min}(H))$. Since the map $t \mapsto t^{-1}\mathcal{U}(\phi^{tH})$ is continuous by (8.17), and the set A_0 is nowhere dense, we learn that $a_0 := t^{-1}\mathcal{U}(\phi^{tH})$ is constant for $t \in (0, T^{\min}(H))$. We are done. $\square$

We are now ready to prove Theorems 6.2(iv) and 8.1.

Proof of Theorem 6.2(iv) Let U be a bounded open set. Assume that $H, K \in \mathcal{H}$ with the following properties: H is independent of time, $H \leq 0$, $T^{\min}(H) > 1$, the support of H is contained in U , and

$$\psi(U) \cap U = \emptyset,$$

for $\psi = \phi^K$. We then would like to deduce

$$(8.23) \quad -\inf H = \mathcal{U}(\varphi) \leq E(\psi).$$

(The equality follows from Proposition 8.8(iv).) By Proposition 8.8(ii),

$$\begin{aligned}\mathcal{U}(\varphi) &\leq \mathcal{U}(\psi \circ \varphi) + E^+(\psi) = \mathcal{U}(\psi \circ \varphi) - \mathcal{U}(\psi) + \mathcal{U}(\psi) + E^+(\psi) \\ &\leq \mathcal{U}(\psi \circ \varphi) - \mathcal{U}(\psi) + E^-(\psi) + E^+(\psi) \leq \mathcal{U}(\psi \circ \varphi) - \mathcal{U}(\psi) + E(\psi).\end{aligned}$$

This implies (8.23) if we can show

$$(8.24) \quad \mathcal{U}(\psi \circ \varphi) = \mathcal{U}(\psi).$$

To prove (8.24), it suffices to show that

$$(8.25) \quad \sigma(\psi \circ \phi_t^H) = \sigma(\psi),$$

for $t \geq 0$. Once (8.25) is established, then we use the continuity of $t \mapsto \mathcal{U}(\psi \circ \phi_t^H) \in \sigma(\psi)$, and the nowhere denseness of the set $\sigma(\psi)$, to deduce (8.24).

For (8.25), observe

$$Fix(\psi \circ \phi_t^H) = Fix(\psi),$$

because $Fix(\psi) \cap U = \emptyset$. Since

$$\phi_t^H(x) = x, \quad \Lambda_{\phi_t^H}(x) = 0, \quad \text{for } x \in Fix(\psi) \subset \mathbb{R}^{2n} \setminus U,$$

we deduce

$$\begin{aligned}\sigma(\psi \circ \phi_t^H) &= \left\{ \Lambda_{\psi \circ \phi_t^H}(x) : x \in Fix(\psi \circ \phi_t^H) \right\} = \left\{ \Lambda_{\psi \circ \phi_t^H}(x) : x \in Fix(\psi) \right\} \\ &= \left\{ (\Lambda_\psi \circ \phi_t^H)(x) + \Lambda_{\phi_t^H}(x) : x \in Fix(\psi) \right\} \\ &= \{ \Lambda_\psi(x) : x \in Fix(\psi) \} = \sigma(\psi),\end{aligned}$$

as desired. □

Proof of Theorem 8.1 It suffices to show that if $T^{min}(H) > 1$, then $E(\varphi) = \|H\|$, where $\varphi = \phi^H$. By Proposition 8.8(ii),

$$(8.26) \quad E(\varphi) \geq \mathcal{U}(\varphi) + \mathcal{U}(\varphi^{-1}).$$

On the other hand, by Exercise 8.1(i) below, we have $\varphi^{-1} = \phi^{-H}$. Since $T^{min}(-H) = T^{min}(H) > 1$, we learn from Proposition 8.8(iv)

$$\mathcal{U}(\varphi) = -\inf H, \quad \mathcal{U}(\varphi^{-1}) = -\inf(-H).$$

From and (8.22) we deduce

$$\|H\| = -\inf(-H) - \inf H = \mathcal{U}(\varphi) + \mathcal{U}(\varphi^{-1}) \leq E(\varphi) \leq \|H\|.$$

This completes the proof. $\square$

Proof of Theorem 8.4 *Step 1.* Let us write Φ_t for the flow of $-\nabla \mathcal{A}_H$. Recall that by the formula (7.13),

$$\Phi_t(x) = e^{-t}x^+ + x^0 + e^tx^- + K_t(x),$$

for a suitable function $K_t : \mathcal{E} \rightarrow \mathcal{E}$ that maps bounded sets into precompact sets. The set $\mathcal{F}$ takes the form

$$\mathcal{F} = \{g(\mathcal{E}^+) : g \in \mathcal{G}\},$$

for a suitable set $\mathcal{G}$ of maps $g : \mathcal{E} \rightarrow \mathcal{E}$. The set $\mathcal{G}$ consists of homeomorphism $g : \mathcal{E} \rightarrow \mathcal{E}$ of the form

$$g(x) = e^{\gamma^+(x)}x^+ + x^0 + e^{\gamma^-(x)}x^- + K(x),$$

for continuous functions $\gamma^\pm : \mathcal{E} \rightarrow \mathbb{R}$ and $K : \mathcal{E} \rightarrow \mathcal{E}$ with the following properties:

- (i) The restriction of functions $\gamma^\pm$ and K to $\mathcal{E}^+$ are 0 outside a bounded set.
- (ii) The functions $\gamma^\pm$ and K map bounded sets into precompact sets.

Step 2. One can show that indeed the set $\mathcal{G}$ is a group of homeomorphism.

Step 3 A crucial property of $\mathcal{G}$ is that for every $g \in \mathcal{G}$, we have

$$(8.27) \quad g(\mathcal{E}^+) \cap (\mathcal{E}^- \oplus \mathcal{E}^0) \neq \emptyset.$$

For this, we need to show that for each $g \in \mathcal{G}$, we can find $x \in \mathcal{E}^+$ such that $P^+g(x) = 0$. For $g(x) = e^{\gamma^+(x)}x^+ + x^0 + e^{\gamma^-(x)}x^- + K(x)$, we need to solve

$$e^{\gamma^+(x)}x + P^+K(x) = 0,$$

in $\mathcal{E}^+$. To express this as a fixed point problem, define the operator

$$T(x) = -e^{-\gamma^+(x)}P^+K(x),$$

so that (8.27) is equivalent to find a fixed point for $T : \mathcal{E}^+ \rightarrow \mathcal{E}^+$. Note that since K is of bounded support in $\mathcal{E}^+$, we can find ℓ such that $T(x) = 0$ for $\|x\| > \ell$. As a result,

$$\{x \in \mathcal{E}^+ : \lambda T(x) = x \text{ for some } \lambda \in [0, 1]\} \subset \{x \in \mathcal{E}^+ : \|x\| \leq \ell\}.$$

On the other hand since T is a compact operator, we may use Schaefer's Fixed Point Theorem to assert that T has a fixed point, proving (8.27).

Step 3. Evidently

$$(8.28) \quad \mathcal{U}(0) \geq \inf_{\mathcal{E}^+} \frac{1}{2} (\|x^+\|^2 - \|x^-\|^2) = \inf_{\mathcal{E}^+} \frac{1}{2} \|x^+\|^2 = 0.$$

Moreover, by (8.27),

$$\mathcal{U}(0) \leq \sup_{x \in \mathcal{E}^- \oplus \mathcal{E}^0} \frac{1}{2} (\|x^+\|^2 - \|x^-\|^2) = 0.$$

This and (8.28) imply that $\mathcal{U}(0) = 0$.

It remains to check that $\mathcal{U}(H) \in \sigma(H)$. The proof of this is similar to the proof of Minimax Theorem of Chapter 7 and is omitted. $\square$

Remark 8.5 As a variation of what we have seen so far, consider the space $\mathcal{D}_{per}$ of symplectic maps $\varphi = \phi^H : T\mathbb{T}^n \rightarrow \mathbb{T}^n \times \mathbb{R}^n$, such that the corresponding Hamiltonian functions

$$H : \mathbb{T}^n \times \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R},$$

are of compact support. Let us assume that H is 1-periodic in time and convex in p . We write L for the Legendre Transform of H in the p -variable. Let us write $\hat{\phi}_t^L$ for the flow of the Euler-Lagrange-Newton's equation

$$\frac{d}{dt} L_v(q, \dot{q}, t) = L_q(q, \dot{q}, t).$$

As in *Mather Theory*, let us consider the following variational problem:

$$\bar{L}(V) = \inf_{\mu} \int L(q, v) \mu(dq, dv),$$

where the infimum is over invariant measures μ of the flow $\hat{\phi}_t^L$ with the rotation number V ;

$$V = \int v \mu(dq, dv).$$

If $x(\cdot) = (q(\cdot), p(\cdot))$ is a 1-periodic orbit of X_H , then $(q(\cdot), \dot{q}(\cdot))$ yields a 1-periodic orbit of the Lagrangian flow $\hat{\phi}_t^L$. Out this, we can build an invariant measure

$$\mu(dq, dv) := \mathbb{1}(t \in [0, 1]) \delta_{(q(t), \dot{q}(t))}(dq, dv) dt,$$

which is of rotation number 0. As a result,

$$\int_0^1 L(q(t), \dot{q}(t), t) dt \geq \bar{L}(0).$$

From this and Remark 8.3(iii) we learn

$$\inf \sigma(H) \geq \bar{L}(0).$$

There is an alternative way of establishing this inequality: According to Mather Theory, many dynamical properties of the flow ϕ_t^H are encoded in the sets of the form

$$\mathcal{L}_P = \{(q, P + u_q(q, t; P)) : q \in \mathbb{T}^n\},$$

where for each $P \in \mathbb{R}^n$, the function $u(\cdot, \cdot; P) : T\mathbb{T}^n \rightarrow \mathbb{R}$ is 1-periodic in t , and solves the Hamilton-Jacobi Equation

$$u_t + H(q, u_q, t) = \bar{H},$$

for a suitable constant $\bar{H} = \bar{H}(P)$. Note that if $h \in \sigma(H)$, then there is a 1-periodic orbit $x(t) = (q(t), p(t))$ of X_H such that

$$h = \int_0^1 \hat{H}(q(t), p(t), t) dt.$$

On the other hand, since H is convex,

$$\hat{H}(q, p, t) = p \cdot H_p(q, p, t) - H(q, p, t) \geq (P + u_q(q, p, t)) \cdot H_p(q, p, t) - H(q, P + u_q(q, p, t), t).$$

From this we learn

$$\begin{aligned} h &\geq \int_0^1 [(P + u_q(q(t), p(t), t)) \cdot H_p(q(t), p(t), t) - H(q(t), P + u_q(q(t), p(t), t), t)] dt \\ &= \int_0^1 [(P + u_q(q(t), p(t), t)) \cdot \dot{q}(t) - H(q(t), P + u_q(q(t), p(t), t), t)] dt \\ &= \int_0^1 [\dot{z}(t) - u_t(q(t), p(t), t) - H(q(t), P + u_q(q(t), p(t), t), t)] dt = -H(P), \end{aligned}$$

where $z(t) = u(q(t), p(t), t)$. As a result,

$$\inf \sigma(H) \geq \sup_P (-\bar{H}(P)) = \hat{L}(0).$$

where $\hat{L}$ denotes the Legendre Transform of $\bar{H}$. It turns out that $\bar{L} = \hat{L}$. More generally, if we define $H^v(q, p, t) = H(q, p, t) - p \cdot v$, then

$$\inf \sigma(H^v) \geq \sup_P (p \cdot v - \bar{H}(P)) = \bar{L}(v).$$

We may define D on $\mathcal{D}_{per}$ in the same way we defined D on $\mathcal{D}$. In view of Proposition 8.8, we may expect

$$D(id, \phi^H) \geq \inf \sigma(H).$$

This turns out to be the case as was demonstrated by Siburg [Si]. $\square$

Remark 8.6 The minimax principle used for the proof of Theorem 8.4 can be used to define Ekeland-Hofer Capacities $(\mathbf{c}_{EH}^k : k \in \mathbb{N})$. First we describe $\mathbf{c}_{EH}^1$. To prepare for the definition of this capacity, we first define a variant of the function $\mathcal{U}$. More precisely, fix an open neighborhood U of the origin, and let $\mathcal{H}^1(U)$ denote the set of nonnegative C^2 Hamiltonian functions for which the following conditions are true:

- H vanishes in U .
- There exist $R > 0$ and $\rho > \pi$, such that for x satisfying $|x| \geq R$, we have $H(x) = \rho|x|^2$.

We write $\mathcal{H}^1$ for the set of Hamiltonian H such that for some open neighborhood U of the origin, we have $H \in \mathcal{H}^1(U)$. We then define $\mathcal{V} : \mathcal{H}^1 \rightarrow [0, \infty)$ by

$$(8.29) \quad \mathcal{V}(H) = \sup_{g \in \mathcal{G}} \inf_{x \in \Gamma} \mathcal{A}_H(g(x)),$$

where the group $\mathcal{G}$ was defined in the proof of Theorem 8.4, and

$$\Gamma = \{x \in \mathcal{E}^+ : \|x\| = 1\}.$$

As in the proof of Theorem 7.2, we can show that $\mathcal{V}(H) < \infty$. For this, it suffices to show that for every $g \in \mathcal{G}$,

$$(8.30) \quad g(\Gamma) \cap \hat{\mathcal{E}} \neq \emptyset,$$

where

$$\hat{\mathcal{E}} = \mathcal{E}^- \oplus \mathcal{E}^0 \oplus \mathbb{R}e^+,$$

with $e^+(t) = e^{2\pi \bar{J}t}(1, 0, \dots, 0)$. Assuming (8.30), we can assert

$$\mathcal{V}(H) \leq \sup_{\hat{\mathcal{E}}} \mathcal{A}_H.$$

Since $H \in \mathcal{H}$, we can find $\rho > \pi$ and a constant c_0 such that

$$H(x) \geq \rho|x|^2 - c_0.$$

As a result, for $x = x^- + x^0 + se^+$,

$$\begin{aligned} \mathcal{A}_H(x) &\leq \pi s^2 - 2^{-1}\|x^-\|^2 - \rho \int_0^1 |x(t)|^2 dt + c_0 \\ &= \pi s^2 - 2^{-1}\|x^-\|^2 - \rho \left[\int_0^1 |x^-(t)|^2 dt + x^0 + s^2 \right] + c_0 \leq c_0, \end{aligned}$$

because $\mathcal{E}^+$, $\mathcal{E}^-$, $\mathcal{E}^0$ are orthogonal in L^2 . This implies that $\mathcal{U}'(H) < \infty$.

Also, it is not hard to show that $\mathcal{V}(H) > 0$. To see this, choose $g_\varepsilon \in \mathcal{G}$ so that $g_\varepsilon(\Gamma) = x_0 + \varepsilon\Gamma$, where $\varepsilon > 0$ and x_0 is chosen so that H vanishes in a neighborhood of x_0 . For such g_ε ,

$$\inf_{g_\varepsilon(\Gamma)} \mathcal{A}_H > 0,$$

for small ε , because $\mathcal{A}_H(x_0) = 0$, $\nabla \mathcal{A}_H(x_0) = 0$, and $D^2 \mathcal{A}_H(x_0) = P^+ - P^-$. $\square$

We now turn our attention to bounding $\|H\|_{\infty,1}$, or $D(\varphi, \psi)$ from above. As a preparation we make some definitions.

Definition 8.4(i) We write

$$\text{supp}(\varphi) = \{x : \varphi(x) \neq 0\},$$

for the support of φ .

(ii) Let us write $\pi_1(x) = \pi(x_1, \dots, x_d) = x_1$ for the projection onto the first coordinate. We say that the bounded open sets $U_1, \dots, U_k$ are *properly separated*, if there exists $\eta \in \mathcal{D}$ such that the convex hulls of $\pi_1(\eta(\bar{U}_1)), \dots, \pi_1(\eta(\bar{U}_k))$ are disjoint intervals.

(iii) We say $\varphi_1, \dots, \varphi_k \in \mathcal{D}$ are *properly separated* if $\text{supp}(\varphi_1), \dots, \text{supp}(\varphi_k)$ are properly separated.

(iv) We define the *proper displacement energy* of a bounded open set U by

$$\hat{\mathbf{e}}(U) = \inf \{ \|H\|_{\infty,1} : U \text{ and } \phi^H(U) \text{ are properly separated} \}.$$

$\square$

We are now ready to state two bounds on $D(\text{id}, \varphi)$.

Theorem 8.5 (Hofer) *For every $\varphi \in \mathcal{D}$,*

$$D(\text{id}, \varphi) \leq 128 \text{diam}(\text{supp}(\varphi)) \|\text{id} - \varphi\|_\infty.$$

Theorem 8.6 (Sikorav) *If $H \in \mathcal{H}$ and $\text{supp}(H) \subseteq U \times [0, 1]$, then*

$$E(\phi^H) \leq 16 \hat{\mathbf{e}}(U).$$

As a preparation for Theorem 8.6, we prove the following variant:

Lemma 8.1 (i) *If $\varphi_1, \dots, \varphi_k \in \mathcal{D}$ are properly separated, then*

$$(8.31) \quad E(\varphi_1 \circ \dots \circ \varphi_k) \leq 2 \max_{1 \leq i \leq k} E(\varphi_i).$$

(ii) *If $\varphi_1, \dots, \varphi_k \in \mathcal{D}$ with $\text{supp}(\varphi_1), \dots, \text{supp}(\varphi_k) \subseteq U$, for a bounded open set U , then*

$$E(\varphi_1 \circ \dots \circ \varphi_k) \leq 16 \hat{\mathbf{e}}(U) + 2 \max_{1 \leq i \leq k} E(\varphi_i).$$

Proof (i) To ease the notation, we simply write $\alpha\beta$ for $\alpha \circ \beta$. Set $U_i = \text{supp}(\varphi_i)$. Take $\eta \in \mathcal{D}$ such that $V_1 = \eta(U_1), \dots, V_k = \eta(U_k)$ are properly separated. Note that if $\psi_i = \eta\varphi_i\eta^{-1}$, then $\text{supp}(\psi_i) = V_i$ and $E(\varphi_1 \dots \varphi_k) = E(\psi_1 \dots \psi_k)$. Choose H_i such that $\psi_i = \phi^{H_i}$. Note that because of 1-periodic orbits, in general $\text{supp}(H_i) \neq \text{supp}(\psi) \times [0, 1] = V_i \times [0, 1]$. So the support of H_i 's may not be disjoint. Let us write $W_i \times [0, 1]$ for the support of H_i . We now claim that we can find $\hat{\tau} \in \mathcal{D}$ such that if $\hat{\psi}_i = \hat{\tau}\psi_i\hat{\tau}^{-1}$, and $\hat{W}_i = \hat{\tau}(W_i)$, then $\hat{W}_1, \dots, \hat{W}_k$ are disjoint. To see this, first choose $r > 0$ sufficiently large so that $\text{supp}(H_j) \subseteq B_r(0) + V_i = \hat{V}_i$. We then use $\hat{\tau} \in \mathcal{D}$ so that $\hat{\tau}(x) = x + a_j$ is simply a translation on V_j . These a_j 's can be chosen so that $\hat{\tau}(\hat{V}_1), \dots, \hat{\tau}(\hat{V}_k)$ are disjoint.

Let us write $\hat{H}_i = H_i \circ \hat{\tau}_i$ and set $H = \hat{H}_1 + \dots + \hat{H}_k$. Since the support of $\hat{H}_1, \dots, \hat{H}_k$ are disjoint, we have

$$\phi^H = \phi^{\hat{H}_1} \dots \phi^{\hat{H}_k} = \hat{\psi}_1 \dots \hat{\psi}_k.$$

As a result,

$$E(\varphi_1 \dots \varphi_k) = E(\hat{\psi}_1 \dots \hat{\psi}_k) = E(\phi^H) \leq \|H\|_{\infty,1}.$$

But

$$\max_x H(x, t) - \min_x H(x, t) \leq 2 \max_{1 \leq i \leq k} \left(\max_x \hat{H}_i(x, t) - \min_x \hat{H}_i(x, t) \right).$$

As a result,

$$\|H\|_{\infty,1} \leq 2 \max_{1 \leq i \leq k} \|\hat{H}_i\|_{\infty,1},$$

completing the proof of (8.31).

(ii) First pick $\varepsilon > 0$ and choose $\eta \in \mathcal{D}$ such that

$$E(\eta) \leq \hat{\mathbf{e}}(U) + \varepsilon, \quad \text{and} \quad U \quad \text{and} \quad \eta(U) \quad \text{are properly separated.}$$

Out of η we build a sequence $\alpha_1, \beta_1, \dots, \alpha_k, \beta_k \in \mathcal{D}$ such that

- (i) For each i , there exist $\tau_i, \hat{\tau}_i \in \mathcal{D}$ such that $\alpha_i = \tau_i \eta \tau_i^{-1}$ and $\beta_i = \hat{\tau}_i \eta \hat{\tau}_i^{-1}$.
- (ii) For each i , the support of α_i and β_i are a suitable translates of U .
- (iii) The maps $\alpha_1, \beta_1, \dots, \alpha_k, \beta_k \in \mathcal{D}$ are properly separated.

Set $\psi_i = \varphi_1 \dots \varphi_i$. We set

$$\hat{\psi}_i = \alpha_i \psi_i \alpha_i^{-1}, \quad \bar{\psi}_i = \beta_i \psi_i^{-1} \beta_i^{-1},$$

so that the maps $\hat{\psi}_1, \bar{\psi}_1, \dots, \hat{\psi}_k, \bar{\psi}_k$, are properly separated and commute. We write

$$\varphi_1 \dots \varphi_k = \psi_k = \psi_k \bar{\psi}_k \hat{\psi}_{k-1} \bar{\psi}_{k-1} \dots \hat{\psi}_1 \bar{\psi}_1 (\bar{\psi}_k \hat{\psi}_{k-1} \bar{\psi}_{k-1} \dots \hat{\psi}_1 \bar{\psi}_1)^{-1}.$$

From this and (8.31) we learn

$$\begin{aligned} E(\varphi_1 \dots \varphi_k) &\leq E(\psi_k \bar{\psi}_k \hat{\psi}_{k-1} \bar{\psi}_{k-1} \dots \hat{\psi}_1 \bar{\psi}_1) + E(\bar{\psi}_k \hat{\psi}_{k-1} \bar{\psi}_{k-1} \dots \hat{\psi}_1 \bar{\psi}_1) \\ &\leq 2 \max_{1 \leq i \leq k} E(\hat{\psi}_i \bar{\psi}_i) + 2 \max_{1 \leq i \leq k} E(\hat{\psi}_i \bar{\psi}_{i-1}). \end{aligned}$$

We note

$$\begin{aligned} E(\hat{\psi}_i \bar{\psi}_i) &\leq E(\alpha_i) + E(\psi_i \alpha_i^{-1} \beta_i \psi_i^{-1}) + E(\beta_i^{-1}) = 2E(\eta) + E(\alpha_i^{-1} \beta_i) \leq 4E(\eta), \\ E(\hat{\psi}_i \bar{\psi}_{i-1}) &\leq E(\alpha_i) + E(\psi_i \alpha_i^{-1} \beta_{i-1} \psi_{i-1}^{-1}) + E(\beta_{i-1}^{-1}) \\ &= 2E(\eta) + E(\psi_i \psi_{i-1}^{-1} \psi_{i-1} \alpha_i^{-1} \beta_{i-1} \psi_{i-1}^{-1}) \\ &= 2E(\eta) + E(\psi_i \psi_{i-1}^{-1}) + E(\psi_{i-1} \alpha_i^{-1} \beta_{i-1} \psi_{i-1}^{-1}) \\ &= 2E(\eta) + E(\varphi_i) + E(\alpha_i^{-1} \beta_{i-1}) \leq 4E(\eta) + E(\varphi_i). \end{aligned}$$

We are done. $\square$

Proof of Theorem 8.6 Pick $\varepsilon > 0$. We write $\phi^H = \varphi_1 \circ \dots \circ \varphi_k$, where $\varphi_i = \phi_{t_i, t_{i+1}}^H$ and $0 < t_1 < \dots < t_k < t_{k+1} = t$ are chosen so that $E(\varphi_i) \leq \varepsilon$. By Lemma 8.1(ii),

$$E(\phi^H) \leq 16 \hat{e}(U) + 2\varepsilon.$$

As send $\varepsilon \rightarrow 0$ to arrive at the desired conclusion. $\square$

An interesting application of Theorem 6.2(iv) is the following regularity of the Poisson Bracket.

Theorem 8.7 *There is a constant c_0 such that such that for every Hamiltonian functions $f, g, F, G : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,*

$$(8.32) \quad \max\{f, g\} \leq \max\{F, G\} + c_0(1 + A^2)(\delta^{2/3} + A\delta^{1/3})$$

where $\delta = \max\{\|F - f\|_\infty, \|G - g\|_\infty\}$, and $A = \|f\|_{C^2} + \|g\|_{C^2}$.

Proof According to Theorem 6.2(iv),

$$t\|g - G\|_\infty \leq \pi r^2 \quad \Rightarrow \quad \phi_t^g(B_r(x)) \cap \phi_t^G(B_r(x)) \neq \emptyset,$$

for every $r, t > 0$ and $x \in \mathbb{R}^{2n}$. As a result, if we fix $x \in \mathbb{R}^{2n}$, $r > 0$, and choose $t = \pi r^2 \|G - g\|_\infty^{-1}$, then we can find $y, y' \in B_r(x)$ such that $\phi_t^G(y) = \phi_t^g(y')$. This in turn implies

$$\begin{aligned} f(\phi_t^g(y')) &= f(\phi_t^G(y)) \leq F(\phi_t^G(y)) + \|F - f\|_\infty \\ &= F(y) + \int_0^t \{F, G\}(\phi_s^G(y)) \, ds + \|F - f\|_\infty \\ &\leq f(y') + \int_0^t \{F, G\}(\phi_s^G(y)) \, ds + 2\|F - f\|_\infty + 2r \, \text{Lip}(f). \end{aligned}$$

Hence,

$$(8.33) \quad t^{-1} \int_0^t \{f, g\}(\phi_s^g(y')) \, ds \leq t^{-1} \int_0^t \{F, G\}(\phi_s^G(y)) \, ds + 2t^{-1} \|F - f\|_\infty + 2t^{-1} r \, Lip(f).$$

Observe

$$\{f, g\}(\phi_s^g(y')) \geq \{f, g\}(y') - s \|\{f, g\}, g\|_\infty.$$

From this and (8.33) we learn

$$\{f, g\}(y') \leq \max\{F, G\} + 2t^{-1} \|F - f\|_\infty + 2t^{-1} r \, Lip(f) + 2^{-1} t \|\{f, g\}, g\|_\infty.$$

From this we deduce

$$\{f, g\}(x) \leq \max\{F, G\} + 2t^{-1} (\|F - f\|_\infty + r \, Lip(f)) + 2^{-1} t \|\{f, g\}, g\|_\infty + r \, Lip(\{f, g\}),$$

for every x . Hence

$$\max\{f, g\} \leq \max\{F, G\} + 2t^{-1} (\|F - f\|_\infty + r \, Lip(f)) + 2^{-1} t \|\{f, g\}, g\|_\infty + r \, Lip(\{f, g\}),$$

provided that $t = \pi r^2 \|G - g\|_\infty^{-1}$. As a result,

$$\begin{aligned} \max\{f, g\} - \max\{F, G\} &\leq cr^{-2} \|G - g\|_\infty \|F - f\|_\infty + cr^{-1} \|G - g\|_\infty \, Lip(f) \\ &\quad + cr^2 \|G - g\|_\infty^{-1} \|\{f, g\}, g\|_\infty + cr \, Lip(\{f, g\}) \\ &\leq cr^{-2} \|G - g\|_\infty \|F - f\|_\infty + cAr^{-1} \|G - g\|_\infty \\ &\quad + cA^3 r^2 \|G - g\|_\infty^{-1} + cA^2 r, \end{aligned}$$

where $A = \|f\|_{C^2} + \|g\|_{C^2}$. In the same fashion we can show

$$\begin{aligned} \min\{f, g\} - \min\{F, G\} &\geq -cr^{-2} \|G - g\|_\infty \|F - f\|_\infty - cAr^{-1} \|G - g\|_\infty \\ &\quad - cA^3 r^2 \|G - g\|_\infty^{-1} - cA^2 r. \end{aligned}$$

We now interchange f with g to deduce

$$\begin{aligned} \min\{g, f\} - \min\{G, F\} &\geq -cr^{-2} \|G - g\|_\infty \|F - f\|_\infty - cAr^{-1} \|F - f\|_\infty \\ &\quad - cA^3 r^2 \|F - f\|_\infty^{-1} - cA^2 r. \end{aligned}$$

Equivalently,

$$\begin{aligned} \max\{f, g\} - \max\{F, G\} &\leq cr^{-2} \|G - g\|_\infty \|F - f\|_\infty + cAr^{-1} \|F - f\|_\infty \\ &\quad + cA^3 r^2 \|F - f\|_\infty^{-1} + cA^2 r. \end{aligned}$$

Hence,

$$\begin{aligned} \max\{f, g\} - \max\{F, G\} &\leq cr^{-2}\|G - g\|_\infty\|F - f\|_\infty + cAr^{-1}\|F - f\|_\infty \\ &\quad + cA^3r^2(\max\{\|F - f\|_\infty, \|G - g\|_\infty\})^{-1} + cA^2r \\ &\leq cr^{-2}\delta^2 + cAr^{-1}\delta + cA^3r^2\delta^{-1} + cA^2r, \end{aligned}$$

where $\delta = \max\{\|F - f\|_\infty, \|G - g\|_\infty\}$. If we choose $r = \delta^{2/3}$, we arrive at

$$\max\{f, g\} - \max\{F, G\} \leq c(1 + A^2)(\delta^{2/3} + A\delta^{1/3}),$$

as desired. $\square$

In Chapter 6, we learned that the period of the shortest periodic orbit of X_H is related to $\|H\|$ and the capacity of the support of H . Also the replacement energy provided another mean to measure the symplectic size of a set. We now examine the relationship between the time it takes to get from a set A to another set B and the Poisson bracket. We first start with a general fact about ODE: Recall $\mathcal{L}_X f$ denotes the Lie derivative of a 0-form f . In Euclidean setting we simply have $\mathcal{L}_X f = X \cdot \nabla f$.

Definition 8.5 Given two sets $A, B \subset M$, and a vector field X on M , we set

$$\begin{aligned} T(A, B; X) &= \inf \{t \geq 0 : \phi_t(A) \cap B \neq \emptyset\} \\ S(A, B; X) &= \inf \left\{ \max_M(L_X f) : f \in C^1, \max_A f \leq 0, \min_B f \geq 1 \right\}, \end{aligned}$$

where ϕ_t denotes the flow of the vector field X . $\square$

Proposition 8.9 (Fathi) We have $T(A, B; X)S(A, B; X) = 1$.

We may also formulate a discrete variant of Proposition 8.9 that was used by Fathi for the proof of this proposition.

Definition 8.5 Let (M, d) be a compact metric space with metric d and assume that $\psi : X \rightarrow X$ is a homeomorphism. Given two sets $A, B \subset M$, we set

$$\begin{aligned} T(A, B; \psi) &= \inf \{n \geq 0 : \psi^n(A) \cap B \neq \emptyset\} \\ S(A, B; \psi) &= \inf \left\{ \max_M(f \circ \psi - f) : f \in C^0, \max_A f \leq 0, \min_B f \geq 1 \right\}. \end{aligned}$$

where ϕ_t denotes the flow of the vector field X . $\square$

Proposition 8.10 (Fathi) assume that A and B are two closed subset of M . Then $T(A, B; \psi)S(A, B; \psi) = 1$.

Proof (*Step 1.*) Set $Lf = f \circ \psi - f$. Take a continuous f with $\max_A f \leq 0$, and $\min_B f \geq 1$. Take any $a \in A$ and $n \geq 0$ such that $\psi^n(a) \in B$. Then

$$1 \leq f(\psi^n(a)) = f(a) + \sum_{i=0}^{n-1} Lf(\psi^i(a)) \leq n \max(Lf).$$

As a result, $T(A, B; \psi)S(A, B; \psi) \geq 1$. (*Step 2.*) Take any continuous function $\Gamma : M \times M \rightarrow [0, \infty)$ with the following properties:

- $\Gamma(x, z) \leq \Gamma(x, y) + \Gamma(y, z)$.
- $\Gamma(x, \psi(x)) = 1$.

Given such Γ , define the operator $\Lambda : C(X) \rightarrow C(X)$ by

$$\Lambda(g)(x) = \min_{y \in M} \{g(y) + \Gamma(y, x)\}.$$

Observe that the properties of Γ guarantee that $\Lambda(g) \circ \psi \leq \Lambda(g) + 1$, or $L\Lambda(g) \leq 1$. Since

$$(\Lambda g \circ \psi)(x) \leq g(x) + \Gamma(x, \psi(x)) = g(x) + 1,$$

we also have $L\hat{\Lambda}g \leq 1$, for

$$\hat{\Lambda}(g) = \min\{g, \Lambda g\}.$$

Now assume that $g \geq 0$, and that $g = 0$ on a set A . Then $\hat{\Lambda}g = 0$ on A as well because $\Lambda g \geq 0$. Now we wish to find a lower bound on $\hat{\Lambda}g$ on the set B in terms of $T(A, B; \psi)$. For this we need to construct a suitable Γ to serve our purposes.

(*Step 3*) We construct Γ with the additional property

$$\Gamma(x, y) \leq T(x, y; \psi),$$

where $T(x, y; \psi) \leq T(\{x\}, \{y\}; \psi)$. In other words, if $\psi^n(x) = y$, then $T(x, y; \psi) \leq n$. This may be achieved by taking a sequence from x to y and penalize it when it deviates from an orbit of ψ . More precisely, take (a large) $k > 0$ and set

$$\Gamma(x, y) = \Gamma_k(x, y) = \inf \left\{ \sum_{i=0}^{\ell-1} d_k(\psi(x_i), x_{i+1}) : x_0, \dots, x_\ell \in M, \text{ with } x_0 = x, x_\ell = y \right\},$$

for $d_k(x, y) = kd(x, y) + 1$. For a large k , the appearance of the term $kd(\psi(x_i), x_{i+1})$ penalize deviation from the orbits of ψ , while the appearance of 1 counts the time it takes to reach y from x . It is not hard to see that Γ possesses all the aforementioned properties.

(*Final Step.*) Let us write $\hat{\Lambda}_k$ for $\hat{\Lambda}$ associated with Γ_k . Recall that $g \geq 0$ is 1 on the set A . We wish to find a lower bound for

$$\bar{t} = \sup_k \min_B \hat{\Lambda}_k g.$$

Pick and $t' > \bar{t}$, and for each $k \in \mathbb{N}$, choose $\ell_k \in \mathbb{N}$, a sequence

$$\mathbf{x}_k = (x_0^k = x, x_1^k, \dots, x_{\ell_k-1}^k, x_{\ell_k}^k = y),$$

such that

$$\sum_0^{\ell_k-1} d_k(\psi(x_i^k), x_{i+1}^k) \leq t'$$

Since $\ell_k \leq A'$, we may choose a subsequence $k_r \rightarrow \infty$ such that $\ell_{k_r} = \ell$ is constant, and that the limit

$$\mathbf{x} = (x_0, \dots, x_\ell) = \lim_{r \rightarrow \infty} \mathbf{x}_{k_r},$$

exists. For this sequence, we certainly have $\psi(x_i) = x_{i+1}$ because $k_r \rightarrow \infty$. Hence

$$\ell = \lim_{r \rightarrow \infty} \sum_0^{\ell_{k_r}-1} d_{k_r}(\psi(x_i^{k_r}), x_{i+1}^{k_r}) \leq t'.$$

As a result $t' \geq T(A, B; \psi)$. This in turn implies $t \geq T(A, B; \psi)$. Hence for large k , we have

$$T(A, B; \psi) \leq \inf_B \hat{\Lambda}_k g,$$

as desired. □

Proof of Proposition 8.9 Take and $f \in C^1$ with $\max_A f \leq 0$, and $\min_B f \geq 1$. Take any $a \in A$ and $t \geq 0$ such that $\phi_t(x) \in B$. Then

$$1 \leq f(\phi_t(a)) = f(a) + \int_0^t L_X f(\phi_s(a)) \, ds \leq t \max(L_X f).$$

As a result, $T(A, B; X)S(A, B; X) \geq 1$.

We first give a short prove for a weaker reverse inequality. Pick any $t_0 < T(A, B; X)$ so that $\phi_{[0, t_0]}(A) \cap B = \emptyset$. As a result

$$(8.34) \quad \phi_{[0, t_0/2]}(A) \cap \phi_{[-t_0/2, 0]}(B) = \emptyset.$$

Take any C^1 function $g : M \rightarrow \mathbb{R}$ and observe

$$(8.35) \quad t^{-1}(g \circ \phi_t - g) = t^{-1} \int_0^t \frac{d}{ds} (g \circ \phi_s) \, ds = t^{-1} \int_0^t L_X (g \circ \phi_s) \, ds = L_X f,$$

where

$$f = t^{-1} \int_0^t g \circ \phi_s \, ds.$$

Choose $t = t_0/2$ and a C^1 function $g : M \rightarrow [0, 1]$ such that $g = 0$ on $\phi_{[0, t_0/2]}(A)$, and $g = 1$ on $\phi_{[-t_0/2, 0]}(B)$. Such a function exists because (8.33) holds. For such g , the corresponding f satisfies $f = 0$ on A and $f = 1$ on B . On the other hand, we have

$$\max(\mathcal{L}_X f) \leq 2t_0^{-1},$$

by (8.36). As a result, $S(A, B; X) \leq t_0/2$. This implies that $T(A, B; X)S(A, B; X) \leq 2$. $\square$

Let us write $\mathcal{H}_1(U, K)$ for the set of C^1 Hamiltonian functions $H : M \rightarrow [0, 1]$ such that $H = 0$ outside U and $H = 1$ on $K \subset U$. We then set

$$\begin{aligned} \hat{T}(A, B; U, K) &= \inf \left\{ t \geq 0 : \phi_t^H(A) \cap B \neq \emptyset \text{ for some } H \in \mathcal{H}_1(U, K) \right\} \\ &= \sup_{H \in \mathcal{H}(U, K)} T(A, B; X_H), \\ \hat{S}(A, B; U, K) &= \inf \left\{ \max_M(L_{X_H} f) : H \in \mathcal{H}(U, K), f \in C^1, \max_A f \leq 0, \min_B f \geq 1 \right\} \\ &= \inf \left\{ \max_M \{H, f\} : H \in \mathcal{H}_1(U, K), f \in C^1, \max_A f \leq 0, \min_B f \geq 1 \right\}, \end{aligned}$$

Proposition 8.11 *We have $\hat{T}(A, B; U, K)\hat{S}(A, B; U, K) = 1$.*

Remark 8.7 We now describe a strategy of Buhevsky, Entrov and Polterovich for finding a lower bound on $\hat{S}$ when $\omega = d\lambda$. Set $\omega_t = d\lambda_t$ with $\lambda_t = \lambda + t f dH$. Since

$$\omega_t^d = \omega^d - t\{H, f\}\omega^d = (1 - t\{H, f\})\omega^d,$$

we learn that ω_t is symplectic so long as

$$t \in [0, \max\{H, f\}^{-1}).$$

Choose a time dependent vector field X_t such that

$$\frac{d\omega_t}{dt} + \mathcal{L}_{X_t}\omega_t = 0.$$

This would be the case if $i_{X_t}\omega_t = Hdf$. If ϕ_t is the flow of X_t , then $\phi_t^*\omega_t = \omega$, or

$$\omega_t = (\phi_t)^{-1*}\omega.$$

Hence for every closed curve Γ ,

$$\int_{\phi_t^{-1}\Gamma} \omega + t \int_{\Gamma} f dH = \int_{\Gamma} \omega.$$

Now the function (f, H) maps the set $A \cap B \cap U \cap K$ to the square $[0, 1]^2$. If $(f, H)(\Gamma)$ is the square with positive orientation, then

$$\int_{\Gamma} f dH = \int_{(f,H)(\Gamma)} q dp = 1.$$

Hence

$$\int_{\phi_t^{-1}\Gamma} \omega + t = \int_{\Gamma} \omega.$$

By choosing Γ for which

$$\int_{\phi_t^{-1}\Gamma} \omega > 0.$$

we get the bound $t \leq \int_{\Gamma} \omega$. This means

$$\max\{H, f\}^{-1} \leq \int_{\Gamma} \omega.$$

□

We can use Theorem 6.2(iv) to get a lower bound for $D(\phi, \psi)$, namely, if

$$\begin{aligned} \hat{D}(\phi, \psi) &= \sup \{ \pi r^2 : \exists a \text{ such that } \phi(B_r(a)) \cap \psi(B_r(a)) = \emptyset \} \\ &\leq \sup \{ \mathbf{e}(U) : U \text{ open, } \phi(U) \cap \psi(U) = \emptyset \}, \end{aligned}$$

then

$$(8.36) \quad \hat{D}(\phi, \psi) \leq D(\phi, \psi).$$

Here is a useful consequence of (8.36).

Proposition 8.12 *For every $\psi, \phi \in \mathcal{D}$,*

$$(8.37) \quad \|\phi - \psi\|_{C^0} \leq (\alpha_{\phi} + \alpha_{\psi}) \left(\sqrt{\pi^{-1} D(\phi, \psi)} \right),$$

where α_{ϕ} denotes the modulus of continuity of ϕ .

Proof Given a , we certainly have

$$\pi r^2 < D(\phi, \psi) \implies \phi(B_r(a)) \cap \psi(B_r(a)) \neq \emptyset.$$

This means that for such $r > 0$, and any a , we can find $x, y \in B_r(a)$ such that $\phi(x) = \psi(y)$. As a result,

$$|\phi(a) - \psi(a)| \leq |\phi(a) - \phi(x)| + |\psi(y) - \psi(a)| \leq (\alpha_\phi + \alpha_\psi)(r).$$

We then maximize over r and minimize over a in this order to deduce (8.37). $\square$

Note that if we restrict D to a C^0 -compact subset of $\mathcal{D}$, then (8.37) means that the uniform metric is dominated by D . Let $\mathcal{K}$ be such a compact set. Define

$$\mathcal{K}_U = \{\varphi \in \mathcal{D} : \text{supp}(\varphi) \subset U, \text{ var} \in \mathcal{K}\}.$$

Then D induces the same topology as uniform norm on $\mathcal{K}_U$ for any bounde U .

Exercise 8.1(i) Show directly that $(\phi_t^H)^{-1} = \phi_t^{-H}$ for a time-independent Hamiltonian function H . Use Proposition 8.1(ii) to draw the same conclusion.

(ii) Show that the limit

$$\hat{D}(\varphi) := \lim_{\ell \rightarrow \infty} \ell^{-1} D(id, \varphi^\ell),$$

exists and equals to $\inf_\ell \ell^{-1} D(id, \varphi^\ell)$. $\square$

9 Generating Function, Twist Map and Arnold's Conjecture

Given a Hamiltonian function $H : M \times [0, \infty) \rightarrow \mathbb{R}$ on a symplectic manifold (M, ω) , we may wonder whether or not the corresponding Hamiltonian vector field $X_H = X_H^\omega$ has T -periodic orbits for a given period T . We have addressed this question in Chapters 2,4-8 already, but now we would like to find some non trivial lower bounds on the number of such periodic orbits. To convince that such a question is natural and important, let us examine this question when the Hamiltonian function is time-independent first. We note that for the autonomous X_H we can even find rest points (or constant orbits) and there is a one-one correspondence between the constant orbits of X_H and the critical points of H . We can appeal to the following classical theories in Algebraic Topology to obtain sharp universal lower bounds on the number of critical points of a smooth function on M where M is a smooth closed manifold. Let us write $Crit(H)$ for the set of critical points of $H : M \rightarrow \mathbb{R}$.

(i) According to Lusternik-Schnirelmann (LS) Theorem,

$$(9.1) \quad \sharp Crit(H) \geq c\ell(M),$$

where $c\ell(M)$ denotes the cuplength of M . (See Appendix D.)

(ii) According to Morse Theory, for a Morse function H ,

$$(9.2) \quad \sharp Crit(H) \geq \sum_k \beta_k(M),$$

where $\beta_k(M)$ denotes the k -th Betti's number of M . (See Appendix F).

According to Arnold's conjecture, the analogs of (9.1) and (9.2) are true for the non-autonomous Hamiltonian functions provided that we count 1-periodic orbits of X_H instead of constant orbits. For the sake of comparison, we may regard (9.1) and (9.2) as a lower bound on the number of 0-periodics orbit when H is 0-periodic in t . In Arnold's conjecture, we replace 0-periodicity with 1-periodicity. To this end, define

$$Per(H) = \{x \in M : \phi_1^H(x) = x\} = Fix(\phi_1^H).$$

Arnold's Conjecture Let (M, ω) be a closed symplectic manifold and let $H : M \times [0, \infty) \rightarrow \mathbb{R}$ be a smooth Hamiltonian function that is 1-periodic in the time variable. Then

$$(9.3) \quad \sharp Per(H) \geq c\ell(M).$$

Moreover, if $\varphi := \phi_1^H$ is non-degenerate in the sense that $\det(d\varphi - id)_x \neq 0$ for every $x \in Fix(\varphi)$, then

$$(9.4) \quad \sharp Per(H) \geq \sum_k \beta_k(M).$$

□

We now describe our strategies for establishing Arnold's conjecture under some additional conditions on M :

(Main Strategy) (1) As in Chapters 5, 7 and 8, a natural way to tackle Arnold's conjecture is to study the set of critical points of $\mathcal{A}_H : \Gamma \rightarrow \mathbb{R}$, where Γ is the space of 1-periodic $x : \mathbb{R} \rightarrow M$ and

$$(9.5) \quad \mathcal{A}_H(x(\cdot)) = \int_w \omega - \int_0^1 H(x(t), t) dt,$$

where $w : \mathbb{D} \rightarrow M$ is any extension of $x : \mathbb{S}^1 \rightarrow M$ to the unit disc $\mathbb{D}$. (Note that since ω is closed, the right-hand side of (9.5) is independent of the extension.) We may try to apply LS and Morse Theory to the functional $\mathcal{A}_H$ in order to get lower bounds on $\sharp \text{Per}(H)$. Of course we cannot apply either Morse Theorem (9.2) or LS Theorem (9.1) to $\mathcal{A}_H$ directly because Γ is neither compact nor finite-dimensional. However, we may attempt to modify the proofs of (9.1) and (9.2) to analyze the critical values of $\mathcal{A}_H$. A standard way to prove either (9.1) or (9.2) is to study the gradient flow associated with $\mathcal{A}_H$ as we did in Chapter 7. This turns out to be plausible in some cases only, most notably when $M = \mathbb{T}^{2n}$ is a torus. In other words, we may establish Arnold's conjecture in the case of a torus by analyzing the gradient flow associated with $\mathcal{A}_H : \mathcal{E} \rightarrow \mathbb{R}$, where $\mathcal{E} = \mathcal{H}^{1/2}$, and

$$\mathcal{A}_H(x(\cdot)) = \mathcal{A}_H(x(\cdot)) = \int_w \omega - \int_0^1 H(x(t), t) dt = \frac{1}{2} \int_{x(\cdot)} \bar{J}x \cdot \dot{x} dt - \int_0^1 H(x(t), t) dt,$$

with $H : \mathbb{R}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}$, 1-periodic in x and t . More precisely, we look at the ODE

$$(9.6) \quad \frac{d}{ds}u = -\nabla \mathcal{A}_H(u),$$

where

$$\mathcal{A}_H(x(\cdot)) = P^+x(\cdot) - P^-x(\cdot) - \mathcal{I}(\nabla H(x(\cdot), \cdot)).$$

Here we have used our notations in Chapter 7. We note that if $x(\cdot)$ is 1-periodic, then $H(x(\cdot), \cdot)$ is also 1-periodic because H is 1-periodic in t . Also note that a solution of (9.6) is a function $u : \mathbb{R}^2 \rightarrow M$ such that u is 1-periodic with respect to the second coordinate. To find critical points of $\mathcal{A}_H$, we would look for solutions of (9.6) such that

$$\lim_{s \rightarrow \infty} u(s, \cdot) := x_\infty(\cdot),$$

exists. This suggests that we may restrict the flow $\Phi_t = \Phi_t^H$ of the ODE (9.6) to those starting loops for which the flow remains bounded for all times. In other words, we are only interested in the set

$$(9.7) \quad \mathcal{E}_\infty = \left\{ x \in \mathcal{E} : \sup_{t \in \mathbb{R}} \|\Phi_t(x)\|_{1/2} < \infty \right\}.$$

It turns out that the set $\mathcal{E}_\infty$ is compact; this allows us to apply LS or Morse Theory for the flow ψ_t restricted to $\mathcal{E}_\infty$. Let us remark that if $H = 0$, then the equation (9.6) is easily solved by

$$\Phi_t^0(x) = e^t P^- x + x_0 + e^{-t} P^+ x.$$

In this case, the equation (9.6) is a linear hyperbolic ODE with $\mathcal{E}^0$ consisting of the rest points, and $\mathcal{E}^\pm$ representing the stable/unstable subspaces. When $H \neq 0$, we have a compact perturbation of this hyperbolic picture.

(2) As we mentioned before, the strategy sketched in (1) works for the torus but not for more general manifolds. Floer adopted another approach by studying the gradient flow in L^2 setting. That is,

$$(9.8) \quad \frac{dw}{ds} = -\partial \mathcal{A}_H(w).$$

Note however this equation as an ODE is not well-posed because $\partial \mathcal{A}_H(w)$ is not defined for $w \in L^2(\mathbb{S}^1; M)$. As we will see in Chapter 10, indeed (9.8) is an elliptic PDE and not well-posed for an initial value problem. However, we may hope to solve this equation for suitable *boundary conditions*; given two loops $x^\pm : \mathbb{S}^1 \rightarrow M$, we may wonder whether or not there exists a solution w of (9.8) such that

$$\lim_{s \rightarrow \pm\infty} w(s, \cdot) = x^\pm(\cdot).$$

(3) In our Strategy (1) and (2), we solve a differential equation (9.8) or (9.6) in an infinite dimensional space. However in the case of the torus or $M = T^*L$, we may reduce the problem by using *generalized generating functions*. In fact, one can show that ϕ_t^H has a type II generating function (recall (8.11)) provided that t is sufficiently small. We then use group property of the flow to write

$$\varphi = \phi_1^H = \psi_1 \circ \cdots \circ \psi_N,$$

where each ψ_i has a generating function. This can be used to build a generalized generating function for φ a la Chaperon. We may establish Arnold's conjecture with the aid of generalized generating functions in some cases. $\square$

We now use our described strategies to establish Arnold's conjecture in two cases: When $M = \mathbb{S}^1 \times [-1, 1]$ is a cylinder or $M = \mathbb{T}^{2n}$ is a torus. For these two cases, we may use Strategy (3) to establish Arnold's conjecture. Recall the definition of Type I generating function as was discussed in Chapter 8 (see (8.8) and (8.9)). If $\varphi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is a symplectic diffeomorphism with $\varphi(q, p) = (Q, P)$, we wish to find scalar-valued function $S(q, Q)$ so that

$$(9.9) \quad \varphi(q, -S_q(q, Q)) = (Q, S_Q(q, Q)).$$

Such S always exists locally under a non-degeneracy assumption as the next Proposition demonstrates.

Proposition 9.1 Let $\varphi : U \rightarrow \mathbb{R}^{2n}$ be a symplectic transformation and assume that at $(q^0, p^0) \in U$,

$$(9.10) \quad \det \frac{\partial Q}{\partial p}(q^0, p^0) \neq 0.$$

Then there exist a neighborhood V of q^0 and $Q^0 = Q(q^0, p^0)$, and a C^1 function $S : V \rightarrow \mathbb{R}$ such that (9.11) holds.

Proof From (9.10) or (8.9), and Implicit Function Theorem, the relation $Q = Q(q, p)$ can be solved for $p = p(q, Q)$ for q and Q near q^0 and Q^0 . We then set $P(q, Q) = P(q, p(q, Q))$. To solve (8.9) for S , we need to verify the solvability criterion

$$P_q + p_Q = 0,$$

regarding P and p as functions of q and Q . This is exactly $d(P \cdot dQ - p \cdot dq) = 0$. $\square$

As our first example, let us study generating functions for the case of a cylinder $M = \mathbb{S}^1 \times [-1, 1]$. Let $\varphi : M \rightarrow M$ be a symplectic (area-preserving) diffeomorphism. Such a transformation was encountered by Poincaré as he used a Poincaré's section to study solutions to Hamiltonian systems. (Here we use the standard symplectic (area) form on M .) To have fixed points of φ , we need some appropriate boundary conditions. As we will see such fixed points exist if φ is a *twist map*. To explain the twist condition, let us assume that if

$$\varphi(\mathbb{S}^1 \times \{\pm 1\}) = (\mathbb{S}^1 \times \{\pm 1\}),$$

then the restriction of φ to the boundary circles are two circle diffeomorphisms for which *rotation numbers* can be defined. Roughly the twist condition means that the map on top and bottom circles twist in opposite directions.

Definition 9.1(i) Regarding $\mathbb{S}^1$ as the interval $[0, 1]$ with $0 = 1$, let $f : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ be an orientation preserving homeomorphism. Its *lift* $F = \ell(f)$ is an increasing map $F : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = F(x) \pmod{1}$, and F can be written as $F(x) = x + g(x)$, for a 1-periodic function $g : \mathbb{R} \rightarrow \mathbb{R}$.

(ii) For f and F as in **(i)**, we define its rotation number

$$(9.11) \quad \rho(f) = \rho(F) = \lim_{n \rightarrow \infty} n^{-1} F^n(x).$$

By a result of Poincaré, the limit exists and is independent of x .

(iii) Let $\varphi : \mathbb{S}^1 \times [-1, 1] \rightarrow \mathbb{S}^1 \times [-1, 1]$, be an orientation preserving homeomorphism. Its *lift* is a homeomorphism $\Phi : \mathbb{R} \times [-1, 1] \rightarrow \mathbb{R} \times [-1, 1]$ such that

$$\varphi(x) = \Phi(x) \pmod{1},$$

and $\Phi = \ell(\varphi)$ can be written as $\Phi(q, p) = (q, 0) + \Psi(q, p)$, for a continuous $\Psi : \mathbb{R} \times [-1, 1] \rightarrow \mathbb{R} \times [-1, 1]$, that is 1-periodic function in q -variable.

(iv) An orientation-preserving diffeomorphism $\varphi : \mathbb{S}^1 \times [-1, 1] \rightarrow \mathbb{S}^1 \times [-1, 1]$ is called a *twist map* if the following conditions are met:

- (i) φ (or equivalently its lift) is area-preserving.
- (ii) If we define $\varphi^\pm$ by $(\varphi^\pm(q), \pm 1) = \varphi(q, \pm 1)$, then $\rho(\varphi^+) < 0 < \rho(\varphi^-)$.

□

Our main result about twist maps is the following result of Poincaré and Birkhoff.

Theorem 9.1 *Any twist map has at least two fixed points.*

To see Poincaré-Birkhoff's theorem within a larger context, let us recall a result of Poincaré for circle homeomorphism:

Proposition 9.2 *Let $f : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ be an orientation preserving homeomorphism. Then the following statements are true for f :*

- (i) *f has a fixed point iff $\rho(f) = 0$.*
- (ii) *$\pm\rho(f) > 0$ iff $\pm(\ell(f)(x) - x) > 0$.*
- (iii) *Let (r, s) be a pair of coprime positive integers. Then f has a (r, s) -periodic orbit (this means that $F^s(x) = F(x) + r$ for $F = \ell(f)$), iff $\rho(f) = r/s$.*

One way to interpret Theorem 9.1 is that since $0 \in (\rho(\varphi^-), \rho(\varphi^+))$, then φ has at least two orbits in the interior of the cylinder that are associated with 0 rotation number, namely fixed points. In fact an analogous result is true for periodic orbits which may be regarded as a variant Theorem 9.2(ii) for the twist maps.

Theorem 9.2 (Birkhoff) *Let $\varphi : \mathbb{S}^1 \times [-1, 1] \rightarrow \mathbb{S}^1 \times [-1, 1]$, be an area and orientation preserving C^1 -diffeomorphism. If $r/s \in (\rho(\varphi^-), \rho(\varphi^+))$ is a rational number with r and s coprime, then φ has at least two (r, s) -periodic orbits in the interior of $\mathbb{S}^1 \times [-1, 1]$.*

Naturally we are led to the following question: How about an irrational $\rho \in (\rho^-, \rho^+)$? Can we find an orbit of φ associated with such ρ ? The answer to this question is affirmative and this is the subject of the *Mather Theory*. For any irrational $\rho \in (\rho^-, \rho^+)$, there exists an invariant set on the cylinder that in some sense has the rotation number ρ . This invariant set q -projects onto either a Cantor-like subset of S^1 or the whole S^1 . The invariant set lies on a graph of a Lipschitz function defined on S^1 . These invariant sets are known as *Aubry-Mather sets*.

Poincaré established Theorem 9.1 provided that φ has a global generating function. Such a generating function exists if φ is a *monotone twist map*. To explain Poincaré's argument, let us formulate a condition on $\Phi = \ell(\varphi)$ that would guarantee the existence of a global generating function $S(q, Q)$ for Φ .

Definition 9.2 A twist map φ or its lift $\Phi(q, p) = (Q(q, p), P(q, p))$ is called *positive (monotone)* if $Q(q, p)$ is increasing in p for every $q \in \mathbb{R}$. We say φ is *negative (monotone)* twist if φ^{-1} is a positive twist. $\square$

Proposition 9.3 *Let Φ be a monotone twist map. Then there exists a C^2 function $S : U \rightarrow \mathbb{R}$ with*

$$U = \{(q, Q) : Q(q, -1) \leq Q \leq Q(q, +1)\}$$

such that

$$\Phi(x, -S_q(q, Q)) = (Q, S_Q(q, Q)).$$

Moreover

$$(9.12) \quad S(q+1, Q+1) = S(q, Q), \quad S_{qQ} < 0.$$

Proof The image of the line segment $\{q\} \times [-1, 1]$ under Φ is a curve γ with parametrization $\gamma(p) = (Q(q, p), P(q, p))$. By the monotonicity, the relation $Q(q, p) = Q$ can be inverted to yield $p = p(q, Q)$ which is increasing in Q . The set $\gamma[-1, 1]$ can be viewed as a graph of the function

$$Q \mapsto P(q, p(q, Q))$$

with $Q \in [Q(q, -1), Q(q, +1)]$. The anti-derivative of this function yields S . This can be geometrically described as the area of the region Δ between the curve $\gamma([-1, 1])$, the line $P = -1$ and the vertical line $\{q\} \times [-1, 1]$. We now apply Φ^{-1} on this region. The line segment $\{Q\} \times [-1, 1]$ is mapped to a curve $\hat{\gamma}([-1, 1])$ which coincides with a graph of a function $q \mapsto p$. Since Φ is area preserving the area of $\Phi^{-1}(\Delta)$ is $S(q, Q)$. From this we deduce that $S_Q = -p$. Here we have used the fact that Φ^{-1} is a (negative) twist map. This is because if we write $\Phi^{-1}(Q, P) = (q(Q, P), p(Q, P))$, then

$$(\Phi^{-1})' = \begin{bmatrix} q_Q & q_P \\ p_Q & p_P \end{bmatrix} = \begin{bmatrix} Q_q & Q_p \\ P_q & P_p \end{bmatrix}^{-1} = \begin{bmatrix} P_p & -Q_p \\ -P_q & Q_q \end{bmatrix}$$

which implies that $q_Q = -Q_p < 0$.

The periodicity (9.12) is an immediate consequence of $\Phi(q+1, p) = \Phi(q, p) + (1, 0)$;

$$\Phi(\{q+1\} \times [-1, 1]) = \Phi(\{q\} \times [-1, 1]) + (1, 0).$$

As for the second assertion in (9.12), recall that $p(q, Q)$ is increasing in Q . Hence

$$S_{qQ} = -p_Q < 0.$$

□

A partial converse to Proposition 9.3 is true, namely if a function S satisfies (9.12), then it generates a map Φ which is area preserving. We don't address the behavior of Φ on the boundary lines and for simplicity assume that S is defined on $\mathbb{R}^2$.

Proposition 9.4 *Let S be a C^2 function satisfying (9.12). Then there exists a C^1 -function Ψ such that*

- (i) $\Phi(q+1, p) = \Phi(q, p) + (1, 0)$
- (ii) $\Psi(q, -S_q(q, Q)) = (Q, S_Q(q, Q))$
- (iii) $\det \Phi' \equiv 1$.

Proof. Since $S_{qQ} < 0$, the function $Q \mapsto -S_q(q, Q)$ is increasing. As a result, $p = -S_q(q, Q)$ can be inverted to yield $Q = Q(q, p)$. We then set

$$P(q, p) = S_Q(q, Q(q, p)) \text{ and } \Phi(q, p) = (Q(q, p), P(q, p)).$$

Evidently (ii) is true and (i) follows from (ii) and (9.12) because $S_q(q+1, Q+1) = S_q(q, Q)$, and $S_Q(q+1, Q+1) = S_Q(q, Q)$. It remains to verify (iii). For this, set $\hat{S}(q, p) = S(q, Q(q, p))$. We have

$$\begin{aligned} \hat{S}_q &= S_q + S_Q Q_q = -p + P Q_q, \\ \hat{S}_p &= S_Q Q_p = P Q_p. \end{aligned}$$

Differentiating again yields

$$\begin{aligned} \hat{S}_{qp} &= -1 + P_p Q_q + P Q_{qp}, \\ \hat{S}_{pq} &= P_q Q_p + P Q_{pq}. \end{aligned}$$

Since $S \in C^2$, we must have $\hat{S}_{qp} = \hat{S}_{pq}$, which yields $P_p Q_q - P_q Q_p = 1$, as desired. □

We now show how the existence of a generating function can be used to prove the existence of fixed points.

Proof of Theorem 9.1 for a monotone twist map Define $L(q) = S(q, q)$. We first argue that a critical point of L corresponds to a fixed point of Φ . Indeed, if $L'(q^0) = 0$, then $S_q(q^0, q^0) + S_Q(q^0, q^0) = 0$. Since $\Phi(q^0, -S_q(q^0, q^0)) = (q^0, S_Q(q^0, q^0))$, we deduce that $\Phi(q^0, y^0) = (q^0, y^0)$ for $y^0 = -S_q(q^0, q^0) = S_Q(q^0, q^0)$. On the other hand, by (9.12), we have that $L(q+1) = L(q)$. Either L is identically constant which yields a continuum of fixed

points for Ψ , or L is not constant. In the latter case, L has at least two distinct critical points, namely a maximizer and minimizer. These yield two distinct critical points of Φ . $\square$

We may wonder whether a similar strategy as in the above proof can be used to Prove Theorem 9.2 when φ is a monotone area-preserving map. Indeed if Φ is a monotone twist map, then we can associated with it a variational principle which is the discrete analog of the Lagrange Variational Principle, as can be seen in the following proposition.

Proposition 9.5 *Let Φ be a monotone twist map with generating function S . Given q and $Q \in \mathbb{R}$, define*

$$L(q_1, q_2, \dots, q_{k-1}; q, Q) = \sum_{j=0}^{n-1} S(q_j, q_{j+1}),$$

with $q_0 = q$, and $q_n = Q$. Then the following statements are true.

- (i) *The point $(q_1, q_2, \dots, q_{k-1})$ is a critical point of $L(\cdot; q, Q)$ iff there exist $p_0, p_1, \dots, p_k$ such that $\Phi^j(q_j, p_j) = (q_{j+1}, p_{j+1})$ for $j = 1, 2, \dots, k-1$.*
- (ii) *The point $(q_0, q_1, q_2, \dots, q_{s-1})$ is a critical point of*

$$K(q_1, q_2, \dots, q_s) = S(q_{s-1}, q_0 + r) + \sum_{j=0}^{s-2} S(q_j, q_{j+1})$$

if and only if there exist $p_0, p_1, p_2, \dots, p_{s-1}$ such $\Phi^j(q_j, p_j) = (q_{j+1}, p_{j+1})$ for $j = 0, \dots, s-1$, with $q_s = q_0 + r$.

Proof We only prove (ii) because (i) can be proved by a verbatim argument. Let $(q_0, \dots, q_{s-1})$ be a critical point and set $q_s = q_0 + r$. We also set $p_j = -S_q(q_j, q_{j+1})$. The result follows because if $P_j = S_Q(q_j, q_{j+1})$, then

$$K_{q_j} = p_j - P_{j-1}$$

for $j = 0, 1, 2, \dots, s-1$ and $\Psi(q_j, p_j) = (q_{j+1}, P_j)$. $\square$

Before we discuss the proof of Theorem 9.1 for general twist maps, let us study an example of a map which is not quite a twist map but still possesses a global generating function and Theorem 9.2 may be applied to guarantee the existence of its periodic orbits.

Example 9.1 (Billiard map in a convex domain). Let C be a strictly bounded convex domain in $\mathbb{R}^2$ and denote its boundary by Γ . Without loss of generality, we assume that the

total length of Γ is 1. First we describe the billiard flow in C . This is the flow associated with the Hamiltonian function $H(q, p) = \frac{1}{2}|p|^2 + V(q)$ where

$$V(q) = \begin{cases} 0 & \text{if } q \in C \\ \infty & \text{if } q \notin C. \end{cases}$$

Here is the interpretation of the corresponding flow: A ball of velocity x starts from a point $a \in C$ and is bounced off the boundary Γ by the law of reflection. This induces a transformation for the hitting location and reflection angle. More precisely, if a trajectory $a+tv$, $t > 0$ hits the boundary at a point $\gamma(q)$ and a post-reflection angle θ , then we write $\gamma(Q)$ and Θ for the location and post-reflection angle of the next reflection. Here q is the length of arc between a reference point $A \in \Gamma$ and $\gamma(q)$ on Γ in positive direction, and θ measures the angle between the tangent at $\gamma(q)$ and the post-reflection velocity vector. We write ψ for the map $(q, \theta) \mapsto (Q, \Theta)$ with $q, Q \in S^1$ and $\theta, \Theta \in [0, \pi]$. It is more convenient to define $p = -\cos \theta$ so that in the (q, p) coordinates, we have a map $\varphi : S^1 \times [-1, 1] \rightarrow S^1 \times [-1, 1]$. As before, we write Φ for its lift. We claim that Φ is a monotone area-preserving map; it is not a twist map because the twist conditions on the boundary lines $p = \pm 1$ are violated. We show this by applying Proposition 9.4. In fact the generating function is simply given by

$$S(q, Q) = -|\gamma(q) - \gamma(Q)|,$$

because

$$\begin{aligned} -S_q(q, Q) &= -\frac{(\gamma(Q) - \gamma(q))}{|\gamma(Q) - \gamma(q)|} \cdot \dot{\gamma}(q) = -\cos \theta, \\ S_Q(q, Q) &= -\frac{(\gamma(Q) - \gamma(q))}{|\gamma(Q) - \gamma(q)|} \cdot \dot{\gamma}(Q) = \cos \Theta, \\ S_{Qq}(q, Q) &= \Theta_q \sin \Theta. \end{aligned}$$

Note that if $\Theta \in (0, \pi)$, then $\sin \Theta > 0$, and Θ is decreasing in q which means that $S_{Qq} < 0$. Here of course we are using the strict convexity. As for the boundary lines, we have $\Phi(q, -1) = (q, -1)$, $\Phi(q, 1) = (q + 1, 1)$. Note that $S(q, Q)$ is defined for (q, Q) satisfying $Q \in [q, q + 1]$. Also note that Φ has no fixed point inside $\mathbb{R} \times (-1, 1)$. It is not hard to show that $\rho^- = \rho(\varphi^-) = 0$ and $\rho^+ = \rho(\varphi^+) = 1$. According to Theorem 9.2, Φ has at least two periodic orbits of type (r, s) whenever r and s are relatively prime and $r < s$. Recall that the function K of Proposition 9.5 is defined for $(q_0, q_1, \dots, q_{s-1})$ provided that $q_{j+1} \in [q_j, q_j + 1]$ for $j = 0, 1, \dots, s-1$ with $q_s = q_0 + r$. This however does not reflect the ordering of the orbit. For our purposes we define K on a smaller set Λ which consists of $(q_0, q_1, \dots, q_{s-1})$ such that there exists $z_0 \leq z_1 \leq \dots \leq z_{rs}$, with $z_{i+s} = z_i + 1$ for $i = 0, 1, \dots, (r-1)s$, and $q_j = z_{jr}$ for $j = 0, 1, \dots, s$. Note that once $(q_0, q_1, \dots, q_{s-1}) \in \Lambda$ is known, then all z_j can be determined. Of course $\mathbf{q} \in \Lambda$ imposes various inequalities between $q_0, q_1, \dots, q_{s-1}$.

On the other hand, we can regard K as a function of $(z_0, z_1, \dots, z_{s-1})$. Also there are only s many independent variables among them, say $z_0, z_1, \dots, z_{s-1}$. So, we now have a function $\hat{K}(z_0, z_1, \dots, z_{s-1}) = K(q_0, q_1, \dots, q_{s-1})$. The advantage of $\hat{K}$ to K is that it has a domain which is much easier to describe, namely

$$\hat{\Lambda} = \{(z_0, z_1, \dots, z_{s-1}) : z_0 \leq z_1 \leq \dots \leq z_{s-1} \leq z_0 + 1\}.$$

Since $K(q_0 + 1, \dots, q_{s-1} + 1) = K(q_0, \dots, q_{s-1})$, we learn that $\hat{K}(z_0 + 1, \dots, z_{s-1} + 1) = \hat{K}(z_0, \dots, z_{s-1})$. Introducing $w_j = z_j - z_{j-1}$, we have that $\mathbf{z} = (z_0, \dots, z_{s-1}) \in \hat{\Lambda}$ if and only if $(z_0, w_1, \dots, w_{s-1})$ belongs to the set of points with

$$0 \leq w_1, w_2, \dots, w_{s-1}, w_1 + w_2 + \dots + w_{s-1} \leq 1.$$

Writing $\bar{K}$ for $\hat{K}$ as a function of $z_0, w_1, \dots, w_{s-1}$, then $\bar{K}$ is defined on

$$\bar{\Lambda} = \left\{ (z_0, w_1, \dots, w_{s-1}) : z_0 \in \mathbb{R}, w_1, \dots, w_{s-1} \geq 0, \sum_1^{s-1} w_j \leq 1 \right\}.$$

Since $\bar{K}(z_0 + 1, w_1, \dots, w_{s-1}) = \bar{K}(z_0, w_1, \dots, w_{s-1})$, $\bar{K}$ is a lift of a function $\bar{k}$ which is defined on the set

$$\lambda = S^1 \times \left\{ (w_1, \dots, w_{s-1}) : w_1, \dots, w_{s-1} \geq 0, \sum_1^{s-1} w_j \leq 1 \right\} \subseteq S^1 \times [0, 1]^{s-1}.$$

Of course $\bar{k}$ has a maximizer and a minimizer. We now argue that a minimizer yields a critical point which is in the interior of λ . To see this, let us assume that to the contrary the minimizer is a point on the boundary. To explain this in its simplest non-trivial case, let us take a boundary point of the form

$$z_{n-1} < z_n = z_{n+1} < z_{n+2}.$$

Recall that $\Phi(q, p(q, Q)) = (Q, P(q, Q))$ where $p(q, Q) = -S_q(q, Q)$ is increasing in Q and $P(q, Q) = S_Q(q, Q)$ is decreasing in q . We now examine several cases:

(i) $y(z_n, z_{n+r}) < Y(z_{n-r}, z_n)$

In this case $\frac{\partial \hat{K}}{\partial z_n} = S_Y(z_{n-r}, z_n) + S_y(z_n, z_{n-r}) > 0$. Hence by decreasing z_n a little bit, we decrease $\hat{K}$. This contradicts the fact that $\mathbf{z}$ is a minimizer.

(ii) $p(z_{n+1}, z_{n+r+1}) > P(z_{n-r+1}, z_{n+1})$

In this case $\frac{\partial \hat{K}}{\partial z_{n+1}} = S_Y(z_{n-r+1}, z_{n+1}) + S_y(z_n, z_{n-r}) < 0$. Hence by increasing z_{n+1} , the value $\hat{K}$ decreases, contradicting the fact that $\hat{\mathbf{z}} = (z_0, \dots, z_{s-1})$ is a minimizer.

(iii) If (i) and (ii) do not occur, then

$$\begin{aligned} Y(z_{n-r}, z_n) &\leq y(z_n, z_{n+r}) = y(z_{n+1}, z_{n+r}) \\ &\leq y(z_{n+1}, z_{n+r+1}) \leq Y(z_{n-r+1}, z_{n-1}) \\ &= Y(z_{n-r+1}, z_n) \leq Y(z_{n-r}, z_n). \end{aligned}$$

Hence $z_{n-r+1} = z_{n-r}$, $z_{n+r+1} = z_{n+r}$ and

$$\begin{aligned} Y(z_{n-r}, z_n) &= y(z_n, z_{n+r}), \\ Y(z_{n-r+1}, z_{n+1}) &= y(z_{n-1}, z_{n+r+1}) \end{aligned}$$

which means that (z_{n-r}, z_n, z_{n+1}) is the x -coordinate of an orbit. This is what we wanted.

In fact the other critical point is a saddle point in λ . To see this, let us examine the problem when $s = 2$ and $r = 1$ which corresponds to a periodic orbit of period 2. In this case, we simply have

$$K(q_1, q_2) = S(q_1, q_2) + S(q_2, q_1 + 1) = 2S(q_1, q_2)$$

which is defined on the set

$$\Lambda = \{(q_1, q_2) : q_1 \leq q_2 \leq q_1 + 1\}.$$

Note that $K(q_1, q_2) = 0$ if $(q_1, q_2) \in \partial\Lambda$ and we always have $K(q_1, q_2) < 0$ if $(q_1, q_2) \in \Lambda^0$. Writing K in terms of q_1 and $w_2 = q_2 - q_1$ yields $\hat{K}(q_1, w_2) = 2S(q_1, q_1 + w_2)$ which is defined for $(q_1, w_2) \in \mathbb{R} \times [0, 1]$. Since $\hat{K}$ is periodic in q_1 , $\hat{K}$ is the lift of $\hat{k} : S^1 \times [0, 1] \rightarrow \mathbb{R}$ and evidently its minimum is attained in the interior of $S^1 \times [0, 1]$. Note that $-\min \hat{k}$ is simply the diameter of the convex set C . We now assert that $\hat{k}$ has a saddle critical point which corresponds to the width of C . To see this, for any $q_1 \in S^1$, we can find $\eta(q_1) = q_2 \in S^1$ such that the tangents at q_1 and $q_2 = \eta(q_1)$ are parallel. Now $-\max_{q_1} S(q_1, \eta(q_1)) = -S(q_1^*, \eta(q_1^*))$ yields the *width* of C . We assert that $(q_1^*, q_2^*) = (q_1^*, \eta(q_1^*))$ is the other critical point of K . Indeed, since $\Theta = \pi - \theta$, we have

$$S_q(q_1, \eta(q_1)) = S_Q(q_1, \eta(q_1))$$

for all q_1 . On the other hand, at a maximizer q_1^* of $S(q_1, \eta(q_1))$, we must have

$$\begin{aligned} 0 &= S_q(q_1^*, \eta(q_1^*)) + S_Q(q_1^*, \eta(q_1^*))\eta'(q_1^*) \\ &= S_q(q_1^*, q_2^*)(1 + \eta'(q_1^*)). \end{aligned}$$

It is not hard to show that in fact $\eta'(q_1^*) > 0$. Hence we must have $S_q = S_Q = 0$ at (q_1^*, q_2^*) . $\square$

We now turn to Theorem 9.2. So far we have a proof in the case of a monotone twist map. Following an idea of Chaperon, we try to express a twist map as a composition of monotone twist maps. To this end, let us define some function spaces.

- (i) $\mathcal{T}$ denotes a space of homeomorphism ψ from cylinder $C = S^1 \times [0, 1]$ onto itself which is an orientation and area preserving diffeomorphism in the interior of C and preserves the boundary circles $S^1 \times \{0\}$ and $S^1 \times \{1\}$. The rotation numbers of ψ restricted to $S^1 \times \{0\}$ and $S^1 \times \{1\}$ are denoted by $\rho_-(\psi)$ and $\rho_+(\psi)$ respectively.
- (ii) $\mathcal{T}^*$ denotes the space of $\psi \in \mathcal{T}$ such that $\rho^-(\psi) \neq \rho^+(\psi)$. $\mathcal{T}^+$ denotes the space of $\psi \in \mathcal{T}$ with $\rho^-(\psi) < \rho^+(\psi)$. $\mathcal{T}^-$ denotes the space of $\psi \in \mathcal{T}$ with $\rho^+(\psi) < \rho^-(\psi)$. $\mathcal{M}^+$ denotes the space of monotone twist maps.

We equip the space of $\mathcal{T}$ with the topology of C^1 -convergence in the interior of C and uniform-convergence up to the boundary. Evidently $\mathcal{T}$ is a topological group with multiplication given by composition. As an example, note that the shear map ξ with lift $(q, p) \mapsto (q + p, p)$ belongs to $\mathcal{T}^+$ whereas $\xi^{-1} = \lambda$ belongs to $\mathcal{T}^-$. We have the following straightforward lemma.

Lemma 9.1 *Every element ψ in the connected component of identity in $\mathcal{T}$ can be written as*

$$(9.13) \quad \psi = \lambda \circ \psi_1 \circ \lambda \circ \psi_2 \circ \cdots \circ \lambda \circ \psi_n$$

with $\psi_1, \psi_2, \dots, \psi_n \in \mathcal{M}^+$.

Proof Evidently there exists an open set U in $\mathcal{T}$ such that $\xi \in U \subseteq \mathcal{M}^+$. As a result, $\text{id} \in \xi^{-1}U = \lambda U =: V$ is an open neighborhood of identity and each $\psi \in V$ can be written as $\psi = \lambda \circ \psi_1$ with $\psi_1 \in \mathcal{M}^+$. We now write Ω for the set of ψ in $\mathcal{T}$ for which the decomposition (9.11) exists with $\psi_1, \psi_2, \dots, \psi_n \in V$. Clearly Ω is open because V is open. If we can show that V is also closed, then we deduce that Ω is the connected component of id in $\mathcal{T}$. To see the closedness of Ω , let $\{\varphi_m\}$ be a convergent sequence in Ω . If $\lim_{m \rightarrow \infty} \varphi_m = \varphi$, then $\lim_{m \rightarrow \infty} \varphi \circ \varphi_m^{-1} = \text{id}$ and, as a result, $\varphi \circ \varphi_m^{-1} \in V$, for large m . Hence there exists $\bar{\psi} \in \mathcal{M}^+$, such that $\varphi \circ \varphi_m^{-1} = \lambda \circ \bar{\psi}$ for a sufficiently large m . That is, $\varphi = \lambda \circ \bar{\psi} \circ \varphi_m$. Since $\varphi_m \in \Omega$, we deduce that $\varphi \in \Omega$, completing the proof of closedness of Ω . $\square$

Proof of Theorem 9.1 Let Ψ be a twist map. On account of Lemma 9.1, there exist monotone twist maps $\Psi_1, \Psi_2, \dots, \Psi_n$ such that

$$\Psi = \Lambda \circ \Psi_1 \circ \Lambda \circ \Psi_2 \circ \cdots \circ \Lambda \circ \Psi_n.$$

Each Ψ_j has a generating function $S_j(q, Q)$ with $S_j : \Gamma_j \rightarrow \mathbb{R}$ with

$$\Gamma_j = \{(q, Q) : A_-^j(q) \leq Q \leq A_+^j(q)\}.$$

Note that the generating function for Λ is $T(q, Q) = -\frac{1}{2}(Q - q)^2$ which is defined on the set

$$\{(q, Q) : q - 1 \leq Q \leq q\}.$$

We now define

$$L(q_0, q_1, \dots, q_{2n-1}) = - \sum_{j=0}^{n-1} \frac{1}{2} (q_{ij} - q_{ij+1})^2 + \sum_{j=0}^{n-2} S_j(q_{2j+1}, q_{2j+2}) + S_n(q_{2n-1}, q_0)$$

on the set Γ which consists of points $q_0, q_1, \dots, q_{2n-1}$ such that for $j = 0, 1, \dots, n-2$

$$-1 \leq q_{2j} - q_{2j+1} \leq 0, \quad A_-^j(q_{2j+1}) \leq q_{2j+2} \leq A_+^j(q_{2j+1})$$

with $q_{2j} = q_0$. Now as in Proposition 8.8, we can show that if $q_0, q_1, \dots, q_{2n-1}$ is a critical point of L , then

$$\Lambda(q_{2j}, p_{2j}) = (q_{2j+1}, p_{2j+1}), \quad \Psi_j(q_{2j+1}, p_{2j+1}) = (q_{2j+2}, p_{2j+2})$$

for $j = 0, 1, \dots, n-1$, where $p_{2j+1} = T_Q(q_{2j}, q_{2j+1}) = q_{2j} - q_{2j+1}$, $p_{2j+2} = S_Q(q_{2j+1}, q_{2j+2})$, $q_{2j} = q_0$ and $p_{2j} = p_0$. Of course, in particular (q_0, p_0) is a fixed point of Ψ . $\square$

Remark 9.1 Following an idea of Alan Weinstein, the functional $\mathcal{A}_H$ may be regarded as a generating function for $\varphi = \phi_1^H$. To explain this, let us write $\hat{\Gamma}$ for the space of C^1 functions $x : [0, 1] \rightarrow \mathbb{R}^{2n}$, and regard $\mathcal{A}_H : \hat{\Gamma} \rightarrow \mathbb{R}$. Write $\bar{\Gamma}$ for the space C^1 functions $\bar{x} = (\bar{q}, p) \in \hat{G}$ such that $\bar{q}(0) = \bar{q}(1) = 0$. Consider the map

$$A : \mathbb{R}^n \times \mathbb{R}^n \times \bar{G} \rightarrow \hat{G}, \quad A(a, b, \bar{q}(\cdot), p(\cdot))(t) = ((1-t)a + tb + \bar{q}(t), p(t)),$$

and define $\mathcal{A} = \mathcal{A}_H \circ A$, so that

$$\mathcal{A}(a, b, \bar{q}(\cdot), p(\cdot)) = \int_0^1 \left[p(t) \cdot \frac{d\bar{q}}{dt}(t) + p(t) \cdot (b - a) - H((1-t)a + tb + \bar{q}(t), p(t), t) \right] dt.$$

We can readily show

$$\frac{\partial \mathcal{A}}{\partial \bar{x}}(a, b, \bar{q}(\cdot), p(\cdot))(t) = -\bar{J}\dot{x}(t) - \nabla H(x(t), t).$$

Hence if $\partial \mathcal{A} / \partial \bar{x} = 0$, at $(a, b, \bar{q}(\cdot), p(\cdot))$, then $\dot{x} = \bar{J}\nabla H(x, t)$, for $x(t) = ((1-t)a + tb + \bar{q}(t), p(t))$. For such a critical path x , we also have

$$\begin{aligned} \frac{\partial \mathcal{A}}{\partial a}(a, b, \bar{q}(\cdot), p(\cdot)) &= - \int_0^1 [p(t) + (1-t)H_q(x(t), t)] dt \\ &= \int_0^1 [-p(t) + (1-t)\dot{p}(t)] dt = -p(0), \\ \frac{\partial \mathcal{A}}{\partial b}(a, b, \bar{q}(\cdot), p(\cdot)) &= \int_0^1 [p(t) - tH_q(x(t), t)] dt \\ &= \int_0^1 [p(t) + t\dot{p}(t)] dt = p(1). \end{aligned}$$

In summary, $\mathcal{A}$ is a generating function of Type I for φ because

$$(9.14) \quad \frac{\partial \mathcal{A}}{\partial \bar{x}}(a, b, \bar{x}(\cdot)) = 0 \quad \Rightarrow \quad \varphi \left(a, \frac{\partial \mathcal{A}}{\partial a}(a, b, \bar{x}(\cdot)) \right) = \left(b, \frac{\partial \mathcal{A}}{\partial b}(a, b, \bar{x}(\cdot)) \right).$$

□

So far we have used generating functions to study various properties of twist maps on cylinders. We now discuss possible generalizations of such global properties for other manifolds. As a start, let us take a symplectic $\varphi : \mathbb{T}^{2n} \rightarrow \mathbb{T}^{2n}$ and wonder how many fixed points it can have. Evidently an irrational translation on $\mathbb{T}^2$ is a symplectic diffeomorphism with no fixed point. This is not in contradiction with Arnold's conjecture because such map does not come from a Hamiltonian vector field on the torus. Arnold's conjecture in the case of a torus was established by Conley and Zehnder. The inequality (9.3) takes the following form.

Theorem 9.3 (Conley-Zehnder) *Let $\varphi : \mathbb{T}^{2n} \rightarrow \mathbb{T}^{2n}$ be the time one map of a Hamiltonian function $H : \mathbb{T}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}$, and assume that H is 1-periodic in time. Then φ has at least $2n + 1$ many fixed points.*

As in the case of cylinder, Theorem 9.3 is an immediate consequence of LS Theorem, if φ possess a global generating function. Though in the case of a torus, we need a generating function of Type II, that was defined in Chapter 8. In Proposition 9.1, we showed that a generating function $S(q, Q)$ of Type I always exists locally provided that $\frac{\partial Q}{\partial p}(q^0, p^0)$ is non-singular. Note that this condition fails for the identity. However generating function of Type II exists trivially for identity. The following proposition can be proved as Proposition 9.1.

Proposition 9.6 *Let $\Phi(q, p) = (Q, P)$ be a symplectic diffeomorphism with*

$$\det Q_q(q^0, p^0) \neq 0.$$

Then there exist a neighborhood U of $Q^0 = Q(q^0, p^0)$ and p^0 , and a C^1 -function $W : U \rightarrow \mathbb{R}$ such that

$$(9.15) \quad W_Q = P, \quad W_p = q.$$

We note that identity transformation has such a generating function with $W(Q, p) = Q \cdot p$. More generally, we may write $\hat{S}(Q, p) = Q \cdot p - V(Q, p)$ with V now satisfying

$$(9.16) \quad P - p = -V_Q, \quad Q - q = V_p,$$

which can be thought of as a discrete analog of a Hamiltonian system where $V(Q, p)$ plays the role of the Hamiltonian function. From LS Theorem and Proposition 9.6 we can readily deduce

Proposition 9.7 *Let $\varphi : \mathbb{T}^{2n} \rightarrow \mathbb{T}^{2n}$ be a symplectic diffeomorphism which is a small perturbation of identity. Then φ has at least $2n + 1$ many fixed points.*

As we have learned from Remark 9.1 any symplectic map coming from a Hamiltonian system has a kind of generalized generating function in the sense of (9.14). The function L that appeared in Proposition 9.4, should be regarded as a finite dimensional approximation of $\mathcal{A}$ of (9.14). Motivated by Proposition 9.5, we formulate a definition for a generalization of the generating functions.

Definition 9.1 Let $\varphi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a symplectic diffeomorphism. We say that a map $u : \mathbb{R}^{2n} \times \mathbb{R}^N \rightarrow \mathbb{R}$ is a *generalized generating function* (in short GGF) for φ if

$$(9.17) \quad u_\xi(Q, p; \xi) = 0 \quad \Rightarrow \quad \varphi(Q + u_p(Q, p; \xi), p) = (Q, p + u_Q(Q, p; \xi)),$$

and there exists a function $B(Q, p; \xi)$ that is a non-degenerate quadratic function in ξ , and satisfy

$$(9.18) \quad \sup_{Q, p, \xi} [|u(Q, p; \xi) - B(Q, p; \xi)| + |\nabla u(Q, p; \xi) - \nabla B(Q, p; \xi)|] < \infty.$$

□

Proposition 9.8 *Let $\varphi = \phi_1^H$ where $H : \mathbb{R}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}$ be a Hamiltonian function that is 1-periodic in all its arguments. Then φ possesses a GGF.*

Proof Set $t_j = j/k$, $\varphi_j = \phi_{t_{j-1}, t_j}^H$, so that we can write $\varphi = \varphi_1 \circ \cdots \circ \varphi_k$. For k sufficiently large, each φ_j has a Type II periodic generating function u^j by Proposition 9.6. This means

$$\varphi_j(Q + u_p^j(Q, p), p) = (Q, p + u_Q^j(Q, p)).$$

We now set $\xi = (Q_1, p_1, Q_2, p_2, \dots, Q_{k-1}, p_{k-1})$, and define u by $u(Q, p; \xi) = \hat{u}(Q, p; \xi) + B(Q, p; \xi)$, where

$$(9.19) \quad \hat{u}(Q, p; \xi) = \sum_{j=1}^k u^j(Q_j, p_{j-1}), \quad B(Q, p; \xi) = \sum_{j=1}^k (Q_j - Q_{j-1}) \cdot p_{j-1},$$

where $p_0 = p$, $Q_k = Q$, and $Q_0 = 0$. Since u^j s are 1-periodic, the function $\bar{u}$ is bounded in C^1 and (9.18) holds. On the other hand, if we write

$$q_{j-1} = Q_j + u_p^j(Q_j, p_{j-1}), \quad P_j = p_{j-1} + u_Q^j(Q_j, p_{j-1}),$$

then $\varphi_j(q_{j-1}, p_{j-1}) = (Q_j, P_j)$, and

$$\begin{aligned} u_{Q_j}(Q, p; \xi) &= p_{j-1} - p_j + u_Q^j(Q_j, p_{j-1}) = P_j - p_j, \\ u_{p_{j-1}}(Q, p; \xi) &= Q_j - Q_{j-1} + u_p^j(Q_j, p_{j-1}) = q_{j-1} - Q_{j-1}. \end{aligned}$$

Hence

$$u_\xi(Q, p; \xi) = 0 \quad \Rightarrow \quad \varphi_j(q_{j-1}, p_{j-1}) = (q_j, p_j),$$

for $j = 1, \dots, k$. This in turn implies (9.17) because $\varphi_j = \phi_{t_{j-1}, t_j}^H$. $\square$

Exercise 9.1(i) Consider the Billiard map in a circle. Determine the generating function. Find periodic orbits and describe the remaining orbits.

(ii) Prove Proposition 9.2.

(iii) Consider the Billiard map in an ellipse. Show that the 2-periodic orbits correspond to the reflection along the axes of symmetry. Show that the 2-periodic orbit associated with the shorter axis of symmetry is a saddle point of the generating function. **Hint:** (q_1^*, q_2^*) maximizers of $S(q_1, \eta(q_1))$ but is a local minimum for $S(q_1^*, q_2)$. $\square$

10 Pseudo-Holomorphic Curves and Gromov Width

In Chapters 5-8, we use the invariance (3.2) of the action $A(\gamma)$ to construct various symplectic capacities. For our constructions, we used the periodic orbits of the Hamiltonian systems played in an essential way. In this chapter, we focus on the invariance of $\int_{\Gamma} \omega$ to analyze symplectic maps. Our main tool will be the *pseudo-holomorphic curves* of Gromov. We give two motivations for the relevance of such complex curves (or rather real surfaces):

1. Observe that if $\Gamma = w(D)$ is a 2-dimensional surface with the parametrization $w : D \rightarrow \mathbb{R}^{2n}$ for a planar domain D , then

$$\int_{\Gamma} \bar{\omega} = \int_D \bar{\omega}(w_s, w_t) ds dt = \int_D \bar{J} w_s \cdot w_t ds dt.$$

This integral is simply $-area(\Gamma)$ when $n = 1$. For $n > 1$, the integral $\int_{\Gamma} \omega$ is not of a definite sign and does not represent any kind of *size* of the surface Γ . However, if w satisfies

$$(10.1) \quad w_s = \bar{J} w_t,$$

then $w_s \cdot w_t = 0$, and

$$(10.2) \quad \int_{\Gamma} (-\bar{\omega}) = \int_D |w_s|^2 ds dt = \int_D (|w_s|^2 |w_t|^2 - (w_s \cdot w_t)^2)^{1/2} ds dt = area(\Gamma).$$

This means that for such surfaces, $-\int_{\Gamma} \bar{\omega}$ is indeed the area of Γ . If we write $w = (u, v)$ with u and v representing the position and momentum, then (10.1) reads as

$$u_s = v_t, \quad v_s = -u_t,$$

which are nothing other than Cauchy-Riemann equations. When (10.1) is satisfied, we say that w is a *holomorphic curve*. Holomorphic curves are not preserved under a symplectic change of coordinates. However, if $\varphi \circ \hat{w} = w$, then

$$(10.3) \quad \hat{w}_s = J(\hat{w}) \hat{w}_t,$$

where

$$J = (d\varphi)^{-1} \circ \bar{J} \circ d\varphi,$$

is an example of an *almost complex structure* and $\hat{w}$ is called a *J-holomorphic curve*. Observe that if we set $\bar{g}(a, b) = a \cdot b$, and define a metric $g = \varphi^* \bar{g}$, then

$$(10.4) \quad \bar{\omega}(a, b) = \bar{\omega}((d\varphi)_x a, (d\varphi)_x b) = \bar{g}(\bar{J}(d\varphi)_x a, (d\varphi)_x b) = (\varphi^* \bar{g})_x(J(x)a, b).$$

We can now repeat our calculation in (10.2) to assert that if $\hat{\Gamma} = \hat{w}(D)$, then

$$\int_{\Gamma} (-\bar{\omega}) = area_g(\hat{\Gamma}),$$

where $area_g$ denotes the area with respect to the metric g .

2. Recall that if $H(x, t)$ is a time dependent Hamiltonian function, then the critical points of the functional

$$\mathcal{A}(x) = \int_0^1 [p \cdot \dot{q} - H(x, t)] dt,$$

corresponds to the 1-periodic orbits of $\dot{x} = \bar{J}\nabla H(x, t)$. For example if $H : \mathbb{T}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}$ is 1-periodic in t , and if we regard $\mathcal{A} : C^1(S^1, \mathbb{R}^{2n}) \rightarrow \mathbb{R}$, then the gradient of $\mathcal{A}$ with respect to the L^2 -inner product is given by

$$\partial\mathcal{A}(x(\cdot)) = -\bar{J}\dot{x}(\cdot) - \nabla H(x(\cdot), t).$$

Observe that the gradient with respect to $\mathcal{H}^{1/2}$ -inner product, namely $\nabla\mathcal{A}$ is related to $\partial\mathcal{A}$ by the formula $\nabla\mathcal{A} = \mathcal{I}\partial\mathcal{A}$. In fact the critical points of $\mathcal{A}$ corresponds to the fixed points of the map $\psi = \phi_1^H$ and following an idea of Floer, we may study such critical points by developing a Morse-type theory for $\mathcal{A}$. Morse Theory may be developed by studying the gradient flow

$$(10.5) \quad \frac{dw}{ds} = -\partial\mathcal{A}(w).$$

Regarding $w : \mathbb{R} \times S^1 \rightarrow \mathbb{R}^{2n}$ as a function of two variables s and t , (10.5) reads as

$$(10.6) \quad w_s = \bar{J}w_t + \nabla H(w, t).$$

This is very different from the corresponding

$$(10.7) \quad \frac{dw}{ds} = -\nabla\mathcal{A}(w) = -\mathcal{I}\partial\mathcal{A}(w);$$

the right-hand side of (10.7) is an integro-differential equation and is well-defined as an ODE, whereas the equation (10.6) is an elliptic PDE and not well-posed as an initial-value problem. Evidently, (10.6) is the same as (10.1) when $H = 0$. The elliptic PDE (10.1) is well-posed for a prescribed $w(\partial D)$. $\square$

Motivated by (10.4), let us give a general definition for almost complex structures. Let M be a C^1 manifold. By an *almost complex structure* on M , we mean a continuous $x \mapsto J_x$ with $J_x : T_x M \rightarrow T_x M$ linear function satisfying $J_x^2 = -id$. The pair (M, J) is called an *almost complex manifold*. If (M, ω) is symplectic, then we say (J, g) and ω are compatible if

$$(10.8) \quad g_x(a, b) = \omega_x(a, J_x b), \quad a, b \in T_x M$$

is a Riemannian metric on M . We write $\mathcal{I}(M, \omega)$ for the compatible pairs (g, J) . We also set

$$\mathcal{G}(M, \omega) = \{g : (g, J) \in \mathcal{I}(M, \omega)\}, \quad \mathcal{J}(M, \omega) = \{J : (g, J) \in \mathcal{I}(M, \omega)\}.$$

Proposition 10.1 (i) Let (M, ω) be symplectic with a Riemannian metric $\hat{g}$. Then $\mathcal{I}(M, \omega)$ is nonempty.

(ii) If $g \in \mathcal{G}(M, \omega)$ and $\dim M = 2n$, then $\mu := (n!)^{-1}\omega^n$ is the Riemannian volume form associated with the Riemannian metric g .

Proof The main observation is that if A is a skew-symmetric matrix, and if $A = SJ$ is its polar decomposition with S symmetric, then orthogonal J inherits skew-symmetry from A and satisfies $J^* = -J$, $J^2 = I$. To apply the polar decomposition, we may choose a reference inner-product/metric.

Fix x . Both $a \mapsto \omega_x(a, \cdot)$ and $a \mapsto \hat{g}_x(a, \cdot)$ are linear isomorphisms between $T_x M$ and $(T_x M)^*$. Hence there exists a linear invertible $A_x : T_x M \rightarrow T_x M$ such that $\omega_x(a, b) = \hat{g}_x(A_x a, b)$. Note that A_x is skew symmetric because

$$\hat{g}(A^* a, b) = \hat{g}(a, Ab) = \hat{g}(Ab, a) = \omega(b, a) = -\omega(a, b) = -\hat{g}(Aa, b).$$

In fact a candidate for J is simply the orthogonal matrix that appears in the polar decomposition of A . More precisely, if we use the unique representation $A_x = S_x J_x$, with $S_x = (A_x A_x^*)^{1/2} = (-A_x^2)^{1/2}$, and J_x orthogonal with respect to $\hat{g}_x$, then $J \in \mathcal{J}(M, \omega)$. To verify this, we build this polar decomposition directly. The idea is that since A is real and skew-symmetric, its eigenvalues appear as $\pm i\lambda$ with λ real. Hence the eigenvalues of $-A^2$ are positive and of even multiplicities. In fact, the matrix $B = A^2$ is negative definite, and if v is an eigenvector of B associated with the eigenvalue $-\lambda^2$ for some positive λ , then $w = \lambda^{-1}Av$ is another eigenvector for B that is orthogonal to v :

$$\hat{g}(v, w) = \lambda^{-1}\hat{g}(v, Av) = \lambda^{-1}\omega(v, v) = 0, \quad Bw = \lambda^{-1}ABv = -\lambda Av = -\lambda^2 w.$$

We now describe S and J on $\{v, w\}$, and hence on the span of $\{v, w\}$:

$$\begin{aligned} Av &= \lambda w, & Sv &= \lambda v, & Jv &= w, \\ Aw &= -\lambda v, & Sw &= \lambda w, & Jw &= -v. \end{aligned}$$

More generally, we can find an orthonormal basis $\{v_1, \dots, v_{2n}\}$ and positive numbers $\lambda_1, \dots, \lambda_n$ such that $Bv_i = -\lambda_i^2 v_i$ for each i . We can readily show that $\{w_i = \lambda_i^{-1}Av_i : i = 1, \dots, 2n\}$ is also an orthonormal set and that $Aw_i = -\lambda_i v_i$, $Bw_i = \lambda_i^2 w_i$. But since A is skew symmetric, w_i and v_i are orthogonal eigenvectors of B associated with the same eigenvalue $-\lambda_i^2$. We relabel our eigenvalues and eigenvectors so that $Av_{2i} = \lambda_i v_{2i-1}$ and $Av_{2i-1} = -\lambda_i v_{2i}$ for $i = 1, \dots, n$. We then define S and J by $Sv_{2i} = \lambda_i v_{2i}$, $Sv_{2i-1} = \lambda_i v_{2i-1}$, $Jv_{2i} = v_{2i-1}$ and $Jv_{2i-1} = -v_{2i}$. It is straightforward to check that $A = SJ$, $J^2 = -id$ and $J^* = -J$. We also have

$$g(a, b) := \omega(a, Jb) = g(Aa, Jb) = g(Sa, b).$$

We are done because $S > 0$.

(ii) First we claim that there exists a local orthonormal frame for $(M, \hat{g})$ of the form $E_n = \{e_1, Je_1, \dots, e_n, Je_n\}$. This frame is constructed inductively; if we already have $2k$ many vector fields $E_k = \{e_1, Je_1, \dots, e_k, Je_k\}$ with $\hat{g}(e_i, e_j) = \delta_{ij}$, $\hat{g}(e_i, Je_j) = 0$ for $i, j = 1, \dots, k$, then locally we can find a new vector field e_{k+1} which is orthogonal to vectors in E_k and $\hat{g}(e_{k+1}, e_{k+1}) = 1$. We then use (3.14) and $J^2 = -id$ to deduce that Je_{k+1} is also orthogonal to E_k and e_{k+1} and that $\hat{g}(Je_{k+1}, Je_{k+1}) = 1$.

From $\omega(e_i, e_j) = \omega(Je_i, Je_j) = 0$ and $\omega(e_i, Je_j) = \delta_{ij}$, we can readily deduce

$$\omega^n(e_1, Je_1, \dots, e_n, Je_n) = n!,$$

and this implies that μ is the Riemannian volume. $\square$

In the prominent work [G], Gromov uses the J -Holomorphic curves to establish his non-squeezing result, namely Theorem 6.2(i). The main ingredient for his proof is an existence result:

Theorem 10.1 (Gromov) *For every $J \in \mathcal{J}(\mathbb{R}^{2n}, \bar{\omega})$ and every $x^0 \in Z^{2n}(1)$, there exists a J -holomorphic $\hat{w} : \mathbb{D} \rightarrow Z^{2n}(1)$ such that $x^0 \in \hat{w}(\mathbb{D})$, $w(\partial\mathbb{D}) \subset \partial Z^{2n}(1)$, and*

$$(10.9) \quad \int_{\hat{w}(\mathbb{D})} (-\bar{\omega}) = \pi.$$

Let us first see how Theorem 10.1 implies the non-squeezing property of symplectomorphisms.

Proof of Theorem 6.2(i) Let $\varphi : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a symplectomorphism such that $\varphi(B_R) \subset Z^{2n}(1)$, where $B_R = \{x : |x| \leq R\}$. Let J be as in (10.3). By Theorem 10.1, we can find $\hat{w} : \mathbb{D} \rightarrow Z^{2n}(1)$ such that $\varphi(0) \in \hat{w}(\mathbb{D})$, and (10.9) is true. Let $w = \varphi^{-1} \circ \hat{w}$, so that w solves (10.1), $0 \in w(\mathbb{D})$, and

$$Area(w(\mathbb{D})) = \int_{w(\mathbb{D})} (-\bar{\omega}) = \int_{\hat{w}(\mathbb{D})} (-\bar{\omega}) = \pi,$$

by (10.9). On the other-hand by a classical theorem of Lelong, we must have

$$\pi = Area(w(\mathbb{D})) \geq Area(B_R) = \pi R^2.$$

Thus, $R \leq 1$. $\square$

Before embarking on the proof of Theorem 10.1, let us develop some feel for the equation (10.3). To ease the notation, let us write w for $\hat{w}$. Also, to avoid the occurring minus sign, we take $J \in \mathcal{J}(\mathbb{R}^{2n}, -\bar{\omega})$, so that instead of (10.3), we now have

$$(10.10) \quad w_t = J(w)w_s.$$

Note that now J as in (10.3) satisfies

$$(10.11) \quad J = (\varphi')^{-1} i \varphi',$$

where $i = -\bar{J}$ is simply the multiplication by i , when we use the identification $\mathbb{R}^{2n} = \mathbb{C}^n$. We now give two interpretations for the equation (10.10).

1. In our first interpretation, we compare (10.10) with its equivalent formulation that we obtain by multiplying both sides of (10.1) by $J(w)$:

$$(10.12) \quad w_t = J(w)w_s \quad - w_s = J(w)w_t.$$

Regarding $dw : T\mathbb{D} \rightarrow T\mathbb{R}^{2n}$, $i : T\mathbb{D} \rightarrow T\mathbb{D}$, and $J : T\mathbb{R}^{2n} \rightarrow T\mathbb{R}^{2n}$, we may rewrite (10.12) as

$$(10.13) \quad dw \circ i = J \circ dw;$$

if we evaluate both sides of (10.13) at the vector fields $\frac{\partial}{\partial s}$ and $\frac{\partial}{\partial t}$, we obtain the equations of (10.12), because

$$i \left(\frac{\partial}{\partial s} \right) = \frac{\partial}{\partial t}, \quad i \left(\frac{\partial}{\partial t} \right) = -\frac{\partial}{\partial s}.$$

The reader may compare (10.13) with (10.11) that may be written as

$$(10.14) \quad i \circ d\varphi = d\varphi \circ J.$$

The advantage of the formulation (10.13) is that it has an obvious generalization to arbitrary manifolds.

Definition 10.1 Let M be a manifold and $J : TM \rightarrow TM$ be an *almost complex structure* i.e. $J^2 = -id$.

(i) We say that $w : \mathbb{D} \rightarrow M$ is a J -holomorphic curve if (10.3) is valid. We also define the operator

$$(10.15) \quad \bar{\partial}_J w = \frac{1}{2}(dw + J \circ dw \circ i),$$

so that the equation (10.13) may be written as $\bar{\partial}_J w = 0$.

(ii) We say that J is a *complex structure* if for every $x \in M$, we can find $U \subset \mathbb{C}^n$ and a diffeomorphism $\varphi : U \rightarrow M$ such that $x \in \varphi(U)$, and (10.14) is valid. $\square$

2. Let us use complex-variable notation to write $z = s + it$ and $\bar{z} = s - it$. We also use the notations

$$\begin{aligned} dz &= ds + idt, & d\bar{z} &= ds - idt, \\ \frac{\partial}{\partial z} &= \frac{1}{2} \left(\frac{\partial}{\partial s} - i \frac{\partial}{\partial t} \right), & \frac{\partial}{\partial \bar{z}} &= \frac{1}{2} \left(\frac{\partial}{\partial s} + i \frac{\partial}{\partial t} \right). \end{aligned}$$

By definition,

$$w_t = i(w_z - w_{\bar{z}}), \quad w_s = (w_z + w_{\bar{z}}).$$

Hence, if $i + J$ is invertible, the (10.10) may be written as

$$(10.16) \quad w_{\bar{z}} = K(w)w_z.$$

where

$$K(w) := (i + J(w))^{-1}(i - J(w)).$$

Proposition 10.2 *If $J \in \mathcal{J}(\mathbb{R}^{2n}, -\bar{\omega})$, then $i + J$ is invertible and $\|K\| < 1$. Moreover, K is symmetric, and $\mathbb{C}$ -antilinear (the latter means that $Ki = -iK$).*

Proof First observe that since $J \in \mathcal{J}(\mathbb{R}^{2n}, -\bar{\omega})$, we know

$$(10.17) \quad a \neq 0 \Rightarrow (ia) \cdot (Ja) = (-\bar{\omega})(a, Ja) > 0.$$

To show that $i + J$ is invertible, note that if to the contrary we can find $a \neq 0$, such that $(i + J)a = 0$, then

$$(ia) \cdot (Ja) = -|ia|^2 < 0,$$

which contradicts (10.17).

To show that $\|K\| < 1$, it suffices to check that $|b| < |a|$ whenever $K(a) = b$ and $a \neq 0$. We have

$$Ka = b \Leftrightarrow (i - J)(a) = (i + J)(b) \Leftrightarrow b - a = iJ(b + a).$$

Clearly, if $a + b = 0$, then $a = b = 0$. Now if $a, a + b \neq 0$, then by (10.17),

$$|b|^2 - |a|^2 = (b - a) \cdot (b + a) = (iJ)(b + a) \cdot (b + a) = -[J(b + a) \cdot i(b + a)] < 0,$$

as desired.

As for the $\mathbb{C}$ -antilinear property,

$$(10.18) \quad (i + J)^{-1}(i - J)i = -i(i + J)^{-1}(i - J),$$

Observe,

$$\begin{aligned} (i + J)(-i)(i + J)^{-1}(i - J) &= (I - Ji)(i + J)^{-1}(i - J) \\ &= -(J^2 + Ji)(i + J)^{-1}(i - J) \\ &= -J(i - J) = -Ji - I \\ &= -Ji + i^2 = (i - J)i, \end{aligned}$$

which implies (10.19).

We may write $K = (I - iJ)^{-1}(I + iJ)$. on the other hand, since $-i = GJ$, for a symmetric G , we learn that $G = iJ$ is symmetric. This implies that K is symmetric. $\square$

Remark 10.1 The equation (10.13), is equivalent to the classical *Beltrami Equation* when $n = 1$. This equation is related to the theory of *quasi-conformal maps* and a generalization of the *Riemann mapping theorem*. To explain this, first observe that (10.13) and (10.14) are equivalent in the case of $n = 1$ by setting $\varphi = w^{-1}$. As we showed in Example 2.2, the set $\mathcal{I}(\mathbb{R}^2, -\bar{\omega})$ consists of pairs (g, J) , with $g(a, b) = Ga \cdot b$, such that

$$(10.19) \quad G = \begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix}, \quad J = \begin{bmatrix} -\beta & -\gamma \\ \alpha & \beta \end{bmatrix},$$

with $\alpha, \gamma > 0$ and $\alpha\gamma - \beta^2 = 1$. Now let us write $\varphi(z) = u + iv$ with $z = s + it$, and evaluate both sides of (10.14) at the vector fields $\frac{\partial}{\partial s}$ and $\frac{\partial}{\partial t}$, with J as in (10.19). We obtain

$$i\varphi_s = -\beta\varphi_s + \alpha\varphi_t, \quad i\varphi_t = -\gamma\varphi_s + \beta\varphi_t.$$

This is equivalent to $\varphi_t = j\varphi_s$ for $j = (\beta + i)/\alpha = \gamma/(\beta - i)$. Equivalently,

$$(10.20) \quad \varphi_{\bar{z}} = \mu(z)\varphi_z, \quad \text{with } \mu = \frac{i - j}{i + j}.$$

This equation is known as the *Beltrami Equation*, and resembles (10.16) because

$$|\mu| = \left| \frac{\beta + (\alpha - 1)i}{\beta + (\alpha + 1)i} \right| < 1,$$

by our assumption $\alpha > 0$. We refer to Appendix D for more information about the Beltrami Equation.

As for the equation (10.16) in dimension 2, observe that when $n = 1$,

$$\begin{aligned} K &= \begin{bmatrix} -\beta & -\gamma - 1 \\ \alpha + 1 & \beta \end{bmatrix}^{-1} \begin{bmatrix} \beta & \gamma - 1 \\ 1 - \alpha & -\beta \end{bmatrix} \\ &= (\alpha + \gamma + 2)^{-1} \begin{bmatrix} \beta & \gamma + 1 \\ -\alpha - 1 & -\beta \end{bmatrix} \begin{bmatrix} \beta & \gamma - 1 \\ 1 - \alpha & -\beta \end{bmatrix} \\ &= (\alpha + \gamma + 2)^{-1} \begin{bmatrix} \gamma - \alpha & -2\beta \\ -2\beta & \alpha - \gamma \end{bmatrix}. \end{aligned}$$

From this, we deduce that in fact the equation (10.16) can be written as

$$(10.21) \quad w_{\bar{z}} = m(w)\bar{w}_z,$$

where $m = (\gamma - \alpha - 2\beta i)/(\alpha + \gamma + 2)$. We note

$$|m|^2 = \frac{(\alpha + \gamma)^2 - 4}{(\alpha + \gamma + 2)^2} = \frac{\alpha + \gamma - 2}{\alpha + \gamma + 2} = 1 - 4(\alpha + \gamma + 2)^{-1} < 1.$$

□

As we discussed in Remark 10.1, the equation (10.16) is a multi-dimensional generalization of the classical Beltrami-equation. Ignoring the boundary condition requirements of Theorem 10.2 we can use the classical transforms of *Cauchy* and *Bewrling* to construct solutions of equation (10.16). We define

$$(10.22) \quad \mathcal{C}(h)(z) = -\frac{1}{\pi} \int_{\mathbb{D}} \frac{h(\zeta)}{\zeta - z} dsdt, \quad \mathcal{B}(h)(z) = \frac{1}{\pi} PV \int_{\mathbb{D}} \frac{h(\zeta)}{(\zeta - z)^2} dsdt,$$

where PV stands for the principle value (see Appendix D). The main property of the Cauchy Transform $\mathcal{C}$ is that $\mathcal{C}(h)_{\bar{z}} = h$ and that its z derivative is the Buerling transform; $\mathcal{C}(h)_z = \mathcal{B}(h)$. Moreover, by *Caldron-Zygmund Theory*, the operator $\mathcal{B}$ is bounded on L^p for every $p \in (1, \infty)$. In fact $\mathcal{B}$ is an isometry on L^2 and its norm $C(p)$ on the L^p converges to 1 as $p \rightarrow 2$. We refer to Theorem D.1 of Appendix D for more details.

Theorem 10.2 *Assume that $\|K\|_{L^\infty} = c_0 < 1$. Let $q_1, \dots, q_n : \mathbb{D} \rightarrow \mathbb{C}$ be n holomorphic functions. Then there exists $p = p(c_0) > 2$ and a function $w = (U_1, \dots, U_n) \in W^{1,p}$, $U_i : \mathbb{D} \rightarrow \mathbb{C}$, of the form*

$$U_i = \mathcal{C}(u_i) + q_i, \quad u_i \in L^p(\mathbb{D}),$$

that solves the equation (10.16) weakly.

Proof We use column vectors to write

$$w = \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, \quad q = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}, \quad \hat{q}(w) = K(w) \begin{bmatrix} q'_1 \\ \vdots \\ q'_n \end{bmatrix}, \quad \hat{\mathcal{B}}(u) = \begin{bmatrix} \mathcal{B}(u_1) \\ \vdots \\ \mathcal{B}(u_n) \end{bmatrix}.$$

Using the definition of $\mathcal{C}$ and $\mathcal{B}$, we can write (10.17) as

$$(10.23) \quad w_{\bar{z}} = u = K(w) \hat{\mathcal{B}}(u) + \hat{q}(w).$$

We wish to invert the operator $I - K(w) \hat{\mathcal{B}}$ for a given vector w . Since $\mathcal{B}$ is an isometry on L^2 , the operator of $\hat{\mathcal{B}}$ is also an isometry. Using Theorem D.1 of the Appendix, we can show that for every $p \in (1, \infty)$, there exists a constant $c(p)$ such that $\lim c(p) = 1$ as $p \rightarrow 2$, and

$$\|\hat{\mathcal{B}}(u)\|_{L^p} \leq c(p) \|u\|_{L^p}.$$

From this we learn

$$\|A(w)\hat{\mathcal{B}}(u)\|_{L^p} \leq c(p)c_0\|u\|_{L^p}.$$

Choose $p > 2$ such that $c(p)c_0 < 1$. For such p ,

$$\left\| (I - A(w)\hat{\mathcal{B}})^{-1}h \right\|_{L^p} \leq (1 - c(p)c_0)^{-1}\|h\|_{L^p}.$$

This in turn implies that if

$$\mathcal{D}(w) = (I - A(w)\hat{\mathcal{B}})^{-1}(\hat{q}(w)),$$

then

$$\|\mathcal{D}(w)\|_{L^p} \leq c_0(1 - c(p)c_0)^{-1}\|\hat{q}\|_{L^p} := c_1(p).$$

By applying Theorem D.1 of Appendix

$$\|\mathcal{C} \circ \mathcal{D}(w)\|_{W^{1,p}} \leq c_2,$$

for some constant $c_2 = c_2(p)$. Then rewrite (10.16) as

$$(10.24) \quad w = (\mathcal{C} \circ \mathcal{D})(w) + q := \mathcal{E}(w).$$

Hence w is a fixed point of the operator $\mathcal{E}$. Set

$$\Gamma_L = \{w : \|w\|_{L^\infty} \leq L\}.$$

We wish to show that $\mathcal{E} : \Gamma_L \rightarrow \Gamma_L$ and it has a fixed point. As a preparation, we first bound the nonlinear $\mathcal{D}$. Since $p > 2$, we may apply Morrey Inequality to deduce

$$(10.25) \quad \|\mathcal{C} \circ \mathcal{D}(w)\|_{C^{1-2/p}} \leq c_2(p)\|\mathcal{C} \circ \mathcal{D}(w)\|_{W^{1,p}} \leq C(p)c_1(p) = c_3(p),$$

where C^α denotes the space of α -Hölder continuous functions. In particular

$$\|\mathcal{C} \circ \mathcal{D}(w)\|_{L^\infty} \leq c_3(p).$$

Setting $L = c_3(p) + \|q\|_{L^\infty}$, we deduce that $\mathcal{E}(L^p) \subseteq \Gamma_L$. In particular, $\mathcal{E}$ maps Γ_L into itself. On the other hand, the bound (D.12) implies that the image of Γ_L under $\mathcal{E}$ is in fact relatively compact. This allows us to use the *Schauder Fixed Point Theorem* to deduce that $\mathcal{E}$ has a fixed point. $\square$

In this chapter, we will describe two approaches for establishing Theorem 10.1. In our first approach, we will follow a work of Sukhov and Tumanov [ST] that treats (10.16) as a generalization of the Beltrami Equation and use Cauchy and Beurling transforms to construct solutions. (We refer to Appendix D for a thorough discussion of these transforms and their use in solving Beltrami-type equations.) The main ingredient of the proof of Theorem 10.16

a la [SK] is a variant of the Cauchy operator that is designed to solve the d-bar problem with a boundary condition. More precisely, given a holomorphic function $Q : \mathbb{D} \rightarrow \mathbb{C}$ that is nonzero inside $\mathbb{D}$, we define

$$(10.26) \quad \begin{aligned} \mathcal{C}^Q(f)(z) &= Q(z) \left(\mathcal{C}(f/Q)(z) + z^{-1} \overline{\mathcal{C}(f/Q)(1/\bar{z})} \right) \\ &= \frac{Q(z)}{2\pi i} \iint_{\mathbb{D}} \left[\frac{f(\zeta)}{Q(\zeta)(\zeta - z)} + \frac{\overline{f(\zeta)}}{\overline{Q(\zeta)}(z\bar{\zeta} - 1)} \right] d\zeta \wedge d\bar{\zeta}. \end{aligned}$$

We note

$$(10.27) \quad \mathcal{C}^Q(f) = Q\mathcal{C}(f/Q) + h^Q,$$

for a function h^Q that is holomorphic inside $\mathbb{D}$. The type of Q we have in mind is

$$(10.28) \quad Q(z) = a_0 \prod_{j=1}^l (z - a_j)^{\alpha_j},$$

with $a_1, \dots, a_l \in \partial\mathbb{D}$ distinct and $\alpha_1, \dots, \alpha_l \in (0, 1]$. We can take a branch of the holomorphic Q that is defined on

$$\Omega_Q = \mathbb{C} \setminus \cup_{\alpha_j < 1} \{ra_j : r > 0\}.$$

The following theorem of Monakhov guarantees that the operator $\mathcal{C}^Q$ and $\mathcal{B}^Q(f) = (\mathcal{C}^Q(f))_z$ satisfy many properties of the Cauchy and Beurling Transforms.

Theorem 10.3 *Let Q be as in (10.27). For every $p \in (p_1, p_2)$ with*

$$p_1 = \max_j (1 - \alpha_j/2)^{-1}, \quad p_2 = 2 \max_j (1 - \alpha_j)^{-1},$$

the operators $\mathcal{C}^Q : L^p(\mathbb{D}) \rightarrow W^{1,p}(\mathbb{D})$, and $\mathcal{B}^Q : L^p(\mathbb{D}) \rightarrow L^p(\mathbb{D})$, are bounded.

In view of (10.27) and Theorem 10.2, we have

$$\mathcal{C}^Q(h)_{\bar{z}} = h, \quad \mathcal{C}^Q(h)_z = \mathcal{B}(h),$$

weakly. We note that when $z \in \mathbb{D}$, then we can write

$$(10.29) \quad \mathcal{C}^Q(f)(z) = \frac{Q(z)}{\sqrt{z}} \iint_{\mathbb{D}} \left[\frac{\sqrt{z}f(\zeta)}{Q(\zeta)(\zeta - z)} + \frac{\overline{\sqrt{z}f(\zeta)}}{\overline{Q(\zeta)}(z\bar{\zeta} - 1)} \right] \frac{d\zeta \wedge d\bar{\zeta}}{2\pi i}.$$

What we learn from this is that $\mathcal{C}^Q(f)(z)$ is a real multiple of $Q(z)/\sqrt{z}$ whenever $|z| = 1$. Hence the boundary behavior of $\mathcal{C}^Q(f)$ is tied to that of $Q(z)/\sqrt{z}$. It is this property of $\mathcal{C}^Q$ that we use later on in the proof of Theorem 10.1.

Proof of Theorem 10.1 Step 1. Without loss of generality, we may assume that $n = 2$. Also, the surface w we wish to construct may embed into a set that is symplectomorphic to $Z^4(R)$ for some R . For our purposes, we choose $T \times \mathbb{C}$ with $T \subset \mathbb{C}$ a triangle with vertices ± 1 and i . We wish to find a transformation $w = (U, V) : \mathbb{D} \rightarrow \mathbb{C}^2$ that solves (10.16) and satisfies the following properties:

- (i) $U : \mathbb{D} \rightarrow T$, with $U(\partial\mathbb{D}) = \partial T$, and $V : \mathbb{D} \rightarrow \mathbb{C}$,
- (ii) $(u^0, v^0) \in w(\mathbb{D})$,
- (iii) $2\text{area}(w(\mathbb{D})) = i \int_{\mathbb{D}} dU \wedge d\bar{U} + dV \wedge d\bar{V} = 2$.

Note that we switched from (10.9) to (iii) because $\text{area}(T) = 1$. Pick $a \in \mathbb{D}$. We search for solutions of the form

$$U = \mathcal{C}^1(u) + q, \quad V = \mathcal{C}^2(v) - \mathcal{C}^2(v)(a) + v^0,$$

for $u, v : \mathbb{D} \rightarrow \mathbb{C}$ in L^p , and a holomorphic function q . The operators $\mathcal{C}^1$ and $\mathcal{C}^2$ are defined as $\mathcal{C}^i(f) = \mathcal{C}^{Q^i}(f)$, where

$$Q^1(z) = e^{3i\pi/4}(z-1)^{1/4}(z+1)^{1/4}(z-i)^{1/2}z^{-1/2}, \quad Q^2(z) = z-1.$$

We observe that Q^2 maps $\partial\mathbb{D}$ to lines parallel to the side of T .

We claim

$$(10.30) \quad z \in \partial\mathbb{D} \implies \mathcal{C}^2(v) \in i\mathbb{R}.$$

To see this, observe that if $|z| = 1$, then $z = 1/\bar{z}$, and

$$\mathcal{C}^{Q^2}v(z) = (z-1)\mathcal{C}(v/Q)(z) + (z-1)\bar{z}\overline{\mathcal{C}(v/Q)(z)} \in i\mathbb{R},$$

because $(z-1)\bar{z} = (1-\bar{z}) = \overline{-z-1}$. This proves (10.30). This in turn implies

$$(10.31) \quad \int_{\mathbb{D}} dV \wedge d\bar{V} = 0.$$

□

Exercise 10.1(i) Consider the form $\alpha = a(q) \cdot dq + p \cdot dq$ in $\mathbb{R}^4$. Use the proof of Proposition 10.1(i) to determine a (g, J) in $\mathcal{I}(\mathbb{R}^4, d\alpha)$. Find φ such that $-\varphi^*\bar{\omega} = d\alpha$. Use this to find another (g, J) in $\mathcal{I}(\mathbb{R}^4, d\alpha)$.

(ii) Find J holomorphic curves for $d\alpha$.

(iii) Show that for any $n \times n$ matrix Γ , there are two positive symmetric matrices S_1 and S_2 , and orthogonal matrix O such that $\Gamma = S_1O = OS_2$.

(iv) Let $\omega(a, b) = Ca \cdot b$ be a symplectic form with

$$C = \begin{bmatrix} 0 & \Gamma \\ -\Gamma^* & 0 \end{bmatrix}.$$

Use the proof of Proposition 10.1(i) to determine a (g, J) in $\mathcal{I}(\mathbb{R}^{2n}, \omega)$.

□

11 Arnold's Conjecture and Floer's Equation

In Chapter 9 we formulated Arnold's Conjecture and described a strategy for tackling this conjecture. Given a symplectic manifold (M, ω) , we write Γ for the set of C^1 loops $x : \mathbb{S}^1 \rightarrow M$. We also write Γ_0 for the set of contractible loops, which is a connected component of Γ that contains constant loops. We wish to define $\mathcal{A}_H : \Gamma \rightarrow \mathbb{R}$ as in (9.5), and study its critical points. As in Chapter 9, we assume that $H : M \times \mathbb{R} \rightarrow \mathbb{R}$ is a C^2 Hamiltonian function that is 1-periodic in t . We set,

$$(11.1) \quad \mathcal{A}_H(x) = \int w^* \omega - \int_0^1 H(x(t), t) dt,$$

where $w : \mathbb{D} \rightarrow M$ is an extension of x , i.e., $w = x$ on $\partial\mathbb{D} = \mathbb{S}^1$. To have $\mathcal{A}_H$ well-defined, we need to make sure that if $w, w' : \mathbb{D} \rightarrow M$ are two surfaces that extend x , then

$$(11.2) \quad 0 = \int_{w(\mathbb{D})} \omega - \int_{w'(\mathbb{D})} \omega = \int_{w''(\mathbb{S}^2)} \omega.$$

where $w'' : \mathbb{S}^2 \rightarrow M$ is a surface we obtain by gluing w and w' across the boundary. Throughout this chapter, we assume that (11.2) holds. Equivalently,

$$(11.3) \quad [\omega](u) = 0,$$

for every closed surface $u : \mathbb{S}^2 \rightarrow M$. Note that (11.3) holds if the second fundamental group $\pi_2(M)$ is trivial, or ω is exact. In particular (11.3) holds for any *aspherical manifold* (i.e. $\pi_n(M) = 0$ for $n \geq 2$) such as torus.

Proposition 11.1 *Assume (11.3) holds. Let U be an open subset of M such that $\bar{U}$ is compact. Consider the map $A : C^1(\mathbb{S}^1; \bar{U}) \rightarrow \mathbb{R}$, given by*

$$A(x) = \int_{\mathbb{S}^1} w^* \omega,$$

where w is any extension of x . Then A has a continuous extension from $C^1(\mathbb{S}^1, M)$ to $\mathcal{H}^1(\mathbb{S}^1, M)$.

Proof When $\omega = d\lambda$ in U , then

$$A(x) = \int_{\mathbb{S}^1} \lambda_x(\dot{x}) dt.$$

Since $(x, v) \rightarrow \lambda_x(v)$ is continuous, $A(x)$ is well-defined and continuous for $x \in \mathcal{H}^1$. In general, we need to make sure that x can be extended to $w : \mathbb{D} \rightarrow U$, with

$$\int_{\mathbb{D}} |dw|^2 dt d\theta \leq c_0 \int_{\mathbb{S}^1} (|x|^2 + |\dot{x}|^2) dt,$$

for a universal constant c_0 . □

From now on, we assume that (11.1) holds. We wish to study the critical points of $\mathcal{A}_H : \Gamma_0 \rightarrow \mathbb{R}$. Given $x \in \Gamma_0$, We may identify

$$T_x \Gamma_0 = \{ \xi : \mathbb{S}^1 \rightarrow TM : \xi(t) \in T_{x(t)} M \}.$$

Given $\xi \in T_x \Gamma_0$, and small $\delta > 0$, we may define $v : \mathbb{S}^1 \times [0, \delta] \rightarrow M$, so that

$$v_s(t, 0) = \xi(t), \quad v(t, 0) = x(t).$$

For example, if M is equipped with a Riemannian metric g , we can set $v(t, s) = \exp_{x(t)}(s\xi(t))$, where $\exp$ denotes the exponential map associated with g . Let us extend $v(\cdot, s) : \mathbb{S}^1 \rightarrow M$, to a surface $w(\cdot, s) : \mathbb{D} \rightarrow M$. In other words, $w : \mathbb{D} \times [0, \delta] \rightarrow M$, with $w = v$ on $\partial\mathbb{D} \times [0, \delta]$. From

$$\int_{\mathbb{D} \times [s_1, s_2]} w^*(d\omega) = 0,$$

and Stokes' formula, we learn

$$(11.4) \quad \int_{\mathbb{D}} w(\cdot, s_2)^* \omega - \int_{\mathbb{D}} w(\cdot, s_1)^* \omega + \int_{s_1}^{s_2} \int_{\mathbb{S}^1} \omega_{w(t,s)}(w_t, w_s)(t, \theta) dt d\theta = 0.$$

This in turn implies

$$\frac{d}{ds} \left[\int_{\mathbb{D}} w(\cdot, s)^* \omega \right] \Big|_{s=0} = - \int_{\mathbb{S}^1} \omega_{x(t)}(\dot{x}(t), \xi(t)) dt.$$

From this we learn that the Gateau derivative of $\mathcal{A}$ is given by

$$\begin{aligned} (d\mathcal{A}_H)_x(\xi) &= - \int_{\mathbb{S}^1} [\omega_{x(t)}(\dot{x}(t), \xi(t)) + (dH)_{x(t)}(\xi(t), t)] dt \\ &= - \int_{\mathbb{S}^1} \omega_{x(t)}(\dot{x}(t) - X_H(x(t), t), \xi(t)) dt \end{aligned}$$

If we take $(g, J) \in \mathcal{I}(M, \omega)$, and define a metric on Γ by

$$\langle\langle \xi, \eta \rangle\rangle_x = \int_0^t g_{x(t)}(\xi(t), \eta(t)) dt,$$

then we can write

$$\begin{aligned} (d\mathcal{A}_H)_x(\xi) &= - \int_{\mathbb{S}^1} g_{x(t)}(J(x(t))(\dot{x}(t) - X_H(x(t))), \xi(t)) dt \\ &= - \langle\langle J(x)(\dot{x} - X_H(x)), \xi \rangle\rangle_x, \end{aligned}$$

or in short,

$$\partial \mathcal{A}_H(x(\cdot)) = -J(x)(\dot{x} - X_H(x)).$$

Clearly, critical points of $\mathcal{A}_H$ correspond to the 1-periodic orbits of X_H . The corresponding gradient flow $\dot{u} = \partial \mathcal{A}_H(u)$, takes the form

$$(11.5) \quad u_s = -J(u)(u_t - X_H(u, t)).$$

We assume that $u : \mathbb{S}^1 \times \mathbb{R} \rightarrow M$ is a C^1 function. We may use the metric g to write

$$(dH)_x(v) =: g_x(\nabla H(x, t), v), \quad X_H(x, t) = J(x)\nabla H(x, t).$$

This yields an equivalent formulation of (11.5) as

$$(11.6) \quad u_s + J(u)u_t + \nabla H(u, t) = 0.$$

Remark 11.1(i) Let (M, α) be a contact manifold, and set

$$\hat{M} = M \times \mathbb{R}, \quad \hat{\alpha}_{(x, \theta)}(v, \tau) = e^\theta \alpha_x(v), \quad \hat{\omega} = d\hat{\alpha},$$

as in Chapter 3. If we choose the Hamiltonian function $H = 1$, $\hat{H}(x, \theta) = e^\theta$, then

$$\mathcal{A}_{\hat{H}}(\hat{x}) = \int_{\mathbb{S}^1} \hat{x}^* \alpha - \int_{\mathbb{S}^1} \hat{H}(\hat{x}) dt,$$

where $\hat{x} : \mathbb{S}^1 \rightarrow M \times \mathbb{R}$. We may consider only those $\hat{x}$ that lies on $M = M \times \{0\}$; $\hat{x}(t) = (x(t), 0)$. For such $\hat{x}$ we have $\mathcal{A}_{\hat{H}}(\hat{x}) = \mathcal{A}_\alpha(x) - 1$, where

$$\mathcal{A}_\alpha(x) = \int_{\mathbb{S}^1} x^* \alpha = \int_{\mathbb{S}^1} \alpha_x(\dot{x}) dt.$$

Again by Stokes formula if $w : \mathbb{S}^1 \times [s_1, s_2] \rightarrow M$ is a surface, then

$$\int_{\mathbb{S}^1} w(\cdot, s_2)^* \alpha - \int_{\mathbb{S}^1} w(\cdot, s_1)^* \alpha = - \int_{\mathbb{S}^1 \times [s_1, s_2]} w^* d\alpha,$$

which is (11.4) in our setting. From this we obtain

$$(d\mathcal{A}_\alpha)_x(\xi) = \int_{\mathbb{S}^1} (d\alpha)_x(\xi, \dot{x}) dt.$$

Note that if X is a vector field such that $X = \xi$ on x , then we can also use Cartan's formula to assert

$$(d\mathcal{A}_\alpha)_x(\xi) = \int_{\mathbb{S}^1} x^* \mathcal{L}_X \alpha = \int_{\mathbb{S}^1} x^* i_X d\alpha = \int_{\mathbb{S}^1} (d\alpha)_x(\xi, \dot{x}) dt.$$

(ii) Using an idea of Gromov, we can regard the solution u of (11.4) as a pseudo-holomorphic curve. We may regard u as a function $u : \mathbb{R}^2 \rightarrow M$ that is 1-periodic in t . We then define $\hat{u} : \mathbb{S}^1 \times \mathbb{R} \rightarrow M \times \mathbb{R}^2$, by $\hat{u}(z) = (u(z), z)$. The equation (11.21) can be rewritten as

$$(11.7) \quad \hat{u}_s + \hat{J}(\hat{u})u_t = 0,$$

where,

$$\hat{J}(\hat{x})(v, \alpha, \beta) = \hat{J}(x, s, t)(v, \alpha, \beta) := (J(x)v + \beta \nabla H(x, t) + \alpha J(x) \nabla H(x, t), -\beta, \alpha),$$

so that

$$\hat{J}(\hat{x})(v, 0, 1) = (J(x)v + \nabla H(x, t), -1, 0).$$

Observe that $(\hat{J})^2 = -id$. □

Our hope is that the solutions of (11.6) can be used to find critical points of $\mathcal{A}_H$. Observe that if u solves (11.6), and $w : \mathbb{D} \times \mathbb{R} \rightarrow M$ is an extension of u , then we may use (11.3) to assert

$$\begin{aligned} \int_{\mathbb{D}} w(\cdot, s_2)^* \omega - \int_{\mathbb{D}} w(\cdot, s_1)^* \omega &= - \int_{s_1}^{s_2} \int_{\mathbb{S}^1} \omega_{w(t,s)}(w_t, w_s)(t, \theta) dt d\theta \\ &= \int_{s_1}^{s_2} \int_{\mathbb{S}^1} g_{w(t,s)}(w_t(t, s), w_t(t, s) - X_H(w(t, s), t)) dt ds. \end{aligned}$$

From this we can readily deduce

$$(11.8) \quad \mathcal{A}_H(w(\cdot, s_2)) - \mathcal{A}_H(w(\cdot, s_1)) = \int_{s_1}^{s_2} \int_{\mathbb{S}^1} |w_t(t, s) - X_H(w(t, s), t)|_{w(t,s)}^2 dt ds.$$

We define

$$(11.9) \quad \mathcal{X}_H = \left\{ u \in C^1(\mathbb{R} \times \mathbb{S}^1) \rightarrow M : (11.5) \text{ holds, } \sup_s |\mathcal{A}_H(u(\cdot, s))| < \infty \right\}.$$

Note that if $u \in \mathcal{X}_H$, then

$$(11.10) \quad \int_{-\infty}^{\infty} \int_{\mathbb{S}^1} |u_t(t, s) - X_H(u(t, s), t)|_{u(t,s)}^2 dt ds < \infty$$

Theorem 11.1 *Assume that H is a C^2 function of compact support, and J is Lipschitz continuous. Then the set $\mathcal{X}_H$ is compact in $W^{2,p}(\mathbb{S}^1 \times \mathbb{R}; M)$ for any $p < \infty$ (and as a consequence in $C^{1,a}(\mathbb{S}^1 \times \mathbb{R}; M)$, for every $a \in (0, 1)$).*

Proof (Step 1) We first study the limit points of $u(\cdot, s)$, as $s \rightarrow \pm\infty$. We may embed M in a Euclidean space $\mathbb{R}^N$ (by Whitney's embedding theorem, we can choose $N = 2 \dim M$). If $\bar{g}$ denotes the standard metric of $\mathbb{R}^N$, then by compactness of M , we can find a constant c_1 such that

$$(11.11) \quad \bar{g}(v, v) \leq c_1 g_x(v, v).$$

Let $u \in \mathcal{X}_H$, so that (11.9) holds. Choose a sequence $s_n \rightarrow \infty$ such that

$$(11.12) \quad \lim_{n \rightarrow \infty} \int_{\mathbb{S}^1} |u_t(t, s_n) - X_H(u(t, s_n), t)|_{u(t, s_n)}^2 dt = 0.$$

Since $H \in C^1$, we deduce

$$c_2 = \sup_n \int_{\mathbb{S}^1} |u_t(t, s_n)|_{u(t, s_n)}^2 dt < \infty.$$

As a consequence of this, the compactness of M , and (11.11),

$$\int_{\mathbb{S}^1} |u_t(t, s_n)|^2 dt \leq c_2 c_1, \quad \sup_n \sup_t |u(t, s_n)| < \infty,$$

These bounds guarantee the boundedness of the sequence $\{u(\cdot, s_n)\}_n$ in $C^{1/2}$. As a consequence, we may choose a further subsequence (again denoted by s_n) such that the uniform limit

$$(11.13) \quad \lim_{n \rightarrow \infty} u(t, s_n) =: x^+(t),$$

exists. From this and (11.11) we deduce

$$\lim_{n \rightarrow \infty} \int_{\mathbb{S}^1} |u_t(t, s_n) - X_H(x^+(t), t)|_{u(t, s_n)}^2 dt = \lim_{n \rightarrow \infty} \int_{\mathbb{S}^1} |u_t(t, s_n) - X_H(x^+(t), t)|^2 dt = 0.$$

Hence $u(\cdot, s_n) \rightarrow x^+(\cdot)$ in $\mathcal{H}^1$. This, (11.12), and Proposition 11.1 imply

$$(11.14) \quad \lim_{n \rightarrow \infty} \mathcal{A}_H(u(t, s_n)) = \mathcal{A}_H(x^+).$$

This in turn implies

$$(11.15) \quad \sup_s \mathcal{A}_H(u(\cdot, s)) = \mathcal{A}_H(x^+) \leq \sup \{ \mathcal{A}_H(x) : x \in \bar{\mathcal{X}}_H \} =: c_3 < \infty.$$

where $\bar{\mathcal{X}}_H$ consists of $x \in \Gamma_0$ such that $\dot{x}(t) = X_H(x(t), t)$. The finiteness of c_3 follows from the compactness of $\bar{\mathcal{X}}_H$ in $\mathcal{H}^1$, and the continuity of $\mathcal{A}_H$. In the same fashion, we can show

$$\inf_s \mathcal{A}_H(u(\cdot, s)) \geq \inf \{ \mathcal{A}_H(x) : x \in \bar{\mathcal{X}}_H \} =: c_4 > -\infty.$$

In summary there are universal constants c_3 and c_4 such that for every $u \in \mathcal{X}_H$, and every $s \in \mathbb{R}$,

$$(11.16) \quad c_4 \leq \mathcal{A}_H(u(\cdot, s)) \leq c_3.$$

(Step 2) We now find a universal constant c_5 such that

$$(11.17) \quad \sup_{u \in \mathcal{X}_H} \sup_{(t,s) \in \mathbb{S}^1 \times \mathbb{R}} |\nabla u(t, s)| = c_5 < \infty.$$

Suppose to the contrary there exist sequences $\{(t_n, s_n)\}_n$ in $\mathbb{S}^1 \times \mathbb{R}$, and $\{u^n\}_n$ in $\mathcal{X}_H$, such that $s_n \rightarrow \infty$, and

$$\lim_{n \rightarrow \infty} |\nabla u^n(t_n, s_n)| = \infty.$$

Note that if $u \in \mathcal{X}_H$, then its s -translate

$$\tau_a u(t, s) = u(t, s + a),$$

is also in $\mathcal{X}_H$. Hence, we may assume that $t_n \rightarrow \bar{t} \in \mathbb{S}^1$, as $n \rightarrow \infty$, and $s_n = 0$ for all n . To ease the notation, let us write f_n for $|\nabla u_n|$, and z_n for $(t_n, 0)$. Pick a sequence $\varepsilon_n > 0$ such that

$$(11.18) \quad \lim_{n \rightarrow \infty} \varepsilon_n f_n(z_n) = \infty.$$

We claim that there exist sequences y_n , and $\delta_n > 0$ such that

$$(11.19) \quad \begin{aligned} |y_n - z_n| &\leq 2\varepsilon_n, & \delta_n &\leq \varepsilon_n, & \varepsilon_n f_n(z_n) &\leq \delta_n f_n(y_n), \\ |z - y_n| &\leq \delta_n & \implies & & f_n(z) &\leq 2f_n(y_n). \end{aligned}$$

To prove this, fix n , and set $a_0 = z_n$. Either the choice $(\delta_n, y_n) = (\varepsilon_n, z_n)$ satisfies (11.18), or else we can find a_1 such that

$$|a_1 - a_0| \leq \varepsilon_n, \quad f_n(a_1) > 2f_n(a_0).$$

Continuing this inductively, imagine that we can find $a_1, \dots, a_k$ such that for $\ell = 1, \dots, k$,

$$|a_\ell - a_{\ell-1}| \leq 2^{1-\ell} \varepsilon_n, \quad f_n(a_\ell) > 2f_n(a_{\ell-1}).$$

Then either we can choose $(y_n, \delta_n) = (a_k, 2^{-k} \varepsilon_n)$, or else there exists a_{k+1} , such that

$$|a_{k+1} - a_k| \leq 2^{-k} \varepsilon_n \implies f_n(a_{k+1}) > 2f_n(a_k).$$

If the existence of such a_k does not terminate, then we have a sequence $\{a_k\}_{k \in \mathbb{Z}^+}$ such that

$$|a_{k+1} - a_k| \leq 2^{-k} \varepsilon_n, \quad f_n(a_k) > 2^k f_n(a_0).$$

This is impossible because if $a_k \rightarrow a$, then by the continuity of f , we must have $f_n(a) = \infty$. As a consequence, for each n , there exists (y_n, δ_n) such that (11.19) holds.

(Step 3) For our purposes, we regard $u^n = u^n(t, s)$ as a function from $\mathbb{R}^2$ into M that is 1-periodic in t . We also set $R_n = |\nabla u^n(y_n)|$. We define $w_n : B(0, \delta_n R_n) \rightarrow M$, by

$$w^n(z) = u^n(y_n + R_n^{-1}z).$$

Observe that by (11.19), we know

$$\delta_n R_n \geq \varepsilon_n f_n(z_n).$$

From this and (11.18) we learn

$$(11.20) \quad \lim_{n \rightarrow \infty} \delta_n R_n = \infty.$$

Moreover by (11.19),

$$(11.21) \quad |\nabla w^n(0)| = 1, \quad |\nabla w^n(z)| \leq 2,$$

for every $z \in B(y_n, \delta_n R_n)$. Furthermore, w^n satisfies

$$(11.22) \quad w_s^n(t, s) + J(w^n(t, s))w_t^n(t, s) + h_n(t, s) = 0,$$

where $h_n(t, s) = R_n^{-1} \nabla H(w^n(t, s), t_n + R_n^{-1}t)$. Observe that by (11.16), (11.6) and (11.7),

$$\begin{aligned} \int_{B(0, \delta_n R_n)} |\nabla w^n|_{w^n}^2 dz &= \int_{B(y_n, \delta_n)} |\nabla u^n|_{u^n}^2 dz \\ &= \int_{B(y_n, \delta_n)} [|u_t^n|_{u^n}^2 + |u_s^n|_{u^n}^2] dz \\ &= \int_{B(y_n, \delta_n)} [|u_t^n|_{u^n}^2 + |u_t^n - X_H(u^n, t)|_{u^n}^2] dz \\ &\leq 3 \int_{B(y_n, \delta_n)} |u_t^n - X_H(u^n, t)|_{u^n}^2 dz + 2 \max |X_H| |B(y_n, \delta_n)| \\ &\leq 3(c_3 - c_4) + 2 \max |X_H| |B(y_n, \delta_n)|. \end{aligned}$$

From this we deduce

$$(11.23) \quad \int_{B(0, \delta_n R_n)} |\nabla w^n|_{w^n}^2 dz \leq c_6,$$

for a constant c_6 .

(Step 4) We wish to study the large n limit of w_n . Let us simply write ∂ for a partial differentiation operator, and write v^n for ∂w^n . (11.22) implies

$$(11.24) \quad v_s^n + \hat{J}^n(\hat{w}^n)v_t^n = g^n,$$

where $g^n = \partial(\hat{J}(w^n))w^n - \partial h_n$.] From $J \in C^1$, $H \in C^2$, and (11.21) we learn

$$\sup_n \|g^n\|_{L^\infty} < \infty.$$

From this and Theorem D.4 below we deduce

$$\sup_n \|v^n\|_{W^{1,p}(B)} < \infty,$$

for every $p > 1$, and every bounded set B (in fact we have $Dv^n \in BMO(B)$). This implies

$$\sup_n \|w^n\|_{W^{2,p}(B)} < \infty.$$

This implies that we can choose a subsequence of w^n that converges strongly to w in $W_{loc}^{1,p}$. We have that $w : \mathbb{R}^2 \rightarrow M$ satisfies

$$(11.25) \quad w_s + J(w)w_t = 0.$$

Moreover, from (11.23), and (11.21) we learn

$$(11.26) \quad |\nabla w| \leq 2, \quad |\nabla w(0)| = 1, \quad 0 < 2 \int_{\mathbb{R}^2} w^* \omega = \int_{\mathbb{R}^2} |\nabla w|^2 dz \leq c_6.$$

(Step 5) Let us define the non-decreasing function $f : [0, \infty) \rightarrow [0, \infty)$

$$f(R) = \int_{B_R(0)} |\nabla w|^2 dz.$$

This function is bounded by c_6 by (11.26). On the other hand, if $\gamma_R(\theta) = w(Re^{2\pi i\theta})$, then

$$f(\infty) = 2\pi \int_0^\infty \int_0^1 |\nabla w(Re^{2\pi i\theta})|^2 R d\theta dR = \int_0^\infty g(R)(2\pi R)^{-1} dR < \infty,$$

where

$$g(R) := \int_0^1 |\dot{\gamma}_R(\theta)|^2 d\theta.$$

From $f(\infty) < \infty$ we learn that there exists a sequence $R_k \rightarrow \infty$, such that $g(R_k) \rightarrow 0$ in large k -limit. As a result,

$$(11.27) \quad \lim_{k \rightarrow \infty} |\gamma_{R_k}(\mathbb{S}^1)| = \int_0^1 |\dot{\gamma}_{R_k}(\theta)| d\theta \leq \lim_{k \rightarrow \infty} \sqrt{g(R_k)} = 0,$$

where $|\gamma|$ represents the length of γ . In words, the length of the boundary curve $\eta_k := \gamma_{R_k}(\mathbb{S}^1) = \partial w(B_{R_k}(0))$ shrinks to 0 in large k limit. Pick a point $a_k \in \gamma_{R_k}(\mathbb{S}^1)$, and take a subsequence of a_k such that $a_k \rightarrow a$. Since $\gamma_{R_k}(\mathbb{S}^1)$ is shrinking, the curves $\gamma_{R_k}(\mathbb{S}^1)$ lie in a small neighborhood of a for a subsequence. We continue to write R_k for such a subsequence. Consider the surface $\Gamma_k := w(B_{R_k}(0))$. As k increases, the surface Γ_k grows, while its boundary η_k tends to a point a . Take a local chart U that contains a and choose a 1-form α such that $\omega = d\alpha$ in U . For large k , the curve $\eta_k = \partial\Gamma_k$ lies in U . We may choose a surface $\hat{\Gamma}_k$ in U such that $\eta_k = \partial\hat{\Gamma}_k$. We have

$$\int_{\Gamma_k} \omega = \int_{\hat{\Gamma}_k} \omega = \int_{\hat{\Gamma}_k} d\alpha = \int_{\eta_k} \alpha,$$

which implies

$$\int_{w(\mathbb{R}^2)} \omega = \lim_{k \rightarrow \infty} \int_{\Gamma_k} \omega = \lim_{k \rightarrow \infty} \int_{\eta_k} \alpha = 0.$$

This is absurd because the left-hand side is positive (which is consequence of $\nabla w(0) \neq 0$). This completes the proof of (11.17).

(Step 6) We now use (11.17) to prove that for every $p \in [1, \infty)$, and every bounded set $U \subset \mathbb{S}^1 \times \mathbb{R}$,

$$(11.28) \quad \sup_{u \in \mathcal{X}_H} \|D^2 u\|_{L^p(U)} < \infty.$$

To see this, observe that if u is a solution of (11.6), and $v = \partial u$ is a partial derivative of u , then v satisfies

$$(11.29) \quad v_s + \bar{J}v_t = -\partial(\nabla H(u, t)) - (\partial\bar{J})v_t =: h,$$

where $\bar{J} = J(u)$. Observe that from our assumptions on H and J , and (11.17), we learn

$$\sup_{u \in \mathcal{X}_H} \sup_{z \in \mathbb{S}^1 \times \mathbb{R}} [|\bar{J}(z)| + |\nabla \bar{J}(z)| + |h(z)|] < \infty.$$

This, (11.29), and Theorem D.4 imply

$$(11.30) \quad \sup_{u \in \mathcal{X}_H} \|\nabla v\|_{L^p(U)} < \infty,$$

for every bounded set U , and every $p \in [1, \infty)$. By Morrey's inequality, the bound (11.30) implies

$$(11.31) \quad C_U := \sup_{u \in \mathcal{X}_H} \|\nabla u\|_{C^a(U)} < \infty,$$

for every $a \in (0, 1)$. Observe that since $u \in \mathcal{X}_H$ implies that $\tau_\theta u \in \mathcal{X}_H$. Hence the constant $C_{U(k)}$ of (11.31) for $U(k) = \mathbb{S}^1 \times [k, k+1]$ is independent of k . From this, and the boundedness of ∇u , we can readily deduce

$$(11.32) \quad \sup_{u \in \mathcal{X}_H} \|\nabla u\|_{C^a(\mathbb{S}^1 \times \mathbb{R})} < \infty,$$

for every $a \in (0, 1)$.

(*Final Step*) The bounds (11.17) and (11.32) imply the precompactness of $\mathcal{X}_H$. These bounds also guarantees that if u^n is a sequence in $\mathcal{X}_H$, then it has a subsequence such that $u^n \rightarrow u$, and $\nabla u^n \rightarrow \nabla u$ locally uniformly. Clearly u solves (11.6). On the other hand,

$$\lim_{n \rightarrow \infty} \mathcal{A}_H(u^n(\cdot, s)) = \mathcal{A}_H(u(\cdot, s)),$$

for every s by Proposition 11.1. Since the bound (11.16) holds for each u^n , and n , the same bound holds for u . This implies the closedness of the set $\mathcal{X}_H$. The proof is now complete. $\square$

Remark 11.2 If we assume that $J \in C^k$, and $H \in C^{k+1}$, then we can show

$$(11.33) \quad \sup_{u \in \mathcal{X}_H} \|u\|_{W^{k+1,p}} < \infty.$$

$\square$

We now focus on the existence of solutions to Floer's equation (11.6). We first formulate an approximation of (11.6) that is easier to study. We fix a large positive $T \in \mathbb{R}$, and consider a function $u : S_T \rightarrow M$, where $S_T := \mathbb{S}^1 \times [-T, T]$. In other words, we replace the infinite cylinder with a finite long cylinder so that we can use the standard elliptic theory in bounded domains to show the existence and global regularity of u . We wish to assume boundary conditions for u so that the action $\mathcal{A}_H(u(\cdot, \pm T))$ is uniformly bounded in T . To guarantee this, we assume that $u(\cdot, \pm T)$ has a $\pm J$ -holomorphic extension to a disk. More precisely, we assume that u solves (11.6) in the interior of S_T , and there are

$$w^\pm : \mathbb{D}^\pm \rightarrow M,$$

such that $\mathbb{D}^\pm$ are two copies of the standard disks, and

$$w^\pm(e^{2\pi i t}) = u(t, \pm T),$$

and $w^\pm$ is a $\pm J$ -holomorphic curve. This means that $w^\pm(\theta_1, \theta_2)$ satisfies

$$(11.34) \quad w_{\theta_2}^\pm \pm J(w^\pm)w_{\theta_1}^\pm = 0,$$

in $\mathbb{D}^\pm$. By gluing the disks $\mathbb{D}^\pm$ to the boundary of ∂S_T , we obtain a capped cylinder that will be denoted by $\hat{S}_T$. More concretely,

$$\hat{S}^T = \partial(\mathbb{D} \times [-T, T]), \quad \mathbb{D}^\pm = \mathbb{D} \times \{\pm T\}.$$

We may use (u, w^-, w^+) to define a map $\hat{u} : \hat{S}_T \rightarrow M$, such that $\hat{u} = u$ on $\mathbb{S}^1 \times (-T, T)$, and

$$\hat{u}(\rho e^{2\pi i t}, \pm T) = w^\pm(\rho e^{\pm 2\pi i t}).$$

Note that for gluing $w^\pm$ disks to the cylinder, we need to reverse the orientation of w^- .

Let us write $\mathcal{B}^T$ for the set of continuous maps $\hat{u} : \hat{S}_T \rightarrow M$, $\hat{u} \in W^{1,p}(\hat{S}_T, M)$, and define $\mathcal{L}_t(\hat{u})$ such that for $z \in S_T$,

$$(\mathcal{L}_T(\hat{u}))_z := (d\hat{u} + J(\hat{u})(du)j + \nabla H(\hat{u}, t) ds - J(\hat{u})\nabla H(\hat{u}, t) dt)_z,$$

where $j(v) = -iv$. Observe that for $z \in S_T$,

$$\begin{aligned} \mathcal{L}(\hat{u}) \left(\frac{\partial}{\partial s} \right) &= u_s + J(u)u_t + \nabla H(u, t), \\ \mathcal{L}(\hat{u}) \left(\frac{\partial}{\partial t} \right) &= u_t - J(u)u_s - J(u)\nabla H(u, t). \end{aligned}$$

For $z \in \mathbb{D}^\pm$, we simply have

$$(\mathcal{L}(\hat{u}))_z = (d\hat{u} + J(\hat{u})(d\hat{u})j)_z.$$

Observe that if $\eta = (du + J(u)(du)j)_z$, and $\zeta = (\nabla H(u, t) ds - J(u)\nabla H(u, t) dt)_z$, then

$$\begin{aligned} J(u(z))\gamma j &= J(u(z))((du)_z + J(u(z))(du)_z j)j = J(u(z))(du)_z j + (du)_z = \gamma, \\ J(u(z))\zeta j &= J(u(z))(\nabla H(u(z), t) ds - J(u(z))\nabla H(u(z), t) dt)j \\ &= -J(u(z))(\nabla H(u(z), t) dt + J(u(z))\nabla H(u(z), t) ds) = \zeta. \end{aligned}$$

We wish to find $\hat{u}$ such that $\mathcal{L}_T(\hat{u}) = 0$. We will interpret $\mathcal{L} = \mathcal{L}_T$ as a section of a suitable vector bundle $\pi : \mathcal{E} \rightarrow \mathcal{B}$. Here $\mathcal{B} = \mathcal{B}^T := W^{1,p}(\hat{S}_T; M)$, which is a Banach manifold with respect to the Banach space $\mathcal{V} := W^{1,p}(\hat{S}_T; \mathbb{R}^{2d})$. We next define the fiber $\mathcal{E}_{\hat{u}}$, for each $\hat{u} \in \mathcal{B}$. The fiber $\mathcal{E}_{\hat{u}}$ is the space of sections of a vector bundle $\Omega^{\hat{u}}$ that is associated with $\hat{u}$. The vector bundle $\Omega^{\hat{u}}$ is over the base $\hat{S}_T$. Indeed, for $z \in \hat{S}_T$, the fiber $\Omega_z^{\hat{u}}$ consists of *linear* maps of the form

$$\beta : T_z \hat{S}_T \rightarrow T_{\hat{u}(z)} M, \quad \text{such that} \quad J(\hat{u}(z))\beta j = \beta.$$

for every $z \in \hat{S}_T \setminus \mathbb{S}^1 \times \{-T, T\}$. Observe that $\hat{u} \in \mathcal{B}^T := W^{1,p}(\hat{S}_T; M)$ yields a pullback bundle $\hat{u}^*TM$ which consists of pairs (z, v) such that $z \in \hat{S}_T$, and $v \in T_{u(z)}M$. We may regard the fiber $\mathcal{E}_u$ as the space of anti-linear forms $\Omega^{0,1}(u^*TM)$.

Let us define

$$\mathcal{X}_H^T = \{\hat{u} \in \mathcal{B}^T : \mathcal{L}_T(\hat{u}) = 0\}.$$

Theorem 11.2 (i) *We have*

$$(11.35) \quad \sup_{T>0} \sup_{\hat{u} \in \mathcal{X}_H^T} \sup_{s \in (-T, T)} \mathcal{A}_H(u(\cdot, s)) < \infty,$$

$$(11.36) \quad \sup_{T>0} \sup_{\hat{u} \in \mathcal{X}_H^T} \sup_{z \in S_T} |\nabla u(z)| < \infty.$$

Moreover, if $\hat{u}^T \in \mathcal{X}_H^T$, for every T , and u^T denotes the restriction of $\hat{u}^T$ to S_T , then $\{u^T : T > 0\}$ is precompact with respect to $W_{loc}^{1,p}(S^T, M)$, and its limit points belong to $\mathcal{X}_H$.

(ii) For every $T > 0$, and every $a \in M$, there exists $\hat{u} \in \mathcal{X}_H^T$ such that $u(0, 0) = a$.

Proof(i) Since $s \mapsto \mathcal{A}_H(u(\cdot, s))$ is non-decreasing, it suffices to find upper bounds on $\pm \mathcal{A}_H(u(\cdot, \pm T))$. Evidently

$$(11.37) \quad \pm \mathcal{A}_H(u(\cdot, \pm T)) \leq \max_{x,t} |H(x, t)| \pm \int_{w^\pm(\mathbb{D})} \omega.$$

On the other hand,

$$\begin{aligned} \pm \int_{w^\pm(\mathbb{D})} \omega &= \pm \int_{\mathbb{D}} \omega(w_t^\pm, w_s^\pm) dt ds = \pm \int_{\mathbb{D}} \omega(w_t^\pm, \mp J(w^\pm)w_t^\pm) dt ds \\ &= - \int_{\mathbb{D}} |w_t^\pm|_{w^\pm}^2 dt ds \leq 0. \end{aligned}$$

This and (11.37) imply (11.35). A repetition of the proof of (11.17) would yield (11.36). As a result, we will have the analog of (11.28) :

$$(11.38) \quad \sup_{T>T_0} \sup_{\hat{u} \in \mathcal{X}_H^T} \|D^2 u\|_{L^p(\mathbb{S}^1 \times (-T_0, T_0))} < \infty,$$

for every $p \in [1, \infty)$. This in turn yields the analog of (11.32):

$$(11.39) \quad \sup_{T>T_0} \sup_{\hat{u} \in \mathcal{X}_H^T} \|\nabla u\|_{C^\alpha(\mathbb{S}^1 \times (-T_0, T_0))} < \infty,$$

for $\alpha \in (0, 1)$. This bound gives the precompactness of the family $\{u^T : T > 0\}$, and the claim that any limit point of this family is in $\mathcal{X}_H$.

(ii) (*Step 1*) For the existence of a solution to $\mathcal{L}_T(U) = \mathcal{L}(U) = 0$, we use Theorem G.5 of the Appendix. To apply this Theorem, let us define $\mathcal{L}(\hat{u}, \theta) = \mathcal{L}_T(\hat{u}, \theta)$ so that for $z \in S_T$,

$$\mathcal{L}(\hat{u}, \theta)_z = (d\hat{u} + J(\hat{u})(d\hat{u})j + \theta(\nabla H(\hat{u}, t) ds - J(\hat{u})\nabla H(\hat{u}, t) dt))_z,$$

and for $z \in \mathbb{D}^\pm$, we simply have

$$\mathcal{L}(\hat{u}, \theta)_z = (d\hat{u} + J(\hat{u})(d\hat{u})j)_z.$$

We write $\mathcal{B}^{T,a}$ for the Banach manifold given by

$$\mathcal{B}^{T,a} = \{\hat{u} \in \mathcal{B}^T : u(0) = a\}.$$

The projection $\pi : \mathcal{B}^{T,a} \rightarrow \mathcal{E}$ is a vector bundle over $\mathcal{B}^{T,a}$. For each θ , we regard $\mathcal{L}(\cdot, \theta) : \mathcal{B}^{T,a} \rightarrow \mathcal{E}$, is a section. Let us define

$$\mathcal{X}_H^{a,T,\theta} = \{\hat{u} \in W^{1,p}(\hat{S}_T; M) : u(0) = a, \mathcal{L}_T(\hat{u}, \theta) = 0\},$$

so that

$$\mathcal{L}^{-1}(0) = \{(\hat{u}, \theta) : \hat{u} \in \mathcal{X}_H^{a,T,\theta}\}.$$

Observe that the bound (11.37) holds uniformly over $\theta \in [0, 1]$ as we replace H with θH . From this, we wish to argue that the bounds we deduce that (11.35) and (11.36) hold uniformly over $\theta \in [0, 1]$ as we replace H with θH . This is indeed the case. For example, in our bubbling-off argument, we now have $\theta_n H$ in place of H in (11.22), for a sequence $\theta_n \in [0, 1]$ such that $\theta_n \rightarrow \bar{\theta}$ in large n -limit. Since the analogs of (11.35) and (11.36) hold uniformly over $\theta \in [0, 1]$, we deduce that the analogs of (11.38) and (11.39) also hold uniformly over $\theta \in [0, 1]$. This in turn implies the compactness of $\mathcal{L}^{-1}(0)$. In summary the condition (i) of Theorem G.4 holds.

(*Step 2*) For condition (ii) of Theorem G.4, we claim that $\mathcal{X}_H^{a,T,0}$ consists of a single element, namely $\hat{u}(z) = a$ for all $z \in \hat{S}_T$. To see this, observe that when $H = 0$, and $\hat{u} : \hat{S}_T \rightarrow M$, then $\hat{u}$ is a J -holomorphic sphere, and must be constant. Indeed by (11.3),

$$0 = \int_{u(S_T) \cup \hat{w}^-(\mathbb{D}) \cup w^+(\mathbb{D})} \omega.$$

Here $\hat{w}^-$ is w^- with the orientation reversed so that $\hat{w}^-$ has an orientation compatible with u . Equivalently,

$$0 = -\frac{1}{2} \int_{S_T} [|u_s|^2 + |u_t|^2] dt ds + \int_{\mathbb{D}} [|\nabla w^+|^2 + |\nabla w^-|^2](t, t') dt dt'.$$

Hence u is a constant, and this constant must be $a = u(0)$.

(Step 3) We next focus on condition (iii) of Theorem G.4. We wish to study the fiberwise differential

$$E_{\hat{u}}\mathcal{L}(\cdot, \theta) : T_{\hat{u}}\mathcal{B}^{T,a} \rightarrow \mathcal{E}_{\hat{u}}.$$

We have

$$T_{\hat{u}}\mathcal{B}^{T,a} = \{\xi : \hat{S}_T \rightarrow TM : \xi(z) \in T_{\hat{u}(z)}M \text{ for } z \in \hat{S}_T \setminus \mathbb{S}^1 \times \{-T, T\}, \xi(0) = 0\}.$$

For $\theta = 0$, we wish to linearize $\mathcal{L}$ at the constant map $\hat{u} = a$. Note

$$D_a\mathcal{L} : T_a\mathcal{B}^{T,a} \rightarrow \mathcal{E}_a,$$

where $T_a\mathcal{B}^{T,a} = \{\xi : \hat{S}_T \rightarrow T_aM : \xi(0) = 0\}$, and $\mathcal{E}_a$ consists of sections $\eta : \hat{S}_T \rightarrow \Omega^a$, where

$$\Omega_z^a = \{\beta : T_z\hat{S}_T \rightarrow T_aM : \beta \text{ is linear and } J(a)\beta j = \beta\}.$$

Given $\xi \in T_a\mathcal{B}^{T,a}$, we simply have

$$(D_a\mathcal{L}_T)\xi = d\xi + J(a) \circ d\xi \circ j \in \mathcal{E}_a.$$

We wish to show that $D_a\mathcal{L}_T : T_a\mathcal{B} \rightarrow \mathcal{E}_a$ is an isomorphism. If $\dim M = d$, then we may represent T_aM as $\mathbb{R}^d \oplus J(a)\mathbb{R}^d$. Treating $J(a)v$ as iv , we may identify T_aM as $\mathbb{C}^d$. To show $D_a\mathcal{L}_T$ is surjective, we need to show that for every L^p function $f : \hat{S}_T \rightarrow \mathbb{C}^d$, we can find $\xi : \hat{S}_T \rightarrow \mathbb{C}^d$ in $W^{1,p}$ such that

$$(11.40) \quad \partial_{\bar{z}}\xi = \xi_t + i\xi_s = f, \quad \xi(0) = 0.$$

This has been shown in Theorem D.1 of Appendix D when the domain $\hat{S}_T$ is replaced with $\mathbb{C}$. We now show how our Theorem D.1 can be used to prove the existence of ξ when the domain is the Riemann Sphere $\mathbb{C} \cup \{\infty\}$. To solve (11.39), we write $f = g + h$ where $g(z) = f\mathbb{1}(|z| \leq 1)$, and $h(z) = f(z)\mathbb{1}(|z| > 1)$. Using the Cauchy's operator (see Appendix D), the function

$$\xi^1(z) = \mathcal{C}(f)(z) = -\frac{1}{\pi} \int_{|\zeta| \leq 1} \frac{f(\zeta)}{\zeta - z} dt ds, \quad \zeta = t + is,$$

satisfies $\partial_{\bar{z}}\xi^1 = g$. It is not hard to see that $\xi^1(\infty) = 0$. So, ξ is well-defined on $\mathbb{C} \cup \{\infty\}$. Similarly, if $\tau(z) = z^{-1}$, then $h \circ \tau$ is defined on $B_1(0)$, and $\xi^2 = \mathcal{C}(\hat{h}) \circ \tau$, for $\hat{h}(z) = -z^2(h \circ \tau)(z)$, satisfies $\partial_{\bar{z}}\xi^2 = h$. We finally set $\xi = \eta - \eta(0)$, with $\eta = \xi^1 + \xi^2$ to solve (11.39).

We now argue that (11.40) has a unique solution. Note that if ξ_1 and ξ_2 are two solutions, then $\zeta = \xi_1 - \xi_2$ is a holomorphic function with $\zeta(0) = 0$, and $\zeta' \in L^p$. We then use mean value formula to deduce that $\zeta' = 0$. This in turn implies that $\zeta = 0$.

(Step 4) It remains to show that $D_u \mathcal{L}_T$ is Fredholm. Given $\xi \in T_u \mathcal{B}$, the corresponding $\eta = (D_u \mathcal{L}_T) \xi$ satisfies $\eta + J(u) \eta j = 0$. Locally, we may express η as $\zeta ds - J(u) \zeta dt$, where

$$\zeta = \xi_s + J(u) \xi_t + (\nabla_\xi J(u)) u_t + \nabla_\xi \nabla H(u, t).$$

Here and below, ∇_ξ denotes the covariant derivative in ξ -direction. Since $\xi \in W^{1,p}$, $J \in C^1$, and $H \in C^2$, the linear operator

$$\xi \mapsto (\nabla_\xi J(u)) u_t + \nabla_\xi \nabla H(u, t),$$

is compact by Rellich's theorem. Hence, it suffices to show

$$\xi \mapsto F(\xi) := \xi_s + J(u) \xi_t,$$

is a Fredholm operator of index 0.

Let us select open sets

$$U_i \subset \bar{U}_i \subset U'_i \subset \bar{U}'_i \subset U''_i \subset \hat{S}_T, \quad i = 1, \dots, \ell$$

such that each U''_i is a local chart, and

$$\hat{S}_T = \cup_{i=1}^\ell U_i.$$

Assuming $M \subset \mathbb{R}^N$, we consider the operator

$$F_i : W_0^{1,p}(U''_i, \mathbb{R}^N) \rightarrow L^p(U''_i, \mathbb{R}^N), \quad F_i(\xi)(z) := \xi_s(z) + \hat{J}(z) \xi_t(z),$$

where $\hat{J}(z) = J(u(z))$. We will see later that the operator F_i is an isomorphism for every i provided that the open sets U''_i is sufficiently small for every i . We then choose a C^1 partition of unity $(\chi_i : i = 1, \dots, \ell)$ subordinate to $\{U_i : i = 1, \dots, \ell\}$. We also choose C^1 -function $\hat{\chi}_i : \hat{S}_T \rightarrow \mathbb{R}$ such that $\hat{\chi}_i(z) = 1$, for $z \in U_i$, and $\hat{\chi}_i(z) = 0$ for $z \notin U''_i$. Consider the operator

$$\mathcal{K}(h) = \sum_{i=1}^\ell \hat{\chi}_i F_i^{-1}(h \chi_i).$$

Note that $h \chi_i \in L^p(U''_i)$, and $F_i^{-1}(h \chi_i) \in W^{1,p}(U''_i)$. However, $\hat{\chi}_i F_i^{-1}(h \chi_i)$ is regarded as a function on $\hat{S}_T$. On the other hand,

$$F(F_i^{-1}(h \chi_i)) = h \chi_i,$$

on U''_i . As a result,

$$F \circ \mathcal{K}(h) = h \sum_{i=1}^\ell \hat{\chi}_i \chi_i + \sum_{i=1}^\ell F(\hat{\chi}_i) F_i^{-1}(h \chi_i) =: h + \mathcal{R}h.$$

To show that the Fredholm index of F is 0, it suffice to show that $\mathcal{R} : L^p(\hat{S}_T) \rightarrow L^p(\hat{S}_T)$ is a compact operator. Observe that since $u \in W^{1,p}$, and $\dim \hat{S}_T = 2$, the function u is bounded. On the other hand

$$\|F_i^{-1}(h\chi_i)\|_{W^{1,p}} \leq c_0 \|h\|_{L^p},$$

for a constant c_0 . Rellich's theorem can now be used to deduce the compactness of $\mathcal{K}$.

(*Final Step*) It remains to show that each F_i is an isomorphism if the open set U_i'' is sufficiently small. Note that if $z_i \in U_i''$, then $F_i = F_i' + F_i''$, where

$$F_i'(\xi) = \xi_s + \hat{J}(z_i)\xi_t.$$

By Morrey's inequality,

$$\|u\|_{C^\alpha} \leq c_1 \|u\|_{W^{1,p}},$$

for $\alpha = 1 - 2/p$. Because of this, we have a bound

$$\sup_i \sup_{z \in U_i''} |\hat{J}(x) - \hat{J}(z_i)| \leq c_2 \sup_i \text{diam}(U_i'')^\alpha,$$

which is small if the sets U_i'' , $i = 1, \dots, \ell$ are small. On the other hand we can show that F_i' is invertible for each i , and

$$\|F_i^{-1}(h)\|_{W_0^{1,p}(U_i'')} \leq c_3 \|h\|_{L^p(U_i'')},$$

for a constant c_3 that is independent of i , so long that the diameters of the sets U_i'' , $i = 1, \dots, \ell$ are uniformly bounded, and the boundaries of these sets are sufficiently nice. As a result, F_i is small perturbation of F_i' , and hence an isomorphism if the set U_i'' is sufficiently small in diameter for every i . In Appendix D we show that F_i' is indeed an isomorphism. $\square$

We are now ready to establish Arnold's conjecture (9.3). For this, we need to apply LS theory for our setting. First observe that we have a natural flow on $\mathcal{X}_H$ that is defined by

$$\Phi_\theta : \mathcal{X}_H \rightarrow \mathcal{X}_H, \quad \Phi_\theta(u)(t, s) = u(t, s + \theta).$$

This flow is gradient-like in the following sense: If

$$\mathcal{V} : \mathcal{X}_H \rightarrow \mathbb{R}, \quad \mathcal{V}(u) = \mathcal{A}_H(u(\cdot, 0)),$$

then

$$\frac{d}{d\theta} \mathcal{V}(\Phi_\theta(u)) = \frac{d}{d\theta} \mathcal{A}_H(u(\cdot, \theta)) \geq 0.$$

To apply LS theory to the flow Φ_θ , we need more:

Theorem 11.3 *If $s \mapsto u(\cdot, s)$ is nonconstant, then*

$$(11.41) \quad \frac{d}{d\theta} \mathcal{V}(\Phi_\theta(u)) > 0.$$

Proof Assume

$$\frac{d}{d\theta} \mathcal{A}_H(u(\cdot, \theta_0)).$$

This means that $x(t) = u(t, \theta_0)$ is a periodic orbit of X_H . Given $u \in \mathcal{X}_H$, define w by $\phi_t(w(t, s)) = u(t, s)$, where $\phi_t = \phi_t^H$ is the flow of the Hamiltonian vector field X_H . Observe

$$\begin{aligned} u_s + J(u)u_t + \nabla H(u, t) &= (d\phi_t)_w w_s + J(\phi_t(w))(d\phi_t)_w w_s \\ &\quad + J(\phi_t(w)) \left(\frac{d\phi_t}{dt} \right) (w) + \nabla H(\phi_t(w), t) \\ &= (d\phi_t)_w \left(w_s + \hat{J}(w, t)w_t \right), \end{aligned}$$

where

$$\hat{J}(w, t) = (d\phi_t)_w^{-1} J(\phi_t(w)) (d\phi_t)_w.$$

On the other hand,

$$\phi_t(w(t, \theta_0)) = u(t, \theta_0) = x(t),$$

which implies that $w(t, \theta_0) = x(0) =: a$ for every t . Hence the set $\{(t, s) : w(t, s) = a\}$ contains a circle. Take a local chart U with $a \in U$. In local coordinates, we assume that a corresponds to 0. We now have a solution to Beltrami type equation such that its zero set contains a segment of a circle. By a theorem of Carleman, w must be constant. Hence $u(t, \theta) = x(t)$ for all θ . $\square$

Theorem 11.3 would allow us to have a generalization of LS theorem to assert the number of 1-periodic orbit is at least $C\ell(\mathcal{X}_H)$. It remains to show

$$(11.42) \quad C\ell(\mathcal{X}_H) \geq C\ell(M).$$

To achieve this, consider the map

$$\tau : \mathcal{X}_H \rightarrow M, \quad \tau(u) = u(0, 0).$$

This map induces a map $\tau^* : H^*(M) \rightarrow H^*(\mathcal{X}_H)$. If we can show that τ^* is injective, then we can deduce (11.42).

11.1 Lagrangian Intersection

Let L and L' be two Lagrangian submanifolds of a symplectic manifold (M, ω) . Consider

$$\Gamma(L, L') = \{ \gamma : [0, t] \rightarrow M : \gamma \in C^1, \gamma(0) \in L, \gamma(t) \in L' \}.$$

Fix γ^0 and write $\Gamma_0(L, L')$ for the connected component $\Gamma(L, L')$ that contains γ^0 . Given $\gamma \in \Gamma_0(L, L')$, there exists a C^1 function $w : [0, t] \times [0, 1] \rightarrow M$ such that $w(\cdot, 0) = \gamma^0(\cdot)$, and $w(\cdot, 1) = \gamma(\cdot)$. We then define $\mathcal{A}_H : \Gamma_0(L, L') \rightarrow \mathbb{R}$ by

$$\mathcal{A}_H(\gamma) = \int_{[0, t] \times [0, 1]} w^* \omega - \int_0^t H(\gamma(\theta), \theta) dt.$$

Since we are assuming (11.3), the function $\mathcal{A}_H$ is well-defined on $\Gamma_0(L, L')$. We are interested in finding a lower bound on the cardinality of the set $\phi_t^H(\Lambda) \cap \Lambda'$.

Example 11.1 Assume that $M = T^*M^0$, and $\omega = d\lambda$. Given C^1 function $g : M^0 \rightarrow \mathbb{R}$, and $Q \in M^0$, we set

$$L = \{(q, (dg)_q) : q \in M^0\}, \quad L' = T_Q^*M^0.$$

The set $\phi_t^H \cap L'$ corresponds to those orbits $x(\theta) = \phi_\theta^H(a)$ such that $a \in L$, and $\phi_t(a) = (Q, P)$ for a suitable $P \in T_Q^*M^0$. $\square$

12 Spectral Invariant Metric

In Chapter 8, we studied the Hofer metric on the group of Hamiltonian diffeomorphism of $\mathbb{R}^{2n}$. In this chapter we discuss the spectral metrics that were introduced and studied by Viterbo [V] in the Euclidean setting. Their extensions to general symplectic manifolds were carried out by Schwartz [S] and Oh [O].

Definition 12.1(i) Let (M, ω) be a closed symplectic manifold, and assume that $[\omega]|_{\pi_2(M)} = 0$. As before, we write $\Gamma = \Gamma(M)$ for the set of C^1 functions $x : \mathbb{S}^1 \rightarrow M$. We write $\mathcal{H}(M)$ for the set of C^2 Hamiltonian functions $H : M \times \mathbb{S} \rightarrow \mathbb{R}$. By the support of H we mean

$$\text{Supp}(H) = \{x \in M : dH(\cdot, t)_x \neq 0 \text{ for some } t \in \mathbb{S}^1\}.$$

Note that if $x \notin \overline{\text{Supp}(H)}$, then $dH(\cdot, t)_y = 0$ for y near x , and all t . Hence $\phi_t^H(x) = x$ for $x \notin \text{Supp}(H)$.

(ii) The action functional $\mathcal{A} : \mathcal{H}(M) \times \Gamma(M) \rightarrow \mathbb{R}$ is defined as $\mathcal{A}(H, x) = \mathcal{A}_H(x)$, where $\mathcal{A}_H$ is given by (9.5).

(iii) We define

$$\mathcal{D}(M) = \{\phi^H := \phi_1^{X_H^\omega} : H \in \mathcal{H}(M)\}.$$

Given $H \in \mathcal{H}(M)$, its Hofer's norm is defined by

$$\|H\| = \int_0^1 \left(\max_{x \in M} H(x, t) - \min_{x \in M} H(x, t) \right) dt.$$

The Hofer's metric D on $\mathcal{D}(M)$ is defined by

$$D(\varphi, \psi) = \inf\{\|H - K\| : \varphi = \phi^H, \psi = \phi^K\}.$$

(iv) We write $\text{Crit}(H)$ for the set of critical points of $\mathcal{A}_H$, and

$$\sigma(H) = \{\mathcal{A}_H(x) : x \in \text{Crit}(H)\}.$$

As we have seen before, $\text{Crit}(H)$ consists 1-periodic solution of the Hamiltonian vector field $X_H = X_H^\omega$. $\square$

Theorem 12.1 *There exists a function*

$$\mathfrak{c} : H^*(M) \setminus \{0\} \times \mathcal{H}(M) \rightarrow \mathbb{R},$$

such that the following properties hold:

- (i) If $\phi^H = \phi^K$, then $\mathfrak{c}(\alpha, H) = \mathfrak{c}(\alpha, K)$.
- (ii) $\mathfrak{c}(\alpha, H) \in \sigma(H)$ for every $(\alpha, H) \in H^*(M) \setminus \{0\} \times \mathcal{H}(M)$.
- (iii) $\mathfrak{c}(\alpha \wedge \beta, H \sharp K) \leq \mathfrak{c}(\alpha, H) + \mathfrak{c}(\beta, K)$.
- (iv) $|\mathfrak{c}(\alpha, H) - \mathfrak{c}(\alpha, K)| \leq \|H - K\|$.
- (v) If H is independent of t , and $T_{\min}(H) > 1$ (equivalently $\mathcal{T}(H) < 1$), then

$$(12.1) \quad \mathfrak{c}(\alpha, H) = c_{LS}(\alpha, H).$$

Definition 11.2(i) On account of property (i), we can also define

$$\mathfrak{c} : H^*(M) \setminus \{0\} \times \mathcal{D}(M) \rightarrow \mathbb{R},$$

so that $\mathfrak{c}(\alpha, \phi^H) = \mathfrak{c}(\alpha, H)$.

(ii) We define a norm-like quantity on $\mathcal{H}(M)$ or $\mathcal{D}(M)$ by

$$\gamma(H) = \gamma(\phi^H) = \mathfrak{c}(v_M, M) - \mathfrak{c}(\mathbb{1}, M),$$

where v_M represents the class of (normalized) volume form, and $\mathbb{1}$ is 0-form 1.

(iii) With the aid of γ , we define a capacity

$$\mathbf{c}_\gamma(A) = \mathbf{c}_\gamma(A; M) = \sup \{ \gamma(\varphi) : \varphi(x) = x \text{ for } x \notin A \}.$$

□

We now design a strategy for finding upper and lower bounds on $\gamma(\varphi)$.

Proposition 12.1 *Let $H, K \in \mathcal{H}(M)$ and $U \subseteq M$ an open set such that $H(x, t) =: h_0(t)$ for $x \in M \setminus U$, and*

$$(12.2) \quad \phi^K(U) \cap U = \emptyset.$$

Then

$$(12.3) \quad \text{Fix}(\phi^H \circ \phi^K) = \text{Fix}(\phi^K \circ \phi^H) = \text{Fix}(\phi^K),$$

$$(12.4) \quad \text{Crit}(H \sharp K) = \text{Crit}(K \sharp H) = \text{Crit}(K),$$

$$(12.5) \quad \gamma(\phi^H \circ \phi^K) = \gamma(\phi^K \circ \phi^H) = \gamma(\phi^K).$$

Proof To ease the notation, we write $\varphi = \phi^H$, $\psi = \phi^K$. Since $\text{Supp}(H) = U \times \mathbb{S}^1$, we have

$$(12.6) \quad x \notin U, \ t \in \mathbb{S}^1 \quad \implies \quad \phi_t^H(x) = x.$$

Because of (12.2), we have $\text{Fix}(\psi) \cap U = \emptyset$. This and (12.6) imply

$$x \in \text{Fix}(\psi) \quad \implies \quad x = \varphi(x) = \psi(x) \quad \implies \quad x = (\varphi \circ \psi)(x) = (\psi \circ \varphi)(x).$$

As a result

$$(12.7) \quad \text{Fix}(\psi) \subseteq \text{Fix}(\psi \circ \varphi), \ \text{Fix}(\varphi \circ \psi).$$

Observe that by (12.2), we have $\psi(U) \subset U^c$, which implies that $(\varphi \circ \psi)(U) = \psi(U)$, and

$$(\varphi \circ \psi)(U) \cap U = \emptyset.$$

Hence we can use the analog of (12.7) to assert

$$(12.8) \quad \text{Fix}(\varphi \circ \psi) \subseteq \text{Fix}(\varphi^{-1} \circ \varphi \circ \psi) = \text{Fix}(\psi).$$

Moreover, since $\varphi(U) = U$, we also have

$$(\psi \circ \varphi)(U) \cap U = \emptyset.$$

Hence an analog of (12.7) can be used to write

$$\text{Fix}(\psi \circ \varphi) \subseteq \text{Fix}(\psi \circ \varphi \circ \varphi^{-1}) = \text{Fix}(\psi).$$

This, (12.8), and (12.7) imply (12.3).

We now turn our attention to (12.4). If $\Theta(\cdot) \in \text{Crit}(K)$, then $\Theta(t) = \phi_t^K(a)$, for some $a \in \text{Fix}(\psi)$, and

$$\phi_t^{K\sharp H}(a) = (\phi_t^K \circ \phi_t^H)(a) = \phi_t^K(a), \quad \phi^{H\sharp K}(a) = (\phi^H \circ \phi^K)(a) = \phi^K(a),$$

because $\text{Fix}(\psi) \cap U = \emptyset$, and $\phi^K(a) \notin U$. As a result,

$$(12.9) \quad \text{Crit}(K) \subseteq \text{Crit}(H\sharp K), \ \text{Crit}(K\sharp H).$$

Conversely, if $\Theta(\cdot) \in \text{Crit}(K\sharp H)$, then $\Theta(t) = \phi_t^{K\sharp H}(a)$, for some $a \in \text{Fix}(\phi^{K\sharp H}) = \text{Fix}(\phi^K)$, and

$$\phi^K(a) = (\phi^{K\sharp H} \circ (\phi^H)^{-1})(a) = \phi^{K\sharp H}(a),$$

because $(\phi^H)^{-1}(a) = a$ which follows from $a \in \text{Fix}(\psi) \subset U^c$. Hence

$$\text{Crit}(K\sharp H) \subseteq \text{Crit}(K).$$

In the same fashion we can show

$$\text{Crit}(H\sharp K) \subseteq \text{Crit}(K).$$

From this and (12.9) we deduce (12.4).

Take $a \in \text{Fix}(\psi)$. Clearly $\phi_t^K(a) \in \text{Fix}(\psi)$. If we set

$$A := \{\phi_s^K(a) : s \in [0, 1]\} \subset M \setminus U.$$

As a result,

$$\begin{aligned} (K\sharp H)(x, t) &= K(x, t) + H((\phi_t^K)^{-1}(x), t) = K(x, t) + h_0(t), \\ (H\sharp K)(x, t) &= H(x, t) + K((\phi_t^H)^{-1}(x), t) = K(x, t) + h_0(t), \end{aligned}$$

for $x \in M \setminus U$. From this we learn that for $\Theta(t) = \phi_t^K(a)$, for some $a \in \text{Fix}(\psi)$,

$$(12.10) \quad \mathcal{A}(K\sharp H, \Theta) = \mathcal{A}(H\sharp K, \Theta) = \mathcal{A}(K, \Theta) - \bar{h}_0.$$

where

$$\bar{h}_0 = \int_0^1 h_0(t) dt.$$

From (12.10) we learn

$$\sigma(K\sharp H) = \sigma(H\sharp K) = \sigma(K) - \bar{h}_0.$$

Clearly we can replace H with sH , for $s \in [0, 1]$:

$$\sigma(K\sharp(sH)) = \sigma(sH\sharp K) = \sigma(K) - s\bar{h}_0.$$

Fix $\alpha \in H^*(M)$. From Theorem 12.1(ii),

$$\mathfrak{c}(\alpha, K\sharp(sH)) \in \sigma(K\sharp(sH)) = \sigma(K) - s\bar{h}_0.$$

By Theorem 12.1(iv), the map $s \mapsto \mathfrak{c}(\alpha, K\sharp(sH)) + s\bar{h}_0$ is continuous, hence constant because the set $\sigma(K)$ is nowhere dense. From this we learn

$$\mathfrak{c}(\alpha, K\sharp H) + \bar{h}_0 = \mathfrak{c}(\alpha, K).$$

This in turn implies that $\gamma(K\sharp H) = \gamma(K)$. In the same fashion we can show $\gamma(H\sharp K) = \gamma(K)$. $\square$

Proposition 12.2 *The following statements hold*

- (i) $\gamma(\varphi) \leq E(\varphi)$.

- (ii) $\gamma(\varphi) = \gamma(\tau^{-1}\varphi\tau)$ for every symplectic diffeomorphism τ .
- (iii) $\gamma(\varphi \circ \psi) \leq \gamma(\varphi) + \gamma(\psi)$.
- (iv) $\gamma(\varphi) = \gamma(\varphi^{-1})$.
- (v) $\gamma(\varphi) \leq 2\mathbf{e}(\text{Supp}(\varphi))$, where $\text{Supp}(\varphi) = \{x : \varphi(x) \neq x\}$.
- (vi) For every open set U , we have $\mathbf{c}_{HZ}(U) \leq \mathbf{c}_\gamma(U) \leq 2\mathbf{e}(U)$.

Proof

(v) We claim

$$(12.11) \quad \varphi = \phi^H, \text{Supp}(H) \subset U, \psi(U) \cap U = \emptyset \implies \gamma(\varphi) \leq 2\gamma(\psi).$$

Indeed by (12.5),

$$\gamma(\varphi) \leq \gamma(\varphi \circ \psi) + \gamma(\psi^{-1}) = \gamma(\psi) + \gamma(\psi^{-1}) = 2\gamma(\psi).$$

On the other hand if $\psi = \phi^K$, then

$$\gamma(\psi) \leq E(\psi) \leq \|K\|.$$

This and (12.9) implies (v).

(vi) The second inequality is equivalent to the claim

$$(12.12) \quad \text{Supp}(\varphi) \subseteq U \implies \gamma(\varphi) \leq 2\mathbf{e}(U).$$

This follows from (v).

For first inequality it suffices to prove this: For every open set U ,

$$H \text{ independent of } t, \text{Supp}(H) \subseteq U, T_{\min}(H) > 1 \implies \|H\| = \gamma(\phi^H).$$

For such H , we apply Theorem 11.1(v) to assert

$$\gamma(\phi^H) = \gamma(H) = \mathbf{c}(\mathbb{1}, H) - \mathbf{c}(v_M, H) = c_{LS}(\mathbb{1}, H) - c_{LS}(v_M, H) = \max H - \min H = \|H\|,$$

as desired. □

We continue with a variant of Proposition 12.2 that was used by Seyfaddini [S] to bound γ from above.

Proposition 12.3 *Let g be a Riemannian metric on M , and write $d : M \times M \rightarrow [0, \infty)$ for the corresponding distance on M . Assume that $K : M \rightarrow \mathbb{R}$ is a Hamiltonian function with $T_{\min}(K) > 1$, and*

$$(12.13) \quad \max_x d(x, \phi^H(x)) < \inf \{d(y, \phi^K(y)) : y \in \text{Supp}(H)\}.$$

Then

$$(12.14) \quad \text{Fix}(\phi^H \circ \phi^K) = \text{Fix}(\phi^K),$$

$$(12.15) \quad \text{Crit}(H \sharp K) = \text{Crit}(K),$$

$$(12.16) \quad \mathfrak{c}(\mathbb{1}, H) \leq 2(\max K - \min K).$$

Proof Since $T_{\min}(K) > 1$, we have

$$(12.17) \quad \text{Fix}(\phi^K) = \{\phi^K(x) = x\} = \{x : (dK)_x = 0\}.$$

Because of (12.13), the right-hand side of (12.13) is strictly positive. As a result,

$$(12.18) \quad \text{Fix}(\phi^K) \cap \text{Supp}(H) = \emptyset.$$

This in turn implies

$$x \in \text{Fix}(\phi^K) \implies x = \phi^K(x) = \phi^H(x) \implies x \in \text{Fix}(\phi^H \circ \phi^K).$$

For the converse, first we claim

$$(12.19) \quad \text{Fix}(\phi^H \circ \phi^K) \cap \text{Supp}(H) = \emptyset,$$

which is a consequence of (12.13), because

$$y \in \text{Fix}(\phi^H \circ \phi^K) \cap \text{Supp}(H) \implies d(\phi^K(y), y) = d(\phi^K(y), (\phi^H \circ \phi^K)(y)) < d(y, \phi^K(y)).$$

Here we have used

$$y \in \text{Fix}(\phi^H \circ \phi^K) \cap \text{Supp}(H) \implies \phi^K(y) \in \text{Supp}(H),$$

because if $\phi^K(y) \notin \text{Supp}(H)$, then

$$\phi^K(y) = (\phi^H \circ \phi^K)(y) = y,$$

which contradicts $y \in \text{Supp}(H)$ and the positivity of the right-hand side of (12.13). Using (12.19), we can argue

$$x \in \text{Fix}(\phi^H \circ \phi^K) \implies \phi^K(x) = ((\phi^H)^{-1} \circ \phi^H \circ \phi^K)(x) = (\phi^H)^{-1}(x) = x.$$

This completes the proof of (12.14). As in the proof of Proposition 12.1, we can use (12.4) to show (12.5).

For (12.16), we certainly have

$$\mathfrak{c}(\mathbb{1}, H) \leq \mathfrak{c}(\mathbb{1}, H\sharp K) + \mathfrak{c}(\mathbb{1}, -K),$$

and

$$\mathfrak{c}(\mathbb{1}, -K) = \max K - \min K,$$

by Theorem 12.1(v). As result, (12.12) would follow if we can show

$$(12.20) \quad \mathfrak{c}(\mathbb{1}, H\sharp K) \leq \max K - \min K.$$

Observe

$$Crit(K) = \{\Theta \in \Gamma(M) : \Theta(t) = a, (dK)_a = 0\}.$$

From this and (12.5), we learn

$$\sigma(H\sharp K) = \{\mathcal{A}_{H\sharp K}(\Theta) : \Theta \in Crit(K)\} = \{-K(a) - h_0 : (dK)_a = 0\},$$

where $h_0 = \int_0^1 \int H(\cdot, t) \omega^n$. From this, we can readily deduce (12.20). $\square$

Example 12.1 Assume that the support of φ is contained in the annulus

$$U = U(r_1, r_2) = \{x \in \mathbb{R}^{2d} : r_1 < |x| < r_2\}.$$

In other words, $\varphi(x) = x$ if $x \notin U$. For the Hamiltonian K we choose

$$K(x) = f(|x|),$$

for a nondecreasing function $f : [0, \infty) \rightarrow \mathbb{R}$. Note that if $h(r) = f'(r)/r$, then the corresponding Hamiltonian ODE is

$$\dot{x} = h(|x|)Jx.$$

We write $x = (z_1, \dots, z_d) \in \mathbb{C}^d$, and $Jx = -(iz_1, \dots, iz_d)$. For $x = (\rho_1 e^{i\theta_1}, \dots, \rho_d e^{i\theta_d})$,

$$\phi_t^K(x) = (\rho_1 e^{i(\theta_1 - h(\rho)t)}, \dots, \rho_d e^{i(\theta_d - h(\rho)t)}),$$

where $\rho = |x| = (\rho_1^2 + \dots + \rho_d^2)^{1/2}$. Here we are using the fact that $|x|$ is conserved. Hence we can find a positive constant c_0 such that

$$\min_{x \in U} |\psi(x) - x| \geq c_0 \min_{\rho \in [r_1, r_2]} [\rho h(\rho)] = c_0 \min_{\rho \in [r_1, r_2]} f'(\rho).$$

The period is $T = 2\pi/h(\rho)$. So we need the condition $h(\rho) < 2\pi$ for $\rho \in [\rho_1, \rho_2]$. So our condition is

$$A := \max_{x \in U} |x - \varphi(x)| < c_0 \min_{\rho \in [r_1, r_2]} f'(\rho), \quad \max_{\rho \in [r_1, r_2]} h(\rho) < 2\pi,$$

and we get the bound

$$\mathfrak{c}(H) \leq 2(f(r_2) - f(r_1)).$$

For example, if we choose f linear with slope ℓ , we get

$$2\pi\rho_1 > \ell > c_0^{-1}A \quad \implies \quad \mathfrak{c}(H) \leq 2\ell(r_2 - r_1).$$

In summary,

$$A < 2c_0\pi\rho_1 \quad \implies \quad \mathfrak{c}(H) \leq 2c_0^{-1}A(r_2 - r_1).$$

□

13 Weinstein's Conjecture and Hofer's Energy

Given a closed contact manifold (M, α) , recall that $\hat{M} = M \times \mathbb{R}$, ξ is the kernel of α , and R denotes the corresponding Reeb vector field. We also have bundle splitting

$$T\hat{M} = TM \times \mathbb{R} = \xi \oplus \zeta,$$

where ζ is the plane bundle that is spanned by R and ∂_s . For our purposes we consider an almost complex structure $\hat{J}_x : T_x M \times \mathbb{R} \rightarrow T_x M \times \mathbb{R}$, of the form

$$\hat{J}_x(v, \theta) = (J_x(\pi_x(v)) + \theta R_x, -\alpha_x(v))$$

for $(v, \theta) \in T_x M \times \mathbb{R}$. Here $\pi(v) = v - \alpha(v)R$ denotes the R -projection onto ξ . Note that J depends on x only and we assume that $J : \xi \rightarrow \xi$ is an almost complex structure is compatible with symplectic bundle $(\xi, d\alpha|_\xi)$. The latter means that

$$g_x(a, a') = (d\alpha)_x(v, J_x(v')), \quad G_x(v, v') = g_x(v, v') + \alpha_x(v)\alpha_x(v'),$$

are metric on ξ_x and $T_x M$ respectively. We also have the metric

$$\hat{g}_{(x,s)}((v, \theta), (v', \theta')) = G_x(v, v') + \theta\theta',$$

on $T\hat{M}$. To simplify the notation, we set

$$\|a\| = \|a\|_J = g(v, v)^{1/2}, \quad \|v\| = \|v\|_J = G(v, v)^{1/2}, \quad \|\hat{v}\|^* = \|\hat{v}\|_J = \hat{g}(\hat{v}, \hat{v})^{1/2}.$$

We write $\mathcal{J}(M, \alpha)$ for the set of such almost complex structure J .

Consider a $\hat{J}$ -pseudoholomorphic curve $\hat{u} = (u, w) : S \rightarrow \hat{M}$, where $S = (S, i)$ is a Riemann surface. We write $z = \tau + it \in S$ for the argument of S . Then u satisfies

$$(13.1) \quad \pi_u(u_\tau) + J_u \pi_u(u_t) = 0, \quad w_\tau - \alpha_u(u_t) = 0, \quad w_t + \alpha_u(u_\tau) = 0.$$

Note that the PDE for w implies

$$(\alpha_u(u_\tau))_\tau + (\alpha_u(u_t))_t = 0.$$

Once this holds, we can find w so that the last two equations for w holds. Hence, (13.1) can be formulated in terms of u as

$$(13.2) \quad \pi_u(u_\tau) + J_u \pi_u(u_t) = 0, \quad (\alpha_u(u_\tau))_\tau + (\alpha_u(u_t))_t = 0.$$

We write $\mathcal{H}(S, J)$ for the set of such $\hat{J}$ -pseudoholomorphic curves.

We also note that for any increasing C^1 function $\eta : \mathbb{R} \rightarrow [0, 1]$, we also know that $\omega^\eta := d(\eta(s)\alpha)$ is symplectic. We write Γ for the set of such functions, and $\bar{\Gamma}$ for the set of non-decreasing functions.

Definition 13.1 Given $J \in \mathcal{J}(M, \alpha)$, and $\hat{u} = (u, w) \in \mathcal{H}(S, J)$, we set

$$(13.3) \quad \mathcal{E}(\hat{u}) = \sup_{\eta \in \Gamma} \int_{\hat{u}(S)} \omega^\eta,$$

for its (Hofer) energy. □

We first try to understand the meaning of $\mathcal{E}(\hat{u}) < \infty$. Observe

$$\begin{aligned} \int_{\hat{u}(S)} \eta \, d\alpha &= \int_S \eta(w)(d\alpha)_u(u_\tau, u_t) \, d\tau dt = \int_S \eta(w)(d\alpha)_u(\pi_u(u_\tau), \pi_u(u_t)) \, d\tau dt \\ &= \int_S \eta(w)(d\alpha)_u(\pi_u(u_\tau), J_u \pi_u(u_\tau)) \, d\tau dt \\ &= \int_S \eta(w) g_u(\pi_u(u_\tau), \pi_u(u_\tau)) \, d\tau dt \\ &= \frac{1}{2} \int_S \eta(w) (\|\pi_u(u_\tau)\|_J^2 + \|\pi_u(u_t)\|_J^2) \, d\tau dt \geq 0. \end{aligned}$$

Moreover,

$$\begin{aligned} \int_{\hat{u}(S)} \eta' \, ds \wedge \alpha &= \int_S \eta'(w) (ds \wedge \alpha)(\hat{u}_\tau, \hat{u}_t) \, d\tau dt \\ &= \int_S \eta'(w) (w_\tau \alpha_u(u_t) - w_t \alpha_u(u_\tau)) \, d\tau dt \\ &= \int_S \eta'(w) (w_\tau^2 + w_t^2) \, d\tau dt = \int_S \eta'(w) (\alpha_u(u_t)^2 + \alpha_u(u_\tau)^2) \, d\tau dt \\ &= \frac{1}{2} \int_S \eta'(w) (|\nabla w|^2 + \alpha_u(u_t)^2 + \alpha_u(u_\tau)^2) \, d\tau dt \geq 0. \end{aligned}$$

This suggests defining

$$\mathcal{E}_1(\hat{u}) := \sup_{\eta \in \Gamma} \int_{\hat{u}(S)} \eta \, d\alpha, \quad \mathcal{E}_2(\hat{u}) := \sup_{\eta \in \Gamma} \int_{\hat{u}(S)} \eta' \, ds \wedge \alpha.$$

Clearly

$$\mathcal{E}(\hat{u}) < \infty \quad \Leftrightarrow \quad \mathcal{E}_1(\hat{u}), \mathcal{E}_2(\hat{u}) < \infty.$$

It is not hard to show that in the definition of $\mathcal{E}_i(\hat{u})$, we can replace Γ with $\bar{\Gamma}$. It is not hard to show that the supremum in the definition of $\mathcal{E}_1(\bar{u})$ is achieved when $\eta = 1$. For $\mathcal{E}_2(\hat{u})$

however the supremum is achieved when $\eta' = \delta_{s_0}$ for a suitable $s_0 \in \mathbb{R}$. To explain this further, let us use coarea formula to write,

$$\begin{aligned} \int_S \eta(w)(d\alpha)_u(u_\tau, u_t) \, d\tau dt &= \int_{\mathbb{R}} \eta(s) \left[\int_{S \cap w^{-1}(s)} |\nabla w|^{-1} (d\alpha)_u(u_\tau, u_t) \, d\sigma \right] ds \\ &=: \int_{w(S)} \eta(s) \Lambda_1(\hat{u})(s) \, ds, \end{aligned}$$

where $d\sigma$ is the length element. Similarly,

$$\begin{aligned} \int_S \eta'(w) (ds \wedge \alpha)(\hat{u}_\tau, \hat{u}_t) \, d\tau dt &= \int_{\mathbb{R}} \eta'(s) \left[\int_{S \cap w^{-1}(s)} |\nabla w|^{-1} (ds \wedge \alpha)(\hat{u}_\tau, \hat{u}_t) \, d\sigma \right] ds \\ &=: \int_{w(S)} \eta'(s) \Lambda_2(\hat{u})(s) \, ds, \end{aligned}$$

Hence

$$(13.4) \quad \mathcal{E}_1(\hat{u}) = \int_{w(S)} \Lambda_1(\hat{u})(s) \, ds, \quad \mathcal{E}_2(\hat{u}) = \text{esssup}_{w(S)} \Lambda_2(\hat{u}).$$

Note that $u \in \mathcal{H}(S, J)$,

$$|\nabla w| = (\alpha_u(u_t)^2 + \alpha_u(u_\tau)^2)^{1/2}.$$

Hence

$$\Lambda_2(\hat{u}) = \int_{S \cap w^{-1}(s)} (\alpha_u(u_t)^2 + \alpha_u(u_\tau)^2)^{1/2} \, d\sigma$$

We now claim

$$(13.5) \quad \Lambda_2(\hat{u}) = \int_{u(S \cap w^{-1}(s))} \alpha.$$

Note however that the right-hand side depends on orientation/parametization of the curve $w^{-1}(s)$. If we do not specify an orientation, then $\Lambda_2(\hat{u})$ equals to the absolute value of the right-hand side of (13.5). In other words, for (13.5), we tacitly are selecting an orientation of $w^{-1}(s)$ so that the right-hand side of (13.5) is non-negative. To see (13.5), let us parametrize $w^{-1}(s)$ by a curve $\gamma : [0, r_0] \rightarrow S$ such that $|\dot{\gamma}| = 1$, and r_0 is the total length of γ . Hence

$$(13.6) \quad \int_{u(S \cap w^{-1}(s))} \alpha = \int_0^{r_0} \alpha_{u(\gamma)}((du)_\gamma(\dot{\gamma})) \, d\theta.$$

Since $\dot{\gamma} \perp \nabla w(\gamma)$, and $(\alpha_u(u_\tau), \alpha_u(u_t)) \perp \nabla w$, we may assume

$$\dot{\gamma} = |\nabla w(\gamma)|^{-1} (\alpha_{u(\gamma)}(u_\tau(\gamma)), \alpha_{u(\gamma)}(u_t(\gamma))).$$

As a result,

$$\alpha_{u(\gamma)}((du)_\gamma(\dot{\gamma})) = |\nabla w(\gamma)|^{-1}(\alpha_u(u_t)^2 + \alpha_u(u_\tau)^2) \circ \gamma = |\nabla w(\gamma)|.$$

From this and (13.6) we deduce (13.5).

Observe that by Stokes' formula

$$\int_{u(S \cap w^{-1}([s_1, s_2]))} d\alpha = \int_{u(S \cap w^{-1}(s_2))} \alpha - \int_{u(S \cap w^{-1}(s_1))} \alpha,$$

for $s_1 < s_2$. From this we learn that

$$s \mapsto \int_{u(S \cap w^{-1}(s))} \alpha,$$

is a non-decreasing function. Moreover, no matter what orientation we choose, we always have

$$\left| \int_{u(S \cap w^{-1}(s_2))} \alpha \right| \leq \left| \int_{u(S \cap w^{-1}(s_1))} \alpha \right| + \mathcal{E}^1(\hat{u}).$$

for every s_1 and s_2 . Hence,

$$(13.7) \quad \Lambda_2(\hat{u})(s_2) \leq \Lambda_2(\hat{u})(s_1) + \mathcal{E}^1(\hat{u}).$$

From this we learn,

$$(13.8) \quad \mathcal{E}^1(\hat{u}) < \infty, \quad \exists s \in \mathbb{R} \text{ such that } \Lambda_2(\hat{u})(s) < \infty \quad \implies \quad \mathcal{E}^2(\hat{u}) < \infty.$$

So far we have seen that the energy functionals $\mathcal{E}^1(\hat{u})$ and $\mathcal{E}^2(\hat{u})$ can be expressed in terms of u . We also know that $\mathcal{E}^2$ can be expressed in terms of w because

$$\int_{\hat{u}(S)} \eta' ds \wedge \alpha = \int_S \eta'(w) |\nabla w|^2 d\tau dt.$$

We now claim that $\mathcal{E}^1(\hat{u})$ can also be expressed in terms of w because

$$(13.9) \quad (d\alpha)_u(u_\tau, u_t) = \Delta w.$$

To see this, observe that for $z = \tau + it$,

$$\begin{aligned} (u^* \alpha)_z &= \alpha_{u(z)}(u_\tau(z)) d\tau + \alpha_{u(z)}(u_t(z)) dt = -w_t(z) d\tau + w_\tau(z) dt, \\ (d(u^* \alpha))_z &= \Delta w(z) d\tau \wedge dt, \\ (u^*(d\alpha))_z &= (d\alpha)_{u(z)}(u_\tau(z), u_t(z)). \end{aligned}$$

This and $u^*(d\alpha) = d(u^*\alpha)$ implies (13.9).

Observe that if S is a closed surface then

$$\int_{\hat{u}(S)} \omega^\eta = 0,$$

because ω^η is exact.

Example 13.1 If $x : \mathbb{R} \rightarrow M$ is a T -periodic orbit of the Reeb vector field R , then $\hat{u}(\tau, t) = (\tau, x(tT))$ is a $\hat{J}$ -holomorphic curve for $S = [0, \infty) \times \mathbb{S}^1$ (regarding $\mathbb{S}^1$ as $[0, 1]$ with $0 = 1$) because $u_\tau = 0$, $u_t = TR(u)$, which implies that $\pi(u_\tau) = \pi(u_t) = 0$, and $\alpha_u(u_\tau) = 0$, $\alpha_u(u_t) = T$. On the other hand if $\mathcal{E}_1(\hat{u}) = 0$, and

$$\mathcal{E}_2(\hat{u}) = T \sup_{\eta \in \Gamma} \int_S \eta'(\tau) d\tau dt = T.$$

More generally take a holomorphic map $h = (w^1, w^2) : S' \rightarrow S$ (with respect to the complex structures on the Riemann surfaces S and S') and set $\hat{u}' = \hat{u} \circ h$. Again $\mathcal{E}^1(\hat{u}') = 0$, $\mathcal{E}^2(\hat{u}') = T$. $\square$

Indeed we have the following partial inverse:

Proposition 13.1 *If $\hat{u} \in \mathcal{H}(J, S)$, and $\mathcal{E}_1(\hat{u}) = 0$, then $u(\mathbb{S})$ is a subset of an orbit of R . If additionally S is simply connected, then there exists a Reeb orbit $\gamma : I \rightarrow M$ and a holomorphic function $h = w + iw'$ such that $\hat{u} = (w, \gamma \circ w')$.*

Proof From $\mathcal{E}_1(\hat{u}) = 0$, we learn that $\pi_u(u_t) = \pi_u(u_\tau) = 0$. Hence $u_\tau, u_t \in \ell_u$. Define

$$A = \{z \in S : \nabla u(z) = 0\} = \{z \in S : \nabla w(z) = 0\}, \quad U = S \setminus A.$$

Then in the open set U , the rank of du is 1. Hence by the Rank Theorem, the set $u(U)$ is a one-dimensional manifold. By Sard's theorem, the 1-dimensional measure of the set $u(A)$ is zero. Hence $u(S)$ is one-dimensional.

Observe that by (13.9), w is harmonic. If S is simply connected, then it has a harmonic conjugate. Let w' be its harmonic conjugate. Set $a = u(0)$, and $U(z) = \psi_{w'(z)}(a)$, where ψ_θ denotes the flow of the Reeb vector field R . Observe that $U(0) = u(0)$, and for every $z = \tau + it$,

$$\begin{aligned} U_\tau(z) &= w'_\tau(z)R_{U(z)} = -w_t(z)R_{U(z)}, & U_t(z) &= w'_t(z)R_{U(z)} = w_\tau(z)R_{U(z)}, \\ u_\tau(z) &= \alpha_{u(z)}(u_\tau)R_{u(z)} = -w_t(z)R_{u(z)}, & u_t(z) &= \alpha_{u(z)}(u_t)R_{u(z)} = w_\tau(z)R_{u(z)}. \end{aligned}$$

We first set $t = 0$, and vary τ . By uniqueness for ODEs, we deduce $u(\tau, 0) = U(\tau, 0)$. We then fix τ and vary t . Again the uniqueness yields $u(\tau, t) = U(\tau, t)$. This completes the proof of (i). $\square$

As our first theorem, we show that the existence of a $\hat{J}$ -holomorphic curve of finite energy implies the existence of a periodic Reeb orbit.

Theorem 13.1 (*Hofer*) *Assume that J is a Lipschitz continuous function. Let $\hat{u} \in \mathcal{H}(S, J) \cap W_{loc}^{1,p}(S)$ for $S = [0, \infty) \times \mathbb{S}^1$ and some $p > 1$. If $0 < \mathcal{E}(\hat{u}) < \infty$, then there exists a sequence $\tau_k \rightarrow \infty$ such that*

$$(13.10) \quad \lim_{k \rightarrow \infty} \hat{u}(\tau_k, e^{2\pi i t/T}) =: x(t),$$

exists and is a Reeb orbit, where $T = \mathcal{E}(\hat{u})$.

Proof (*Step 1*) We first show that for every $C_0 > 0$,

$$(13.11) \quad \sup \mathcal{X}(C_0) := \sup \left\{ \sup_{z \in S} |\nabla \hat{u}(z)| : \hat{u} \in \mathcal{H}(S, J), \|\hat{u}\|_{W_{loc}^{1,p}(S)} + \mathcal{E}(\hat{u}) \leq C_0 \right\} < \infty.$$

Since $\|\hat{u}\|_{W^{1,p}(S)} \leq C_0$ for $\hat{u} \in \mathcal{X}(C_0)$, we may use the elliptic regularity (see Appendix D) to assert that for every bounded set $A \subset S$, there exists a constant $c_1(A) = c_1(A, C_0)$ such that

$$(13.12) \quad \sup_{\hat{u} \in \mathcal{X}(C_0)} \sup_{z \in A} |\nabla \hat{u}(z)| \leq c_1(A).$$

We need to replace the bounded set A with S . Suppose to the contrary there exist sequences $\{(t_n, \tau_n)\}_n$ in $\mathbb{S}^1 \times \mathbb{R}$, and $\{\hat{u}^n\}_n$ in $\mathcal{X}(C_0)$, such that $\tau_n \rightarrow \infty$, and

$$\lim_{n \rightarrow \infty} |\nabla \hat{u}^n(t_n, \tau_n)| = \infty.$$

To ease the notation, let us write f_n for $|\nabla \hat{u}^n|$, and z_n for (τ_n, t_n) . As in Chapter 12, pick a sequence $\varepsilon_n > 0$ such that

$$\lim_{n \rightarrow \infty} \varepsilon_n f_n(z_n) = \infty.$$

As we showed in Chapter 12, there exist sequences y_n , and $\delta_n > 0$ such that

$$(13.13) \quad \begin{aligned} |y_n - z_n| &\leq 2\varepsilon_n, & \delta_n &\leq \varepsilon_n, & \varepsilon_n f_n(z_n) &\leq \delta_n f_n(y_n), \\ |z - y_n| &\leq \delta_n & \implies & & f_n(z) &\leq 2f_n(y_n). \end{aligned}$$

For our purposes, we regard $\hat{u}^n = u^n(\tau, t)$ as a function from $\mathbb{R}^2$ into $\hat{M}$ that is 1-periodic in t . We also set $R_n = |\nabla \hat{u}^n(y_n)|$. We define $v_n : B(0, \delta_n R_n) \rightarrow M$, by

$$v^n(z) = \hat{u}^n(y_n + R_n^{-1}z).$$

Observe that by (13.12), we know

$$\delta_n R_n \geq \varepsilon_n f_n(z_n).$$

From this and (13.13) we learn

$$(13.14) \quad \lim_{n \rightarrow \infty} \delta_n R_n = \infty.$$

Moreover by (13.13),

$$(13.15) \quad |\nabla v^n(0)| = 1, \quad |\nabla v^n(z)| \leq 2,$$

for every $z \in B(y_n, \delta_n R_n)$. Furthermore, v^n satisfies

$$(13.16) \quad v_\tau^n(z) + \hat{J}(v^n(z))v_t^n(z) = 0.$$

Recall that $|y_n| \rightarrow \infty$, and $\delta_n \rightarrow 0$. If needed, we may switch to a subsequence so that two successive y'_n 's are at least one unit apart. Hence, without loss of generality, we may assume

$$|y_{n+1} - y_n| > 1, \quad \delta_n \leq 1/2.$$

Under these conditions, the discs $(B(y_n, \delta_n) : n \in \mathbb{N})$ are disjoint. Let us write $\|\cdot\|_a$ for the norm associated to the norm at the base point $a \in M$ (the metric associated with J). The norm associated with $\hat{J}$ is denoted by $\|\cdot\|_a^*$, $a \in \hat{M}$. Observe that the expression

$$\sum_n \int_{B(0, \delta_n R_n)} (\|\pi_{v^n}(v_\tau^n)\|_{v^n}^2 + \|\pi_{v^n}(v_t^n)\|_{v^n}^2) dz$$

equals to

$$\sum_n \int_{B(y_n, \delta_n)} [\|\pi_{u^n} u_\tau^n\|_{u_n}^2 + \|\pi_{u^n} u_t^n\|_{u_n}^2] dz \leq \int_S [\|\pi_{u^n} u_\tau^n\|_{u_n}^2 + \|\pi_{u^n} u_t^n\|_{u_n}^2] dz \leq C_0.$$

Here for the first inequality, we have used the disjointness of the discs, and for the second inequality we are choosing $\eta = 1$ in (13.3). From this we learn that there exists a subsequence (not labeled) so that

$$(13.17) \quad \lim_{n \rightarrow \infty} \int_{B(0, \delta_n R_n)} (\|\pi_{v^n}(v_\tau^n)\|_{v^n}^2 + \|\pi_{v^n}(v_t^n)\|_{v^n}^2) dz = 0.$$

We wish to study the large n limit of v^n . As in Chapter 12, if we write ∂ for a partial differentiation operator, and write $\tilde{v}^n$ for ∂v^n , then (13.16) implies

$$(13.18) \quad \tilde{v}_\tau^n + \hat{J}^n(v^n)\tilde{v}_t^n = g^n,$$

where $g^n = \partial(\hat{J}(v^n))v^n$. From the Lipschitzness of J , and (13.11) we learn

$$\sup_n \|g^n\|_{L^\infty} < \infty.$$

From this and Theorem D.4 below we deduce

$$\sup_n \|\tilde{v}^n\|_{W^{1,p}(B)} < \infty,$$

for every $p > 1$, and every bounded set B (in fact we have $D\tilde{v}^n \in BMO(B)$). This implies

$$\sup_n \|v^n\|_{W^{2,p}(B)} < \infty.$$

This implies that we can choose a subsequence of v^n that converges strongly to v in $W_{loc}^{1,p}$. We have that $v : \mathbb{R}^2 \rightarrow \hat{M}$ satisfies

$$(13.19) \quad v_\tau + \hat{J}(v)v_t = 0.$$

From (13.3) we learn

$$\int_B (\|\pi_v(v_\tau)\|_v^2 + \|\pi_v(v_t)\|_v^2) dz = 0,$$

from every bounded domain B . From this and (13.15) we learn

$$(13.20) \quad \|\nabla v\| \leq 2, \quad \|\nabla v(0)\| = 1, \quad \int_{\mathbb{R}^2} (\|\pi_v(v_\tau)\|_v^2 + \|\pi_v(v_t)\|_v^2) dz = 0.$$

Moreover, if $v^n = (\bar{w}^n, \bar{u}^n)$ and $v = (\bar{w}, \bar{u})$, then $\mathcal{E}_2(\hat{u}) \leq C_0$ yields

$$\frac{1}{2} \int_{B(0, \delta_n R_n)} \eta'(\bar{w}^n) (|\nabla \bar{w}^n|^2 + \alpha_{\bar{u}^n}(\bar{u}_t^n)^2 + \alpha_{\bar{u}^n}(\bar{u}_\tau^n)^2) d\tau dt \leq C_0,$$

which in turns implies

$$(13.21) \quad \frac{1}{2} \int_{\mathbb{R}^2} \eta'(\bar{w}) (|\nabla \bar{w}|^2 + \alpha_{\bar{u}}(\bar{u}_t)^2 + \alpha_{\bar{u}}(\bar{u}_\tau)^2) d\tau dt \leq C_0.$$

From this and (13.20) we deduce

$$\mathcal{E}_1(v) = 0, \quad \mathcal{E}_2(v) < \infty.$$

By (13.9), the function $\bar{w}$ is harmonic. Hence $\nabla \bar{w}$ is harmonic. This, (13.20), and Liouville's theorem implies that $\nabla \bar{w}$ is a constant: $\bar{w}(\tau, t) = c_0 + c_1\tau + c_2t$, for som constants c_0 , c_1 and c_2 . By (13.21), we also know

$$|\nabla \bar{w}|^2 \int_{\mathbb{R}^2} \eta'(\bar{w}) d\tau dt < \infty.$$

This is impossible unless $\nabla \bar{w} = 0$. This in turn implies that $\alpha_{\bar{u}}(\bar{u}_t) = \alpha_{\bar{u}}(\bar{u}_\tau) = 0$. Since $\bar{u}_t, \bar{u}_\tau \in \ell_{\bar{u}}$, we deduce that $d\bar{u} = 0$. This contradicts $\|\nabla v(0)\| = 1$. As a result, (13.10) holds.

(Step 2) We replace S with $[0, \infty) \times \mathbb{R}$, and regard $\hat{u} = (u, w) : [0, \infty) \times \mathbb{R} \rightarrow \hat{M}$, as a 1-periodic in t . Given $r > 0$, we set

$$\hat{u}^r(\tau, t) = (u(\tau + r, t), w(\tau + r, t) - w(r, 0)), \quad (\tau, t) \in [-r, \infty) \times \mathbb{R} =: S_r,$$

Clearly $\hat{u}^r$ is 1-periodic in t , and

$$\hat{u}^r(0, 0) = (u(r, 0), 0), \quad \mathcal{E}(\hat{u}^r) = \mathcal{E}(\hat{u}).$$

On account of (13.10), we also have

$$\sup_{r \geq 0} \sup_{z \in \mathbb{R}^2} |\nabla \hat{u}^r(z)| < \infty.$$

As in (11.28), we use the elliptic regularity to assert that for every bounded $U \subset \mathbb{R}^2$, and every $p \in [1, \infty)$, there exists a constant $C_1(U, p)$ such that

$$U \subset S_r \implies \|D^2 \hat{u}^r\|_{L^p(U)} \leq C_1(U, p).$$

Hence $\hat{u}^r$ is precompact in $W_{loc}^{2,p}$. Let $\bar{u}$ be a limit point $\hat{u}^r$. That is, for some sequence $r_k \rightarrow \infty$, we have

$$\tilde{u} = \lim_{k \rightarrow \infty} \hat{u}^{r_k}.$$

Then $\tilde{u} = (\bar{u}, \bar{w}) : \mathbb{R}^2 \rightarrow \hat{M}$ is Lipschitz, $\tilde{u} \in W_{loc}^{2,p}$, $\tilde{u}$ is 1-periodic in t , and

$$\mathcal{E}(\tilde{u}) = \mathcal{E}(\hat{u}).$$

We claim that $\mathcal{E}^1(\tilde{u}) = 0$. Indeed for every bounded interval I ,

$$\begin{aligned} \int_{I \times [0,1]} (d\alpha)_{\bar{u}}(\bar{u}_\tau, \bar{u}_t) \, d\tau dt &= \lim_{k \rightarrow \infty} \int_{I \times [0,1]} (d\alpha)_{\bar{u}^{r_k}}(\bar{u}_\tau^{r_k}, \bar{u}_t^{r_k}) \, d\tau dt \\ &= \lim_{k \rightarrow \infty} \int_{(I+r_k) \times \mathbb{S}^1} (d\alpha)_u(u_\tau, u_t) \, d\tau dt = 0, \end{aligned}$$

because

$$\int_S (d\alpha)_u(u_\tau, u_t) \, d\tau dt < \infty.$$

As a result, $u_\tau, u_t \in \ell_{\bar{u}}$, and

$$\mathcal{E}(\tilde{u}) = \mathcal{E}_2(\bar{u}).$$

From (13.9), we learn that $\bar{w}$ is harmonic. We now claim that $\nabla \bar{w}$ is constant. Observe that if $\bar{w}'$ is a harmonic conjugate for w , then $F = \bar{w} + i\bar{w}' : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic and 1-periodic in t . From the periodicity of $\bar{w}$, we deduce that $\bar{w}'(\tau, t+1) = \bar{w}'(\tau, t) + c$, for a constant c . Without loss of generality, we may assume that $c > 0$. We use the map $\pi : \mathbb{R} \rightarrow [0, c]$, $p(t) = t \pmod{c}$ to project $\mathbb{R}$ onto a circle $\mathbb{S} = [0, c]/(0 = c)$. Hence F is a lift of a map from $f : \mathbb{R} \times \mathbb{S}^1 \rightarrow \mathbb{R} \times \mathbb{S}$, where $f = \bar{w} + i\bar{w}''$, with $\bar{w}'' = \bar{w} \pmod{c}$. The map is holomorphic with respect to the complex structure i . In fact, this map can be regarded as a holomorphic curve as in (Step 1), where the contact manifold (M, α) is replaced with the one dimensional contact manifold $(\mathbb{S}, dt)$, where dt is the (non-exact) standard 1 form of the circle. Moreover, the corresponding Hofer energy $\mathcal{E}(f) \in (0, \infty)$, with $\mathcal{E}^1(f) = 0$ and $\mathcal{E}^2(f) \in (0, \infty)$, with the latter given by

$$\mathcal{E}^2(f) = \sup_{\eta \in \Gamma} \int_{f(\mathbb{S})} \eta'(\tau) d\tau \wedge dt.$$

This means that the bound (13.11) holds for f . In other words, $\nabla \bar{w}$ is uniformly bounded. But ∇w is a harmonic function. Hence $\nabla \bar{w}$ is constant by Liouville's theorem. Since $\bar{w}$ is periodic in t , we must have $\bar{w}(\tau, t) = c_1\tau + c_2$ for constants c_1 and c_2 . Note

$$\int_{f(\mathbb{S})} \eta'(\tau) d\tau \wedge dt = \int_{\mathbb{R}} \int_0^c \eta'(c_1\tau + c_2) d\tau \wedge dt = |c_1|^{-1}(\eta(\infty) - \eta(0))c.$$

Hence $\mathcal{E}(f) = |c_1|^{-1}c = T$. This completes the proof. $\square$

As we emntioned in the beginning of this chapter, we can state our results purely in terms of w :

Definition 13.2 Given $w : S \rightarrow \mathbb{R}$, define

$$\begin{aligned} \mathcal{E}(w) &= \int_S |\Delta w| dz + \sup_{\zeta \in P} \int_S \zeta(w) |\nabla w|^2 dz, \\ \mathcal{B}(w) &= \int_S |\Delta w| dz + \sup_{\zeta \in P} \left(\int_S \zeta(w) |\nabla w|^2 dz \right)^{1/2}, \end{aligned}$$

where P is the set of probability densities: $\zeta : \mathbb{R} \rightarrow [0, \infty)$, with $\int \zeta(\tau) d\tau = 1$. $\square$

Proposition 13.2 Let $S = [0, \infty) \times \mathbb{S}^1$ with $\mathbb{S}^1 = [0, 1]/0 = 1$. Then there exists a constant C_0 such that

$$(13.22) \quad \sup_{z \in S} |\nabla w(z)| \leq C_0 \mathcal{B}(w).$$

Moreover, $\bar{w} : \mathbb{R} \times \mathbb{S}^1 \rightarrow \mathbb{R}$ is a limit point of the family

$$(w^r(\tau, t) = w(\tau + r, t) : r \in [0, \infty)), \quad w^r : [-r, \infty) \times \mathbb{S}^1 \rightarrow \mathbb{R},$$

with respect to L_{loc}^∞ , then $w(\tau, t) = c_0\tau + c_1$, for a constant c_1 , and $c_0 = \mathcal{E}(w)$.

The proof of Proposition 13.2 is the same as the proof of Theorem 13.1. The key point is the scale invariance of $\mathcal{E}(w)$. More precisely, regarding w as a map $w : \mathbb{R}^2 \rightarrow \mathbb{R}$, that is 1-periodic in t , and writing $\hat{w}(z) = w(z_0 + k^{-1}z)$, then

$$\int_{B_{k\delta}(z_0)} |\Delta \hat{w}| dz = \int_{B_\delta(z_0)} |\Delta w| dz, \quad \int_{B_{k\delta}(z_0)} \zeta(\hat{w}) |\nabla \hat{w}|^2 dz = \int_{B_\delta(z_0)} \zeta(w) |\nabla w|^2 dz.$$

We now examine the question of the existence of a holomorphic curve of finite Hofer Energy. For this, we replace an infinite cylinder with a finite cylinder, and introduce boundary conditions. A natural candidate for the boundary conditions would be capping up the finite cylinder by two disks and search for a holomorphic sphere. As in Chapter 11, we choose standard holomorphic/antiholomorphic disks for the caps. Let us write $S_r = [0, r] \times \mathbb{S}^1$ for our finite cylinder. The following Proposition, will help us to design a strategy for constructing a family of finite cylinder with uniformly bounded Hofer energies:

Proposition 13.3 *Let $\hat{u} \in \mathcal{H}(S_r, J)$ and $\eta \in \Gamma$. Then*

$$(13.23) \quad \mathcal{E}(\hat{u}) \leq (c_0(\eta)^{-1} + c_1(\eta)^{-1}) E(\eta),$$

where

$$E(\eta) := \int_{\hat{u}(S_r)} d(\eta\alpha), \quad c_0(\eta) := \inf \eta, \quad c_1(\eta) := \sup_{s_0 \in \mathbb{R}} \inf_{s \in [s_0, s_0+1]} \eta'(s).$$

Proof Evidently

$$(13.24) \quad c_0(\eta) \mathcal{E}^1(\hat{u}) \leq \int_{\hat{u}(S_r)} \eta d\alpha \leq E(\eta).$$

On the other hand, if $c'_1 < c_1$, then we can find a point $s_0 \in \mathbb{R}$ such that

$$\inf_{s \in [s_0, s_0+1]} \eta'(s) \geq c'_1.$$

As a result,

$$\begin{aligned} E(\eta) &\geq \int_{\hat{u}(S_r)} \eta' ds \wedge \alpha \geq c'_1 \int_{\hat{u}(S) \cap (M \times [s_0, s_0+1])} ds \wedge \alpha \\ &= c'_1 \int_{s_0}^{s_0+1} \left[\int_{u(M \cap w^{-1}(s))} \alpha \right] ds, \end{aligned}$$

where for the last equality we used coarea formula as in (13.4), and (13.5). Hence, there exists $s_1 \in [s_0, s_0 + 1]$ such that

$$E(\eta) \geq c'_1 \int_{u(M \cap w^{-1}(s_1))} \alpha.$$

This and (13.7) imply

$$\mathcal{E}^2(\hat{u}) \leq (c'_1)^{-1} E(\eta) + \mathcal{E}^1(\hat{u}).$$

This, and (13.24) imply (13.23). $\square$

We are now ready to tackle Weinstein's conjecture when (M, α) is symplectically fillable. Let us assume that (W, Ω) is a symplectic manifold, $Y \subset W$, and (U, Ω) is a neighborhood of Y that is isomorphic to $(\hat{M}, d\hat{\alpha})$, where $\hat{M} = M \times (-\varepsilon, \varepsilon)$, and $\hat{\alpha} = e^s \alpha$. We then replace U with $(M \times [-L, L], d(\eta_L \alpha))$, for a suitable $\eta_L = e^\zeta_L$. On account of Proposition 13.3, the function $\zeta_L : [-L, L] \rightarrow [-\varepsilon, \varepsilon]$ is chosen so that

$$\inf_L c_0(\zeta_L), \inf_L c_1(\zeta_L) > 0.$$

For the lower bound on c_1 , we may choose ζ_L as $z_L(s) = s + L - \varepsilon$ for $s \in [L - 1, L]$, and $z_L(s) = s - L + \varepsilon$ for $s \in [-L, -L + 1]$.

13.1 Feral Curves and Invariant Measures

Recall that the triplet (M, α, ω) is a *framed Hamiltonian manifold* if $\ell_x(\omega) \cap \xi_x(\alpha) = \{0\}$ for every $x \in M$. Also the corresponding *Reeb vector field* $R = R^\alpha$ is the unique $R \in \ell_x$ such that $\alpha(R) = 1$. As before, the R -projection onto ξ is denoted by π . Let $J : \xi \rightarrow \xi$ be a complex structure that is compatible with the symplectic bundle (ξ, ω) . As before we define the metrics

$$g(a, a') = \omega(a, Ja'), \quad G(a, a') = g(a, a') + \alpha(a)\alpha(a').$$

Definition 13.3 By a J -holomorphic curve we mean $u : S \rightarrow M$ such that (13.2) holds. Equivalently, $\hat{u} = (u, w) : S \rightarrow M \times \mathbb{R}$ such that (13.1) holds. We define

$$\mathcal{E}^1(u) = \int_{u(S)} \omega = \frac{1}{2} \int_S (\|\pi_u(u_\tau)\|_J^2 + \|\pi_u(u_t)\|_J^2) d\tau dt.$$

When $\mathcal{E}^1(\hat{u}) < \infty$, and $\mathcal{E}^2(\hat{u}) = \infty$, we say that u is a *feral* J -holomorphic curve. $\square$

As we have seen in Theorem 3.1, $\mathcal{E}(u) < \infty$ can be used to produce a periodic orbit. As we have seen in Proposition 13.1, $\mathcal{E}^1(u) < \infty$ forces $u(S)$ to be asymptotically (as w gets large) to be close to a union of Reeb orbits, whereas $\mathcal{E}^2(u) < \infty$ guarantees periodic orbits. Following the work of Fish and Hofer [FH] and Prasad [P], We now discuss how feral curves can be used to produce invariant sets and measures that are not associated with periodic orbits.

Note that for a J -holomorphic curve $\hat{u} = (u, w) : S \rightarrow M \times \mathbb{R}$, with $\mathcal{E}^1(\hat{u}) < \infty$, and a sequence $s_k \rightarrow \infty$, we have

$$\lim_{k \rightarrow \infty} \int_{\hat{u}(S) \cap ((s_k, s_k+1) \times M)} \omega = 0.$$

Define $\theta_a : M \times \mathbb{R} \rightarrow M \times \mathbb{R}$ by $\theta_a(s, x) = (s - a, x)$, and set

$$U_k = \theta_{s_k}(\hat{u}(S) \cap (s_k, s_k + 1)) \subset M \times (0, 1).$$

Evidently,

$$(13.25) \quad \lim_{k \rightarrow \infty} \int_{U_k} \omega = 0.$$

We wish to use the J -holomorphic curves $(U_k : k \in \mathbb{N})$ to produce an invariant measure for Reeb vector field R .

Proposition 13.4 (*Prasad*) *For every J -holomorphic $\hat{u} : S \rightarrow \hat{M} := M \times (0, 1)$, with $\hat{u}(\partial S) \subset \partial \hat{M}$, and any 2-form β that is supported in $\hat{M}$,*

$$(13.26) \quad \left| \int_{\hat{u}(S)} \mathcal{L}_{\partial_\tau} \beta \right| + \left| \int_{\hat{u}(S)} \mathcal{L}_R \beta \right| \leq 5 \|\beta\|_{C^1} \mathcal{E}^1(\hat{u}) \sigma(\hat{u}(S)),$$

where σ denotes the 2-dimensional surface measure.

Proof We simply write $\langle \cdot, \cdot \rangle_J$ for the inner production associated with $\hat{J}$, and $\|\cdot\|_J$ for the corresponding metric. Note that since β is supported in $\hat{M}$, we have

$$(13.27) \quad \int_{\hat{u}(S)} \mathcal{L}_R \beta = \int_{\hat{u}(S)} (i_R d\beta + d i_R \beta) = \int_{\hat{u}(S)} i_R d\beta, \quad \int_{\hat{u}(S)} \mathcal{L}_{\partial_\tau} \beta = \int_{\hat{u}(S)} i_{\partial_\tau} d\beta.$$

We may write $\beta = \gamma + d\tau \wedge \zeta$, where γ is a 2-form on M , and ζ is a 1-form on M . We have

$$d\beta = d\gamma - d\tau \wedge d\zeta, \quad i_R d\beta = i_R d\gamma + d\tau \wedge i_R d\zeta, \quad i_{\partial_\tau} d\beta = i_{\partial_\tau} d\gamma - d\zeta.$$

Hence, for $\hat{v} = v + a\partial_\tau$,

$$\begin{aligned} i_R d\beta(\hat{v}, \hat{J}\hat{v}) &= d\gamma(R, \pi(v), J\pi(v)) + (d\tau \wedge i_R d\zeta)(\hat{v}, \hat{J}\hat{v}) \\ &= d\gamma(R, \pi(v), J\pi(v)) + a d\zeta(R, J\pi(v)) - \alpha(v) d\zeta(R, v), \\ i_{\partial_\tau} d\beta(\hat{v}, \hat{J}\hat{v}) &= d\gamma(\partial_\tau, \alpha(v)R + \pi(v), aR + J\pi(v)) - d\zeta(\alpha(v)R + \pi(v), aR + J\pi(v)). \end{aligned}$$

As a result,

$$\begin{aligned} |i_R d\beta(\hat{v}, \hat{J}\hat{v})| &\leq \|d\gamma\|_{C^0} \|R\|_J \|v\|_J^2 + \|d\zeta\|_{C^0} [|a| + |\alpha(v)|] \|R\|_J \|v\|_J \leq 3\|\beta\|_{C^1} \|\hat{v}\|_J^2, \\ |i_{\partial_\tau} d\beta(\hat{v}, \hat{J}\hat{v})| &\leq \|d\gamma\|_{C^0} \|v\|_J^2 + \|d\zeta\|_{C^0} \|v\|_J \|\hat{v}\|_J \leq 2\|\beta\|_{C^1} \|\hat{v}\|_J^2. \end{aligned}$$

This and (13.27) imply (13.26). $\square$

Writing $U_k = \hat{u}(S_k)$, we apply (13.26) to assert

$$(13.28) \quad \lim_{k \rightarrow \infty} \sigma(U_k)^{-1} \int_{U_k} \mathcal{L}_{\partial_\tau} \beta = \lim_{k \rightarrow \infty} \sigma(U_k)^{-1} \int_{U_k} \mathcal{L}_R \beta = 0.$$

We wish to use this to produce an invariant measure that is supported in a set that is approximated by U_k , in large k limit. Using the metric associated with J , we can put the metric on the space of 2-vectors on $\hat{M}$, and a dual metric, on the space of 2-forms as in the proof of Proposition 13.25. Given a 2 dimensional surface U in $\hat{M}$, and a 2-form γ , we may write

$$(13.29) \quad \int_U \gamma = \int \langle \gamma, N_U \rangle d\sigma_U,$$

where

- σ_U is the U -restriction of the 2-dimensional Hausdorff measure associate with $\|\cdot\|_J$.
- $N_U : U \rightarrow \Lambda_2(\hat{M})$, is an orientation of U , so that $\|N_U(x)\| = 1$.
- $\langle \cdot, \cdot \rangle$ is the pairing between 2-forms and 2-vectors.

On account of (13.28) and (13.29), we have

$$(13.30) \quad \lim_{k \rightarrow \infty} \int \langle \mathcal{L}_R \beta, d\mu_k \rangle = \lim_{k \rightarrow \infty} \int \langle \mathcal{L}_{\partial_\tau} \beta, d\mu_k \rangle = 0,$$

for every 2-form β , where

$$d\mu_k = \sigma(U_k)^{-1} N_{U_k} d\sigma_{U_k} =: N_{U_k} d\nu_k.$$

One popular interpretation is to regard the 2-vector valued measure μ_k as a current a la de Rham, and study its weak limits. Since $\|N_{U_k}\| = 1$, we know that $\|\mu_k\| = \nu_k$ is a probability measure. Let us write (ν, μ) for a limit point of the sequence $(\nu_k, \mu_k)_{k \in \mathbb{N}}$.

From

$$\left| \int \langle \zeta, N_{U_k} \rangle d\nu_k \right| \leq \int \|\zeta\| d\nu_k,$$

we deduce

$$\left| \int \langle \zeta, d\mu \rangle \right| \leq \int \|\zeta\| \, d\nu,$$

for every 2-form ζ . From this we deduce that μ is absolutely continuous with respect to ν . Moreover, if $d\mu = N \, d\nu$, then $\|N\| \leq 1$, ν -a.e.

Theorem 13.2 (*Prasad*) *The measure $\nu = \nu^0 \times m$, where ν^0 is R -invariant measure, and m is the Lebesgue measure on $\mathbb{R}$.*

Proof By Proposition 13.3, we have

$$\int \langle \mathcal{L}_{\partial_\tau} \beta, d\mu \rangle = \int \langle \mathcal{L}_R \beta, d\mu \rangle = 0,$$

for every 2-form β . This in turn implies

$$\frac{d}{dt} \int \langle \psi_t^* \beta, d\mu \rangle = \frac{d}{dt} \int \langle \varphi_t^* \beta, d\mu \rangle = 0,$$

where ψ and φ are the flows of R and ∂_τ respectively. □

A Vector Fields and Differential Forms

Lemma A.1 (*De Rham*) Let M be a compact connected orientable k -dimensional manifold and let α be a k -form with $\int_M \alpha = 0$. Then α is exact.

Proof Let $\{U_1, \dots, U_r\}$ be a finite cover of M with each U_i diffeomorphic to a simply connected subset of $\mathbb{R}^k$. To ease the notation, we assume that $r = 2$. Choose φ_1 and φ_2 with $\varphi_1 + \varphi_2 = 1$, $\varphi_1, \varphi_2 \geq 0$, $\varphi_1, \varphi_2 \in C^1$ and $\text{supp } \varphi_i \subseteq U_i$ for $i = 1, 2$. We can readily construct a k -form β such that $\text{supp } \beta \subseteq U_1 \cap U_2$ and $\int_M \beta = 1$. Define $\gamma = \varphi_1 \alpha - c\beta$ with $c = \int_M \varphi_1 \alpha$. Then $\text{supp } \gamma \subseteq U_1$ and $\int_{U_1} \gamma = 0$. By Poincaré's lemma, there exists a form $\hat{\gamma}$ such that $d\hat{\gamma} = \gamma$. Similarly, if $\tau = \varphi_2 \alpha + c\beta$, then $\text{supp } \tau \subseteq U_2$ and $\int_{U_2} \tau = \int_M (1 - \varphi_1) \alpha + c\beta = \int_M \alpha = 0$. Hence, we can apply Poincaré's lemma to find $\hat{\tau}$ with $d\hat{\tau} = \tau$. We now have

$$d(\hat{\gamma} + \hat{\tau}) = \varphi_1 \alpha - c\beta + \varphi_2 \alpha + c\beta = \alpha.$$

□

Lemma A.2 Let X be a C^1 vector field of M and write ϕ_t^X for its flow. If $\varphi : N \rightarrow M$ is a diffeomorphism, then $\psi_t = \varphi^{-1} \circ \phi_t^X \circ \varphi$, is the flow of the vector field $\varphi^* X := (d\varphi)^{-1} X \circ \varphi$.

Proof We have

$$\begin{aligned} \frac{d\psi_t}{dt}(x) &= (d\varphi^{-1})_{\phi_t^X \circ \varphi(x)} (X \circ \phi_t^X \circ \varphi)(x) = (d\varphi^{-1})_{\varphi \circ \psi_t(x)} (X \circ \varphi \circ \psi_t^X)(x) \\ &= (d\varphi)_{\psi_t(x)}^{-1} (X \circ \varphi)(\psi_t(x)), \end{aligned}$$

which means

$$\psi_t = \phi_t^{(d\varphi)^{-1} X \circ \varphi}.$$

□

The vector field $\varphi^* X$ is called the *pull-back* of X . Also, if Y is a vector field on N , we define its φ -*push-forward* by

$$(\varphi_* Y)(y) = (d\varphi)_{\varphi^{-1}(y)} Y(\varphi^{-1}(y)).$$

We define the *Lie derivative* of a vector field Y with respect to another vector field X by

$$(A.1) \quad [X, Y] = \mathcal{L}_X Y := \lim_{t \rightarrow 0} t^{-1} ((\phi_t^X)^* Y - Y).$$

Using

$$(\phi_t^X)^* (\phi_s^X)^* Y = (\phi_{s+t}^X)^* Y,$$

we can readily show

$$(A.2) \quad \frac{d}{dt} (\phi_t^X)^* Y = \mathcal{L}_X (\phi_t^X)^* Y = (\phi_t^X)^* \mathcal{L}_X Y.$$

In particular $Z(x, t) = (\phi_t^X)^* Y$ satisfies

$$Z_t = [X, Z] = \mathcal{L}_X Z.$$

Lemma A.3 *For every 1-form α and vector fields X and Y , we have*

$$(A.3) \quad d\alpha(X, Y) = \mathcal{L}_X(\alpha(Y)) - \mathcal{L}_Y(\alpha(X)) - \alpha([X, Y]).$$

Proof To ease the notation, we write ϕ_t for ϕ_t^X . By definition

$$(\phi_t^* \alpha)(\phi_t^* Y) = \alpha(Y) \circ \phi_t.$$

We now differentiate both sides with respect to t and set $t = 0$. We obtain

$$(\mathcal{L}_X \alpha)(Y) + \alpha([X, Y]) = \mathcal{L}_X(\alpha(Y)).$$

By Cartan's formula, the left hand side equals

$$(i_X d\alpha)(Y) + d(\alpha(X))(Y) + \alpha([X, Y]) = d\alpha(X, Y) + \mathcal{L}_Y(\alpha(X)) + \alpha([X, Y]),$$

which implies (A.3). □

Given a sub-bundle ξ of TM of dimension m , we may wonder whether or not there exists a foliation of M that consists of submanifolds N such that for each $x \in N$, we have $T_x N = \xi_x$. If such a foliation exists locally, we say that ξ is *integrable*. According to *Frobenius Theorem* the sub-bundle ξ is integrable iff for every vector fields $X, Y \in \xi$, we have $[X, Y] \in \xi$.

We are particularly interested in the case of $\xi_x = \ker \alpha_x$, for a 1-form α . For example, if $M = \mathbb{R}^k$ and $\alpha = u \cdot dx$, then $\xi_x = u^\perp$ consists of vectors that are perpendicular to u . If $k = 3$, $D = B_1(0)$ is the unit disk, and $w : D \rightarrow M$ parametrizes a surface with $T_x w(D) = u^\perp$, then by Stokes' theorem,

$$\begin{aligned} 0 &= \int_{w(\gamma)} \alpha = \int_{w(\Gamma)} d\alpha = \int_{\Gamma} (d\alpha)_{w(s_1, s_2)}(w_{s_1}(s_1, s_2), w_{s_2}(s_1, s_2)) ds_1 ds_2 \\ &= \int_{\Gamma} [(w_{s_1} \times w_{s_2}) \cdot (\nabla \times u)(w)] ds_1 ds_2. \end{aligned}$$

for every open subset $\Gamma \subset D$ with $\partial\Gamma = \gamma$. By varying Γ and using the assumption that u is parallel to $w_{s_1} \times w_{s_2}$, we learn that $(\nabla \times u) \cdot u \equiv 0$. More generally we have the following consequence of Lemma A.3.

Lemma A.4 *Let α be a non-degenerate 1-form on M and set $\xi = \ker \alpha$. The following statements are equivalents:*

- (i) *The sub-bundle ξ is integrable.*
- (ii) *For every vector fields $X, Y \in \xi$, we have $d\alpha(X, Y) = 0$.*
- (iii) *$\alpha \wedge d\alpha = 0$.*

Proof The equivalence of (i) and (ii) is an immediate consequence of (A.3). We now show that (ii) implies (iii). Assume (ii). Take any vector fields X, Y in ξ and any vector field R such that $\alpha(R) = 1$. Define $\pi(v) = v - \alpha(v)R$ so that π_x is the $R(x)$ -projection onto ξ_x . We have

$$\begin{aligned} (\alpha \wedge d\alpha)(v_1, v_2, v_3) &= \alpha(v_1)d\alpha(v_2, v_3) - \alpha(v_2)d\alpha(v_1, v_3) + \alpha(v_3)d\alpha(v_1, v_2) \\ &= \alpha(v_1)[\alpha(v_2)d\alpha(R, \pi(v_3)) + \alpha(v_2)d\alpha(\pi(v_2), R)] \\ &\quad - \alpha(v_2)[\alpha(v_1)d\alpha(R, \pi(v_3)) + \alpha(v_3)d\alpha(\pi(v_1), R)] \\ &\quad + \alpha(v_3)[\alpha(v_1)d\alpha(R, \pi(v_2)) + \alpha(v_2)d\alpha(\pi(v_1), R)] \\ &= 0, \end{aligned}$$

which means that $\alpha \wedge d\alpha = 0$.

Conversely, if (iii) is true and X, Y are any two vector fields in ξ , then

$$0 = (\alpha \wedge d\alpha)(R, X, Y) = \alpha(R)d\alpha(X, Y) - \alpha(X)d\alpha(R, Y) + \alpha(Y)d\alpha(R, X) = d\alpha(X, Y),$$

as desired. □

B Sobolev Inequality

It is well-known that $\mathcal{H}^{1/2}$ is a subset of the space of functions of *bounded mean oscillation* (BMO). In particular $\mathcal{H}^{1/2} \subset L^p$ for all $p \geq 2$. We will first provide an elementary proof for this when $p < 3$.

Lemma B.1 *For every $p \in [2, 3)$, there exists a constant $c_0 = c_0(p)$ such that*

$$\left(\int_0^1 |x(t)|^p dt \right)^{1/p} \leq c_0(p) \|x\|.$$

Proof We identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$ and write $-i$ for $\bar{J}$. Hence $x(t) = \sum_k e^{2\pi kt\bar{J}} x_k$ can be rewritten as $\sum_k e^{-2\pi kti} x_k$ with $x_k \in \mathbb{C}^n$. Since it suffices to establish the inequality for each component, we may assume without loss of generality that $n = 1$.

We now find an expression of $\|x\|$ that involves the function $x(\cdot)$ directly and does not involve its Fourier coefficients. We claim

$$(B.1) \quad \int_0^{2\pi} \int_0^{2\pi} \frac{|x(e^{i\theta}) - x(e^{i\varphi})|^2}{|e^{i\theta} - e^{i\varphi}|^2} d\theta d\varphi = 4\pi^2 \sum_k |k| |x_k|^2.$$

This follows from a direct calculation; the left-hand side equals

$$\begin{aligned} & \int_0^{2\pi} \int_0^{2\pi} |e^{i\theta} - e^{i\varphi}|^{-2} \left| \sum_k x_k e^{-ik\theta} - \sum_k x_k e^{-ik\varphi} \right|^2 d\theta d\varphi \\ &= \int_0^{2\pi} \int_0^{2\pi} |1 - e^{i\tau}|^{-2} \left| \sum_k x_k (1 - e^{-ik\tau}) e^{-ik\theta} \right|^2 d\theta d\tau \\ &= 2\pi \int_0^{2\pi} |1 - e^{i\tau}|^{-2} \sum_k |x_k|^2 |1 - e^{-ik\tau}|^2 d\tau = (2\pi)^2 \sum_k |k| |x_k|^2, \end{aligned}$$

because

$$\int_0^{2\pi} \frac{|1 - e^{-ik\tau}|^2}{|1 - e^{i\tau}|^2} d\tau = \int_0^{2\pi} (e^{-i(k-1)\tau} + \dots + 1)(e^{i(k-1)\tau} + \dots + 1) d\tau = 2\pi.$$

Let us write $\Lambda(x)$ for the left-hand side of (B.1). To simplify the notation write $y(\theta) = x(e^{i\theta})$. We have that for constants c_1 and c_2 ,

$$(B.2) \quad \int_0^{2\pi} \int_0^{2\pi} |y(\theta) - y(\varphi)|^2 \mathbb{1}(|\theta - \varphi| < l^{-1}) d\theta d\varphi \leq c_1 l^{-2} \Lambda(x) = c_2 l^{-2} \|x\|^2.$$

Given $l \geq 1$, define $z^l(t) = l \int_0^{l^{-1}} x(t + \alpha) d\alpha$. By (B.2),

$$\begin{aligned} \int_0^1 |x(t) - z^l(t)|^2 dt &= \int_0^1 \left| l \int_0^{l^{-1}} (x(t) - x(t + \alpha)) d\alpha \right|^2 dt \\ &\leq l \int_0^1 \int_0^{l^{-1}} |x(t) - x(t + \alpha)|^2 dt d\alpha \\ &\leq c_2 l^{-1} \|x\|^2. \end{aligned}$$

From this we deduce that for every $l \geq 1$,

$$x = z^l + w^l$$

with

$$\|w^l\|_0^2 \leq c_2 l^{-1} \|x\|^2, \quad \|z^l\|_{L^\infty} \leq l \|x\|,$$

because

$$|z^l(t)| \leq l \int_0^1 |x(t)| \, dt \leq l \|x\|_0.$$

Hence we apply Chebyshev's inequality to assert

$$|\{t : |x(t)| > 2l\|x\|\}| \leq |\{t : |w^l(t)| > l\|x\|\}| \leq \frac{\|w^l\|_0^2}{l^2 \|x\|^2} \leq c_2 l^{-3},$$

whenever $l \geq 1$. On the other hand

$$\begin{aligned} \int_0^1 |x(t)|^p \, dt &\leq 1 + \int_0^\infty p l^{p-1} |\{x > l\}| \, dl \\ &\leq 1 + c_3 \|x\|^3 \int_1^\infty p l^{p-1} l^{-3} \, dl \\ &= 1 + c_4(p) \|x\|^3 \end{aligned}$$

with $c_4(p) < \infty$ whenever $p < 3$. Finally, we replace x with λx , $\lambda > 0$ to deduce

$$\|x\|_{L^p}^p \leq \lambda^{-p} + c_4(p) \lambda^{3-p} \|x\|^3.$$

Minimizing the right-hand side over $\lambda > 0$ yields the desired inequality. □

We define the BMO norm and seminorm $[\cdot]_{1/2}$ as follow,

$$\begin{aligned} \|y\|_*^2 &= \sup_{\varepsilon > 0} \sup_t \int_0^1 \int_0^1 |y(t + \varepsilon\theta) - y(t + \varepsilon\tau)|^2 \, d\theta d\tau, \\ \|y\|_{**}^2 &= \sup_{\varepsilon > 0} \sup_t \int_0^1 \int_0^1 |y(t + \varepsilon\theta) - y_\varepsilon(t)|^2 \, d\theta, \\ [y]_{1/2}^2 &= \int_0^1 \int_0^1 \frac{|y(t) - y(s)|^2}{|t - s|^2} \, ds dt' \end{aligned}$$

where

$$y_\varepsilon(t) = \int_0^1 y(t + \varepsilon\theta) \, d\theta.$$

Proposition B.1 *We have*

$$(B.3) \quad \|y\|_{**} \leq \|y\|_* \leq [y]_{1/2}.$$

Proof We simply have,

$$\begin{aligned} \int_0^1 \int_0^1 |y(t + \varepsilon\theta) - y(t + \varepsilon\tau)|^2 d\theta d\tau &= \varepsilon^{-2} \int_t^{t+\varepsilon} \int_t^{t+\varepsilon} |y(\theta) - y(\tau)|^2 d\theta d\tau \\ &\leq \int_t^{t+\varepsilon} \int_t^{t+\varepsilon} \frac{|y(\theta) - y(\tau)|^2}{|\theta - \tau|^2} d\theta d\tau \\ &\leq [y]_{1/2}^2, \end{aligned}$$

as desired. $\square$

By a celebrated inequality of John and Nirenburg, we can bound the L^p norm of a function by its BMO norm.

C Degree Theory

We first review the classical Brouwer degree theory. Consider triplets (f, U, y) with $U \subseteq \mathbb{R}^d$ open and bounded, $f : \bar{U} \rightarrow \mathbb{R}^d$ continuous and $y \notin f(\partial U)$. We now would like to assign an integer $\deg(f, U, y)$ to (f, U, y) that, in some sense, counts the solutions to the equation $f(x) = y$, $x \in U$, with a sign. This degree satisfies the following properties:

- (i) If $V \subseteq U$ and $f^{-1}(\{y\}) \subseteq V$, then $\deg(f, V, y) = \deg(f, U, y)$.
- (ii) For a constant a , $\deg(f + a, U, y + a) = \deg(f, U, y)$.
- (iii) If $U \cap V = \emptyset$ and $y \notin f(\partial U) \cup f(\partial V)$, then $\deg(f, U \cup V, y) = \deg(f, U, y) + \deg(f, V, y)$.
- (iv) If $f : \bar{U} \times [0, 1] \rightarrow \mathbb{R}^d$ is continuous with $y \notin f(\partial U, t)$ for every $t \in [0, 1]$, then

$$\deg(f(\cdot, 1), U, y) = \deg(f(\cdot, 0), U, y).$$

- (v) If $\deg(f, U, y) \neq 0$, then $f(x) = y$ has a solution in U .
- (vi) $\deg(id, B, 0) = 1$ where $B = \{x : |x| < 1\}$.

It turns out that the above properties determine “deg” uniquely. Indeed one can show that for any triplet as above, we can find (g, U, y) with g smooth and f and g homotopic. As for $g \in C^1$, deg is defined by

$$\deg(g, U, y) = \sum_{x \in f^{-1}\{y\}} \text{sgn}(\det g'(x)).$$

In the same fashion, we can define the degree of a continuous map between manifolds. Given two compact manifolds M and N , and a C^1 map $f : N \rightarrow M$, we say $x \in N$ is *regular*

if df_x is invertible. We say $x \in M$ is a *regular value* if $f^{-1}\{x\}$ consists of regular points. By inverse mapping theorem, it is not hard to show that if x is a regular value, then $f^{-1}\{x\}$ is finite. For such a value we may define the degree by

$$(C.1) \quad \deg_x(f) = \sum_{y \in f^{-1}\{x\}} \epsilon_y,$$

where $\epsilon_y = \pm 1$ according to whether df_x preserves or reverses orientation. In the Euclidean case $\epsilon_y = \text{sgn det } D_x f$. The degree of a continuous $f : N \rightarrow M$ defined to be the degree of a C^1 function $g : N \rightarrow M$ that is sufficiently close to f . As we will see in Lemma C.1, this is well-defined.

Lemma C.1 (i) *If $f : N \rightarrow M$ is a C^1 function and Ω is a volume form with $\Omega > 0$, $\int_M \Omega = 1$, then for every regular value $x \in M$,*

$$\deg_x f = \int_N f^* \Omega.$$

(ii) *The degree is invariant under homotopies consisting of C^1 -maps.*

(iii) *Any two C^1 -maps $f, g : N \rightarrow M$ that are sufficiently C^0 -close are homotopic via C^1 -maps.*

(iv) *Let X be an orientable manifold with $\partial X = N$ and let $F : X \rightarrow M$ be a continuous map. Then the degree of $f = F|_N$ is zero.*

Proof (i) Let x be a regular value and assume $f^{-1}\{x\} = \{x_1, \dots, x_k\}$. Find an open neighborhood V of x such that $f^{-1}(V) = U_1 \cup \dots \cup U_k$ with $U_1, \dots, U_k$, open and disjoint, $x_i \in U_i$ for $i = 1, \dots, k$, and $f|_{U_i} : U_i \rightarrow V$ a diffeomorphism for every i . We now take an n -form α with support in V such that $\int_V \alpha = 1$. By Lemma A.1, we may find an $(n-1)$ -form β such that $\Omega = \alpha + d\beta$. We now have

$$\begin{aligned} \int_N f^* \Omega &= \int_N f^* \alpha + \int_N df^* \beta = \int_N f^* \alpha \\ &= \sum_{i=1}^k \int_{U_i} f^* \alpha = \sum_{i=1}^k \epsilon_{x_i}. \end{aligned}$$

(ii) Clearly degree is C^1 -continuous and locally constant.

(iii) Put a Riemannian metric on M . Given $x \in N$, we may find a geodesic curve connecting $f(x)$ to $g(x)$. We now define $\psi(t, x) = \gamma(t; f(x), g(x))$ where $\gamma(t; a, b)$ is defined to be a point

on the geodesic connecting a to b with $\gamma(0; a, b) = a$, $\gamma(1; a, b) = b$. This can be done smoothly for a, b sufficiently close.

(iv) Let Ω be a volume form on M with $\int_M \Omega = 1$, $\Omega > 0$. We then have

$$\deg(f) = \int_N f_* \Omega = \int_{\partial X} f_* \Omega = \int_X d(F_* \Omega) = \int_X F_* d\Omega = 0.$$

□

The *Leray–Schander Theory* allows us to have a similar notion of degree for functions of the form $f = I + L : \bar{U} \rightarrow \mathcal{E}$ with $\mathcal{E}$ a Banach space, U a bounded open subset of $\mathcal{E}$, and L a compact operator. Again we wish to define $\deg(f, U, y)$ provided that $y \notin f(\partial U)$. To do so, first we find a sequence $L_m : \bar{U} \rightarrow \mathcal{E}$ such that the range of L_m , denoted by $\mathcal{E}_m(\bar{U})$, is a subset of finite dimensional space $\mathcal{E}_m$, and

$$\lim_{m \rightarrow \infty} \sup_{x \in \bar{U}} \|L_m(x) - L(x)\| = 0.$$

We then set

$$(C.2) \quad \deg(f, U, y) = \deg(f_m, \mathcal{E}_m \cap U, y),$$

for large m , where $f_m = I + L_m : \mathcal{E}_m \cap \bar{U} \rightarrow \mathcal{E}_m$. For this to work, we need to check that $y \notin f_m(\partial U)$ for sufficiently large m . By Exercise C(ii), the set $f(\partial U)$ is closed. Since $x \notin f(\partial U)$, we have $\text{dist.}(x, f(\partial U)) = \delta > 0$. Then we find m_0 such that if $m > m_0$, then $\text{dist.}(x, f_m(\partial U)) \geq \delta/2$. Hence the right-hand side of (A.2) is well-defined by Lemma C.2 below.

Lemma C.2 *Let U be an open bounded subset of $\mathbb{R}^n = \mathbb{R}^{n_1} \oplus \mathbb{R}^{n_2}$. Consider $f(x) = x + L(x)$ with $L : \bar{U} \rightarrow \mathbb{R}^{n_1}$, $f : \bar{U} \rightarrow \mathbb{R}^n$. If $y \notin f(\partial U)$ and $y \in \mathbb{R}^{n_1}$, then*

$$\deg(f, U, y) = \deg(f|_{U_1}, U_1, y)$$

where $U_1 = U \cap \mathbb{R}^{n_1}$.

Proof We may assume $f \in C^1(U)$ and $y = 0$. Let us take two continuous functions φ_1, φ_2 with $\varphi_i : \mathbb{R}^{n_i} \rightarrow \mathbb{R}$, $\int \varphi_i dy_i = 1$, for $i = 1, 2$, and both φ_1, φ_2 have support near 0. Set $\varphi(y_1, y_2) = \varphi_1(y_1)\varphi_2(y_2)$ and $\omega = \varphi(y_1, y_2)dy_1dy_2$ is a volume form of total volume 1. We then have

$$\deg(f, U, y) = \int_U f^* \omega,$$

by Lemma A.1. But $\det(I_n + \nabla L) = \det\left(I_{n_1} + \frac{\partial L}{\partial x_1}\right)$. As a result,

$$\deg(f, U, y) = \int \varphi_1(x_1 + L(x))\varphi_2(x_2) \det\left(I_{n_1} + \frac{\partial L}{\partial x_1}\right) dx_1 dx_2.$$

We may send φ_2 to δ_0 to yield

$$\begin{aligned}\deg(f, U, y) &= \int \varphi_1(x_1 + L(x)) \det \left(I_{n_1} + \frac{\partial L}{\partial x_1} \right) dx_1 \\ &= \deg(f|_{U_1}, U_1, 0).\end{aligned}$$

□

Exercise C Let $\mathcal{E}$ be a Banach space and $K : \Omega \rightarrow \mathcal{E}$ be a compact operator with Ω a bounded closed subset of X .

- (i) Show that K is a uniform limit of finite dimensional transformations. **Hint:** Cover the compact set $\overline{K(\Omega)}$ by finitely many open balls, and use a partition of unity.
- (ii) Show that $I + K$ maps closed sets to closed sets.

□

D Cauchy and Beurling Transforms

A classical way of solving the Laplace and Poisson equation in an bounded open subset of $\mathbb{R}^k$ with regular boundary is by first finding its Green's function. That is a function $G : \bar{U} \times \bar{U} \rightarrow \mathbb{R}$, such that

$$\begin{cases} \Delta_x G(x, y) = \delta_y(dx), & x \in U, \\ G(x, y) = 0, & x \in \partial U. \end{cases}$$

Once such G is found, we then use Green's identity to derive the following identity:

$$(D.1) \quad u(x) = \int_U \Delta u(y) G(x, y) dy + \int_{\partial U} u(y) \frac{\partial G}{\partial n}(x, y) dy,$$

where $\partial G / \partial n$ denotes the normal derivative of G . Once the Green's function G and the *Poisson's function* $\partial G / \partial n$ are known, then we can use (D.1) to solve the PDE

$$\begin{cases} \Delta u(x) = f(x), & x \in U, \\ u(x) = g(x), & x \in \partial U. \end{cases}$$

for the given f and g . In $\mathbb{C}$ an analogous representation formula can be derived that in turn can be used to solve the celebrated *d-bar and Beltrami equations*.

Proposition D.1 (*Cauchy-Pompeiu Formula*) Let $U \subset \mathbb{C}$ a bounded domain with a boundary that is positively oriented and parametrized by a curve γ . Then, for every C^1 function $f : \mathbb{C} \rightarrow \mathbb{C}$,

$$(D.2) \quad \begin{aligned} f(z) &= \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta + \frac{1}{2\pi i} \iint_U \frac{f_{\bar{\zeta}}(\zeta)}{\zeta - z} d\zeta \wedge d\bar{\zeta}, \\ f(z) &= -\frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{\bar{\zeta} - \bar{z}} d\bar{\zeta} + \frac{1}{2\pi i} \iint_U \frac{f_{\zeta}(\zeta)}{\bar{\zeta} - \bar{z}} d\zeta \wedge d\bar{\zeta}. \end{aligned}$$

Proof We only derive the first formula in (D.2) because the second identity can be established by a verbatim argument. Write $\zeta = s + it$ and pick any C^1 function $g = u + iv$. Note

$$\frac{i}{2}(d\zeta \wedge d\bar{\zeta}) = ds \wedge dt.$$

Using the Green's formula,

$$(D.3) \quad \begin{aligned} \oint_{\gamma} g(\zeta) d\zeta &= \oint_{\gamma} (u ds - v dt) + i(u dt + v ds) = \iint_U [-(u_t + v_s) + i(u_s - v_t)] ds dt \\ &= i \iint_U [(u_s - v_t) + i(u_t + v_s)] ds dt = i \iint_U (g_s + i g_t) ds dt \\ &= 2i \iint_U g_{\bar{\zeta}} ds dt. \end{aligned}$$

A more compact version of the above calculation is

$$\oint_{\gamma} g(\zeta) d\zeta = \iint_U dg \wedge d\zeta = \iint_U g_{\bar{\zeta}} d\bar{\zeta} \wedge d\zeta = 2i \iint_U g_{\bar{\zeta}} ds dt.$$

We now take $z \in U$, choose $\varepsilon > 0$ so small that $B_{\varepsilon}(z) \subset U$, set $g(\zeta) = f(\zeta)/(\zeta - z)$ and replace U with $U/B_{\varepsilon}(z)$ in (D.3) to deduce

$$\frac{1}{2\pi i} \oint_{|\zeta-z|=\varepsilon} \frac{f(\zeta)}{\zeta - z} d\zeta = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{\pi} \iint_U \frac{f_{\bar{\zeta}}(\zeta)}{\zeta - z} ds dt.$$

It remains to show that the right-hand side converges to $f(z)$ as $\varepsilon \rightarrow 0$. For this it suffices to check

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi i} \oint_{|\zeta-z|=\varepsilon} \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = 0.$$

This is an immediate consequence of the Lipschitzness of f that implies the boundedness of the integrand. $\square$

Given a C^1 function $h : \bar{U} \rightarrow \mathbb{C}$, we wish to find $w : \bar{U} \rightarrow \mathbb{C}$ such that $w_{\bar{z}} = h(z)$ in U . Indeed if w is a solution, then by (D.2), $w = \mathcal{C}(h) + \Gamma(h)$, where the Cauchy operators Γ and $\mathcal{C}$ are defined by

$$\Gamma(g) = \frac{1}{2\pi i} \int_{\gamma} \frac{g(\zeta)}{\zeta - z} d\zeta, \quad \mathcal{C}(h) = -\frac{1}{\pi} \iint_U \frac{h(\zeta)}{\zeta - z} dsdt.$$

This means that a solution can be expressed as $w = \mathcal{C}(h) + q$ for a holomorphic function q . Put this differently, what we learn from (D.2) is

$$(D.4) \quad \frac{\partial}{\partial \bar{z}} \mathcal{C}(h) = h, \quad \frac{\partial}{\partial z} \bar{\mathcal{C}}(h) = h,$$

where

$$\bar{\mathcal{C}}(h) = -\frac{1}{\pi} \iint_U \frac{h(\zeta)}{\bar{\zeta} - \bar{z}} dsdt.$$

With the aid of the Cauchy operator/transform $\mathcal{C}$, we can also solve the Beltrami's equation

$$(D.5) \quad \varphi_{\bar{z}} = \mu \varphi_z.$$

This is an elliptic PDE only if $|\mu(z)| < 1$ for all z . We may set $h = \varphi_{\bar{z}}$ and express φ as $\varphi = \mathcal{C}(h) + q$, for a holomorphic function q . Observe that if φ satisfies (D.5), then

$$h = \varphi_{\bar{z}} = \mu \varphi_z = \mu \mathcal{B}(h) + \mu q',$$

where $\mathcal{B}$ is the *Beurling's operator*:

$$\mathcal{B}(h) = \frac{\partial}{\partial z} \mathcal{C}(h).$$

This means that the function h satisfies

$$(D.6) \quad (I - \mu \mathcal{B})h = \mu q'.$$

In other words, for every holomorphic q , we first solve (D.6) for h and using this $h = h^q$, we have a solution for (D.5) in the form $\varphi = q + \mathcal{C}(h^q)$. Note that $q = 0$ yields the trivial solution $\varphi = 0$. For a nontrivial example, search for a solution of the form $\varphi = \mathcal{C}(h) + 1$, where h solves

$$(I - \mu \mathcal{B})h = \mu.$$

This or more generally (D.6) can be solve if $\|\mu \mathcal{B}\| < 1$, for a suitable operator norm. As we will see below, if we take the operator norm with respect to the L^2 space, then $\|\mathcal{B}\| = 1$ and if $\sup_z \|\mu(z)\| < 1$, then we can invert $I - \mu \mathcal{B}$ and solve (D.6) in L^2 .

The operator $\mathcal{B}$ is the complex-variable analog of the *Hilbert Transform*; from (D.4) we can readily deduce

$$\mathcal{B}(f_{\bar{z}}) = f_z + Q,$$

for some holomorphic Q . It turns out that if we assume $U = \mathbb{C}$ and that f and its first derivative vanish at infinity, then $Q = 0$ (because a bounded entire function is constant), and we simply have $\mathcal{B}(f_{\bar{z}}) = f_z$. If we write

$$\mathcal{F}(h)(\eta) = \hat{h}(\eta) = \int \exp(2\pi i \operatorname{Re}(z\bar{\eta})) h(z) dz = \int \exp(\pi i(z\bar{\eta} + \bar{z}\eta)) h(z) dz,$$

for the Fourier Transform of h , then we have

$$\mathcal{F}(f_z)(\eta) = -\frac{i\bar{\eta}}{2} \mathcal{F}(f)(\eta), \quad \mathcal{F}(f_{\bar{z}})(\eta) = -\frac{i\eta}{2} \mathcal{F}(f)(\eta).$$

Hence

$$\mathcal{F}(\mathcal{B}(h)) = \frac{\bar{\eta}}{\eta} \mathcal{F}(h)(\eta).$$

From this and Plancherel's equation we deduce that

$$(D.7) \quad \|\mathcal{B}(h)\|_{L^2} = \|h\|_{L^2}.$$

In fact $\mathcal{B}$ is a singular operator that may be defined by

$$\mathcal{B}(h)(z) = \frac{1}{\pi} PV \int \frac{h(\zeta)}{(\zeta - z)^2} dsdt = \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|\zeta - z| > \varepsilon} \frac{h(\zeta)}{(\zeta - z)^2} dsdt,$$

and is also bound on L^p for $p \in (1, \infty)$ by *Calderon-Zygmund Theory*.

Theorem D.1 *Define the operators $\mathcal{C}$ and $\mathcal{B}$ on smooth functions by*

$$(D.8) \quad \mathcal{C}(h)(z) = -\frac{1}{\pi} \int_{\mathbb{D}} \frac{h(\zeta)}{\zeta - z} dsdt, \quad \mathcal{B}(h)(z) = \frac{1}{\pi} PV \int_{\mathbb{D}} \frac{h(\zeta)}{(\zeta - z)^2} dsdt.$$

Then for every $p \in (1, \infty)$, there exists constants $C(p)$ and $C'(p)$ such that $\lim_{p \rightarrow 2} C'(p) = 1$ as $p \rightarrow 2$, and

$$\|\mathcal{C}(h)\|_{W^{1,p}} \leq C(p) \|h\|_{L^p}, \quad \|\mathcal{B}(h)\|_{L^p} \leq C'(p) \|h\|_{L^p}.$$

Writing $\mathcal{C} : L^p(\mathbb{C}) \rightarrow W^{1,p}(\mathbb{C})$ and $\mathcal{B} : L^p(\mathbb{C}) \rightarrow L^p(\mathbb{C})$ for their extensions, we have

$$\mathcal{C}(h)_{\bar{z}} = h, \quad \mathcal{C}(h)_z = \mathcal{B}(h),$$

weakly.

Remark D.1(i) Assuming that for some $p_0 > 2$, there exists a constant c_0 such that

$$\|\mathcal{B}(h)\|_{L^{p_0}} \leq c_0 \|h\|_{L^{p_0}},$$

for every h , then we can use (D.7) and *Riesz-Thorin Interpolation Theorem* to assert that if

$$\frac{1}{p_\theta} = \frac{\theta}{2} + \frac{1-\theta}{p_0},$$

then

$$\|\mathcal{B}(h)\|_{L^{p_\theta}} \leq c_0^{1-\theta} \|h\|_{L^{p_\theta}}.$$

This allows us to choose $C'(p_\theta) = c_0^{1-\theta}$, for $\theta \in [0, 1]$, which enjoys the property

$$\lim_{p \rightarrow 2} C'(p) = 1.$$

(ii) Clearly there are many solutions to (D.5). In fact if $\mu = 0$, then (D.5) simply requires that φ to be holomorphic and if we also specify $\varphi(\mathbb{D})$, then we still have many solutions by the Riemann Mapping Theorem. Even if μ is nonzero, we can still require $\varphi(\mathbb{D})$ to be a simply connected domain $U \neq \mathbb{C}$ and solve (D.5) provided that $\|\mu\|_{L^\infty} < 1$. Indeed if φ is any solution to (D.5) and $\varphi(\mathbb{D}) = V$, then by the Riemann Mapping Theorem, we can find a holomorphic function $f : V \rightarrow \mathbb{C}$ such that $f(V) = U$. Now $\tilde{\varphi} = f \circ \varphi$ does the job because

$$\tilde{\varphi}_{\bar{z}} = (f' \circ \varphi) \varphi_{\bar{z}}, \quad \tilde{\varphi}_z = (f' \circ \varphi) \varphi_z.$$

This may be seen from

$$d\tilde{\varphi} = f_\varphi d\varphi + f_{\bar{\varphi}} d\bar{\varphi} = (f' \circ \varphi) d\varphi = (f' \circ \varphi)(\varphi_z dz + \varphi_{\bar{z}} d\bar{z}).$$

(iii) Observe that if $q, Q : \mathbb{D} \rightarrow \mathbb{C}$ are two holomorphic functions with $Q \neq 0$ anywhere inside $\mathbb{D}$, then $f = Q\mathcal{C}(g/Q) + q$ would solve the equation $f_{\bar{z}} = g$. $\square$

After solving the Beltami Equation, we may wonder whether the solution φ of (D.5) is a homeomorphism. One strategy for verifying the invertibility of φ is to derive an equation for its inverse, verify its solvability, and show that the solution is indeed the inverse of φ . As we have seen in Remark 10.1, the inverse w solves

$$(D.9) \quad w_{\bar{z}} = m(w) \overline{w_z},$$

with $|m| < 1$.

Theorem D.2 Assume that $\|m\|_{L^\infty} = c_0 < 1$. Then there exists $p = p(c_0) > 2$ and a function $w \in W^{1,p}$ that solves the equation (D.9) weakly.

Proof The method we described above would not work and need to be modified. Indeed if we set $v = w_{\bar{z}}$, we certainly have $w = \mathcal{C}(v) + q$ for a holomorphic q . On the other hand

$$(D.10) \quad v = m(w)\bar{\mathcal{B}}(v) + m(w)\bar{q}',$$

which cannot be solved as before because m depends on the unknown. To get around this, observe that for given w , the operator $I - m(w)\bar{\mathcal{B}}$ is invertible because $\|m\|_{L^\infty} = c_0 < 1$. Let us define

$$\mathcal{D}(w) = (I - m(w)\bar{\mathcal{B}})^{-1}(m(w)\bar{q}')$$

Then we use (D.10) to rewrite (D.9) as

$$(D.11) \quad w = (\mathcal{C} \circ \mathcal{D})(w) + q := \mathcal{E}(w).$$

Hence w is a fixed point of the operator $\mathcal{E}$. To show that $\mathcal{E}$ has a fixed point, we first decide on its domain of definition. Set

$$\Gamma_L = \{w : \|w\|_{L^\infty} \leq L\}.$$

We wish to show that $\mathcal{E} : \Gamma_L \rightarrow \Gamma_L$ and it has a fixed point. As a preparation, we first bound the nonlinear $\mathcal{D}$. Let $C'(p)$ be as in Theorem D.1. Choose $p > 2$ such that $c_0 C'(p) < 1$. By Theorem D.1,

$$\left\| (I - m(w)\bar{\mathcal{B}})^{-1} h \right\|_{L^p} \leq (1 - c_0 C'(p))^{-1} \|h\|_{L^p}.$$

This in turn implies,

$$\|\mathcal{D}(w)\|_{L^p} \leq c_0 (1 - c_0 C'(p))^{-1} \|q'\|_{L^p} := c_1(p).$$

By applying Theorem D.1 again we learn

$$\|\mathcal{C} \circ \mathcal{D}(w)\|_{W^{1,p}} \leq C(p) c_1.$$

Since $p > 2$, we may apply Morrey Inequality to deduce

$$(D.12) \quad \|\mathcal{C} \circ \mathcal{D}(w)\|_{C^{1-2/p}} \leq c_2(p) \|\mathcal{C} \circ \mathcal{D}(w)\|_{W^{1,p}} \leq C(p) c_1(p) = c_3(p),$$

where C^α denotes the space of α -Hölder continuous functions. In particular

$$\|\mathcal{C} \circ \mathcal{D}(w)\|_{L^\infty} \leq c_3(p).$$

Setting $L = c_3(p) + \|q'\|_{L^\infty}$, we deduce that $\mathcal{E}(L^p) \subseteq \Gamma_L$. In particular, $\mathcal{E}$ maps Γ_L into itself. On the other hand, the bound (D.12) implies that the image of Γ_L under $\mathcal{E}$ is in fact compact. This allows us to use the *Schauder Fixed Point Theorem* to deduce that $\mathcal{E}$ has a fixed point. $\square$

Remark D.1 Given $h : \mathbb{D} \rightarrow \mathbb{C}$, consider

$$v(z) = (\mathcal{C}h)(z) - \overline{(\mathcal{C}h)(1/z)} - 2i \operatorname{Im}((\mathcal{C}(h)(1)).$$

Note

$$\overline{(\mathcal{C}h)(1/z)} = \frac{z}{\pi} \int_{\mathbb{D}} \frac{\bar{h}(\zeta)}{1 - z\bar{\zeta}} dsdt,$$

is holomorphic inside $\mathbb{D}$. Hence

$$v_{\bar{z}} = h,$$

in $\mathbb{D}$. On the other hand, $v(1) = 0$, and

$$z \in \partial\mathbb{D} \implies v(z) \in i\mathbb{R}.$$

□

Remark D.2 By chain rule,

$$(h \circ \tau)_z = (h_z \circ \tau)k_z + (h_{\bar{z}} \circ \tau)(\bar{\tau})_z, \quad (h \circ \tau)_{\bar{z}} = (h_z \circ \tau)\tau_{\bar{z}} + (h_{\bar{z}} \circ \tau)(\bar{\tau})_{\bar{z}}.$$

As an example, when $\tau(z) = z^{-1}$,

$$(h \circ \tau)_{\bar{z}} = -(\bar{z})^{-2}(h_{\bar{z}} \circ \tau).$$

Recall that if

$$h(z) := \mathcal{C}(g)(z) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{g(\zeta)}{\zeta - z} dsdt, \quad \zeta = s + it,$$

we have $\partial_{\bar{z}}h = g$. Hence,

$$\partial_{\bar{z}}(\mathcal{C}(g) \circ \tau)(z) = -(\bar{z})^{-2} (g \circ \tau).$$

□

We continue with some local regularity estimates. Consider the multi-dimensional Beltrami equation

$$(D.13) \quad u_{\bar{z}}(z) = \mu(z)u_z(z) + f(z),$$

where $u, f : \mathbb{D} \rightarrow \mathbb{C}^n$, and $\mu : \mathbb{D} \rightarrow \mathbb{C}^{n \times n}$ satisfies

$$(D.14) \quad \sup_{z \in \mathbb{D}} \|\mu(z)\| = m < 1.$$

We assume that $u \in W^{k,p}$, and we wish to obtain local estimates for u in terms of μ and f .

Theorem D.3 *assume that u satisfies (D.13), and (D.14) holds. Then for every $r \in (0, 1)$, and any $p \geq 2$ such that $C'(p)m < 1$, there exists a constant $C'' = C''(r, p)$ such that*

$$(D.15) \quad \|Du\|_{L^p(B)} \leq C''(\|u\|_{L^1(B)} + \|f\|_{L^p(B)}),$$

where $B = B_0(r)$. More generally, for $k \geq 1$, there exists a constant C_k'' such that

$$(D.16) \quad \|D^k u\|_{L^p(B)} \leq C_k''(\|u\|_{L^p(B)} + \|f\|_{W^{k-1,p}(B)} \|\mu\|_{W^{k-1,\infty}(B)}).$$

Proposition D.2 *For every $r \in (0, 1)$, there exists a constant $C_k(r)$ such that for every holomorphic $q : \mathbb{D} \rightarrow \mathbb{C}$,*

$$(D.17) \quad \sup_{z \in B_r(0)} |q^{(k)}(z)| \leq C_k(r) \|q\|_{L^1(\mathbb{D})}.$$

Proof Since q is harmonic, we have

$$q(z) = \frac{1}{\pi r_0^2} \int_{B_{r_0}(z)} q(\zeta) d\zeta,$$

for every $z \in B_r(0)$, and $r_0 = 1 - r$, so that $B_{r_0}(z) \subset \mathbb{D}$. From this it is clear that (D.15) holds for $k = 0$. On the other hand, by Cauchy's formula,

$$q^{(k)}(z) = \frac{(-1)^k k!}{2\pi i} \int_{\partial B_{r_1}(0)} \frac{d\zeta}{(\zeta - z)^{k+1}},$$

where $r_1 \in (r, 1)$. From this, we learn that there exists a constant $c_k = c_k(r_1)$, such that

$$\sup_{z \in B_r(0)} |q^{(k)}(z)| \leq c_k \|q\|_{L^\infty(B_{r_1}(0))} \leq c_k C_0(r_1) \|q\|_{L^1(\mathbb{D})},$$

as desired. □

Proof of Theorem D.3 As before, if $h = u_{\bar{z}}$, then $u = \mathcal{C}h + q$, for a holomorphic function q . Let $B = B(0, r_0)$, with $r_0 < 1$. Observe that since $\mathcal{C}h = h * \zeta$, for a kernel $\zeta \in L^1(B)$, we can readily show

$$(D.18) \quad \|\mathcal{C}h\|_{L^p(B)} \leq \|\zeta\|_{L^1(B)}^{p-1} \|h\|_{L^p(B)},$$

for $p \geq 1$. As a consequence

$$(D.19) \quad \|q\|_{L^p(B)} \leq \|u\|_{L^p(B)} + c_1 \|h\|_{L^p(B)},$$

for a constant $c_1 = c_1(p)$. From this and (D.17) we learn

$$(D.20) \quad \|q^{(k)}\|_{L^\infty(B)} \leq c(k) (\|u\|_{L^1(B)} + c_1 \|h\|_{L^1(B)}),$$

for a constant $c(k)$. This and Theorem D1 yield

$$(D.21) \quad \begin{aligned} \|u_z\|_{L^p(B)} &= \|\mathcal{B}h + q'\|_{L^p(B)} \leq \|\mathcal{B}h\|_{L^p(B)} + \|q'\|_{L^p(B)} \\ &\leq c_2 (\|h\|_{L^p(B)} + \|u\|_{L^1(B)}), \end{aligned}$$

From this and $u_{\bar{z}} = h$ we learn,

$$(D.22) \quad \|Du\|_{L^p(B)} \leq c_3 (\|h\|_{L^p(B)} + \|u\|_{L^1(B)}).$$

On the other hand, (D.13) can be written as

$$h = (I - \mu\mathcal{B})^{-1}f.$$

We use Theorem D1 to assert,

$$\|h\|_{L^p(B)} \leq (1 - \|\mu\mathcal{B}\|_{L^p(B)})^{-1} \|f\|_{L^p(B)} \leq (1 - C'(p)m)^{-1} \|f\|_{L^p(B)},$$

for any $p \geq 2$ such that

$$C'(p)m < 1.$$

This and (D.22) yield (D.15)

If ∂ represents a partial differentiation operator, then applying ∂ to both sides of (D.13) yields

$$(D.23) \quad w_{\bar{z}} = \mu w_z + g,$$

where $w = \partial u$, and $g = \partial f + (\partial\mu)u$. Applying (D.20) to w yields

$$\begin{aligned} \|Dw\|_{L^p(B)} &\leq c_3 (\|g\|_{L^p(B)} + \|w\|_{L^1(B)}) \\ &\leq c_4 (\|Df\|_{L^p(B)} + \|Du\|_{L^1(B)} + \|u\|_{L^p(B)} + \|D\mu\|_{L^\infty(B)}). \end{aligned}$$

This yields (D.16) for $k = 2$. Continuing inductively, we can readily establish (D.16). $\square$

We can also deduce Theorem D.3 directly from Calderon-Zygmund Inequality using Laplace operator as opposed to the Beurling operator. To explain this, use write $z = s + it$, and consider

$$(D.24) \quad u_s(z) + J(z)u_t(z) = g(z),$$

or equivalently

$$(D.25) \quad u_t(z) - J(z)u_s(z) = -(Jg)(z),$$

where $u : \mathbb{D} \rightarrow \mathbb{R}^{2n}$.

Theorem D.4 *Given open balls B' and B with $\bar{B}' \subset B$, and $p > 2$, there exists a constant $C = C(p, B', B)$ such that*

$$(D.26) \quad \|u\|_{W^{1,p}(B)} \leq C \|J\|_{W^{1,p}(\mathbb{D})} (\|g\|_{L^p(\mathbb{D})} + \|u\|_{L^p(\mathbb{D})}),$$

Proof (step 1) Differentiate (D.24) and (D.25) with respect to s and t respectively:

$$u_{ss} + (Ju_t)_s = g_s, \quad u_{tt} - (Ju_s)_t = -(Jg)_t.$$

As a result,

$$\Delta u = g_s - (Ju_t)_s + (Ju_s)_t - (Jg)_t = g_s + (J_t u)_s - (J_s u)_t - (Jg)_t = \operatorname{div} Y,$$

where $Y = (g + J_t u, -J_s u - Jg)$. Choose w such that $\Delta w = Y$, and

$$\Delta \operatorname{div} w = \operatorname{div} Y, \quad \Delta(u - \operatorname{div} w) = 0.$$

Given a closed ball B' in the interior of B , we apply Calderon-Zygmund Theorem to assert that for every $r \in (1, \infty)$,

$$(D.27) \quad \|w\|_{W^{2,r}(B')} \leq c_1 \|Y\|_{L^r(\mathbb{D})}.$$

For this we simply extend Y to $\mathbb{R}^{2n}$ by setting $Y = 0$ outside $\mathbb{D}$, and choose $w = Y * K$, where K is the kernel of Δ^{-1} (which is a constant multiple of $|x|^{2-2n}$, when $n > 1$). Also

$$\begin{aligned} \|u - \operatorname{div} w\|_{L^r(B')} &\leq \|u\|_{L^r(B)} + \|\operatorname{div} w\|_{L^r(B)} \leq \|u\|_{L^r(B)} + c_2 \|w\|_{W^{1,r}(B)} \\ &\leq \|u\|_{L^r(B)} + c_3 \|Y\|_{L^r(B)}. \end{aligned}$$

Note that if $\Delta h = 0$, then by mean-value property

$$\begin{aligned} \nabla h(x) &= |B_\theta(0)|^{-1} \int_{B_\theta(0)} \nabla h(x+y) dy = |B_\theta(0)|^{-1} \int_{\partial B_\theta(0)} h(x+y) \frac{y}{|y|} dy \\ &= \theta^{-1} |B_1(0)|^{-1} \int_{\partial B_1(0)} h(x+\theta y) \frac{y}{|y|} dy. \end{aligned}$$

Hence

$$\nabla h(x) = \rho^{-1} \int_{\theta_0}^{\theta_0+\rho} |B_r(0)|^{-1} \int_{\partial B_r(0)} h(x+y) \frac{y}{|y|} dy d\theta.$$

Using this, it is not hard to deduce

$$\|h\|_{W^{1,r}(B')} \leq c_4 \|h\|_{L^r(B)}.$$

Hence, for a given closed ball B' in the interior of B ,

$$\|u - \operatorname{div} w\|_{W^{1,r}(B')} \leq c_5(\|u\|_{L^r(B)} + \|Y\|_{L^r(B)}),$$

by Proposition D.4. From this and (D.27) we deduce

$$(D.28) \quad \|u\|_{W^{1,r}(B')} \leq c_5(\|u\|_{L^r(B)} + \|Y\|_{L^r(B)}).$$

(Step 2) We next choose $\alpha, \beta > 1$ such that $\alpha^{-1} + \beta^{-1} = 1$. Observe that by Hölder's inequality,

$$\left[\int (FG)^r dz \right]^{1/r} \leq \left[\int F^{r\alpha} dz \right]^{1/(r\alpha)} \left[\int G^{r\beta} dz \right]^{1/(r\beta)}.$$

Hence

$$(D.29) \quad \|Y\|_{L^r(B)} \leq 2\|J\|_{L^\infty(B)} \|g\|_{L^r(B)} + 2\|J\|_{W^{1,r\alpha}(B)} \|u\|_{L^{r\beta}(B)}.$$

We choose (α, β) so that $\alpha r = p$. Note that $p > 2$ implies that $W^{1,p}$ dominates L^∞ -norm. Also, if we write q for $r\beta$, then

$$r^{-1} = p^{-1} + q^{-1},$$

and by (D.30) and (D.26),

$$(D.30) \quad \|u\|_{W^{1,r}(B')} \leq c_6\|J\|_{W^{1,p}(B)} (\|g\|_{L^p(B)} + \|u\|_{L^q(B)}),$$

because $r \leq p$. On the other hand, by Sobolev's inequality,

$$\|u\|_{L^Q(B'')} \leq c_7\|u\|_{W^{1,r}(B')},$$

where

$$Q^{-1} = \begin{cases} r^{-1} - 2^{-1}, & \text{if } r < 2, \\ 0, & \text{if } r \geq 2. \end{cases}$$

In particular,

$$(D.31) \quad \|u\|_{L^Q(B'')} \leq c_8\|J\|_{W^{1,p}(B)} (\|g\|_{L^p(B)} + \|u\|_{L^q(B)}).$$

Note that the map $q \mapsto Q = Q(q)$, satisfies

$$Q^{-1} = q^{-1} + (p^{-1} - 2^{-1}),$$

so long as $Q > 0$. Since $p > 2$, or $p^{-1} - 2^{-1} < 0$, for each q , there exists $\ell \in \mathbb{N}$ such that $Q^\ell(q) = \infty$ (or $1/Q^\ell(q) = 0$). In other words, we start from (D.30) for the initial choice

$q = p$, $r = p/2$, which yields $Q(p)^{-1} = (2p^{-1} - 2^{-1})^+$. Then apply (??) for the choice $Q(p)$. We continue this till $Q = \infty$, which yields $r = p$. $\square$

Fix a matrix $J_0 \in \mathbb{R}^{d \times d}$, such that $J_0^2 = -I$. For $z = (s, t)$, we write

$$\bar{\partial} = \partial_{\bar{z}} = \partial_s + J_0 \partial_t, \quad \partial = \partial_z = \partial_s - J_0 \partial_t.$$

Given a bounded open set $U \subset \mathbb{R}^2$ with C^1 boundary, we write $\mathcal{H}^1(U) = W^{1,2}(U)$ for the Sobolev space of functions $u : \mathbb{R}^2 \rightarrow \mathbb{R}^{2d}$ with the norm $\|u\|_{L^2(U)} + \|\nabla u\|_{L^2(U)}$. If $C_0^\infty(U)$ denotes the set of smooth functions with compact support strictly inside U , then the closure of $C_0^\infty(U)$ with respect to $\mathcal{H}^1(U)$ is denoted by $H_0^1(U)$. The space $H_0^1(U)$ represents the space of $u \in H^1(U)$, which vanishes on ∂U . By the classical Poincare-Sobolev type inequality we can show that there exists a constant $c_0 = c_0(U)$ such that $\|u\|_{L^2(U)} \leq c_0 \|\nabla u\|_{L^2(U)}$, for every $u \in H_0^1(U)$. Hence, the space $H_0^1(U)$ can be equipped with an inner product

$$\langle u, v \rangle = \int_U \nabla u \cdot \nabla v \, dx,$$

which yields a norm that is equivalent to $\mathcal{H}^1(U)$ -norm. The dual of $H_0^1(U)$ is denoted by $H^{-1}(U)$. The Laplace problem $-\Delta u = f$ in U , with the boundary condition $u = 0$ on ∂U can be solved for $f \in H^{-1}(U)$. As a weak formulation, we require that for every $v \in H_0^1(U)$,

$$(D.32) \quad \langle u, v \rangle = \langle f, v \rangle,$$

where $\langle \cdot, \cdot \rangle$ on the right hand side represents the pairing between $H_0^1(U)$, and $H^{-1}(U)$. Indeed the equation (D.32) has a unique solution by Riesz Representation Theorem. In other words

$$-\Delta : H_0^1(U) \rightarrow H^{-1}(U),$$

is an isomorphism. We next examine the dbar problem

$$(D.33) \quad \bar{\partial} u = h,$$

in U , with $u = 0$ on ∂U . If we assume that $h \in L^2(U)$, then by applying $-\partial$ to both sides of (D.33) we deduce

$$-\Delta u = -\partial h := f,$$

with $f \in H^{-1}(U)$. There exists a unique solution of this equation (or equivalently (D.33)) that is given by

$$u = \Delta^{-1}(\partial h) =: \bar{\partial}^{-1} h,$$

which belongs to $H^1(U)$. This means that

$$\bar{\partial}^{-1} : L^2(U) \rightarrow H_0^1(U),$$

is an isomorphism.

More generally, if $r, r' > 1$ with

$$\frac{1}{r} + \frac{1}{r'} = 1,$$

we write $W^{-1,r'}(U)$ for the dual of the Banach space $W_0^{1,r}$.

Theorem D.5 *The operator $-\Delta : W_0^{1,p} \rightarrow W^{-1,p'}(U)$ is an isomorphism.*

To prepare for the proof of Theorem D.5, we first define Campanato spaces. Let $U \subset \mathbb{R}^d$ with a smooth boundary. Given $\lambda > 0$ and $p \geq 1$, we set

$$\|u\|_{\mathcal{C}^{\lambda,p}(U)}^p = \sup_{x \in U} \sup_{r > 0} r^{-\lambda} \int_{U(x,r)} |u(y) - u_{x,r}|^p dy,$$

where $U(x, r) = B_r(x) \cap U$, and

$$u_{x,r} = |U(x, r)|^{-1} \int_{U(x,r)} u(y) dy.$$

We write $\mathcal{C}^{\lambda,p}(U)$ for the set of measurable maps $u : U \rightarrow \mathbb{R}$ such that $\|u\|_{\mathcal{C}^{\lambda,p}(U)} < \infty$. We can readily show that $\|\cdot\|_{\mathcal{C}^{\lambda,p}(U)}$ is a semi-norm, and $\mathcal{C}^{\lambda,p}(U)$, modulo constants is a Banach space. According to a theorem of Campanato, $\mathcal{C}^{d+p\alpha,p}(U)$ is isomorphic to the Hölder space $\mathcal{C}^\alpha(U)$ for every $\alpha = 0$. The space $\text{TCa}^{d,p}$ is known as BMO (Bounded Mean Oscillation), and is independent of p . The following generalization of Schauder estimate holds true:

Theorem D.6 *For every $\lambda \geq d+2$, there exists a constant $C = C(U, \lambda)$ such that if $u \in \mathcal{H}_0^1$ satisfies $-\nabla \Delta u = \text{div } f$. Then*

$$(D.34) \quad \|\nabla u\|_{\mathcal{C}^{\lambda,2}(U)} \leq C \|f\|_{\mathcal{C}^{\lambda,2}(U)}.$$

In particular, we have that if $\mathcal{T}(f) = \nabla u$, then

$$(D.35) \quad \|\mathcal{T}(f)\|_{\text{BMO}(U)} \leq C \|f\|_{\text{BMO}(U)} \leq C \|f\|_{L^\infty(U)}.$$

Note that we also know

$$(D.36) \quad \|\mathcal{T}(f)\|_{L^2(U)} \leq \|f\|_{L^2(U)}.$$

According to Stampacchia's interpolation theorem, we can interpolate and deduce that there exists a constant $C(p)$ such that

$$(D.37) \quad \|\mathcal{T}(f)\|_{L^p(U)} \leq C(p) \|f\|_{L^p(U)}.$$

From this we can readily deduce Theorem D.5. Observe that if $\bar{\partial}u = f$, then $\Delta u = \partial f$. Hence

$$\bar{\partial}^{-1} = \partial \circ \Delta^{-1}.$$

Hence

$$\|u\|_{W_0^{1,p}(U)} \leq C(p)\|f\|_{L^p(U)}.$$

This implies that $\bar{\partial}$ is an isomorphism between $W_0^{1,p}(U)$ and $L^p(U)$, for every $p \in [2, \infty)$.

E Lusternik-Schnirelmann Theory

The purpose of the Lusternik-Schnirelmann (LS) Theory is finding a universal lower bound for the number of the critical points of $f : M \rightarrow \mathbb{R}$. By “universal”, we mean a lower bound that depends on M only.

Definition E1(i) Let M be a compact metric space. Its cuplength is defined by

$$cl(M) = \min \{k : \exists \text{ contractible open sets } U_1, \dots, U_k \text{ with } M = U_1 \cup \dots \cup U_k\}.$$

By a contractible U , we mean that the inclusion map $j : U \rightarrow M$, is *nullhomotopic*.

(ii) For a closed smooth manifold M , we define

$$\begin{aligned} Cl(M) &= \min \left\{ k : \begin{array}{l} \exists \text{ closed forms } \alpha_1, \dots, \alpha_k \text{ with } \deg \alpha_1, \dots, \deg \alpha_k \geq 1 \\ \text{and } \alpha_1 \wedge \dots \wedge \alpha_k \text{ is exact} \end{array} \right\} \\ &= \max \left\{ k : \begin{array}{l} \exists \text{ closed forms } \beta_1, \dots, \beta_{k-1} \text{ with } \deg \beta_1, \dots, \deg \beta_{k-1} \geq 1 \\ \text{and } \beta_1 \wedge \dots \wedge \beta_{k-1} \text{ is not exact} \end{array} \right\}. \end{aligned}$$

□

Proposition E.1 Let M be a closed smooth manifold. Then $cl(M) = Cl(M)$.

Theorem E.1 Let M be a closed smooth manifold. Then for every C^1 function $H : M \rightarrow \mathbb{R}$,

$$(E.1) \quad \#Crit(H) \geq cl(M).$$

As the definitions of $Cl(M)$ and $cl(M)$ may suggest, there are two possible strategies to tackle Theorem E.1: Using open covers or differential forms. We start with the former.

First Proof of Theorem E.1 Step 1. We wish to find a universal lower bound for the number of the critical points of $f : M \rightarrow \mathbb{R}$. When M is a smooth manifold and f is C^2 , the Minimax Principle of Theorem 7.3 suggests a way to achieve this: find a collection $\mathcal{F}$ of

subsets of M that is closed under the flow $\Phi_t, t > 0$ of ∇f . (Here we are taking a Riemannian metric g on M so that we can talk about ∇f). The only problem is that we do not want this $\mathcal{F}$ to depend on f . This forces us to be more generous with sets in $\mathcal{F}$. In fact we consider collection that satisfy two properties:

- (1) For every $A \in \mathcal{F}$ and every homeomorphism $\varphi : M \rightarrow M$, we have that $\varphi(A) \in \mathcal{F}$.
- (2) For every $A \in \mathcal{F}$, and B satisfying $A \subset B$, we have that $B \in \mathcal{F}$.

We have two immediate consequences of (1) and (2).

- In view of Theorem 7.3, we have the following straightforward consequence of (1): $c(f, \mathcal{F})$, defined by

$$(E.2) \quad c(f, \mathcal{F}) = \inf_{A \in \mathcal{F}} \sup_{x \in A} f(x),$$

is a critical value of f .

- For very continuous $f : M \rightarrow \mathbb{R}$,

$$(E.3) \quad c(f, \mathcal{F}) = \inf \{r : M_f(r) \in \mathcal{F}\},$$

where $M_f(r) = \{x : f(x) \leq r\}$.

To verify (E.3), call its right-hand side $\hat{c}$. Since $\sup_{M_f(r)} f \leq r$, we have $c \leq \hat{c}$. For the reverse inequality, pick $a > c = c(f, \mathcal{F})$ so that we can find $A \in \mathcal{F}$ with $\sup_A f < a$. This means that $A \subseteq M_f(a)$, and by (2), we have $M_f(a) \in \mathcal{F}$. So, $\hat{c} \leq a$, completing the proof of (E.3).

We can readily construct examples of collections $\mathcal{F}$ for which (1) and (2) are true. However, for proving Theorem E.1, we need to find collections $\mathcal{F}_k \subset \mathcal{F}_{k-1} \subset \cdots \subset \mathcal{F}_1$, for which

$$(E.4) \quad c_1(f) := c(f, \mathcal{F}_1) < \cdots < c_k(f) := c(f, \mathcal{F}_k).$$

We have an obvious candidates for $\mathcal{F}_1$ and $\mathcal{F}_k$, namely

$$\mathcal{F}_1 = \{A : A \subseteq M\}, \quad \mathcal{F}_k = \{M\},$$

so that $c_1(f) = \min f$ and $c_k(f) = \max f$. To develop an intuition for these collections, let us set

$$i(A) = \max \{j : A \in \mathcal{F}_j\},$$

so that instead of finding $\mathcal{F}_1, \dots, \mathcal{F}_k$, we search for a function $i : 2^M \rightarrow \mathbb{N}$ such that

(1) $A \subseteq B$ implies $i(A) \leq i(B)$,

(2) $i(\varphi(A)) = i(A)$ for every homeomorphism φ .

We think of $i(\cdot)$ as a way of ranking subsets of M to measure their topological complexities. Clearly $i(M) = k$.

Step 2. In addition to the conditions (1) and (2) of Step 1, we make three more requirements on i (or equivalently $\mathcal{F}_k$) to guarantee that $c_j(f)$ s are distinct. Namely

(3) $i(A) = 1$ for A finite.

(4) If $i(A) = j$, then there exists open set $U \supseteq A$ with $i(U) = i(A)$.

(5) $i(A \cup B) \leq i(A) + i(B)$.

We now claim that if there exists $i : 2^M \rightarrow \mathbb{N}$ such that (1)-(5) are valid, then

$$(E.5) \quad \sharp Crit(f) \geq i(M).$$

To prove this, it suffices to show

$$(E.6) \quad \sharp Crit(f) < \infty \quad \Rightarrow \quad c_1(f) < \cdots < c_k(f),$$

or equivalently

$$(E.7) \quad c_j(f) = c_{j+1}(f) \quad \text{for some } j < k \quad \Rightarrow \quad \sharp Crit(f) = \infty.$$

To see this, assume that to the contrary $\sharp Crit(f) < \infty$ and

$$\{x : f(x) = c\} \cap Crit(f) = \{x_1, \dots, x_\ell\} =: A_0,$$

where $c = c_j(f) = c_{j+1}(f)$. Let U_0 be an open neighborhood of A_0 with $i(A_0) = i(U_0) = 1$. We may choose U_0 so that

$$U_0 \cap Crit(f) = A_0.$$

Now if Φ is the time-1 map of $-\nabla f$, then there exists $\varepsilon > 0$ such that

$$\Phi(M_f(c + \varepsilon) \setminus U_0) \subseteq M_f(c - \varepsilon),$$

because $(M_f(c + \varepsilon) \setminus U_0) \cap Crit(f) = \emptyset$. As a result,

$$\begin{aligned} i(M_f(c_{j+1} + \varepsilon)) &= i(M_f(c + \varepsilon)) \leq i(M_f(c + \varepsilon) \setminus U_0) + i(U_0) = i(M_f(c + \varepsilon) \setminus U_0) + 1 \\ &= i(M_f(c + \varepsilon) \setminus U_0) + 1 = i(\Phi(M_f(c + \varepsilon) \setminus U_0)) + 1 \\ &\leq i(M_f(c - \varepsilon)) + 1 = i(M_f(c_j - \varepsilon)) + 1 < j + 1, \end{aligned}$$

which is impossible. Hence $\sharp \text{Crit}(f) < \infty$ cannot be true and (E.6) is true.

Final Step. We now would like to construct a function $i : 2^X \rightarrow \mathbb{N}$ for which (1)-(5) are valid. Indeed

$$i(A) = \min \{k : \exists \text{ contractible open sets } U_1, \dots, U_k \text{ with } A \subseteq U_1 \cup \dots \cup U_k\},$$

does the job. This and (E.5) complete the proof of (E.1) because for this choice of i , we have $i(M) = \text{cl}(M)$. $\square$

From the first proof of (E.1) we learned that there is an axiomatic way of relating the topological complexity of the underlying manifold M to $\text{Crit}(f)$ via the function $i : 2^M \rightarrow \mathbb{N}$, satisfying the conditions (1)-(5). Though this can be avoided; the following direct (and rather nonconstructive) argument has its own merits.

Second Proof of Theorem E.1 Let $f : M \rightarrow \mathbb{R}$ be a C^2 function and write Φ_t for the flow of $-\nabla f$, where ∇ is defined with respect to a fixed Riemannian metric on M . To establish (E.1), we may assume that $\text{Crit}(f) = \{x_1, \dots, x_\ell\}$ is finite. Note

$$\lim_{t \rightarrow \pm\infty} \Phi_t(a) =: x_{\pm\infty}(a) \in \text{Crit}(f)$$

always exists. Pick contractible open neighborhood V_i of x_i and observe

$$M = \cup_{t \geq 0} \cup_{i=1}^{\ell} \Phi_t(V_i).$$

By the compactness of M , we may find $T > 0$ such that

$$M = \cup_{t \in [0, T]} \cup_{i=1}^{\ell} \Phi_t(V_i).$$

Hence $M = U_1 \cup \dots \cup U_\ell$, where

$$U_i = \cup_{t \in [0, T]} \Phi_t(V_i).$$

Since each V_i is contractible relative to M , each U_i is also contractible relative to M . As a result, $\text{cl}(M) \leq \ell$, by the definition. We are done. $\square$

We now develop a new way of finding critical values of a function f associated with certain forms that is very much the reason behind the definition $\text{Cl}(M)$. To prepare for this, we review some basic facts about *relative de Rham cohomology*.

Let M be a manifold and assume that N is a submanifold of M . Let us write $\Omega^r(M)$ for the space of r -forms on M and set

$$\Omega^r(M, N) = \Omega^r(M) \oplus \Omega^{r-1}(N),$$

for the space of *relative* r -forms. We define an exterior differentiation operator

$$\hat{d} = \hat{d}^r : \Omega^r(M, N) \rightarrow \Omega^{r+1}(M, N),$$

by the formula $\hat{d}(\alpha, \theta) = (d\alpha, j^*\alpha - d\theta)$, where $j : N \rightarrow M$ is the inclusion map (so that j^* is the restriction map). Clearly $\hat{d} \circ \hat{d} = 0$. We define the space of closed and exact forms by

$$\begin{aligned} Z^r(M) &= \ker d^{r+1}, & B^r(M) &= d^r(\Omega^r(M)), \\ Z^r(M, N) &= \ker \hat{d}^{r+1}, & B^r(M, N) &= d^r(\Omega^r(M, N)). \end{aligned}$$

The *(relative) de Rham cohomology classes* are defined by

$$H^r(M) = Z^r(M)/B^r(M), \quad H^r(M, N) = Z^r(M, N)/B^r(M, N).$$

Similarly, we write

$$H^*(M) = \bigoplus_{r=0}^{\infty} H^r(M), \quad H^*(M, N) = \bigoplus_{r=0}^{\infty} H^r(M, N).$$

Remark E.1 Note that if $(\alpha, \theta) \in Z^r(M, N)$, then $\alpha \in Z^r(M)$, and $j^*\alpha = d\theta$. In other word, a form α is closed relative to N , if α is closed and its restriction to N is exact. Take a neighborhood U of N in M such that N is a retract of U . Hence there exists a continuous map $h : U \rightarrow N$ with $h(a) = a$ for $a \in N$. Now take a continuous $\zeta : M \rightarrow [0, 1]$ such that $\zeta|_N = 1, \zeta|_{M \setminus N} = 0$, and set $\tau = \zeta(h^*\theta)$. We have

$$(\alpha, \theta) - \hat{d}(\tau, 0) = (\alpha - d\tau, \theta - j^*\tau) = (\alpha - d\tau, 0) =: (\alpha', 0),$$

with $j^*\alpha' = j^*\alpha - dj^*\tau = j^*\alpha - \theta = 0$. Hence if α comes from a closed form relative to N , then we can modify this form by means of subtracting an exact form and obtain a closed form for which its N -restriction is 0. $\square$

We now develop some techniques that would eventually yield the inequality

$$(E.8) \quad \sharp \text{Crit}(f) \geq C\ell(M).$$

This and Proposition E.1 imply Theorem E.1. Recall $M_f(a) = \{x : f(x) \leq a\}$.

Theorem E.2 Pick $a, b \in \mathbb{R}$ with $a < b$ and let $f : M \rightarrow \mathbb{R}$ be a C^2 function.

- (i) If f has no critical value in $[a, b]$, then $H^*(M_f(b), M_f(a)) = 0$.
- (ii) For every nonzero $\alpha \in H^*(M_f(b), M_f(a))$, define

$$(E.9) \quad c_{LS}(f, \alpha) = \inf \{s \in [a, b] : j_s^*\alpha \neq 0\},$$

where $j_s : M_f(s) \hookrightarrow M_f(b)$ is the inclusion map. Then the value $c_{LS}(f, \alpha)$ is critical for f .

Theorem E.2 guarantees the existence of critical points but for (E.9) we need to figure under what condition different forms lead to distinct critical points.

Theorem E.3 *Let $0 \neq \alpha \in H^*(M_f(b), M_f(a))$ and $\beta \in H^*(M_f(b))$ with $\deg \beta \geq 1$.*

(i) *We have $c_{LS}(f, \alpha \wedge \beta) \geq c_{LS}(f, \alpha)$.*

(ii) *If $c := c_{LS}(f, \alpha \wedge \beta) = c_{LS}(f, \alpha)$, then*

$$\#\{x : f(x) = c, (df)_x = 0\} = \infty.$$

Proof of (E.9) Let $f : M \rightarrow \mathbb{R}$ be a C^2 function with $\#Crit(f) < \infty$. Take closed forms $\beta_1, \dots, \beta_{k-1}$ such that $\deg \beta_j \geq 1$ for $j \in \{1, \dots, k-1\}$, and $\beta_1 \wedge \dots \wedge \beta_{k-1}$ is not exact. Set $c_j = c_{LS}(f, \eta_j)$, where η_j 's are defined by

$$\eta_1 = 1, \quad \eta_2 = \eta_1 \wedge \beta_1 = \beta_1, \quad \dots, \quad \eta_k = \eta_{k-1} \wedge \beta_{k-1} = \beta_1 \wedge \dots \wedge \beta_{k-1}.$$

By Theorems E.2 and E.3, each c_j is a critical value and $c_1 < \dots < c_k$, proving (E.9). $\square$

F Morse Theory

Let M be a smooth manifold. Given a C^2 function $f : M \rightarrow \mathbb{R}$, recall

$$Crit(f) = \{x \in M : (df)_x = 0\}.$$

Definition F1(i) Given $v, w \in T_x M$, take a smooth curve $\gamma = \gamma^{v,w} : (-1, 1)^2 \rightarrow M$, such that $\gamma_s(0, 0) = v, \gamma_t(0, 0) = w$, where γ_s and γ_t denote the partial derivatives of γ with respect to the first and second arguments respectively. Given and $x \in Crit(f)$, we define $Hess(f)_x : T_x M \times T_x M \rightarrow \mathbb{R}$ by

$$Hess(f)_x(v, w) = \frac{\partial^2}{\partial s \partial t} (f \circ \gamma^{v,w})(0, 0).$$

Clearly the linear map $Hess(f)_x$ is symmetric.

(ii) We define the *Morse index* and the *nullity* of f at a critical point x by

$$\begin{aligned} ind_f(x) = ind(x) &= \max \left\{ \dim V : V \text{ linear subspace of } T_x M, (Hess(f)_x)|_{V \times V} < 0 \right\}, \\ n_f(x) &= \dim \{v \in T_x M : Hess(f)_x(v, v) = 0\}. \end{aligned}$$

(iii) We say a function $f : M \rightarrow \mathbb{R}$ is Morse if $n_f(x) = 0$ for every $x \in Crit(f)$.

(iv) Given a C^2 function $f : M \rightarrow \mathbb{R}$, we set

$$Crit_k(f) = \{x \in Crit(f) : ind_f(x) = k\}.$$

(v) We write $H_k(M)$ for the k -th homology group of M . Its rank is called the k -th Betti number and it is denoted by $\beta_k(M)$. $\square$

Theorem F.1 (Morse) For any Morse function $f : M \rightarrow \mathbb{R}$,

$$\sharp\{x : (df)_x = 0, m_x(f) = k\} \geq \beta_k(M).$$

Definition F2(i) Let (M, g) be a closed Riemannian manifold and let $f : M \rightarrow \mathbb{R}$. We write ∇f for the gradient of f with respect to g : $(df)_x(v) = g(v, \nabla f)$. We also write $\psi_t = \psi_t^f$ for the flow of the vector field $-\nabla f$.

(ii) Given $x, y \in Crit(f)$, we set

$$\begin{aligned} W_x^s(f) &:= \{a \in M : \lim_{t \rightarrow \infty} \psi_t(a) = x\} \\ W_x^u(f) &:= \{a \in M : \lim_{t \rightarrow -\infty} \psi_t(a) = x\} \\ M(x, y) &:= W_x^u(f) \cap W_y^s(f) = \{a \in M : \lim_{t \rightarrow \infty} \psi_t(a) = y, \lim_{t \rightarrow -\infty} \psi_t(a) = x\}. \end{aligned}$$

Generically W_x^u, W_x^s and $M(x, y)$ are manifolds with

$$\dim W_x^u = ind(x), \quad \dim W_x^s = \dim M - ind(x), \quad \dim M(x, y) = ind(x) - ind(y).$$

(iii) The group $(\mathbb{R}, +)$ is acting on $M(x, y)$ by $\psi_t : M(x, y) \rightarrow M(x, y)$. This action is free (there is no fixed point), hence the quotient

$$\hat{M}(x, y) = M(x, y)/\mathbb{R},$$

is also a manifold of dimension $ind(x) - ind(y) - 1$. When $ind(x) - ind(y) = 1$, one can show that $M(x, y)$ is compact, hence a finite set. We set $m(x, y) = \sharp M(x, y)$. We also write $\bar{m}(x, y) \in \mathbb{Z}_2$ for mod 2 representation of $m(x, y)$. (When $ind(x) - ind(y) > 1$, the manifold $M(x, y)$ is not compact and may have boundary.)

(iv) We write $C_k = C_k(f)$ for the $\mathbb{Z}_2$ -span of $Crit_k(f)$. This is a vector space of expressions of the form

$$\sum_{z \in Crit_k(f)} t_z z,$$

with $t_z \in \mathbb{Z}_2$.

(v) We define $d : C_k \rightarrow C_{k-1}$ by

$$dx = \sum \{ \bar{m}(x, y)y : y \in \text{Crit}_{k-1} \},$$

for every $x \in \text{Crit}_k(f)$. □

Example F1(i) Take $M = \mathbb{T}$ be the circle and $h : \mathbb{T} \rightarrow \mathbb{R}$ be the height function. Then there are two critical points x_0 and x_1 of indices 0 and 1. The set $\hat{M}(x_1, x_0)$ consists of two points, representing two semicircle going from x_1 to x_0 .

(ii) Let h be as in Part (i), and define $f : \mathbb{T}^2 \rightarrow \mathbb{R}$ by $f(x, y) = h(x) + h(y)$. This function has a max x_2 , a min x_0 , and two saddle points x_1 and $\hat{x}_1$ of index 1. We have

$$m(x_2, x_1) = m(x_2, \hat{x}_1) = m(x_1, x_0) = m(\hat{x}_1, x_0) = 2.$$

On the other hand $\hat{M}(x_2, x_0)$ consists of four open intervals. The boundary points of these intervals represent broken orbits (each a union of two orbits from x_2 to x_0 via either x_1 or $\hat{x}_1$). □

Proposition F.1 *We have $d^2 = 0$.*

Proof For $x \in \text{Crit}_k(f)$, we have

$$d^2x = \sum \{ \bar{m}(x, y)\bar{m}(y, z)z : y, z \in \text{Crit}(f), \text{ind}(x) - \text{ind}(y) = \text{ind}(y) - \text{ind}(z) = 1 \}.$$

The expression $\sum_y m(x, y)m(y, z)$ represents the number of broken orbits from x to z , and these orbits are in one-to-one correspondence with the boundary points of the one dimensional manifold $\bar{M}(x, z)$. Since this number is even, we learn that $\bar{m}(x, y)\bar{m}(y, z) = 0$. □

G Fredholm Theory

Definition G.1(i) Let V_1 and V_2 be two Banach spaces. We write $\mathcal{L}(V_1, V_2)$ for the space of bounded linear maps $L : V_1 \rightarrow V_2$.

(i) Given $L \in \mathcal{L}(V_1, V_2)$, we write $\mathbf{N}(L)$ and $\mathbf{R}(L)$ for the null space and the range of L respectively. We say L is *Fredholm* if

$$\dim \mathbf{N}(L) < \infty, \quad \text{codim } \mathbf{R}(L) = \dim (V_2 / \mathbf{R}(L)) < \infty.$$

We set

$$\text{ind } L := \dim \mathbf{N}(L) - \text{codim } \mathbf{R}(L),$$

for the *index* of L . The space of Fredholm operators is denoted by $\mathcal{F}(V_1, V_2)$.

(ii) We say that S is compact if the set $S(B)$ is precompact in V_2 for the unit ball B of V_1 .

(iii) Let $K : V_2 \rightarrow V_1$ be a bounded linear operator. We say K is a semi-inverse of L if $S := KL - I_{V_1}$ and $S' := LK - I_{V_2}$ are compact operators. Here I_V is the identity map of V . We say K is a pseudo-inverse if S and S' are of finite ranks. $\square$

Theorem G.1 (i) *A map L is Fredholm iff L has a pseudo-inverse (semi-inverse).*

(ii) *For any bounded linear compact map $S : V \rightarrow V$, the map $I + S$ is Fredholm with $\text{ind}(I + S) = 0$.*

(iii) *If L is Fredholm, and S is compact, then $L + S$ is also Fredholm with $\text{ind}(L + S) = \text{ind}(L)$.*

Proposition G.1 (i) *For two Fredholm operators L_1 and L_2 , we have $\text{ind}(L_1 L_2) = \text{ind}(L_1) + \text{ind}(L_2)$.*

(ii) *If K is a semi-inverse of L , then $\text{ind}(K) + \text{ind}(L) = 0$.*

Clearly (i) and Theorem G.1(ii) imply (ii).

Proposition G.2 *Let $L : V_1 \rightarrow V_2$ be a Fredholm map. Then there exists $\varepsilon > 0$ such that if $A : V_1 \rightarrow V_2$ is a linear map with $\|A\| < \varepsilon$, then $L + A$ is also a Fredholm map. Moreover*

$$\text{ind}(L + A) = \text{ind}(L).$$

Proof Let L be Fredholm, and let K be a pseudo-inverse of K . Hence

$$K(L + A) = I + KA + S,$$

for a finite rank operator S . Choose $\varepsilon = \|K\|^{-1}$, and assume that $\|A\| < \varepsilon$, so that $(I + KA)^{-1}$ is well-defined. We have,

$$\hat{K}(L + A) := (I + KA)^{-1}K(L + A) = I + (I + KA)^{-1}S =: I + \hat{S},$$

with $\hat{S}$ finite rank. Also observe

$$\mathbf{N}(L + A) \subseteq \mathbf{N}(\hat{K}(L + A)) = \mathbf{N}(I + \hat{S}),$$

which is of finite dimension because $I + \hat{S}$ is Fredholm. Similarly, since

$$(L + A)K = I + AK + S',$$

for a finite rank operator S' , we have

$$(L + A)\hat{K}' := (L + A)K(I + AK)^{-1} = I + S'(I + AK)^{-1} =: I + \hat{S}',$$

with S' finite rank. Hence

$$\mathbf{R}(I + \hat{S}') = \mathbf{R}((L + A)\hat{K}') \subset \mathbf{R}(L + A).$$

As a result,

$$\text{codim} \mathbf{R}(L + A) \leq \text{codim} \mathbf{R}(I + \hat{S}'),$$

which is finite because $\hat{S}'$ is finite rank. As a result, $L + A$ is Fredholm.

Clearly $(I + KA)^{-1}$ is Fredholm, because it is invertible. Hence $\hat{K}$ is invertible. Observe

$$0 = \text{ind}(I + \hat{S}) = \text{ind}(\hat{K}) + \text{ind}(L + A) = \text{ind}(K) + \text{ind}(L + A).$$

As a result,

$$\text{ind}(L + A) = -\text{ind}(K) = \text{ind}(L),$$

by Proposition G.1(ii). □

What we learn from Proposition G.1 is that the set $\mathcal{F} = \mathcal{F}(V_1, V_2)$ is open, and that on each (pathwise) connected component of $\mathcal{F}$, the index is constant. We simply write $\mathcal{F}(V)$ for $\mathcal{F}(V_1, V_2)$ when $V = V_1 = V_2$. In fact one can show that the set

$$\mathcal{F}_n(V_1, V_2) = \{L \in \mathcal{F}(V_1, V_2) : \text{ind}(L) = n\},$$

is pathwise connected for every $n \in \mathbb{Z}$. Clearly, $\mathcal{F}_0(V_1, V_2)$ consists of all invertible bounded linear maps, and their compact perturbations.

We now focus on solvability problems of nonlinear equations. We first recall some basic definitions, and classical results in finite dimension.

Definition G.2 Let M and N be to C^1 manifolds, and let $f : M \rightarrow N$ be a C^1 function. We say $x \in M$ is a regular point of f if $(df)_x : T_x M \rightarrow T_{f(x)} N$ is onto. We say $x \in M$ is critical if x is not regular. Note that $\dim \mathbf{N}(df)_x = \dim M - \dim N$. We say $y \in N$ is a regular value if either $y \notin f(M)$, or every point in $f^{-1}(y)$ is regular. $\square$

Theorem G.2 Let f be as in Definition G.2. Assume that $f \in C^k$, with $k > \dim M - \dim N$.

(i) (Sard) The set of regular values of f is of full measure (with respect to the volume measure) in N .

(ii) The set of regular values of f is residual in N .

(iii) (Implicit Function Theorem) If y is a regular value, and $f^{-1}(y)$ is not empty, then $f^{-1}(y)$ is a submanifold of dimension $\dim M - \dim N$, with $T_x(f^{-1}(y)) = \mathbf{N}(df)_x$, for every $x \in f^{-1}(y)$.

Proof(ii) Write C for the set of critical points, which is a closed set in M . Let $\{U_i : i \in \mathbb{N}\}$ be a countable base of bounded open sets. Observe that the set $C \cap \bar{U}_i$ is compact. Since $f(C)$ is of 0 measure by part (i), the set $N \setminus f(C \cap \bar{U}_i)$ is open and dense in N . From this we deduce that $N \setminus f(C)$ is residual. $\square$

Definition G.3 Let V be a Banach space. We say that $\mathcal{U}$ is a Banach manifold with respect to V , if $\mathcal{U}$ is locally diffeomorphic to open subsets of V . Assume that $\mathcal{U}_1$ and $\mathcal{U}_2$ are two C^1 Banach manifolds with respect to V_1 and V_2 , and $G : \mathcal{U}_1 \rightarrow \mathcal{U}_2$ is a C^1 function. We then say that G is Fredholm, if $(dG)_x \in \mathcal{F}(V_1, V_2)$ for every $x \in \mathcal{U}_1$. We write $\mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$ for the set of C^1 Fredholm maps. Regular points and values of G are defined as in the finite dimension. $\square$

Example G.1(i) Let V_1 and V_2 be two finite dimensional vector spaces, and let $B : V_1 \rightarrow V_2$ be a linear map. Observe

$$\text{ind}(B) = \dim \mathbf{N}(B) - \text{codim} \mathbf{R}(B) = \dim \mathbf{N}(B) - \dim V_2 + \dim \mathbf{R}(B) = \dim V_1 - \dim V_2,$$

where we have used the rank-nullity theorem for the last equality.

(ii) Let V, V_1, V_2 be a Banach spaces, with $\dim V_1, \dim V_2 < \infty$. Consider the bounded linear operator

$$L : V \times V_1 \rightarrow V \times V_2, \quad L(a, b) = (a, Aa + Bb),$$

for linear $A : V \rightarrow V_2$, $B : V_1 \rightarrow V_2$. We may write $L = L_0 + S$, where $L_0(a, b) = (a, 0)$, $S(a, b) = (0, Aa + Bb)$. Evidently

$$\mathbf{N}(L_0) = V_1, \quad (V \times V_2)/\mathbf{R}(L_0) = V_2.$$

Hence L_0 is Fredholm with $\text{ind}(L_0) = \dim V_1 - \dim V_2$. Since S is of finite rank, we learn that L is Fredholm with $\text{ind}(L) = \dim V_1 - \dim V_2$. Note

$$\mathbf{N}(L) = \{0\} \times \mathbf{N}(B).$$

Also, using the isomorphism $(a, Bb) \mapsto (a, Aa + Bb)$, we learn

$$\mathbf{R}(L) \equiv V \times \mathbf{R}(B), \quad (V \times V_2)/\mathbf{R}(L) \equiv V_2/\mathbf{R}(B).$$

Hence

$$\text{ind}(L) = \dim \mathbf{N}(B) - \text{codim } \mathbf{R}(B),$$

which is consistent with part (i).

(iii) Let V, V_1, V_2 be as in part (ii), and assume that $\mathcal{U}$, $\mathcal{U}_1$, and $\mathcal{U}_2$ are manifolds with respect to V , V_1 , and V_2 respectively. Consider the C^1 map $G : \mathcal{U} \times \mathcal{U}_1 \rightarrow \mathcal{U}_2$, that is given by $G(x, y) = (x, f(x, y))$. Observe

$$(dG)_{(x,y)} : T_x \mathcal{U} \times T_y \mathcal{U}_1 \rightarrow T_x \mathcal{U} \times T_y \mathcal{U}_2, \quad (dG)(a, b) = (a, Aa + Bb),$$

where $A = f_x$, $B = f_y$ represent the partial derivatives of f . By part (ii), the map G is Fredholm of index

$$(G.1) \quad \text{ind}(dG) = \dim V_1 - \dim V_2.$$

Observe that (x, y) is a regular point of G iff y is a regular point for $f(x, \cdot)$. Also (x, z) is a regular value for G if z is a regular value of $f(x, \cdot)$. Moreover, we may use the implicit function theorem to assert that if (x, z) is a regular value, and the set

$$G^{-1}(x, z) = \{(x, y) : f(x, y) = z\} = \{x\} \times f(x, \cdot)^{-1}(z),$$

is a submanifold of dimension (G.1). □

Theorem G.3 Assume that $\mathcal{U}_1$ and $\mathcal{U}_2$ are Banach manifolds with respect to Banach spaces W_1 and W_2 . Assume that $G \in \mathcal{F}(\mathcal{U}_1, \mathcal{U}_2)$, and $\mathcal{U}_1$ is connected (so that $\text{ind}(G) := \text{ind}(dG)_x$ is independent of x). Assume that $G \in C^k$ for some $k > \text{ind}(G)$.

(i) (Smale) The map G is locally proper. That is, if Z is compact in $\mathcal{U}_2$, and $G(a) \in Z$, then $G^{-1}(Z) \cap \mathcal{V}$ is compact for some neighborhood $\mathcal{V}$ of a . Moreover, G is locally closed.

(ii) (Smale) Assume that W_1 is second countable. Then the set of regular values of G is residual in $\mathcal{U}_2$.

(iii) (Implicit Function Theorem) If y is a regular value, and $G^{-1}(y)$ is not empty, then $G^{-1}(y)$ is a submanifold of dimension $\text{ind}(G)$.

Proof(i) Write C for the set of critical points of G . Fix a countable base of open sets $\{\Omega_i : i \in \mathbb{N}\}$. Pick $a \in \mathcal{U}_1$, and set $L = (dG)_a$. Write $V_1 = \mathbf{N}(L)$, $V_2 = W_2/\mathbf{R}(L)$. We may represent W_1 and W_2 as $V \times V_1$ and $V \times V_2$, for a suitable Banach space V . We may regard $L : V \times V_1 \rightarrow V \times V_2$, with $L(v, v_1) = (K(v), 0)$, for an invertible $K : V \rightarrow V$. Consider local charts $\mathcal{W}_1$ and $\mathcal{W}_2$ for $\mathcal{U}_1$ and $\mathcal{U}_2$ such that $a \in \mathcal{U}_1$, and $G(a) \in \mathcal{U}_2$. Without loss of generality, we may assume that $\mathcal{W}_1 \subset V \times V_1$, and $\mathcal{W}_2 \subset V \times V_2$, $\mathcal{W}_1 \subset \Omega_i$, for some $i \in \mathbb{N}$, and $a = 0$, $G(a) = 0$. Write $G(x, y) = (g(x, y), h(x, y))$. Recall that $(dG)_0 = L$. Define $H(x, y) = (g(x, y), y)$. Since $(dH)_0$ is invertible, we can invert H near 0. We then set

$$G' := G \circ H^{-1} : \mathcal{W}'_1 \rightarrow \mathcal{W}'_2, \quad G'(x, y) = (x, f(x, y)),$$

with $f = h \circ H^{-1}$. Note that the map G' is proper: If $G'(x_n, y_n)$ is convergent, then x_n is convergent, and since $\{y_n : n \in \mathbb{N}\}$ is a subset of a bounded set, it has a convergent subsequence. From this we learn that the restriction of G to $\mathcal{W}_1$ is proper. Similarly G' is closed: If A is closed in $\mathcal{W}_1$, then $G'(A)$ is closed. Indeed if $(x_n, h(x_n, y_n))$ converges to (x, z) , for some sequence $(x_n, y_n) \in A$, then $x_n \rightarrow x$, and since y_n belongs to a bounded subset of the finite dimensional space $\mathcal{V}_1$, we may assume that $y_n \rightarrow y$. Since $(x, y) \in A$ by closedness of A , we are done.

(ii) We repeat the proof of Theorem G.2(ii) to assert that the set $\mathcal{U}_2 \setminus G(\bar{\mathcal{W}}_1 \cap C)$ is open and dense.

(iii) Let y be a regular value. We wish to show that $G^{-1}(y)$ is a manifold of dimension $\text{ind}(G)$ if $G^{-1}(y)$ is nonempty. It suffices to verify this locally, which is a consequence of (i), and Example G.1(i) $\square$

Theorem G.4 *Let $\mathcal{U}$ be a Banach manifold with respect to V , and let W be a Banach space. Let $F : \mathcal{U} \times [0, 1] \rightarrow W$, be a C^2 function, and assume*

- (i) *The set $F^{-1}(0)$ is compact in $\mathcal{U} \times [0, 1]$.*
- (ii) *The set $F^{-1}(0) \cap (\mathcal{U} \times \{0\}) = \{x_0\}$ is a singleton.*
- (iii) *The differential $F_x(x_0, 0) : V \rightarrow W$ is invertible.*
- (iv) *For every $(x, \theta) \in F^{-1}(0)$, we have $F_x(x, \theta) \in \mathcal{F}_0(V, W)$.*

Then for every $\theta \in [0, 1]$, there exists $x(\theta) \in \mathcal{U}$ such that $F(x(\theta), \theta) = 0$.

Proof (Step 1) Observe that for $(x, \theta) \in F^{-1}(0)$,

$$\hat{L} \begin{bmatrix} v \\ r \end{bmatrix} = Lv + ra,$$

where $\hat{L} = DF(x, \theta)$, $L := F_x(x, \theta) \in \mathcal{F}_0(V, W)$, and $a := F_\theta(x, \theta) \in W$. There are two cases to consider.

(1) If $a \in \mathbf{R}(L)$, then we can find $b \in V$ such that $Lb = -a$. Hence

$$\mathbf{N}(\hat{L}) = \{(rb + v, r) : v \in \mathbf{N}(L), r \in \mathbb{R}\}, \quad \mathbf{R}(\hat{L}) = \mathbf{R}(L).$$

Hence $\text{ind}(\hat{L}) = \text{ind}(L) + 1$.

(2) If $a \notin \mathbf{R}(L)$, then

$$\mathbf{N}(\hat{L}) = \{(v, 0) : v \in \mathbf{N}(L)\}, \quad \mathbf{R}(\hat{L}) = \mathbf{R}(L) \oplus \mathbb{R}a.$$

Hence $\text{ind}(\hat{L}) = \text{ind}(L) + 1$.

In summary, we always have $\text{ind}(\hat{L}) = 1$, and $\hat{L} \in \mathcal{F}_1(V, W)$.

(Step 2) Let

$$\mathcal{U}_0 = \{(x, \theta) \in \mathcal{U} \times [0, 1] : DF(x, \theta) \in \mathcal{F}_1(V, W)\}.$$

Since $F \in C^1$, we learn that $\mathcal{U}_0$ is an open neighborhood of $F^{-1}(0)$. For each $y \in F^{-1}(0)$, we can find an open set $\mathcal{U}(y) \subset \mathcal{U}_0$ such that the restriction of F to $\mathcal{U}(y)$ is closed (also proper). By compactness of $F^{-1}(0)$, we find a finite collection $\mathcal{U}^1, \dots, \mathcal{U}^\ell$ such that

$$F^{-1}(0) \subset \mathcal{U}^1 \cup \dots \cup \mathcal{U}^\ell =: \mathcal{V},$$

and the restriction F^j of F to each $\mathcal{U}^j$ is closed. The set R_j of the regular values of F^j is residual in W . We write $\bar{F}$ for the restriction of F to $\mathcal{V}$. The map $\bar{F}$ is Fredholm of index 1, and the set

$$R = R_1 \cap \dots \cap R_\ell,$$

of the regular values of $\bar{F}$ is residual. Note that $\partial\mathcal{V}$ is closed, and since $\bar{F}$ is closed, we learn that $\bar{F}(\partial\mathcal{V})$ is a closed set. Clearly $0 \notin \bar{F}(\partial\mathcal{V})$. Set

$$r_0 = \inf \{\|z\| : z \in \bar{F}(\partial\mathcal{V})\}.$$

Also, we can find open $\mathcal{U}_0$ in $\mathcal{U}$ such that $x_0 \in \mathcal{U}_0$, and the restriction of $F(\cdot, 0)$ to $\mathcal{U}_0$ is a diffeomorphism. We set

$$r_1 = \inf \{F(x, 0) : x \in \mathcal{U}_0\}.$$

Now take $y \in R$, and assume that $\|y\| < r_1 \wedge r_0$. Then $\bar{F}^{-1}(y)$ is a one-dimensional manifold such that

$$\bar{F}^{-1}(y) \cap (\mathcal{U} \times \{0\}) =: \{\bar{x}(y)\},$$

is a singleton, and

$$\partial\bar{F}^{-1}(y) \subset \mathcal{U} \times \{0, 1\}.$$

Since $\bar{F}^{-1}(y)$ is a union of circles and line segments, there must be a subset $\Gamma(y)$ of $\bar{F}^{-1}(y)$ that is homeomorphic to a line segment such that

$$\Gamma(y) \cap (\mathcal{U} \times \{0\}) =: \{\bar{x}(y)\}.$$

By the connectedness of $\Gamma(y)$, there exists $x(y, \theta) \in \mathcal{V}$ such that $\bar{F}(x(y, \theta), \theta) = y$. Choose y_n as above so that $y_n \rightarrow 0$. Then by the properness of $\bar{F}$, we can find a subsequence of $x(y_n, \theta)$ that converges. Its limit $x(\theta)$ satisfies $F(x(\theta), \theta) = 0$. $\square$

We next state a generalization of Theorem G.4 that can be established in a similar fashion. Let us assume that $\pi : \mathcal{E} \rightarrow \mathcal{B}$ is a vector bundle with $\mathcal{B}$ a Banach manifold with respect to Banach space V , and $\mathcal{E}$ fibers are isomorphic to a Banach vector space W . Observe that if $S : \mathcal{B} \rightarrow \mathcal{E}$ is a section is a C^1 -section, from $\pi \circ S = id$ we learn

$$(d\pi)_{X(x)} \circ (dS)_x = I : T_x \mathcal{B} \rightarrow T_x \mathcal{B},$$

where I denotes the identity. For example, when $\mathcal{E} = \mathcal{B} \times W$, we simply have $S(x) = (x, X(x))$, with $X : \mathcal{B} \rightarrow W$, and $(dS)_x = (I, (dX)_x)$. Because of this, we may define a *fiberwise differential* of S , which is the analog of $(dX)_x$. To do so, observe that $T_{(x,v)} \mathcal{E}$ is isomorphic to $T_x \mathcal{B} \times \mathcal{E}_x$, and let us write $\hat{\pi}_x : T_{(x,v)} \mathcal{E} \rightarrow \mathcal{E}_x$ for the projection onto the second component. Finally we define the fiberwise derivative

$$E_x S := \hat{\pi}_x \circ (dS)_x : T_x \mathcal{B} \rightarrow \mathcal{E}_x.$$

Theorem G.5 *Let $G : \mathcal{B} \times [0, 1] \rightarrow \mathcal{E}$, be a C^2 function such that $G(\cdot, \theta)$ is a section of $\mathcal{E}$ for every $\theta \in [0, 1]$. Assume*

- (i) *The set $G^{-1}(0)$ is compact in $\mathcal{B} \times [0, 1]$.*
- (ii) *The set $G^{-1}(0) \cap (\mathcal{B} \times \{0\}) = \{(x_0, 0)\}$ is a singleton.*
- (iii) *The differential $E_{x_0} G(\cdot, 0) : T_{x_0} \mathcal{B} \rightarrow \mathcal{E}_{x_0}$ is invertible.*
- (iv) *For every $(x, \theta) \in G^{-1}(0)$, we have $E_x G(\cdot, \theta) \in \mathcal{F}_0(T_x \mathcal{B}, \mathcal{E}_x)$,*

where $0 = 0_{\mathcal{E}}$ denotes the 0-section of $\mathcal{E}$, and $E_x G(\cdot, \theta)$ denotes the fiberwise differential of $G(\cdot, \theta)$. Then for every $\theta \in [0, 1]$, there exists $x(\theta) \in \mathcal{B}$ such that $F(x(\theta), \theta) = 0$.

Observe that when $\mathcal{E} = \mathcal{B} \times W$, we may write $G(x, \theta) = (x, F(x, \theta))$, we are in the setting of Theorem G.4. We also recall that by a classical theorem of Kuiper, all vector bundles over Hilbert spaces are trivial. This is also the case if $W = L^p(\Lambda)$ for manifold Λ (see [Mi]).

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HERIOT-WATT UNIVERSITY

FINAL YEAR PROJECT

**Discrete notions of curvature on graphs:
Ollivier Ricci Curvature and
non-convergence of the Ollivier Ricci Flow**

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supervised by
Dr. Alessandro Sisto

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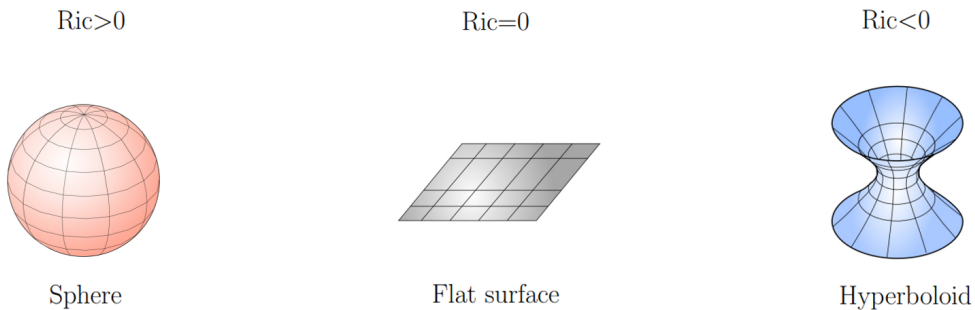
Abstract

This dissertation focuses on the Ollivier Ricci curvature on a graph. This is a discrete notion of edge-based curvature that can be formulated in terms of an optimal transport problem. The first part of the report introduces the definition and discusses some of its main properties. The second part of the report focuses on the Ollivier Ricci Flow, an iterative process which uses the values of the curvature to try to find a set of edge weights that make the curvature of the graph constant. We present a graph for which this algorithm doesn't converge and explain experimentally and theoretically the reason why. Finally, we study the robustness of the Ollivier Ricci curvature and the Ollivier Ricci Flow to noisy data inputs using simulations in Python.

1 Introduction

Complex networks are now fundamental objects in many scientific fields. They can be found in everything from financial markets, roads and layout of cities, drug discovery, social networks and more. These types of datasets can be mathematically phrased as a set of nodes, items, connected together by edges. Significant effort has been put to develop novel and powerful techniques to analyse these objects. A prominent approach has been to use geometrical information of the network, such as curvature, to understand its underlying structure.

Curvature is usually a notion found on the context of differential manifolds. There are several notions of curvature that can be defined at a point, some of them being more complex and sophisticated than others, but they all aim to quantify the deviation of an object from being flat. Typically the values are defined to be large when the object deviates strongly from its flat counterpart, and to be close to zero when it is nearly flat. In the image below there is an example of surfaces with positive, zero and negative Ricci curvature.



Discrete spaces don't have a differential structure, but recent research has shown that the concept of curvature can be transferred to the discrete setting by using *analogies* with the continuous case. There are multiple notions of discrete curvature that have been defined in the last few years, each of them revealing different properties of the network. This dissertation focuses on the Ollivier Ricci Curvature. This is an edge based curvature that can be formulated in terms of an optimal transportation problem. This curvature has received special attention due to its success in identifying community structures within a graph. Broadly speaking, edges with positive curvature tend to represent edges within a cluster, while negatively curved edge tend to be the *bridges* within communities. This property potentially allows to identify communities within a graph using intrinsic properties of the network only, without the need of any embedding techniques.

The first part of this report focuses on the Ollivier Ricci Curvature. We provide its definition and explain its connection to the Ricci Curvature defined in manifolds. We then expose some experimental results obtained regarding regular graphs with constant curvature. On the second part we introduce the Ollivier Ricci Flow, an iterative process that aims to *flatten* the curvature of a graph. We explain the methodology behind the algorithm and show an example of the Ollivier Ricci Flow implemented in a real-world dataset. We then focus on the study of the convergence of this algorithm. We prove experimentally and theoretically that the algorithm applied to a graph consisting of two triangles glued together by an edge doesn't converge. To conclude the dissertation, we provide some potential future lines of research.

2 Ollivier Ricci Curvature

2.1 Construction

Definition 1: Let $G(V, E)$ be a connected graph. A *probability distributions* μ on G can be understood as a function $\mu : V \rightarrow [0, 1]$ with

$$\sum_{x \in V} \mu(x) = 1$$

Let $G = (V, E)$ be a finite undirected graph with vertex set V , edge set E , and with no multiple edges or self loops. The Ollivier Ricci Curvature is an edge based curvature that can be formulated in terms of the optimal transport problem between probabilities distributions around adjacent nodes. The probability distribution centered at a node x is defined as:

$$m_x^\alpha(y) = \begin{cases} \alpha & \text{if } y = x \\ \frac{(1-\alpha)\psi(\omega_{xy})}{\sum_{z \in N_x} \psi(\omega_{zx})} & \text{if } y \in N_x \\ 0 & \text{Otherwise} \end{cases} \quad (1)$$

where:

- N_x (neighborhood of x) is the set of nodes that are adjacent to x .
- $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is a one-to-one function in terms of the edge weight. For an unweighted graph, this term can be disregarded (gets cancelled out with the normalization factor). For a weighted graph, ψ determines how much does the edge weight impact the mass distribution at node x . Intuitively, if we look at the edge weight as a measure of proximity between adjacent nodes, we would want to put less mass on the nodes that are more *far away* from x (connected with a high edge weight). Therefore, ψ needs to be a decreasing function for our purposes. This term will be of special importance when implementing the Ricci Flow described in Section 3.
- The parameter $\alpha \in [0, 1]$ is called the *idleness* as can be seen as the probability of a random walker to stay at x . As we would see in the upcoming sections, this parameter will play an important role on the convergence of the Ollivier Ricci Flow.

We can look at the probability measure defined in (1) as a mass distribution, which needs to be transported to the mass distribution defined around the other adjacent node. We thus produce a transport plan ξ that satisfies the following conditions:

$$\begin{aligned} \sum_{l \in N_y} \xi_{xy}(k, l) &= m_x^\alpha(k) \\ \sum_{k \in N_x} \xi_{xy}(k, l) &= m_y^\alpha(l) \end{aligned}$$

The first condition ensures that the total amount send from node k to all of the nodes in the neighbor of y is equal to the amount initially at k . The second condition ensures that the total amount send from all the nodes neighboring x to a node l in the neighborhood of y , is be equal to the final amount at l . Let Π be the set of all transport plans from $m_x^\alpha(v)$ to $m_y^\alpha(v)$. The distance between the probability distribution can be defined as

$$W[m_x^\alpha, m_y^\alpha] = \inf_{\xi \in \Pi} \sum_{k \in N_x, l \in N_y} d(k, l) \xi_{xy}(k, l) \quad (2)$$

Where W is the Wasserstein distance (also known as the Earth-mover distance). Finally, the Ricci curvature of an edge xy is defined as [1]:

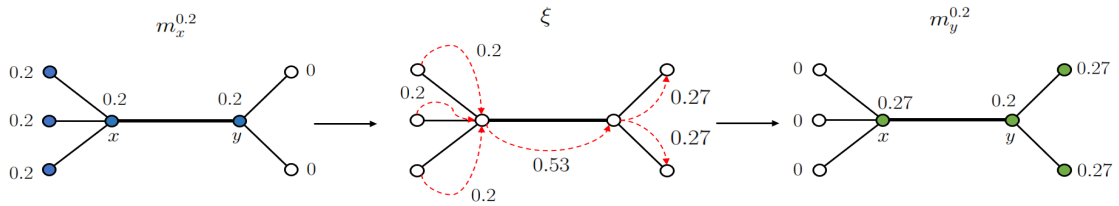
$$k_{xy}^\alpha = 1 - \frac{W[m_x^\alpha, m_y^\alpha]}{d(x, y)} \quad (3)$$

The Wasserstein distance can also be expressed as a linear programming problem, where ρ_{ij} is the proportion of mass originally at x_i transported to y_j :

$$\begin{aligned} \min_x \quad & \sum_{j=1}^n \sum_{i=1}^m d(x_i, y_j) \rho_{ij} m_x^\alpha(x_i) \\ \text{s.t.} \quad & \sum_{j=1}^n \rho_{ij} = 1 \quad \forall i \\ & \sum_{i=1}^m \rho_{ij} m_x^\alpha(x_i) = m_y^\alpha(y_j) \quad \forall j \\ & 0 \leq \rho_{ij} \leq 1 \end{aligned} \quad (4)$$

Note that (4) is bounded from below as $W[m_x^\alpha, m_y^\alpha] \geq 0$ and is not an unfeasible as in a connected graph there always exist a path between two nodes x_i and y_j . Therefore, by the *Fundamental Theorem of Linear Programming*, there exists an optimal solution to (4) (but it doesn't need to be unique). Solving this optimization problem can be very computationally expensive, especially for large graphs with high number of edges, as is the case of real-world social networks. Therefore, in practice different approximations are used to reduce the complexity of the calculations.

Example: Let's consider the following unweighted graph. We want to calculate the curvature along edge xy . The mass distributions around node x and y is represented in the diagram below (left and right diagram, respectively):



As this is a simple graph, it's quite easy to find the optimal transport plan ξ between m_x^α and m_y^α . This consists on moving all the mass from the nodes adjacent to x (apart from node y) to x and leaving 0.27 mass there. We then transport the rest of the mass, 0.53, along edge xy and leave 0.2 mass at node y . Finally, we equidistribute the remaining mass between the rest of the neighbours of node y .

Therefore,

$$\begin{aligned} W[m_x^{0.2}, m_y^{0.2}] &= 0.2 \cdot 3 + 0.53 + 0.27 \cdot 2 = 1.4 \\ k_{xy}^{0.2} &= 1 - 1.4 = -0.4 \end{aligned}$$

2.2 Connection to Ricci Curvature in a manifold

In this subsection, we provide an intuition on the connection between the Ollivier Ricci Curvature and the Ricci Curvature defined in Riemannian manifolds. We will not provide the formal definition of Ricci Curvature in a manifold, but we will introduce one of its main characterisations: Let x and y denote two close points in a manifold $\mathcal{M}$ of dimension n , and let $d(x,y)$ be the distance along a geodesic γ joining these two points. We can then take a vector $\mathbf{v} \in T_x\mathcal{M}$ and parallel transport it along γ to a vector $\mathbf{v}' \in T_y\mathcal{M}$. Informally, to parallel transport a vector along a curve means to transport it without changing its magnitude and direction (We would need to introduce the concept of Levi-Civita connection for a formal definition). We then project x along $\mathbf{v}$ back to the manifold again to obtain x' and project y along $\mathbf{v}'$ to obtain y' , and calculate the distance between these two points along a geodesic. We repeat this procedure for every possible direction of v and v' , constructing an infinitesimal balls $B_\epsilon(x)$ and $B_\epsilon(y)$ centered at x and y respectively. It is well known that the average distance between x and y' is $d(x,y)(1 - \epsilon^2 \frac{\text{Ric}(\mathbf{v},\mathbf{v})}{2n} + \mathcal{O}(\epsilon^3 + \epsilon^2 d(x,y)))$. [2]

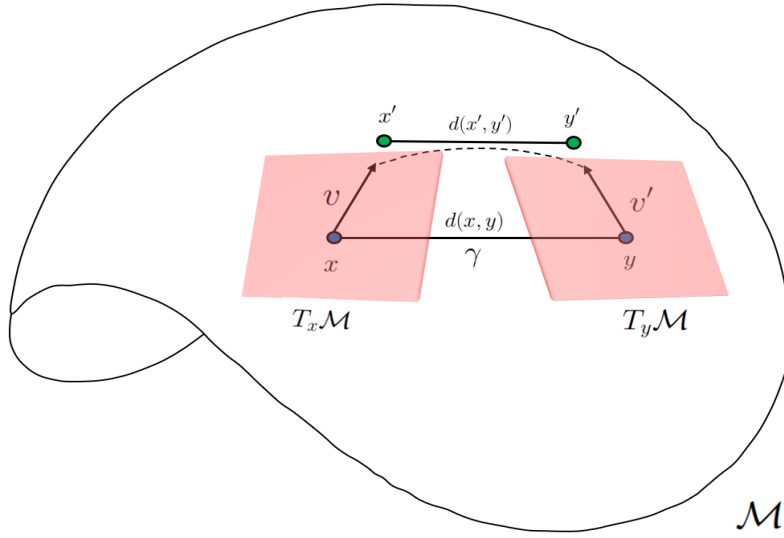


Figure 1: Ricci Curvature on a manifold

In the Ollivier Ricci Curvature, we replace the the idea of infinitesimal balls $B_\epsilon(x)$ and $B_\epsilon(y)$ with the mass distributions m_x^α and m_y^α . The distance between these masses distributions is then computed using the Wasserstein distance from optimal transport theory, and is compared to the distance between the centers of the ball, that is, $d(x,y)$ in formula (3). Therefore, the connection between the Ricci curvature in a manifold and the Ollivier Ricci Curvature is that both notions reflect the fact that when the the curvature is positive, the average distance between respective points in small metric ball is smaller than the distance between its centers (and the opposite for negative curvature).

$$\text{Average distance } d(x', y') = d(x, y)(1 - \epsilon^2 \frac{\text{Ric}(\mathbf{v}, \mathbf{v})}{2n}) + \mathcal{O}(\epsilon^3 + d(x, y)\epsilon^2)$$

$$k_{xy}^\alpha = 1 - \frac{W[m_x^\alpha, m_y^\alpha]}{d(x, y)}$$

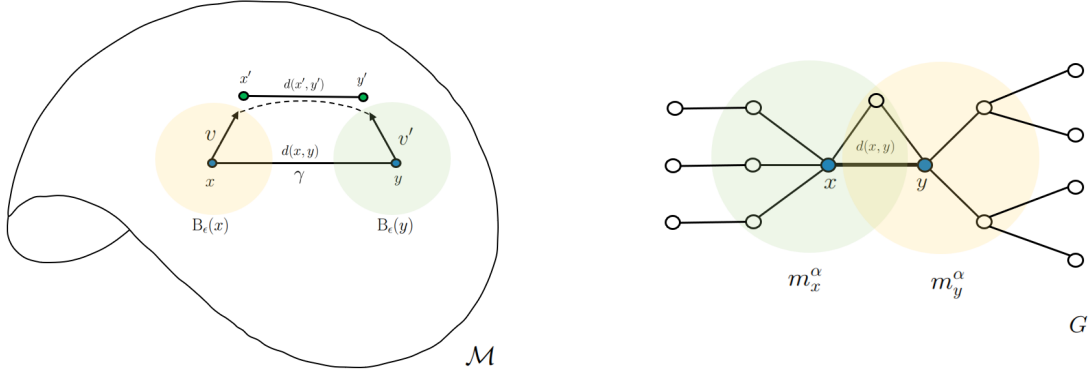


Figure 2: Comparison between Ricci Curvature defined on manifolds (left) and Ricci Curvature defined on graphs (right)

From this construction we can expect that graphs with a clique-like structure will have a positive curvature as the mass distribution around adjacent nodes will highly overlap, so $W[m_x^\alpha, m_y^\alpha]$ will be small and, hence, often less than 1. By contrast, graphs with a tree-like structure will have a negative curvature as we will have to transport large amounts of mass to go from m_x^α to m_y^α , so $W[m_x^\alpha, m_y^\alpha]$ will tend to be larger than 1. On the figure below we can see a summary of these properties.

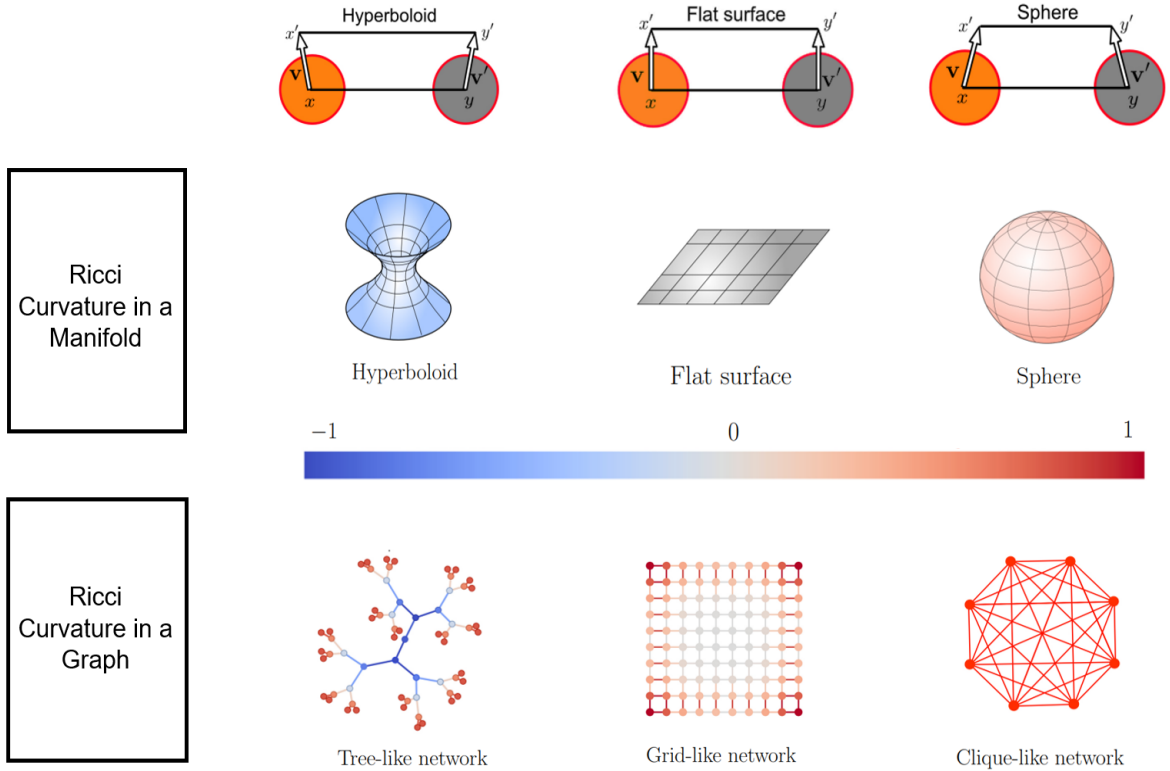


Figure 3: Ricci Curvature on manifolds vs Ricci Curvature on graphs [3]

2.3 Graphs with constant curvature

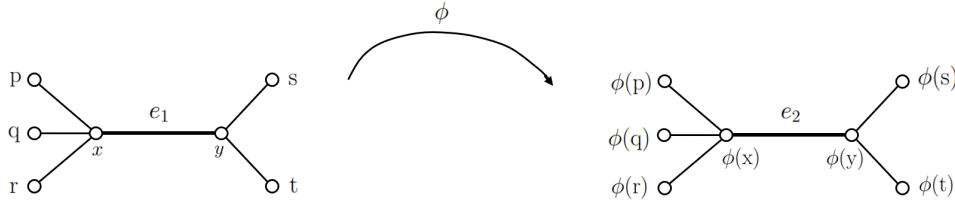
In this subsection we study graphs of constant curvature, that is, graphs whose all edges have the same value of curvature. The study of these graphs is of special interest as their shape is unchanged under the Ricci Flow, as we will describe in the next section. In this dissertation, we have written down the proof for the case of edge-transitive graphs, cycles and complete graphs. The following definitions were used on the proofs:

Definition 2: The girth of an undirected graph is the length of a shortest cycle contained in the graph.

Definition 3: A graph G is edge-transitive if for all pairs (e_1, e_2) there exists an element $\psi \in \text{Aut}(G)$ such that $\psi(e_1) = e_2$.

Proposition 1 Edge transitive graphs have constant curvature

Proof: Let ϕ be an automorphism of G , $\phi : V(G) \rightarrow V(G)$, such that $xy \in E(G)$ if and only if $\phi(x)\phi(y) \in E(G)$.



By definition, ϕ preserves node adjacency. That is, if $s \sim x$ if and only if $\phi(s) \sim \phi(x)$. As a consequence, $\deg(x) = \deg(\phi(x))$ and the mass distribution around node x , is the same as the mass distribution around $\phi(x)$. So $m_x^\alpha = m_{\phi(x)}^\alpha$. In addition, as the node adjacency is preserved, the shortest path between x and y is the same as the shortest path between $\phi(x)$ and $\phi(y)$. As a consequence, $d(x, y) = d(\phi(x), \phi(y))$. Putting these statements together, it follows that $W[m_x^\alpha, m_y^\alpha] = W[m_{\phi(x)}^\alpha, m_{\phi(y)}^\alpha]$.

$$\begin{aligned} W[m_x^\alpha, m_y^\alpha] &= \inf_{\xi \in \Pi} \sum_{k \in N_x, l \in N_y} d(k, l) \xi_{xy}(k, l) = \\ &= \inf_{\xi \in \Pi} \sum_{k \in N_{\phi(x)}, l \in N_{\phi(y)}} d(\phi(k), \phi(l)) \xi_{\phi(x)\phi(y)}(\phi(k), \phi(l)) = W[m_{\phi(x)}^\alpha, m_{\phi(y)}^\alpha] \end{aligned}$$

As there exists an automorphism ϕ that maps any two pairs of adjacent nodes to each other, we then conclude that the curvature in every edge will be the same.

Proposition 2 Complete graphs have positive constant curvature of $k_{xy}^\alpha = 1 - \left| \alpha - \frac{1-\alpha}{N-1} \right|$.

Proof: Let G be a complete graph with N nodes. By definition, all the nodes in a complete graph are connected to each other. Hence, the mass distribution around two adjacent nodes will overlap. In particular,

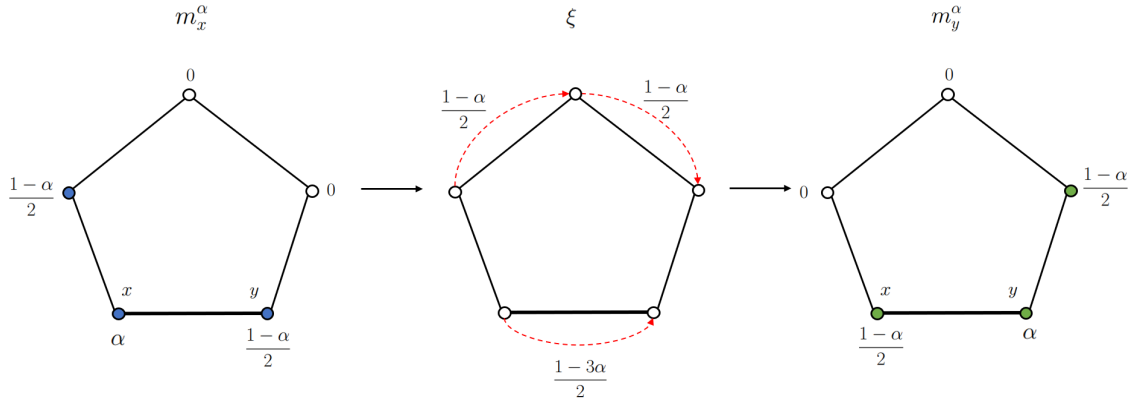
$$m_x^\alpha(v) = \begin{cases} \alpha & \text{if } v = x \\ \frac{1-\alpha}{N-1} & \text{otherwise} \end{cases} \quad m_y^\alpha(v) = \begin{cases} \alpha & \text{if } v = y \\ \frac{1-\alpha}{N-1} & \text{otherwise} \end{cases}$$

Hence, to go from $m_x^\alpha(y)$ to $m_x^\alpha(v)$ we just need to move $\left| \alpha - \frac{1-\alpha}{N-1} \right|$ along the edge xy . Therefore,

$$W[m_x^\alpha, m_y^\alpha] = \left| \alpha - \frac{1-\alpha}{N-1} \right| \rightarrow k_{xy}^\alpha = 1 - \left| \alpha - \frac{1-\alpha}{N-1} \right| > 0, \forall \alpha$$

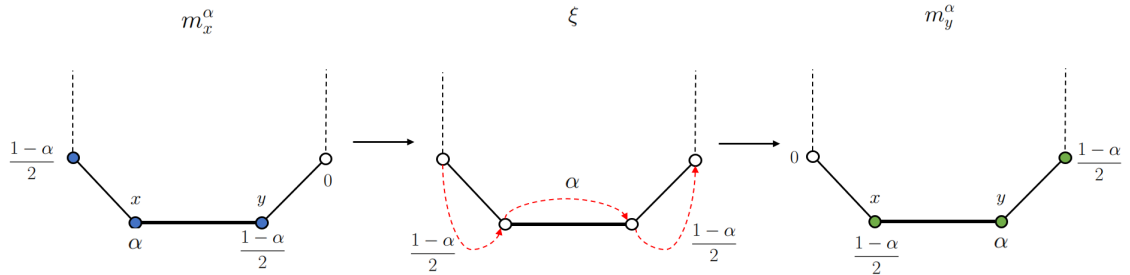
Proposition 3 Cycle graphs C_n of n vertices have constant curvature. In particular, for $n < 6$ the curvature $k_{xy}^\alpha > 0 \forall xy \in E(C_n)$, and for $n \geq 6$, $k_{xy}^\alpha = 0 \forall xy \in E(C_n)$.

Proof: Let's first consider cycles C_n for $n < 6$. The mass distribution around node x and y for $n=5$ is represented in the diagram below, along with the optimal transport plan ξ . For $n \neq 5$, the mass distribution would be the same, but the optimal transport plan would move $\frac{1-\alpha}{2}$ along $n-3$ edges.



$$W[m_x^\alpha, m_y^\alpha] = \left| \frac{1-3\alpha}{2} \right| + \frac{1-\alpha}{2} \cdot (n-3) > 0, \forall \alpha$$

For C_n for $n < 6$, the mass distribution around node x and y is represented in the diagram below, along with the optimal transport plan ξ . As the girth of the graph is larger than 3, the optimal transport plan ξ just shifts every mass through one edge.



$$W[m_x^\alpha, m_y^\alpha] = \frac{1-\alpha}{2} + \alpha + \frac{1-\alpha}{2} = 1$$

$$k_{xy}^\alpha = 1 - 1 = 0$$

All the edges xy of C_n will have the same transport plan ξ (as cycles are edge-transitive graphs), so the curvature will be the same in all the edges.

2.3.1 Regular Graphs of constant curvature

We have experimentally found regular connected graphs of constant curvature up to 11 vertices. We have used 2 different methodologies depending on the number of vertices and degree of the graph:

- **Graphs up to 10 vertices and degree 5:** We have used a database that contains the full list of regular graphs with those characteristics [4]. We have extracted their adjacency list and uploaded it into Python. We have created a Python function that returns those graphs whose edges have the same curvature. The Python code can be found in Appendix II.
- **Graphs with 11 vertices and 10 vertices with degree larger than 5:** The adjacency list mentioned above is not available for graphs larger than 10 vertices and degree 5. Therefore, we generated the regular graphs using a uniformly random generator algorithm in Python [5] and extracted the ones with constant curvature. To do that, we first need to determine the minimum number of iterations needed to make sure we have generated most of the graphs in the sample space.

Let X be the number of distinct graphs generated after n iterations. In particular,

- Let $X_i = 1$ if a graph G is generated at least once
- Let $X_i = 0$ otherwise

The algorithm we used generates graphs asymptotically uniform, we can express the expected value X as follows:

$$E(X) = E\left(\sum_{i=1}^m X_i\right) = \sum_{i=1}^m \mathbf{P}(X_i = 1) = \sum_{i=1}^m \left(1 - \left(\frac{m-1}{m}\right)^n\right) = m \left(1 - \left(\frac{m-1}{m}\right)^n\right)$$

To get $E(X)$ we then need to determine the size of the sample space m . We know that there are 265 and 266 distinct graphs with 11 vertices and degree 4 and 6, respectively (from source [4]). Nevertheless, the sample space m is larger than that because the algorithm is generating isomorphic graphs as well. Therefore, to calculate the size of the sample space m , we have calculated the average number of isomorphic graphs that exist for every graph G . To do that, we have used the VF2 Algorithm in Python to calculate the size of the automorphism group of a random generated graph G , over 1000 iterations, to obtain the average size of $|\text{Aut}(G)|$. Therefore, the sample space m can be approximated as:

$$m \approx \text{Number of distinct graphs} \times \sum_{i=1}^{1000} \frac{|\text{Aut}(G)|}{1000}$$

Putting these results together, for $E(x) \geq 0.9$ we need at least 1100 iterations. We now can check for the graphs of constant curvature over those generated.

The results containing all regular graphs of constant curvature can be found in Appendix I.

Notice that we have found a regular graph of constant curvature for every combination of possible vertex and degrees. As future work, it would be interesting to study if this is the case for larger regular graphs. The methodology used above using random graphs generators can also be used to find graphs of constant curvature for higher number of vertices. For the purpose of this dissertation, we have only analyzed up to 11 vertices due to the high computational cost of calculating the Ollivier Ricci Curvature, mainly computing the Wasserstein distance, for such big number of graphs.

Throughout this analysis, we have also found the following interesting results:

1. Regular graph with negative constant curvature: All the graphs listed in Appendix I have positive constant curvature. This is because for small connected graphs, the amount of mass we need to transport from adjacent mass distribution is not enough to have $W[m_x^\alpha, m_y^\alpha] > 1$. The left graph in Figure 4 is one example of a graph with constant negative curvature. Our intuition is that as girth of the regular graph increases, more examples of constant negative curvature will appear.
2. Regular graph with constant curvature but not edge transitive: All the graphs listed in Appendix I are edge transitive. We have only found one example of a regular graph with constant curvature which is not edge transitive. This graph is shown in Figure 4 on the right.

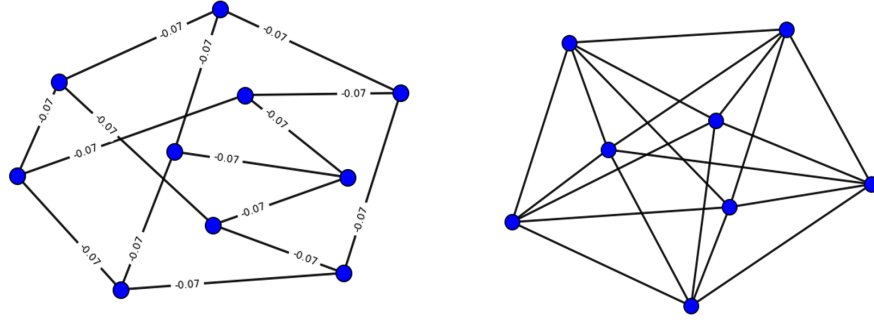


Figure 4: Graph with constant negative curvature (left) and graph with constant curvature which is not edge-transitive (right)

3 Ollivier Ricci Flow

The Ollivier Ricci Flow is an iterative process that modifies the edge weights to try to flatten the curvature of the graph. Broadly speaking, edges with positive curvature will shrink (the edge weight will decrease) and the edges with negative curvature will expand (the edge weight will increase). This property has found important applications in community detection, as nodes within the same community will get closer to each other, while moving away from nodes belonging to other communities. The Ollivier Ricci Flow has also been used in other contexts, such as to study the fragility of financial markets through the analysis of the correlation between the robustness of a graph and its curvature [6], to study information propagation on a graph and its applications to the over-squashing phenomenon of Graph Neural Networks [7], to study the internet topology [8], and many more.

To give an intuitive example on the idea behind the algorithm, we have implemented the Ollivier Ricci Flow to the *Zachary's Karate Club* dataset [9]. This is a famous social network used broadly in network analysis. It is well-known that this dataset contains two communities. In the image below the weights and curvature are analyzed before and after implementing the Ricci Flow. We are considering the unweighted version of the data, so before implementing the Ricci Flow, all the weights are 1 (lower left diagram) and there is a mixture of curvatures ranging from -2 to 1 (upper left diagram). After the Ricci Flow, all the edge curvatures have converged to 0 (upper right diagram), while the weights now range from 0 to 7 (lower right diagram). Notice that the weight of the edges that initially had a negative curvature has increased after the Ricci Flow (expanded), while the others have decreased (shrunk). The two communities can be clearly identified after the Ricci Flow from the lower right diagram, and correspond to the true values of the data. This is just an intuitive example on some properties of the algorithm, a more in depth analysis will follow in the next sections.

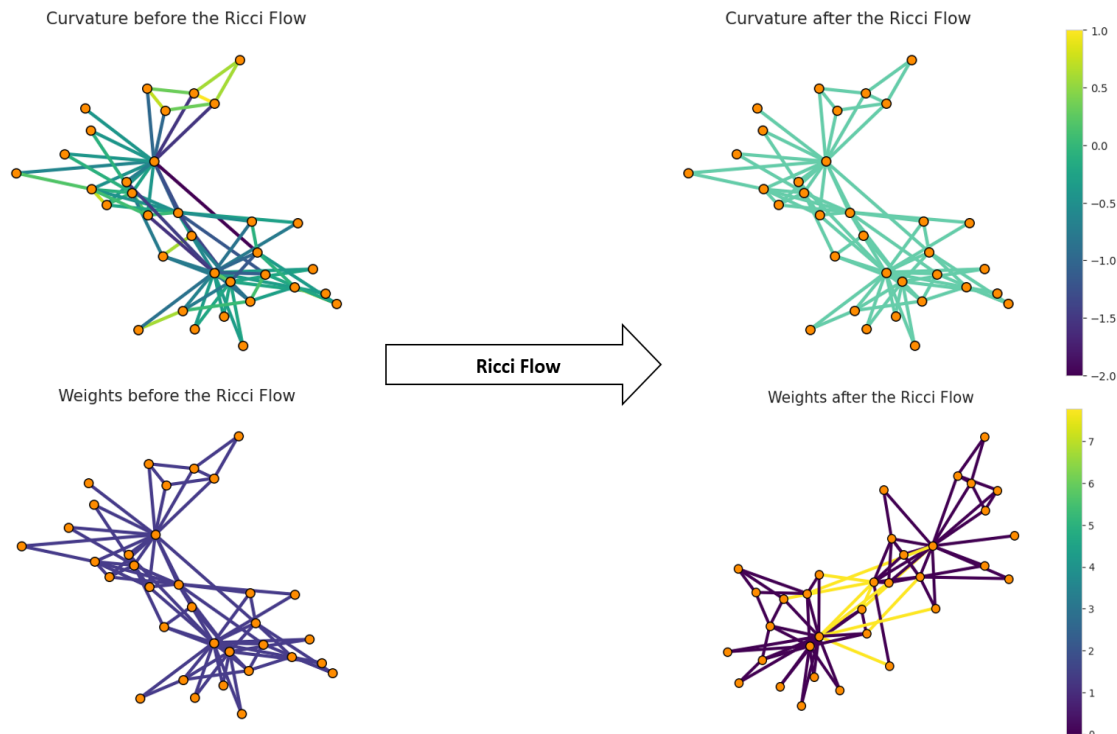


Figure 5: Implementation of the Ricci Flow to the *Zachary's Karate Club* network. The value of the edge weights and curvature can be seen from the colour-bars before (left) and after (right) the Ricci Flow

3.1 Construction

There are different approaches used for implementing the Ricci Flow. In this thesis, we would follow the methodology described in the paper: *Community detection on networks with Ricci flow* [10]. In particular, we followed the Python code that they made available on Github and documented the steps of their computation. The key idea is that at every iteration of the Ricci Flow, the mass distribution is re-defined using the normalized weights from the previous iteration, and the curvature is re-calculated using formula 3. The new weights of every edge j are then updated using $w_j^{t+1} = (1 - k_j^t) \cdot w_j^t$. We then normalize the weights to add up to N and repeat the process. A more detailed explanation of the algorithm is provided below:

Algorithm 1 Mass distribution around a node x

Input: Node x , Edge weights $\mathbf{w}$ of G , idleness α , parameter p

```

1: function Mass Distribution( $x, \mathbf{w}, \alpha, p$ ) ▷ Array of weights of all edges
2: for each vertex  $y$  in  $V(G)$  do
3:   if  $y$  in  $N_x$  then
4:     if  $y=x$  then
5:        $m_x^\alpha(y) = \alpha$ 
6:     else if  $y \neq x$  then
7:        $m_x^\alpha(y) = \frac{(1-\alpha)e^{-w_{xy}^p}}{\sum_{z \in N_x} e^{-w_{zx}^p}}$ 
8:     end if
9:   else if then
10:     $m_x^\alpha(y) = 0$ 
11:   end if
12: end for
13: return  $m_x^\alpha$ 

```

Algorithm 2 Ricci Curvature single edge

Input: Adjacent nodes x and y , Edge weights $\mathbf{w}$ of G , idleness α , parameter p

```

1: function Ricci Curvature( $x, y, \mathbf{w}, \alpha, p$ ) ▷ Curvature of the edge  $xy$ 
2:  $m_x^\alpha =$  Call Mass Distribution( $x, \mathbf{w}, \alpha, p$ )
3:  $m_y^\alpha =$  Call Mass Distribution( $y, \mathbf{w}, \alpha, p$ )
4:  $k_{xy} = 1 - \frac{W[m_x^\alpha, m_y^\alpha]}{w(x,y)}$  ▷ The Wasserstein distance is calculated using the Sinkhorn library [11].
5: return Ricci Curvature  $xy$ 

```

Algorithm 3 Ricci Flow

Input: Undirected graph G , Number of Iterations $\mathcal{I}$, idleness α , parameter p

```

1: for  $t \leftarrow 1$  to  $\mathcal{I}$  do
2:   for each  $j$  in  $E(G)$  do
3:      $k_j^t =$  Call Ricci Curvature ( $x, y, \mathbf{w}^t, \alpha, p$ ) ▷  $xy$  are the nodes adjacent to edge  $j$ 
4:      $w_j^{t+1} = (1 - \epsilon \cdot k_j^t) \cdot w_j^t$  ▷  $\epsilon$  is the step size
5:   end for
6:   for each  $j$  in  $E(G)$  do
7:      $w_j^{t+1} = \frac{w_j^t \times |E(G)|}{\sum_{j \in E(G)} w_j^t}$  ▷ Normalize edge weights used for the next iteration
8:   end for
9: end for
10: return Final weights

```

3.2 Convergence of Ricci Flow

The convergence of the Ricci Flow is of great importance when applying the algorithm to real-world datasets. Even though a number of experimental analyses have been carried out, no theoretical guarantees about its convergence has been obtained yet. This is a particularly difficult task due to the following factors:

1. In general, there is no explicit formula for the Wasserstein distance between masses distributions around adjacent edges and, hence, for the curvature of an edge.
2. The normalization factor implemented at the end of every iteration of the Ricci Flow avoids expansion of the weights to infinity, but makes the analysis significantly more complicated. This is because even though the Wasserstein distance between masses distribution around adjacent nodes may decrease from one iteration to another, the weight of the edge can still increase if the sum of the total Wasserstein distance from all the edges of the graph has also decreased. Therefore, because of the normalization factor, the weight of an edge at the end of one iteration doesn't depend only on the local neighborhood around the edge, but on the global structure of the graph.
3. The shortest path between two nodes can change as the weight of the edges changes, making difficult (or impossible) to track what is the optimal path between two nodes at a given time.

In this section, we will analyze the convergence of the Ricci Flow using a simplified version of the mass distribution around a node x described in Algorithm 1. We will explain how this simplification mitigates some of the difficulties in the analysis outlined above. As stated in Algorithm 1, the mass distribution is dependent of a parameter p which quantifies how much we want to disregard the neighbours that are *far away* (have a large edge weight). Mainly, the mass distribution has the following form:

$$m_x^\alpha(y) = \begin{cases} \alpha & \text{if } y = x \\ \frac{(1-\alpha)e^{-w_{xy}^p}}{\sum_{z \in N_x} e^{-w_{zx}^p}} & \text{if } y \in N_x \\ 0 & \text{Otherwise} \end{cases} \quad (5)$$

As the value of p grows, $\psi(w_{xy}) = e^{-w_{xy}^p}$ becomes steeper around $x=1$. For our analysis, we will approximate this behaviour using the step function below:

$$\phi(w_{xy}) = \begin{cases} 1 & \text{if } w_{xy} \leq 1 \\ 0 & \text{Otherwise} \end{cases} \quad (6)$$

Therefore, the mass distribution around node x that we would be considering is:

$$m_x^\alpha(y) = \begin{cases} \alpha & \text{if } y = x \\ \frac{(1-\alpha)\phi(w_{xy})}{\sum_{z \in N_x} \phi(w_{xz})} & \text{if } y \in N_x \text{ and } \sum_{z \in N_x} \phi(w_{xz}) \neq 0 \\ \frac{(1-\alpha)}{|N_x|} & \text{if } y \in N_x \text{ and } \sum_{z \in N_x} \phi(w_{xz}) = 0 \\ 0 & \text{Otherwise} \end{cases} \quad (7)$$

That is, if the edges around node x all have weight smaller than 1 or all weight larger than 1, we equidistribute the mass. If some edges have weight larger than 1 and some smaller than one, then we equidistribute the mass between the edges with weight smaller than 1 only.

Conceptually, with mass distribution (7), we are completely disregarding the neighbours that have edge weight larger than 1. Thus, this may not be best choice in real-world applications, but is convenient for theoretical analysis.

Let w_j^t be the weight of edge j at iteration t . From Algorithm 3,

$$w_j^{t+1} = (1 - \epsilon \cdot k_j^t) \cdot w_j^t \quad (8)$$

and the curvature of edge j at iteration t , k_j^t , is

$$k_j^t = 1 - \frac{W[m_x^\alpha, m_y^\alpha]}{w_j^t} \quad (9)$$

where x and y are the nodes adjacent to edge j and ϵ is the step size, which we would set equal to 1. Combining expression 8 and 9 together, leads to

$$w_j^{t+1} = W[m_x^\alpha, m_y^\alpha] \quad (10)$$

Let $\mathbf{w}^t$ be an n -dimensional vector, where a $n=|E(G)|$, containing the weights of all edges at iteration t . Now, for a fixed α , we can define a $n \times n$ matrix A whose entry A_{jk} is the mass transported around edge k when computing the optimal Wasserstein distance between the masses distributions adjacent to edge j . Note that we can define the matrix A because the mass distribution described in expression (7) only depends on the edge weight of the neighboring edges being larger or smaller than 1, but not on the value of the weight itself. So the coefficients A_{jk} are not depend on $\mathbf{w}$. Then, we can rewrite the non-normalized weights at iterations $t+1$ as

$$\mathbf{w}^{t+1} = A\mathbf{w}^t$$

If sequential weights have the same optimal transport plan, they will share the same matrix A , so we can express the weights at iteration t as

$$\mathbf{w}^t = A^t \mathbf{w}^0$$

where $\mathbf{w}^0$ are the initial weights associated with A . Realistically, this will only happen in small graphs. We have not proved that this formulation is possible for a general graph, but it certainly is for the small graph that we will be analyzing in the next section. In the case where this formulation is feasible, the step function in expression (7) we applied to the mass distribution has allowed us to write down this problem a system of piecewise linear maps. This has made the problem significantly easier to tackle, as we can now use linear algebra techniques to investigate the convergence. As a future work, it would be interesting to prove if this formulation is correct for a general graph. Assuming the formulation above is feasible for a graph G , we can divide the parameter space into different regions following the below procedure.

1. If the mass distribution around adjacent nodes changes, the optimal transportation plan will definitely change and, hence, the entries of the matrix A too. The mass distribution will change if and only if the weight of an edge changes from smaller than 1 to larger than 1, or viceversa. Therefore, every matrix A has associated a set of conditions which specify if the weight of every edge should be smaller or larger than 1. These set of conditions can be written as the intersection of n half spaces, so every region associated with a matrix A can be viewed as a convex polytope. The sample space Ω will be the union of finitely many convex polytopes. In particular, for a graph with n edges there would be at most 2^n different combinations, so Ω will be the union of at most 2^n convex polytopes. In the diagram below there is a representation of such space.

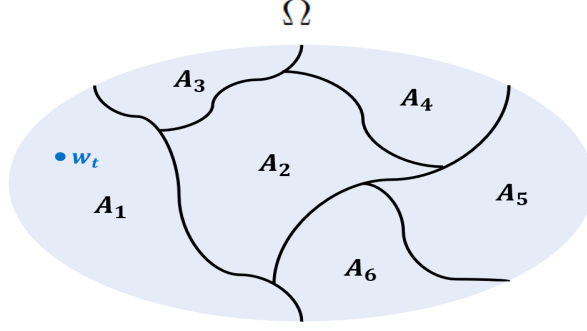


Figure 6: Representation of the sample space Ω , Every region A_i is a convex polytope that determine which edge weights are bigger or smaller than 1

2. Given that $\mathbf{w}^t$ lies inside one of the regions, let's say A_1 , the coefficients of the matrix can still change if the optimal transport plan changes. This can happen, for example, if the shortest path between two nodes has changed with respect to the previous iteration. Hence, every polytope is subdivided by different regions determined by the set of optimal transport plans, as seen below:

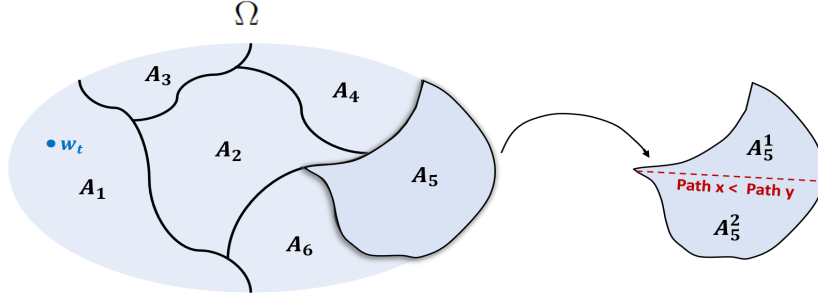


Figure 7: Every region in Ω is subdivided by the different determined by the set of optimal transportation plans

With this construction in mind we are now ready to study how the weights $\mathbf{w}^t$ change throughout the Ricci Flow. To do that, we need to find an answer to the following question: Does there exist a t such that $\mathbf{w}^t = A^t \mathbf{w}^0$ ($\mathbf{w}^0$ being the initial weights associated with A) will remain in the same region A or $\mathbf{w}^t$ will keep *jumping* between regions in Ω ? We will try to find an answer to this question by looking at the direction along the largest eigenvector associated with A . Assuming that A is diagonalizable, we can compute the eigendecomposition of A as below, where D is an $n \times n$ diagonal matrix whose entries are the eigenvalues $\{\lambda_i\}_{i=0}^n$, and P is an $n \times n$ matrix whose columns are the eigenvectors.

$$\mathbf{w}^t = P D^t P^{-1} \mathbf{w}^0$$

Let $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be the set of normalized eigenvectors. We can then express the weights $\mathbf{w}$ in the basis of $\mathcal{B}$:

$$[\mathbf{w}^t]_{\mathcal{B}} = D^t [\mathbf{w}^0]_{\mathcal{B}}$$

where $[\mathbf{w}^0]_{\mathcal{B}} = \sum_{i=0}^n a_i \mathbf{v}_i$, for some $a_i \in \mathbb{C}$. Then,

$$[\mathbf{w}^t]_{\mathcal{B}} = D^t [\mathbf{w}^0]_{\mathcal{B}} = a_0 \lambda_0^t \mathbf{v}_0 + a_1 \lambda_1^t \mathbf{v}_1 + \dots + a_n \lambda_n^t \mathbf{v}_n$$

Normalizing,

$$[\mathbf{w}_N^t]_{\mathcal{B}} = \frac{a_0 \lambda_0^t \mathbf{v}_0 + a_1 \lambda_1^t \mathbf{v}_1 + \dots + a_n \lambda_n^t \mathbf{v}_n}{\sqrt{a_0^2 \lambda_0^{2t} + a_1^2 \lambda_1^{2t} + \dots + a_n^2 \lambda_n^{2t}}}$$

Let λ_k be the eigenvalues with the largest value in modulus and $\mathbf{v}_k$ its associated eigenvector. Then, as $t \rightarrow \infty$

$$[\mathbf{w}_N^t]_{\mathcal{B}} \rightarrow \mathbf{v}_k$$

If the largest eigenvalue λ_k has multiplicity $1 < m < n$, with associated eigenvectors $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$, then as $t \rightarrow \infty$,

$$[\mathbf{w}_N^t]_{\mathcal{B}} \rightarrow \frac{a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_m \mathbf{v}_m}{\sqrt{a_1^2 + a_2^2 + \dots + a_m^2}} \quad (11)$$

In summary, given $\mathbf{w}^t$ in region A, as $t \rightarrow \infty$ the weights will convergence to the eigenvector with the biggest eigenvalue (or a combination of eigenvectors if the eigenvalue has multiplicity larger than 1). We then just need to change $[\mathbf{w}_N^t]_{\mathcal{B}}$ to the canonical basis using the matrix of eigenvectors P and normalize to add up to n . If $\mathbf{w}^t$ still belongs to the same region A, then there is a possibility that the algorithm has converged. This is still not sufficient to prove convergence because in equation (10) we are not normalizing the weights after every iteration. So even if the eigenvector $\mathbf{v}_k$ lies inside the same polytope as $\mathbf{w}^0$, due to the normalization, some weights may swap from being smaller than 1 to larger than 1, or viceversa, moving to another polytope and changing the matrix A . Nevertheless, this approach is sufficient to prove non-convergence if we identify a bouncing between two regions, as we will describe in the following section.

3.2.1 Non-Convergence of the Ollivier Ricci Flow

So far, there are no examples in literature of non-convergence of the Ollivier Ricci Flow. We believe that there are two scenarios that can cause the weights to diverge; Firstly, the normalization factor can create a bouncing between edges being larger and smaller than 1 and, secondly, a bouncing can occur between different optimal paths joining two nodes. In this section, we provide an example of non-convergence due to the first reason stated.

Theorem 1: Let $G(V,E)$ be an undirected and unweighted graph consisting of two triangles glued together along an edge, as illustrated in Figure 8. The Ollivier Ricci flow with the mass distribution defined in (7) converges for $\frac{1}{3} \leq \alpha < \frac{1}{2}$ and doesn't converge for $\alpha < \frac{1}{3}$.

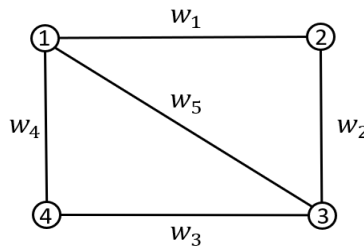


Figure 8: Graph G from Theorem 1

We will experimentally and theoretically prove Theorem 1 in the following sections. The labelling of the nodes and the edges of G illustrated in Figure 8 is the one we will refer to throughout the next sections.

3.2.2 Experimental Analysis

We have used the Python code provided in [10] to investigate the convergence of the Ricci Flow for graph G . The function I have coded to generate the following graphs can be found in Appendix III. The experimental analysis has been very important to get an intuition of how the algorithm

behaves, which has been very useful when constructing the theoretical poof of Theorem 1 in the next section.

In Figure 9, we observe that the Ricci Flow diverges for $\alpha = \frac{1}{4}$ and converges for $\alpha = \frac{1}{3}$ and $\alpha = \frac{1}{2}$. In addition, for $\alpha = \frac{1}{3}$ the weight of the exterior edges converges to $\frac{15}{13}$ and the weight of the interior edge converges to $\frac{5}{13}$ in just one iteration, and for $\alpha = \frac{1}{2}$ the weight of the exterior edges converges to $\frac{5}{4}$ and the weight of the interior edge converges to 0. We would prove all these results theoretically in the next section.

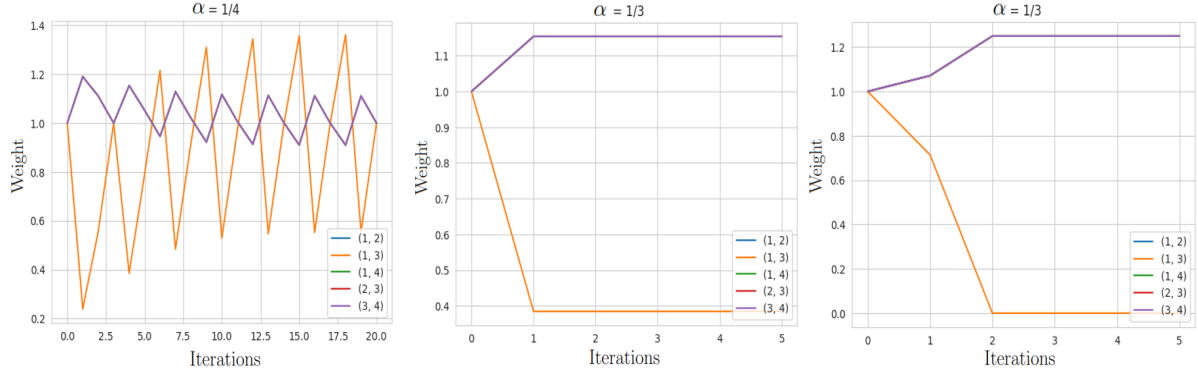


Figure 9: Graphs of the edge weights in every iteration of the Ricci Flow for $\alpha = \frac{1}{4}, \alpha = \frac{1}{3}, \alpha = \frac{1}{2}$

We have also plotted the change in the weights of the exterior edges along the Ricci Flow for different values of α in a 3D plot as seen in Figure 10 below. It can be clearly seen how the weights bounce for $\alpha < \frac{1}{3}$ and converge otherwise.

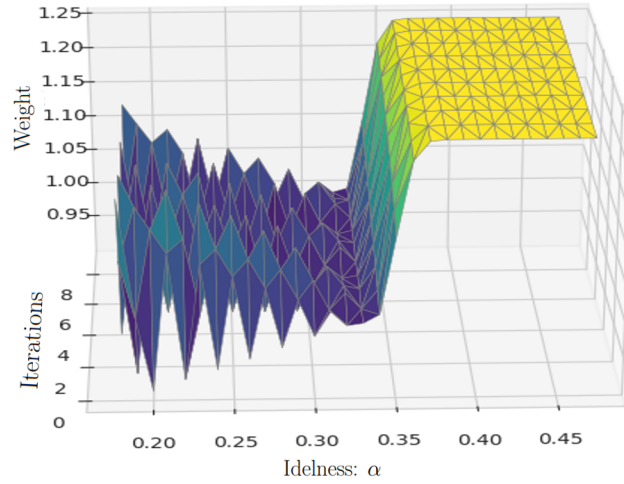


Figure 10: 3D plot of the weights of the exterior edge along iterations of the Ricci Flow for different values of α

3.2.3 Proof of Theorem 1

We will proof Theorem 1 using the approach that we described in Section 3.2. This graph only has two types of edges: the exterior and the interior ones. Therefore, we can divide the sample space Ω on three convex polytopes A_1 , A_2 and A_3 , defined as below:

$$\mathbf{w}^t \in A_1 \text{ if } \mathbf{w}_1 = \mathbf{w}_2 = \mathbf{w}_3 = \mathbf{w}_4 = \mathbf{w}_5 = 1$$

$$\mathbf{w}^t \in A_2 \text{ if } \mathbf{w}_5 \leq 1 \text{ and } \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4 > 1$$

$$\mathbf{w}^t \in A_3 \text{ if } \mathbf{w}_5 > 1 \text{ and } \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4 \leq 1$$

Each one of these regions is subdivided into smaller sub-regions determined by the set of optimal transport plans, as we explained in Section 3.2. In particular, for a fixed α , when $\mathbf{w}_t \in A_3$ there are two possible optimal paths from node 1 to node 3: Through the diagonal (edge 5) or through the exterior edges (edge 1 and 2, or edge 3 and 4), as can be seen in Figure 11 below. We will discuss these subregions in more detail in the next pages.

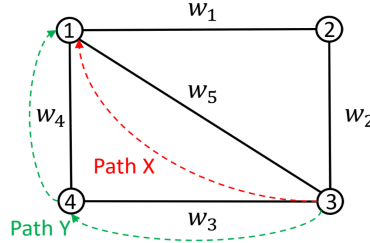


Figure 11: Graph G with labelling of nodes and edges. There are two possible optimal paths joining node 1 and node 2: Path X and Path Y

We call the first path *Path X* and the second one *Path Y*. Therefore, putting these together we get to the following representation of the sample space Ω for G:

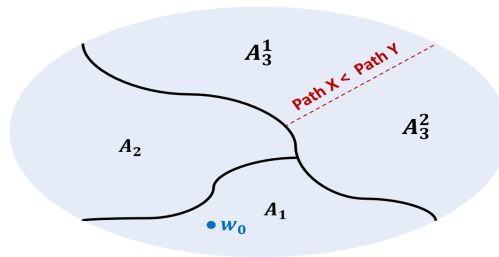
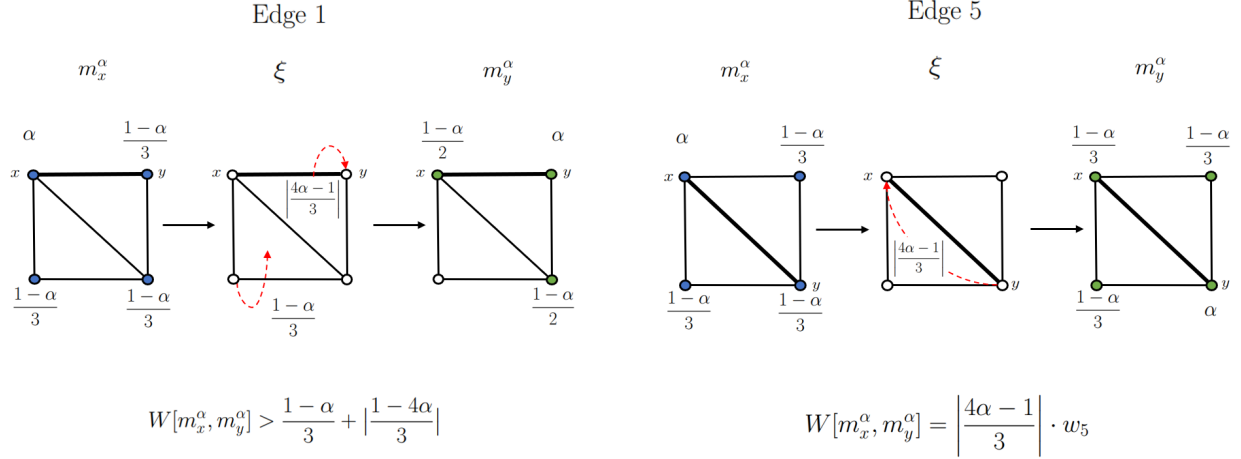


Figure 12: Representation of the sample space Ω for the graph G

As G is an unweighted graph, $\mathbf{w}^0$ initially lies in region A_1 . The next step is to calculate the Wasserstein distance between adjacent nodes. For edges 1,2,3,4 (external edges), there are 3 optimal transportation plans depending on the value of α . In particular, for $\alpha < \frac{1}{7}$, $\frac{1}{7} < \alpha < \frac{1}{2}$ and $\alpha > \frac{1}{2}$. To avoid considering these cases separately, we will find a common lower bound for the Wasserstein distance and we will then prove that this bound is enough to make $w_5^1 < 1$ and $w_1^1 = w_2^1 = w_3^1 = w_4^1 > 1$. The figure and table below outlines the steps used to obtain the lower bound in the Wasserstein distance for edges 1,2,3,4 and the step used to obtain the optimal transportation plan in edge 5.



Optimal transport plan edges 1,2,3,4

1. As m_y^α does not contain any mass at node 4, we will have to move $\frac{1-\alpha}{3}$ mass from node 4 in m_x^α . Depending on the value of α , the optimal transport plan will move the mass to node 1 or to node 3.
 2. Regardless of the outcome of the last step, the optimal transport plan will have to move at least $|\frac{1-\alpha}{3} - \alpha| = |\frac{1-4\alpha}{3}|$ mass to node 2 (the difference between the two masses distributions at node 2).
 3. Then, $W[m_x^\alpha, m_y^\alpha] > \frac{1-\alpha}{3} + \left| \frac{1-4\alpha}{3} \right|$
-

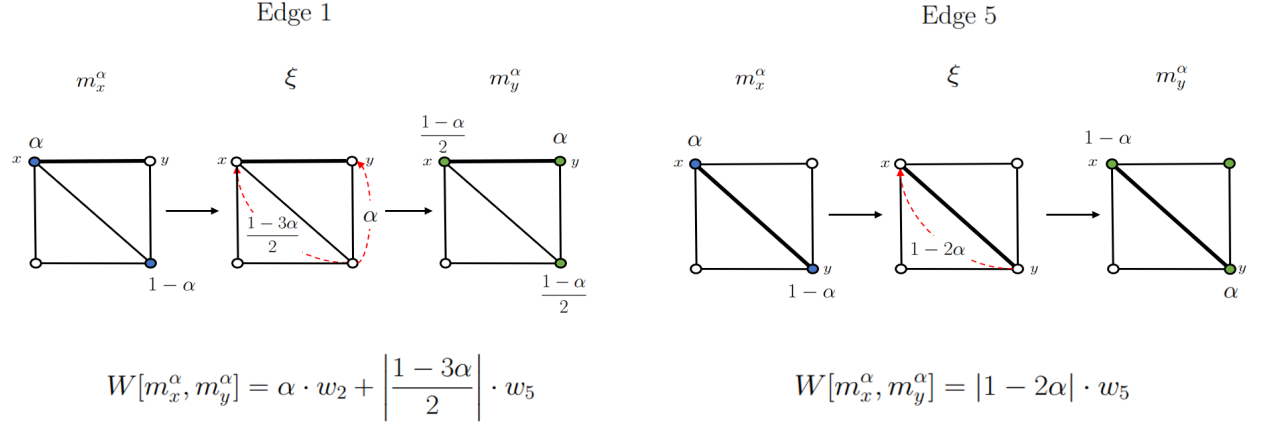
Optimal transport plan edge 5

1. We need to rearrange the remaining mass between node 1 and 3. As the weight of all the edges is the same, the optimal transport plan will move the mass along edge 5. Notice that the optimal transport plan is the same for all values of α .
 2. Then, $W[m_x^\alpha, m_y^\alpha] = \left| \frac{1-4\alpha}{3} \right|$
-

To obtain the values of the weight after the first iteration, we need to normalize the Wasserstein distances obtained above. One can check that for edge 5,

$$w_5^1 < \frac{5 \left| \frac{1-4\alpha}{3} \right|}{\left| \frac{1-4\alpha}{3} \right| + 4 \left(\frac{1-\alpha}{3} + \left| \frac{1-\alpha}{3} \right| \right)} = \frac{5 \left| \frac{1-4\alpha}{3} \right|}{5 \left| \frac{1-4\alpha}{3} \right| + 4 \left(\frac{1-\alpha}{3} \right)} < 1, \forall \alpha$$

And, by consequence, we will have that the weight of all the other edges will be larger than 1. Hence, we have moved to the A_2 region. In the diagram below one can see the masses distributions for edge 1 and edge 5 respectively in A_2 . As a reminder, here we are using the mass distribution defined in expression (7), so we only put mass along the edges with weight smaller than 1 (or equidistribute if all the adjacent edges have weight larger than 1). That is the reason why in the masses distribution in Edge 5 we don't put mass in node 2 or 4. We claim that for $\alpha < \frac{1}{2}$ (the case that we are considering in Theorem 1), there is one optimal transport plan in A_2 . The table below outlines the reason behind it.



Optimal transport plan edges 1,2,3,4

1. For $\alpha < \frac{1}{2}$, in m_x^α there is more mass in node 3 than in node 1, hence the optimal plan to obtain α mass in node 2 is to transport α mass from node 3 to node 2.
2. Then we just need to rearrange the remaining mass between node 1 and 3. As the weight of the interior edge is smaller than the weight at the exterior edge, the optimal plan is to move the difference of masses along edge 5.

Optimal transport plan edge 5

1. We just need to rearrange the difference in masses between node 1 and 3. As the weight of the interior edge is smaller than the weight at the exterior edge, the optimal plan is to move the mass along edge 5.

Using these optimal transports, we can construct the matrix A_2 as follows:

$$\begin{pmatrix} w_1^t \\ w_2^t \\ w_3^t \\ w_4^t \\ w_5^t \end{pmatrix} = \begin{pmatrix} 0 & \alpha & 0 & 0 & \left| \frac{1-3\alpha}{2} \right| \\ 0 & 0 & \alpha & 0 & \left| \frac{1-3\alpha}{2} \right| \\ 0 & 0 & 0 & \alpha & \left| \frac{1-3\alpha}{2} \right| \\ \alpha & 0 & 0 & 0 & \left| \frac{1-3\alpha}{2} \right| \\ 0 & 0 & 0 & 0 & |1-2\alpha| \end{pmatrix}^t \begin{pmatrix} w_1^0 \\ w_2^0 \\ w_3^0 \\ w_4^0 \\ w_5^0 \end{pmatrix} \quad (12)$$

The eigendecomposition of A_2 is

$$A_2 = \begin{pmatrix} -1 & 1 & i & -i & \frac{1}{2} \\ 1 & 1 & -1 & -1 & \frac{1}{2} \\ -1 & 1 & -i & i & \frac{1}{2} \\ 1 & 1 & 1 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -\alpha & 0 & 0 & 0 & 0 \\ 0 & \alpha & 0 & 0 & 0 \\ 0 & 0 & -i\alpha & 0 & 0 \\ 0 & 0 & 0 & i\alpha & 0 \\ 0 & 0 & 0 & 0 & 1-2\alpha \end{pmatrix} \begin{pmatrix} -1 & 1 & i & -i & \frac{1}{2} \\ 1 & 1 & -1 & -1 & \frac{1}{2} \\ -1 & 1 & -i & i & \frac{1}{2} \\ 1 & 1 & 1 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The biggest eigenvalue is $1-2\alpha$ for $\alpha < \frac{1}{3}$, α for $\alpha > \frac{1}{3}$ and $1-2\alpha = \alpha$ for $\alpha = \frac{1}{3}$. We will analyze these three cases separately.

Case 1: $\alpha < \frac{1}{3}$. The eigenvector associated with the largest eigenvalue is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1)^T$. Changing to the canonical basis using the matrix of eigenvectors above, we get that the weights converge to $(0, 0, 0, 0, 1)^T$. This doesn't belong to the region A_2 anymore. As $w_5 > w_1, w_2, w_3, w_4$, it belongs to A_3 . Nevertheless, we still need to determine if the weights will be in the A_3^1 or A_3^2 region. That is, whether the optimal path is *Path X* or *Path Y* from Figure 11.

Lemma 2: Given that the weights of graph G are such that $w_5 < w_1, w_2, w_3, w_4$, the resulting weights obtained by applying A_2 are such that the *Path X* is always the optimal path between node 1 and 3.

Proof: To prove that *Path X* is the optimal path between node 1 and 3, we need to prove that the resulting weights after applying A_2 are such that $w_5^t < w_1^t + w_2^t$. Due to normalization we know that at the end of the previous iteration $w_1^t + w_2^t + w_3^t + w_4^t + w_5^t = 5$ and due to the symmetry of the graph we have that $w_1^t = w_2^t = w_3^t = w_4^t$. Putting these two results together, we get that $w_5 + 4w_1 = 5$. So, $w_5^t < w_1^t + w_2^t \rightarrow w_5^t < 2w_1 \rightarrow w_5^t < \frac{5}{3}$. Hence, to prove that $w_5^t < w_1^t + w_2^t$ we need to prove that $w_5^t < \frac{5}{3}$. We don't know the exact values of the weight from the previous iteration, so we would have to work with general values instead. Let $\mathbf{w}^{t-1} = (1 + \delta, 1 + \delta, 1 + \delta, 1 + \delta, 1 - 4\delta)^T$ where $0 < \delta < \frac{1}{4}$. (Note that as $\mathbf{w}^{t-1} \in A_2, w_5 < 1$ and $w_1, w_2, w_3, w_4 > 1$). We need to prove that for all $0 < \delta < \frac{1}{4}$ and $0 < \alpha < \frac{1}{3}$, the resulting weights after applying A_2 to $\mathbf{w}^{t-1}$ are such that $w_5 < \frac{5}{3}$.

$$\begin{pmatrix} 0 & \alpha & 0 & 0 & \frac{1-3\alpha}{2} \\ 0 & 0 & \alpha & 0 & \frac{1-3\alpha}{2} \\ 0 & 0 & 0 & \alpha & \frac{1-3\alpha}{2} \\ \alpha & 0 & 0 & 0 & \frac{1-3\alpha}{2} \\ 0 & 0 & 0 & 0 & 1-2\alpha \end{pmatrix} \begin{pmatrix} 1+\delta \\ 1+\delta \\ 1+\delta \\ 1+\delta \\ 1-4\delta \end{pmatrix} = \begin{pmatrix} \alpha(\delta+1) + \frac{1-3\alpha}{2}(1-4\delta) \\ \alpha(\delta+1) + \frac{1-3\alpha}{2}(1-4\delta) \\ \alpha(\delta+1) + \frac{1-3\alpha}{2}(1-4\delta) \\ \alpha(\delta+1) + \frac{1-3\alpha}{2}(1-4\delta) \\ (1-2\alpha)(1-4\delta) \end{pmatrix}$$

Normalizing, we get that $w_5^t = 5 \frac{(1-2\alpha)(1-4\delta)}{\alpha(9\delta-4) - 3(\delta-1)}$. Plotting this inequality on Python (Figure 18 below), we indeed get that $w_5^t = 5 \frac{(1-2\alpha)(1-4\delta)}{\alpha(9\delta-4) - 3(\delta-1)} < \frac{5}{3}, \forall \alpha \in (0, \frac{1}{3}), \forall \delta \in (0, \frac{1}{4})$, as required.

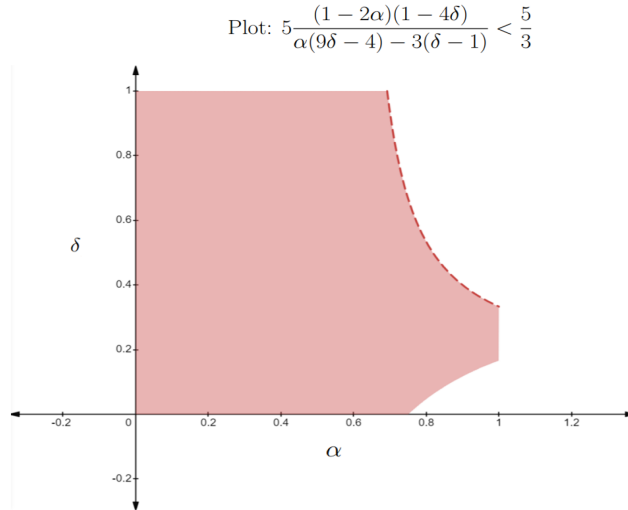
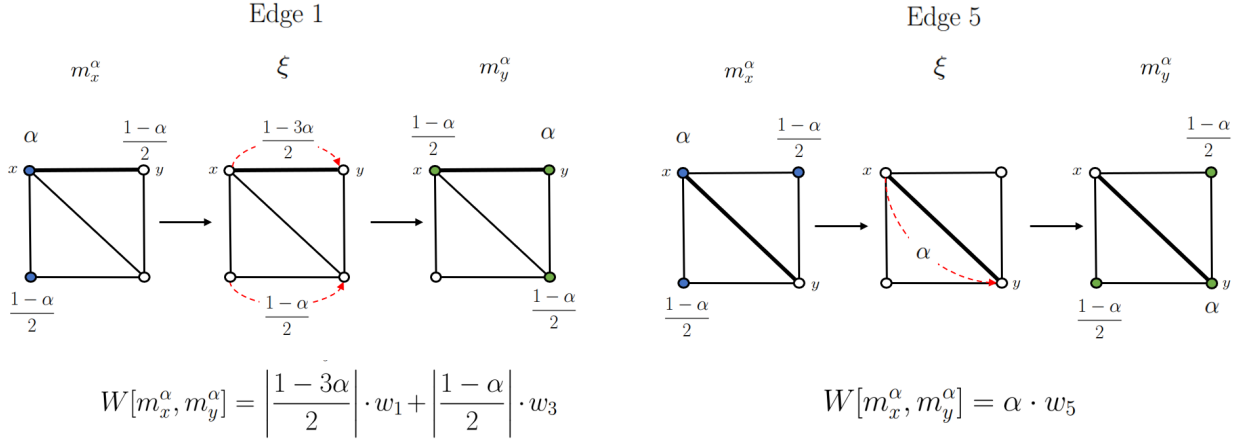


Figure 13: Region satisfying the inequality

In conclusion, $\mathbf{w}^t$ has moved from the polytope A_2 to the polytope A_3^1 . Therefore, we need to compute the new matrix A_3^1 following the same procedure as before. Now, $w_5 > 1$ and $w_1, w_2, w_3, w_4 < 1$, so the mass distribution is as follows:



On the table below there is a justification of the optimal transport plan.

Optimal transport plan edges 1,2,3,4	Optimal transport plan edge 5
- Starting at the distribution around x, we need to move all the mass from node 4. As node 3 needs the same mass contained in node 4, the optimal transport plan is to move the mass from node 4 to node 3 along edge 3. - We then move the extra mass from node 2 to node 1 along edge 1.	To move from m_x^α to m_y^α we only need to redistribute the mass in node 1 and 3. As we have proved in Lemma 2 that Path X is the optimal path, we obtain that the optimal transportation plan is just to move α along edge 5.

Using these transportation plans, we can construct the matrix A_3^1 as follows:

$$\begin{pmatrix} w_1^t \\ w_2^t \\ w_3^t \\ w_4^t \\ w_5^t \end{pmatrix} = \begin{pmatrix} \frac{1-3\alpha}{2} & 0 & \frac{1-\alpha}{2} & 0 & 0 \\ 0 & \frac{1-3\alpha}{2} & 0 & \frac{1-\alpha}{2} & 0 \\ \frac{1-\alpha}{2} & 0 & \frac{1-3\alpha}{2} & 0 & 0 \\ 0 & \frac{1-\alpha}{2} & 0 & \frac{1-3\alpha}{2} & 0 \\ 0 & 0 & 0 & 0 & \alpha \end{pmatrix}^t \begin{pmatrix} w_1^0 \\ w_2^0 \\ w_3^0 \\ w_4^0 \\ w_5^0 \end{pmatrix}$$

Its eigendecomposition is

$$\begin{pmatrix} 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} \alpha & 0 & 0 & 0 & 0 \\ 0 & -\alpha & 0 & 0 & 0 \\ 0 & 0 & -\alpha & 0 & 0 \\ 0 & 0 & 0 & 1-\alpha & 0 \\ 0 & 0 & 0 & 0 & 1-\alpha \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

As $\alpha < \frac{1}{3}$, the biggest eigenvalue is $1 - \alpha$ with associated eigenvectors $\left\{ (1, 0, 1, 0, 0)^T, (0, 1, 0, 1, 0)^T \right\}$. To find the weights where these eigenvectors converge to, we need to find the coefficients a_1 and a_2 from the expression 11. We don't know the exact values of $[w^0]_{\mathcal{B}}$ (as we don't the exact values of the weights after the 1st iteration), but due to the symmetry of the graph $w_1^0 = w_2^0 = w_3^0 = w_4^0$, hence $a_1 = a_2$, and $[\mathbf{w}]_{\mathcal{B}}$ converges to $(0, 0, 0, 1, 1)^T$. Changing to the canonical basis and normalizing, we get that weights converge to $(\frac{5}{4}, \frac{5}{4}, \frac{5}{4}, \frac{5}{4}, 0)^T$. This doesn't belong to the polytope A_3^1 anymore as clearly $w_5 < w_1, w_2, w_3, w_4$. In fact, $\mathbf{w}_t$ is on the region A_2 again.

In conclusion, for $\alpha < \frac{1}{3}$, $\mathbf{w}_t$ will keep bouncing between A_2 and A_3^1 and the Ollivier Ricci Flow will not converge.

Case 2: $\frac{1}{3} < \alpha < \frac{1}{2}$ The eigenvectors associated with the largest eigenvalue $|\alpha|$ are $\left\{ (-1, 1, -1, 1, 0)^T, (1, 1, 1, 1, 0)^T, (i, -1, -i, 1, 0)^T, (-i, -1, i, 1, 0)^T \right\}$. As before, we need to find the coefficients a_1, a_2, a_3, a_4 respectively from expression 11. By the symmetry of the graph ($w_1^t = w_2^t = w_3^t = w_4^t$), we get that $a_1, a_3, a_4 = 0$, and $a_2 = \frac{2w_1^0}{w_5^0}$, where w_i^0 are the initial weights associated with polytope A_2 . Hence, the weights $[\mathbf{w}^t]_{\mathcal{B}}$ will converge to $(0, \frac{2w_1^0}{w_5^0}, 0, 0, 0)^T$. Changing to the canonical basis and normalizing, we get that the weights converge to $(\frac{5}{4}, \frac{5}{4}, \frac{5}{4}, \frac{5}{4}, 0)^T$. This is still in the A_2 polytope, but this is not enough to prove convergence as we have to make sure that the contribution of the other eigenvectors on the first iterations is not enough to make the weight $\mathbf{w}_t$ change polytope due to normalization. We will prove that this is not the case by manipulating the inequalities associated to the matrix A_2 . Assume that at iteration t the normalized weights $\mathbf{w}_t$ satisfy that $\mathbf{w}_t \in A_2$. This leads to the following set of inequalities:

$$\begin{cases} 4w_5^t < w_1^t + w_2^t + w_3^t + w_4^t \end{cases} \quad (13)$$

$$\begin{cases} 4w_1^t > w_2^t + w_3^t + w_4^t + w_5^t \end{cases} \quad (14)$$

$$\begin{cases} 4w_2^t > w_1^t + w_3^t + w_4^t + w_5^t \end{cases} \quad (15)$$

$$\begin{cases} 4w_3^t > w_1^t + w_2^t + w_4^t + w_5^t \end{cases} \quad (16)$$

$$\begin{cases} 4w_4^t > w_1^t + w_2^t + w_3^t + w_5^t \end{cases} \quad (17)$$

We have to prove that those inequalities are still satisfied throughout the iterations. We will start by analyzing inequality 13. We want to prove that

$$4w_5^{t+1} < w_1^{t+1} + w_2^{t+1} + w_3^{t+1} + w_4^{t+1} \quad (18)$$

We can re-write the weights at the iteration $t+1$ in terms of the weights at iteration t by applying the matrix A_2 to $\mathbf{w}_t$. Substituting the result into inequality 18 we get:

$$4|1 - 2\alpha|w_5^t < \alpha(w_1^t + w_2^t + w_3^t + w_4^t) + 2|1 - 3\alpha|w_5^t$$

Rearranging, we get that

$$4\left(|1 - 2\alpha| - \frac{|1 - 3\alpha|}{2}\right)w_5^t < \alpha(w_1^t + w_2^t + w_3^t + w_4^t)$$

We can now multiply both sides of the first inequality from 13 by $|1 - 2\alpha| - \frac{|1 - 3\alpha|}{2}$. Noticing that $|1 - 2\alpha| - \frac{|1 - 3\alpha|}{2} < \alpha$ for all $\alpha > \frac{1}{3}$ we get that

$$4\left(|1 - 2\alpha| - \frac{|1 - 3\alpha|}{2}\right)w_5^t < \left(|1 - 2\alpha| - \frac{|1 - 3\alpha|}{2}\right)(w_1^t + w_2^t + w_3^t + w_4^t) \leq \alpha(w_1^t + w_2^t + w_3^t + w_4^t), \quad \forall \alpha > \frac{1}{3}$$

Therefore, we have proved that inequality 18 is satisfied on every iteration. The other inequalities can be proven analogously, the procedure for inequality 14 is outlined below. We want to proof that:

$$4w_1^{t+1} > w_2^{t+1} + w_3^{t+1} + w_4^{t+1} + w_5^{t+1}$$

Applying the matrix A_2 and substituting the result into the equation, we get that

$$4\left(\alpha \cdot w_2^t + \left|\frac{1-3\alpha}{2}\right|w_5^t\right) > \alpha(w_1^t + w_3^t + w_4^t) + 2|1-3\alpha|w_5^t + |1-2\alpha|w_5^t$$

Rearranging,

$$\alpha \cdot w_1^t < \alpha(4w_2^t - w_3^t - w_4^t) - |1-2\alpha|w_5^t$$

One can check that $1-2\alpha < \alpha \ \forall \alpha > \frac{1}{3}$. Using the fact that $w_1^t = w_2^t = w_3^t = w_4^t > w_5^t$, we conclude that inequality 18 is satisfied in every iteration.

In conclusion, for $\alpha > \frac{1}{3}$, the weights after the Ricci Flow will converge to $(\frac{5}{4}, \frac{5}{4}, \frac{5}{4}, \frac{5}{4}, 0)^T$. This value coincides with the one obtained experimentally in Section 3.2.2.

Case 3: $\alpha = \frac{1}{3}$ When $\alpha = \frac{1}{3}$, all the eigenvalues have the same magnitude. Let $[\mathbf{w}^0]_{\mathcal{B}} = \sum_{i=0}^n a_i \mathbf{v}_i$ be the initial weights associated to A_2 on the basis of eigenvectors. As $t \rightarrow \infty$, $[\mathbf{w}^0]_{\mathcal{B}}$ will converge to:

$$[\mathbf{w}_N^t]_{\mathcal{B}} \rightarrow \frac{a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n}{\sqrt{a_1^2 + a_2^2 + \dots + a_n^2}} = \frac{[\mathbf{w}^0]_{\mathcal{B}}}{\sqrt{a_1^2 + a_2^2 + \dots + a_n^2}}$$

Which is just the normalized initial weights. Hence, the weights will remain constant through every iteration. The initial weights in A_2 , $\mathbf{w}^0$, are the weights after the first iteration in A_1 (as the weights move from A_1 to A_2 in just one iteration). So we just need to study the final weights in A_1 to find the exact values where the weights converge to. We will not calculate this value in the dissertation because the procedure is long and analogous to the techniques already used throughout the proof. Broadly speaking, we will have to construct the 3 versions of the matrix A_1 depending on the value of α (as $\alpha < \frac{1}{7}$, $\frac{1}{7} < \alpha < \frac{1}{2}$ and $\alpha > \frac{1}{2}$ have different optimal transport plans). We then just need to apply A_1 to the initial weights and normalize.

In conclusion, for $\alpha = \frac{1}{3}$ the edge weights will converge after 1 iteration. This is consistent with the experimental results, as in the middle graph of Figure 9 one can see that the weights remain constant after the first iteration.

To summarize our analysis, we get that the weights after Ricci Flow the weights will converge/-diverge as follows:

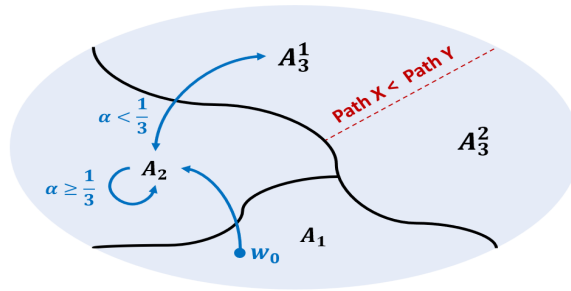


Figure 14: Summary of the convergence of the Ricci Flow for G

These theoretical results are consistent with the experimental results obtained in section 3.2.2.

3.2.4 Interesting points and future work on the convergence of the Ricci Flow

Along the experimental and theoretical analysis we have come across some points that would be interesting to investigate in a future work.

1. Throughout out the theoretical analysis we have noticed that the weights only have a direction towards eigenvectors associated with real and positive eigenvalues. Negative and complex eigenvalues could create another type of bouncing on the edge weights that could prevent convergence. We believe that there is no feasible optimal transport plan that will give rise to this situation, but it would be interesting to understand theoretically the reason why.
2. In this dissertation, we have only proved the non-convergence of the Ricci Flow for the specific graph G shown in Figure 8. Nevertheless, this is a behaviour that we have seen in other graphs containing a high number of triangles. In Figure 15 there are some examples of non-convergence of the Ricci Flow that we came across during the experimental analysis. An interesting future work will be to study if it is possible to construct an arbitrarily large graph exhibiting this phenomena.

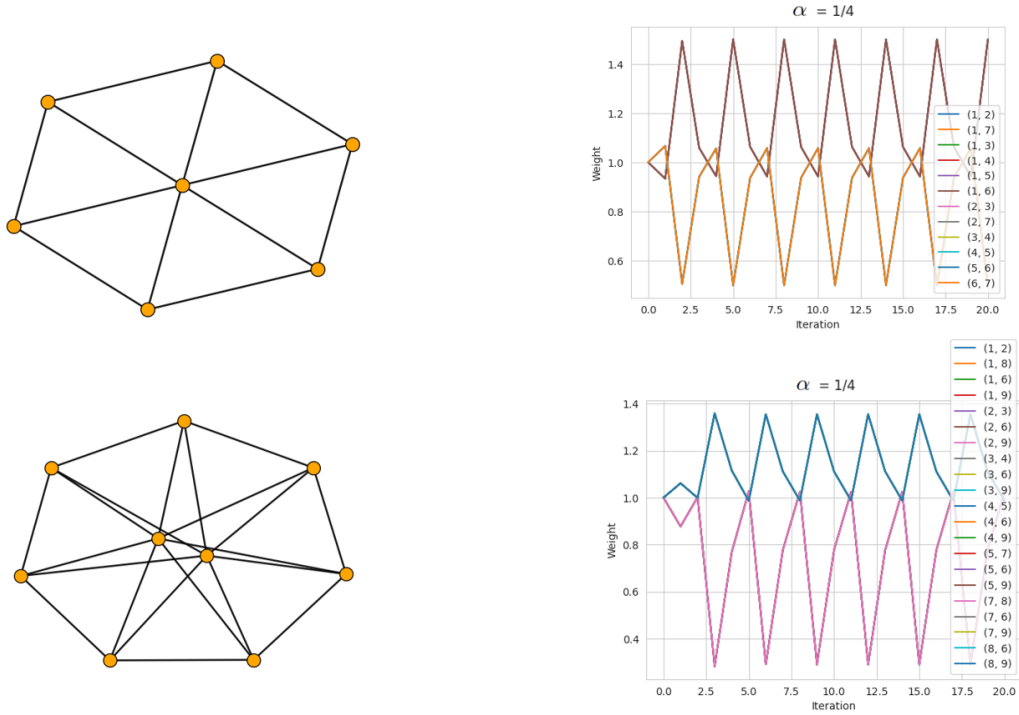


Figure 15: Other examples of non-convergence of Ollivier Ricci Flow

3. We have only proved non-convergence when using the mass distribution defined in 7, which is not the one used for practical purposes as we discussed earlier. Experimentally, we were able to check that using the continuous mass distribution from Algorithm 1, the Ricci Flow applied to G doesn't converge for $p > 5$. This is because for large values of p , the continuous mass distribution defined in Algorithm 1 is *similar enough* to the step function we have been working on (7). It could be interesting to extend our proof for the continuous version of the mass distribution. Our proof could be adapted by adding a small portion of mass δ along the edges with negative curvature and analyzing the convergence using the same techniques. This approach will only work as long as we can express the problem as a system of piecewise linear maps, which as we already discussed, we were not able to prove for the general case.

4 Robustness of Ollivier Ricci Curvature and Ricci Flow

In order to be able to use the Ollivier Ricci Curvature and Ollivier Ricci Flow in real-world applications it is essential for the algorithms to be robust to noise in the input data. To study the robustness of these algorithms we will apply Gaussian noise to the weights of an unweighted graph and study how sensible the algorithms are to those changes. We decided to use Gaussian noise as this represents small random fluctuations in the data that may arise in real-world datasets. In this case, we will use a truncated version of the normal distribution as we don't want the weights to be negative, nor to be larger than the number of edges in the graph. In particular, the noise will follow the below normal distribution with mean 0 and standard deviation σ .

$$\text{Noise}(\sigma) \sim -0.9 \leq \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{x}{\sigma})^2} \leq |E(G)|$$

In order to evaluate the robustness of the algorithms, we will express the problem in terms of a classification problem and look at the accuracy. That is, when working with the Ollivier Ricci Curvature, we will first compute the curvature of the edges without the noise. The edges with positive curvature will belong to one group (call it 0) and the edges with negative curvature will belong to the other group (call it 1). We will then apply the noise to the input weights of the graph and look how many edges have been classified correctly. We will repeat this procedure for different values of standard deviation σ . Our intuition is that as the standard deviation increases, the accuracy will decrease. For the Ollivier Ricci Flow, the methodology is the same, but we define one group as being the set of edges with weight smaller than 1 after the implementation of the Ricci Flow (call it 0) and the other group as the set of edges with weight larger or equal than 1 after the implementation of Ricci Flow (call it 1). Finally, the accuracy of the algorithm is defined as follows:

$$\text{Accuracy} = \frac{\text{Number of correct classifications}}{\text{Total number of classifications}}$$

Notice that this is a simplification of the problem, as we are not considering *how much* has the weight of a noisy edge changed with the Ricci Flow compared to its non noisy counterpart. We are only analyzing if the general behaviour of the algorithms (positive/negative curvature and shrunked/expanded edges) is the same.

The dataset that we have used to run our experiments is a random generated network from Python following a Watts–Strogatz model. This model generates graphs with small-world properties. These are types of graphs with small clustering coefficients and that have been found to represent well the dynamics of social networks [12]. The graph we have used contains 200 nodes and 1000 edges. We are not working with larger graphs due to the computational costs of the Ricci Flow.

In the diagram below illustrates the confusion matrix for the Ricci Curvature (right) and Ricci Flow (left) when a Gaussian noise with standard deviation 4 is applied to the initial weights. The algorithm classified most of the edges with initial positive curvature in the wrong group, indicating that these could be the more unstable edges. In the other hand, the Ricci Flow managed to classify most of the edges that initially shrunked (weight smaller than 1) correctly. Overall, the confusion matrices indicate that the Ricci Flow has managed to classify most of the edges in the correct group, while the Ricci Curvature seems to be more sensible to noisy data.

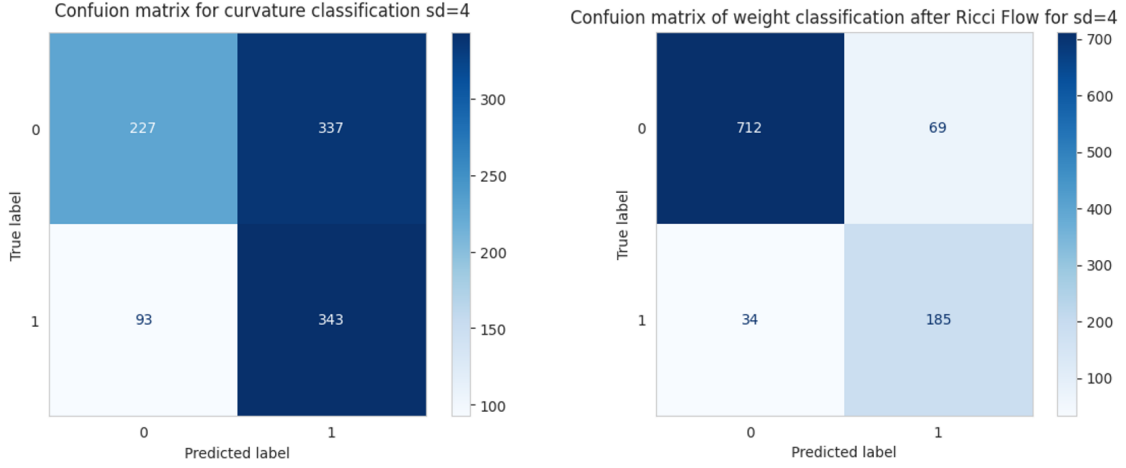


Figure 16: Accuracy in terms of the standard deviation for the Ricci Curvature and Ricci Flow

We then studied how the accuracy of the classification changed as we increase the value of the standard deviation σ . For our experiments, we have used the code outlined in Section 3.1. We have set the parameters $\alpha = 0.2$ and $p = 1$. The results can be seen in Figure 17.

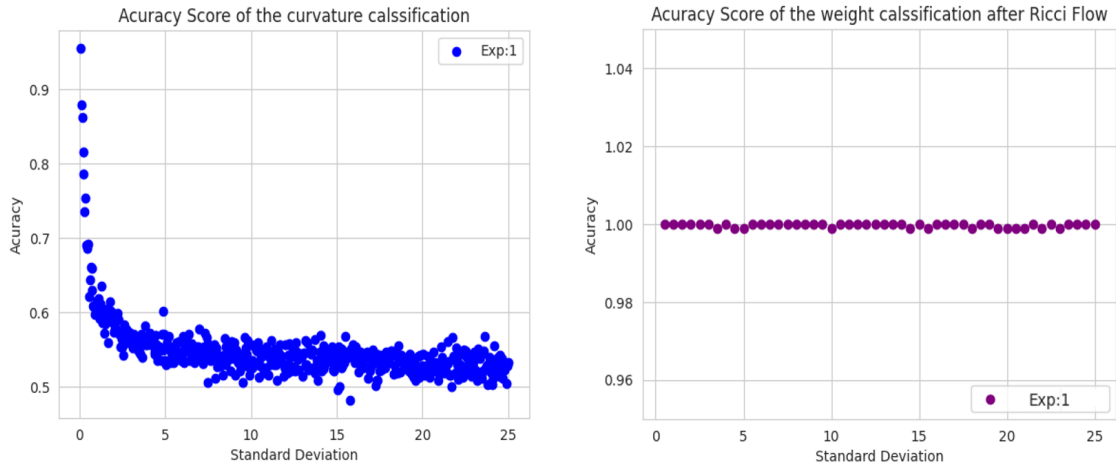


Figure 17: Accuracy in terms of the standard deviation for the Ricci Curvature and Ricci Flow

The accuracy of the Ricci Curvature quickly decreased as the value of the standard deviation increases, while the accuracy of the Ricci Flow is maintained consistently very close to 1. Notice that the accuracy of the curvature tends to 0.5. This is because we are only working with two groups (positive/negative curved edges), so for large values of σ the algorithm is obtaining the same accuracy as if it was guessing the values at random. It is interesting that even though the Ricci Flow is working with noisy values of curvature, it still manages to recover high values of accuracy.

In conclusion, the Ricci Flow seems to be a robust algorithm to noisy data, classifying most of the edges correctly even for large values of the standard deviation. In the other hand, the Ricci curvature appears to be a lot more sensible to changes in the input data. This should be considered when working with these notion of real-world applications.

5 Conclusion

In this report we studied the Ollivier Ricci Curvature and the Ollivier Ricci Flow.

We first studied the definition and properties of the Ollivier Ricci Curvature, and its connection to the Ricci Curvature defined in a manifold. We were able to understand how this notion give us information about the structure of a graph and how can this be used in network analysis applications. We were able to experimentally find regular graphs with constant curvature up to and including 11 vertices using random generator models in Python.

We then studied the Ollivier Ricci Flow. We were able to understand the algorithm used in the research paper *Community detection on networks with Ricci flow* [10] and explore potential applications to community detection. We then proved experimentally and theoretically that the Ollivier Ricci Flow doesn't converge by giving a counterexample. Finally, we implemented Python simulations to understand how robust are those notions to noisy dataset, a fundamental property to be understood for real-world applications. Our results showed that the Ricci Flow appears to be a robust algorithm, even for large value of noise, while the curvature seems to be more sensible to changes in the input data. A more in depth analysis should be carried out with different types of network and different parameters of the Ricci Flow to understand better this behaviour.

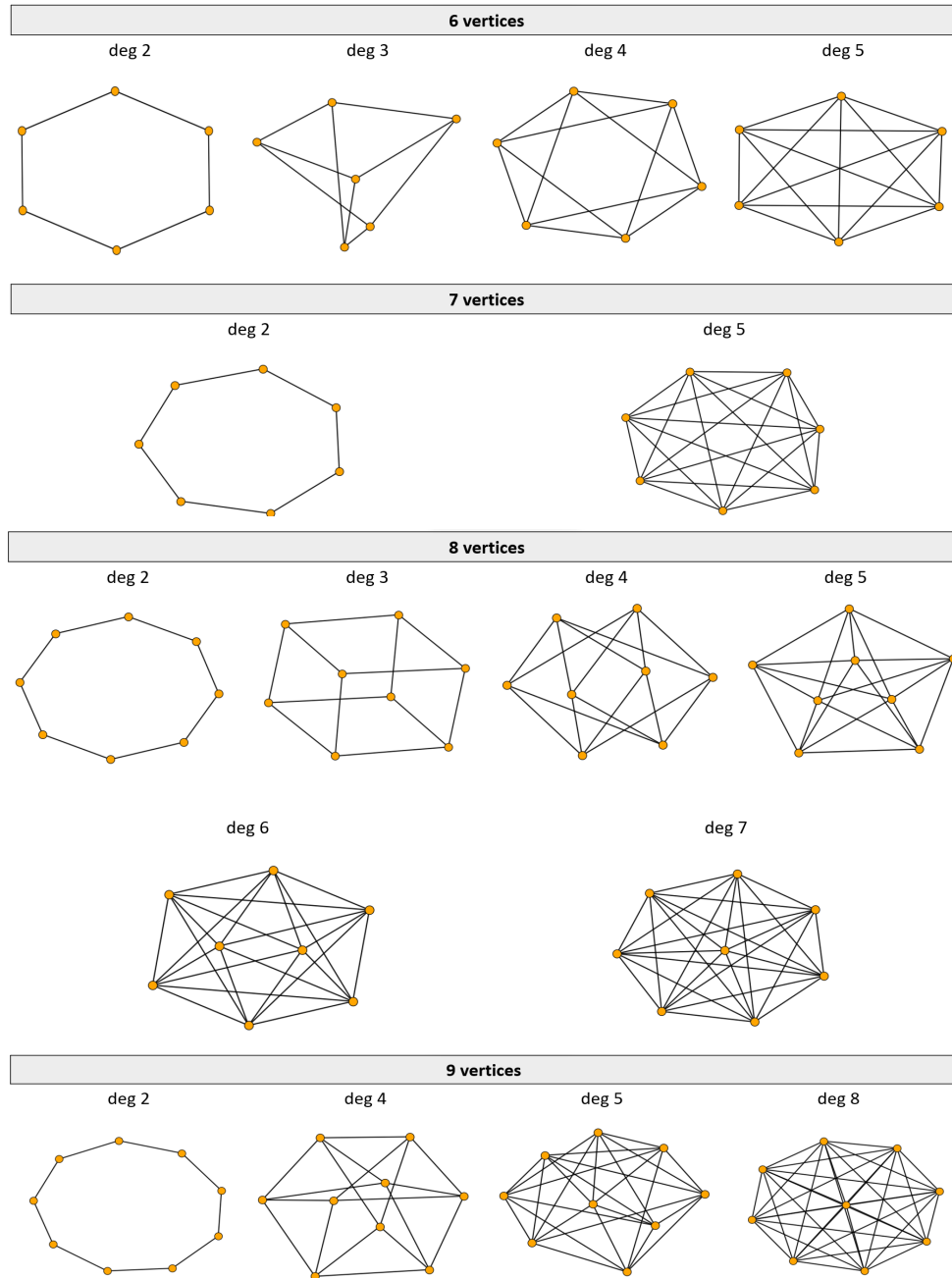
This project also gave rise to numerous future lines of research that we described in Section 3.2.4.

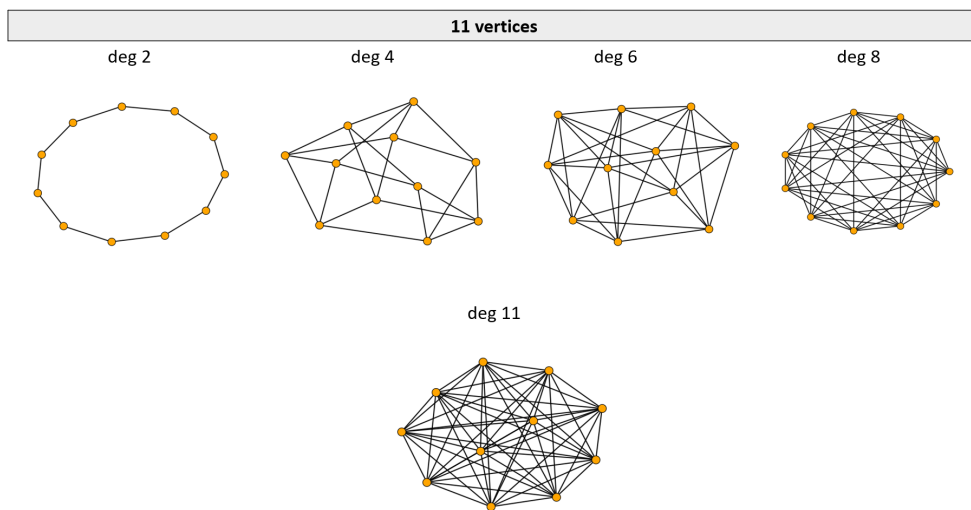
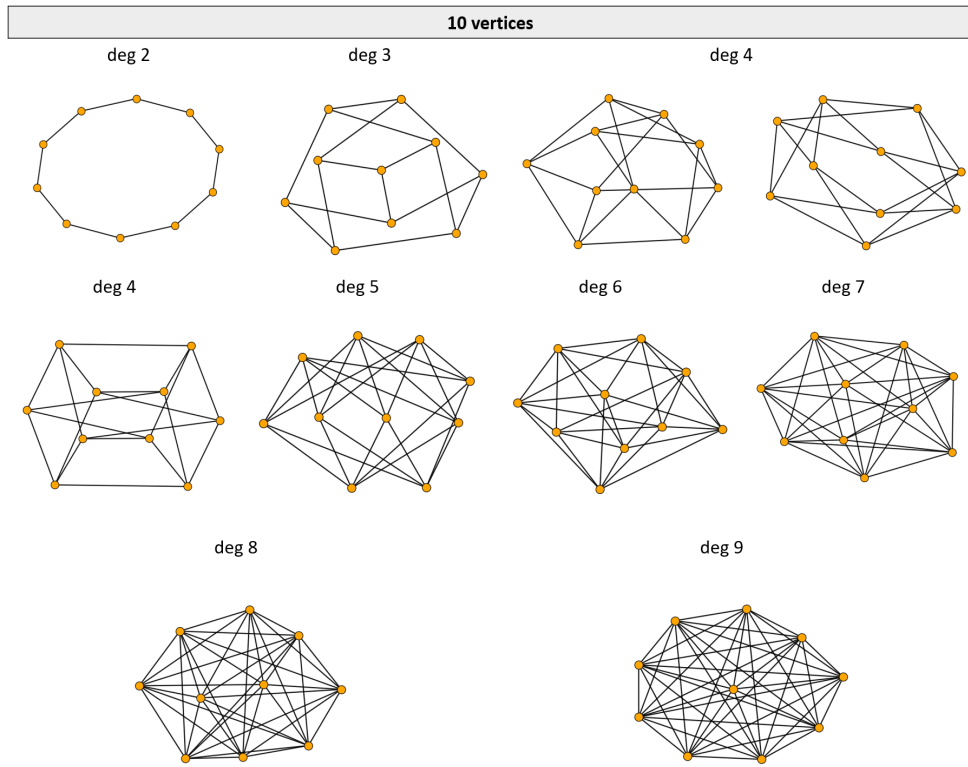
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6 Appendix

6.1 Appendix I: Regular graphs of constant curvature up to 11 vertices





6.2 Appendix II: Random graphs with constant curvature

```

    "Detect regular graphs with constant curvature"

#Clean file
e=open("/content/drive/MyDrive/adj21.txt", "r+")
for line in e:
    if line[0].isdigit() == True:
        edge[cont].append(line)
    else:
        if line[0] == 'G':
            print(cont)
        else:
            cont=cont+1

#Compute curvature
for i in range(0,size,2):
    sum=0
    G = nx.parse_adjlist(edge[i], nodetype=int) #create graph with given edge list
    for n1,n2 in list(G.edges()):
        orc = OllivierRicci(G, alpha=0.2) #compute ricci curvature
        orc.compute_ricci_curvature()
        G_orc = orc.G.copy()
        sum=sum + (G_orc[n1][n2]["ricciCurvature"])

x=[e for e in G.edges]
dif1=abs(sum - G_orc[x[0][0]][x[0][1]]["ricciCurvature"]*len(x))
#if the sum of the curvature of the edges is the same as the curvature of an
#edge times the number of edges then
#the graph is a potential candidate
if ((dif1 < 0.01) == True and (dif2 < 0.01) == True) :
    print("Found= ",i, len(x), sum, G_orc[x[0][0]][x[0][1]]["ricciCurvature"])

```

6.3 Appendix III: Experimental analysis of non-convergence

```

    "How the weight changes for every iteration of the Ricc Flow"

#Initialize Ricci Flow
G1=G #make a copy of the original graph
orf = OllivierRicci(G, weight="weight", alpha=0.1, proc=4, verbose="INFO")
number_of_lists = G.number_of_edges()
flow = [[] for i in range(number_of_lists)]
iter=20 #set number of iterations
num= [ ]
for i in range(iter):
    orf.set_verbose("ERROR")
    orf.compute_ricci_flow(iterations=i)
    G_rf = orf.G.copy()
    weights = nx.get_edge_attributes(G_rf, "weight").values()
    weight_list=list(weights)
    weights=np.array(weight_list)
    for j in range(number_of_lists):
        flow[j].append(weights[j])
    num.append(i)
    i=0
    G=G1 #reset G to the original graaph
#Plot results
for n1,n2 in list(G.edges()):
    plt.plot(num, flow[i], label=(n1, n2))
    i=i+1
plt.xlabel('Iteration')

```



```
plt.ylabel('Weight')
plt.legend(loc='lower right')
```

```
"3D Plot of the change in the weights throughout the Ricci Flow in function
of the idleness"

number_of_lists = G.number_of_edges()
flow = [ ]
iter=10 #Set number of iterations
p=0.1 #set alpha
num= [ ]
para=[ ]
G1=G #copy of the original graph
for j in range(14):
    orf = OllivierRicci(G, alpha=p, proc=4, verbose="INFO")
    for i in range(iter):
        orf.set_verbose("ERROR")
        orf.compute_ricci_flow(iterations=i)
        G_rf = orf.G.copy()
        weights = nx.get_edge_attributes(G_rf, "weight").values()
        weight_list=list(weights)
        weights=np.array(weight_list)
        flow.append(weights[3]) #choose to plot a specific edge (i.e edge 3)
        para.append(p)
        num.append(i)
    G=G1 #reset to original graph
    p=p+1/20

#plot results
plt.rcParams["figure.figsize"] = [7, 7]
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_trisurf(para,num,flow, cmap='viridis', edgecolor='none');
ax.set_ylabel('Iteration', fontsize=10)
ax.set_xlabel('Idleness: p', fontsize=10)
ax.set_zlabel('Weight', fontsize=10)
ax.view_init(elev=35., azimuth=-45, roll=0)
plt.show()
```

6.4 Appendix IV: Robustness analysis

```
"Robustness analysis using normal distribution truncated"

#Initialize graph using Watts Strogatz model
G = nx.watts_strogatz_graph(200,10, 0.2, seed=None)
Gcc = sorted(nx.connected_components(G), key=len, reverse=True)
G = G.subgraph(Gcc[0])
pos = nx.spring_layout(G)
nx.draw_networkx(G, pos, node_color='green', alpha=0.5, font_size =0, node_size=
20)

G0=G
iter=10
desv=0
value=[ ] #value of standard derviation
acuracy1=[ ]
for k in range(0,1):
    G1=G0
    i=0
    x=[ ]
```

```

desv=desv+0.1
value.append(desv)
X = get_truncated_normal(mean=0, sd=desv, low=-0.9, upp=number_of_lists ) #get
truncated normal

number_of_lists = G1.number_of_edges()
x.append(X.rvs(number_of_lists))
weight=np.full((1,number_of_lists), 1)
weight=np.ones(number_of_lists) #all the weights are set up to 1 as unweighted
graph

weight= x[0]+weight #weights with noise
flow = [[] for i in range(number_of_lists)]
num= [ ]
for n1,n2 in list(G1.edges()): #update weights with the noise on the grapg
    G1.edges[(n1,n2)]['weight'] = weight[0][i]
    i=i+1
orf = OllivierRicci(G1, weight="weight", alpha=0.2, base=math.e, exp_power=1,
proc=4, verbose="INFO") #initialize
Ricci Flow

for i in range(iter):
    G1=G0
    orf.set_verbose("ERROR")
    orf.compute_ricci_flow(iterations=i)
    G_rf = orf.G.copy()
    weights = nx.get_edge_attributes(G_rf, "weight").values()
    weight_list=list(weights)
    weights=np.array(weight_list)
    for j in range(number_of_lists):
        flow[j].append(weights[j])
    num.append(i)
predicted=[ ]
for i in range(0,number_of_lists): #classify in groups 0 or 1
    if flow[i][iter-1] <= 1:
        predicted.append(0)
    else:
        predicted.append(1)
accuracy1.append(metrics.accuracy_score(actuals, predicted))

#Plot Results
plt.scatter(value,accuracy1, label="Step Function", c='purple')
plt.title("Acuracy Score of the weight calssification after Ricci Flow")
plt.xlabel('Standard Deviation')
plt.ylabel('Acuracy')
plt.ylim(0.95,1.05)
plt.legend(loc='lower right')

```

Vertex operator algebras, fusion rules and modular transformations

Yi-Zhi Huang

Abstract

We discuss a recent proof by the author of a general version of the Verlinde conjecture in the framework of vertex operator algebras and the application of this result to the construction of modular tensor category structure on the category of modules for a vertex operator algebra.

0 Introduction

One of the most important discoveries by physicists in two-dimensional conformal field theory is the famous relation between the fusion rules and the action of the modular transformation $\tau \mapsto -1/\tau$ on the space of vacuum characters. It states that this action of the modular transformation diagonalizes the matrices formed by the fusion rules. This relation was first conjectured by Verlinde [V] based on a comparison between the fusion algebra of a rational conformal field theory and an algebra arising from the study of the genus-one part of the theory. Assuming the axioms for rational conformal field theories, Moore and Seiberg [MS1] proved this Verlinde conjecture by deriving a fundamental set of polynomial equations. Moore and Seiberg [MS2] also observed that the genus-zero part of these polynomial equations is analogous to braided tensor categories. The theory of complete sets of such polynomial equations was then called “modular tensor category” which was first suggested by I. Frenkel. Later, this notion of modular tensor category was reformulated precisely using the language of tensor categories (see, for example, [T] and [BK] for expositions and references on modular tensor categories).

Given a modular tensor category, we have fusion rules and an action of modular transformations. It is not difficult to show (see, for example, [BK]) that for the fusion rules and the action of the modular transformation $\tau \mapsto -1/\tau$ given in this way, the Verlinde conjecture holds. But this version of the Verlinde conjecture is not the original one because the action of the modular transformation is the one constructed from the modular tensor category, not the one on the space of the vacuum characters of the corresponding conformal field theory. The missing piece is an identification of the two actions of the same modular transformation $\tau \mapsto -1/\tau$, or equivalently, is a mathematical construction of the modular tensor category associated to a rational conformal field theory. Moreover, the starting point of the original work [MS1] and [MS2] of Moore and Seiberg is the axioms for rational conformal field theories, which are actually even more difficult to prove than the Verlinde conjecture. It is therefore desirable to formulate and prove a general and mathematical version of the Verlinde conjecture which should give precise and natural conditions on the vertex operator algebras (chiral algebras) such that the Verlinde conjecture holds for these algebras.

Recently, the author was able to formulate and prove such a general and mathematical version of the Verlinde conjecture. Using this result, the author has also proved the rigidity and nondegeneracy of the semisimple braided tensor category structure constructed by Lepowsky and the author on the category of modules for the vertex operator algebra. In particular, the modular tensor category structure on the category of modules for the vertex operator algebra is mathematically constructed. See [H7] and [H9] for details and see also [H8] for an announcement of the results. In the present paper, we shall discuss this general and mathematical version of the Verlinde conjecture, its proof and its application to the proofs of the rigidity and nondegeneracy of the braided tensor category structure mentioned above.

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1 Vertex operator algebras and fusion rules

Vertex (operator) algebras were introduced in 1986 by Borchers [B] in connection with representations of affine Lie algebras and the moonshine module

for the Monster finite simple group, which was conjectured to exist by Conway and Norton [CN] and constructed using vertex operators by Frenkel, Lepowsky and Meurman [FLM1] [FLM2]. These algebras are essentially equivalent to chiral algebras (see, for example, [MS2]) in physics, which were first studied systematically by Belavin, Polyakov and Zamolodchikov [BPZ] in 1984, though without the name chiral algebra. Here we explain briefly the basic concepts in the theory of vertex operator algebras.

A vertex operator algebra is a graded vector space $V = \coprod_{n \in \mathbb{Z}} V_{(n)}$ equipped with a vertex operator map $Y : V \otimes V \rightarrow V((z))$ (the space of formal Laurent series in z with finitely many negative power terms), a vacuum $\mathbf{1} \in V$ and a conformal element $\omega \in V$. These data must satisfy the following axioms which are very natural from the point of view of conformal field theory in physics: One formulation of the main axioms is: For $u_1, u_2, v \in V$, $v' \in V' = \coprod_{n \in \mathbb{Z}} V_{(n)}^*$, the series

$$\begin{aligned} &\langle v', Y(u_1, z_1)Y(u_2, z_2)v \rangle \\ &\langle v', Y(u_2, z_2)Y(u_1, z_1)v \rangle \\ &\langle v', Y(Y(u_1, z_1 - z_2)u_2, z_2)v \rangle \end{aligned}$$

are absolutely convergent in the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, respectively, to a common rational function in z_1 and z_2 with the only possible poles at $z_1, z_2 = 0$ and $z_1 = z_2$. Other axioms include:

$$\dim V_{(n)} < \infty$$

for $n \in \mathbb{Z}$,

$$V_{(n)} = 0$$

when n is sufficiently negative (these are called grading-restriction conditions);

$$Y(\mathbf{1}, z) = 1, \quad \lim_{z \rightarrow 0} Y(u, z)\mathbf{1} = u;$$

let $L(n) : V \rightarrow V$ be defined by $Y(\omega, z) = \sum_{n \in \mathbb{Z}} L(n)z^{-n-2}$, then

$$[L(m), L(n)] = (m - n)L(m + n) + \frac{c}{12}(m^3 - m)\delta_{m+n,0}$$

(c is called the central charge of V),

$$\frac{d}{dz}Y(u, z) = Y(L(-1)u, z)$$

and

$$L(0)u = nu \quad \text{for } u \in V_{(n)}$$

(n is called the weight of u and is denoted $\text{wt } u$). For $u \in V$, we write $Y(u, z) = \sum_{n \in \mathbb{Z}} u_n z^{-n-1}$. Then for homogeneous $u \in V$, the maps $u_n : V \rightarrow V$ have weights $\text{wt } u - n - 1$, that is, they map $V_{(m)}$ to $V_{(\text{wt } u - n - 1 + m)}$.

For vertex operator algebras, we have the following important condition which was first introduced by Zhu [Z]:

C_2 -cofiniteness condition: Let V be a vertex operator algebra and $C_2(V)$ be the subspace of V spanned by elements of the form $u_{-2}v$ for $u, v \in V$. Then we say that V is C_2 -cofinite or satisfies the C_2 -cofiniteness condition if $\dim V/C_2(V) < \infty$.

In other words, the C_2 -cofiniteness condition says that the coefficients of the terms in the first power of z in $Y(u, z)v$ for all $u, v \in V$ span the whole space V except for a finite-dimensional subspace. This condition is an easy consequence of the condition that the spaces of genus-one conformal blocks are finite-dimensional. The author showed in [H6] that together with some other conditions, the C_2 -cofiniteness condition also implies the finiteness of the dimensions of the spaces of genus-one conformal blocks. In practice, it is much easier to verify this condition than to prove the finiteness of the dimensions of the spaces of genus-one conformal blocks. For vertex operator algebras associated to the Wess-Zumino-Novikov-Witten models, vertex operator algebras associated to minimal models, lattice vertex operator algebras and the moonshine module vertex operator algebra, the C_2 -cofiniteness condition was stated to hold in Zhu's paper [Z] and was verified by Dong-Li-Mason [DLM] (see also [AN] for the case of minimal models).

A V -module can be defined simply as a $\mathbb{C}$ -graded vector space $W = \coprod_{n \in \mathbb{C}} W_{(n)}$ equipped with a vertex operator map $Y_W : V \otimes W \rightarrow W[[z, z^{-1}]]$ satisfying all those axioms for V which still make sense. We also need the notion of $\mathbb{N}$ -gradable weak V -module. An $\mathbb{N}$ -gradable weak V -module is an $\mathbb{N}$ -graded vector space $W = \coprod_{n \in \mathbb{N}} W_{[n]}$ and a vertex operator map

$$\begin{aligned} Y : V \otimes W &\rightarrow W[[z, z^{-1}]] \\ u \otimes w &\mapsto Y(u, z)w = \sum_{n \in \mathbb{Z}} u_n z^{-n-1} w \end{aligned}$$

satisfying all axioms for V -modules except that the condition $L(0)w = nw$ for $w \in W_{(n)}$ is replaced by $u_k w \in W_{[m-k-1+n]}$ for $u \in V_{(m)}$ and $w \in W_{[n]}$.

For V -modules W_1, W_2 and W_3 , an intertwining operator of type $\binom{W_3}{W_1 W_2}$ is a linear map $\mathcal{Y} : W_1 \otimes W_2 \rightarrow W_3\{z\}$, where $W_3\{z\}$ is the space of all series in complex powers of z with coefficients in W_3 , satisfying all those axioms for V which still make sense. For example, for $u \in V, w_1 \in W_1, w_2 \in W_2$ and $w'_3 \in W'_3 = \coprod_{n \in \mathbb{C}} (W_3)_{(n)}^*$,

$$\begin{aligned} & \langle w'_3, Y_{W_3}(u, z_1) \mathcal{Y}(w_1, z_2) w_2 \rangle \\ & \langle w'_3, \mathcal{Y}(w_1, z_2) Y_{W_2}(u, z_1) w_2 \rangle \\ & \langle w'_3, \mathcal{Y}(Y_{W_1}(u, z_1 - z_2) w_1, z_2) w_2 \rangle \end{aligned}$$

are absolutely convergent in the regions $|z_1| > |z_2| > 0, |z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, respectively, to a common (multivalued) analytic function in z_1 and z_2 with the only possible singularities (branch points) at $z_1, z_2 = 0$ and $z_1 = z_2$. Also

$$\frac{d}{dz} \mathcal{Y}(w_1, z) = \mathcal{Y}(L(-1)w_1, z).$$

We denote the space of intertwining operator of type $\binom{W_3}{W_1 W_2}$ by $\mathcal{V}_{W_1 W_2}^{W_3}$. The dimension of $\mathcal{V}_{W_1 W_2}^{W_3}$ is the fusion rule $N_{W_1 W_2}^{W_3}$.

2 Fusing and braiding isomorphisms

In the definitions of vertex operator algebra, module and intertwining operator, we see that the main axioms are all about products and iterates of vertex operators or products and iterates of one vertex operator and one intertwining operator. It is natural to expect that products and iterates of intertwining operators should have similar properties. Indeed, it was proved by the author in [H1] and [H5] that for vertex operator algebras satisfying suitable finiteness and reductivity conditions, intertwining operators satisfy associativity and commutativity properties. These properties give fusing and braiding isomorphisms.

Note that the associativity for intertwining operators is a strong version of the operator product expansion of “chiral vertex operators” (which is equivalent to intertwining operators for vertex operator algebras). In fact, in the important work [MS1] and [MS2] of Moore and Seiberg, in addition to other axioms for rational conformal field theories, the operator product expansion of chiral vertex operators was a basic assumption. In physics, though there were calculations for particular operators in particular examples, operator

product expansion was not a mathematical theorem and calculations based on it are by no means simple. The associativity for intertwining operators proved in [H1] and [H5] under suitable conditions is in fact a strong version of operator product expansion for intertwining operators in the following sense: The associativity states that the product of two intertwining operators is equal to the iterate of two other intertwining operators in a suitable region. If we expand the intertwining operator inside the other intertwining operator in the iterate, we immediately obtain the operator product expansion of intertwining operators. On the other hand, in [H1], the author proved associativity for intertwining operators under the assumption that the convergence and extension property and some other conditions hold. Operator product expansion for chiral vertex operators (intertwining operators) assumed in [MS1] and [MS2] actually implies the convergence and extension property. So the operator product expansion together with some other conditions implies associativity for intertwining operators. Since the associativity immediately gives the operator product expansion, but on the other hand, only together with some other conditions the operator product expansion gives the associativity, we see that the associativity is indeed stronger than the operator product expansion.

Let V be a simple vertex operator algebra of central charge c and V' the contragredient module of V as a V -module. In the remaining part of the present paper, we shall always assume that V satisfies the following conditions:

1. $V_{(n)} = 0$ for $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ and V' is isomorphic to V .
2. Every $\mathbb{N}$ -gradable weak V -module is completely reducible.
3. V is C_2 -cofinite, that is, $\dim V/C_2(V) < \infty$.

Note that finitely generated $\mathbb{N}$ -gradable weak V -modules are what naturally appear in the proofs of the theorems on genus-zero and genus-one correlation functions. Thus Condition 2 is natural and necessary because the Verlinde conjecture concerns V -modules, not finitely generated $\mathbb{N}$ -gradable weak V -modules. For vertex operator algebras associated to affine Lie algebras (Wess-Zumino-Novikov-Witten models) and vertex operator algebras associated to the Virasoro algebra (minimal models), Condition 2 can be verified easily by reformulating the corresponding complete reducibility results

in terms of the representation theory of affine Lie algebras and the Virasoro algebra.

From [DLM], we know that there are only finitely many inequivalent irreducible V -modules. Let $\mathcal{A}$ be the set of equivalence classes of irreducible V -modules. We denote the equivalence class containing V by e . For each $a \in \mathcal{A}$, we choose a representative W^a of a . Note that the contragredient module of an irreducible module is also irreducible (see [FHL]). So we have a map

$$\begin{aligned} ' : \mathcal{A} &\rightarrow \mathcal{A} \\ a &\mapsto a'. \end{aligned}$$

From [AM] and [DLM], we know that irreducible V -modules are in fact graded by rational numbers. Thus for $a \in \mathcal{A}$, there exist $h_a \in \mathbb{Q}$ such that $W^a = \coprod_{n \in h_a + \mathbb{N}} W_{(n)}^a$.

Let $\mathcal{V}_{a_1 a_2}^{a_3}$ for $a_1, a_2, a_3 \in \mathcal{A}$ be the space of intertwining operators of type $\left(\begin{smallmatrix} W^{a_3} \\ W^{a_1} W^{a_2} \end{smallmatrix} \right)$ and $N_{a_1 a_2}^{a_3}$ for $a_1, a_2, a_3 \in \mathcal{A}$ the fusion rule, that is, the dimension of the space of intertwining operators of type $\left(\begin{smallmatrix} W^{a_3} \\ W^{a_1} W^{a_2} \end{smallmatrix} \right)$. For any $\mathcal{Y} \in \mathcal{V}_{a_1 a_2}^{a_3}$, we know from [FHL] that for $w_{a_1} \in W^{a_1}$ and $w_{a_2} \in W^{a_2}$

$$\mathcal{Y}(w_{a_1}, x)w_{a_2} \in x^{\Delta(\mathcal{Y})}W^{a_3}[[x, x^{-1}]], \quad (2.1)$$

where

$$\Delta(\mathcal{Y}) = h_{a_3} - h_{a_1} - h_{a_2}.$$

From [GN], [L], [AN], [H6], we also know that the fusion rules $N_{a_1 a_2}^{a_3}$ for $a_1, a_2, a_3 \in \mathcal{A}$ are all finite. For $a \in \mathcal{A}$, let $\mathcal{N}(a)$ be the matrix whose entries are $N_{aa_1}^{a_2}$ for $a_1, a_2 \in \mathcal{A}$, that is,

$$\mathcal{N}(a) = (N_{aa_1}^{a_2}).$$

We also need matrix elements of fusing and braiding isomorphisms. In the proof of the Verlinde conjecture, we need to use several bases of one space of intertwining operators. We shall use $p = 1, 2, 3, 4, 5, 6, \dots$ to label different bases. For $p = 1, 2, 3, 4, 5, 6, \dots$ and $a_1, a_2, a_3 \in \mathcal{A}$, let $\{\mathcal{Y}_{a_1 a_2; i}^{a_3; (p)} \mid i = 1, \dots, N_{a_1 a_2}^{a_3}\}$, be a basis of $\mathcal{V}_{a_1 a_2}^{a_3}$. For $a_1, \dots, a_6 \in \mathcal{A}$, $w_{a_1} \in W^{a_1}$, $w_{a_2} \in W^{a_2}$, $w_{a_3} \in W^{a_3}$, and $w'_{a_4} \in (W^{a_4})'$, using the differential equations satisfied by the series

$$\langle w'_{a_4}, \mathcal{Y}_{a_1 a_5; i}^{a_4; (1)}(w_{a_1}, x_1) \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}(w_{a_2}, x_2) w_{a_3} \rangle \big|_{x_1^n = e^{n \log z_1}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}}$$

and

$$\langle w'_{a_4}, \mathcal{Y}_{a_6 a_3; k}^{a_4; (3)} (\mathcal{Y}_{a_1 a_2; l}^{a_6; (4)} (w_{a_1}, x_0) w_{a_2}, x_2) w_{a_3} \rangle \big|_{x_0^n = e^{n \log(z_1 - z_2)}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}},$$

it was proved in [H5] that these series are convergent in the regions $|z_1| > |z_2| > 0$ and $|z_2| > |z_1 - z_2| > 0$, respectively. Note that for any $a_1, a_2, a_3, a_4, a_5, a_6 \in \mathcal{A}$, $\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}$ and $\mathcal{Y}_{a_6 a_3; l}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_2; k}^{a_6; (4)}$ are bases of $\mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}$ and $\mathcal{V}_{a_6 a_3}^{a_4} \otimes \mathcal{V}_{a_1 a_2}^{a_6}$, respectively. The associativity of intertwining operators proved and studied in [H1], [H4] and [H5] says that there exist

$$F(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}, \mathcal{Y}_{a_6 a_3; l}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_2; k}^{a_6; (4)}) \in \mathbb{C}$$

for $a_1, \dots, a_6 \in \mathcal{A}$, $i = 1, \dots, N_{a_1 a_5}^{a_4}$, $j = 1, \dots, N_{a_2 a_3}^{a_5}$, $k = 1, \dots, N_{a_6 a_3}^{a_4}$, $l = 1, \dots, N_{a_1 a_2}^{a_6}$ such that

$$\begin{aligned} & \langle w'_{a_4}, \mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} (w_{a_1}, x_1) \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)} (w_{a_2}, z_2) w_{a_3} \rangle \big|_{x_1^n = e^{n \log z_1}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}} \\ &= \sum_{a_6 \in \mathcal{A}} \sum_{k=1}^{N_{a_6 a_3}^{a_4}} \sum_{l=1}^{N_{a_1 a_2}^{a_6}} F(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}, \mathcal{Y}_{a_6 a_3; l}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_2; k}^{a_6; (4)}) \cdot \\ & \quad \cdot \langle w'_{a_4}, \mathcal{Y}_{a_6 a_3; k}^{a_4; (3)} (\mathcal{Y}_{a_1 a_2; l}^{a_6; (4)} (w_{a_1}, z_1 - z_2) w_{a_2}, z_2) w_{a_3} \rangle \big|_{x_0^n = e^{n \log(z_1 - z_2)}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}} \end{aligned} \quad (2.2)$$

when $|z_1| > |z_2| > |z_1 - z_2| > 0$, for $a_1, \dots, a_5 \in \mathcal{A}$, $w_{a_1} \in W^{a_1}$, $w_{a_2} \in W^{a_2}$, $w_{a_3} \in W^{a_3}$, $w'_{a_4} \in (W^{a_4})'$, $i = 1, \dots, N_{a_1 a_5}^{a_4}$ and $j = 1, \dots, N_{a_2 a_3}^{a_5}$. The numbers

$$F(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}, \mathcal{Y}_{a_6 a_3; k}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_2; l}^{a_6; (4)})$$

are the matrix elements of the fusing isomorphism, that is, these numbers together give a matrix which represents a linear isomorphism, called fusing isomorphism, from

$$\coprod_{a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}$$

to

$$\coprod_{a_1, a_2, a_3, a_4, a_6 \in \mathcal{A}} \mathcal{V}_{a_6 a_3}^{a_4} \otimes \mathcal{V}_{a_1 a_2}^{a_6}.$$

By the commutativity of intertwining operators proved and studied in [H2], [H4] and [H5], there exist

$$B^{(r)}(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}, \mathcal{Y}_{a_2 a_6; l}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_3; k}^{a_6; (4)}) \in \mathbb{C}$$

for $r \in \mathbb{Z}$, $a_1, \dots, a_6 \in \mathcal{A}$, $i = 1, \dots, N_{a_1 a_5}^{a_4}$, $j = 1, \dots, N_{a_2 a_3}^{a_5}$, $k = 1, \dots, N_{a_2 a_6}^{a_4}$, $l = 1, \dots, N_{a_1 a_3}^{a_6}$, such that the analytic extension of the single-valued analytic function

$$\langle w'_{a_4}, \mathcal{Y}_{a_1 a_5; i}^{a_4; (1)}(w_{a_1}, x_1) \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}(w_{a_2}, x_2) w_{a_3} \rangle \big|_{x_1^n = e^{n \log z_1}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}}$$

on the region $|z_1| > |z_2| > 0$, $0 \leq \arg z_1, \arg z_2 < 2\pi$ along the path

$$t \mapsto \left(\frac{3}{2} - \frac{e^{(2r+1)\pi i t}}{2}, \frac{3}{2} + \frac{e^{(2r+1)\pi i t}}{2} \right)$$

to the region $|z_2| > |z_1| > 0$, $0 \leq \arg z_1, \arg z_2 < 2\pi$ is

$$\sum_{a_6 \in \mathcal{A}} \sum_{k=1}^{N_{a_2 a_6}^{a_4}} \sum_{l=1}^{N_{a_1 a_3}^{a_6}} B^{(r)}(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}; \mathcal{Y}_{a_2 a_6; k}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_3; l}^{a_6; (4)}) \cdot \langle w'_{a_4}, \mathcal{Y}_{a_2 a_6; k}^{a_4; (3)}(w_{a_2}, z_1) \mathcal{Y}_{a_1 a_3; l}^{a_6; (4)}(w_{a_1}, z_2) w_{a_3} \rangle \big|_{x_1^n = e^{n \log z_1}, x_2^n = e^{n \log z_2}, n \in \mathbb{Q}}.$$

The numbers

$$B^{(r)}(\mathcal{Y}_{a_1 a_5; i}^{a_4; (1)} \otimes \mathcal{Y}_{a_2 a_3; j}^{a_5; (2)}; \mathcal{Y}_{a_2 a_6; k}^{a_4; (3)} \otimes \mathcal{Y}_{a_1 a_3; l}^{a_6; (4)})$$

are the matrix elements of the braiding isomorphism.

We need an action of S_3 on the space

$$\mathcal{V} = \coprod_{a_1, a_2, a_3 \in \mathcal{A}} \mathcal{V}_{a_1 a_2}^{a_3}.$$

For $a_1, a_2, a_3 \in \mathcal{A}$, we have isomorphisms $\Omega_{-r} : \mathcal{V}_{a_1 a_2}^{a_3} \rightarrow \mathcal{V}_{a_2 a_1}^{a_3}$ and $A_{-r} : \mathcal{V}_{a_1 a_2}^{a_3} \rightarrow \mathcal{V}_{a_1 a_3'}^{a_2'}$ for $r \in \mathbb{Z}$ (see [HL2]). For $a_1, a_2, a_3 \in \mathcal{A}$, $\mathcal{Y} \in \mathcal{V}_{a_1 a_2}^{a_3}$, we define

$$\begin{aligned} \sigma_{12}(\mathcal{Y}) &= e^{\pi i \Delta(\mathcal{Y})} \Omega_{-1}(\mathcal{Y}) \\ &= e^{-\pi i \Delta(\mathcal{Y})} \Omega_0(\mathcal{Y}), \\ \sigma_{23}(\mathcal{Y}) &= e^{\pi i h_{a_1}} A_{-1}(\mathcal{Y}) \\ &= e^{-\pi i h_{a_1}} A_0(\mathcal{Y}). \end{aligned}$$

Then the actions σ_{12} and σ_{23} of (12) and (23) on $\mathcal{V}$ defined above generate a left action of S_3 on $\mathcal{V}$.

We now choose a basis $\mathcal{Y}_{a_1 a_2; i}^{a_3}$, $i = 1, \dots, N_{a_1 a_2}^{a_3}$, of $\mathcal{V}_{a_1 a_2}^{a_3}$ for each triple $a_1, a_2, a_3 \in \mathcal{A}$. For $a \in \mathcal{A}$, we choose $\mathcal{Y}_{ea; 1}^a$ to be the vertex operator Y_{W^a}

defining the module structure on W^a and we choose $\mathcal{Y}_{ae;1}^a$ to be the intertwining operator defined using the action of σ_{12} ,

$$\begin{aligned}\mathcal{Y}_{ae;1}^a(w_a, x)u &= \sigma_{12}(\mathcal{Y}_{ea;1}^a)(w_a, x)u \\ &= e^{xL(-1)}\mathcal{Y}_{ea;1}^a(u, -x)w_a \\ &= e^{xL(-1)}Y_{W^a}(u, -x)w_a\end{aligned}$$

for $u \in V$ and $w_a \in W^a$. Since V' as a V -module is isomorphic to V , we have $e' = e$. From [FHL], we know that there is a nondegenerate invariant bilinear form $(\cdot, \cdot)$ on V such that $(\mathbf{1}, \mathbf{1}) = 1$. We choose $\mathcal{Y}_{aa';1}^e = \mathcal{Y}_{aa';1}^{e'}$ to be the intertwining operator defined using the action of σ_{23} by

$$\mathcal{Y}_{aa';1}^{e'} = \sigma_{23}(\mathcal{Y}_{ae;1}^a),$$

that is,

$$(u, \mathcal{Y}_{aa';1}^{e'}(w_a, x)w'_a) = e^{\pi i h_a} \langle \mathcal{Y}_{ae;1}^a(e^{xL(1)}(e^{-\pi i} x^{-2})^{L(0)} w_a, x^{-1})u, w'_a \rangle$$

for $u \in V$, $w_a \in W^a$ and $w'_a \in (W^a)'$. Since the actions of σ_{12} and σ_{23} generate the action of S_3 on $\mathcal{V}$, we have

$$\mathcal{Y}_{a'a;1}^e = \sigma_{12}(\mathcal{Y}_{aa';1}^e)$$

for any $a \in \mathcal{A}$. When $a_1, a_2, a_3 \neq e$, we choose $\mathcal{Y}_{a_1 a_2; i}^{a_3}$, $i = 1, \dots, N_{a_1 a_2}^{a_3}$, to be an arbitrary basis of $\mathcal{V}_{a_1 a_2}^{a_3}$. Note that for each element $\sigma \in S_3$, $\{\sigma(\mathcal{Y})_{a_1 a_2; i}^{a_3} \mid i = 1, \dots, N_{a_1 a_2}^{a_3}\}$, is also a basis of $\mathcal{V}_{a_1 a_2}^{a_3}$.

3 Modular Invariance

We discuss modular invariance briefly in this section. Let $q_\tau = e^{2\pi i \tau}$ for $\tau \in \mathbb{H}$ ($\mathbb{H}$ is the upper-half plane). We consider the q_τ -traces of the vertex operators Y_{W^a} for $a \in \mathcal{A}$ on the irreducible V -modules W^a of the following form:

$$\text{Tr}_{W^a} Y_{W^a}(e^{2\pi i z L(0)} u, e^{2\pi i z}) q_\tau^{L(0) - \frac{c}{24}} \quad (3.1)$$

for $u \in V$ (recall that c is the central charge of V). In [Z], under some conditions slightly different from (mostly stronger than) those we assume in this paper, Zhu proved that these q -traces are independent of z , are absolutely convergent when $0 < |q_\tau| < 1$ and can be analytically extended to analytic

functions of τ in the upper-half plane. We shall denote the analytic extension of (3.1) by

$$E(\mathrm{Tr}_{W^a} Y_{W^a}(e^{2\pi iz L(0)} u, e^{2\pi iz}) q_\tau^{L(0) - \frac{c}{24}}).$$

In [Z], under his conditions mentioned above, Zhu also proved the following modular invariance property: For

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z}),$$

let $\tau' = \frac{a\tau+b}{c\tau+d}$. Then there exist unique $A_{a_1}^{a_2} \in \mathbb{C}$ for $a_1, a_2 \in \mathcal{A}$ such that

$$\begin{aligned} E\left(\mathrm{Tr}_{W^{a_1}} Y_{W^{a_1}} \left(e^{\frac{2\pi iz}{c\tau+d} L(0)} \left(\frac{1}{c\tau+d} \right)^{L(0)} u, e^{\frac{2\pi iz}{c\tau+d}} \right) q_{\tau'}^{L(0) - \frac{c}{24}} \right) \\ = \sum_{a_2 \in \mathcal{A}} A_{a_1}^{a_2} E(\mathrm{Tr}_{W^{a_2}} Y_{W^{a_2}}(e^{2\pi iz L(0)} u, e^{2\pi iz}) q_\tau^{L(0) - \frac{c}{24}}) \end{aligned}$$

for $u \in V$. In [DLM], Dong, Li and Mason, among many other things, improved Zhu's results above by showing that the results of Zhu above also hold for vertex operator algebras satisfying the conditions (slightly weaker than what) we assume in this paper. In particular, for

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in SL(2, \mathbb{Z}),$$

there exist unique $S_{a_1}^{a_2} \in \mathbb{C}$ for $a_1 \in \mathcal{A}$ such that

$$\begin{aligned} E\left(\mathrm{Tr}_{W^{a_1}} Y_{W^{a_1}} \left(e^{-\frac{2\pi iz}{\tau} L(0)} \left(-\frac{1}{\tau} \right)^{L(0)} u, e^{-\frac{2\pi iz}{\tau}} \right) q_{-\frac{1}{\tau}}^{L(0) - \frac{c}{24}} \right) \\ = \sum_{a_2 \in \mathcal{A}} S_{a_1}^{a_2} E(\mathrm{Tr}_{W^{a_2}} Y_{W^{a_2}}(e^{2\pi iz L(0)} u, e^{2\pi iz}) q_\tau^{L(0) - \frac{c}{24}}) \end{aligned}$$

for $u \in V$. When $u = \mathbf{1}$, we see that the matrix $S = (S_{a_1}^{a_2})$ actually acts on the space of spanned by the vacuum characters $\mathrm{Tr}_{W^a} q_\tau^{L(0) - \frac{c}{24}}$ for $a \in \mathcal{A}$.

4 The Verlinde conjecture and consequences

In [H7], the author proved the following general version of the Verlinde conjecture in the framework of vertex operator algebras (cf. Section 3 in [V] and Section 4 in [MS1]):

Theorem 4.1 *Let V be a vertex operator algebra satisfying the following conditions:*

1. $V_{(n)} = 0$ for $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$ and V' is isomorphic to V as a V -module.
2. Every $\mathbb{N}$ -gradable weak V -module is completely reducible.
3. V is C_2 -cofinite, that is, $\dim V/C_2(V) < \infty$.

Then for $a \in \mathcal{A}$,

$$F(\mathcal{Y}_{ae;1}^a \otimes \mathcal{Y}_{a'a;1}^e; \mathcal{Y}_{ea;1}^a \otimes \mathcal{Y}_{aa';1}^e) \neq 0$$

and

$$\sum_{a_1, a_3 \in \mathcal{A}} (S^{-1})_{a_4}^{a_1} N_{a_1 a_2}^{a_3} S_{a_3}^{a_5} = \delta_{a_4}^{a_5} \frac{(B^{(-1)})^2 (\mathcal{Y}_{a_4 e;1}^{a_4} \otimes \mathcal{Y}_{a'_2 a_2;1}^e; \mathcal{Y}_{a_4 e;1}^{a_4} \otimes \mathcal{Y}_{a'_2 a_2;1}^e)}{F(\mathcal{Y}_{a_2 e;1}^{a_2} \otimes \mathcal{Y}_{a'_2 a_2;1}^e; \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a_2 a'_2;1}^e)}. \quad (4.1)$$

In particular, the matrix S diagonalizes the matrices $\mathcal{N}(a_2)$ for all $a_2 \in \mathcal{A}$.

The main work in the proof of this theorem is to prove the following formulas derived by Moore and Seiberg [MS1] from the axioms of rational conformal field theories: For $a_1, a_2, a_3 \in \mathcal{A}$,

$$\begin{aligned} & \sum_{i=1}^{N_{a_1 a_2}^{a_3}} \sum_{k=1}^{N_{a'_1 a_3}^{a'_2}} F(\mathcal{Y}_{a_2 e;1}^{a_2} \otimes \mathcal{Y}_{a'_3 a_3;1}^e; \mathcal{Y}_{a'_1 a_3;k}^{a_2} \otimes \mathcal{Y}_{a_2 a'_3;i}^{a'_1}). \\ & \quad \cdot F(\mathcal{Y}_{a'_1 a_3;k}^{a_2} \otimes \sigma_{123}(\mathcal{Y}_{a_2 a'_3;i}^{a'_1}); \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a'_1 a_1;1}^e) \\ & = N_{a_1 a_2}^{a_3} F(\mathcal{Y}_{a_2 e;1}^{a_2} \otimes \mathcal{Y}_{a'_2 a_2;1}^e; \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a_2 a'_2;1}^e) \end{aligned} \quad (4.2)$$

and

$$\begin{aligned} & \sum_{a_4 \in \mathcal{A}} S_{a_1}^{a_4} (B^{(-1)})^2 (\mathcal{Y}_{a_4 e;1}^{a_4} \otimes \mathcal{Y}_{a'_2 a_2;1}^e; \mathcal{Y}_{a_4 e;1}^{a_4} \otimes \mathcal{Y}_{a'_2 a_2;1}^e) (S^{-1})_{a_4}^{a_3} \\ & = \sum_{i=1}^{N_{a_1 a_2}^{a_3}} \sum_{k=1}^{N_{a'_1 a_3}^{a'_2}} F(\mathcal{Y}_{a_2 e;1}^{a_2} \otimes \mathcal{Y}_{a'_3 a_3;1}^e; \mathcal{Y}_{a'_1 a_3;k}^{a_2} \otimes \mathcal{Y}_{a_2 a'_3;i}^{a'_1}). \\ & \quad \cdot F(\mathcal{Y}_{a'_1 a_3;k}^{a_2} \otimes \sigma_{123}(\mathcal{Y}_{a_2 a'_3;i}^{a'_1}); \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a'_1 a_1;1}^e). \end{aligned} \quad (4.3)$$

The proof of the first formula (4.2) uses mainly the works of Lepowsky and the author [HL1]–[HL4] and of the author [H1] [H2] [H4] and [H5] on the tensor product theory, intertwining operator algebras and the construction of genus-zero chiral conformal field theories. The main technical result used is the associativity for intertwining operators proved in [H1] and [H5] for vertex operator algebras satisfying the three conditions stated in the theorem.

As is discussed in Section 2, assuming the convergence and extension property and some other conditions, the associativity for intertwining operators was proved in [H1]. In [H3], the commutativity for intertwining operators was proved using the associativity. In Lemma 4.1 in [H6], using the associativity and commutativity proved in [H1] and [H2], respectively, it was shown that one can analytically extend products or iterates of two intertwining operators to

$$M^2 = \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1, z_2 \neq 0, z_1 \neq z_2\}.$$

Moreover, the construction of the map $(\Psi_{P(z_1, z_2)}^{(1)})^{-1}$ in (14.65) in [H1] can be reinterpreted as a proof of the fact that any four point correlation function can be obtained in this way. For multipoint correlation functions, the corresponding results can be proved using the same method or using the results for four point correlation functions above. This result shows that assuming the convergence and extension property and some other conditions, products or iterates of intertwining operators can be analytically extended to multi-valued analytic functions defined on the moduli spaces of pointed genus-zero Riemann surfaces.

The convergence and extension property was proved in [H1] under conditions weaker than the conditions we assume above. The proof is based on the existence of systems of differential equations and the regularity of the singular points of these systems. Since we know only the existence, not the explicit forms, of these systems of differential equations, many powerful tools available for Knizhnik-Zamolodchikov equations in the Wess-Zumino-Novikov-Witten models are not available here anymore. However, the theory of intertwining operator algebras developed in [H3], [H4] and [H5] allows us to reduce the use of these differential equations to a minimum. For example, we do need the regularity of the singular points of these systems. But only the regularity of some ordinary differential equations induced from these systems are needed and such regularity can be proved easily.

Now using the associativity for intertwining operators repeatedly to express the correlation functions obtained from products of three suitable inter-

twining operators as linear combinations of the correlation functions obtained from iterates of three intertwining operators in two ways, we obtain a formula for the matrix elements of the fusing isomorphisms. Then using some properties of the matrix elements of the fusing isomorphisms and their inverses proved in [H7], we obtain the first formula (4.2).

The proof of the second formula (4.3) mainly uses the results obtained in [H6] on the convergence and analytic extensions of the q_τ -traces of products of what we call “geometrically-modified intertwining operators”, the genus-one associativity, and the modular invariance of the space of these analytic extensions of the q_τ -traces, where, as in Section 3, $q_\tau = e^{2\pi\tau}$. In [Z], in addition to the convergence and modular invariance of q_τ -traces of vertex operators, Zhu also proved the convergence and modular invariance of the space of the q_τ -traces of products of more than one vertex operators acting on modules. In [DLM], Dong, Li and Mason generalized Zhu’s result to the case of twisted modules. In [M], Miyamoto generalized Zhu’s result to the case of products of one intertwining operator and arbitrarily many vertex operators. However, these results of Zhu, Dong-Li-Mason and Miyamoto do not give all genus-one correlation functions and thus are not enough for the construction conformal field theories. More specifically, the problem of proving the convergence, the modular invariance and duality properties for q_τ -traces of products of *more than one* intertwining operator was still open at that time. This problem is equivalent to the problem of constructing all the chiral genus-zero correlation functions and establishing all the desired properties.

The difficulty in the case of more than one intertwining operator is that the method of Zhu, further developed by Dong-Li-Mason and Miyamoto, cannot be adapted or generalized to this case, because there is no commutator formula for general intertwining operators. Here by commutator formulas for general intertwining operators, we mean a formula for

$$\mathcal{Y}_1(w_1, x_1)\mathcal{Y}_2(w_2, x_2) - \mathcal{Y}_3(w_2, x_2)\mathcal{Y}_4(w_1, x_1)$$

where $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{Y}_3$ and $\mathcal{Y}_4$ are suitable intertwining operators. There is no such formula, even in the case of abelian intertwining operator algebras in the sense of Dong and Lepowsky [DL]. Without such commutator formula, one cannot prove a recurrence formula needed in the method of Zhu, Dong-Li-Mason and Miyamoto. Even the generalized commutator formula of Dong and Lepowsky for abelian intertwining operator algebras cannot be used to obtain such recurrence relations.

In [H6], the author solved these previously open problems. These problems actually form one of the main obstructions to a new range of applications of the theory of vertex operator algebras. It is the solution to these problems together with the results on the duality properties of genus-zero correlation functions discussed in the preceding section that allowed us to prove (4.3). The method in [H6] is completely different from the one in [Z], [DLM] and [M] in the case of products of more than one operators: Instead of the nonexistent commutator formula, the author used the associativity and commutativity for intertwining operators and the method of analytic extensions. The lack of a commutator formula was one of the subtle difficulties which was overcome in [H6].

To prove the second formula (4.3), using the results proved in [H6], we introduce two operators α and β on the space of linear maps from $\coprod_{a \in \mathcal{A}} W^a \otimes (W^a)'$ to the space of genus-one two-point correlation functions obtained from the analytic extensions of the q_τ -traces of iterates of suitable “geometrically-modified intertwining operators” introduced in [H6]: α is induced from the translation of one of the points by -1 and β is induced from the translation of one of the points by τ . The work in [H6] is crucial in proving that these operators are well-defined and have the desired properties. The matrix S also acts on the same space of linear maps and it is easy to prove the relation

$$S\alpha S^{-1} = \beta.$$

We then use the genus-one associativity and other properties of the genus-one correlation functions in [H6] to calculate α and β explicitly in terms of the matrix elements of the fusing and braiding isomorphisms. Substituting the explicit expressions of α and β into this relation, we obtain (4.3).

As in [MS1], the conclusions of the theorem follow immediately from (4.2) and (4.3).

All the consequences of the Verlinde conjecture derived by physicists now hold for vertex operator algebras satisfying the conditions in the theorem. In particular, we have the following Verlinde formula for fusion rules: For $a \in \mathcal{A}$, $S_e^a \neq 0$ and

$$N_{a_1 a_2}^{a_3} = \sum_{a_4 \in \mathcal{A}} \frac{S_{a_1}^{a_4} S_{a_2}^{a_4} S_{a_4}^{a_3'}}{S_e^{a_4}} \quad (4.4)$$

(cf. Section 3 in [V]). We also have the following well-known formula: For $a_1, a_2 \in \mathcal{A}$,

$$S_{a_1}^{a_2} = \frac{S_e^e((B^{(-1)})^2(\mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_1a_1;1}^e; \mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_1a_1;1}^e))}{F(\mathcal{Y}_{a_1e;1}^{a_1} \otimes \mathcal{Y}_{a'_1a_1;1}^e; \mathcal{Y}_{ea_1;1}^{a_1} \otimes \mathcal{Y}_{a_1a'_1;1}^e)F(\mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_2a_2;1}^e; \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a_2a'_2;1}^e)}. \quad (4.5)$$

The formula (4.5) and certain properties of the matrix elements of the fusing and braiding isomorphisms proved in [H7] imply that the matrix $(S_{a_1}^{a_2})$ is symmetric.

5 Rigidity, nondegeneracy and modular tensor categories

A tensor category with tensor product bifunctor $\boxtimes$ and unit object V is rigid if for every object W in the category, there are right and left dual objects W^* and *W together with morphisms $e_W : W^* \boxtimes W \rightarrow V$, $i_W : V \rightarrow W \boxtimes W^*$, $e'_W : W \boxtimes {}^*W \rightarrow V$ and $i'_W : V \rightarrow {}^*W \boxtimes W$ such that the compositions of the morphisms in the sequence

$$\begin{array}{ccccccc} W & \longrightarrow & V \boxtimes W & \xrightarrow{i_W \boxtimes I_W} & (W \boxtimes W^*) \boxtimes W & \longrightarrow & \\ & & & & & & \\ & \longrightarrow & W \boxtimes (W^* \boxtimes W) & \xrightarrow{I_W \boxtimes e_W} & W \boxtimes V & \longrightarrow & W \end{array}$$

and three similar sequences are equal to the identity I_W on W . The rigidity is a standard notion in the theory of tensor categories. A rigid braided tensor category together with a twist (a natural isomorphism from the category to itself satisfying natural conditions) is called a ribbon category. A semisimple ribbon category with finitely many inequivalent irreducible objects is a modular tensor category if the following nondegeneracy condition or modularity is satisfied: The $m \times m$ matrix formed by the traces of the morphism $c_{W_i W_j} \circ c_{W_j W_i}$ in the ribbon category for irreducible modules $W_1, \dots, W_m$ is invertible. See [T] and [BK] for details of the notions in the theory of modular tensor categories.

Using the results discussed in the proceeding section, we obtain the following result:

Theorem 5.1 *Let V be a simple vertex operator algebra satisfying the following conditions:*

1. $V_{(n)} = 0$ for $n < 0$, $V_{(0)} = \mathbb{C}\mathbf{1}$, $W_{(0)} = 0$ for any irreducible V -module which is not isomorphic to V .
2. Every $\mathbb{N}$ -gradable weak V -module is completely reducible.
3. V is C_2 -cofinite, that is, $\dim V/C_2(V) < \infty$.

Then the braided tensor category structure on the category of V -modules constructed in [HL1]–[HL4], [H1] and [H5] is rigid, has a natural structure of ribbon category and satisfies the nondegeneracy condition. In particular, the category of V -modules has a natural structure of modular tensor category.

Note that Condition 1 implies that V' is isomorphic to V as a V -module. Thus Condition 1 in the theorem is slightly stronger than Condition 1 in Theorem 4.1.

We now discuss the proof of this theorem. We take both the left and right dual of a V -module W to be the contragredient module W' of W . Since our tensor category is semisimple, to prove the rigidity, we need only discuss irreducible modules. For $a \in \mathcal{A}$, using the universal property for the tensor product module $(W^a)' \boxtimes W^a$, we know that there exists a unique module map $\hat{e}_a : (W^a)' \boxtimes W^a \rightarrow V$ such that

$$\bar{\hat{e}}_a(w'_a \boxtimes w_a) = \mathcal{Y}_{a'a;1}^e(w'_a, 1)w_a$$

for $w_a \in W^a$ and $w'_a \in (W^a)'$, where $\bar{\hat{e}}_a : \overline{(W^a)' \boxtimes W^a} \rightarrow \bar{V}$ is the natural extension of $\hat{e}_a$. Similarly, we have a module map from $W^a \boxtimes (W^a)'$ to V . Since $W^a \boxtimes (W^a)'$ is completely reducible and the fusion rule $N_{W^a(W^a)'}^V$ is 1, there is a V -submodule of $W^a \boxtimes (W^a)'$ which is isomorphic to V under the module map from $W^a \boxtimes (W^a)'$ to V . Thus we obtain a module map $i_a : V \rightarrow W^a \boxtimes (W^a)'$ which maps V bijectively to this submodule of $W^a \boxtimes (W^a)'$. Now

$$\begin{array}{ccccccc} W^a & \longrightarrow & V \boxtimes W^a & \xrightarrow{i_a \boxtimes I_{W^a}} & (W^a \boxtimes (W^a)') \boxtimes W^a & \longrightarrow & \\ & & \longrightarrow & W^a \boxtimes ((W^a)' \boxtimes W^a) & \xrightarrow{I_{W^a} \boxtimes \hat{e}_a} & W^a \boxtimes V & \longrightarrow W_a \end{array} \quad (5.1)$$

is a module map from an irreducible module to itself. So it must be the identity map multiplied by a number. One can calculate this number explicitly and it is equal to

$$F(\mathcal{Y}_{ae;1}^a \otimes \mathcal{Y}_{a'a;1}^e; \mathcal{Y}_{ea;1}^a \otimes \mathcal{Y}_{aa';1}^e).$$

From Theorem 4.1, this number is not 0. Let

$$e_a = \frac{1}{F(\mathcal{Y}_{ae;1}^a \otimes \mathcal{Y}_{a'a;1}^e; \mathcal{Y}_{ea;1}^a \otimes \mathcal{Y}_{aa';1}^e)} \hat{e}_a.$$

Then the map obtained from (5.1) by replacing $\hat{e}_a$ by e_a is the identity. Similarly, we can prove that all the other maps in the definition of rigidity are also equal to the identity. So the tensor category is rigid.

For any $a \in \mathcal{A}$, we define the twist on W^a to $e^{2\pi i h_a}$. Then it is easy to verify that the rigid braided tensor category with this twist is a ribbon category.

To prove the nondegeneracy, we use the formula (4.5). Now it is easy to calculate in the tensor category the trace of $c_{W^{a_2}, W^{a_1}} \circ c_{W^{a_1}, W^{a_2}}$ for $a_1, a_2 \in \mathcal{A}$, where $c_{W^{a_1}, W^{a_2}} : W^{a_1} \boxtimes W^{a_2} \rightarrow W^{a_2} \boxtimes W^{a_1}$ is the braiding isomorphism. The result is

$$\frac{((B^{(-1)})^2(\mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_1a_1;1}^e; \mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_1a_1;1}^e))}{F(\mathcal{Y}_{a_1e;1}^{a_1} \otimes \mathcal{Y}_{a'_1a_1;1}^e; \mathcal{Y}_{ea_1;1}^{a_1} \otimes \mathcal{Y}_{a_1a'_1;1}^e) F(\mathcal{Y}_{a_2e;1}^{a_2} \otimes \mathcal{Y}_{a'_2a_2;1}^e; \mathcal{Y}_{ea_2;1}^{a_2} \otimes \mathcal{Y}_{a_2a'_2;1}^e)}.$$

By (4.5), this is equal to

$$\frac{S_{a_1}^{a_2}}{S_e^e}$$

which form an invertible matrix. So the semisimple balanced rigid tensor category is nondegenerate. Thus the tensor category is modular. Details will be given in [H9].

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A FRAMEWORK FOR FRACTIONAL HARDY INEQUALITIES

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Abstract. We provide a general framework for fractional Hardy inequalities. Our framework covers, for instance, fractional inequalities related to the Dirichlet forms of some Lévy processes, and weighted fractional inequalities on irregular open sets.

1. Introduction

The objective of the present paper is to study inequalities of the general form

$$(1.1) \quad \int_D \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) \leq c \int_D \int_{D \cap B(x, R\delta_x)} \frac{|u(x) - u(y)|^p}{\phi(\delta_x)\delta_x^d} \mu(dy) \mu(dx) \quad c, R > 0,$$

on metric measure spaces (X, ρ, μ) , with the emphasis on $X = \mathbf{R}^d$ equipped with the Euclidean distance and the Lebesgue measure. We write $\delta_x = \text{dist}(x, X \setminus D)$ and $D \subset X$ is a possibly irregular open set. The function $\phi: (0, \infty) \rightarrow (0, \infty)$ is a ‘perturbation’ of a power function $t \mapsto t^\eta$ for some $\eta \in \mathbf{R}$, and the exponent p satisfies $0 < p < \infty$.

Our main result, Theorem 5 in §3, brings together two so-far distinct lines along which the fractional Hardy inequality has been generalised: one of them related to the function ϕ , and the other to the regularity of the open set $D \subset X$. Let us present the Euclidean version of our main result here; Theorem 1 below is a combination of Theorem 5 and Propositions 6 and 10.

Theorem 1. *Let $0 < p < \infty$, $H \in (0, 1]$ and $\eta \in \mathbf{R}$. Suppose that $D \neq \emptyset$ is a proper κ -plump open set in $\mathbf{R}^d$ and $\phi: (0, \infty) \rightarrow (0, \infty)$ is a function so that either condition (T) or condition (F) holds:*

- (T) $\overline{\dim}_A(\partial D) < d - \eta$, D is unbounded, and $\phi \in \text{WUSC}(\eta, 0, H^{-1})$;
- (F) $\underline{\dim}_A(\partial D) > d - \eta$, D is bounded or ∂D is unbounded, and $\phi \in \text{WLSC}(\eta, 0, H)$.

Then inequality (1.1) holds for all measurable functions u for which the left hand side is finite.

For the definitions of plumpness and Assouad dimensions $\overline{\dim}_A$ and $\underline{\dim}_A$, we refer to Section 2; the classes WUSC and WLSC are defined in Definition 3.1. Let

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us remark that inequality (1.1) fails for nonzero constant functions. This means that under the assumptions of Theorem 1, the integral $\int_D \phi(\delta_x)^{-1} dx$ must be divergent, on the other hand, it shows that the assumption of the left hand side of (1.1) being finite is essential. The relatively simple proof of our main result is a refinement of the techniques in [7] where, e.g., Hardy inequalities (1.3) with $\beta = 0$ in case of bounded Lipschitz domains are established.

Theorems 1 and 5 provide a general framework for fractional Hardy inequalities, as they allow for both general open sets D and functions ϕ simultaneously. In the sequel, we will state separate corollaries in each of these directions to make the exposition of our framework transparent. First, instead of considering classes WUSC and WLSC, we will confine ourselves to the well-known *regularly varying* functions. Let us remind that ϕ is called *regularly varying* at origin (resp. infinity) with index η , if

$$\frac{\phi(\lambda x)}{\phi(x)} \rightarrow \lambda^\eta$$

when $x \rightarrow 0_+$ (resp. when $x \rightarrow \infty$) for every $\lambda > 0$. In the following corollary the geometry of the underlying domain $\mathbf{R}^d \setminus \{0\}$ is particularly simple.

Corollary 2. *Suppose that $\phi: (0, \infty) \rightarrow (0, \infty)$ is a regularly varying function at origin of index ρ_0 , a regularly varying function at infinity of index ρ_∞ , and is bounded and bounded away from zero on every compact subset of $(0, \infty)$. Suppose that either $0 < \rho_0, \rho_\infty < d$ or $\rho_0, \rho_\infty > d$. Let $0 < p < \infty$. Then there exists a constant $c = c(\phi, d, p)$ such that inequality*

$$(1.2) \quad \int_{\mathbf{R}^d \setminus \{0\}} \frac{|u(x)|^p}{\phi(|x|)} dx \leq c \int_{\mathbf{R}^d \setminus \{0\}} \int_{\mathbf{R}^d \setminus \{0\}} \frac{|u(x) - u(y)|^p}{\phi(|x - y|)|x - y|^d} dy dx$$

holds for every measurable function u for which the left hand side is finite.

A proof of this corollary can be found in §3.4. Inequalities like (1.2) have been studied in [15] for weights of more general (but also more complicated) form and $p > 1$, and in [14, 15, 22] in the one-dimensional case. The forms appearing on the right hand side of (1.2) for $p = 2$ (and for more general domains) are, at least for some functions ϕ , the Dirichlet forms of certain Lévy processes, which are being extensively studied, see e.g. [13, 20, 33] and [5, Section 4.1].

To discuss our results for irregular open sets, we confine ourselves to weighted fractional Hardy inequalities in $\mathbf{R}^d$, i.e., we consider the function $\phi(t) = t^{sp-\beta}$ with $d + sp \geq 0$, in which case inequality (1.1) yields

$$(1.3) \quad \int_D \frac{|u(x)|^p}{\delta_x^{sp}} \delta_x^\beta dx \leq c \int_D \int_D \frac{|u(x) - u(y)|^p}{|x - y|^{d+sp}} \delta_x^\beta dy dx.$$

An open set $D \subset \mathbf{R}^d$ is said to admit (s, p, β) -Hardy inequality, if inequality (1.3) holds for all functions $u \in C_0^\infty(D)$ (i.e., smooth with compact support in D) with c independent of u . There has been recent interest in $(s, p, 0)$ -Hardy inequalities in connection with the boundary regularity of an open set D , we refer to [10, 16, 17, 18]. In another direction, the sharp constants for fractional Hardy-type inequalities on general domains are obtained in [27], where the distance is replaced with an averaged pseudo distance. In [9] these results are further refined; let us also mention the other related papers [4, 8, 11, 12, 34].

The non-fractional counterpart of inequality (1.3) has also been studied. Namely, the following weighted (p, β) -Hardy inequality, with $c > 0$,

$$(1.4) \quad \int_D \frac{|u(x)|^p}{\delta_x^p} \delta_x^\beta dx \leq c \int_D |\nabla u(x)|^p \delta_x^\beta dx$$

holds for every $u \in C_0^\infty(D)$ if D is a bounded Lipschitz domain, $1 < p < \infty$, and $\beta < p - 1$, [32]. More generally, an open set admits a (p, β) -Hardy inequality if the complement $D^c = \mathbf{R}^d \setminus D$ is either sufficiently ‘thin’ or ‘fat’. For instance, an open set D admits a $(p, 0)$ -Hardy inequality if D^c is $(1, p)$ -uniformly fat and $1 < p < \infty$, [26]. The $(1, p)$ -fatness of D^c is also known to be sufficient for certain (p, β) -Hardy inequalities, [24, 37]. A deeper understanding of the ‘thin’ vs. ‘fat’ dichotomy is reached in an independent recent study [23], where an open set $D \subset X$ is shown to admit a (p, β) -Hardy inequality if $D^c = X \setminus D$ sufficiently thin or fat, measured in terms of upper and lower Assouad dimension ($\overline{\dim}_A$ and $\underline{\dim}_A$), respectively. We also refer to [21].

Our framework covers an Assouad dichotomy result for fractional (s, p, β) -Hardy inequalities with $X = \mathbf{R}^d$, see the following Corollary 3.

Corollary 3. *Let p, s, β be real numbers so that $0 < p < \infty$ and $d + sp \geq 0$. Suppose $D \neq \emptyset$ is a proper κ -plump open set in $\mathbf{R}^d$ so that either condition (T) or condition (F) holds:*

- (T) $\overline{\dim}_A(\partial D) < d - sp + \beta$ and D is unbounded;
- (F) $\underline{\dim}_A(\partial D) > d - sp + \beta$, and D is bounded or ∂D is unbounded.

Then D admits an (s, p, β) -Hardy inequality, i.e., inequality (1.3) holds for every $u \in C_0^\infty(D)$ with a constant $c > 0$ independent of u .

This corollary follows from Theorem 1 with the aid of Example 3.3. As an illustration of this result, we may consider the domain $D \subset \mathbf{R}^2$ bounded by the Koch snowflake. It is a domain with a property $\underline{\dim}_A(\partial D) = \log 4 / \log 3$. Since D is also κ -plump, Corollary 3 does apply. In the ‘thin case’ we may, e.g., consider the unbounded domain $G := \mathbf{R}^d \setminus \overline{D}$. Now G is κ -plump and it satisfies $\overline{\dim}_A(\partial G) = \log 4 / \log 3$. Let us note that [7, Theorem 1.1], apart from the case (T2) for $\alpha > 1$, is a special case of Corollary 3. We would also like to note that we do not need to assume the positivity of s in Corollary 3.

Let us comment on the cases (T) and (F) in Corollary 3. Focusing on the case (T) first, recall that $\overline{\dim}_A(\partial D) = d - 1$ for a Lipschitz domain D . The unboundedness of D cannot be removed, at least if $0 < s < 1$, in which case a bounded Lipschitz domain satisfies an $(s, p, 0)$ -Hardy inequality if and only if $sp > 1$, [7]. Certain non-homogeneous $(s, p, 0)$ -Hardy inequalities remain valid for John domains D with $\overline{\dim}_A(\partial D) < d - sp$, [18]. Therein (T) with $\beta = 0$ is formulated in terms of a certain Aikawa dimension which equals to the upper Assouad dimension in Euclidean spaces, see [25]. Recalling that John domains are both bounded and κ -plump, we may conclude that our framework provides a far-reaching generalisation of the aforementioned non-homogeneous results to the case of unbounded open sets.

Moving on to the case (F) with ‘fat’ boundary, let us first formulate an illustrative, but more restrictive, version of Corollary 3. We refer to §4.4 for the relevant definitions.

Corollary 4. *Let p, s, β be real numbers so that $1 < p < \infty$, $0 < sp - \beta < d$, and $d + sp \geq 0$. Suppose D is a κ -plump open set in $\mathbf{R}^d$ such that ∂D is $(s - \beta/p, p)$ -uniformly fat (-locally uniformly fat, if D is bounded). Then D admits an (s, p, β) -Hardy inequality.*

This corollary is a consequence of Corollary 3 and Propositions 8 and 9. Unlike in the case of inequality (1.4) with $\beta = 0$ and $s = 1$, the (s, p) -uniform fatness of ∂D (let alone D^c) is not a sufficient condition for an open set D to admit an $(s, p, 0)$ -Hardy inequality (at least) in the case of $0 < sp \leq 1$. This ‘non-local obstruction’ is recognised and addressed in [16]. It affects the fractional Hardy inequalities studied in [10], where D^c is assumed to be (s, p) -uniformly fat and, as a conclusion, on the right hand side of (1.3) one then has integration over $\mathbf{R}^d \times \mathbf{R}^d$.

Suppose that D is an open set whose boundary is (s, p) -uniformly fat (locally uniformly fat, if D is bounded). It is an interesting question, what additional conditions are sufficient for D to admit an $(s, p, 0)$ -Hardy inequality. To this end, we improve [16, Corollary 1.4] where uniformity, [29, 36], of a domain D is shown to be a sufficient additional condition. Indeed, by Corollary 4, we may replace uniformity with the significantly weaker κ -plumpness. Let us remark that [16, Theorem 4.1], stated in terms of a ‘visibility condition on the boundary’, still covers some other cases where our results do not apply, e.g., certain domains with outward cusps.

The structure of this paper is as follows. In §2 we define both the lower and upper Assouad dimension, and the notion of κ -plumpness. We also present other basic notation. Our main result is Theorem 5, stated and proven in §3. There we also define classes WLSC and WUSC of functions ϕ and a condition $\text{DC}(a, \gamma, d)$ for open sets D . The latter condition is further clarified in §4 and §5, where we study the cases of ‘fat’ and ‘thin’ boundaries in terms of uniform fatness, and the lower and upper Assouad dimension.

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2. Assouad dimensions and plumpness

We recall the lower and upper Assouad dimensions of a set $\emptyset \neq E \subset \mathbf{R}^d$, [19]. The lower Assouad dimension measures the ‘fatness’ of a set E , whereas the upper one measures how ‘thin’ a set E is. The upper Assouad dimension is often called *Assouad dimension*, a notion tracing back to [2] and even [6]. We refer to [19, 28] for further information and other results.

Definition 2.1. Consider all $\lambda \geq 0$ for which there is $C > 0$ so that, if $0 < r < R < 2 \text{diam}(E)$ and $x \in E$, then at least $C(R/r)^\lambda$ balls—centred in E and of radius r —are needed to cover $B(x, R) \cap E$. The supremum of all such λ is called the *lower Assouad dimension* of E and it is denoted by $\underline{\dim}_A(E)$.

Definition 2.2. Consider all $\lambda \geq 0$ for which there is $C > 0$ so that, if $0 < r < R < 2 \text{diam}(E)$ and $x \in E$, then we can cover $E \cap B(x, R)$ by at most $N \leq C(R/r)^\lambda$

balls $B_1, \dots, B_N$ such that each B_j is centred in E and has radius r . We call the infimum of all such λ the *upper Assouad dimension of E* , and write it as $\overline{\dim}_A(E)$.

We also recall a geometric notion from [36]. See also [30].

Definition 2.3. A set $A \subset \mathbf{R}^d$ is κ -*plump* with $\kappa \in (0, 1)$ if, for each $0 < r < \text{diam}(A)$ and each $x \in \bar{A}$, there is $z \in \bar{B}(x, r)$ such that $B(z, \kappa r) \subset A$.

Here is other notation; (X, ρ, μ) is a metric measure space, and we denote

$$\delta_x = \text{dist}(x, D^c)$$

with $D^c = X \setminus D$. The open ball centred at $x \in X$ and of radius $r > 0$ is denoted by $B(x, r) \subset X$. The boundary of set A is written as ∂A , $\bar{A}$ denotes the closure of A , and $|A|$ is the Lebesgue measure of a measurable set $A \subset \mathbf{R}^d$. For a proper open set $D \subset \mathbf{R}^d$, we fix its Whitney decomposition $\mathcal{W}(D)$, and write $\mathcal{W}_m(D)$ for the family of Whitney cubes with side length 2^{-m} , $m \in \mathbf{Z}$. If $Q \in \mathcal{W}(D)$, then

$$(2.1) \quad \text{diam}(Q) \leq \text{dist}(Q, \partial D) \leq 4 \text{diam}(Q).$$

For other properties of Whitney cubes we refer to [35, VI.1].

3. Main result

We state and prove our main result. For definition of conditions $\text{DC}(a, \gamma, d)$, $\text{WLSC}(\eta, 0, H)$ and $\text{WUSC}(\eta, 0, H)$, we refer to §3.1 and §3.2. The proof of Theorem 5 is taken up in §3.3.

Theorem 5. Suppose that a proper open set $D \subset X$ satisfies $\text{DC}(a, \gamma, d)$ with $a \in (0, \infty) \setminus \{1\}$. Moreover, suppose that for some $H \in (0, 1]$, either $a \in (0, 1)$, $\eta + \gamma - d > 0$ and $\phi \in \text{WLSC}(\eta, 0, H)$, or $a > 1$, $\eta + \gamma - d < 0$ and $\phi \in \text{WUSC}(\eta, 0, H^{-1})$. Then for any $0 < p < \infty$ there exist constants c and $R > 0$ such that

$$(3.1) \quad \int_D \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) \leq c \int_D \int_{D \cap B(x, R\delta_x)} \frac{|u(x) - u(y)|^p}{\phi(\delta_x)\delta_x^d} \mu(dy) \mu(dx)$$

for all measurable functions u for which the left hand side is finite.

3.1. Assumptions on a function ϕ . We adopt the notion of a global weak lower (or upper) scaling condition (WLSC or WUSC for short) from [5, Section 3]. In our case, the middle parameter in $\text{WLSC}(\cdot, 0, \cdot)$ and $\text{WUSC}(\cdot, 0, \cdot)$ is always zero and could be therefore omitted, but we prefer to keep the original notation. We formulate these conditions in an equivalent way, which is more convenient for our purposes than the original formulation.

Definition 3.1. Let $\eta \in \mathbf{R}$ and $H \in (0, 1]$. We say that a function $\phi: (0, \infty) \rightarrow (0, \infty)$ satisfies global $\text{WLSC}(\eta, 0, H)$ (resp., $\text{WUSC}(\eta, 0, H^{-1})$) and write $\phi \in \text{WLSC}(\eta, 0, H)$ ($\phi \in \text{WUSC}(\eta, 0, H^{-1})$), if

$$(3.2) \quad \phi(st) \geq Ht^\eta \phi(s), \quad s > 0,$$

for every $t \geq 1$ (resp., for every $t \in (0, 1]$).

Remark 3.2. If the domain D in Theorem 5 is bounded, then it suffices to assume (3.2) for all $s, st < \text{diam}(D)$.

Example 3.3. Function $\phi(x) = x^\eta$ satisfies $\text{WLSC}(\eta, 0, 1)$ and $\text{WUSC}(\eta, 0, 1)$ whenever $\eta \in \mathbf{R}$.

Example 3.4. Suppose that $\phi: (0, \infty) \rightarrow (0, \infty)$ is a regularly varying function at origin of index ρ_0 , a regularly varying function at infinity of index ρ_∞ , and is bounded and bounded away from zero on every compact subset of $(0, \infty)$. If $\rho_0 > \eta$ and $\rho_\infty > \eta$, then $\phi \in \text{WLSC}(\eta, 0, H)$ for some $H \in (0, 1]$, and if $\rho_0 < \eta$ and $\rho_\infty < \eta$, then $\phi \in \text{WUSC}(\eta, 0, H^{-1})$ for some $H \in (0, 1]$. These follow from Potter's theorem [3, Theorem 1.5.6].

We note that if, say, $a < 1$, $\rho_0 \geq \eta$, $\rho_\infty \geq \eta$, and if $\eta + \gamma - d > 0$ and the assumptions on domain in Theorem 5 hold, then also the assertion (3.1) holds. Indeed, for every $\varepsilon > 0$ function ϕ satisfies $\text{WLSC}(\eta - \varepsilon, 0, H_\varepsilon)$ with some constant $H_\varepsilon \in (0, 1]$, hence by taking $\varepsilon > 0$ small enough we still have $(\eta - \varepsilon) + \gamma - d > 0$.

To have more concrete examples, let us note that functions

$$\phi_1(x) = x^\alpha + x^\beta, \quad \phi_2(x) = x^\eta(1 + |\log x|)^\beta$$

are regularly varying both at the origin (of indices $\min(\alpha, \beta)$ and η , respectively) and at infinity (of indices $\max(\alpha, \beta)$ and η , respectively).

Example 3.5. Functions ϕ are not confined to regularly varying functions. Indeed, $\phi(x) = x^\eta e^x$ satisfies $\text{WLSC}(\eta, 0, 1)$, but is not regularly varying at infinity.

3.2. Assumption $\text{DC}(a, \gamma, d)$ on open sets. In what follows we assume that D is an open set in a metric measure space (X, ρ, μ) .

Definition 3.6. We say that D satisfies a *domain condition* $\text{DC}(a, \gamma, d)$ where $\gamma \in \mathbf{R}$, $d > 0$, $a > 0$, $a \neq 1$, if there exist $M > 0$ and (possibly empty) families $\mathcal{B}^{(n)} = \{B_j^{(n)}\}$ of subsets of D indexed by $n \in \mathbf{Z}$ such that the following conditions (B1)–(B4) hold.

(B1) $D = \cup_{j,n} B_j^{(n)}$ and each $x \in D$ belongs to at most M sets $B_j^{(n)}$.

(B2) For any $B_j^{(n)}$ we have

$$M^{-1}a^n \leq \delta_x \leq Ma^n, \quad x \in B_j^{(n)}, \quad M^{-1}a^{nd} \leq \mu(B_j^{(n)}) \leq Ma^{nd}.$$

(B3) For any $B_j^{(n)}$ and any integer $k > M$, there exists a nonempty finite set $V(B_j^{(n)}, k)$ of indices so that, for each $i \in V(B_j^{(n)}, k)$,

$$\sup\{\rho(x, y) : x \in B_j^{(n)} \text{ and } y \in B_i^{(n+k)}\} \leq \begin{cases} Ma^n, & \text{if } a < 1, \\ Ma^{n+k}, & \text{if } a > 1. \end{cases}$$

(B4) For each $n \in \mathbf{Z}$ and $k > M$,

$$\sup_i \sum_{j: i \in V(B_j^{(n)}, k)} \frac{1}{\#V(B_j^{(n)}, k)} \leq Ma^{k\gamma}.$$

Remark 3.7. The slightly technical Definition 3.6 allows a unified treatment of fractional Hardy inequalities in different cases. In the Euclidean spaces, we usually take Whitney cubes of roughly the same size as the sets $B_j^{(n)}$ and then conditions (B1) and (B2) are immediate, we refer to §4.1 and §5.1 for examples. Formulating the definition using these general sets rather than Whitney cubes gives us some flexibility, see Examples 3.8 and 3.9 for an illustration. The last two conditions (B3) and (B4) specify the relationship between the layers

$$D_n := D \cap \{x : M^{-1}a^n \leq \delta_x \leq Ma^n\}$$

and

$$D_{n+k} := D \cap \{x : M^{-1}a^{n+k} \leq \delta_x \leq Ma^{n+k}\}$$

for large values of k . Namely, in a ‘neighbourhood’ of each $B_j^{(n)} \subset D_n$ there should be ‘sufficiently many’ sets $B_i^{(n+k)} \subset D_{n+k}$; the number γ describes this quantitatively.

Below we provide some illustrative examples of sets satisfying condition $\text{DC}(a, \gamma, d)$. In the two examples $X = \mathbf{R}^d$ with the Euclidean distance, in which case $\delta_x = \text{dist}(x, \partial D)$ for all $x \in D$. Moreover, μ is the Lebesgue measure.

Example 3.8. Set $D = \mathbf{R}^d \setminus \{0\}$ satisfies condition $\text{DC}(a, \gamma, d)$ with $a = 2$, $\gamma = 0$ and

$$M = 2 \vee (1 - 2^{-d})|B(0, 1)| \vee \frac{1}{(1 - 2^{-d})|B(0, 1)|}.$$

Indeed, one may take $B_1^{(n)} := B(0, 2^n) \setminus B(0, 2^{n-1})$. That is, for each n there is exactly one set $B_j^{(n)}$, namely one with $j = 1$. Then $V(B_j^{(n)}, k) = \{1\}$ in (B3).

Example 3.9. Set $D = \mathbf{R}^d \setminus \{0\}$ satisfies condition $\text{DC}(a, \gamma, d)$ with $a = \frac{1}{2}$, $\gamma = 0$ and

$$M = 2 \vee (1 - 2^{-d})|B(0, 1)| \vee \frac{1}{(1 - 2^{-d})|B(0, 1)|}.$$

Indeed, one may take $B_1^{(n)} := B(0, 2^{-n}) \setminus B(0, 2^{-(n-1)})$ and $V(B_j^{(n)}, k) = \{1\}$ in (B3).

3.3. Proof of Theorem 5. Let us write

$$q = 2^{p+1}M^{4+2|\eta|}H^{-1}a^{k(\eta+\gamma-d)}, \quad R = 1 + M^2(1 \vee a^k), \quad S = 2^{p+1}a^{-kd}M^{d+1},$$

where $k > M$ is chosen such that $q < 1$ and $a^k \vee a^{-k} > M^2$.

We fix a function u for which the left hand side of (3.1) is finite, and define a set

$$F = \left\{ x \in D : |u(x)|^p > S\delta_x^{-d} \int_{D \cap B(x, R\delta_x)} |u(x) - u(y)|^p \mu(dy) \right\}.$$

Let us first observe that, for $x \in D \setminus F$,

$$(3.3) \quad \frac{|u(x)|^p}{\phi(\delta_x)} \leq S \int_{D \cap B(x, R\delta_x)} \frac{|u(x) - u(y)|^p}{\phi(\delta_x)\delta_x^d} \mu(dy).$$

Note that if the set F were empty, we would be already done.

At this stage we fix n and claim that, for $x \in F \cap B_j^{(n)}$ and $i \in V(B_j^{(n)}, k)$, we have

$$(3.4) \quad \mu \left(\left\{ y \in B_i^{(n+k)} : \frac{1}{2}|u(x)| \leq |u(y)| \leq \frac{3}{2}|u(x)| \right\} \right) \geq \frac{1}{2}\mu(B_i^{(n+k)}).$$

Suppose (3.4) fails. By our choice of R and conditions (B2) and (B3), $B_i^{(n+k)} \subset D \cap B(x, R\delta_x)$. Thus, we have

$$\begin{aligned} \int_{D \cap B(x, R\delta_x)} |u(x) - u(y)|^p \mu(dy) &\geq \int_{B_i^{(n+k)}} |u(x) - u(y)|^p \mu(dy) \\ &\geq \frac{1}{2}\mu(B_i^{(n+k)}) \cdot 2^{-p}|u(x)|^p \geq 2^{-p-1}a^{kd}M^{-d-1}\delta_x^d|u(x)|^p = S^{-1}\delta_x^d|u(x)|^p, \end{aligned}$$

which contradicts $x \in F$. Thus inequality (3.4) holds as claimed.

Let us record the following estimates for $B_i^{(n+k)} \in \mathcal{B}^{(n+k)}$ and $B_j^{(n)} \in \mathcal{B}^{(n)}$. By condition (B2), $\mu(B_j^{(n)}) \leq M^2 a^{-kd} \mu(B_i^{(n+k)})$, moreover, for $x \in B_j^{(n)}$ and $y \in B_i^{(n+k)}$ it holds $M^2 a^{-k} \delta_y \geq \delta_x \geq M^{-2} a^{-k} \delta_y$. Hence, by condition (3.2)

$$\phi(\delta_x) = \phi\left(\delta_y \frac{\delta_x}{\delta_y}\right) \geq H \left(\frac{\delta_x}{\delta_y}\right)^\eta \phi(\delta_y) \geq H M^{-2|\eta|} a^{-k\eta} \phi(\delta_y).$$

Here we need to ensure that $\frac{\delta_x}{\delta_y} < 1$ in the case when $a > 1$ and that $\frac{\delta_x}{\delta_y} > 1$ in the case when $a < 1$. But these are satisfied since, by assumption, $a^k \vee a^{-k} > M^2$, i.e., k is large enough. By the above estimate and inequality (3.4) we obtain

$$\begin{aligned} \int_{F \cap B_j^{(n)}} \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) &\leq \mu(B_j^{(n)}) \sup_{x \in F \cap B_j^{(n)}} \frac{|u(x)|^p}{\phi(\delta_x)} \\ &\leq \frac{2^{p+1} M^2 a^{-kd}}{\#V(B_j^{(n)}, k)} \sum_{i \in V(B_j^{(n)}, k)} \int_{B_i^{(n+k)}} \frac{|u(y)|^p}{H M^{-2|\eta|} a^{-k\eta} \phi(\delta_y)} \mu(dy). \end{aligned}$$

By summing over all j and applying condition (B4),

$$\begin{aligned} &\sum_j \int_{F \cap B_j^{(n)}} \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) \\ &\leq 2^{p+1} M^{2+2|\eta|} H^{-1} a^{k(\eta-d)} \sup_i \sum_{j: i \in V(B_j^{(n)}, k)} \frac{1}{\#V(B_j^{(n)}, k)} \sum_i \int_{B_i^{(n+k)}} \frac{|u(y)|^p}{\phi(\delta_y)} \mu(dy) \\ &\leq 2^{p+1} M^{3+2|\eta|} H^{-1} a^{k(\eta+\gamma-d)} \sum_i \int_{B_i^{(n+k)}} \frac{|u(y)|^p}{\phi(\delta_y)} \mu(dy), \end{aligned}$$

and after summing over all n

$$\int_F \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) \leq q \int_D \frac{|u(y)|^p}{\phi(\delta_y)} \mu(dy).$$

Recall that $q < 1$. Hence, by finiteness of the left hand side of (3.1),

$$\int_F \frac{|u(x)|^p}{\phi(\delta_x)} \mu(dx) \leq \frac{q}{1-q} \int_{D \setminus F} \frac{|u(y)|^p}{\phi(\delta_y)} \mu(dy).$$

This estimate and inequality (3.3) finish the proof. $\square$

3.4. Proof of Corollary 2. We use Potter's theorem [3, Theorem 1.5.6] to replace $\phi(\delta_x)$ by $c\phi(|x-y|)$ in the denominator, with $c = c(R, \phi)$. The assumption $0 < \rho_0, \rho_\infty$ is used here. The result follows now from Theorem 5 and Examples 3.4, 3.8 and 3.9. $\square$

4. Fat boundary

We prove a domain condition while assuming that the boundary of the open set is sufficiently 'fat' in terms of the lower Assouad dimension. Then we study the relation between lower Assouad dimension and uniform fatness.

Proposition 6. *Suppose $D \neq \emptyset$ is a proper κ -plump open set in $\mathbf{R}^d$ such that D is bounded or ∂D is unbounded. Then D satisfies $\text{DC}(a, \lambda, d)$ if $a = 1/2$ and either*

$0 < \lambda < \underline{\dim}_A(\partial D)$ or $\lambda = 0$. Moreover, the associated constant M depends only on d, κ, λ and the constant C appearing in (F1) below.

Under the assumptions of Proposition 6, the following two conditions hold.

- (F1) There is a constant $C > 0$ as follows. Let $0 < r < R < 2 \operatorname{diam}(\partial D)$ and $x \in \partial D$. Suppose that $B_1, \dots, B_N$ is a cover of $B(x, R) \cap \partial D$ by balls $B_j = B(\omega_j, r)$ with $\omega_j \in \partial D$ for $j = 1, \dots, N$. Then $N \geq C(R/r)^\lambda$.
- (F2) for each $0 < r < \operatorname{diam}(D)$ and each $x \in \partial D$, there is $z \in \bar{B}(x, r)$ so that $B(z, \kappa r) \subset D$.

4.1. Construction of families $\mathcal{B}^{(n)}$. We define a constant

$$(4.1) \quad \tau = \left(\frac{15\sqrt{d}}{\kappa} \right)^d > 1.$$

For a given $n \in \mathbf{Z}$ and $a \in \{\frac{1}{2}, 2\}$, we define

$$\mathcal{B}^{(n)} := \mathcal{B}_{1/2}^{(n)}, \quad \mathcal{B}_a^{(n)} := \{B_j^{(n)}\} := \{Q \in \mathcal{W}(D) : \tau^{-1} \leq a^{-nd}|Q| \leq \tau\}.$$

Recall that $\mathcal{W}(D)$ stands for a Whitney decomposition of D . In particular, by inequalities (2.1), for any $x \in B_j^{(n)} \in \mathcal{B}_a^{(n)}$,

$$\tau^{-1/d}a^n \leq \delta_x = \operatorname{dist}(x, \partial D) \leq 5\sqrt{d}\tau^{1/d}a^n.$$

Observe also that a given Whitney cube $Q \in \mathcal{W}(D)$ may belong to at most $1 + 2d^{-1} \log_2 \tau$ families $\mathcal{B}^{(n)} = \mathcal{B}_a^{(n)}$ indexed by $n \in \mathbf{Z}$. Let us denote by $x_j^{(n)}$ the midpoint of $B_j^{(n)}$. For later purposes we fix, once and for all, any point $y_j^{(n)} \in \partial D$ for which

$$|x_j^{(n)} - y_j^{(n)}| = \operatorname{dist}(x_j^{(n)}, \partial D).$$

4.2. Families $V(B_j^{(n)}, k)$ for k large. If D is unbounded, we construct families $V(B_j^{(n)}, k)$ for $k > 3$. If D is bounded, then we construct these families for $k > 3 \vee \log_2(5\tau^{1/d})$.

Let us fix $B_j^{(n)} \in \mathcal{B}^{(n)}$, and define $E := B(y_j^{(n)}, 2^{-n}) \cap \partial D$. By the $5r$ -covering theorem, see for instance [31, p. 23], there are points $\omega_1, \dots, \omega_N \in E$ such that the balls $B_m := B(\omega_m, 2^{-n-k})$ are disjoint and E is covered by the union of balls $5B_m$, $m = 1, \dots, N$. Let us estimate the number $N =: N_j^{(n,k)}$ of these balls;

Lemma 7. We have $N_j^{(n,k)} \geq C5^{-\lambda}\tau^{-\lambda/d}2^{k\lambda}$.

Proof. First consider the case when D is unbounded. Since $k > 3$, we find that

$$r := 5 \cdot 2^{-n-k} < 2^{-n} =: R.$$

Recall that the balls $5B_m = B(\omega_m, r)$ cover the set $E = B(y_j^{(n)}, R) \cap \partial D$. By condition (F1), we find that $N \geq C(R/r)^\lambda = C5^{-\lambda}2^{k\lambda}$. The bounded case is similar, and we use the facts that $k > 3 \vee \log_2(5\tau^{1/d})$ and $\operatorname{diam}(\partial D) \geq \operatorname{diam}(D)$. $\square$

The next step is to use the plumpness condition (F2) in order to locate a sufficiently large cube inside each B_m . Namely, for each $m = 1, \dots, N$, there is $z_m \in \bar{B}(\omega_m, 2^{-n-k}/3)$ such that

$$B(z_m, \kappa 2^{-n-k}/3) \subset D.$$

Let us consider a Whitney cube $Q_m \in \mathcal{W}(D)$ for which $z_m \in Q_m$. By inequalities (2.1), we have $Q_m \subset B_m$. Moreover,

$$\kappa 2^{-n-k}/3 \leq \text{dist}(z_m, \partial D) \leq 5 \text{diam}(Q_m) \leq 5 \text{dist}(z_m, \partial D) \leq 5 \cdot 2^{-n-k}/3.$$

Hence, by our definition (4.1) of τ , we obtain

$$\tau^{-1} \leq 2^{d(n+k)} |Q_m| \leq \tau.$$

That is, cube $Q_m \subset B_m$ belongs to $\mathcal{B}^{(n+k)}$. Since the balls B_m , $m = 1, \dots, N$, are disjoint, also the cubes Q_m are disjoint. Hence, the indexing set

$$V(B_j^{(n)}, k) = \{i: B_i^{(n+k)} = Q_m \text{ for some } m = 1, \dots, N_j^{(n,k)}\}$$

contains exactly $N_j^{(n,k)}$ indices.

4.3. Proof of Proposition 6. We focus on conditions (B3) and (B4), as the remaining conditions are clearly satisfied. Let us fix $B_j^{(n)}$ and k large enough so that $V(B_j^{(n)}, k)$ is defined. Let us consider $i \in V(B_j^{(n)}, k)$, and two given points $x \in B_j^{(n)}$ and $y \in B_i^{(n+k)}$. Using the notation above, we have $B_i^{(n+k)} = Q_m \subset B_m$ for some $m = 1, \dots, N_j^{(n,k)}$. Thus,

$$\begin{aligned} |x - y| &\leq |x - x_j^{(n)}| + |x_j^{(n)} - y_j^{(n)}| + |y_j^{(n)} - \omega_m| + |\omega_m - y| \\ &< \text{diam}(B_j^{(n)}) + \text{dist}(x_j^{(n)}, \partial D) + 2^{-n} + 2^{-n-k} \leq 8\sqrt{d}\tau^{1/d}2^{-n}. \end{aligned}$$

This is condition (B3). A particular consequence of this estimate is the following. We fix a cube $B_i^{(n+k)}$ and a point y therein. Then, if $B_j^{(n)} \in \mathcal{B}^{(n)}$ is such that $i \in V(B_j^{(n)}, k)$,

$$B_j^{(n)} \subset B(y, 8\sqrt{d}\tau^{1/d}2^{-n}).$$

Since the interiors of cubes in $\mathcal{B}^{(n)}$ are disjoint, we find that there are at most

$$\frac{(16\sqrt{d}\tau^{1/d}2^{-n})^d}{\tau^{-1}2^{-nd}} = (16\sqrt{d})^d \tau^2$$

cubes $B_j^{(n)}$ subject to the conditions above. By using this fact, we may now deduce the remaining estimate as follows; For a fixed i ,

$$\sum_{j: i \in V(B_j^{(n)}, k)} \frac{1}{\#V(B_j^{(n)}, k)} = \sum_{j: i \in V(B_j^{(n)}, k)} \frac{1}{N_j^{(n,k)}} \leq (16\sqrt{d})^d \tau^{2+\lambda/d} C^{-1} 5^\lambda 2^{-k\lambda}.$$

This is condition (B4). □

4.4. Lower Assouad dimension and uniform fatness. We provide a useful connection between the lower Aikawa dimension and (local) uniform fatness. For further discussion, we refer to [19]. Uniform fatness is usually defined in terms of Riesz capacities, [1, 26]. In case of closed sets, there is an equivalent definition—in terms of Hausdorff content—that we adopt. This equivalence is based on the self-improving properties of closed uniformly fat sets, [16].

Recall that the λ -Hausdorff content of a set $E \subset \mathbf{R}^d$ is

$$\mathcal{H}_\infty^\lambda(E) = \inf \left\{ \sum_{i=1}^\infty r_i^\lambda : E \subset \bigcup_{i=1}^\infty B(x_i, r_i), r_i > 0 \right\}.$$

As is easily seen, we may allow also finite coverings in the infimum above. Let $1 < p < \infty$ and $0 < s < d/p$. We say that the boundary ∂D is (s, p) -uniformly fat, if there is $d - sp < \lambda \leq d$ and a constant $C > 0$ such that

$$(4.2) \quad \mathcal{H}_\infty^\lambda(B(x, R) \cap \partial D) \geq CR^\lambda$$

for all $x \in \partial D$ and $R > 0$.

Note that ∂D and D have to be unbounded if the boundary is (s, p) -uniformly fat. Remark 2.3 in [19] shows that $\underline{\dim}_A(\partial D)$ is the supremum of all $\lambda \geq 0$ for which (4.2) holds for every $x \in \partial D$ and $0 < R < \text{diam}(\partial D)$. Below, for the convenience of the reader, we provide a detailed treatment of certain consequences of this statement—that are needed for Corollary 4.

Proposition 8. *Suppose D is an open set in $\mathbf{R}^d$ so that ∂D is (s, p) -uniformly fat for $1 < p < \infty$ and $0 < s < d/p$. Then D satisfies condition (F1) for some $d - sp < \lambda \leq d$ and, as a consequence, we have a strict inequality $\underline{\dim}_A(\partial D) > d - sp$.*

Proof. By assumption, there is $d - sp < \lambda \leq d$ and $C > 0$ such that (4.2) holds for all $x \in \partial D$ and $R > 0$. Let us fix $x \in \partial D$ and $0 < r < R$. Suppose that $B_1, \dots, B_N$ is a cover of $B(x, R) \cap \partial D$ by balls $B_j = B(\omega_j, r)$ with $\omega_j \in \partial D$. Then, by (4.2),

$$Nr^\lambda = \sum_{j=1}^N r^\lambda \geq \mathcal{H}_\infty^\lambda(B(x, R) \cap \partial D) \geq CR^\lambda.$$

Thus, $N \geq C(R/r)^\lambda$, as required. $\square$

As we have observed, uniform fatness is a convenient notion in case of unbounded open sets. In case of a bounded open set D in $\mathbf{R}^d$, it is natural to assume that ∂D is (s, p) -locally uniformly fat. That is, there is $d - sp < \lambda \leq d$ and a constant $C > 0$ such that inequality (4.2) holds for all $x \in \partial D$ and $0 < R < 2 \text{diam}(\partial D) < \infty$.

The following result is analogous to Proposition 8.

Proposition 9. *Let D be a bounded open set in $\mathbf{R}^d$ such that the boundary ∂D is (s, p) -locally uniformly fat for $1 < p < \infty$ and $0 < s < d/p$. Then D satisfies condition (F1) for some $d - sp < \lambda \leq d$ and, as a consequence, we have a strict inequality $\underline{\dim}_A(\partial D) > d - sp$.*

Example 4.1. Consider the Koch snowflake domain $D \subset \mathbf{R}^2$. It is a bounded κ -plump domain and ∂D is (s, p) -locally uniformly fat whenever $1 < p < \infty$ and $2 - \log 4 / \log 3 < sp < 2$, see e.g. [16].

5. Thin boundary

The main result in this section is the following.

Proposition 10. *Let $D \neq \emptyset$ be an unbounded κ -plump open set in $\mathbf{R}^d$, $D \neq \mathbf{R}^d$. Then D satisfies condition DC(a, λ, d) for $a = 2$ and $\lambda > \underline{\dim}_A(\partial D)$. The associated constant M depends only on d, κ, λ and the constant C appearing in (T1) below.*

Before the proof, let us clarify the assumptions. Under the assumptions of Proposition 10, the following two conditions (T1) and (T2) hold.

- (T1) there is a constant $C > 0$ as follows. Assuming that $0 < r < R$ and $x \in \partial D$, there is a cover of $B(x, R) \cap \partial D$ by using balls $B(\omega_j, r)$ with $\omega_j \in \partial D$,

$j = 1, \dots, N$, such that the number of these balls satisfies inequality $N \leq C(R/r)^\lambda$.
 (T2) for each $0 < r$ and each $x \in \partial D$, there is a point $z \in \bar{B}(x, r)$ such that $B(z, \kappa r) \subset D$.

We note that in Definition 2.2 the restriction $R < 2 \operatorname{diam}(E)$ may be removed, resulting in no such restriction in (T1).

5.1. Construction of families $\mathcal{B}^{(n)}$. Let $\tau > 1$ be defined by (4.1). For $n \in \mathbf{Z}$, we define

$$\mathcal{B}^{(n)} := \mathcal{B}_2^{(n)} := \{B_j^{(n)}\} := \{Q \in \mathcal{W}(D) : \tau^{-1} \leq 2^{-nd}|Q| \leq \tau\}.$$

For properties of cubes in these families and definition of $y_j^{(n)}$, see §4.1. Recall also that $\mathcal{W}(D)$ is a Whitney decomposition of D , we refer to §2.

5.2. Families $V(B_j^{(n)}, k)$ with $k > 0$. Let us fix a cube $B_j^{(n)} \in \mathcal{B}^{(n)}$. By condition (T2), there is a point $z_j^{(n)} \in \bar{B}(y_j^{(n)}, 2^{n+k})$ such that

$$B(z_j^{(n)}, \kappa 2^{n+k}) \subset D.$$

Observe how the unboundedness of D is visible here, as $k > 0$ is arbitrary. Let $Q_j \in \mathcal{W}(D)$ be a Whitney cube such that $z_j^{(n)} \in Q_j$. Then

$$\kappa \cdot 2^{n+k} \leq \operatorname{dist}(z_j^{(n)}, \partial D) \leq 5 \operatorname{diam}(Q_j) \leq 5 \operatorname{dist}(z_j^{(n)}, \partial D) \leq 5 \cdot 2^{n+k}.$$

By definition of (4.1) of τ , we then have $Q_j = B_i^{(n+k)} \in \mathcal{B}^{(n+k)}$ for some index i . We define

$$V(B_j^{(n)}, k) = \{i\}.$$

5.3. Proof of Proposition 10. We need an auxiliary estimate analogous to [19, Lemma 4.3]; condition (T1) is our primary tool. For $m \in \mathbf{Z}$, $\omega \in \partial D$, and $R > 0$, we denote

$$\mathcal{W}_m(D; B(\omega, R)) = \{Q \in \mathcal{W}_m(D) : Q \subset B(\omega, R)\}.$$

Lemma 11. Let $0 < 2^{-m} \leq R$, where $m \in \mathbf{Z}$, and let $\lambda > \overline{\dim}_A(\partial D)$. Then for every $\omega \in \partial D$,

$$\#\mathcal{W}_m(D; B(\omega, R)) \leq C(14\sqrt{d})^{d+\lambda} \left(\frac{R}{2^{-m}} \right)^\lambda,$$

where C is as in condition (T1).

Proof. Suppose $B_1, \dots, B_N$ is a cover of $B(\omega, 6\sqrt{d}R) \cap \partial D$ by balls $B_j = B(\omega_j, 2^{-m})$ that are centred in ∂D , see condition (T1). Consider a cube $Q \in \mathcal{W}_m(D; B(\omega, R))$, and fix a point $y_Q \in \partial D$ such that $|x_Q - y_Q| = \operatorname{dist}(x_Q, \partial D)$. Here x_Q denotes the midpoint of Q . By inequalities (2.1) and the fact that $Q \subset B(\omega, R)$,

$$|y_Q - \omega| \leq |y_Q - x_Q| + |x_Q - \omega| < 5 \operatorname{diam}(Q) + R \leq 6\sqrt{d}R.$$

By the covering property, there is $j = j(Q)$ such that $y_Q \in B_j$. We can infer that

$$\mathcal{W}_m(D; B(\omega, R)) = \bigcup_{j=1}^N \mathcal{Q}_j,$$

where $\mathcal{Q}_j = \{Q \in \mathcal{W}_m(D; B(\omega, R)) : y_Q \in B_j\}$. Let $Q \in \mathcal{Q}_j$. Then, for every $x \in Q$,

$$\begin{aligned} |x - \omega_j| &\leq |x - x_Q| + |x_Q - y_Q| + |y_Q - \omega_j| \\ &< \text{diam}(Q) + \text{dist}(x_Q, \partial D) + 2^{-m} \leq 7\sqrt{d}2^{-m}. \end{aligned}$$

Since the interiors of cubes in the family $\mathcal{Q}_j$ are disjoint, there are at most

$$\frac{|B(\omega_j, 7\sqrt{d}2^{-m})|}{2^{-md}} \leq (14\sqrt{d})^d$$

cubes in this family. Hence,

$$\#\mathcal{W}_m(D; B(\omega, R)) \leq \sum_{j=1}^N \#\mathcal{Q}_j \leq (14\sqrt{d})^d N \leq C \left(\frac{6\sqrt{d}R}{2^{-m}} \right)^\lambda \cdot (14\sqrt{d})^d.$$

This concludes the proof. $\square$

We are ready to prove the main result in this section.

Proof of Proposition 10. The properties (B1) and (B2) are clear. In order to verify condition (B3), let us fix $k > 0$ and a cube $B_j^{(n)}$. Consider $i \in V(B_j^{(n)}, k)$, and points $x \in B_j^{(n)}$ and $z \in B_i^{(n+k)}$. Then, by the construction above,

$$\begin{aligned} |z - x| &\leq |z - z_j^{(n)}| + |z_j^{(n)} - y_j^{(n)}| + |y_j^{(n)} - x_j^{(n)}| + |x_j^{(n)} - x| \\ &\leq \text{diam}(B_i^{(n+k)}) + 2^{n+k} + \text{dist}(x_j^{(n)}, \partial D) + \text{diam}(B_j^{(n)}) < 8\sqrt{d}\tau^{1/d}2^{n+k}. \end{aligned}$$

This is condition (B3).

In order to verify the last condition (B4), we fix cubes $B_i^{(n+k)}$ and $B_j^{(n)}$ such that $i \in V(B_j^{(n)}, k)$. Then $\#V(B_j^{(n)}, k) = 1$. Moreover,

$$B_j^{(n)} \subset B(y_i^{(n+k)}, 13\sqrt{d}\tau^{1/d}2^{n+k}).$$

Indeed, for any $x \in B_j^{(n)}$,

$$|x - y_i^{(n+k)}| \leq |x - x_i^{(n+k)}| + |x_i^{(n+k)} - y_i^{(n+k)}| < 13\sqrt{d}\tau^{1/d}2^{n+k}.$$

We still need another auxiliary estimate, namely, if $m \in \mathbf{Z}$ is such that $B_j^{(n)} \in \mathcal{W}_m(D)$, then $\tau^{-1/d} \leq 2^{m+n} \leq \tau^{1/d}$. We can finally proceed as follows

$$\begin{aligned} \sum_{j: i \in V(B_j^{(n)}, k)} \frac{1}{\#V(B_j^{(n)}, k)} &= \#\{j: i \in V(B_j^{(n)}, k)\} \\ &= \sum_m \#\{j: i \in V(B_j^{(n)}, k) \text{ and } B_j^{(n)} \in \mathcal{W}_m(D)\} \\ &\leq \sum_m \#\mathcal{W}_m(D; B(y_i^{(n+k)}, 13\sqrt{d}\tau^{1/d}2^{n+k})) \\ &\leq \sum_m C(14\sqrt{d})^{d+\lambda} \left(\frac{13\sqrt{d}\tau^{1/d}2^{n+k}}{2^{-m}} \right)^\lambda, \end{aligned}$$

where m ranges over indices $-n - \log_2 \tau^{1/d} \leq m \leq -n + \log_2 \tau^{1/d}$. This yields condition (B4). $\square$

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Interior C^2 estimates to hessian-type equations on $\mathbb{S}^n$

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Abstract. In this article, we study global C^2 estimates to smooth admissible solutions of the Hessian-type equations on $\mathbb{S}^n$ and then obtain a unique smooth admissible solution up to a constant. As a special case, the Hessian-type equations contain a Monge-Ampère equation arising in designing a reflecting surface in geometric optics.

1. Introduction

In this paper, we consider interior C^2 estimates the following problem

$$\sigma_k(K_{ij}) = f(x), \quad \text{on} \quad \mathbb{S}^n, \quad (1.1)$$

where

$$K_{ij} = \frac{2uu_{ij} + (u^2 - |\nabla u|^2)\delta_{ij}}{u^2 + |\nabla u|^2}, \quad (1.2)$$

σ_k is the k -th elementary symmetric function which will be defined in section 2, u_{ij} is the second-order derivative with orthogonal frames on $\mathbb{S}^n$. The quantity K_{ij} is referred to as intensity density by Oliker [1]. Equations (1.1) and (1.2) have intimate relation with the reflector designing problem in geometric optics. Many mathematicians have done excellent works in this field. For $k=n$, it is an equation of Monge-Ampère type. Wang [2] and Guan-Wang [3] obtained the existence and uniqueness of smooth solutions up to multiplication of positive constants under the condition $\int_{\mathbb{S}^n} f(x)dx = \int_{\mathbb{S}^n} g(x)dx$. When $k=1$, it is an equation of quasi-linear type. Oliker [1] obtained the existence and uniqueness of this special case.

In this article, we consider the general σ_k case of this reflector designing problem. As for the counterpart of this problem, Christoffel-Minkowski problems have been considered by many mathematicians and received considerable results, such as in [4], [5], [6] and the references therein.



In [4], Guan-Lin-Ma considered the existence of convex Weingarten hypersurfaces. In [5], Caffarelli-Nirenberg-Spruck studied the problem of finding embedded Weingarten hypersurfaces by using the continuity method.

In this paper, we establish the C^2 regularity, existence and uniqueness of the admissible solution to equation (1.1) and (1.2). A solution u to the equation (1.1) and (1.2) is called admissible if the eigenvalues of the intensity density K_{ij} are in the convex cone Γ_k which the definition can be seen in section 2. Admissible solutions are natural since they guarantees the ellipticity of the equation. The problem (1.1) and (1.2) has been studied in [7] where logarithmic gradient estimates were obtained.

It is proved in [7] that if $0 < u \in C^3(\mathbb{S}^n)$ is an admissible solution to equation (1.1) and (1.2), where

$f \in C^1(\mathbb{S}^n)$ is a positive function, then there holds the logarithmic gradient estimates

$$\sup_{\mathbb{S}^n} |\nabla \log u| \leq C_0, \text{ where } C_0 = C_0(n, k, \sup_{\mathbb{S}^n} |f|, \sup_{\mathbb{S}^n} |\nabla f|).$$

This article is a continuation of the paper [7] which further establish the $C^{2,\alpha}$ estimates and existence and uniqueness of problem (1.1) and (1.2). Our main result of this article is the following

Theorem 1.1. (Interior C^2 estimates) Let $u \in C^4(S^n)$ be an admissible solution to equation (1.1)

and (1.2), $f(x) \in C^2(S^n)$ be a nonnegative function. Then we have the interior C^2 estimates

$$\sup_{x \in B_{\frac{r}{2}}} |\nabla^2 u| \leq C_0, \text{ where } C_0 = C_0(n, k, r, |f|, |\nabla f|, |\nabla^2 f|).$$

Remark 1.2. Since $\mathbb{S}^n$ is compact, by using the finite covering theorem and **Theorem 1.1**, we indeed obtain global C^2 estimates.

By using the logarithmic gradient estimates in [7] and Theorem 1.1, we have the following theorem concerning the existence and uniqueness of the σ_k -reflector designing problem.

Theorem 1.3. Suppose $f(x)$ is a C^∞ positive function on $\mathbb{S}^n$. Then there exists a C^∞ admissible solution to equation (1.1) and (1.2), and the solution is unique up to multiplication of positive constants.

The approach of proving **Theorem 1.3** is the continuity method as illustrated in Gilbarg-Trudinger [8]. The key step here is to establish a-priori estimates for solution to (1.1) and (1.2) up to the second order derivatives and then obtain $C^{2,\alpha}$ estimates by Evans-Krylov Theorem [8].

The article is organized as follows. In section 2, we recall definition and some basic properties of the k -th elementary symmetric functions. We derive the interior C^2 apriori estimates in section 3. In the last section, we obtain the existence and uniqueness of the smooth admissible solution to equation (1.1) and (1.2) by using the continuity method.

2. Preliminaries

In this section, we give some preliminaries on the definitions and properties of elementary symmetric functions which will be frequently used in the following sections.

Definition 2.1. For any $k=1, \dots, n$, and for any $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, we define the k -th elementary symmetric function as

$$\sigma_k(\lambda) = \sum_{1 \leq i_1 < i_2 < \dots < i_k} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_k}. \quad (2.1)$$

We also set $\sigma_0 = 1$ and $\sigma_k = 0$ for $k > n$. For $1 \leq k \leq n$, we define Γ_k is a cone in $\mathbb{R}^n$ by

$$\Gamma_k = \{\lambda \in \mathbb{R}^n : \sigma_1(\lambda) > 0, \dots, \sigma_k(\lambda) > 0\}. \quad (2.2)$$

Obviously Γ_k contains the positive cone $\Gamma_n = \{\lambda \in \mathbb{R}^n : \lambda_i > 0, i=1, 2, \dots, n\}$ and Γ_k is symmetric in the sense that if $\lambda \in \Gamma_k$, then any permutation of λ also lies in Γ_k . In fact, Γ_k is a convex cone.

Let us denote by $\sigma_k(\lambda | i)$ the sum of the terms of $\sigma_k(\lambda)$ not containing the factor λ_i . Then the following identities hold.

Proposition 2.2. For any $k=0, 1, \dots, n$, $i=1, 2, \dots, n$ and $\lambda \in \mathbb{R}^n$, there hold the following identities.

$$\frac{\partial \sigma_{k+1}}{\partial \lambda_i}(\lambda) = \sigma_k(\lambda | i), \quad (2.3)$$

$$\sigma_{k+1}(\lambda) = \sigma_{k+1}(\lambda | i) + \lambda_i \sigma_k(\lambda | i), \quad (2.4)$$

$$\sum_{i=1}^n \sigma_k(\lambda | i) = (n-k) \sigma_k(\lambda), \quad (2.5)$$

$$\sum_{i=1}^n \lambda_i \sigma_k(\lambda | i) = (k+1) \sigma_{k+1}(\lambda). \quad (2.6)$$

Proposition 2.3. If $\lambda \in \Gamma_k$ for $k \in \{1, 2, \dots, n\}$, then we have $\sigma_h(\lambda | i) > 0$, for any $h \in \{0, 1, \dots, k-1\}$ and $i \in \{1, 2, \dots, n\}$.

Proposition 2.4. Set $F(W) = \left(\frac{\sigma_k}{C_n^k}\right)^{\frac{1}{k}}(W)$, where $W = \{W_{ij}\}$ is a symmetric matrix. If the eigenvalues of W belong to Γ_k , then the matrix $\left\{\frac{\partial F}{\partial W_{ij}}\right\}$ is positive definite and

$$\sum_{i=1}^n \frac{\partial F}{\partial W_{ii}} \geq 1. \quad (2.7)$$

Moreover, F is concave on the convex cone Γ_k , that is for any $(\eta_{ij}) \in \mathbb{R}^{n \times n}$

$$\sum_{i,j,k,l=1}^n \frac{\partial^2 F}{\partial W_{ij} \partial W_{kl}}(W) \eta_{ij} \eta_{kl} \leq 0 \quad (2.8)$$

Proposition 2.2 is standard which can be checked directly. For the proof of **Proposition 2.3** and **Proposition 2.4**, the readers can refer to [9] for example.

3. Interior C^2 estimates

In this section, we prove the Interior C^2 estimates of Theorem 1.1. We follow the method in [10].

Set $v = \log u$, that is, $u = e^v$.

By direct calculation, we obtain

$$u_i = e^v v_i, \quad (3.1)$$

$$u_{ij} = e^v (v_{ij} + v_i v_j), \quad (3.2)$$

$$u^2 + |\nabla u|^2 = e^{2v} (1 + |\nabla v|^2). \quad (3.3)$$

Therefore the equation (1.1) and (1.2) can be written as $\sigma_k(W_{ij}) = \left(\frac{1+|\nabla v|^2}{2}\right)^k f(x)$,

where

$$W_{ij} = v_{ij} + v_i v_j + \frac{1-|\nabla v|^2}{2} \delta_{ij}. \quad (3.4)$$

Let $F(W_{ij}) = \left(\frac{\sigma_k(W_{ij})}{C_n^k}\right)^{\frac{1}{k}}$, $F^{ij} = \frac{\partial F}{\partial W_{ij}}$, then we get

$$F(W_{ij}) = (C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}}. \quad (3.5)$$

To obtain interior C^2 estimates, let us consider an auxiliary function

$$G(x, \xi) = \eta^2(W_{\xi\xi} + \frac{1}{2}|\nabla v|^2), \quad (3.6)$$

where $0 \leq \eta \leq 1$ is a cut-off function such that $\eta = 1$ in $B_{\frac{r}{2}}$ and $\eta = 0$ outside B_r , and also

$|\nabla \eta| < C \frac{\sqrt{\eta}}{r}$ and $|\nabla^2 \eta| < \frac{C}{r^2}$. Without loss of generality, we assume $r = 1$, since we can make a scaling

for general r and the proof is similar.

Suppose (x_0, ξ_0) is the maximal point of the auxiliary function G , by rotating the coordinate system suitably, we may assume that $\{W_{ij}(x_0)\}$ is diagonal and $\xi_0 = 1$. Therefore $\{F^{ij}(x_0)\}$ is diagonal. At x_0 ,

by taking the first derivative of $\log G$, we have

$$(\log G)_i = \frac{2\eta_i}{\eta} + \frac{W_{11,i} + v_p v_{pi}}{W_{11} + \frac{1}{2} |\nabla v|^2} = 0. \quad (3.7)$$

By taking the second derivative of (3.6) once more and using (3.7), we yield

$$\begin{aligned} (\log G)_{ij} &= \frac{2\eta_{ij}}{\eta} - \frac{2\eta_i \eta_j}{\eta^2} + \frac{W_{11,ij} + 2v_p v_{pij} + 2v_{pi} v_{pj}}{W_{11} + \frac{1}{2} |\nabla v|^2} - \frac{(W_{11,i} + v_p v_{pi})(W_{11,j} + v_p v_{pj})}{W_{11} + \frac{1}{2} |\nabla v|^2} \\ &= \frac{W_{11,ij} + v_p v_{pij} + v_{pi} v_{pj}}{W_{11} + \frac{1}{2} |\nabla v|^2} + \frac{2\eta_{ij}}{\eta} - \frac{6\eta_i \eta_j}{\eta^2} \leq 0. \end{aligned} \quad (3.8)$$

Using the positivity of F^{ij} , we multiplying by F^{ij} on the both sides of the above inequality and get

$$\begin{aligned} 0 &\geq F^{ij} (\log G)_{ij} = F^{ii} (\log G)_{ii} \\ &= \frac{1}{W_{11} + \frac{1}{2} |\nabla v|^2} (F^{ii} W_{11,ii} + F^{ii} v_p v_{pii} + F^{ii} v_{pi}^2) + F^{ii} \left(\frac{2\eta_{ii}}{\eta} - \frac{6\eta_i^2}{\eta^2} \right) \\ &=: \frac{1}{W_{11} + \frac{1}{2} |\nabla v|^2} I + II. \end{aligned} \quad (3.9)$$

To compute the term I , we first recall the following formulas on $\mathbb{S}^n$ which can be referred to [1] or [4] for example:

$$v_{ij} = v_{ji}, \quad (3.10)$$

$$v_{ijk} = v_{ikj} - v_k \delta_{ij} + v_j \delta_{ik}, \quad (3.11)$$

$$v_{ijkl} = v_{ijlk} + v_{ik} \delta_{jl} - v_{il} \delta_{jk} + v_{kj} \delta_{il} + v_{jl} \delta_{ik}. \quad (3.12)$$

By using these formulas, we deduce that

$$W_{ij} = W_{ji}, \quad (3.13)$$

and

$$W_{11,ii} = W_{ii,11} - 2v_i v_{ii1} + v_p v_{p11} + 2v_1 v_{1ii} - p v_p v_{pii} + 2v_{11} - 2v_{ii} + v_{p1}^2 - v_{pi}^2 \quad (3.14)$$

Therefore, we obtain

$$\begin{aligned} I &= F^{ii} W_{11,ii} + F^{ii} v_p v_{pii} + F^{ii} v_{pi}^2 \\ &= F^{ii} W_{ii,11} - 2F^{ii} v_i v_{ii1} + \sum_i F^{ii} v_p v_{p11} + 2v_1 F^{ii} v_{1ii} \\ &\quad + 2v_{11} \sum_i F^{ii} - 2F^{ii} v_{ii} + \sum_i F^{ii} v_{p1}^2 \\ &= F^{ii} W_{ii,11} - 2F^{ii} v_i v_{11i} + \sum_i F^{ii} v_p v_{11p} + 2v_1 F^{ii} v_{ii1} + 2v_{11} \sum_i F^{ii} \end{aligned}$$

$$-2F^{ii}v_{ii} + \sum_i F^{ii}v_{p1}^2 - 2F^{ii}v_i^2 + v_1^2 \sum_i F^{ii} + |\nabla v|^2 \sum_i F^{ii}. \quad (3.15)$$

By (3.4), it follows that

$$v_{ijk} = W_{ij,k} - v_j v_{ik} - v_i v_{jk} + v_p v_{pk} \delta_{ij}. \quad (3.16)$$

Then we calculate the term I as follows:

$$\begin{aligned} I &= F^{ii}W_{ii,11} - 2F^{ii}v_i W_{11i} + F^{ii}v_p W_{11p} + 2v_1 F^{ii}W_{ii1} \\ &\quad - 2F^{ii}v_i v_p v_{ip} + \sum_i F^{ii}v_p v_q v_{pq} + 2v_{11} \sum_i F^{ii} - 2F^{ii}v_{ii} \\ &\quad + \sum_i F^{ii}v_{p1}^2 - 2F^{ii}v_i^2 + v_1^2 \sum_i F^{ii} + |\nabla v|^2 \sum_i F^{ii}. \end{aligned} \quad (3.17)$$

Taking first derivative of equation (3.5), we obtain

$$F^{ii}W_{ii,1} = [(C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}}]_1. \quad (3.18)$$

Taking second derivative of equation (3.5) once more and using the concavity property of $F(W_{ij})$

which is the inequality (2.8) in **Proposition 2.4**, we obtain

$$F^{ii}W_{ii,11} \geq [(C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}}]_{11}. \quad (3.19)$$

By (3.7), (3.18) and (3.19), it follows that

$$\begin{aligned} I &\geq (C_n^k)^{-\frac{1}{k}} \left[\frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}} \right]_{11} - 2F^{ii}v_i \left[-\frac{2\eta_i}{\eta} (W_{11} + \frac{|\nabla v|^2}{2}) - v_p v_{ip} \right] \\ &\quad + F^{ii}v_p \left[-\frac{2\eta_i}{\eta} (W_{11} + \frac{|\nabla v|^2}{2}) - v_q v_{qp} \right] + 2(C_n^k)^{-\frac{1}{k}} v_1 \left[\frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}} \right]_1 \\ &\quad - 2F^{ii}v_i v_p v_{ip} + \sum_{i,p,q} F^{ii}v_p v_q v_{pq} + 2v_{11} \sum_i F^{ii} - 2F^{ii}v_{ii} + \sum_i F^{ii}v_{p1}^2 \\ &\quad - 2F^{ii}v_i^2 + v_1^2 \sum_i F^{ii} + |\nabla v|^2 \sum_i F^{ii} \\ &= [(C_n^k)^{-\frac{1}{k}} f^{\frac{1}{k}} + \sum_i F^{ii} v_{p1}^2 + \frac{2}{k} (C_n^k)^{-\frac{1}{k}} f^{\frac{1}{k}-1} f_1 v_p v_{p1} \\ &\quad - [2(C_n^k)^{-\frac{1}{k}} f^{\frac{1}{k}} \frac{\eta_p}{\eta} v_p - 4F^{ii} \frac{\eta_i}{\eta} v_i + 2 \sum_i F^{ii} \frac{\eta_p}{\eta} v_p] (W_{11} + \frac{|\nabla v|^2}{2}) \\ &\quad + [(C_n^k)^{-\frac{1}{k}} f^{\frac{1}{k}} + \sum_i F^{ii}] |\nabla v|^2 - [(C_n^k)^{-\frac{1}{k}} f^{\frac{1}{k}} + \sum_i F^{ii}] v_1^2 + (C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} (f^{\frac{1}{k}})_{11} \end{aligned} \quad (3.20)$$

Using the condition on the cut-off function η and the inequality (2.7) in **Proposition 2.4**, we get

$$\begin{aligned} I &\geq \frac{1}{2} \sum_i F^{ii} \sum_p v_{p1}^2 - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1) W_{11} - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1) \\ &\geq \frac{1}{2} \sum_i F^{ii} v_{11}^2 - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1) W_{11} - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \sum_i F^{ii} [W_{11} - v_1^2 - \frac{1}{2} (1 - |\nabla v|)^2] \\
&\geq \frac{1}{2} \sum_i F^{ii} W_{11}^2 - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1) W_{11} - \frac{C}{\sqrt{\eta}} (\sum_i F^{ii} + 1).
\end{aligned} \tag{3.21}$$

For the term II , we use the condition on the cut-off function η and the inequality (2.7) in **Proposition 2.4** to obtain

$$II = F^{ii} \left(\frac{2\eta_{ii}}{\eta} - \frac{6\eta_{ii}^2}{\eta^2} \right) \geq -C \sum_i F^{ii}. \tag{3.22}$$

Now multiplying $\eta^4 (W_{11} + \frac{|\nabla v|^2}{2})$ on the both sides of (3.9) and using the expressions of I and II , we derive that

$$\begin{aligned}
0 &\geq \eta^4 I + \eta^4 (W_{11} + \frac{|\nabla v|^2}{2}) II \\
&\geq \frac{1}{2} \eta^4 \sum_i F^{ii} W_{11}^2 - C \eta^{\frac{7}{2}} (\sum_i F^{ii} + 1) W_{11} - C \eta^{\frac{7}{2}} (\sum_i F^{ii} + 1) - C \eta^4 (W_{11} + C) \sum_i F^{ii} \\
&\geq \frac{1}{4} \sum_i F^{ii} (\eta^2 W_{11})^2 - C (\sum_i F^{ii} + 1).
\end{aligned} \tag{3.23}$$

That is

$$C \geq \sum_i F^{ii} [(\eta^2 W_{11})^2 - C]. \tag{3.24}$$

Noting that $\sum_i F^{ii} \geq 1$ by inequality (2.7) in **Proposition 2.4**, we deduce that $\eta^2 W_{11} \leq C$ and then

$G \leq C$. Therefore, $|\nabla^2 v|$ and $|\nabla^2 u|$ is bounded. We thus complete the proof of **Theorem 1.1**.

4. Proof of Theorem 1.3

In this section, we turn to prove the existence and uniqueness of the smooth admissible solutions to equations (1.1) and (1.2).

By setting $v = \log u$, we can normalize the equations (1.1) as in section 3

$$\left(\frac{\sigma_k}{C_k^n} \right)^{\frac{1}{k}} (v_{ij} + v_i v_j + \frac{(1 - |\nabla v|^2) \delta_{ij}}{2}) = (C_n^k)^{-\frac{1}{k}} \frac{(1 + |\nabla v|^2)}{2} f^{\frac{1}{k}}, \tag{4.1}$$

which can also be written in the form

$$F(W_{ij}) = g(x, \nabla v). \tag{4.2}$$

We first get $C^{2,\alpha}$ estimates. By using the logarithmic gradient estimate in [7], we can get the Harnack estimates by the path integral

$$\sup_{x \in S^n} v(x) \leq C \inf_{x \in S^n} v(x). \tag{4.3}$$

Since the equation (1.1) is homogeneous, without loss of generality, we may assume the solution v satisfies $\inf_{x \in S^n} v(x) = 1$. Hence we know that $\sup_{x \in S^n} v(x) \leq C$. Since we can get gradient estimate by

interpolation, we obtain $\|v(x)\|_{C^2(S^n)} \leq C$. Since the operator F is concave with respect to the variable

D^2v which is by (2.8) in **Proposition 2.4**, we know the Hölder norms of second order derivatives of solutions are bounded by the C^2 norm of solutions by the Evans-Krylov estimates (see Gilbarg-Trudinger [8]). Hence we know that

$$\|v(x)\|_{C^{2+\alpha}} \leq C \quad \text{for some } 0 < \alpha < 1. \quad (4.4)$$

4.1. Existence of admissible solutions

For $0 \leq t \leq 1$, consider the family of equations

$$F(W_{ij}) = \frac{1-t}{2} + t(C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}}. \quad (4.5)$$

Let us define a set

$$\Theta = \{t \mid 0 \leq t \leq 1, \text{ the equation (4.5) has positive admissible solution}\}$$

Obviously, $u \equiv 1$ is the solution of equation (4.5) when $t = 0$. Therefore the set $\Theta \neq \emptyset$.

We derive the linearized operator of the Equation (4.5) at v_0 . Consider the equation

$$G(x, v, \nabla v, \nabla^2 v) = \sigma_k^{-\frac{1}{k}}(W_{ij}) - \frac{1-t}{2} - t(C_n^k)^{-\frac{1}{k}} \frac{(1+|\nabla v|^2)}{2} f^{\frac{1}{k}}. \quad (4.6)$$

If we set $v = v_0 + sW$, then we see that v is admissible when s is small enough.

Taking derivative with respect to s , we derive the linearized operator of equation (4.5) is:

$$F^{ij}W_{ij} + 2F^{ij}v_iW_j - F^{ii}v_sW_s - t(C_n^k)^{-\frac{1}{k}}v_sW_s f^{\frac{1}{k}} = 0. \quad (4.7)$$

Since the linearized operator is surjective, the openness of Θ comes from the implicit function Theorem, Schauder theory and the Fredholm alternative [8]. The set Θ is also closed since we have established $C^{2,\alpha}$ estimates (4.4) of the admissible solution to equation (4.5). By using the smoothness of $f(x)$, we easily derive that the admissible solution is smooth by using the bootstrap argument.

So we get the existence part of **Theorem 1.3**.

4.2. Uniqueness of admissible solution

As the equation (1.1) and (4.1) are equivalent, we prove the uniqueness of equation (1.1) up to multiplications of constants. We prove this fact by using a contradiction argument. Suppose u_1 and u_2 are two smooth admissible solutions to equation (1.1) and (1.2). Suppose $u_2 > u_1$ somewhere on $\mathbb{S}^n$.

We can take $t \geq 1$, such that

$$tu_1 \geq u_2 \text{ on } \mathbb{S}^n; \quad tu_1 = u_2 \text{ at some point } P \in \mathbb{S}^n. \quad (4.8)$$

From the homogeneity of the equation, we get

$$F(tu_1) = F\left(\frac{2uu_{ij} + (u^2 - |\nabla u|^2)\delta_{ij}}{u^2 + |\nabla u|^2}\right)\bigg|_{u=tu_1} = F(u_1) = f(x) = F(u_2). \quad (4.9)$$

Hence

$$\tilde{L}(tu_1 - u_2) = 0 \quad (4.10)$$

Here, $\tilde{L}$ is the linearized operator of equation (1.1) and (1.2), that is

$$\begin{aligned} \tilde{L}(W) = & \frac{F^{ij}}{(u^2 + |\nabla u|^2)^2} \{ (u^2 + |\nabla u|^2) [2u_{ij}W + 2uW_{ij} + (2uW - 2u_s W_s) \delta_{ij}] \\ & - [2uu_{ij} + (u^2 - |\nabla u|^2) \delta_{ij}] (2uW + 2u_s W_s) \}, \end{aligned} \quad (4.11)$$

where u takes some admissible function between tu_1 and u_2 and $W = tu_1 - u_2$. The strong maximum principle implies that $tu_1 - u_2 \equiv 0$ on $\mathbb{S}^n$. We hence obtain the uniqueness and finish the proof of

Theorem 1.3.

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